IDEA StatiCa Connection Reports

26016 Concourse Viewing Platform Date: 13.11.2026

The steel connections at radius 8.5 m were verified with IDEA StatiCa Connection 26.0 (CBFEM) for the ULS combinations of this report and were found to be satisfactory.

Joint	Location	Plates (strain %, limit 5)	Bolts %	Welds %	Status	Page
L1-R8-A_000	Level 1, radius 8.5 m, 0°	0.0	13.6	73.3	OK	2
L1-R8-A_080	Level 1, radius 8.5 m, 80°	0.0	13.4	62.5	OK	13
L1-R8-A_120	Level 1, radius 8.5 m, 120°	0.0	13.0	74.7	OK	24
L1-R8-A_160	Level 1, radius 8.5 m, 160°	0.0	13.7	76.5	OK	35
L1-R8-A_200	Level 1, radius 8.5 m, 200°	0.0	13.3	64.2	OK	46
L1-R8-A_240	Level 1, radius 8.5 m, 240°	0.0	13.1	72.0	OK	57
L1-R8-A_320	Level 1, radius 8.5 m, 320°	0.0	11.7	63.8	OK	68
L1-R8-B_040	Level 1, radius 8.5 m, 40°	0.2	18.1	73.9	OK	79
L1-R8-B_280	Level 1, radius 8.5 m, 280°	0.6	17.8	93.8	OK	91
L1-R8-S_020	Level 1, radius 8.5 m, 20°	0.4	29.2	98.2	OK	103
L1-R8-S_100	Level 1, radius 8.5 m, 100°	0.2	28.6	98.2	OK	118
L1-R8-S_140	Level 1, radius 8.5 m, 140°	0.1	22.3	98.2	OK	133
L1-R8-S_180	Level 1, radius 8.5 m, 180°	0.2	28.5	98.3	OK	148
L1-R8-S_220	Level 1, radius 8.5 m, 220°	0.1	21.8	98.1	OK	163
L1-R8-S_300	Level 1, radius 8.5 m, 300°	0.2	25.7	98.2	OK	178
L1-R8-S_340	Level 1, radius 8.5 m, 340°	0.1	26.9	98.1	OK	193
L1-R8-T_060	Level 1, radius 8.5 m, 60°	1.1	23.4	98.9	OK	208
L1-R8-T_260	Level 1, radius 8.5 m, 260°	1.2	29.3	99.0	OK	224
L2-R8-A_000	Level 2, radius 8.5 m, 0°	0.0	0.9	18.0	OK	240
L2-R8-A_020	Level 2, radius 8.5 m, 20°	0.0	0.6	17.5	OK	253
L2-R8-A_080	Level 2, radius 8.5 m, 80°	0.0	1.3	21.1	OK	266
L2-R8-A_100	Level 2, radius 8.5 m, 100°	0.0	0.9	20.2	OK	279
L2-R8-A_120	Level 2, radius 8.5 m, 120°	0.0	0.9	20.2	OK	292
L2-R8-A_140	Level 2, radius 8.5 m, 140°	0.0	0.9	20.5	OK	305
L2-R8-A_160	Level 2, radius 8.5 m, 160°	0.0	0.9	20.1	OK	318
L2-R8-A_180	Level 2, radius 8.5 m, 180°	0.0	0.9	19.8	OK	331
L2-R8-A_200	Level 2, radius 8.5 m, 200°	0.0	0.9	16.7	OK	344
L2-R8-A_220	Level 2, radius 8.5 m, 220°	0.0	0.9	16.7	OK	357
L2-R8-A_240	Level 2, radius 8.5 m, 240°	0.0	1.5	19.2	OK	370
L2-R8-A_300	Level 2, radius 8.5 m, 300°	0.0	0.6	18.2	OK	383
L2-R8-A_320	Level 2, radius 8.5 m, 320°	0.0	0.9	17.8	OK	396
L2-R8-A_340	Level 2, radius 8.5 m, 340°	0.0	0.9	17.9	OK	409
L2-R8-B_040	Level 2, radius 8.5 m, 40°	0.0	8.7	37.0	OK	422
L2-R8-B_060	Level 2, radius 8.5 m, 60°	0.0	5.6	55.4	OK	438
L2-R8-B_260	Level 2, radius 8.5 m, 260°	0.0	4.9	46.1	OK	453
L2-R8-B_280	Level 2, radius 8.5 m, 280°	0.0	8.0	36.2	OK	468

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring

Name	SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

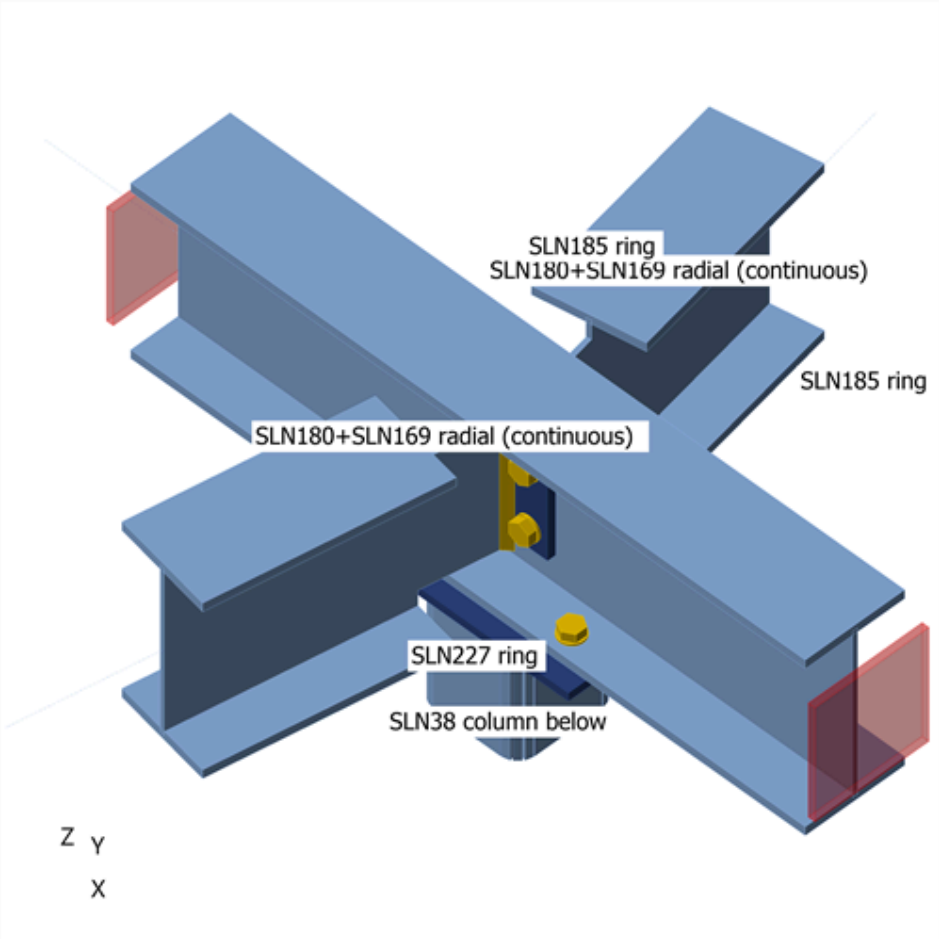
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN180+SLN169 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN38 column below	2 - SHS150x150x10	0.0	90.0	180.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN180+SLN169 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN180+SLN169 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN38 column below / begin		Position	0
SLN185 ring / end		Position	0
SLN227 ring / end		Position	0



SLN180+SLN169 radial (continuous) - SLN38 column below - SLN185 ring - SLN227 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.3	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	18.9	-0.2	-25.8	0.0	0.0	0.0
C LC_103	SLN38 column below / Begin	133.1	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	7.8	0.5	-24.0	0.0	0.0	0.0
	SLN227 ring / End	7.5	-0.1	-24.2	0.0	0.0	0.0
C LC_104	SLN38 column below / Begin	132.3	0.0	-5.0	0.0	-5.9	0.0
	SLN185 ring / End	21.3	0.6	-24.0	0.0	0.0	0.0
	SLN227 ring / End	21.0	-0.1	-24.2	0.0	0.0	0.0
C LC_105	SLN38 column below / Begin	132.9	3.5	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	11.8	0.4	-24.0	0.0	0.0	0.0
	SLN227 ring / End	7.8	-0.2	-24.2	0.0	0.0	0.0
C LC_106	SLN38 column below / Begin	133.1	-3.5	-0.8	0.0	-3.1	0.0
	SLN185 ring / End	8.5	0.5	-24.0	0.0	0.0	0.0
	SLN227 ring / End	11.7	-0.1	-24.2	0.0	0.0	0.0
C LC_107	SLN38 column below / Begin	4.0	0.0	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-10.3	0.2	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-10.3	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN38 column below / Begin	2.3	0.0	-8.4	0.0	-5.8	0.0
	SLN185 ring / End	12.5	0.1	-0.5	0.0	0.0	0.0
	SLN227 ring / End	12.5	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN38 column below / Begin	3.6	7.0	0.0	0.0	-0.2	0.0
	SLN185 ring / End	-4.7	-0.1	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-11.6	-0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN38 column below / Begin	3.9	-7.0	0.0	0.0	-0.1	0.0
	SLN185 ring / End	-11.0	0.0	-0.5	0.0	0.0	0.0
	SLN227 ring / End	-4.2	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN38 column below / Begin	90.6	0.0	-0.6	0.0	-2.3	0.0
	SLN185 ring / End	-2.3	0.5	-16.2	0.0	0.0	0.0
	SLN227 ring / End	-2.5	-0.1	-16.3	0.0	0.0	0.0
C LC_112	SLN38 column below / Begin	89.0	0.0	-8.9	0.0	-7.8	0.0
	SLN185 ring / End	21.8	0.4	-16.2	0.0	0.0	0.0
	SLN227 ring / End	21.7	0.0	-16.3	0.0	0.0	0.0
C LC_113	SLN38 column below / Begin	90.2	7.0	-0.5	0.0	-2.2	0.0
	SLN185 ring / End	4.0	0.2	-16.2	0.0	0.0	0.0
	SLN227 ring / End	-3.2	-0.2	-16.3	0.0	0.0	0.0
C LC_114	SLN38 column below / Begin	90.5	-7.0	-0.5	0.0	-2.1	0.0
	SLN185 ring / End	-2.4	0.3	-16.2	0.0	0.0	0.0
	SLN227 ring / End	4.3	0.0	-16.3	0.0	0.0	0.0
C LC_115	SLN38 column below / Begin	11.3	0.0	-0.1	0.0	-0.4	0.0
	SLN185 ring / End	-8.7	0.2	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-8.7	0.0	-1.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN38 column below / Begin	9.6	0.0	-8.5	0.0	-5.9	0.0
	SLN185 ring / End	14.2	0.1	-1.7	0.0	0.0	0.0
	SLN227 ring / End	14.2	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN38 column below / Begin	10.9	7.0	-0.1	0.0	-0.3	0.0
	SLN185 ring / End	-3.1	0.0	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-9.9	-0.1	-1.8	0.0	0.0	0.0
C LC_118	SLN38 column below / Begin	11.2	-7.0	0.0	0.0	-0.2	0.0
	SLN185 ring / End	-9.3	0.0	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-2.6	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN38 column below / Begin	141.6	0.0	-0.8	0.0	-3.4	0.0
	SLN185 ring / End	10.9	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	10.5	-0.1	-25.8	0.0	0.0	0.0
C LC_120	SLN38 column below / Begin	141.2	0.0	-5.0	0.0	-6.1	0.0
	SLN185 ring / End	24.5	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	24.2	-0.2	-25.8	0.0	0.0	0.0
C LC_121	SLN38 column below / Begin	141.7	3.5	-0.8	0.0	-3.3	0.0
	SLN185 ring / End	14.9	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	10.9	-0.2	-25.8	0.0	0.0	0.0
C LC_122	SLN38 column below / Begin	141.8	-3.5	-0.8	0.0	-3.3	0.0
	SLN185 ring / End	11.5	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	14.8	-0.1	-25.8	0.0	0.0	0.0
C LC_123	SLN38 column below / Begin	107.7	0.0	-0.7	0.0	-2.7	0.0
	SLN185 ring / End	3.7	0.5	-19.4	0.0	0.0	0.0
	SLN227 ring / End	3.5	-0.1	-19.5	0.0	0.0	0.0
C LC_124	SLN38 column below / Begin	106.8	0.0	-9.0	0.0	-8.2	0.0
	SLN185 ring / End	28.2	0.6	-19.4	0.0	0.0	0.0
	SLN227 ring / End	28.0	-0.1	-19.5	0.0	0.0	0.0
C LC_125	SLN38 column below / Begin	107.7	7.0	-0.6	0.0	-2.6	0.0
	SLN185 ring / End	10.2	0.3	-19.4	0.0	0.0	0.0
	SLN227 ring / End	2.9	-0.2	-19.5	0.0	0.0	0.0
C LC_126	SLN38 column below / Begin	108.0	-7.0	-0.6	0.0	-2.5	0.0
	SLN185 ring / End	3.8	0.3	-19.4	0.0	0.0	0.0
	SLN227 ring / End	10.4	0.0	-19.5	0.0	0.0	0.0
C LC_127	SLN38 column below / Begin	28.3	0.0	-0.2	0.0	-0.8	0.0
	SLN185 ring / End	-2.7	0.3	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-2.7	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN38 column below / Begin	27.5	0.0	-8.6	0.0	-6.3	0.0
	SLN185 ring / End	20.5	0.3	-4.9	0.0	0.0	0.0
	SLN227 ring / End	20.6	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN38 column below / Begin	28.4	7.0	-0.2	0.0	-0.7	0.0
	SLN185 ring / End	3.1	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-3.9	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN38 column below / Begin	28.7	-7.0	-0.1	0.0	-0.6	0.0
	SLN185 ring / End	-3.2	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	3.6	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN38 column below / Begin	141.7	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	16.6	0.6	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN227 ring / End	16.2	-0.2	-25.8	0.0	0.0	0.0
C LC_141	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.1	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.1	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.6	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.5	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.2	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN38 column below / Begin	28.3	0.0	-0.1	0.0	-0.5	0.0
	SLN185 ring / End	6.4	0.1	-4.9	0.0	0.0	0.0
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	18.2	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	17.8	-0.1	-25.8	0.0	0.0	0.0
C LC_146	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	20.4	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	20.1	-0.2	-25.8	0.0	0.0	0.0
C LC_147	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.4	0.6	-25.6	0.0	0.0	0.0
	SLN227 ring / End	18.8	-0.2	-25.8	0.0	0.0	0.0
C LC_148	SLN38 column below / Begin	141.8	0.0	-0.8	0.0	-3.2	0.0
	SLN185 ring / End	19.2	0.5	-25.6	0.0	0.0	0.0
	SLN227 ring / End	19.1	-0.2	-25.8	0.0	0.0	0.0
C LC_151	SLN38 column below / Begin	91.2	0.0	-1.7	0.0	-7.0	0.0
	SLN185 ring / End	6.6	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.4	-0.1	-11.5	0.0	0.0	0.0
C LC_152	SLN38 column below / Begin	79.2	0.0	0.8	0.0	3.3	0.0
	SLN185 ring / End	21.0	0.5	-14.2	0.0	0.0	0.0
	SLN227 ring / End	20.7	-0.1	-14.2	0.0	0.0	0.0
C LC_153	SLN38 column below / Begin	141.4	0.0	-0.8	0.0	-3.4	0.0
	SLN185 ring / End	19.3	0.6	-21.0	0.0	0.0	0.0
	SLN227 ring / End	19.0	-0.2	-20.7	0.0	0.0	0.0
C LC_154	SLN38 column below / Begin	90.5	0.0	-1.7	0.0	-6.9	0.0
	SLN185 ring / End	6.7	0.4	-11.7	0.0	0.0	0.0
	SLN227 ring / End	6.5	-0.1	-11.5	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.6 < 100%	OK
Welds	73.3 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN180+SLN169 radial (continuous)-bfl 1	12.7	C LC_120	149.8	0.0	45.1	OK
SLN180+SLN169 radial (continuous)-tfl 1	12.7	C LC_153	72.2	0.0	0.0	OK
SLN180+SLN169 radial (continuous)-w 1	7.2	C LC_125	243.2	0.0	29.8	OK
SLN38 column below	10.0	C LC_120	168.8	0.0	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_120	28.8	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_135	23.5	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_120	166.7	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_120	27.4	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_121	21.7	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_120	123.6	0.0	0.0	OK
End plate column below SLN38 column below	12.0	C LC_108	135.7	0.0	17.5	OK
End plate SLN227 ring	12.0	C LC_109	35.7	0.0	5.9	OK
End plate SLN185 ring	12.0	C LC_125	35.5	0.0	5.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

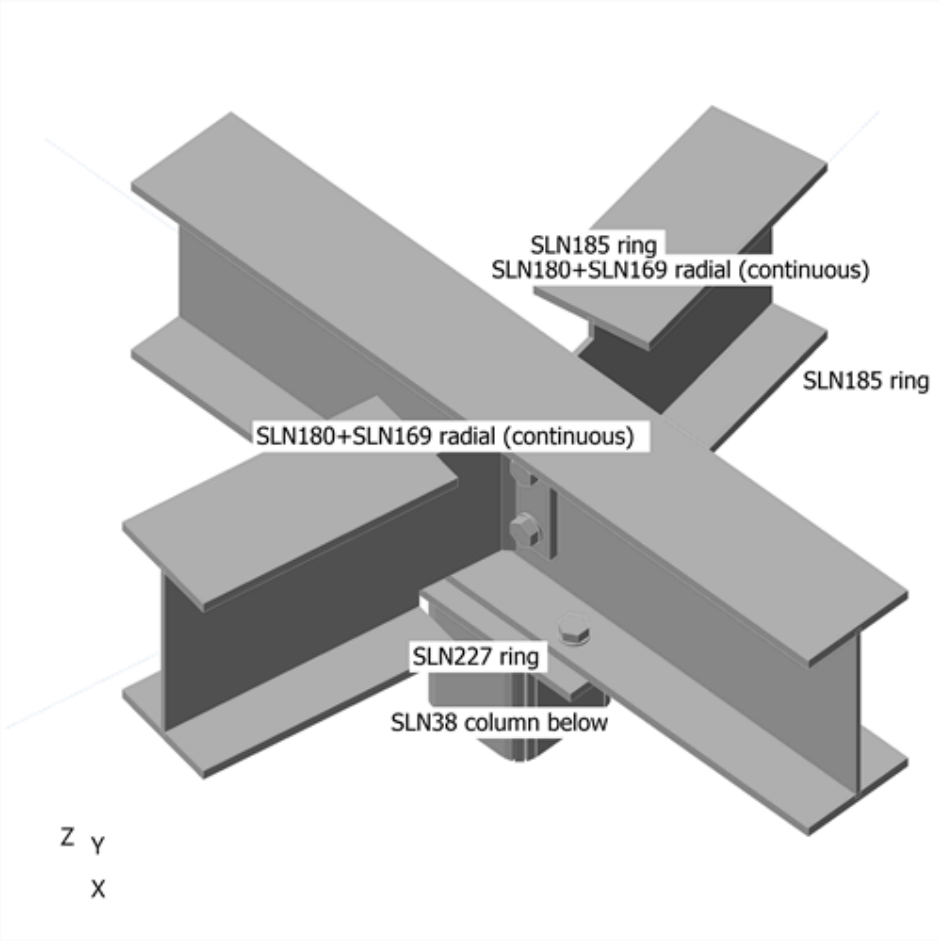
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

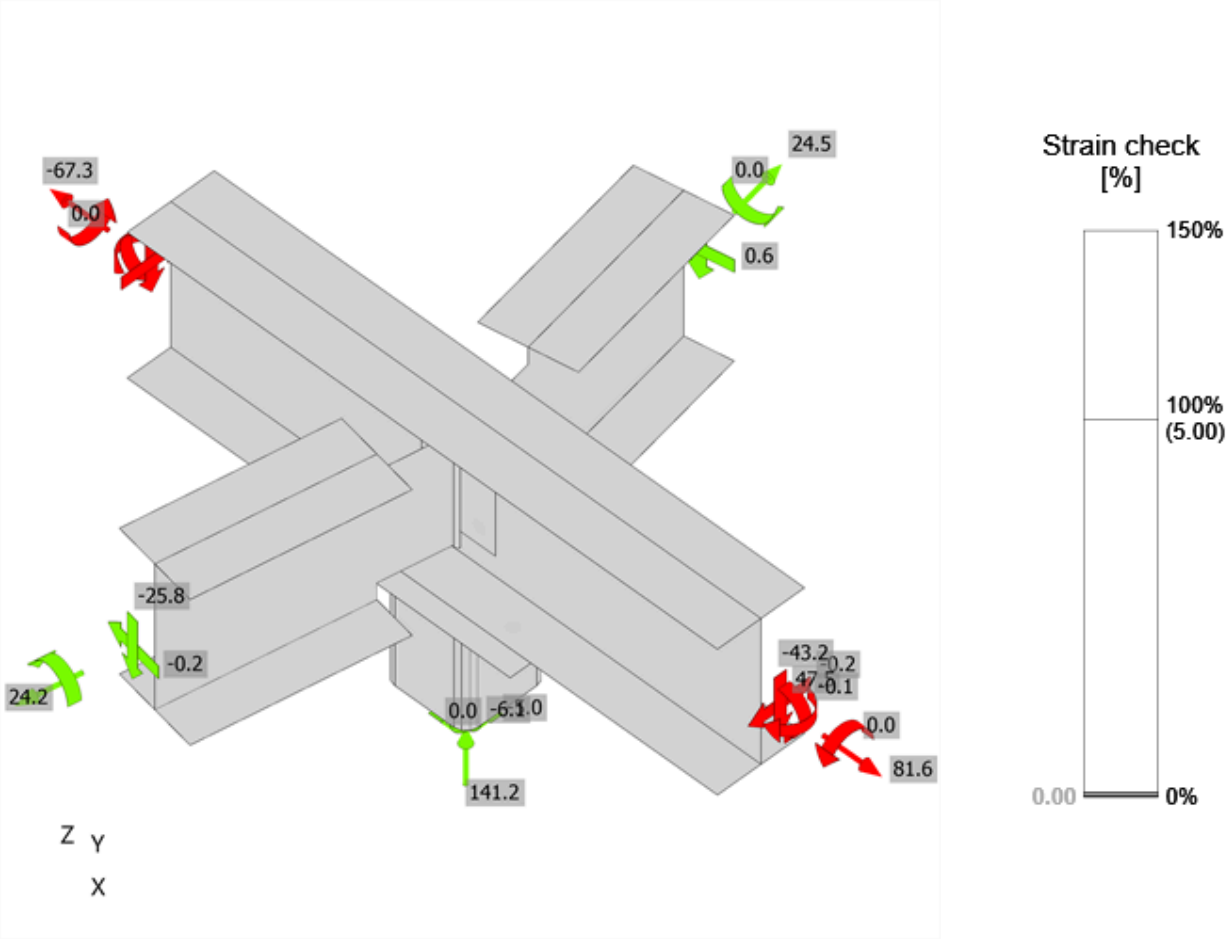
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN38 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

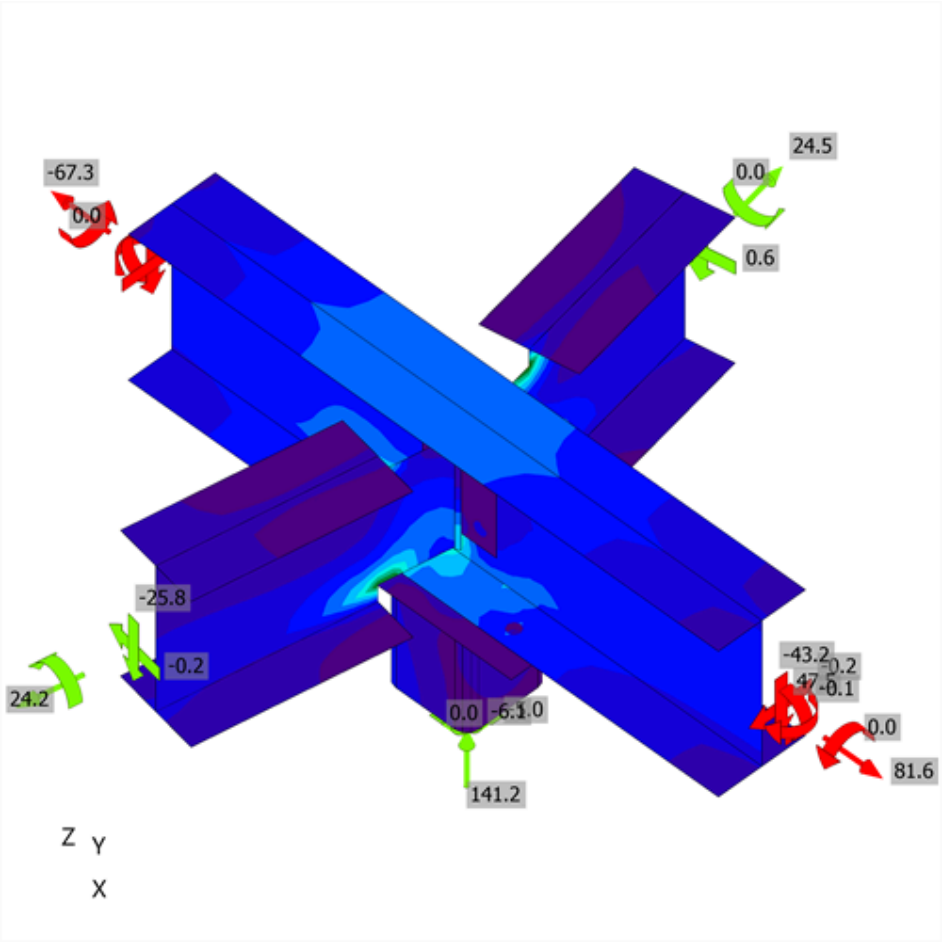
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

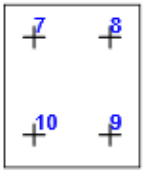


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_124	0.4	4.9	173.9	0.3	5.2	5.4	OK	OK
	B3	M20 8.8 - 1	C LC_108	19.2	2.5	98.1	13.6	2.7	12.4	OK	OK
	B4	M20 8.8 - 1	C LC_108	19.2	2.5	98.0	13.6	2.7	12.4	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.4	4.8	173.9	0.3	5.1	5.4	OK	OK
	B7	M20 8.8 - 1	C LC_124	0.2	0.3	123.8	0.1	0.5	0.4	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.0	0.7	0.5	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.9	0.5	123.8	1.3	0.5	1.5	OK	OK
	B10	M20 8.8 - 1	C LC_125	1.8	0.8	123.8	1.3	0.9	1.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN38 column below	SLN38 column below	◢ 4.2 ◣	521	C LC_120	272.8	0.0	-134.9	-136.9	-0.4	67.4	37.2	OK	OK
		◢ 4.2 ◣	521	C LC_120	296.7	0.0	-149.8	147.9	0.4	73.3	41.1	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_125	97.7	0.0	5.8	7.9	-55.7	24.1	14.7	OK	OK
SLN180+SLN169 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_122	94.9	0.0	4.6	3.2	-54.6	23.5	13.9	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_122	82.6	0.0	-22.0	-17.9	42.3	20.4	12.5	OK	OK
		◢ 4.2 ◣	149	C LC_122	98.2	0.0	-18.3	24.5	-50.0	24.3	15.2	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_121	100.0	0.0	-18.6	-22.9	51.9	24.7	15.8	OK	OK
		◢ 4.2 ◣	149	C LC_121	81.9	0.0	-22.3	19.8	-41.0	20.2	11.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_080
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring

Name	SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

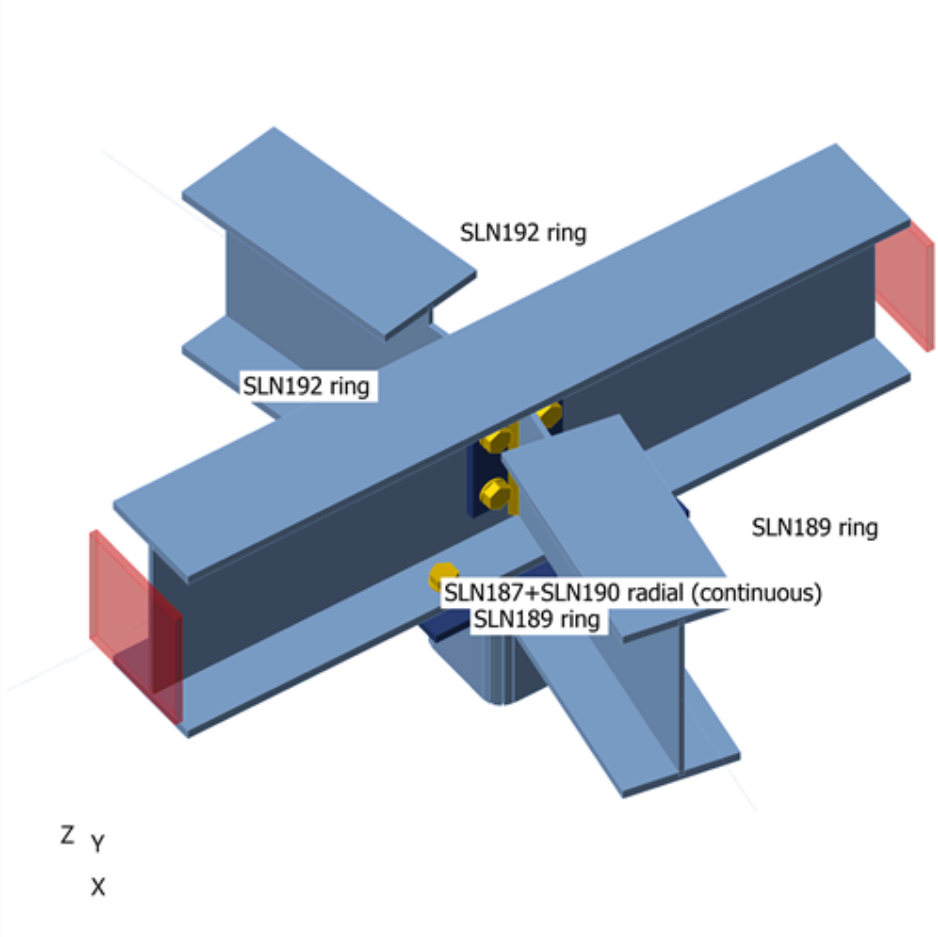
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN187+SLN190 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN42 column below	2 - SHS150x150x10	0.0	90.0	-100.0	0	0	0
SLN189 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN187+SLN190 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN187+SLN190 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN42 column below / begin		Position	0
SLN189 ring / end		Position	0
SLN192 ring / end		Position	0



SLN187+SLN190 radial (continuous) - SLN42 column below - SLN189 ring - SLN192 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN42 column below / Begin	133.1	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	28.0	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN42 column below / Begin	124.6	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	17.5	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	17.3	0.1	-24.0	0.0	0.0	0.0
C LC_104	SLN42 column below / Begin	124.4	-3.4	-1.1	0.0	-1.9	0.0
	SLN189 ring / End	21.2	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	17.1	0.2	-24.0	0.0	0.0	0.0
C LC_105	SLN42 column below / Begin	123.6	0.6	-4.5	0.0	-4.1	0.0
	SLN189 ring / End	30.1	-0.2	-24.2	0.0	0.0	0.0
	SLN192 ring / End	30.3	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN42 column below / Begin	124.9	0.0	-0.4	0.0	-1.6	0.0
	SLN189 ring / End	14.8	-0.1	-24.2	0.0	0.0	0.0
	SLN192 ring / End	14.5	0.2	-24.0	0.0	0.0	0.0
C LC_107	SLN42 column below / Begin	3.2	0.0	0.0	0.0	0.0	0.0
	SLN189 ring / End	-7.7	0.0	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-7.9	-0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN42 column below / Begin	2.8	-6.9	-1.5	0.0	-1.0	0.0
	SLN189 ring / End	-1.6	0.1	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-8.6	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN42 column below / Begin	1.3	1.2	-8.2	0.0	-5.4	0.0
	SLN189 ring / End	14.8	0.0	-0.5	0.0	0.0	0.0
	SLN192 ring / End	15.8	-0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN42 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	-11.6	-0.1	-0.5	0.0	0.0	0.0
	SLN192 ring / End	-11.4	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN42 column below / Begin	84.5	0.0	-0.2	0.0	-1.0	0.0
	SLN189 ring / End	5.6	0.0	-16.4	0.0	0.0	0.0
	SLN192 ring / End	5.4	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN42 column below / Begin	84.1	-6.9	-1.7	0.0	-1.9	0.0
	SLN189 ring / End	12.2	0.0	-16.4	0.0	0.0	0.0
	SLN192 ring / End	4.8	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN42 column below / Begin	82.5	1.2	-8.5	0.0	-6.4	0.0
	SLN189 ring / End	29.0	-0.1	-16.4	0.0	0.0	0.0
	SLN192 ring / End	29.8	0.0	-16.2	0.0	0.0	0.0
C LC_114	SLN42 column below / Begin	85.1	0.0	-0.3	0.0	-1.2	0.0
	SLN189 ring / End	1.3	-0.1	-16.4	0.0	0.0	0.0
	SLN192 ring / End	1.2	0.2	-16.2	0.0	0.0	0.0
C LC_115	SLN42 column below / Begin	10.1	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	-5.5	0.0	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-5.6	0.0	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN42 column below / Begin	9.7	-6.9	-1.5	0.0	-1.0	0.0
	SLN189 ring / End	0.7	0.1	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-6.3	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN42 column below / Begin	8.1	1.2	-8.2	0.0	-5.5	0.0
	SLN189 ring / End	17.1	0.0	-1.8	0.0	0.0	0.0
	SLN192 ring / End	18.1	-0.1	-1.7	0.0	0.0	0.0
C LC_118	SLN42 column below / Begin	10.7	0.0	-0.1	0.0	-0.3	0.0
	SLN189 ring / End	-9.3	-0.1	-1.8	0.0	0.0	0.0
	SLN192 ring / End	-9.2	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN42 column below / Begin	132.9	0.0	-0.4	0.0	-1.5	0.0
	SLN189 ring / End	20.9	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	20.7	0.2	-25.6	0.0	0.0	0.0
C LC_120	SLN42 column below / Begin	132.7	-3.4	-1.1	0.0	-2.0	0.0
	SLN189 ring / End	24.7	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	20.5	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN42 column below / Begin	132.1	0.6	-4.5	0.0	-4.2	0.0
	SLN189 ring / End	33.8	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	34.0	0.2	-25.6	0.0	0.0	0.0
C LC_122	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.6	0.0
	SLN189 ring / End	18.1	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	17.8	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN42 column below / Begin	101.1	0.0	-0.3	0.0	-1.1	0.0
	SLN189 ring / End	12.4	0.0	-19.6	0.0	0.0	0.0
	SLN192 ring / End	12.2	0.1	-19.4	0.0	0.0	0.0
C LC_124	SLN42 column below / Begin	100.7	-6.9	-1.7	0.0	-2.1	0.0
	SLN189 ring / End	19.2	0.0	-19.6	0.0	0.0	0.0
	SLN192 ring / End	11.7	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN42 column below / Begin	99.5	1.2	-8.5	0.0	-6.6	0.0
	SLN189 ring / End	36.4	-0.2	-19.6	0.0	0.0	0.0
	SLN192 ring / End	37.2	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN42 column below / Begin	101.2	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	7.9	-0.1	-19.6	0.0	0.0	0.0
	SLN192 ring / End	7.7	0.2	-19.4	0.0	0.0	0.0
C LC_127	SLN42 column below / Begin	26.7	0.0	-0.1	0.0	-0.2	0.0
	SLN189 ring / End	1.3	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	1.2	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN42 column below / Begin	26.3	-6.9	-1.5	0.0	-1.2	0.0
	SLN189 ring / End	7.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	0.6	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN42 column below / Begin	25.1	1.2	-8.3	0.0	-5.7	0.0
	SLN189 ring / End	24.5	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	25.5	-0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN42 column below / Begin	26.8	0.0	-0.1	0.0	-0.4	0.0
	SLN189 ring / End	-2.8	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	-2.6	0.1	-4.9	0.0	0.0	0.0
C LC_135	SLN42 column below / Begin	133.0	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	28.4	-0.1	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.8	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.9	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	9.1	-0.1	-5.0	0.0	0.0	0.0
	SLN192 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN42 column below / Begin	26.6	0.0	0.0	0.0	-0.1	0.0
	SLN189 ring / End	8.6	0.0	-5.0	0.0	0.0	0.0
	SLN192 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	27.7	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_146	SLN42 column below / Begin	133.0	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	28.3	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN42 column below / Begin	133.0	0.0	-0.3	0.0	-1.4	0.0
	SLN189 ring / End	29.1	-0.2	-25.8	0.0	0.0	0.0
	SLN192 ring / End	28.8	0.3	-25.6	0.0	0.0	0.0
C LC_148	SLN42 column below / Begin	133.1	0.0	-0.4	0.0	-1.4	0.0
	SLN189 ring / End	26.8	-0.1	-25.8	0.0	0.0	0.0
	SLN192 ring / End	26.4	0.1	-25.6	0.0	0.0	0.0
C LC_151	SLN42 column below / Begin	81.1	0.0	-1.3	0.0	-5.3	0.0
	SLN189 ring / End	12.6	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.3	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN42 column below / Begin	78.3	0.0	0.9	0.0	3.6	0.0
	SLN189 ring / End	28.5	-0.1	-14.1	0.0	0.0	0.0
	SLN192 ring / End	28.3	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN42 column below / Begin	133.4	0.0	-0.4	0.0	-1.7	0.0
	SLN189 ring / End	28.1	-0.1	-20.6	0.0	0.0	0.0
	SLN192 ring / End	27.7	0.2	-21.1	0.0	0.0	0.0
C LC_154	SLN42 column below / Begin	80.3	0.0	-1.3	0.0	-5.1	0.0
	SLN189 ring / End	12.7	-0.1	-11.4	0.0	0.0	0.0
	SLN192 ring / End	12.4	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.4 < 100%	OK
Welds	62.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN187+SLN190 radial (continuous)-bfl 1	12.7	C LC_121	128.1	0.0	38.2	OK
SLN187+SLN190 radial (continuous)-tfl 1	12.7	C LC_153	65.5	0.0	0.0	OK
SLN187+SLN190 radial (continuous)-w 1	7.2	C LC_124	230.9	0.0	26.8	OK
SLN42 column below	10.0	C LC_125	141.9	0.0	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_121	29.2	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_121	135.2	0.0	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_121	29.0	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_147	134.1	0.0	0.0	OK
End plate column below SLN42 column below	12.0	C LC_109	126.9	0.0	17.5	OK
End plate SLN189 ring	12.0	C LC_124	34.4	0.0	3.3	OK
End plate SLN192 ring	12.0	C LC_124	34.9	0.0	5.7	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

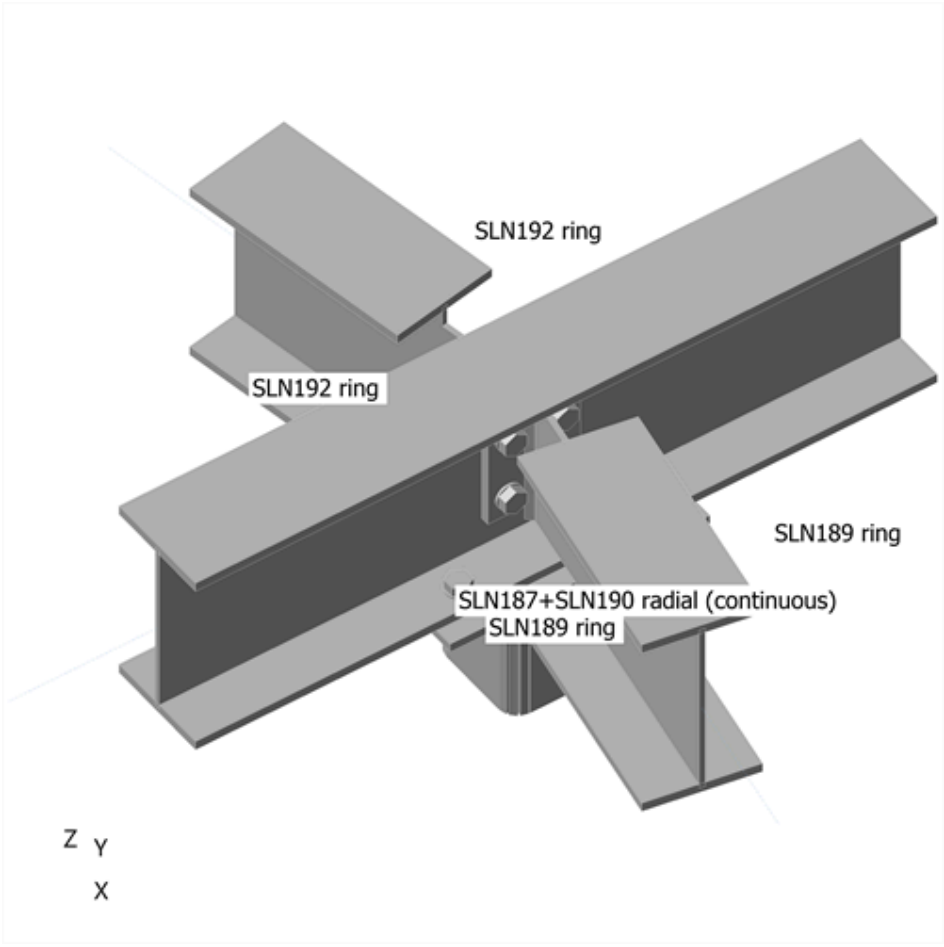
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

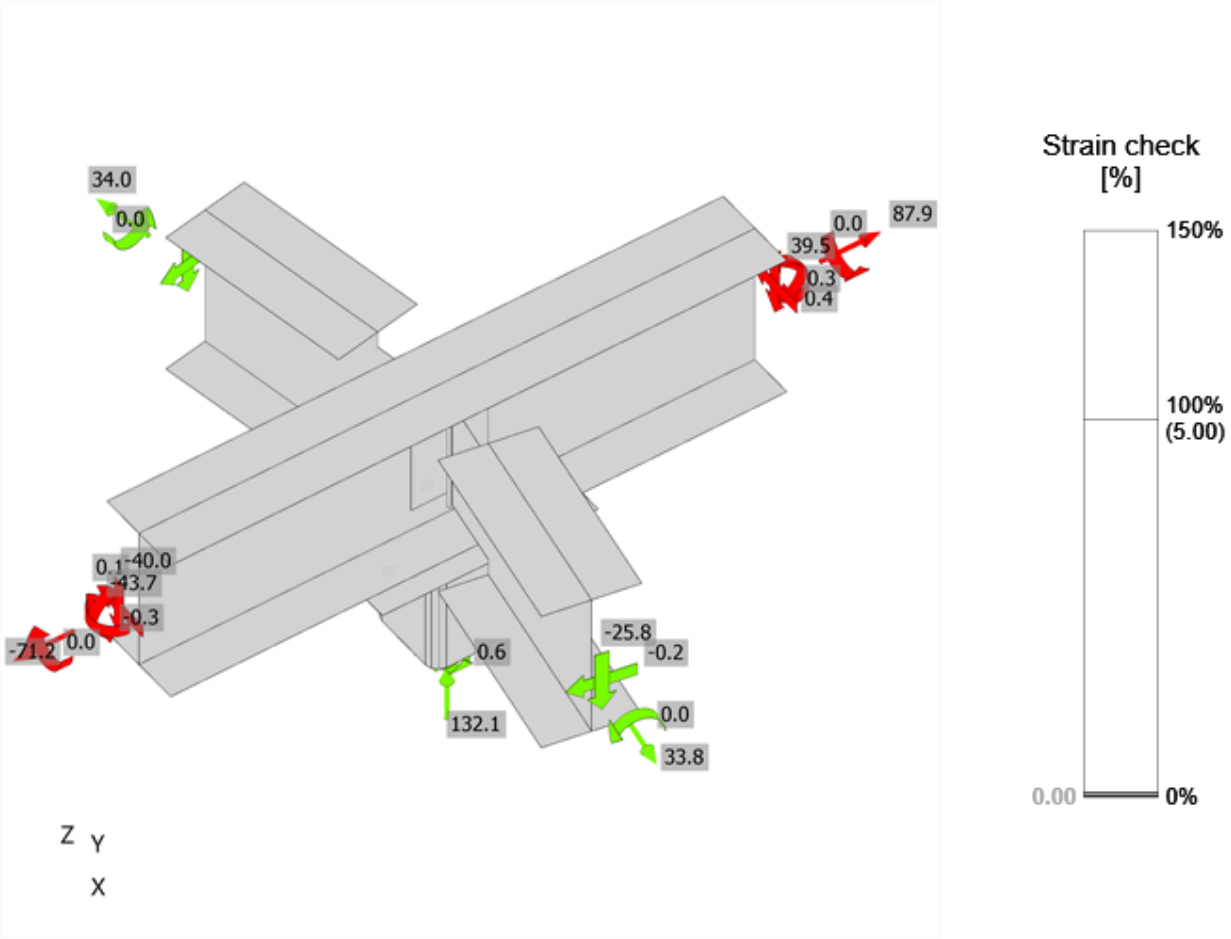
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN42 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

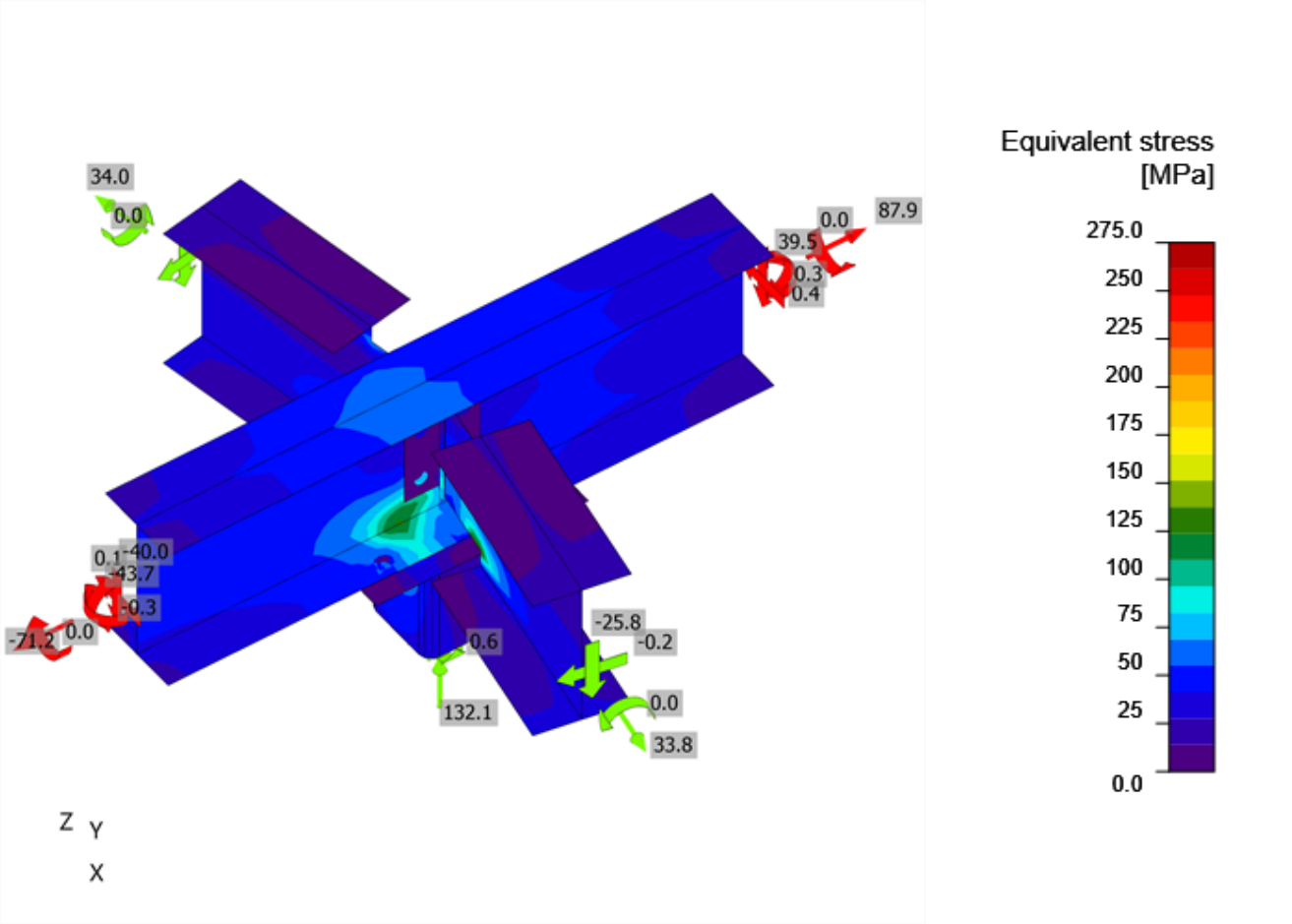
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

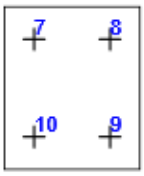


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_125	0.4	4.2	171.0	0.3	4.5	4.7	OK	OK
	B3	M20 8.8 - 1	C LC_109	18.9	2.7	91.6	13.4	2.9	12.4	OK	OK
	B4	M20 8.8 - 1	C LC_109	17.7	2.3	110.5	12.5	2.5	11.4	OK	OK
	B5	M20 8.8 - 1	C LC_125	0.4	4.6	176.2	0.3	4.9	5.1	OK	OK
	B7	M20 8.8 - 1	C LC_124	0.5	0.2	72.9	0.4	0.3	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_124	0.7	0.3	62.0	0.5	0.4	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_124	1.8	0.5	65.3	1.3	0.5	1.5	OK	OK
	B10	M20 8.8 - 1	C LC_124	1.7	0.7	123.8	1.2	0.8	1.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN42 column below	SLN42 column below	◢ 4.2 ◣	521	C LC_121	237.0	0.0	-115.7	-113.8	-35.9	58.6	32.1	OK	OK
		◢ 4.2 ◣	521	C LC_121	252.9	0.0	-121.3	123.5	34.3	62.5	34.7	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_120	94.2	0.0	10.8	-13.2	52.4	23.3	14.9	OK	OK
SLN187+SLN190 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_121	89.3	0.0	27.6	35.3	-34.0	22.1	14.8	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_120	79.4	0.0	-20.8	-17.4	40.7	19.6	12.1	OK	OK
		◢ 4.2 ◣	149	C LC_120	94.5	0.0	-18.6	23.6	-48.0	23.3	14.6	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_121	90.8	0.0	-17.7	-23.7	45.6	22.4	13.9	OK	OK
		◢ 4.2 ◣	149	C LC_121	76.8	0.0	-21.0	16.9	-39.1	19.0	11.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_120
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring

Name	SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

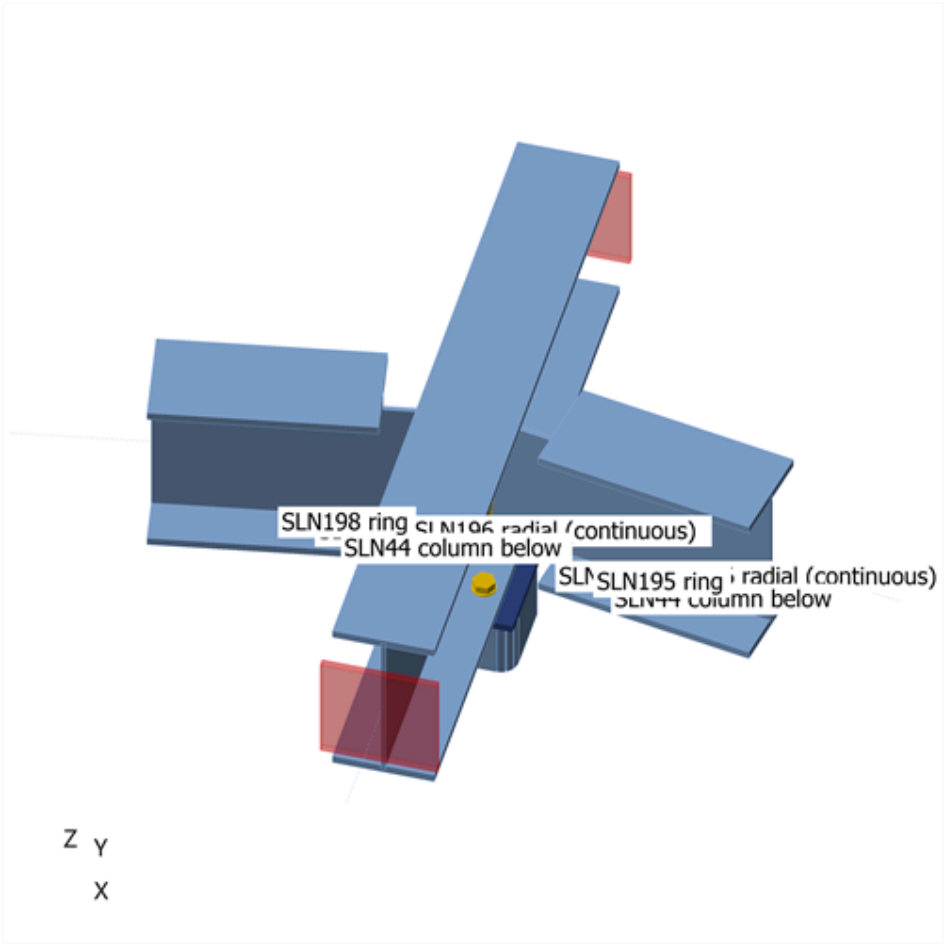
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN194+SLN196 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN44 column below	2 - SHS150x150x10	0.0	90.0	-60.0	0	0	0
SLN195 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN194+SLN196 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN194+SLN196 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN44 column below / begin		Position	0
SLN195 ring / end		Position	0
SLN198 ring / end		Position	0



SLN194+SLN196 radial (continuous) - SLN44 column below - SLN195 ring - SLN198 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.8	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	27.7	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.8	0.2	-26.0	0.0	0.0	0.0
C LC_103	SLN44 column below / Begin	132.3	3.0	-2.9	0.0	-4.9	0.0
	SLN195 ring / End	21.7	-0.2	-24.2	0.0	0.0	0.0
	SLN198 ring / End	25.0	0.2	-24.4	0.0	0.0	0.0
C LC_104	SLN44 column below / Begin	132.6	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	16.3	-0.1	-24.2	0.0	0.0	0.0
	SLN198 ring / End	16.2	0.2	-24.4	0.0	0.0	0.0
C LC_105	SLN44 column below / Begin	132.0	-1.8	-4.5	0.0	-5.8	0.0
	SLN195 ring / End	29.5	-0.2	-24.2	0.0	0.0	0.0
	SLN198 ring / End	27.7	0.2	-24.4	0.0	0.0	0.0
C LC_106	SLN44 column below / Begin	132.7	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	14.9	-0.1	-24.2	0.0	0.0	0.0
	SLN198 ring / End	15.2	0.2	-24.4	0.0	0.0	0.0
C LC_107	SLN44 column below / Begin	3.1	6.1	-4.2	0.0	-3.0	0.0
	SLN195 ring / End	-0.3	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	5.7	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN44 column below / Begin	3.7	0.0	-0.1	0.0	-0.2	0.0
	SLN195 ring / End	-9.3	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	-9.3	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN44 column below / Begin	2.3	-3.5	-7.3	0.0	-5.0	0.0
	SLN195 ring / End	14.2	0.0	-0.5	0.0	0.0	0.0
	SLN198 ring / End	10.6	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN44 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN195 ring / End	-11.0	-0.1	-0.5	0.0	0.0	0.0
	SLN198 ring / End	-10.9	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN44 column below / Begin	89.5	6.1	-4.8	0.0	-5.3	0.0
	SLN195 ring / End	13.4	-0.2	-16.3	0.0	0.0	0.0
	SLN198 ring / End	19.6	0.1	-16.5	0.0	0.0	0.0
C LC_112	SLN44 column below / Begin	90.1	0.0	-0.6	0.0	-2.5	0.0
	SLN195 ring / End	3.8	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	3.7	0.2	-16.5	0.0	0.0	0.0
C LC_113	SLN44 column below / Begin	88.7	-3.5	-7.8	0.0	-7.2	0.0
	SLN195 ring / End	28.2	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	24.6	0.1	-16.5	0.0	0.0	0.0
C LC_114	SLN44 column below / Begin	90.2	0.0	-0.6	0.0	-2.5	0.0
	SLN195 ring / End	1.7	-0.1	-16.3	0.0	0.0	0.0
	SLN198 ring / End	2.0	0.2	-16.5	0.0	0.0	0.0
C LC_115	SLN44 column below / Begin	10.4	6.1	-4.3	0.0	-3.2	0.0
	SLN195 ring / End	2.0	-0.1	-1.8	0.0	0.0	0.0
	SLN198 ring / End	8.0	0.0	-1.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN44 column below / Begin	11.0	0.0	-0.1	0.0	-0.4	0.0
	SLN195 ring / End	-7.0	0.0	-1.8	0.0	0.0	0.0
	SLN198 ring / End	-7.1	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN44 column below / Begin	9.6	-3.5	-7.3	0.0	-5.2	0.0
	SLN195 ring / End	16.5	0.0	-1.8	0.0	0.0	0.0
	SLN198 ring / End	12.9	0.0	-1.8	0.0	0.0	0.0
C LC_118	SLN44 column below / Begin	11.1	0.0	-0.1	0.0	-0.4	0.0
	SLN195 ring / End	-8.8	-0.1	-1.8	0.0	0.0	0.0
	SLN198 ring / End	-8.6	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN44 column below / Begin	141.2	3.0	-3.0	0.0	-5.1	0.0
	SLN195 ring / End	25.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.5	0.2	-26.0	0.0	0.0	0.0
C LC_120	SLN44 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	19.8	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	19.6	0.2	-26.0	0.0	0.0	0.0
C LC_121	SLN44 column below / Begin	140.9	-1.8	-4.5	0.0	-6.1	0.0
	SLN195 ring / End	33.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	31.3	0.3	-26.0	0.0	0.0	0.0
C LC_122	SLN44 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	18.2	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	18.5	0.2	-26.0	0.0	0.0	0.0
C LC_123	SLN44 column below / Begin	107.2	6.1	-4.9	0.0	-5.7	0.0
	SLN195 ring / End	20.4	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	26.7	0.2	-19.7	0.0	0.0	0.0
C LC_124	SLN44 column below / Begin	107.5	0.0	-0.7	0.0	-2.9	0.0
	SLN195 ring / End	10.7	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	10.5	0.2	-19.7	0.0	0.0	0.0
C LC_125	SLN44 column below / Begin	106.7	-3.5	-7.9	0.0	-7.7	0.0
	SLN195 ring / End	35.5	-0.2	-19.5	0.0	0.0	0.0
	SLN198 ring / End	31.8	0.2	-19.7	0.0	0.0	0.0
C LC_126	SLN44 column below / Begin	107.4	0.0	-0.7	0.0	-2.9	0.0
	SLN195 ring / End	8.3	-0.1	-19.5	0.0	0.0	0.0
	SLN198 ring / End	8.7	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN44 column below / Begin	28.1	6.1	-4.4	0.0	-3.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	15.1	0.1	-5.0	0.0	0.0	0.0
C LC_128	SLN44 column below / Begin	28.4	0.0	-0.2	0.0	-0.8	0.0
	SLN195 ring / End	-0.2	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	-0.3	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN44 column below / Begin	27.6	-3.5	-7.4	0.0	-5.6	0.0
	SLN195 ring / End	23.8	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	20.1	0.0	-5.0	0.0	0.0	0.0
C LC_130	SLN44 column below / Begin	28.3	0.0	-0.2	0.0	-0.8	0.0
	SLN195 ring / End	-2.2	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	-2.0	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	28.1	-0.2	-25.8	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / End	28.2	0.2	-26.0	0.0	0.0	0.0
C LC_141	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.9	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN198 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN44 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN195 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN198 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.5	0.3	-26.0	0.0	0.0	0.0
C LC_146	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	27.2	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	27.1	0.2	-26.0	0.0	0.0	0.0
C LC_147	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN195 ring / End	28.8	-0.2	-25.8	0.0	0.0	0.0
	SLN198 ring / End	28.7	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN44 column below / Begin	141.5	0.0	-0.9	0.0	-3.6	0.0
	SLN195 ring / End	26.6	-0.1	-25.8	0.0	0.0	0.0
	SLN198 ring / End	26.8	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN44 column below / Begin	90.2	0.0	-1.8	0.0	-7.2	0.0
	SLN195 ring / End	12.4	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.4	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN44 column below / Begin	78.6	0.0	0.7	0.0	3.0	0.0
	SLN195 ring / End	28.4	-0.2	-14.1	0.0	0.0	0.0
	SLN198 ring / End	28.4	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN44 column below / Begin	140.9	0.0	-0.9	0.0	-3.7	0.0
	SLN195 ring / End	27.8	-0.2	-21.1	0.0	0.0	0.0
	SLN198 ring / End	27.9	0.2	-20.9	0.0	0.0	0.0
C LC_154	SLN44 column below / Begin	89.6	0.0	-1.8	0.0	-7.1	0.0
	SLN195 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN198 ring / End	12.5	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.0 < 100%	OK
Welds	74.7 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN194+SLN196 radial (continuous)-bfl 1	12.7	C LC_121	149.0	0.0	44.7	OK
SLN194+SLN196 radial (continuous)-tfl 1	12.7	C LC_153	71.9	0.0	0.0	OK
SLN194+SLN196 radial (continuous)-w 1	7.2	C LC_123	251.8	0.0	36.5	OK
SLN44 column below	10.0	C LC_121	168.2	0.0	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_121	29.0	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_122	20.2	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_121	138.2	0.0	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_121	28.9	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_122	20.7	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_147	138.0	0.0	0.0	OK
End plate column below SLN44 column below	12.0	C LC_109	130.5	0.0	17.0	OK
End plate SLN195 ring	12.0	C LC_123	31.4	0.0	6.1	OK
End plate SLN198 ring	12.0	C LC_123	34.2	0.0	3.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

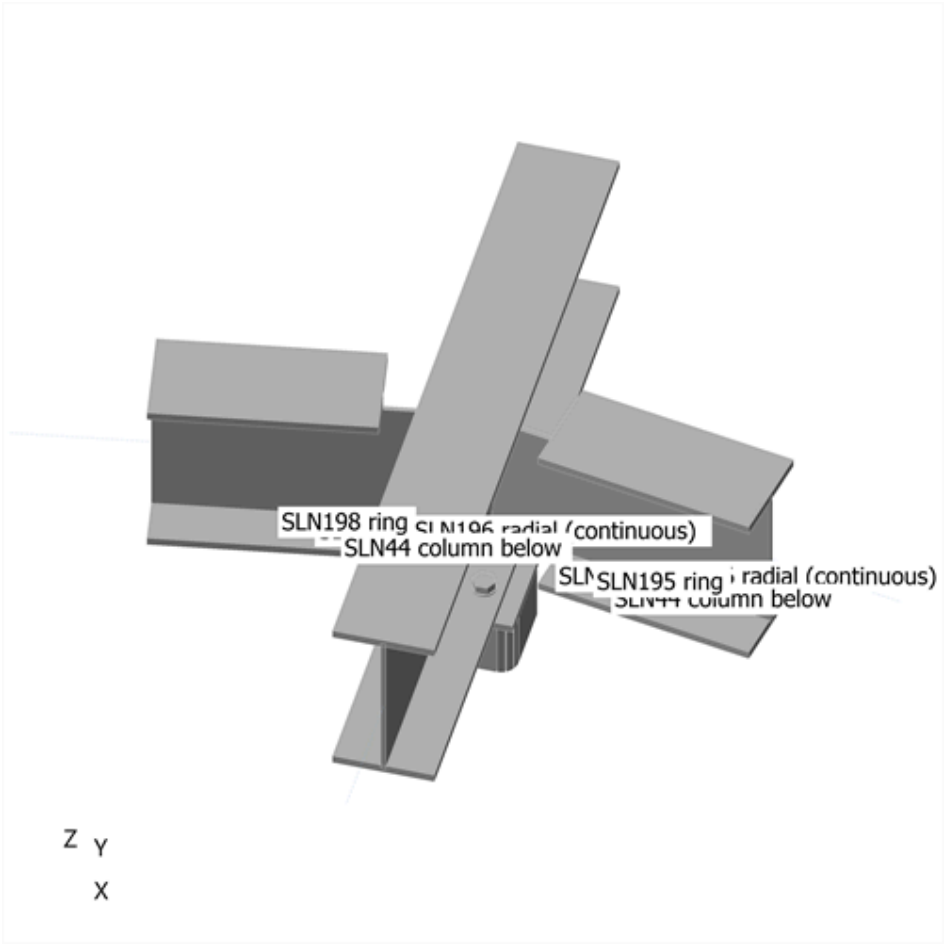
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

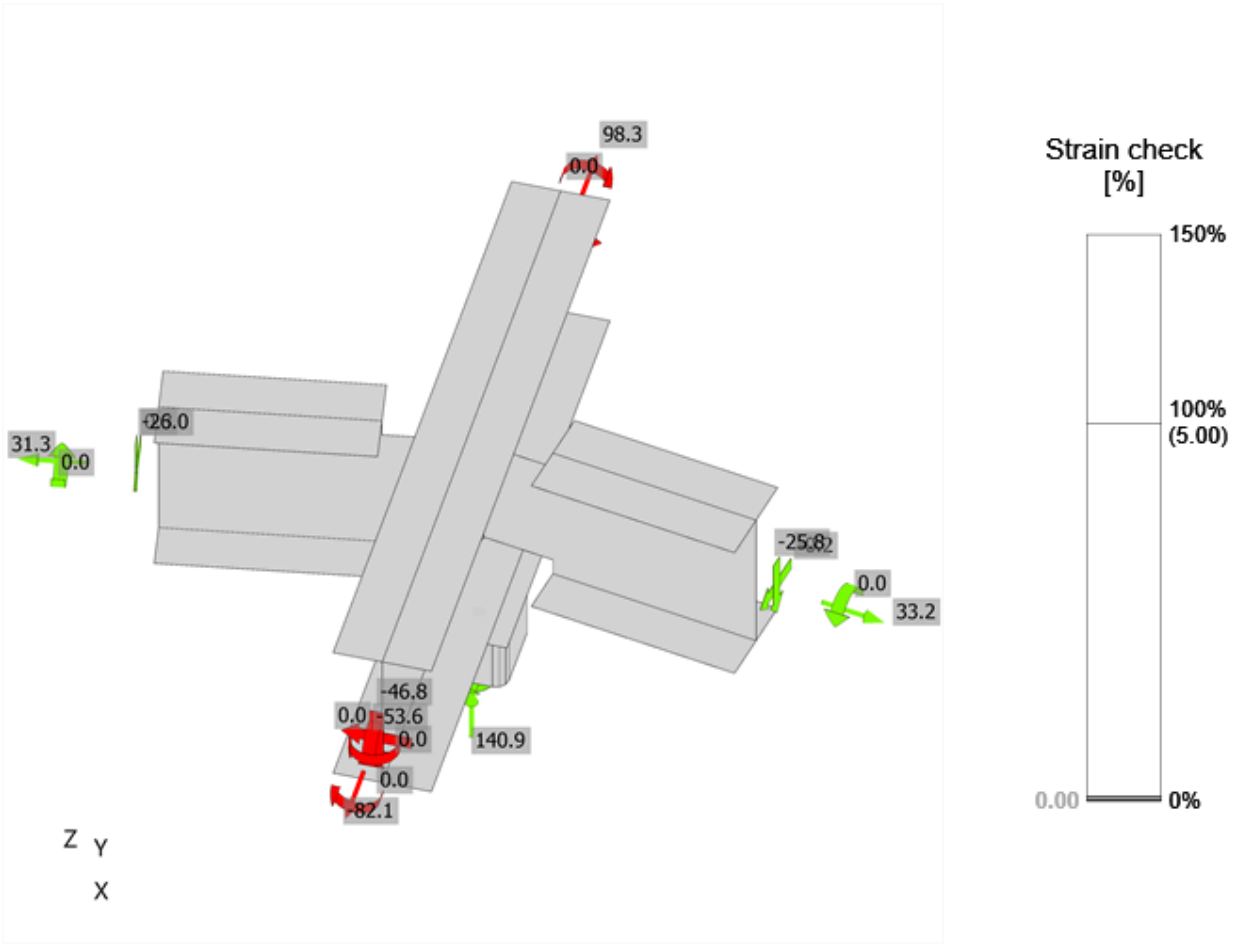
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN44 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

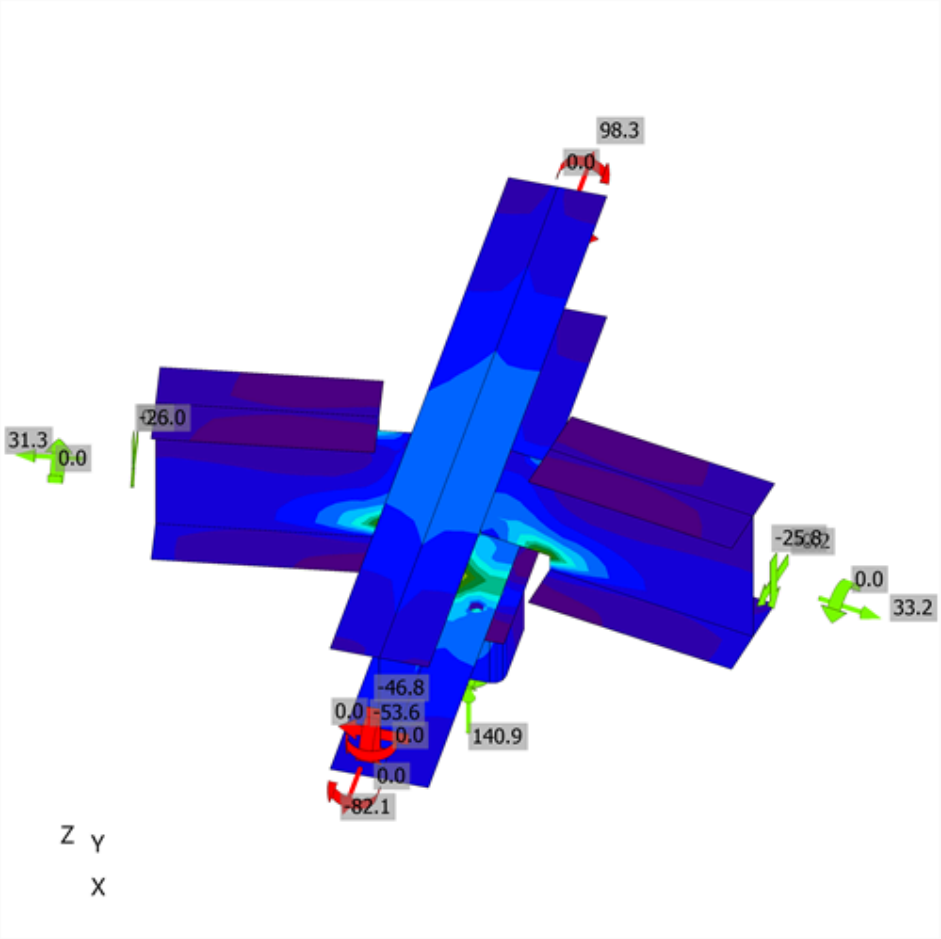
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

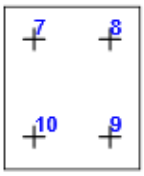


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_125	0.4	5.1	186.9	0.3	5.4	5.6	OK	OK
	B3	M20 8.8 - 1	C LC_109	14.9	1.8	156.4	10.5	1.9	9.5	OK	OK
	B4	M20 8.8 - 1	C LC_109	18.3	2.8	82.0	13.0	3.4	12.3	OK	OK
	B5	M20 8.8 - 1	C LC_123	0.3	4.8	196.0	0.2	5.1	5.3	OK	OK
	B7	M20 8.8 - 1	C LC_123	0.5	0.3	62.7	0.3	0.4	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_123	0.6	0.4	69.9	0.5	0.6	0.8	OK	OK
	B9	M20 8.8 - 1	C LC_123	1.7	0.4	67.8	1.2	0.4	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.7	0.9	123.8	1.2	1.3	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN44 column below	SLN44 column below	◢ 4.2 ◣	521	C LC_121	278.1	0.0	-137.0	-138.6	18.0	68.7	36.8	OK	OK
		◢ 4.2 ◣	521	C LC_121	302.5	0.0	-151.4	149.7	-21.2	74.7	40.7	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_119	93.1	0.0	27.3	-34.6	38.0	23.0	13.7	OK	OK
SLN194+SLN196 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_119	99.5	0.0	12.2	15.0	-55.0	24.6	15.8	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_121	79.1	0.0	-22.5	-18.2	39.8	19.5	11.7	OK	OK
		◢ 4.2 ◣	149	C LC_121	95.1	0.0	-17.8	24.4	-48.1	23.5	14.7	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_119	102.1	0.0	-18.4	-24.7	52.5	25.2	16.0	OK	OK
		◢ 4.2 ◣	149	C LC_119	86.0	0.0	-24.0	20.1	-43.3	21.3	12.7	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_160
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring

Name	SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

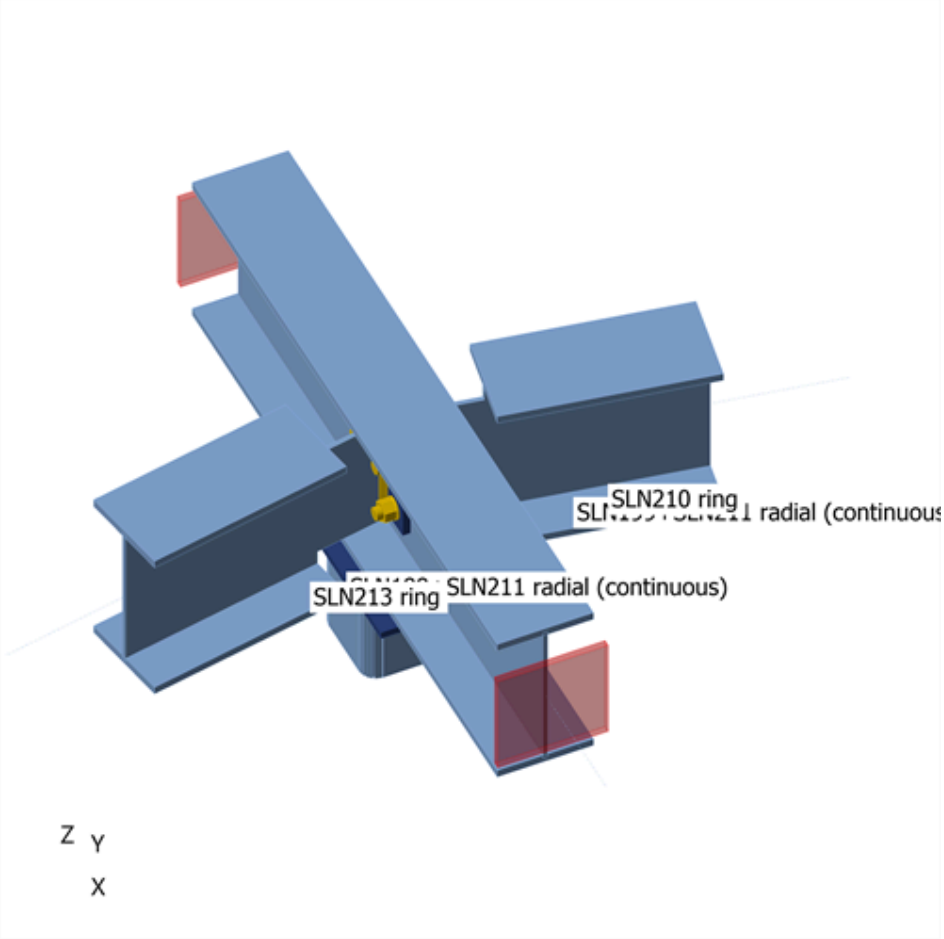
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN199+SLN211 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN46 column below	2 - SHS150x150x10	0.0	90.0	-20.0	0	0	0
SLN210 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN199+SLN211 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN199+SLN211 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN46 column below / begin		Position	0
SLN210 ring / end		Position	0
SLN213 ring / end		Position	0



SLN199+SLN211 radial (continuous) - SLN46 column below - SLN210 ring - SLN213 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.6	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN46 column below / Begin	131.9	1.2	-4.8	0.0	-6.0	0.0
	SLN210 ring / End	27.9	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	28.9	0.2	-24.1	0.0	0.0	0.0
C LC_104	SLN46 column below / Begin	132.7	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	15.6	-0.1	-24.0	0.0	0.0	0.0
	SLN213 ring / End	15.3	0.2	-24.1	0.0	0.0	0.0
C LC_105	SLN46 column below / Begin	132.5	-3.3	-2.3	0.0	-4.4	0.1
	SLN210 ring / End	24.2	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	20.4	0.2	-24.1	0.0	0.0	0.0
C LC_106	SLN46 column below / Begin	132.7	0.0	-0.8	0.0	-3.4	0.0
	SLN210 ring / End	16.0	-0.2	-24.0	0.0	0.0	0.0
	SLN213 ring / End	16.0	0.1	-24.1	0.0	0.0	0.0
C LC_107	SLN46 column below / Begin	2.2	2.4	-7.9	0.0	-5.4	0.0
	SLN210 ring / End	10.7	0.1	-0.5	0.0	0.0	0.0
	SLN213 ring / End	13.1	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN46 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-9.8	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-10.0	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN46 column below / Begin	3.3	-6.6	-2.9	0.0	-2.1	0.0
	SLN210 ring / End	4.4	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-2.3	0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN46 column below / Begin	3.7	0.0	0.0	0.0	-0.2	0.0
	SLN210 ring / End	-10.1	0.0	-0.5	0.0	0.0	0.0
	SLN213 ring / End	-10.1	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN46 column below / Begin	88.7	2.4	-8.5	0.0	-7.6	0.0
	SLN210 ring / End	24.8	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	27.1	0.1	-16.2	0.0	0.0	0.0
C LC_112	SLN46 column below / Begin	90.3	0.0	-0.6	0.0	-2.5	0.0
	SLN210 ring / End	3.0	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	2.7	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN46 column below / Begin	89.7	-6.6	-3.5	0.0	-4.3	0.1
	SLN210 ring / End	18.2	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	11.2	0.2	-16.2	0.0	0.0	0.0
C LC_114	SLN46 column below / Begin	90.1	0.0	-0.6	0.0	-2.4	0.0
	SLN210 ring / End	3.0	-0.1	-16.2	0.0	0.0	0.0
	SLN213 ring / End	3.1	0.0	-16.2	0.0	0.0	0.0
C LC_115	SLN46 column below / Begin	9.6	2.4	-8.0	0.0	-5.6	0.0
	SLN210 ring / End	13.1	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	15.4	0.0	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN46 column below / Begin	11.2	0.0	-0.1	0.0	-0.4	0.0
	SLN210 ring / End	-7.6	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	-7.8	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN46 column below / Begin	10.6	-6.6	-3.0	0.0	-2.3	0.0
	SLN210 ring / End	6.7	0.0	-1.7	0.0	0.0	0.0
	SLN213 ring / End	0.0	0.1	-1.7	0.0	0.0	0.0
C LC_118	SLN46 column below / Begin	11.0	0.0	-0.1	0.0	-0.3	0.0
	SLN210 ring / End	-7.9	-0.1	-1.7	0.0	0.0	0.0
	SLN213 ring / End	-7.8	-0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN46 column below / Begin	140.9	1.2	-4.8	0.0	-6.2	0.0
	SLN210 ring / End	31.4	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	32.5	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN46 column below / Begin	141.3	0.0	-0.9	0.0	-3.7	0.0
	SLN210 ring / End	19.0	-0.1	-25.6	0.0	0.0	0.0
	SLN213 ring / End	18.6	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN46 column below / Begin	141.3	-3.3	-2.3	0.0	-4.6	0.1
	SLN210 ring / End	27.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	23.9	0.3	-25.6	0.0	0.0	0.0
C LC_122	SLN46 column below / Begin	141.4	0.0	-0.9	0.0	-3.6	0.0
	SLN210 ring / End	19.4	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	19.4	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN46 column below / Begin	106.6	2.4	-8.6	0.0	-8.1	0.0
	SLN210 ring / End	32.0	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	34.3	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN46 column below / Begin	107.4	0.0	-0.7	0.0	-2.9	0.0
	SLN210 ring / End	9.8	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	9.4	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN46 column below / Begin	107.4	-6.6	-3.6	0.0	-4.8	0.1
	SLN210 ring / End	25.3	-0.2	-19.4	0.0	0.0	0.0
	SLN213 ring / End	18.2	0.2	-19.4	0.0	0.0	0.0
C LC_126	SLN46 column below / Begin	107.6	0.0	-0.7	0.0	-2.8	0.0
	SLN210 ring / End	9.8	-0.1	-19.4	0.0	0.0	0.0
	SLN213 ring / End	9.9	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN46 column below / Begin	27.5	2.4	-8.1	0.0	-6.0	0.0
	SLN210 ring / End	20.2	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	22.6	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN46 column below / Begin	28.4	0.0	-0.2	0.0	-0.8	0.0
	SLN210 ring / End	-0.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	-1.1	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN46 column below / Begin	28.3	-6.6	-3.1	0.0	-2.7	0.0
	SLN210 ring / End	13.8	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	7.0	0.1	-4.9	0.0	0.0	0.0
C LC_130	SLN46 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN210 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	-1.0	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.3	-0.2	-25.6	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.9	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN46 column below / Begin	28.3	0.0	-0.1	0.0	-0.6	0.0
	SLN210 ring / End	8.7	-0.1	-4.9	0.0	0.0	0.0
	SLN213 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.9	-0.3	-25.6	0.0	0.0	0.0
	SLN213 ring / End	28.7	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	26.8	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	26.4	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	28.3	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.8	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN46 column below / Begin	141.5	0.0	-0.9	0.0	-3.5	0.0
	SLN210 ring / End	27.3	-0.2	-25.6	0.0	0.0	0.0
	SLN213 ring / End	27.3	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN46 column below / Begin	90.1	0.0	-1.8	0.0	-7.3	0.0
	SLN210 ring / End	12.4	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.3	0.1	-11.6	0.0	0.0	0.0
C LC_152	SLN46 column below / Begin	79.0	0.0	0.8	0.0	3.0	0.0
	SLN210 ring / End	28.4	-0.2	-14.1	0.0	0.0	0.0
	SLN213 ring / End	28.2	0.2	-14.2	0.0	0.0	0.0
C LC_153	SLN46 column below / Begin	141.2	0.0	-0.9	0.0	-3.8	0.0
	SLN210 ring / End	28.0	-0.2	-20.8	0.0	0.0	0.0
	SLN213 ring / End	27.7	0.2	-20.8	0.0	0.0	0.0
C LC_154	SLN46 column below / Begin	89.6	0.0	-1.8	0.0	-7.2	0.0
	SLN210 ring / End	12.6	-0.1	-11.6	0.0	0.0	0.0
	SLN213 ring / End	12.4	0.1	-11.6	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.7 < 100%	OK
Welds	76.5 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN199+SLN211 radial (continuous)-bfl 1	12.7	C LC_119	151.8	0.0	45.9	OK
SLN199+SLN211 radial (continuous)-tfl 1	12.7	C LC_153	72.6	0.0	0.0	OK
SLN199+SLN211 radial (continuous)-w 1	7.2	C LC_125	245.2	0.0	35.1	OK
SLN46 column below	10.0	C LC_119	174.4	0.0	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_119	28.6	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_122	20.1	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_145	135.1	0.0	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_119	29.0	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_120	20.3	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_119	141.6	0.0	0.0	OK
End plate column below SLN46 column below	12.0	C LC_107	133.6	0.0	18.0	OK
End plate SLN210 ring	12.0	C LC_125	33.6	0.0	2.0	OK
End plate SLN213 ring	12.0	C LC_125	32.9	0.0	5.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

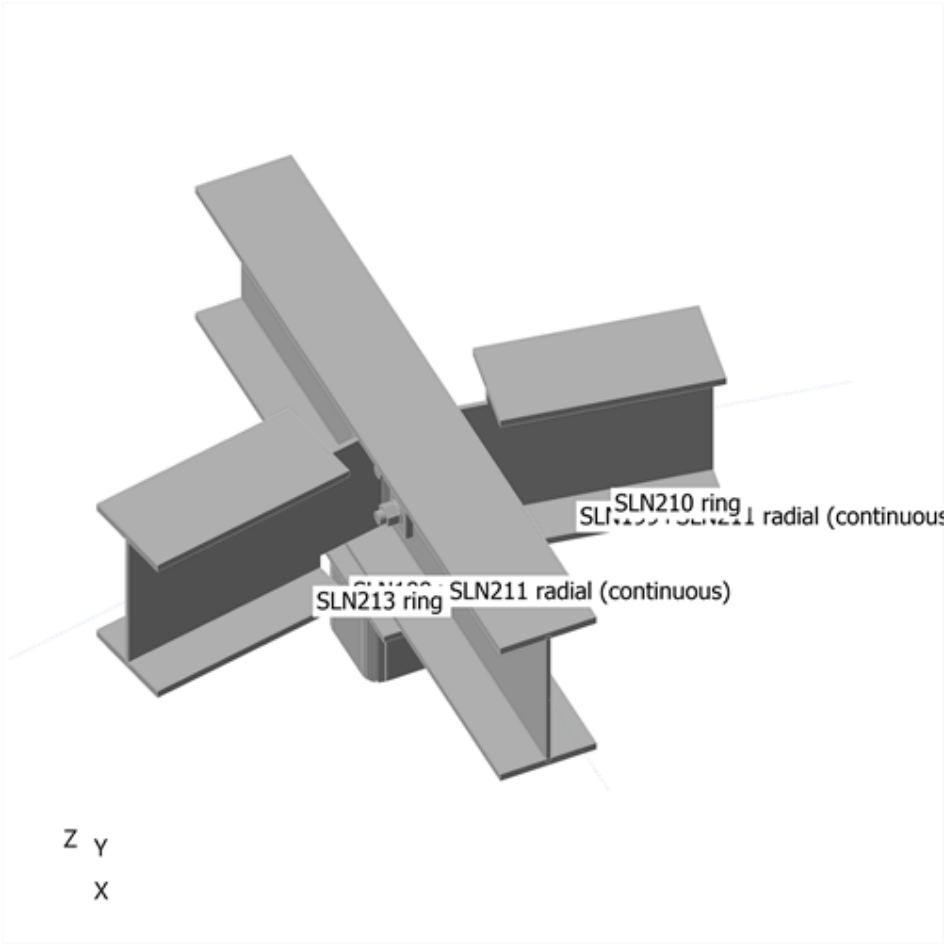
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

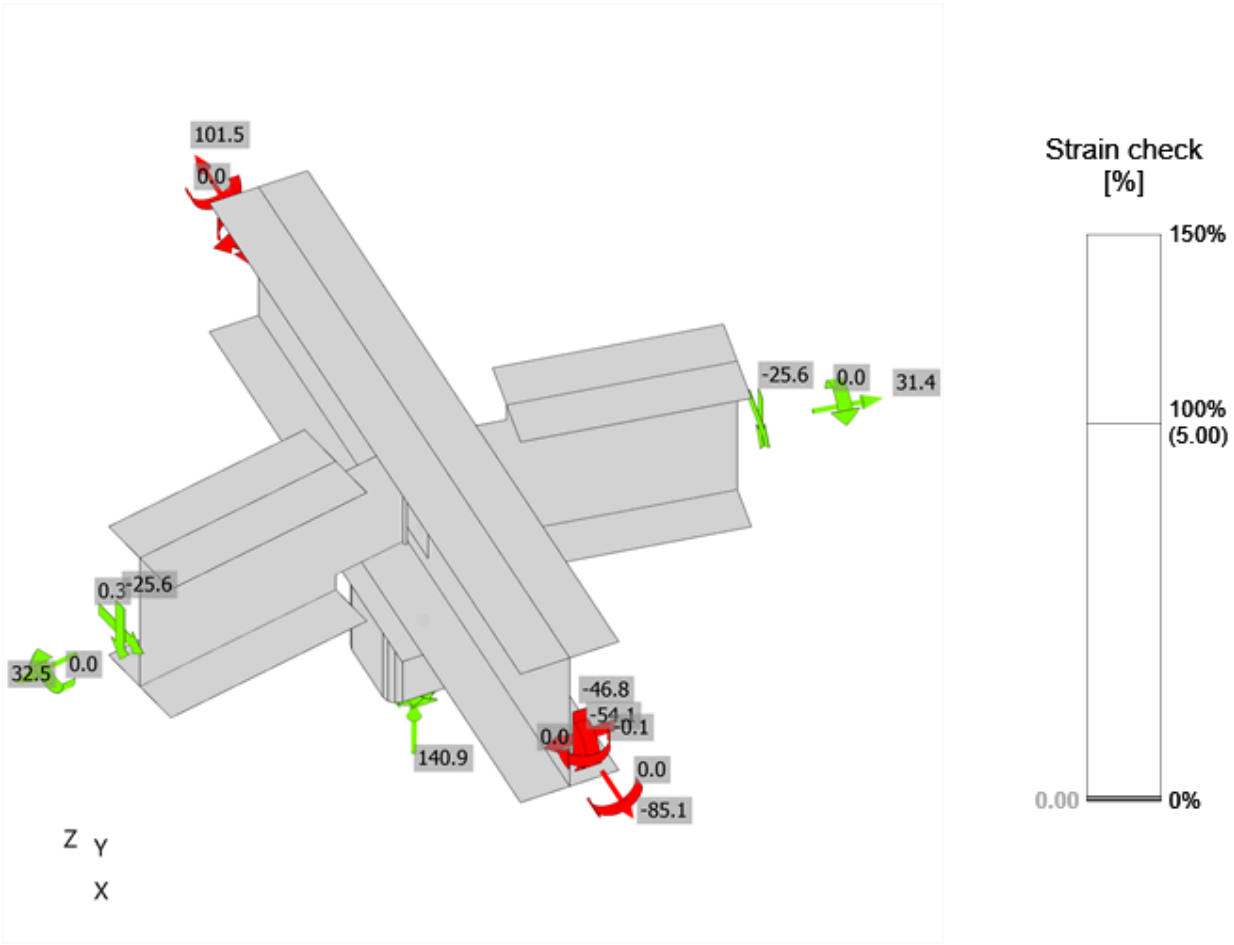
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN46 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

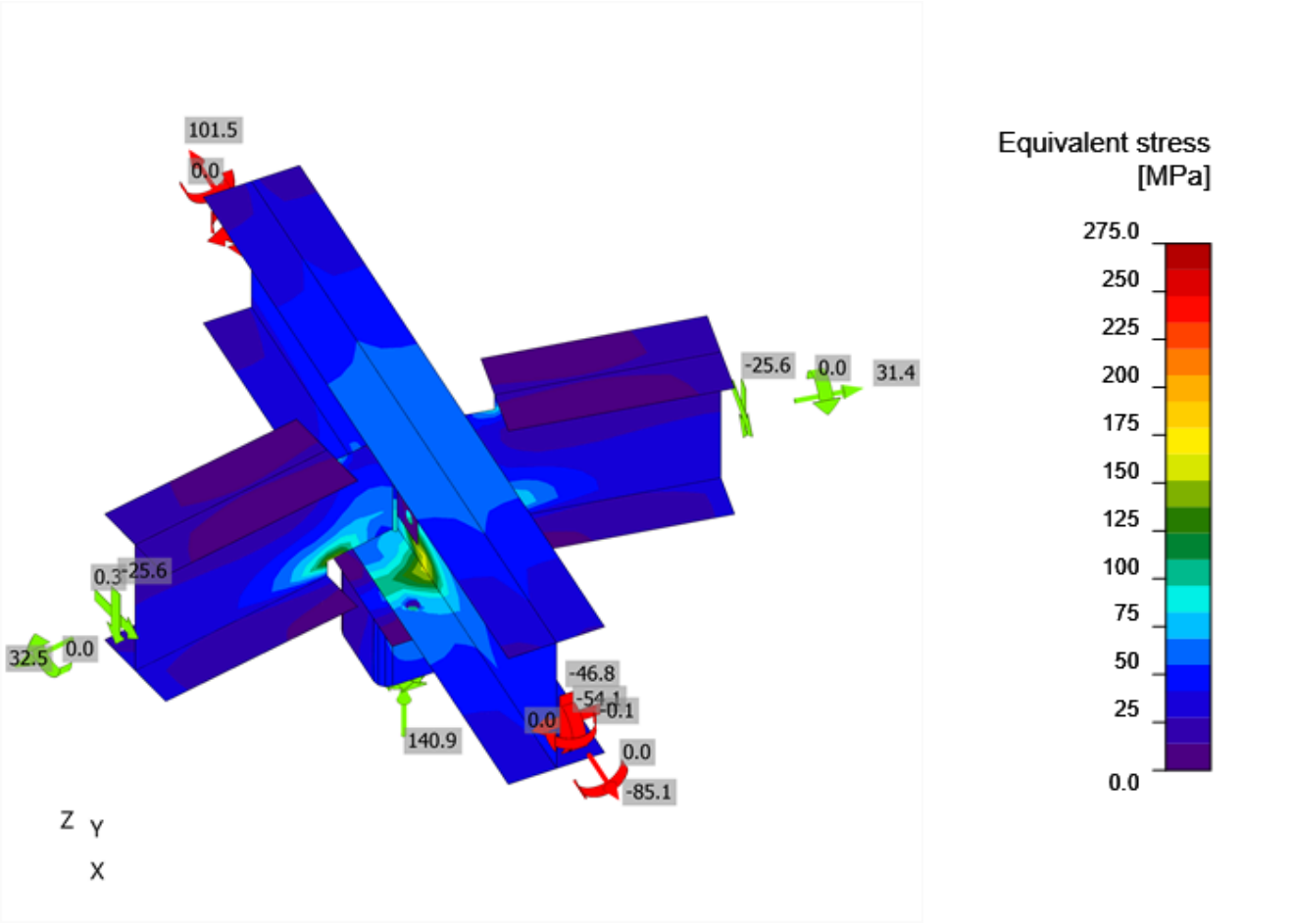
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

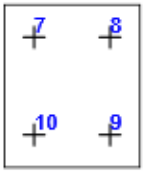


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.6	4.4	172.8	0.4	4.7	5.0	OK	OK
	B3	M20 8.8 - 1	C LC_107	19.3	2.8	85.9	13.7	3.3	12.7	OK	OK
	B4	M20 8.8 - 1	C LC_107	16.9	2.1	129.6	12.0	2.2	10.8	OK	OK
	B5	M20 8.8 - 1	C LC_123	0.4	5.1	181.3	0.3	5.4	5.6	OK	OK
	B7	M20 8.8 - 1	C LC_123	0.2	0.3	123.8	0.2	0.5	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_151	0.1	0.5	123.8	0.1	0.8	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_125	1.7	0.4	63.0	1.2	0.5	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_125	1.7	0.9	123.8	1.2	1.2	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN46 column below	SLN46 column below	◢ 4.2 ◣	521	C LC_119	285.3	0.0	-139.0	-137.0	-43.6	70.5	38.9	OK	OK
		◢ 4.2 ◣	521	C LC_119	309.6	0.0	-148.9	150.8	42.6	76.5	42.6	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_119	94.6	0.0	27.3	-35.6	38.3	23.4	14.6	OK	OK
SLN199+SLN211 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_119	93.3	0.0	11.3	15.1	-51.3	23.0	15.2	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_121	80.7	0.0	-22.5	-19.0	40.5	19.9	12.1	OK	OK
		◢ 4.2 ◣	149	C LC_121	95.7	0.0	-17.2	22.9	-49.3	23.6	15.0	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_119	97.7	0.0	-18.2	-24.6	49.7	24.1	15.2	OK	OK
		◢ 4.2 ◣	149	C LC_119	80.4	0.0	-22.6	18.4	-40.6	19.9	11.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_200
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring

Name	SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

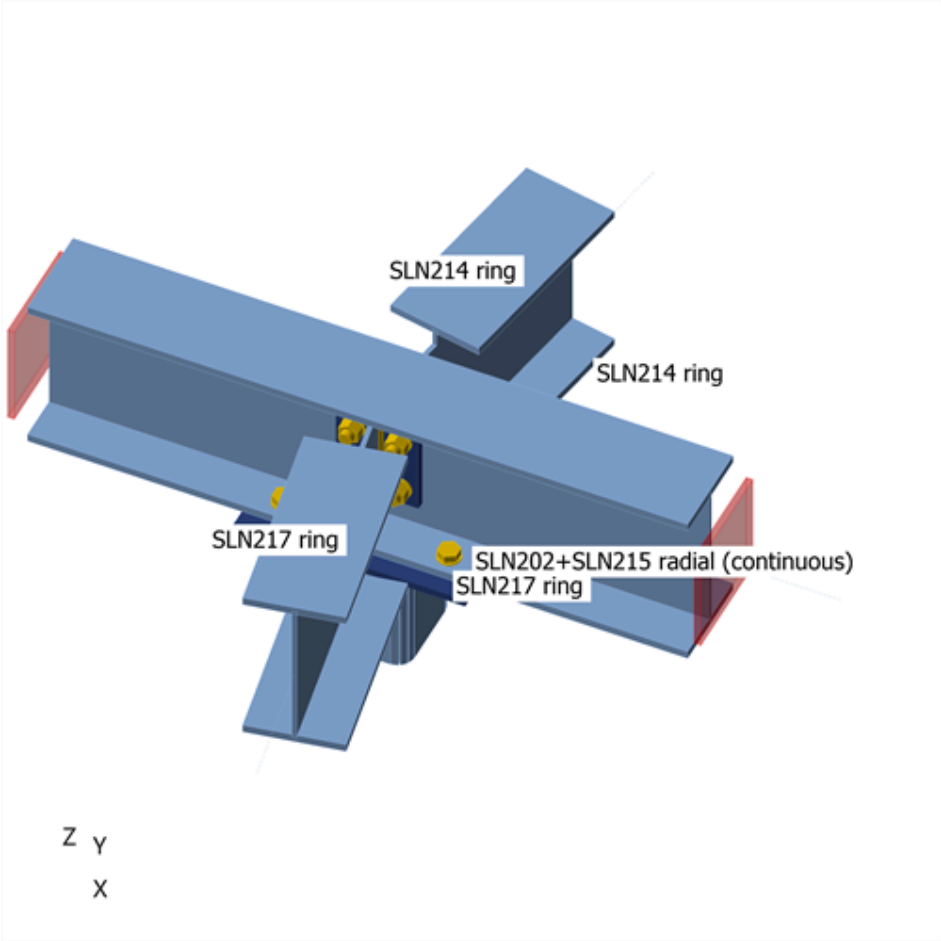
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN202+SLN215 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN48 column below	2 - SHS150x150x10	0.0	90.0	20.0	0	0	0
SLN214 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN202+SLN215 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN202+SLN215 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN48 column below / begin		Position	0
SLN214 ring / end		Position	0
SLN217 ring / end		Position	0



SLN202+SLN215 radial (continuous) - SLN48 column below - SLN214 ring - SLN217 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.7	0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	27.7	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN48 column below / Begin	122.4	-1.2	-4.4	0.0	-4.4	0.0
	SLN214 ring / End	29.0	-0.2	-24.0	0.0	0.0	0.0
	SLN217 ring / End	27.6	0.3	-24.0	0.0	0.0	0.0
C LC_104	SLN48 column below / Begin	123.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	15.3	-0.1	-24.0	0.0	0.0	0.0
	SLN217 ring / End	15.2	0.2	-24.0	0.0	0.0	0.0
C LC_105	SLN48 column below / Begin	123.4	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	16.0	-0.1	-24.0	0.0	0.0	0.0
	SLN217 ring / End	15.7	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN48 column below / Begin	122.9	3.3	-1.9	0.0	-2.8	0.0
	SLN214 ring / End	20.5	-0.3	-24.0	0.0	0.0	0.0
	SLN217 ring / End	23.9	0.1	-24.0	0.0	0.0	0.0
C LC_107	SLN48 column below / Begin	1.3	-2.4	-7.9	0.0	-5.2	0.0
	SLN214 ring / End	13.1	0.2	-0.5	0.0	0.0	0.0
	SLN217 ring / End	10.7	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN48 column below / Begin	3.7	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	-10.0	0.0	-0.5	0.0	0.0	0.0
	SLN217 ring / End	-10.1	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN48 column below / Begin	3.3	0.0	0.0	0.0	-0.1	0.0
	SLN214 ring / End	-10.1	0.0	-0.5	0.0	0.0	0.0
	SLN217 ring / End	-10.2	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN48 column below / Begin	2.4	6.6	-2.9	0.0	-1.9	0.0
	SLN214 ring / End	-2.2	-0.1	-0.5	0.0	0.0	0.0
	SLN217 ring / End	4.5	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN48 column below / Begin	81.7	-2.4	-8.2	0.0	-6.4	0.0
	SLN214 ring / End	27.2	0.0	-16.2	0.0	0.0	0.0
	SLN217 ring / End	24.7	0.2	-16.2	0.0	0.0	0.0
C LC_112	SLN48 column below / Begin	84.2	0.0	-0.4	0.0	-1.5	0.0
	SLN214 ring / End	2.7	-0.1	-16.2	0.0	0.0	0.0
	SLN217 ring / End	2.5	0.2	-16.2	0.0	0.0	0.0
C LC_113	SLN48 column below / Begin	83.8	0.0	-0.3	0.0	-1.3	0.0
	SLN214 ring / End	3.1	-0.1	-16.2	0.0	0.0	0.0
	SLN217 ring / End	2.8	0.1	-16.2	0.0	0.0	0.0
C LC_114	SLN48 column below / Begin	82.8	6.6	-3.2	0.0	-3.1	0.0
	SLN214 ring / End	11.3	-0.2	-16.2	0.0	0.0	0.0
	SLN217 ring / End	18.0	0.0	-16.2	0.0	0.0	0.0
C LC_115	SLN48 column below / Begin	8.1	-2.4	-7.9	0.0	-5.3	0.0
	SLN214 ring / End	15.4	0.1	-1.7	0.0	0.0	0.0
	SLN217 ring / End	13.1	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN48 column below / Begin	10.5	0.0	-0.1	0.0	-0.3	0.0
	SLN214 ring / End	-7.8	0.0	-1.7	0.0	0.0	0.0
	SLN217 ring / End	-7.9	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN48 column below / Begin	10.1	0.0	0.0	0.0	-0.2	0.0
	SLN214 ring / End	-7.9	0.0	-1.7	0.0	0.0	0.0
	SLN217 ring / End	-8.0	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN48 column below / Begin	9.2	6.6	-2.9	0.0	-2.0	0.0
	SLN214 ring / End	0.1	-0.1	-1.7	0.0	0.0	0.0
	SLN217 ring / End	6.7	-0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN48 column below / Begin	130.8	-1.2	-4.4	0.0	-4.5	0.0
	SLN214 ring / End	32.6	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	31.2	0.3	-25.6	0.0	0.0	0.0
C LC_120	SLN48 column below / Begin	131.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	18.7	-0.1	-25.5	0.0	0.0	0.0
	SLN217 ring / End	18.5	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN48 column below / Begin	131.6	0.0	-0.5	0.0	-2.0	0.0
	SLN214 ring / End	19.5	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	19.0	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN48 column below / Begin	131.2	3.3	-1.9	0.0	-2.9	0.0
	SLN214 ring / End	24.0	-0.3	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN48 column below / Begin	98.6	-2.4	-8.2	0.0	-6.7	0.0
	SLN214 ring / End	34.4	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	31.8	0.3	-19.4	0.0	0.0	0.0
C LC_124	SLN48 column below / Begin	100.2	0.0	-0.4	0.0	-1.7	0.0
	SLN214 ring / End	9.4	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	9.2	0.2	-19.4	0.0	0.0	0.0
C LC_125	SLN48 column below / Begin	100.1	0.0	-0.4	0.0	-1.5	0.0
	SLN214 ring / End	10.0	-0.1	-19.3	0.0	0.0	0.0
	SLN217 ring / End	9.5	0.1	-19.4	0.0	0.0	0.0
C LC_126	SLN48 column below / Begin	99.4	6.6	-3.2	0.0	-3.4	0.0
	SLN214 ring / End	18.3	-0.3	-19.3	0.0	0.0	0.0
	SLN217 ring / End	25.1	0.1	-19.4	0.0	0.0	0.0
C LC_127	SLN48 column below / Begin	25.0	-2.4	-7.9	0.0	-5.5	0.0
	SLN214 ring / End	22.6	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	20.2	0.2	-4.9	0.0	0.0	0.0
C LC_128	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.5	0.0
	SLN214 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	-1.3	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN48 column below / Begin	26.5	0.0	-0.1	0.0	-0.4	0.0
	SLN214 ring / End	-1.0	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	-1.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN48 column below / Begin	25.8	6.6	-2.9	0.0	-2.2	0.0
	SLN214 ring / End	7.0	-0.2	-4.9	0.0	0.0	0.0
	SLN217 ring / End	13.8	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	28.1	-0.2	-25.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / End	27.9	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.5	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.7	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN48 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
	SLN217 ring / End	8.8	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	28.8	-0.3	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.6	0.3	-25.6	0.0	0.0	0.0
C LC_146	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	26.5	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	26.4	0.1	-25.6	0.0	0.0	0.0
C LC_147	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.9	0.0
	SLN214 ring / End	27.4	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	27.0	0.2	-25.6	0.0	0.0	0.0
C LC_148	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-1.8	0.0
	SLN214 ring / End	28.0	-0.2	-25.5	0.0	0.0	0.0
	SLN217 ring / End	28.0	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN48 column below / Begin	80.1	0.0	-1.3	0.0	-5.4	0.0
	SLN214 ring / End	12.3	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.3	0.1	-11.8	0.0	0.0	0.0
C LC_152	SLN48 column below / Begin	77.3	0.0	0.8	0.0	3.3	0.0
	SLN214 ring / End	28.3	-0.2	-14.5	0.0	0.0	0.0
	SLN217 ring / End	28.1	0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN48 column below / Begin	131.7	0.0	-0.5	0.0	-2.1	0.0
	SLN214 ring / End	27.8	-0.2	-21.5	0.0	0.0	0.0
	SLN217 ring / End	27.6	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN48 column below / Begin	79.3	0.0	-1.3	0.0	-5.3	0.0
	SLN214 ring / End	12.5	-0.1	-11.9	0.0	0.0	0.0
	SLN217 ring / End	12.4	0.1	-11.8	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.3 < 100%	OK
Welds	64.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN202+SLN215 radial (continuous)-bfl 1	12.7	C LC_119	128.4	0.0	39.0	OK
SLN202+SLN215 radial (continuous)-tfl 1	12.7	C LC_153	64.0	0.0	0.0	OK
SLN202+SLN215 radial (continuous)-w 1	7.2	C LC_126	235.7	0.0	29.4	OK
SLN48 column below	10.0	C LC_119	143.3	0.0	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_119	28.7	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_120	19.9	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_145	135.8	0.0	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_119	28.8	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_120	20.1	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_119	140.7	0.0	0.0	OK
End plate column below SLN48 column below	12.0	C LC_107	132.9	0.0	17.4	OK
End plate SLN214 ring	12.0	C LC_126	34.4	0.0	7.7	OK
End plate SLN217 ring	12.0	C LC_126	34.8	0.0	1.1	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

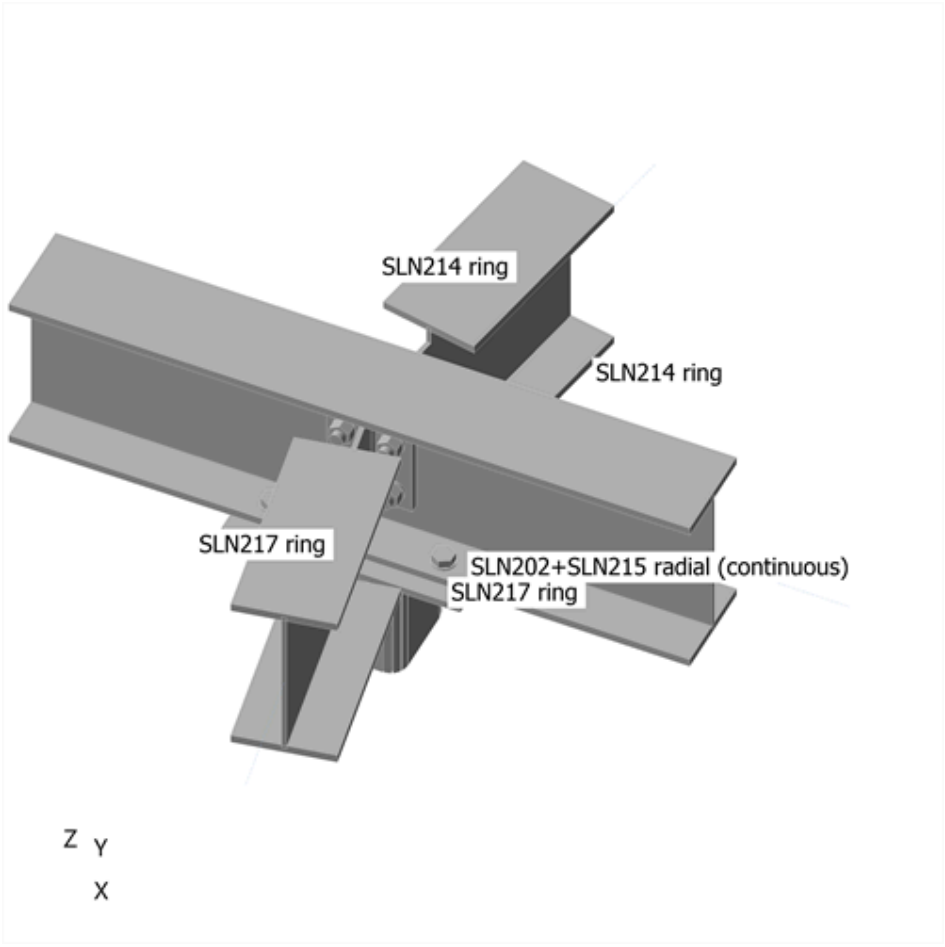
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

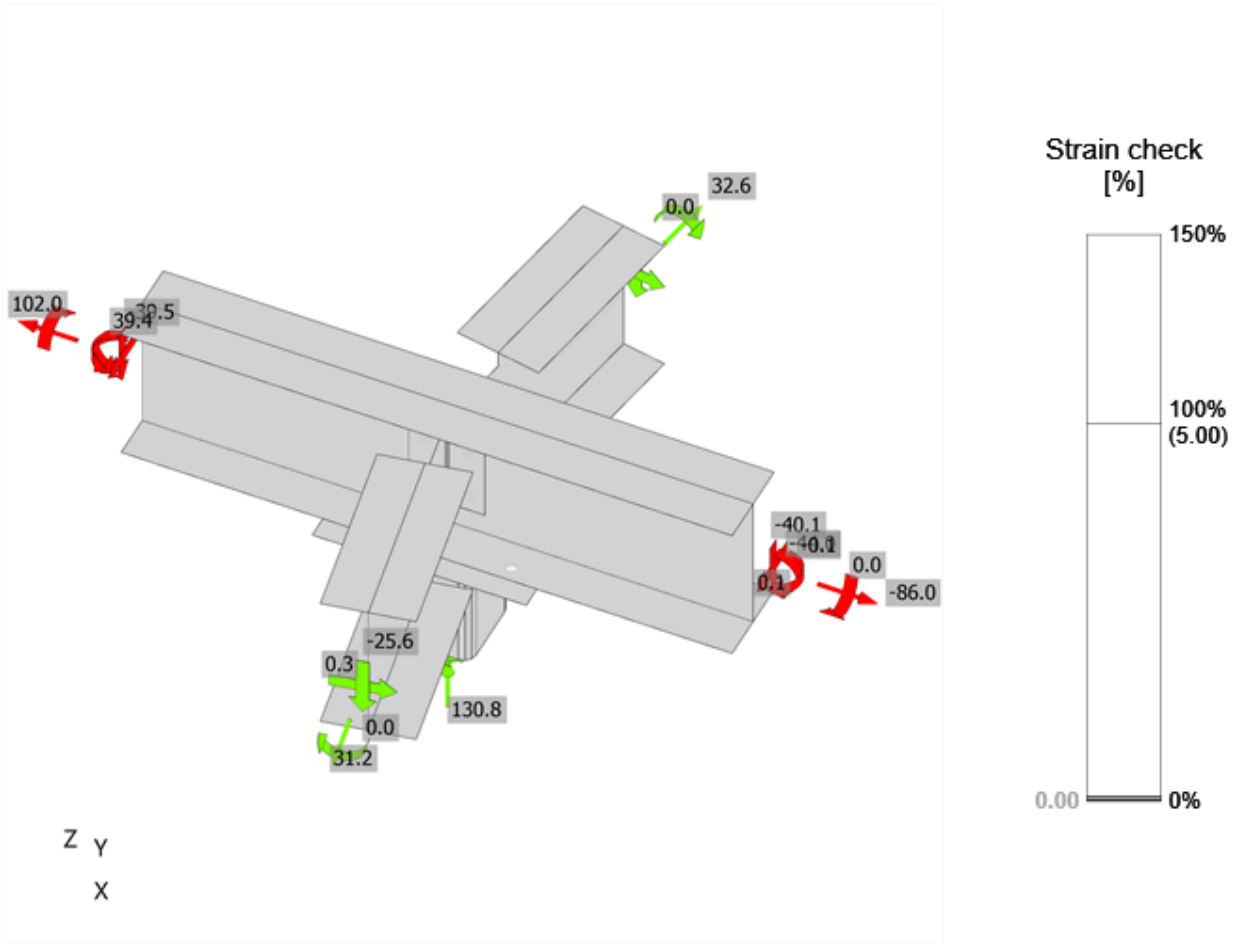
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN48 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

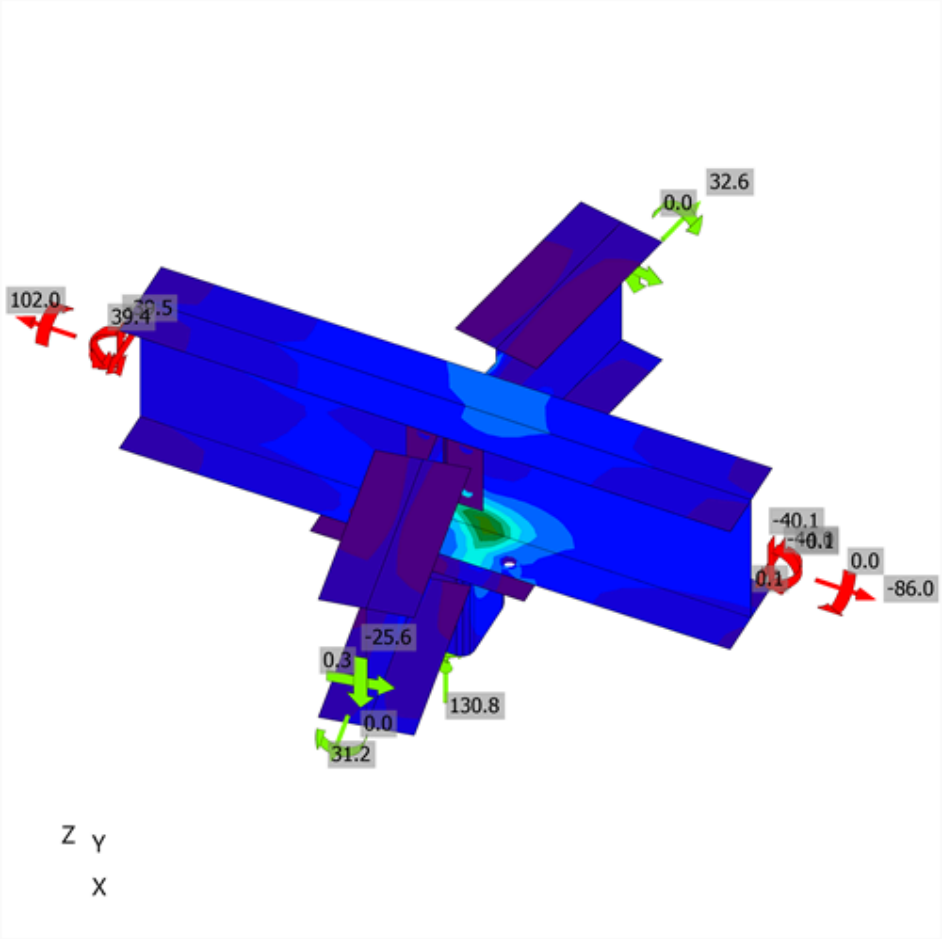
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

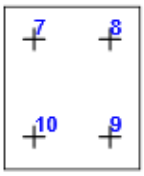


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_123	0.3	4.7	181.5	0.2	5.0	5.1	OK	OK
	B3	M20 8.8 - 1	C LC_107	16.4	2.1	130.9	11.6	2.2	10.5	OK	OK
	B4	M20 8.8 - 1	C LC_107	18.8	2.8	86.2	13.3	3.2	12.5	OK	OK
	B5	M20 8.8 - 1	C LC_126	0.2	4.3	189.8	0.2	4.5	4.6	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.5	0.2	63.8	0.4	0.3	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.7	0.3	62.8	0.5	0.5	0.7	OK	OK
	B9	M20 8.8 - 1	C LC_126	1.8	0.4	72.9	1.3	0.5	1.4	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.7	0.8	123.8	1.2	1.0	1.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN48 column below	SLN48 column below	◢ 4.2 ◣	521	C LC_119	243.3	0.0	-117.5	-121.4	19.8	60.1	32.0	OK	OK
		◢ 4.2 ◣	521	C LC_119	260.0	0.0	-131.7	127.7	-20.8	64.2	34.7	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_122	86.7	0.0	31.5	26.0	-38.7	21.4	12.3	OK	OK
SLN202+SLN215 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	90.1	0.0	11.9	-9.1	50.8	22.3	14.3	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_119	71.6	0.0	-19.4	-15.1	36.8	17.7	10.8	OK	OK
		◢ 4.2 ◣	149	C LC_119	84.9	0.0	-16.8	22.9	-42.2	21.0	13.3	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	94.9	0.0	-18.9	-24.7	47.7	23.4	14.9	OK	OK
		◢ 4.2 ◣	149	C LC_122	78.9	0.0	-20.3	16.1	-41.0	19.5	12.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_240
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring

Name	SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

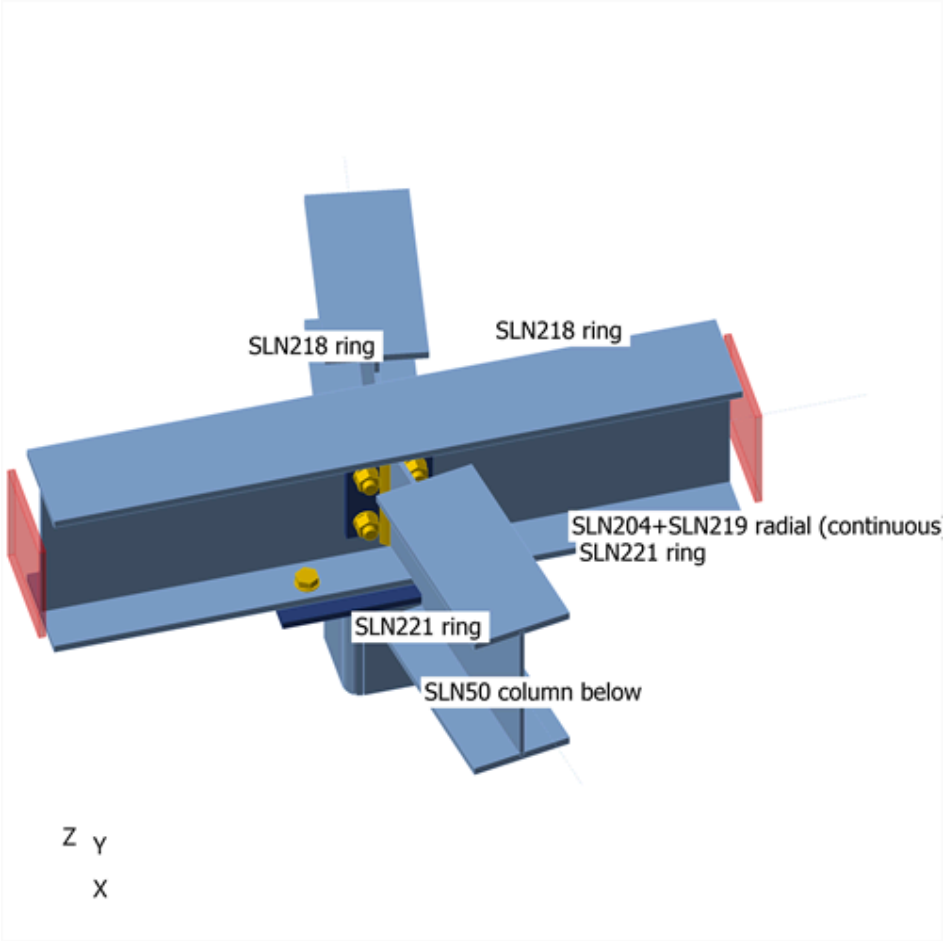
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN204+SLN219 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN50 column below	2 - SHS150x150x10	0.0	90.0	60.0	0	0	0
SLN218 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN204+SLN219 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN204+SLN219 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN50 column below / begin		Position	0
SLN218 ring / end		Position	0
SLN221 ring / end		Position	0



SLN204+SLN219 radial (continuous) - SLN50 column below - SLN218 ring - SLN221 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.7	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.8	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	27.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	27.8	0.2	-25.7	0.0	0.0	0.0
C LC_103	SLN50 column below / Begin	133.0	-3.0	-2.8	0.0	-4.5	0.0
	SLN218 ring / End	24.6	-0.1	-24.5	0.0	0.0	0.0
	SLN221 ring / End	21.8	0.2	-24.1	0.0	0.0	0.0
C LC_104	SLN50 column below / Begin	133.4	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	15.6	-0.1	-24.5	0.0	0.0	0.0
	SLN221 ring / End	16.1	0.1	-24.1	0.0	0.0	0.0
C LC_105	SLN50 column below / Begin	133.5	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	14.8	-0.2	-24.5	0.0	0.0	0.0
	SLN221 ring / End	15.0	0.1	-24.1	0.0	0.0	0.0
C LC_106	SLN50 column below / Begin	132.9	1.8	-4.4	0.0	-5.5	0.0
	SLN218 ring / End	27.3	-0.2	-24.5	0.0	0.0	0.0
	SLN221 ring / End	29.6	0.1	-24.1	0.0	0.0	0.0
C LC_107	SLN50 column below / Begin	2.7	-6.1	-4.2	0.0	-2.9	0.0
	SLN218 ring / End	5.7	0.2	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-0.3	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN50 column below / Begin	3.6	0.0	0.0	0.0	-0.2	0.0
	SLN218 ring / End	-9.8	0.0	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-9.9	0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN50 column below / Begin	3.8	0.0	-0.1	0.0	-0.3	0.0
	SLN218 ring / End	-10.9	-0.1	-0.5	0.0	0.0	0.0
	SLN221 ring / End	-11.0	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN50 column below / Begin	2.5	3.5	-7.3	0.0	-5.0	0.0
	SLN218 ring / End	10.5	0.0	-0.5	0.0	0.0	0.0
	SLN221 ring / End	14.2	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN50 column below / Begin	89.7	-6.1	-4.7	0.0	-4.9	0.0
	SLN218 ring / End	19.3	0.1	-16.5	0.0	0.0	0.0
	SLN221 ring / End	13.5	0.2	-16.3	0.0	0.0	0.0
C LC_112	SLN50 column below / Begin	90.6	0.0	-0.5	0.0	-2.2	0.0
	SLN218 ring / End	3.0	-0.1	-16.5	0.0	0.0	0.0
	SLN221 ring / End	3.2	0.1	-16.3	0.0	0.0	0.0
C LC_113	SLN50 column below / Begin	90.8	0.0	-0.6	0.0	-2.3	0.0
	SLN218 ring / End	1.7	-0.2	-16.5	0.0	0.0	0.0
	SLN221 ring / End	1.8	0.0	-16.3	0.0	0.0	0.0
C LC_114	SLN50 column below / Begin	89.5	3.5	-7.8	0.0	-7.0	0.0
	SLN218 ring / End	24.3	-0.2	-16.5	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.0	-16.3	0.0	0.0	0.0
C LC_115	SLN50 column below / Begin	10.1	-6.1	-4.2	0.0	-3.0	0.0
	SLN218 ring / End	8.0	0.2	-1.8	0.0	0.0	0.0
	SLN221 ring / End	2.0	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN50 column below / Begin	11.0	0.0	-0.1	0.0	-0.3	0.0
	SLN218 ring / End	-7.5	0.0	-1.8	0.0	0.0	0.0
	SLN221 ring / End	-7.7	0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN50 column below / Begin	11.2	0.0	-0.1	0.0	-0.4	0.0
	SLN218 ring / End	-8.7	-0.1	-1.8	0.0	0.0	0.0
	SLN221 ring / End	-8.8	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN50 column below / Begin	9.9	3.5	-7.3	0.0	-5.2	0.0
	SLN218 ring / End	12.8	0.0	-1.8	0.0	0.0	0.0
	SLN221 ring / End	16.5	0.0	-1.7	0.0	0.0	0.0
C LC_119	SLN50 column below / Begin	141.9	-3.0	-2.9	0.0	-4.6	0.0
	SLN218 ring / End	28.2	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	25.4	0.2	-25.7	0.0	0.0	0.0
C LC_120	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	18.9	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	19.5	0.1	-25.7	0.0	0.0	0.0
C LC_121	SLN50 column below / Begin	142.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	18.1	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	18.3	0.1	-25.7	0.0	0.0	0.0
C LC_122	SLN50 column below / Begin	141.9	1.8	-4.4	0.0	-5.7	0.0
	SLN218 ring / End	30.8	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	33.2	0.2	-25.7	0.0	0.0	0.0
C LC_123	SLN50 column below / Begin	107.6	-6.1	-4.8	0.0	-5.3	0.0
	SLN218 ring / End	26.5	0.0	-19.7	0.0	0.0	0.0
	SLN221 ring / End	20.6	0.2	-19.5	0.0	0.0	0.0
C LC_124	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.5	0.0
	SLN218 ring / End	9.7	-0.1	-19.7	0.0	0.0	0.0
	SLN221 ring / End	10.0	0.2	-19.5	0.0	0.0	0.0
C LC_125	SLN50 column below / Begin	108.1	0.0	-0.6	0.0	-2.6	0.0
	SLN218 ring / End	8.4	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	8.4	0.0	-19.5	0.0	0.0	0.0
C LC_126	SLN50 column below / Begin	107.5	3.5	-7.9	0.0	-7.4	0.0
	SLN218 ring / End	31.4	-0.2	-19.7	0.0	0.0	0.0
	SLN221 ring / End	35.6	0.1	-19.5	0.0	0.0	0.0
C LC_127	SLN50 column below / Begin	28.0	-6.1	-4.3	0.0	-3.4	0.0
	SLN218 ring / End	15.2	0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.1	0.2	-4.9	0.0	0.0	0.0
C LC_128	SLN50 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN218 ring / End	-0.9	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	-0.9	0.1	-4.9	0.0	0.0	0.0
C LC_129	SLN50 column below / Begin	28.5	0.0	-0.2	0.0	-0.7	0.0
	SLN218 ring / End	-2.0	-0.2	-5.0	0.0	0.0	0.0
	SLN221 ring / End	-2.2	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN50 column below / Begin	27.9	3.5	-7.4	0.0	-5.5	0.0
	SLN218 ring / End	20.0	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	23.8	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	27.8	-0.2	-26.1	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN221 ring / End	28.3	0.2	-25.7	0.0	0.0	0.0
C LC_141	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.9	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.9	0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.5	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.5	0.0	-5.0	0.0	0.0	0.0
	SLN221 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN50 column below / Begin	28.5	0.0	-0.1	0.0	-0.5	0.0
	SLN218 ring / End	8.9	-0.1	-5.0	0.0	0.0	0.0
	SLN221 ring / End	9.0	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.0	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.2	-25.7	0.0	0.0	0.0
C LC_146	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.6	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	27.3	0.1	-25.7	0.0	0.0	0.0
C LC_147	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.2	0.0
	SLN218 ring / End	26.4	-0.1	-26.1	0.0	0.0	0.0
	SLN221 ring / End	26.7	0.1	-25.7	0.0	0.0	0.0
C LC_148	SLN50 column below / Begin	142.3	0.0	-0.8	0.0	-3.1	0.0
	SLN218 ring / End	28.3	-0.2	-26.1	0.0	0.0	0.0
	SLN221 ring / End	28.9	0.2	-25.7	0.0	0.0	0.0
C LC_151	SLN50 column below / Begin	90.1	0.0	-1.7	0.0	-7.0	0.0
	SLN218 ring / End	12.2	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.5	0.1	-11.7	0.0	0.0	0.0
C LC_152	SLN50 column below / Begin	79.3	0.0	0.8	0.0	3.3	0.0
	SLN218 ring / End	28.0	-0.2	-14.0	0.0	0.0	0.0
	SLN221 ring / End	28.3	0.1	-13.9	0.0	0.0	0.0
C LC_153	SLN50 column below / Begin	141.2	0.0	-0.8	0.0	-3.3	0.0
	SLN218 ring / End	27.5	-0.2	-20.7	0.0	0.0	0.0
	SLN221 ring / End	27.9	0.2	-20.6	0.0	0.0	0.0
C LC_154	SLN50 column below / Begin	89.5	0.0	-1.7	0.0	-6.9	0.0
	SLN218 ring / End	12.3	-0.1	-11.6	0.0	0.0	0.0
	SLN221 ring / End	12.6	0.1	-11.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	13.1 < 100%	OK
Welds	72.0 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN204+SLN219 radial (continuous)-bfl 1	12.7	C LC_122	150.1	0.0	44.7	OK
SLN204+SLN219 radial (continuous)-tfl 1	12.7	C LC_153	72.5	0.0	0.0	OK
SLN204+SLN219 radial (continuous)-w 1	7.2	C LC_123	244.0	0.0	36.4	OK
SLN50 column below	10.0	C LC_122	165.1	0.0	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_122	28.9	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_121	20.7	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_122	136.9	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_122	29.0	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_120	20.0	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_122	132.8	0.0	0.0	OK
End plate column below SLN50 column below	12.0	C LC_110	130.9	0.0	17.2	OK
End plate SLN218 ring	12.0	C LC_123	33.9	0.0	1.5	OK
End plate SLN221 ring	12.0	C LC_123	30.6	0.0	5.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

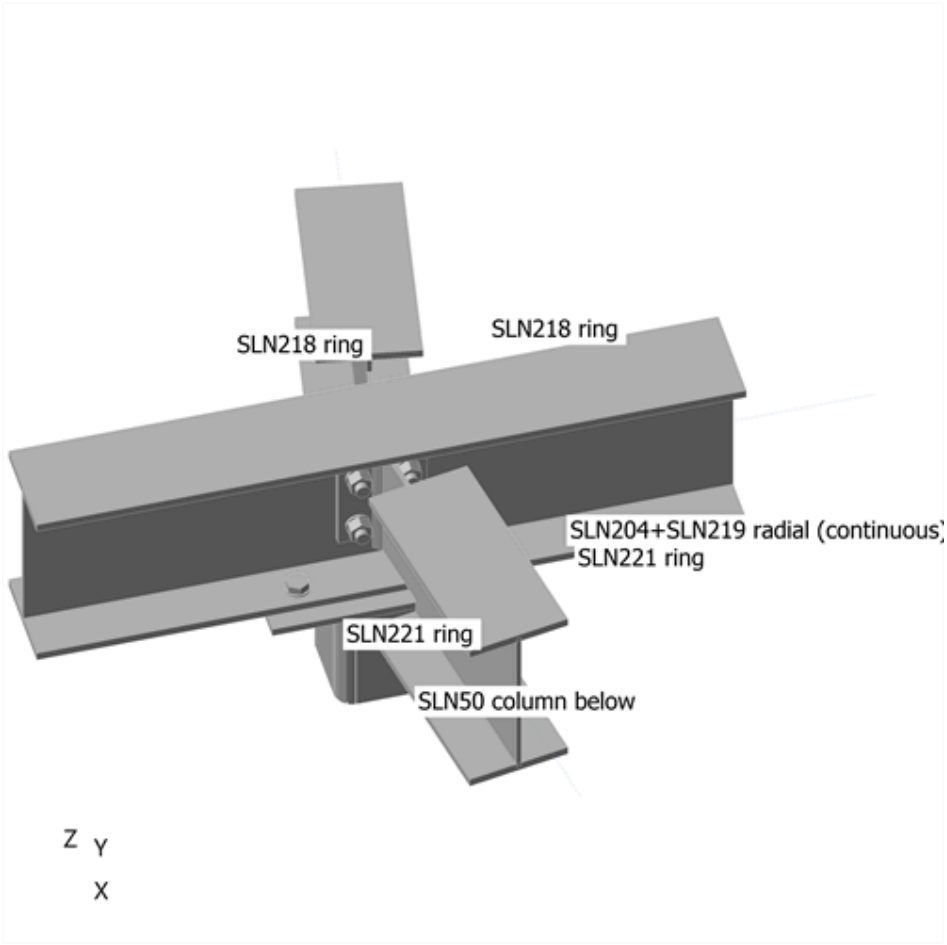
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

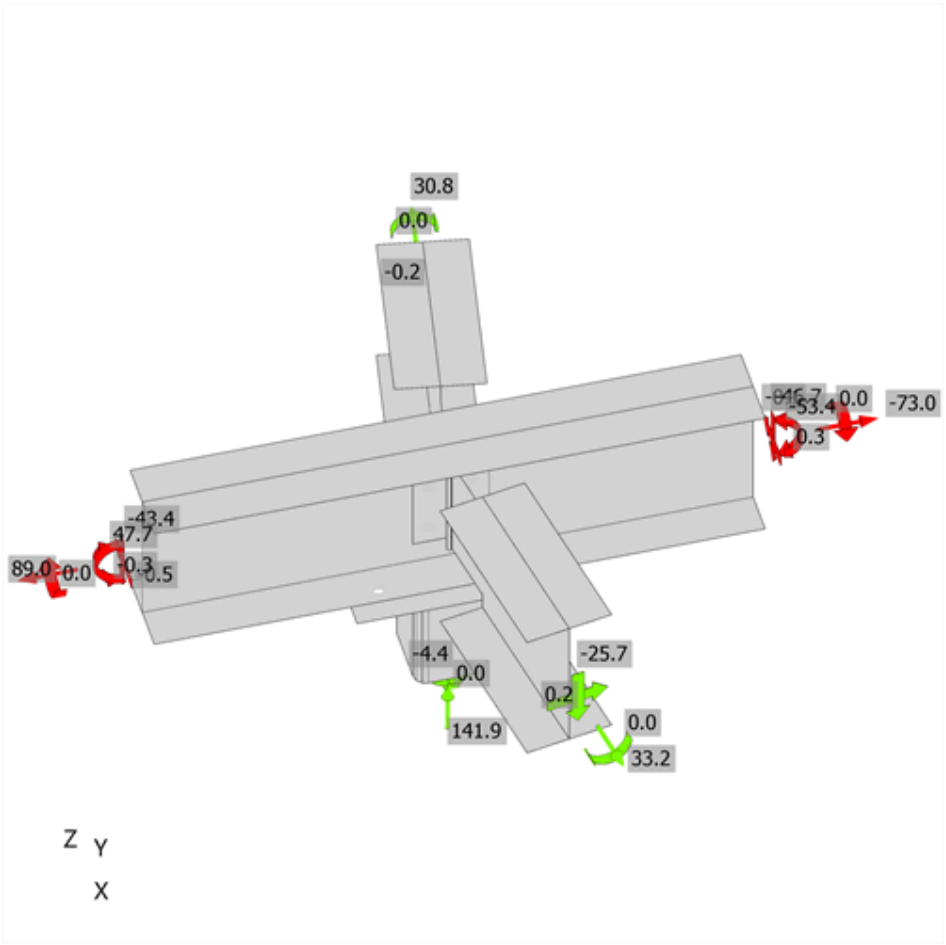
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN50 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

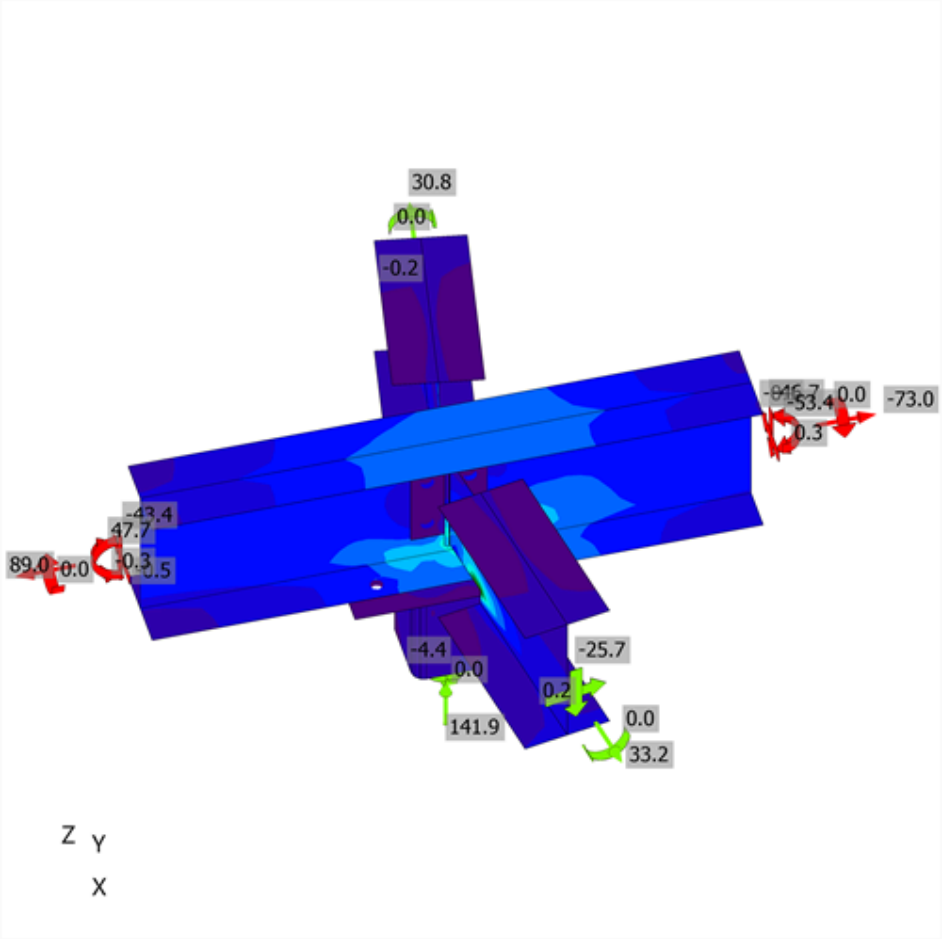
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

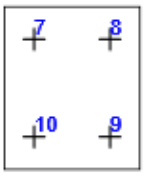


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_123	0.2	4.8	196.7	0.2	5.1	5.2	OK	OK
	B3	M20 8.8 - 1	C LC_110	18.4	2.8	81.9	13.1	3.4	12.3	OK	OK
	B4	M20 8.8 - 1	C LC_110	14.9	1.8	156.4	10.6	2.0	9.5	OK	OK
	B5	M20 8.8 - 1	C LC_126	0.4	5.0	186.8	0.3	5.4	5.6	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.4	0.3	123.8	0.3	0.5	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.6	0.4	123.8	0.4	0.7	0.8	OK	OK
	B9	M20 8.8 - 1	C LC_123	1.7	0.4	63.8	1.2	0.4	1.3	OK	OK
	B10	M20 8.8 - 1	C LC_123	1.6	0.9	123.8	1.2	1.3	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN50 column below	SLN50 column below	◢ 4.2 ◣	521	C LC_122	268.1	0.0	-133.1	-134.3	0.0	66.2	36.8	OK	OK
		◢ 4.2 ◣	521	C LC_122	291.3	0.0	-146.5	145.3	2.1	72.0	40.4	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_119	92.9	0.0	9.7	7.2	-52.8	22.9	14.6	OK	OK
SLN204+SLN219 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_119	90.3	0.0	30.9	-24.2	42.6	22.3	12.9	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_119	81.7	0.0	-21.7	-17.5	42.0	20.2	12.4	OK	OK
		◢ 4.2 ◣	149	C LC_119	96.8	0.0	-18.0	24.2	-49.3	23.9	15.1	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_122	89.3	0.0	-16.4	-22.8	45.3	22.1	14.0	OK	OK
		◢ 4.2 ◣	149	C LC_122	77.2	0.0	-21.7	17.6	-39.0	19.1	11.6	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-A_320
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring

Name	SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

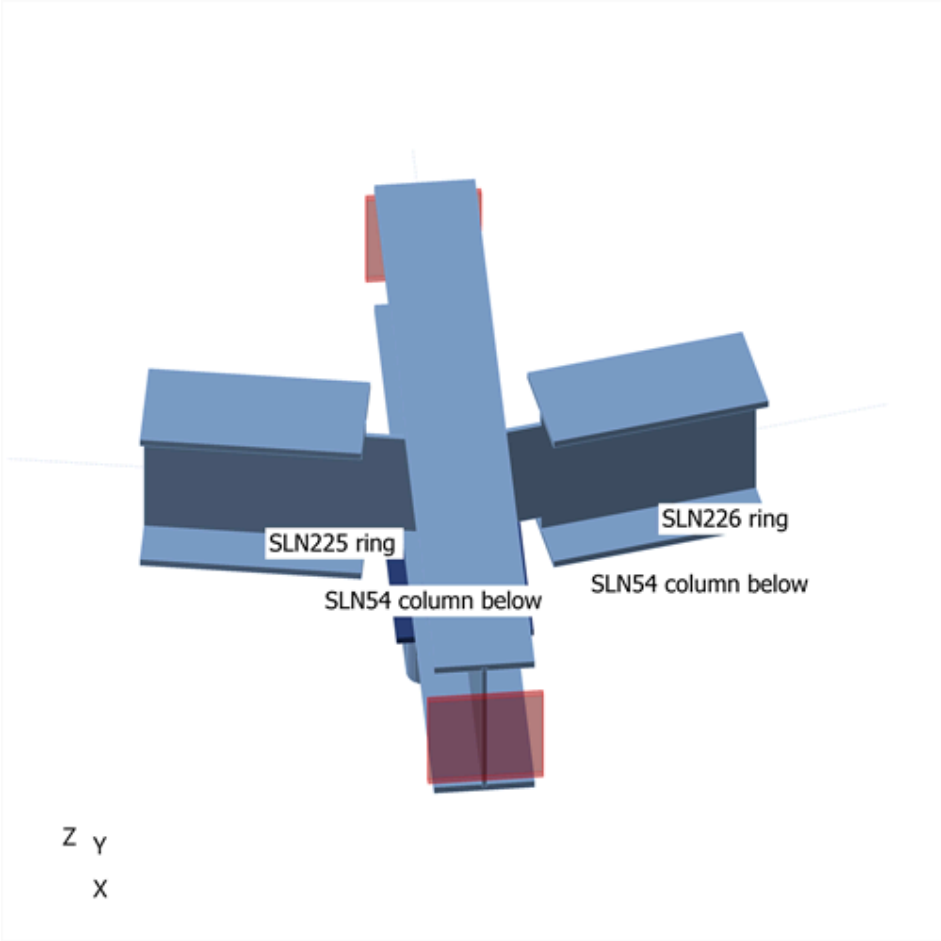
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN208+SLN224 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN54 column below	2 - SHS150x150x10	0.0	90.0	140.0	0	0	0
SLN225 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN208+SLN224 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN208+SLN224 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN54 column below / begin		Position	0
SLN225 ring / end		Position	0
SLN226 ring / end		Position	0



SLN208+SLN224 radial (continuous) - SLN54 column below - SLN225 ring - SLN226 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.3	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.3	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	19.2	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.9	0.2	-25.6	0.0	0.0	0.0
C LC_103	SLN54 column below / Begin	124.1	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	8.2	-0.4	-23.9	0.0	0.0	0.0
	SLN226 ring / End	7.8	0.2	-24.0	0.0	0.0	0.0
C LC_104	SLN54 column below / Begin	123.2	2.2	-3.6	0.0	-3.6	0.0
	SLN225 ring / End	17.5	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	19.6	0.1	-24.0	0.0	0.0	0.0
C LC_105	SLN54 column below / Begin	124.2	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	7.4	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	7.1	0.1	-24.0	0.0	0.0	0.0
C LC_106	SLN54 column below / Begin	123.1	-2.7	-3.1	0.0	-3.3	0.0
	SLN225 ring / End	18.7	-0.5	-23.9	0.0	0.0	0.0
	SLN226 ring / End	15.7	0.2	-24.0	0.0	0.0	0.0
C LC_107	SLN54 column below / Begin	3.5	0.0	-0.1	0.0	-0.2	0.0
	SLN225 ring / End	-10.1	-0.1	-0.5	0.0	0.0	0.0
	SLN226 ring / End	-10.1	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN54 column below / Begin	1.8	4.5	-6.4	0.0	-4.3	0.0
	SLN225 ring / End	5.7	0.0	-0.5	0.0	0.0	0.0
	SLN226 ring / End	10.0	-0.1	-0.5	0.0	0.0	0.0
C LC_109	SLN54 column below / Begin	3.8	0.0	0.0	0.0	-0.1	0.0
	SLN225 ring / End	-11.7	-0.1	-0.5	0.0	0.0	0.0
	SLN226 ring / End	-11.9	0.0	-0.5	0.0	0.0	0.0
C LC_110	SLN54 column below / Begin	1.6	-5.4	-5.4	0.0	-3.6	0.0
	SLN225 ring / End	8.0	0.0	-0.5	0.0	0.0	0.0
	SLN226 ring / End	2.8	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN54 column below / Begin	84.4	0.0	-0.3	0.0	-1.3	0.0
	SLN225 ring / End	-2.0	-0.3	-16.1	0.0	0.0	0.0
	SLN226 ring / End	-2.3	0.2	-16.2	0.0	0.0	0.0
C LC_112	SLN54 column below / Begin	82.7	4.5	-6.7	0.0	-5.3	0.0
	SLN225 ring / End	14.8	-0.3	-16.1	0.0	0.0	0.0
	SLN226 ring / End	19.0	0.0	-16.2	0.0	0.0	0.0
C LC_113	SLN54 column below / Begin	84.6	0.0	-0.3	0.0	-1.2	0.0
	SLN225 ring / End	-3.6	-0.4	-16.1	0.0	0.0	0.0
	SLN226 ring / End	-3.9	0.0	-16.2	0.0	0.0	0.0
C LC_114	SLN54 column below / Begin	82.5	-5.4	-5.6	0.0	-4.7	0.0
	SLN225 ring / End	17.1	-0.4	-16.1	0.0	0.0	0.0
	SLN226 ring / End	11.7	0.2	-16.2	0.0	0.0	0.0
C LC_115	SLN54 column below / Begin	10.3	0.0	-0.1	0.0	-0.3	0.0
	SLN225 ring / End	-8.5	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	-8.6	0.1	-1.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN54 column below / Begin	8.7	4.5	-6.4	0.0	-4.3	0.0
	SLN225 ring / End	7.4	0.0	-1.7	0.0	0.0	0.0
	SLN226 ring / End	11.7	-0.1	-1.7	0.0	0.0	0.0
C LC_117	SLN54 column below / Begin	10.6	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	-10.1	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	-10.3	0.0	-1.7	0.0	0.0	0.0
C LC_118	SLN54 column below / Begin	8.5	-5.4	-5.4	0.0	-3.7	0.0
	SLN225 ring / End	9.6	-0.1	-1.7	0.0	0.0	0.0
	SLN226 ring / End	4.5	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN54 column below / Begin	132.1	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	11.2	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	10.8	0.2	-25.6	0.0	0.0	0.0
C LC_120	SLN54 column below / Begin	131.6	2.2	-3.6	0.0	-3.7	0.0
	SLN225 ring / End	20.7	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	22.7	0.2	-25.6	0.0	0.0	0.0
C LC_121	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.7	0.0
	SLN225 ring / End	10.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	10.2	0.1	-25.6	0.0	0.0	0.0
C LC_122	SLN54 column below / Begin	131.4	-2.7	-3.1	0.0	-3.4	0.0
	SLN225 ring / End	21.8	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.8	0.2	-25.6	0.0	0.0	0.0
C LC_123	SLN54 column below / Begin	100.4	0.0	-0.4	0.0	-1.4	0.0
	SLN225 ring / End	4.1	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	3.7	0.2	-19.4	0.0	0.0	0.0
C LC_124	SLN54 column below / Begin	99.3	4.5	-6.7	0.0	-5.5	0.0
	SLN225 ring / End	21.0	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	25.3	0.1	-19.4	0.0	0.0	0.0
C LC_125	SLN54 column below / Begin	100.7	0.0	-0.3	0.0	-1.4	0.0
	SLN225 ring / End	2.5	-0.4	-19.3	0.0	0.0	0.0
	SLN226 ring / End	2.2	0.0	-19.4	0.0	0.0	0.0
C LC_126	SLN54 column below / Begin	99.1	-5.4	-5.7	0.0	-4.9	0.0
	SLN225 ring / End	23.3	-0.5	-19.3	0.0	0.0	0.0
	SLN226 ring / End	17.8	0.3	-19.4	0.0	0.0	0.0
C LC_127	SLN54 column below / Begin	26.3	0.0	-0.1	0.0	-0.4	0.0
	SLN225 ring / End	-2.5	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	-2.6	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN54 column below / Begin	25.3	4.5	-6.5	0.0	-4.5	0.0
	SLN225 ring / End	13.6	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	18.0	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN54 column below / Begin	26.7	0.0	-0.1	0.0	-0.4	0.0
	SLN225 ring / End	-4.0	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	-4.3	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN54 column below / Begin	25.0	-5.4	-5.5	0.0	-3.9	0.0
	SLN225 ring / End	15.8	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	10.6	0.2	-4.9	0.0	0.0	0.0
C LC_135	SLN54 column below / Begin	132.2	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	16.5	-0.6	-25.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / End	16.2	0.2	-25.6	0.0	0.0	0.0
C LC_141	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.2	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.1	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN54 column below / Begin	26.3	0.0	-0.1	0.0	-0.2	0.0
	SLN225 ring / End	6.5	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.5	0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.2	-0.1	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.2	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN54 column below / Begin	26.3	0.0	0.0	0.0	-0.2	0.0
	SLN225 ring / End	6.5	-0.2	-4.9	0.0	0.0	0.0
	SLN226 ring / End	6.4	0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	18.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.0	0.1	-25.6	0.0	0.0	0.0
C LC_146	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	20.0	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	19.9	0.2	-25.6	0.0	0.0	0.0
C LC_147	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	18.4	-0.5	-25.4	0.0	0.0	0.0
	SLN226 ring / End	18.3	0.1	-25.6	0.0	0.0	0.0
C LC_148	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.6	0.0
	SLN225 ring / End	20.0	-0.6	-25.4	0.0	0.0	0.0
	SLN226 ring / End	19.6	0.2	-25.6	0.0	0.0	0.0
C LC_151	SLN54 column below / Begin	81.5	0.0	-1.3	0.0	-5.2	0.0
	SLN225 ring / End	6.5	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.3	0.1	-11.9	0.0	0.0	0.0
C LC_152	SLN54 column below / Begin	77.6	0.0	0.9	0.0	3.5	0.0
	SLN225 ring / End	21.0	-0.4	-13.9	0.0	0.0	0.0
	SLN226 ring / End	20.7	0.1	-14.1	0.0	0.0	0.0
C LC_153	SLN54 column below / Begin	132.3	0.0	-0.4	0.0	-1.8	0.0
	SLN225 ring / End	19.2	-0.5	-20.6	0.0	0.0	0.0
	SLN226 ring / End	19.0	0.2	-21.0	0.0	0.0	0.0
C LC_154	SLN54 column below / Begin	80.7	0.0	-1.2	0.0	-5.0	0.0
	SLN225 ring / End	6.6	-0.4	-11.7	0.0	0.0	0.0
	SLN226 ring / End	6.5	0.1	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	11.7 < 100%	OK
Welds	63.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN208+SLN224 radial (continuous)-bfl 1	12.7	C LC_120	127.2	0.0	37.1	OK
SLN208+SLN224 radial (continuous)-tfl 1	12.7	C LC_153	64.7	0.0	0.0	OK
SLN208+SLN224 radial (continuous)-w 1	7.2	C LC_126	215.8	0.0	32.7	OK
SLN54 column below	10.0	C LC_120	138.5	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_122	28.2	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_121	22.8	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_122	158.2	0.0	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_120	27.1	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_119	21.3	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_122	123.1	0.0	0.0	OK
End plate column below SLN54 column below	12.0	C LC_108	117.2	0.0	15.4	OK
End plate SLN225 ring	12.0	C LC_126	30.1	0.0	1.3	OK
End plate SLN226 ring	12.0	C LC_124	28.3	0.0	0.9	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

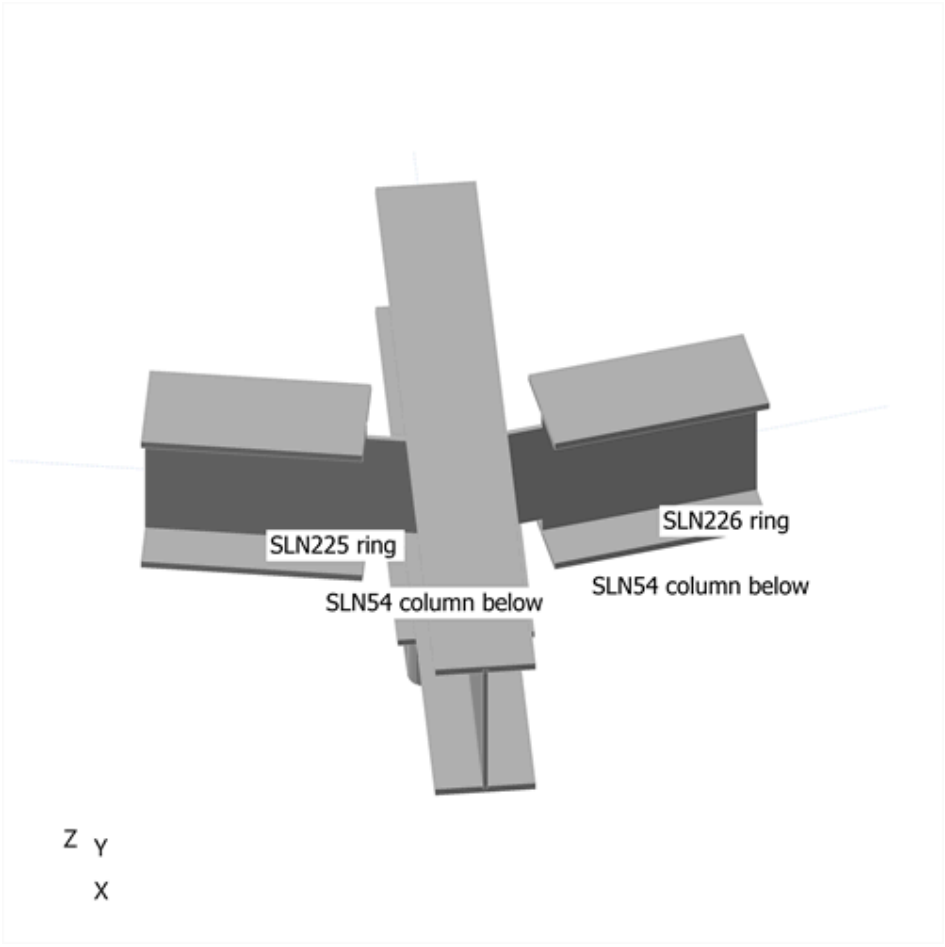
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

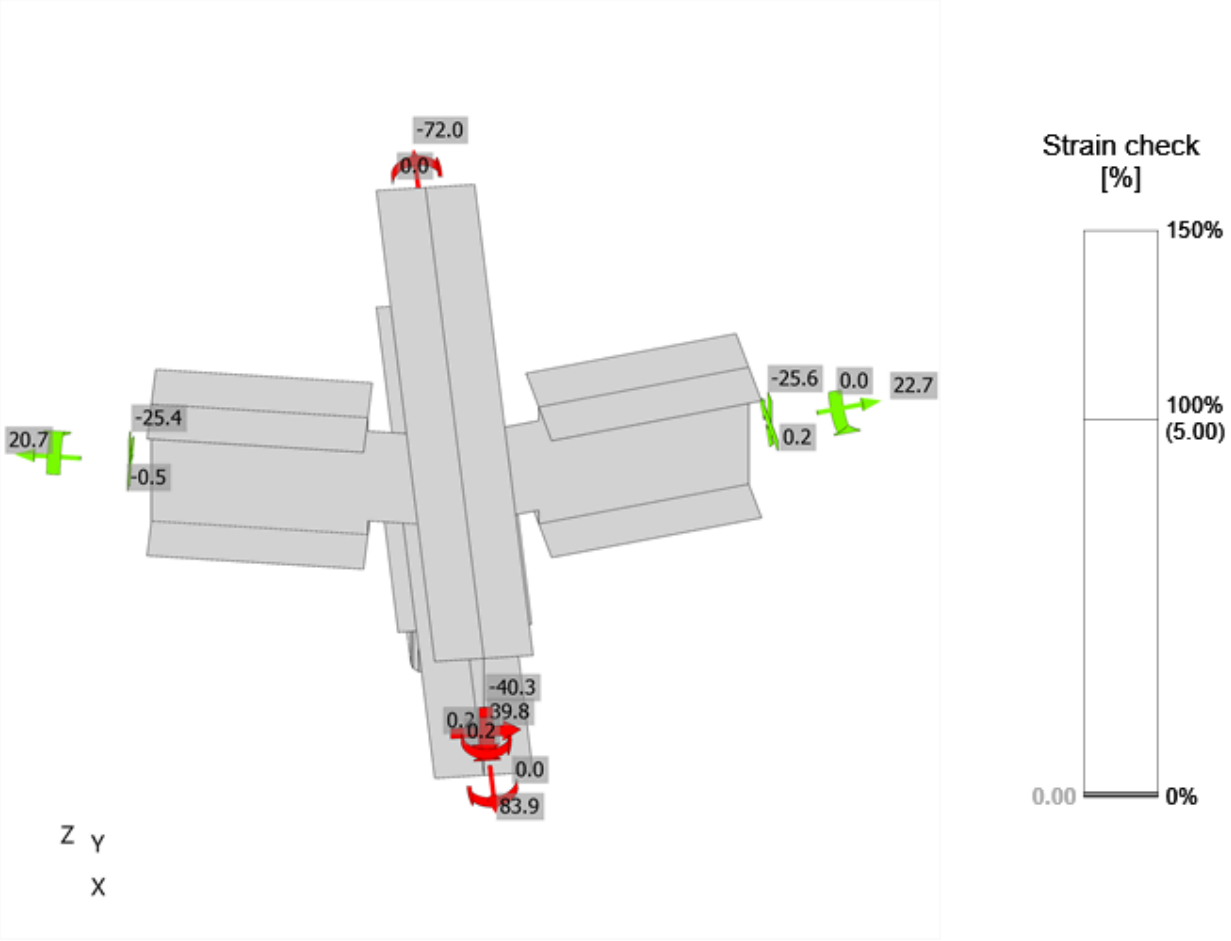
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN54 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

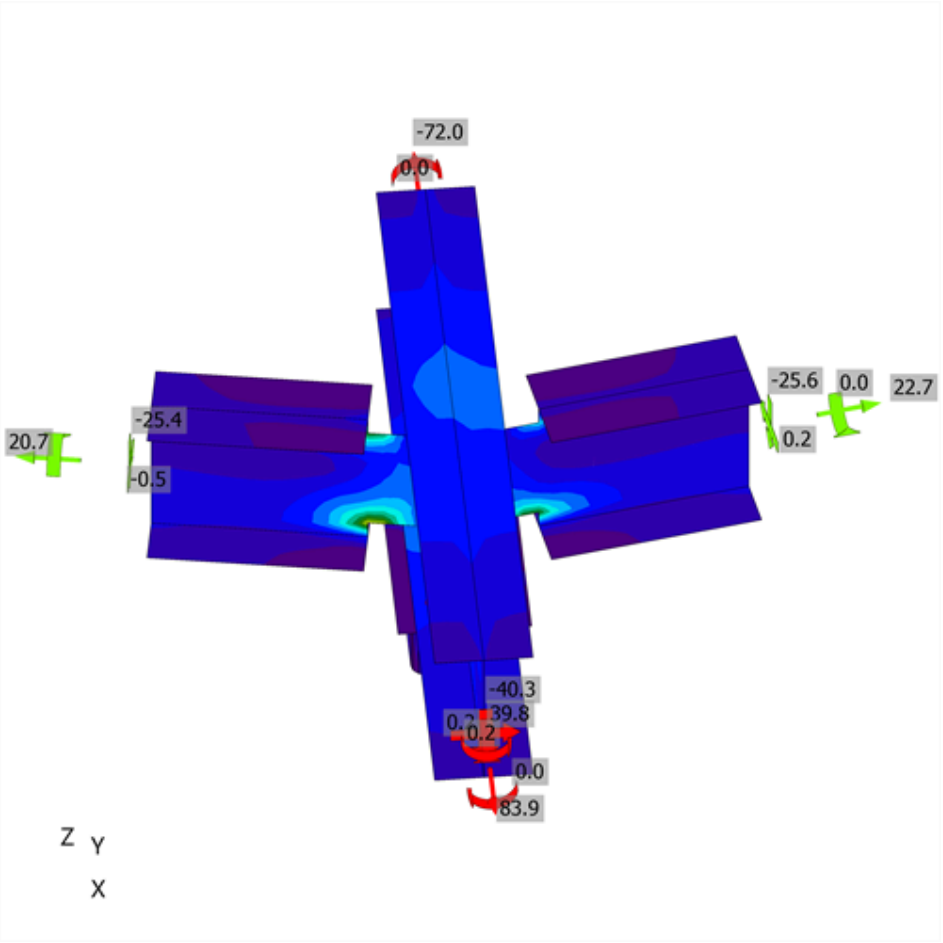
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

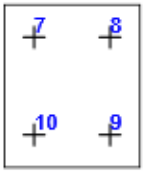


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_126	0.2	4.6	200.1	0.1	4.9	5.0	OK	OK
	B3	M20 8.8 - 1	C LC_108	16.5	2.8	79.4	11.7	3.5	11.3	OK	OK
	B4	M20 8.8 - 1	C LC_110	15.0	2.7	77.0	10.6	3.5	10.4	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.3	4.7	192.0	0.2	5.0	5.2	OK	OK
	B7	M20 8.8 - 1	C LC_126	0.4	0.2	63.3	0.3	0.4	0.5	OK	OK
	B8	M20 8.8 - 1	C LC_126	0.6	0.3	66.7	0.4	0.5	0.6	OK	OK
	B9	M20 8.8 - 1	C LC_126	1.4	0.4	62.3	1.0	0.4	1.1	OK	OK
	B10	M20 8.8 - 1	C LC_126	1.5	0.8	123.8	1.0	1.2	1.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN54 column below	SLN54 column below	◢ 4.2 ◣	521	C LC_120	243.0	0.0	-119.0	-115.8	-39.6	60.1	32.5	OK	OK
		◢ 4.2 ◣	521	C LC_120	258.3	0.0	-121.9	125.1	40.3	63.8	35.0	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_122	83.8	0.0	8.0	4.3	-48.0	20.7	13.1	OK	OK
SLN208+SLN224 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_120	85.2	0.0	7.9	-4.8	48.8	21.1	13.2	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_122	71.0	0.0	-17.9	-14.5	36.9	17.5	10.5	OK	OK
		◢ 4.2 ◣	149	C LC_122	90.6	0.0	-18.6	23.2	-45.7	22.4	14.5	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_120	90.4	0.0	-17.7	-23.7	45.4	22.3	14.1	OK	OK
		◢ 4.2 ◣	149	C LC_120	74.4	0.0	-19.3	15.0	-38.6	18.4	11.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-B_040
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring

Name	SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

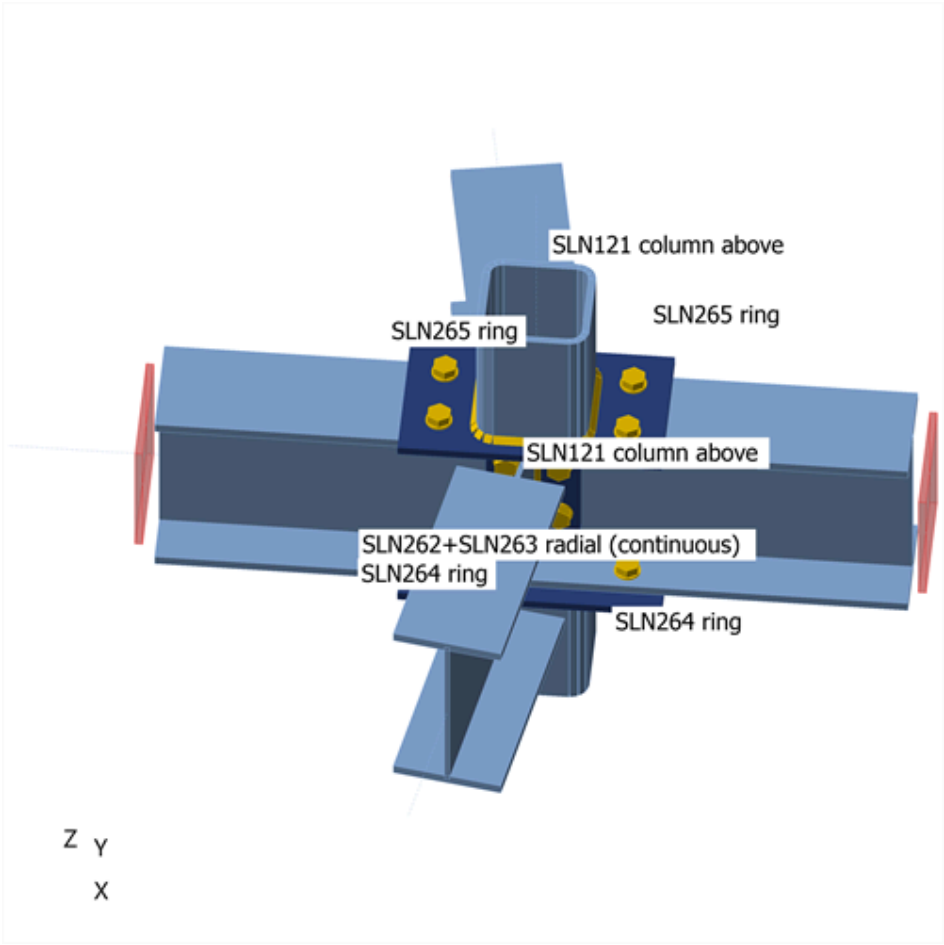
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN262+SLN263 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN40 column below	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN264 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN262+SLN263 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN262+SLN263 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN40 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN264 ring / end		Position	0
SLN265 ring / end		Position	0



SLN262+SLN263 radial (continuous) - SLN40 column below - SLN121 column above - SLN264 ring - SLN265 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.8	0.1	-5.0	0.0	0.0	0.0
C LC_102	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.6	0.1
	SLN264 ring / End	27.5	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.8	0.3	-26.0	0.0	0.0	0.0
C LC_103	SLN40 column below / Begin	220.8	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-61.8	0.1	1.7	-0.1	-1.8	0.2
	SLN264 ring / End	16.0	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	16.5	0.3	-24.4	0.0	0.0	0.0
C LC_104	SLN40 column below / Begin	227.1	-2.5	-3.2	0.0	-1.9	0.9
	SLN121 column above / End	-62.1	-0.2	1.4	-0.2	-0.9	-0.9
	SLN264 ring / End	28.1	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	25.4	0.3	-24.4	0.0	0.0	0.0
C LC_105	SLN40 column below / Begin	221.1	2.9	-2.8	0.0	-2.1	-1.0
	SLN121 column above / End	-62.1	0.3	1.6	-0.1	-1.5	1.0
	SLN264 ring / End	24.3	0.1	-24.0	0.0	0.0	0.0
	SLN265 ring / End	27.7	0.2	-24.4	0.0	0.0	0.0
C LC_106	SLN40 column below / Begin	226.8	0.1	0.1	0.0	0.2	-0.4
	SLN121 column above / End	-61.8	0.1	1.4	-0.1	-1.1	0.4
	SLN264 ring / End	14.5	0.2	-24.0	0.0	0.0	0.0
	SLN265 ring / End	14.9	0.2	-24.4	0.0	0.0	0.0
C LC_107	SLN40 column below / Begin	7.2	0.1	-0.1	0.0	-0.4	-0.3
	SLN121 column above / End	-5.1	0.1	0.1	-0.1	-0.1	0.3
	SLN264 ring / End	-8.7	0.1	-0.5	0.0	0.0	0.0
	SLN265 ring / End	-8.6	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN40 column below / Begin	19.6	-5.0	-6.2	0.0	-3.5	2.0
	SLN121 column above / End	-5.6	-0.5	-0.5	-0.1	1.5	-2.0
	SLN264 ring / End	11.9	0.2	-0.5	0.0	0.0	0.0
	SLN265 ring / End	6.3	0.0	-0.5	0.0	0.0	0.0
C LC_109	SLN40 column below / Begin	7.6	5.8	-5.4	0.0	-3.8	-1.9
	SLN121 column above / End	-5.7	0.5	0.0	0.1	0.3	1.8
	SLN264 ring / End	4.7	0.0	-0.5	0.0	0.0	0.0
	SLN265 ring / End	10.8	-0.2	-0.5	0.0	0.0	0.0
C LC_110	SLN40 column below / Begin	19.0	0.1	0.2	0.0	0.7	-0.5
	SLN121 column above / End	-5.0	0.1	-0.4	0.0	1.3	0.5
	SLN264 ring / End	-12.1	-0.1	-0.5	0.0	0.0	0.0
	SLN265 ring / End	-11.6	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN40 column below / Begin	149.7	0.1	-0.1	0.0	-0.5	-0.3
	SLN121 column above / End	-43.5	0.1	1.2	-0.1	-1.3	0.3
	SLN264 ring / End	4.0	0.2	-16.2	0.0	0.0	0.0
	SLN265 ring / End	4.4	0.3	-16.5	0.0	0.0	0.0
C LC_112	SLN40 column below / Begin	162.1	-5.0	-6.3	0.0	-3.7	2.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN121 column above / End	-43.9	-0.5	0.6	-0.2	0.4	-2.0
	SLN264 ring / End	25.8	0.3	-16.2	0.0	0.0	0.0
	SLN265 ring / End	20.3	0.2	-16.5	0.0	0.0	0.0
C LC_113	SLN40 column below / Begin	150.1	5.8	-5.5	0.0	-4.0	-1.9
	SLN121 column above / End	-44.0	0.5	1.0	0.0	-0.8	1.9
	SLN264 ring / End	18.5	0.0	-16.2	0.0	0.0	0.0
	SLN265 ring / End	24.8	0.0	-16.5	0.0	0.0	0.0
C LC_114	SLN40 column below / Begin	161.5	0.1	0.1	0.0	0.6	-0.6
	SLN121 column above / End	-43.4	0.1	0.7	-0.1	0.1	0.6
	SLN264 ring / End	0.7	0.0	-16.2	0.0	0.0	0.0
	SLN265 ring / End	1.3	0.2	-16.5	0.0	0.0	0.0
C LC_115	SLN40 column below / Begin	20.9	0.1	-0.1	0.0	-0.3	-0.3
	SLN121 column above / End	-9.6	0.1	0.2	-0.1	-0.1	0.3
	SLN264 ring / End	-6.5	0.1	-1.7	0.0	0.0	0.0
	SLN265 ring / End	-6.3	0.2	-1.8	0.0	0.0	0.0
C LC_116	SLN40 column below / Begin	33.3	-5.0	-6.2	0.0	-3.5	2.0
	SLN121 column above / End	-10.1	-0.5	-0.4	-0.1	1.5	-2.0
	SLN264 ring / End	14.2	0.2	-1.7	0.0	0.0	0.0
	SLN265 ring / End	8.6	0.1	-1.8	0.0	0.0	0.0
C LC_117	SLN40 column below / Begin	21.3	5.8	-5.4	0.0	-3.8	-1.9
	SLN121 column above / End	-10.2	0.5	0.1	0.1	0.3	1.9
	SLN264 ring / End	7.0	0.0	-1.7	0.0	0.0	0.0
	SLN265 ring / End	13.1	-0.1	-1.8	0.0	0.0	0.0
C LC_118	SLN40 column below / Begin	32.7	0.1	0.2	0.0	0.8	-0.5
	SLN121 column above / End	-9.5	0.1	-0.3	0.0	1.3	0.5
	SLN264 ring / End	-9.9	-0.1	-1.7	0.0	0.0	0.0
	SLN265 ring / End	-9.4	0.1	-1.8	0.0	0.0	0.0
C LC_119	SLN40 column below / Begin	233.2	0.1	-0.1	0.0	-0.5	-0.3
	SLN121 column above / End	-65.9	0.1	2.1	-0.2	-2.4	0.3
	SLN264 ring / End	19.3	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	19.8	0.3	-26.0	0.0	0.0	0.0
C LC_120	SLN40 column below / Begin	239.2	-2.5	-3.2	0.0	-2.1	1.0
	SLN121 column above / End	-65.6	-0.2	1.4	-0.2	-0.9	-1.0
	SLN264 ring / End	31.7	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	29.0	0.3	-26.0	0.0	0.0	0.0
C LC_121	SLN40 column below / Begin	233.2	2.9	-2.8	0.0	-2.3	-1.0
	SLN121 column above / End	-65.6	0.3	1.6	-0.1	-1.5	1.0
	SLN264 ring / End	28.0	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	31.4	0.2	-26.0	0.0	0.0	0.0
C LC_122	SLN40 column below / Begin	239.2	0.1	0.0	0.0	0.1	-0.4
	SLN121 column above / End	-65.8	0.1	1.8	-0.1	-1.7	0.4
	SLN264 ring / End	17.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	18.1	0.3	-26.0	0.0	0.0	0.0
C LC_123	SLN40 column below / Begin	174.3	0.1	-0.2	0.0	-0.8	-0.5
	SLN121 column above / End	-51.6	0.1	1.9	-0.2	-2.6	0.5
	SLN264 ring / End	10.5	0.3	-19.4	0.0	0.0	0.0
	SLN265 ring / End	11.1	0.3	-19.7	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN40 column below / Begin	186.3	-5.0	-6.4	0.0	-4.1	2.1
	SLN121 column above / End	-50.9	-0.5	0.7	-0.2	0.3	-2.1
	SLN264 ring / End	33.0	0.2	-19.4	0.0	0.0	0.0
	SLN265 ring / End	27.4	0.3	-19.7	0.0	0.0	0.0
C LC_125	SLN40 column below / Begin	174.3	5.8	-5.6	0.0	-4.5	-2.0
	SLN121 column above / End	-51.1	0.5	1.1	0.0	-0.9	1.9
	SLN264 ring / End	25.8	0.0	-19.4	0.0	0.0	0.0
	SLN265 ring / End	32.2	0.1	-19.7	0.0	0.0	0.0
C LC_126	SLN40 column below / Begin	186.2	0.1	0.1	0.0	0.3	-0.6
	SLN121 column above / End	-51.4	0.1	1.3	-0.1	-1.0	0.6
	SLN264 ring / End	7.2	0.0	-19.4	0.0	0.0	0.0
	SLN265 ring / End	7.8	0.2	-19.7	0.0	0.0	0.0
C LC_127	SLN40 column below / Begin	45.5	0.1	-0.1	0.0	-0.6	-0.4
	SLN121 column above / End	-17.7	0.1	0.9	-0.1	-1.5	0.4
	SLN264 ring / End	0.0	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	0.3	0.2	-5.0	0.0	0.0	0.0
C LC_128	SLN40 column below / Begin	57.5	-5.0	-6.3	0.0	-3.9	2.1
	SLN121 column above / End	-17.1	-0.5	-0.3	-0.1	1.4	-2.1
	SLN264 ring / End	21.5	0.2	-4.9	0.0	0.0	0.0
	SLN265 ring / End	15.8	0.1	-5.0	0.0	0.0	0.0
C LC_129	SLN40 column below / Begin	45.5	5.8	-5.6	0.0	-4.3	-1.9
	SLN121 column above / End	-17.3	0.5	0.1	0.1	0.2	1.9
	SLN264 ring / End	14.3	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	20.5	-0.1	-5.0	0.0	0.0	0.0
C LC_130	SLN40 column below / Begin	57.4	0.1	0.1	0.0	0.5	-0.5
	SLN121 column above / End	-17.5	0.1	0.3	0.0	0.1	0.5
	SLN264 ring / End	-3.4	-0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	-2.9	0.1	-5.0	0.0	0.0	0.0
C LC_135	SLN40 column below / Begin	241.8	0.0	0.1	0.0	0.3	-0.1
	SLN121 column above / End	-65.7	0.0	1.5	-0.1	-1.1	0.1
	SLN264 ring / End	27.9	0.3	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.2	0.3	-26.0	0.0	0.0	0.0
C LC_141	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.5	0.1	-4.9	0.0	0.0	0.0
	SLN265 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_142	SLN40 column below / Begin	52.9	0.0	0.0	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_143	SLN40 column below / Begin	52.9	0.0	0.0	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0
	SLN265 ring / End	9.0	0.1	-5.0	0.0	0.0	0.0
C LC_144	SLN40 column below / Begin	52.9	0.0	0.1	0.0	0.2	0.0
	SLN121 column above / End	-17.4	0.0	0.3	0.0	-0.1	0.0
	SLN264 ring / End	8.6	0.1	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN265 ring / End	8.6	0.1	-5.0	0.0	0.0	0.0
C LC_145	SLN40 column below / Begin	237.0	0.0	0.0	0.0	-0.1	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.7	0.1
	SLN264 ring / End	26.5	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	27.0	0.2	-26.0	0.0	0.0	0.0
C LC_146	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.7	0.0	1.7	-0.1	-1.5	0.1
	SLN264 ring / End	28.5	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.6	0.3	-26.0	0.0	0.0	0.0
C LC_147	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.5	0.1
	SLN264 ring / End	28.3	0.1	-25.6	0.0	0.0	0.0
	SLN265 ring / End	28.7	0.3	-26.0	0.0	0.0	0.0
C LC_148	SLN40 column below / Begin	237.1	0.0	0.0	0.0	0.0	-0.1
	SLN121 column above / End	-65.8	0.0	1.7	-0.1	-1.7	0.1
	SLN264 ring / End	26.8	0.2	-25.6	0.0	0.0	0.0
	SLN265 ring / End	26.9	0.2	-26.0	0.0	0.0	0.0
C LC_151	SLN40 column below / Begin	190.0	0.0	-0.7	0.0	-2.9	-0.1
	SLN121 column above / End	-66.3	0.0	3.4	-0.1	-6.0	0.1
	SLN264 ring / End	12.3	0.2	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.5	0.2	-11.3	0.0	0.0	0.0
C LC_152	SLN40 column below / Begin	125.3	0.0	1.3	0.0	5.3	-0.1
	SLN121 column above / End	-16.6	0.0	-2.6	-0.1	7.6	0.1
	SLN264 ring / End	28.2	0.1	-14.0	0.0	0.0	0.0
	SLN265 ring / End	28.3	0.2	-14.0	0.0	0.0	0.0
C LC_153	SLN40 column below / Begin	235.4	0.0	-0.1	0.0	-0.3	-0.1
	SLN121 column above / End	-65.0	0.0	1.7	-0.1	-1.8	0.1
	SLN264 ring / End	27.7	0.2	-20.4	0.0	0.0	0.0
	SLN265 ring / End	27.9	0.3	-20.3	0.0	0.0	0.0
C LC_154	SLN40 column below / Begin	190.1	0.0	-0.7	0.0	-2.8	-0.1
	SLN121 column above / End	-67.0	0.0	3.4	-0.1	-6.0	0.1
	SLN264 ring / End	12.4	0.1	-11.3	0.0	0.0	0.0
	SLN265 ring / End	12.6	0.2	-11.3	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	18.1 < 100%	OK
Welds	73.9 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN262+SLN263 radial (continuous)-bfl 1	12.7	C LC_120	190.8	0.0	54.5	OK
SLN262+SLN263 radial (continuous)-tfl 1	12.7	C LC_152	231.7	0.0	22.9	OK
SLN262+SLN263 radial (continuous)-w 1	7.2	C LC_124	275.4	0.2	46.7	OK
SLN40 column below	10.0	C LC_120	186.3	0.0	0.0	OK
SLN121 column above	10.0	C LC_152	193.3	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_120	28.9	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_122	19.9	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_135	141.1	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_120	29.2	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_122	20.7	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_120	146.7	0.0	0.0	OK
End plate column below SLN40 column below	12.0	C LC_120	146.1	0.0	48.1	OK
End plate column above SLN121 column above	12.0	C LC_152	184.9	0.0	23.3	OK
End plate SLN264 ring	12.0	C LC_125	87.8	0.0	38.3	OK
End plate SLN265 ring	12.0	C LC_124	93.1	0.0	40.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

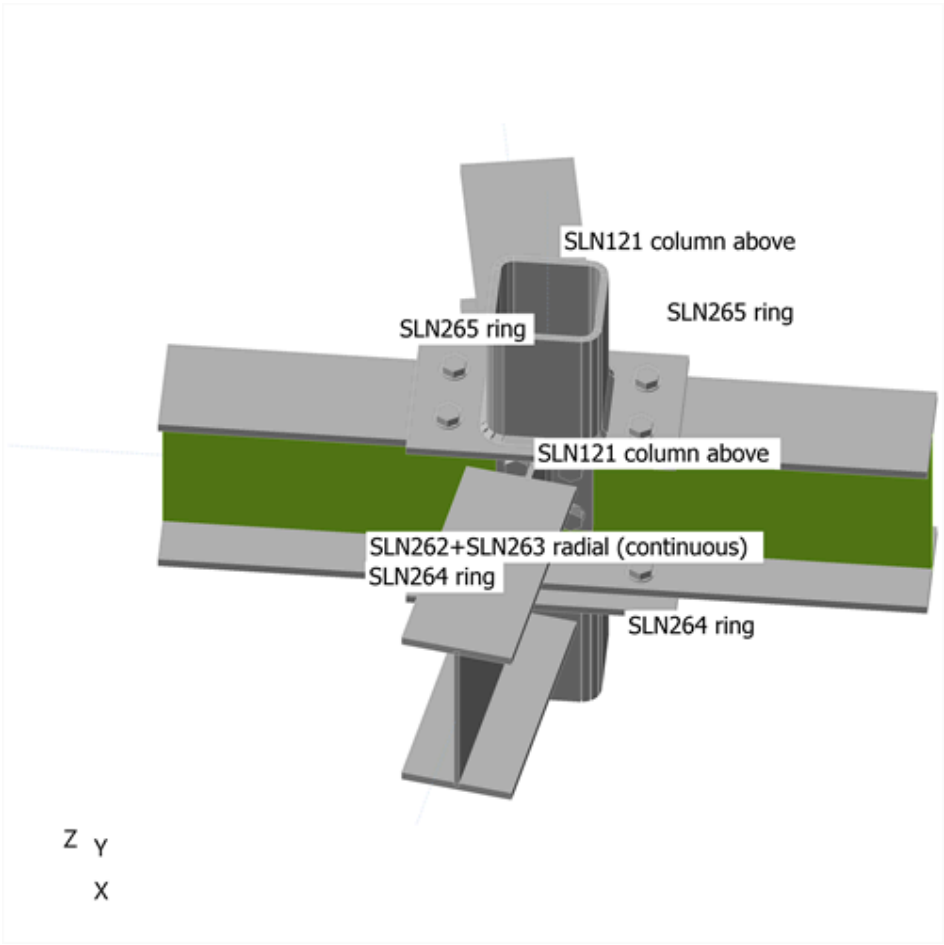
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

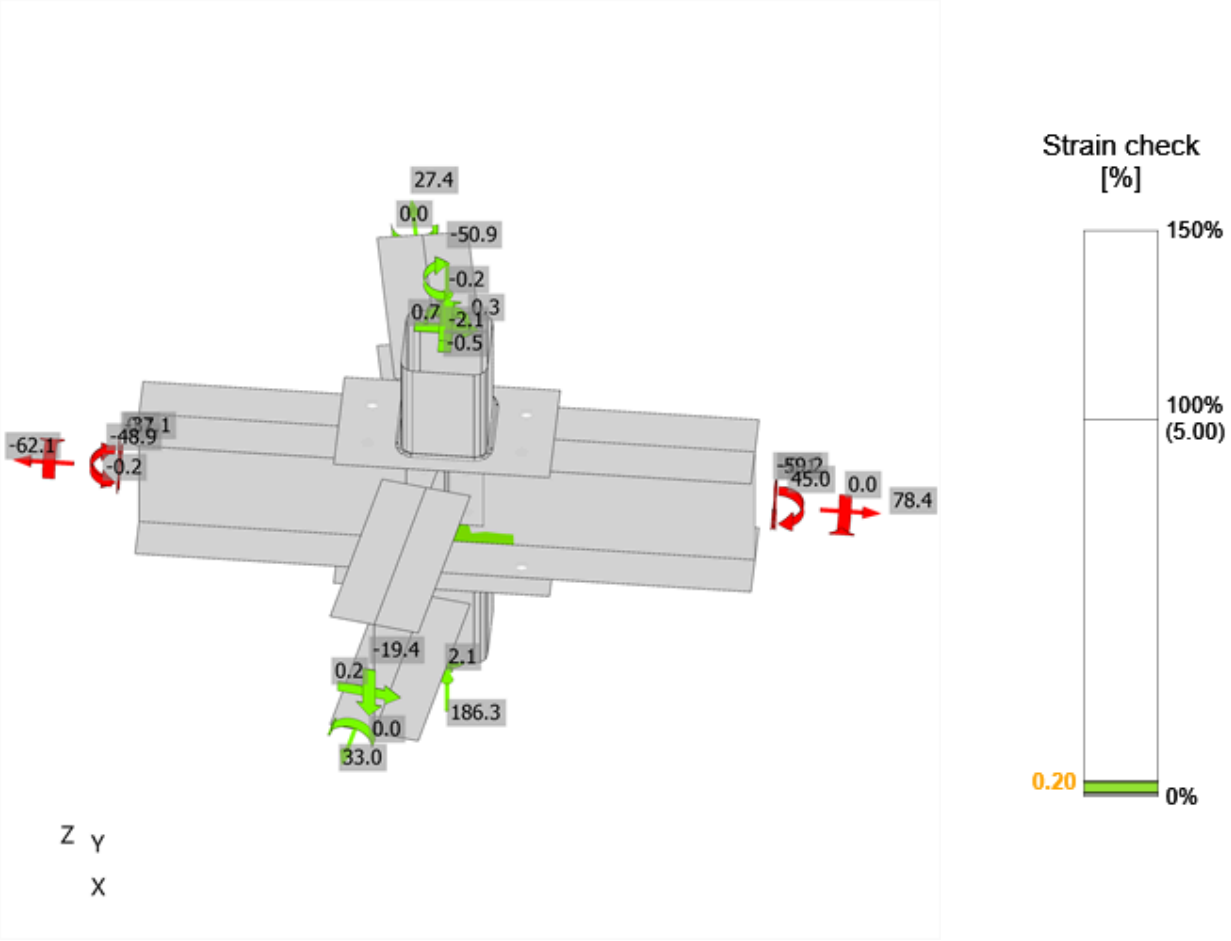
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN40 column below	150	C LC_154	0	5	0.0	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK

Symbol explanation

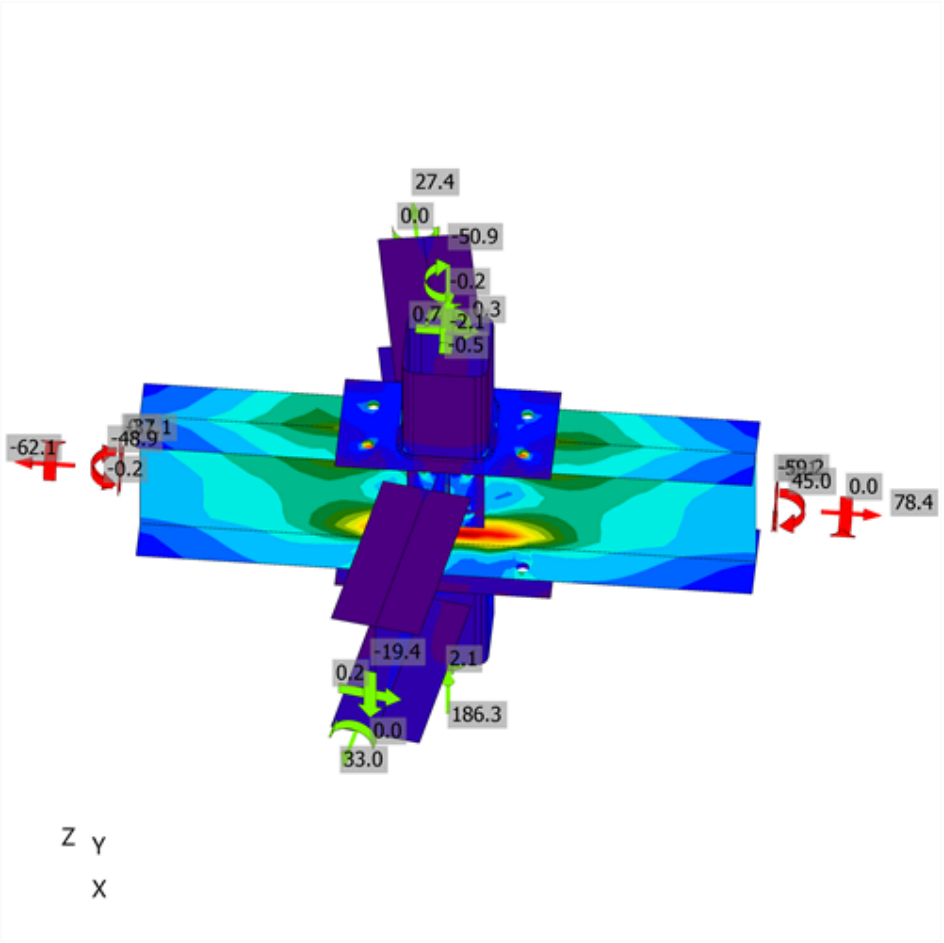
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

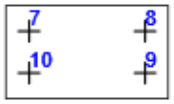
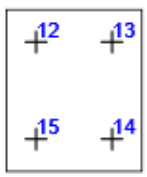


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_124	4.3	5.1	199.1	3.1	5.4	7.6	OK	OK
	B3	M20 8.8 - 1	C LC_108	10.9	2.3	156.4	7.7	2.4	7.9	OK	OK
	B4	M20 8.8 - 1	C LC_109	15.3	2.4	156.4	10.9	2.5	10.3	OK	OK
	B5	M20 8.8 - 1	C LC_124	0.4	6.3	170.1	0.3	6.7	6.9	OK	OK
	B7	M20 8.8 - 1	C LC_125	7.5	3.2	156.4	5.3	3.4	7.2	OK	OK
	B8	M20 8.8 - 1	C LC_152	25.5	3.3	94.1	18.1	3.5	16.4	OK	OK
	B9	M20 8.8 - 1	C LC_152	25.0	3.4	88.3	17.7	3.9	16.3	OK	OK
	B10	M20 8.8 - 1	C LC_124	7.9	3.4	156.4	5.6	3.6	7.6	OK	OK
	B12	M20 8.8 - 1	C LC_128	5.0	0.9	72.9	3.5	1.2	3.5	OK	OK
	B13	M20 8.8 - 1	C LC_124	4.8	1.1	63.5	3.4	1.7	3.6	OK	OK
	B14	M20 8.8 - 1	C LC_128	4.2	0.9	72.3	3.0	1.0	3.1	OK	OK
	B15	M20 8.8 - 1	C LC_124	3.9	1.2	69.7	2.8	1.3	3.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN40 column below	SLN40 column below	◢ 5.6 ◣	521	C LC_120	258.3	0.0	-130.0	-128.7	6.1	63.8	35.2	OK	OK
		◢ 5.6 ◣	521	C LC_120	266.1	0.0	-132.1	133.4	3.0	65.8	36.7	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	521	C LC_154	134.0	0.0	-66.4	-66.3	-10.8	33.1	17.9	OK	OK
		◢ 5.6 ◣	521	C LC_152	146.8	0.0	104.1	-51.9	29.7	36.3	22.4	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_125	283.9	0.0	-9.2	9.3	163.5	70.1	47.4	OK	OK
SLN262+SLN263 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_124	299.2	0.0	-10.2	-9.4	-172.4	73.9	50.0	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_125	206.0	0.0	-56.5	-48.4	103.7	50.9	30.5	OK	OK
		◢ 4.2 ◣	149	C LC_125	240.8	0.0	-45.8	59.1	-123.0	59.5	35.7	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_124	257.7	0.0	-53.5	-64.8	130.3	63.7	37.7	OK	OK
		◢ 4.2 ◣	149	C LC_124	210.8	0.0	-55.4	47.9	-107.2	52.1	31.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-B_280
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring

Name	SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

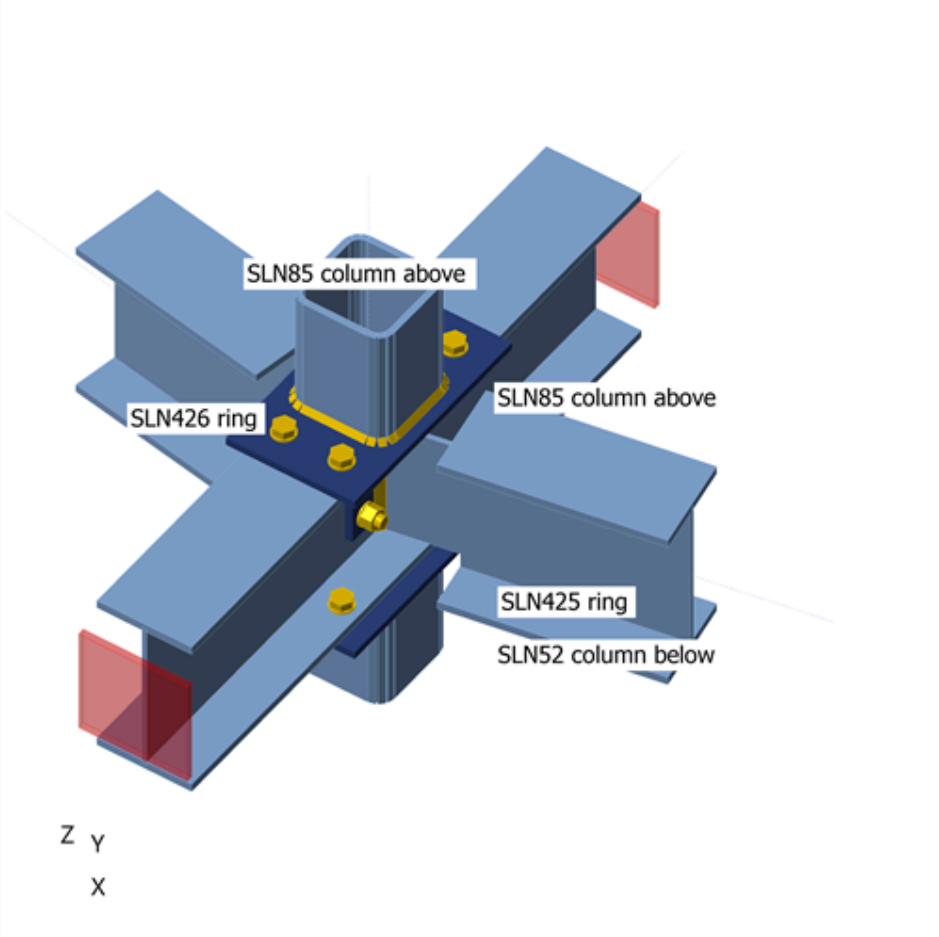
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN206+SLN222 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN52 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN85 column above	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	-180.0	0.0	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN206+SLN222 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN206+SLN222 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN52 column below / begin		Position	0
SLN85 column above / end		Position	0
SLN425 ring / end		Position	0
SLN426 ring / end		Position	0



SLN206+SLN222 radial (continuous) - SLN52 column below - SLN85 column above - SLN425 ring - SLN426 ring, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_102	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.3	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	0.0
C LC_103	SLN52 column below / Begin	222.6	0.0	-0.1	0.0	-0.3	-0.2
	SLN85 column above / End	-64.4	0.1	1.9	0.0	-2.0	0.3
	SLN425 ring / End	16.5	-0.1	-23.9	0.0	0.0	0.0
	SLN426 ring / End	16.8	-0.2	-23.9	0.0	0.0	0.0
C LC_104	SLN52 column below / Begin	230.0	3.8	-0.6	0.0	-0.1	-1.6
	SLN85 column above / End	-64.4	0.4	1.6	0.1	-1.0	1.6
	SLN425 ring / End	20.8	-0.1	-23.9	0.0	0.0	0.0
	SLN426 ring / End	16.6	-0.2	-23.9	0.0	0.0	0.0
C LC_105	SLN52 column below / Begin	226.4	-0.1	0.0	0.0	0.0	0.4
	SLN85 column above / End	-64.2	-0.1	1.8	0.1	-1.6	-0.4
	SLN425 ring / End	13.8	-0.2	-23.9	0.0	0.0	0.0
	SLN426 ring / End	14.3	-0.3	-23.9	0.0	0.0	0.0
C LC_106	SLN52 column below / Begin	226.0	-0.6	-4.1	0.0	-2.7	0.2
	SLN85 column above / End	-64.6	0.0	1.7	0.1	-1.4	-0.1
	SLN425 ring / End	29.7	-0.2	-23.9	0.0	0.0	0.0
	SLN426 ring / End	30.5	-0.2	-23.9	0.0	0.0	0.0
C LC_107	SLN52 column below / Begin	6.6	0.1	-0.1	0.0	-0.4	-0.6
	SLN85 column above / End	-5.6	0.1	0.2	-0.1	-0.3	0.6
	SLN425 ring / End	-8.5	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-8.9	0.1	-0.5	0.0	0.0	0.0
C LC_108	SLN52 column below / Begin	21.2	7.7	-1.2	0.0	0.0	-3.3
	SLN85 column above / End	-5.5	0.8	-0.5	0.0	1.6	3.3
	SLN425 ring / End	-1.1	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-9.4	0.0	-0.5	0.0	0.0	0.0
C LC_109	SLN52 column below / Begin	14.1	-0.2	0.0	0.0	0.2	0.7
	SLN85 column above / End	-5.1	-0.2	-0.2	0.1	0.7	-0.7
	SLN425 ring / End	-12.3	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-11.7	-0.2	-0.5	0.0	0.0	0.0
C LC_110	SLN52 column below / Begin	13.3	-1.3	-8.2	0.0	-5.3	0.2
	SLN85 column above / End	-6.0	-0.1	-0.1	0.0	0.8	-0.2
	SLN425 ring / End	15.0	-0.1	-0.5	0.0	0.0	0.0
	SLN426 ring / End	16.2	0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN52 column below / Begin	150.4	0.1	-0.1	0.0	-0.5	-0.5
	SLN85 column above / End	-45.4	0.1	1.4	-0.1	-1.6	0.6
	SLN425 ring / End	4.4	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	4.3	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN52 column below / Begin	165.1	7.7	-1.2	0.0	-0.1	-3.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN85 column above / End	-45.3	0.8	0.7	0.0	0.3	3.3
	SLN425 ring / End	12.3	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	3.9	-0.1	-16.2	0.0	0.0	0.0
C LC_113	SLN52 column below / Begin	157.9	-0.2	0.0	0.0	0.1	0.7
	SLN85 column above / End	-45.0	-0.2	1.1	0.2	-0.6	-0.7
	SLN425 ring / End	0.2	-0.1	-16.2	0.0	0.0	0.0
	SLN426 ring / End	0.9	-0.3	-16.2	0.0	0.0	0.0
C LC_114	SLN52 column below / Begin	157.1	-1.3	-8.2	0.0	-5.4	0.3
	SLN85 column above / End	-45.8	-0.1	1.1	0.1	-0.4	-0.2
	SLN425 ring / End	28.9	-0.2	-16.2	0.0	0.0	0.0
	SLN426 ring / End	30.2	-0.1	-16.2	0.0	0.0	0.0
C LC_115	SLN52 column below / Begin	20.5	0.1	-0.1	0.0	-0.3	-0.6
	SLN85 column above / End	-10.2	0.1	0.3	-0.1	-0.3	0.6
	SLN425 ring / End	-6.3	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-6.7	0.1	-1.7	0.0	0.0	0.0
C LC_116	SLN52 column below / Begin	35.1	7.7	-1.2	0.0	0.0	-3.3
	SLN85 column above / End	-10.2	0.8	-0.4	0.0	1.6	3.3
	SLN425 ring / End	1.2	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-7.2	0.0	-1.7	0.0	0.0	0.0
C LC_117	SLN52 column below / Begin	28.0	-0.2	0.1	0.0	0.2	0.7
	SLN85 column above / End	-9.8	-0.2	-0.1	0.1	0.7	-0.7
	SLN425 ring / End	-10.1	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-9.5	-0.2	-1.7	0.0	0.0	0.0
C LC_118	SLN52 column below / Begin	27.2	-1.3	-8.2	0.0	-5.2	0.2
	SLN85 column above / End	-10.6	-0.1	-0.1	0.0	0.7	-0.2
	SLN425 ring / End	17.3	-0.2	-1.7	0.0	0.0	0.0
	SLN426 ring / End	18.5	0.1	-1.7	0.0	0.0	0.0
C LC_119	SLN52 column below / Begin	234.9	0.0	-0.1	0.0	-0.5	-0.2
	SLN85 column above / End	-68.4	0.1	2.1	0.0	-2.3	0.2
	SLN425 ring / End	19.9	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	20.3	-0.2	-25.5	0.0	0.0	0.0
C LC_120	SLN52 column below / Begin	242.3	3.8	-0.7	0.0	-0.3	-1.6
	SLN85 column above / End	-68.2	0.4	1.7	0.1	-1.3	1.7
	SLN425 ring / End	24.2	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	20.1	-0.3	-25.5	0.0	0.0	0.0
C LC_121	SLN52 column below / Begin	238.8	-0.1	0.0	0.0	-0.1	0.5
	SLN85 column above / End	-68.5	-0.1	2.2	0.2	-2.3	-0.4
	SLN425 ring / End	17.0	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	17.6	-0.3	-25.5	0.0	0.0	0.0
C LC_122	SLN52 column below / Begin	238.1	-0.6	-4.2	0.0	-2.9	0.2
	SLN85 column above / End	-68.1	0.0	1.7	0.1	-1.3	-0.1
	SLN425 ring / End	33.3	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	34.2	-0.3	-25.5	0.0	0.0	0.0
C LC_123	SLN52 column below / Begin	175.1	0.1	-0.2	0.0	-0.9	-0.4
	SLN85 column above / End	-53.4	0.1	1.8	0.0	-2.2	0.5
	SLN425 ring / End	11.3	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	11.4	-0.1	-19.3	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN52 column below / Begin	189.8	7.7	-1.3	0.0	-0.5	-3.3
	SLN85 column above / End	-53.0	0.8	1.0	0.0	-0.1	3.4
	SLN425 ring / End	19.2	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	10.8	-0.2	-19.3	0.0	0.0	0.0
C LC_125	SLN52 column below / Begin	182.9	-0.2	0.0	0.0	-0.1	0.8
	SLN85 column above / End	-53.6	-0.2	1.9	0.2	-2.1	-0.7
	SLN425 ring / End	6.6	-0.1	-19.3	0.0	0.0	0.0
	SLN426 ring / End	7.3	-0.3	-19.3	0.0	0.0	0.0
C LC_126	SLN52 column below / Begin	181.5	-1.3	-8.3	0.0	-5.9	0.2
	SLN85 column above / End	-52.9	0.0	1.1	0.1	-0.3	-0.2
	SLN425 ring / End	36.3	-0.2	-19.3	0.0	0.0	0.0
	SLN426 ring / End	37.6	-0.2	-19.3	0.0	0.0	0.0
C LC_127	SLN52 column below / Begin	45.2	0.1	-0.2	0.0	-0.7	-0.5
	SLN85 column above / End	-18.2	0.1	0.7	-0.1	-1.0	0.5
	SLN425 ring / End	0.5	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	0.3	0.1	-4.9	0.0	0.0	0.0
C LC_128	SLN52 column below / Begin	59.8	7.7	-1.3	0.0	-0.4	-3.4
	SLN85 column above / End	-17.9	0.9	-0.1	0.0	1.1	3.4
	SLN425 ring / End	8.1	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	-0.3	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN52 column below / Begin	52.9	-0.2	0.0	0.0	0.0	0.7
	SLN85 column above / End	-18.4	-0.2	0.8	0.1	-0.9	-0.7
	SLN425 ring / End	-3.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	-3.0	-0.2	-4.9	0.0	0.0	0.0
C LC_130	SLN52 column below / Begin	51.5	-1.3	-8.3	0.0	-5.7	0.2
	SLN85 column above / End	-17.8	0.0	0.0	0.0	0.8	-0.2
	SLN425 ring / End	24.7	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	25.9	0.0	-4.9	0.0	0.0	0.0
C LC_135	SLN52 column below / Begin	243.9	0.0	0.1	0.0	0.4	0.1
	SLN85 column above / End	-68.3	0.0	1.8	0.1	-1.3	0.0
	SLN425 ring / End	27.7	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	28.1	-0.3	-25.5	0.0	0.0	0.0
C LC_141	SLN52 column below / Begin	53.4	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.6	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_142	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.7	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	0.0
C LC_143	SLN52 column below / Begin	53.4	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.2	0.0
	SLN425 ring / End	8.4	-0.1	-4.9	0.0	0.0	0.0
	SLN426 ring / End	8.5	-0.1	-4.9	0.0	0.0	0.0
C LC_144	SLN52 column below / Begin	53.5	0.0	0.1	0.0	0.2	0.0
	SLN85 column above / End	-18.0	0.0	0.4	0.0	-0.1	0.0
	SLN425 ring / End	8.9	0.0	-4.9	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN426 ring / End	9.0	-0.1	-4.9	0.0	0.0	0.0
C LC_145	SLN52 column below / Begin	239.2	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.0	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.5	-0.3	-25.5	0.0	0.0	0.0
C LC_146	SLN52 column below / Begin	239.4	0.0	0.0	0.0	0.1	0.1
	SLN85 column above / End	-68.3	0.0	2.0	0.1	-1.8	0.0
	SLN425 ring / End	27.5	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	0.0
C LC_147	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.0	0.1
	SLN85 column above / End	-68.4	0.0	2.0	0.1	-1.9	0.0
	SLN425 ring / End	26.1	-0.2	-25.5	0.0	0.0	0.0
	SLN426 ring / End	26.4	-0.2	-25.5	0.0	0.0	0.0
C LC_148	SLN52 column below / Begin	239.3	0.0	0.0	0.0	0.1	0.1
	SLN85 column above / End	-68.3	0.0	1.9	0.1	-1.7	0.0
	SLN425 ring / End	28.5	-0.1	-25.5	0.0	0.0	0.0
	SLN426 ring / End	28.8	-0.3	-25.5	0.0	0.0	0.0
C LC_151	SLN52 column below / Begin	191.9	0.0	-0.7	0.0	-2.8	0.0
	SLN85 column above / End	-68.2	0.0	3.5	0.1	-6.1	0.0
	SLN425 ring / End	12.2	-0.1	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.4	-0.2	-11.9	0.0	0.0	0.0
C LC_152	SLN52 column below / Begin	127.3	0.0	1.3	0.0	5.4	0.2
	SLN85 column above / End	-18.0	0.0	-2.5	0.1	7.6	-0.1
	SLN425 ring / End	27.8	-0.1	-14.0	0.0	0.0	0.0
	SLN426 ring / End	28.0	-0.2	-14.1	0.0	0.0	0.0
C LC_153	SLN52 column below / Begin	238.7	0.0	0.0	0.0	-0.2	0.1
	SLN85 column above / End	-67.6	0.0	1.9	0.1	-1.9	0.0
	SLN425 ring / End	27.4	-0.1	-20.6	0.0	0.0	0.0
	SLN426 ring / End	27.7	-0.3	-21.1	0.0	0.0	0.0
C LC_154	SLN52 column below / Begin	192.1	0.0	-0.7	0.0	-2.7	0.0
	SLN85 column above / End	-68.9	0.0	3.6	0.1	-6.1	0.0
	SLN425 ring / End	12.3	-0.1	-11.5	0.0	0.0	0.0
	SLN426 ring / End	12.5	-0.2	-11.9	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.6 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	17.8 < 100%	OK
Welds	93.8 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN206+SLN222 radial (continuous)-bfl 1	12.7	C LC_120	185.4	0.0	51.6	OK
SLN206+SLN222 radial (continuous)-tfl 1	12.7	C LC_124	256.8	0.0	17.0	OK
SLN206+SLN222 radial (continuous)-w 1	7.2	C LC_124	276.3	0.6	40.4	OK
SLN52 column below	10.0	C LC_122	197.4	0.0	0.0	OK
SLN85 column above	10.0	C LC_108	192.7	0.0	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_122	28.9	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_121	20.0	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_135	133.7	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_122	29.3	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_121	20.5	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_122	144.3	0.0	0.0	OK
End plate column below SLN52 column below	12.0	C LC_122	149.7	0.0	52.2	OK
End plate column above SLN85 column above	12.0	C LC_152	183.4	0.0	22.8	OK
End plate SLN426 ring	12.0	C LC_124	120.2	0.0	36.2	OK
End plate SLN425 ring	12.0	C LC_128	132.9	0.0	31.2	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

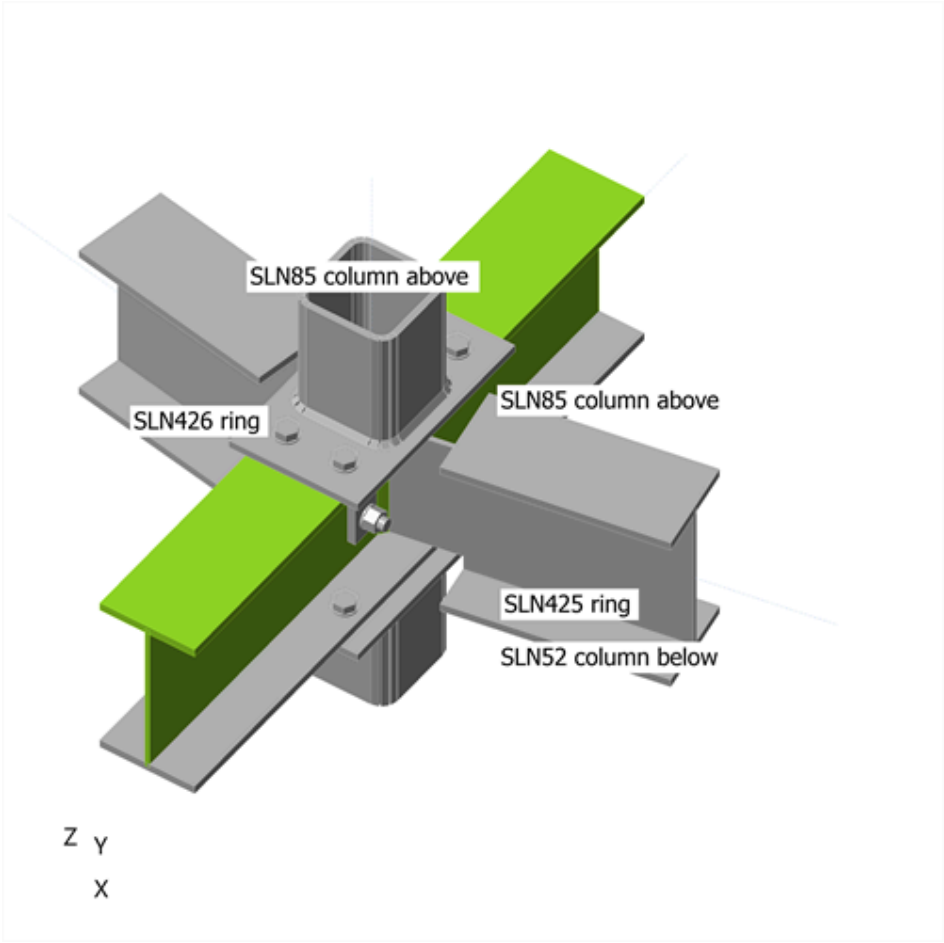
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

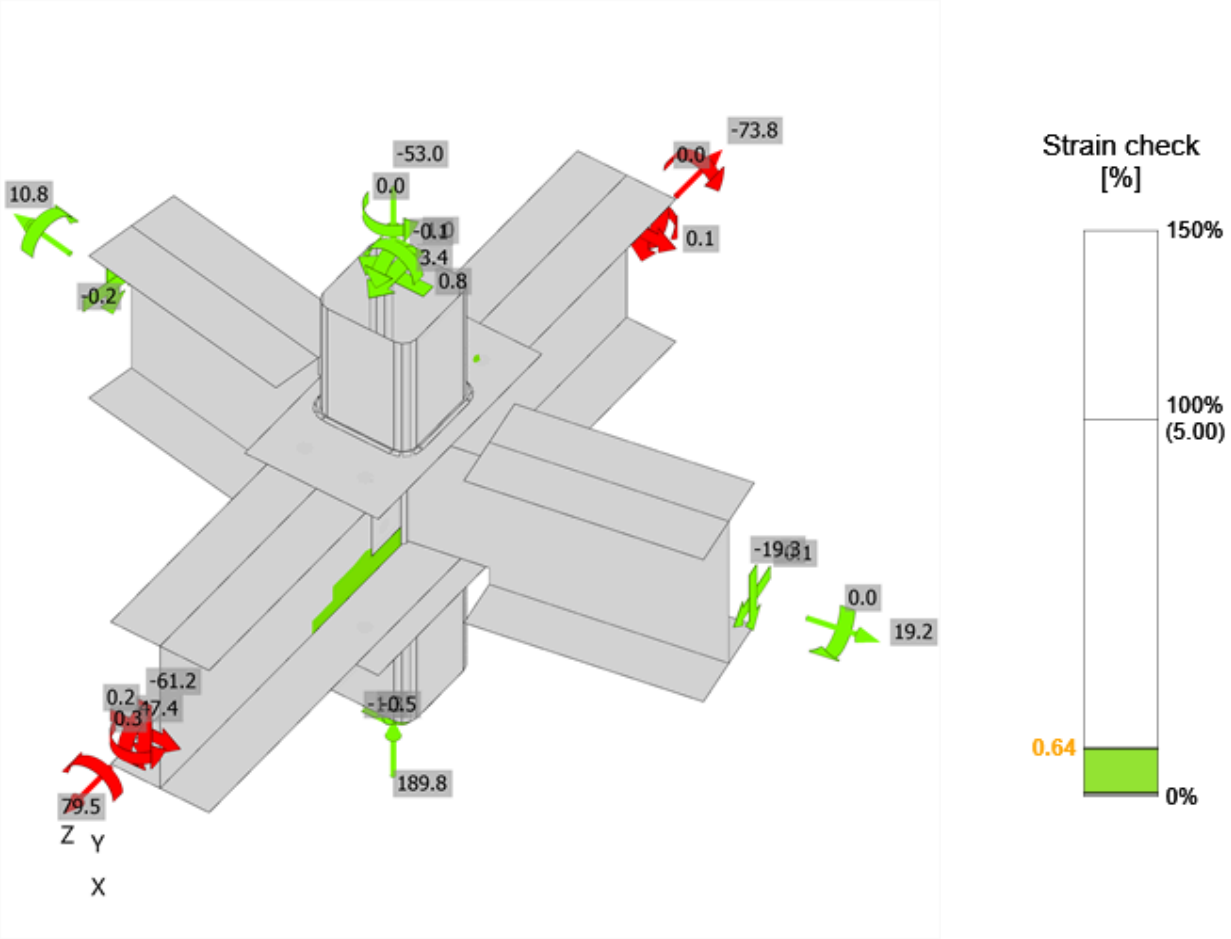
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN52 column below	150	C LC_154	0	5	0.0	OK
SLN85 column above	150	C LC_154	0	5	0.0	OK

Symbol explanation

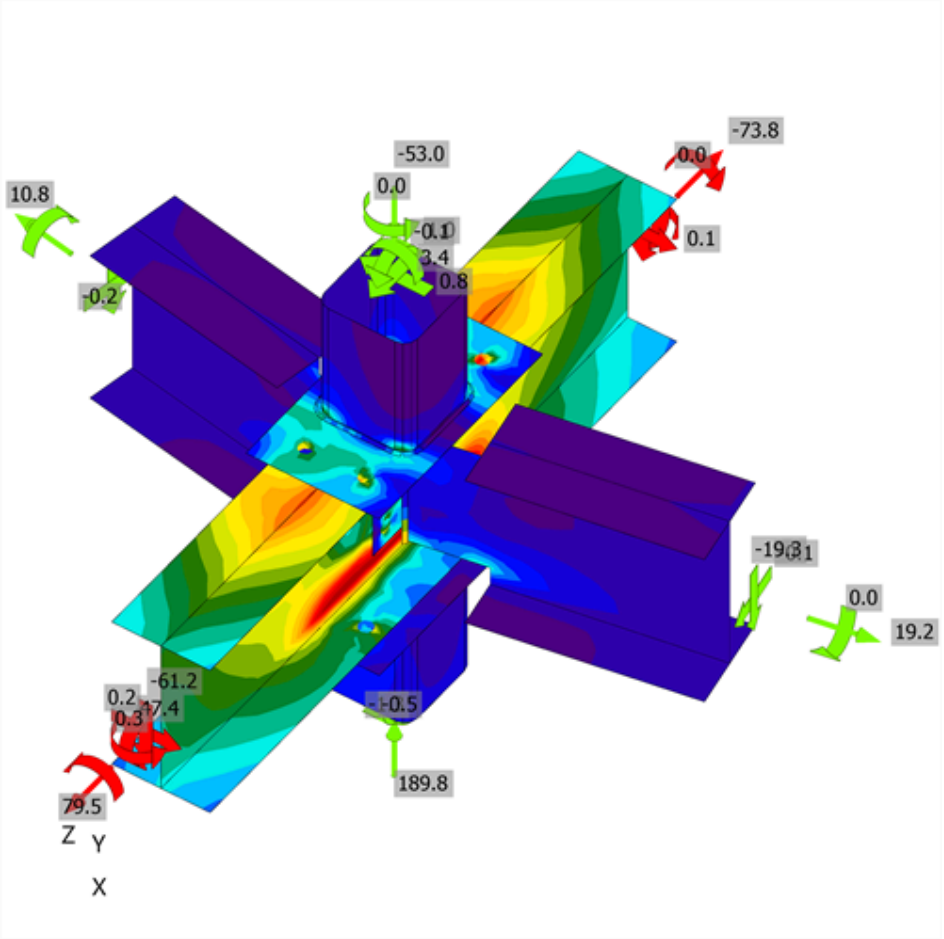
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

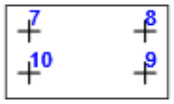
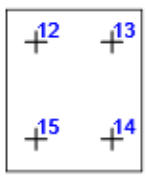


Strain check, C LC_124



Equivalent stress, C LC_124

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_122	0.8	5.8	177.0	0.5	6.2	6.6	OK	OK
	B3	M20 8.8 - 1	C LC_110	13.7	2.1	123.0	9.7	2.2	9.1	OK	OK
	B4	M20 8.8 - 1	C LC_110	13.3	2.2	91.6	9.4	2.4	9.1	OK	OK
	B5	M20 8.8 - 1	C LC_124	8.5	4.1	176.2	6.0	4.3	8.7	OK	OK
	B7	M20 8.8 - 1	C LC_124	15.0	4.8	156.4	10.6	5.1	12.7	OK	OK
	B8	M20 8.8 - 1	C LC_152	24.5	3.4	88.1	17.4	3.8	16.0	OK	OK
	B9	M20 8.8 - 1	C LC_152	25.2	3.3	94.5	17.8	3.5	16.2	OK	OK
	B10	M20 8.8 - 1	C LC_151	3.5	3.2	112.2	2.5	3.4	5.2	OK	OK
	B12	M20 8.8 - 1	C LC_128	7.1	1.3	70.1	5.0	1.7	5.0	OK	OK
	B13	M20 8.8 - 1	C LC_124	7.1	1.4	68.9	5.0	2.1	5.1	OK	OK
	B14	M20 8.8 - 1	C LC_128	6.8	1.5	71.0	4.8	1.6	5.0	OK	OK
	B15	M20 8.8 - 1	C LC_128	6.6	1.4	72.9	4.7	1.5	4.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate column below SLN52 column below	SLN52 column below	◢ 5.6 ◣	521	C LC_122	271.2	0.0	-131.9	-133.3	31.0	67.0	36.1	OK	OK
		◢ 5.6 ◣	520	C LC_122	278.0	0.0	-137.9	136.5	-27.8	68.7	37.7	OK	OK
End plate column above SLN85 column above	SLN85 column above	◢ 5.6 ◣	521	C LC_154	138.2	0.0	-67.8	-69.6	1.1	34.2	18.5	OK	OK
		◢ 5.6 ◣	521	C LC_154	147.2	0.0	-74.9	73.1	-1.8	36.4	19.8	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_108	293.5	0.0	37.0	-36.5	164.1	72.5	39.4	OK	OK
SLN206+SLN222 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_124	379.8	0.0	-43.8	-41.5	-213.8	93.8	61.8	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_124	241.8	0.0	-61.1	-51.5	124.8	59.7	35.9	OK	OK
		◢ 4.2 ◣	149	C LC_124	286.7	0.0	-58.0	71.9	-145.3	70.8	41.7	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_121	177.7	0.0	-33.7	-43.9	90.7	43.9	27.1	OK	OK
		◢ 4.2 ◣	149	C LC_121	157.4	0.0	-41.4	34.9	-80.4	38.9	24.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_020
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut

Name	SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

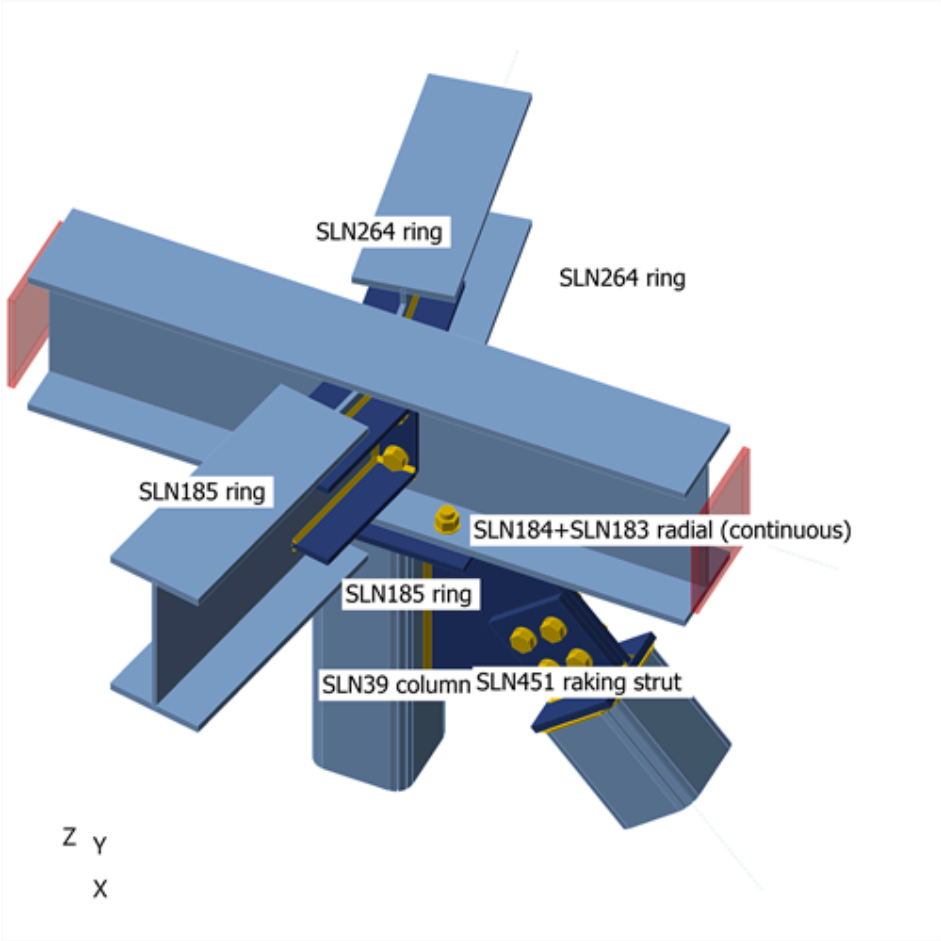
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN184+SLN183 radial (continuous)	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN39 column below	2 - SHS150x150x10	0.0	90.0	-160.0	0	0	0
SLN185 ring	1 - UB254x146x43	100.0	0.0	0.0	0	0	0
SLN264 ring	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN451 raking strut	2 - SHS150x150x10	20.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN184+SLN183 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN184+SLN183 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN39 column below / begin		Position	0
SLN185 ring / begin		Position	0
SLN264 ring / end		Position	0
SLN451 raking strut / end		Position	0



SLN184+SLN183 radial (continuous) - SLN39 column below - SLN185 ring - SLN264 ring - SLN451 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN39 column below / Begin	31.6	0.1	0.3	0.0	1.4	-0.2
	SLN185 ring / Begin	-6.4	-0.2	-4.9	0.0	0.0	0.4
	SLN264 ring / End	8.7	0.0	-4.9	0.0	0.0	0.1
	SLN451 raking strut / End	0.5	0.1	-1.0	0.1	0.0	0.2
C LC_102	SLN39 column below / Begin	145.7	0.2	1.4	0.0	5.7	-1.0
	SLN185 ring / Begin	-19.3	-0.5	-25.6	0.0	0.0	1.6
	SLN264 ring / End	27.5	0.2	-25.6	0.0	0.0	0.5
	SLN451 raking strut / End	-9.5	0.3	-1.0	0.2	0.0	0.9
C LC_103	SLN39 column below / Begin	118.0	0.2	1.4	0.0	5.5	-0.9
	SLN185 ring / Begin	-7.9	-0.7	-24.0	0.0	0.0	1.8
	SLN264 ring / End	15.9	0.1	-24.0	0.0	0.0	0.5
	SLN451 raking strut / End	-34.1	0.2	-0.8	0.2	0.0	0.8
C LC_104	SLN39 column below / Begin	159.4	-1.0	-2.6	0.0	2.9	-0.7
	SLN185 ring / Begin	-21.3	-0.4	-24.0	0.0	0.0	1.5
	SLN264 ring / End	28.2	0.4	-24.0	0.0	0.0	0.9
	SLN451 raking strut / End	19.9	0.3	-1.3	0.3	0.0	0.5
C LC_105	SLN39 column below / Begin	138.4	3.8	-0.1	0.0	4.3	-1.9
	SLN185 ring / Begin	-11.9	-0.4	-24.0	0.0	0.0	1.3
	SLN264 ring / End	24.2	0.2	-24.0	0.0	0.0	0.5
	SLN451 raking strut / End	-5.2	0.3	-1.1	0.0	0.0	2.0
C LC_106	SLN39 column below / Begin	129.7	0.3	1.4	0.0	5.8	-1.0
	SLN185 ring / Begin	-8.3	-0.5	-24.0	0.0	0.0	1.4
	SLN264 ring / End	14.7	0.1	-24.0	0.0	0.0	0.3
	SLN451 raking strut / End	-20.8	0.6	-0.9	0.1	0.0	0.9
C LC_107	SLN39 column below / Begin	-25.5	0.0	0.2	0.0	0.7	0.1
	SLN185 ring / Begin	10.3	-0.2	-0.5	0.0	0.0	0.6
	SLN264 ring / End	-8.7	0.1	-0.5	0.0	0.0	0.2
	SLN451 raking strut / End	-40.7	-0.2	-0.3	0.0	0.0	0.0
C LC_108	SLN39 column below / Begin	49.4	-2.5	-7.7	0.0	-4.6	0.4
	SLN185 ring / Begin	-12.5	-0.1	-0.5	0.0	0.0	0.2
	SLN264 ring / End	11.9	0.2	-0.5	0.0	0.0	0.6
	SLN451 raking strut / End	56.5	0.1	-1.2	0.2	0.0	-0.5
C LC_109	SLN39 column below / Begin	10.4	7.1	-2.9	0.0	-1.9	-2.1
	SLN185 ring / Begin	4.7	0.1	-0.5	0.0	0.0	-0.2
	SLN264 ring / End	4.7	0.0	-0.5	0.0	0.0	-0.1
	SLN451 raking strut / End	10.4	0.1	-0.9	-0.3	0.0	2.4
C LC_110	SLN39 column below / Begin	-5.1	0.1	0.3	0.0	1.3	-0.3
	SLN185 ring / Begin	11.0	0.0	-0.5	0.0	0.0	0.0
	SLN264 ring / End	-12.1	-0.1	-0.5	0.0	0.0	-0.3
	SLN451 raking strut / End	-18.0	0.6	-0.6	-0.1	0.0	0.3
C LC_111	SLN39 column below / Begin	60.4	0.1	1.0	0.0	4.0	-0.5
	SLN185 ring / Begin	2.3	-0.6	-16.2	0.0	0.0	1.6
	SLN264 ring / End	4.0	0.1	-16.2	0.0	0.0	0.5
	SLN451 raking strut / End	-50.3	0.0	-0.5	0.1	0.0	0.5
C LC_112	SLN39 column below / Begin	137.9	-2.3	-6.9	0.0	-1.2	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN185 ring / Begin	-21.8	-0.3	-16.2	0.0	0.0	1.1
	SLN264 ring / End	25.8	0.4	-16.2	0.0	0.0	1.0
	SLN451 raking strut / End	50.4	0.3	-1.5	0.3	0.0	0.0
C LC_113	SLN39 column below / Begin	97.9	7.2	-2.0	0.0	1.5	-2.7
	SLN185 ring / Begin	-4.1	-0.2	-16.2	0.0	0.0	0.7
	SLN264 ring / End	18.3	0.1	-16.2	0.0	0.0	0.2
	SLN451 raking strut / End	3.0	0.3	-1.2	-0.2	0.0	2.9
C LC_114	SLN39 column below / Begin	81.7	0.2	1.1	0.0	4.6	-0.9
	SLN185 ring / Begin	2.5	-0.3	-16.2	0.0	0.0	0.9
	SLN264 ring / End	0.9	0.0	-16.2	0.0	0.0	0.0
	SLN451 raking strut / End	-26.3	0.8	-0.8	0.0	0.0	0.8
C LC_115	SLN39 column below / Begin	-17.5	0.0	0.3	0.0	1.0	0.0
	SLN185 ring / Begin	8.7	-0.3	-1.7	0.0	0.0	0.8
	SLN264 ring / End	-6.5	0.1	-1.7	0.0	0.0	0.2
	SLN451 raking strut / End	-40.8	-0.2	-0.5	0.1	0.0	0.1
C LC_116	SLN39 column below / Begin	57.7	-2.5	-7.6	0.0	-4.2	0.3
	SLN185 ring / Begin	-14.2	-0.1	-1.7	0.0	0.0	0.3
	SLN264 ring / End	14.2	0.2	-1.7	0.0	0.0	0.7
	SLN451 raking strut / End	56.8	0.1	-1.5	0.2	0.0	-0.5
C LC_117	SLN39 column below / Begin	18.6	7.1	-2.8	0.0	-1.5	-2.2
	SLN185 ring / Begin	3.0	0.0	-1.7	0.0	0.0	-0.1
	SLN264 ring / End	7.0	0.0	-1.7	0.0	0.0	-0.1
	SLN451 raking strut / End	10.6	0.1	-1.2	-0.3	0.0	2.5
C LC_118	SLN39 column below / Begin	3.1	0.1	0.4	0.0	1.6	-0.4
	SLN185 ring / Begin	9.4	0.0	-1.7	0.0	0.0	0.1
	SLN264 ring / End	-9.9	-0.1	-1.7	0.0	0.0	-0.2
	SLN451 raking strut / End	-18.0	0.7	-0.8	-0.1	0.0	0.4
C LC_119	SLN39 column below / Begin	131.0	0.2	1.4	0.0	5.6	-0.9
	SLN185 ring / Begin	-10.9	-0.7	-25.6	0.0	0.0	1.8
	SLN264 ring / End	19.2	0.1	-25.6	0.0	0.0	0.6
	SLN451 raking strut / End	-27.7	0.2	-0.8	0.2	0.0	0.8
C LC_120	SLN39 column below / Begin	164.4	-1.0	-2.5	0.0	3.0	-0.7
	SLN185 ring / Begin	-24.5	-0.5	-25.6	0.0	0.0	1.6
	SLN264 ring / End	31.8	0.3	-25.6	0.0	0.0	0.8
	SLN451 raking strut / End	15.2	0.3	-1.3	0.3	0.0	0.6
C LC_121	SLN39 column below / Begin	145.1	3.8	-0.1	0.0	4.4	-2.0
	SLN185 ring / Begin	-15.0	-0.5	-25.6	0.0	0.0	1.4
	SLN264 ring / End	27.8	0.2	-25.6	0.0	0.0	0.5
	SLN451 raking strut / End	-7.6	0.3	-1.1	0.0	0.0	2.0
C LC_122	SLN39 column below / Begin	139.0	0.3	1.4	0.0	5.9	-1.0
	SLN185 ring / Begin	-11.4	-0.5	-25.6	0.0	0.0	1.5
	SLN264 ring / End	17.9	0.1	-25.6	0.0	0.0	0.3
	SLN451 raking strut / End	-19.2	0.6	-0.9	0.1	0.0	0.9
C LC_123	SLN39 column below / Begin	86.4	0.1	1.0	0.0	4.1	-0.6
	SLN185 ring / Begin	-3.8	-0.6	-19.4	0.0	0.0	1.6
	SLN264 ring / End	10.5	0.1	-19.4	0.0	0.0	0.6
	SLN451 raking strut / End	-37.4	0.0	-0.5	0.1	0.0	0.6

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN39 column below / Begin	147.8	-2.3	-6.8	0.0	-0.9	-0.3
	SLN185 ring / Begin	-28.1	-0.4	-19.4	0.0	0.0	1.4
	SLN264 ring / End	33.1	0.4	-19.4	0.0	0.0	0.9
	SLN451 raking strut / End	41.0	0.3	-1.5	0.4	0.0	0.0
C LC_125	SLN39 column below / Begin	111.2	7.3	-2.0	0.0	1.7	-2.8
	SLN185 ring / Begin	-10.3	-0.3	-19.4	0.0	0.0	1.0
	SLN264 ring / End	25.7	0.1	-19.4	0.0	0.0	0.2
	SLN451 raking strut / End	-1.8	0.3	-1.2	-0.2	0.0	3.0
C LC_126	SLN39 column below / Begin	100.5	0.2	1.2	0.0	4.8	-0.9
	SLN185 ring / Begin	-3.6	-0.4	-19.4	0.0	0.0	1.0
	SLN264 ring / End	7.3	0.0	-19.4	0.0	0.0	0.1
	SLN451 raking strut / End	-23.2	0.8	-0.8	0.1	0.0	0.8
C LC_127	SLN39 column below / Begin	8.5	0.0	0.3	0.0	1.1	-0.1
	SLN185 ring / Begin	2.7	-0.3	-4.9	0.0	0.0	0.8
	SLN264 ring / End	0.0	0.1	-4.9	0.0	0.0	0.3
	SLN451 raking strut / End	-27.9	-0.2	-0.5	0.1	0.0	0.1
C LC_128	SLN39 column below / Begin	67.7	-2.5	-7.5	0.0	-4.0	0.3
	SLN185 ring / Begin	-20.5	-0.2	-4.9	0.0	0.0	0.7
	SLN264 ring / End	21.5	0.2	-4.9	0.0	0.0	0.6
	SLN451 raking strut / End	47.4	0.1	-1.5	0.3	0.0	-0.4
C LC_129	SLN39 column below / Begin	31.9	7.1	-2.7	0.0	-1.3	-2.3
	SLN185 ring / Begin	-3.1	-0.1	-4.9	0.0	0.0	0.2
	SLN264 ring / End	14.3	0.0	-4.9	0.0	0.0	-0.1
	SLN451 raking strut / End	5.7	0.1	-1.2	-0.3	0.0	2.6
C LC_130	SLN39 column below / Begin	21.8	0.1	0.4	0.0	1.7	-0.4
	SLN185 ring / Begin	3.3	-0.1	-4.9	0.0	0.0	0.2
	SLN264 ring / End	-3.4	-0.1	-4.9	0.0	0.0	-0.2
	SLN451 raking strut / End	-14.9	0.7	-0.8	0.0	0.0	0.4
C LC_135	SLN39 column below / Begin	150.0	0.3	1.4	0.0	5.8	-1.0
	SLN185 ring / Begin	-16.6	-0.6	-25.6	0.0	0.0	1.9
	SLN264 ring / End	27.9	0.3	-25.6	0.0	0.0	0.7
	SLN451 raking strut / End	-7.0	0.4	-1.0	0.2	0.0	1.1
C LC_141	SLN39 column below / Begin	31.1	0.1	0.3	0.0	1.4	-0.2
	SLN185 ring / Begin	-6.1	-0.2	-4.9	0.0	0.0	0.5
	SLN264 ring / End	8.5	0.0	-4.9	0.0	0.0	0.1
	SLN451 raking strut / End	-0.3	0.1	-1.0	0.1	0.0	0.2
C LC_142	SLN39 column below / Begin	32.2	0.1	0.3	0.0	1.4	-0.2
	SLN185 ring / Begin	-6.6	-0.1	-4.9	0.0	0.0	0.4
	SLN264 ring / End	9.0	0.1	-4.9	0.0	0.0	0.2
	SLN451 raking strut / End	1.2	0.1	-1.0	0.1	0.0	0.2
C LC_143	SLN39 column below / Begin	31.8	0.1	0.3	0.0	1.4	-0.2
	SLN185 ring / Begin	-6.4	-0.1	-4.9	0.0	0.0	0.4
	SLN264 ring / End	8.9	0.1	-4.9	0.0	0.0	0.1
	SLN451 raking strut / End	0.7	0.1	-1.0	0.1	0.0	0.2
C LC_144	SLN39 column below / Begin	31.5	0.1	0.3	0.0	1.4	-0.2
	SLN185 ring / Begin	-6.3	-0.2	-4.9	0.0	0.0	0.4
	SLN264 ring / End	8.6	0.0	-4.9	0.0	0.0	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN451 raking strut / End	0.3	0.1	-1.0	0.1	0.0	0.2
C LC_145	SLN39 column below / Begin	143.5	0.2	1.4	0.0	5.7	-1.0
	SLN185 ring / Begin	-18.2	-0.7	-25.6	0.0	0.0	1.7
	SLN264 ring / End	26.5	0.1	-25.6	0.0	0.0	0.4
	SLN451 raking strut / End	-12.5	0.3	-1.0	0.2	0.0	0.9
C LC_146	SLN39 column below / Begin	147.9	0.2	1.4	0.0	5.7	-1.0
	SLN185 ring / Begin	-20.4	-0.4	-25.6	0.0	0.0	1.5
	SLN264 ring / End	28.6	0.3	-25.6	0.0	0.0	0.6
	SLN451 raking strut / End	-6.5	0.3	-1.0	0.2	0.0	0.9
C LC_147	SLN39 column below / Begin	146.3	0.2	1.4	0.0	5.8	-1.0
	SLN185 ring / Begin	-19.6	-0.5	-25.6	0.0	0.0	1.6
	SLN264 ring / End	28.1	0.2	-25.6	0.0	0.0	0.5
	SLN451 raking strut / End	-8.7	0.3	-1.0	0.2	0.0	0.9
C LC_148	SLN39 column below / Begin	145.1	0.2	1.4	0.0	5.7	-1.0
	SLN185 ring / Begin	-19.0	-0.6	-25.6	0.0	0.0	1.7
	SLN264 ring / End	26.9	0.1	-25.6	0.0	0.0	0.5
	SLN451 raking strut / End	-10.2	0.3	-1.0	0.2	0.0	0.9
C LC_151	SLN39 column below / Begin	91.5	0.2	0.0	0.0	-0.2	-0.7
	SLN185 ring / Begin	-6.6	-0.4	-11.7	0.0	0.0	1.1
	SLN264 ring / End	12.3	0.2	-11.3	0.0	0.0	0.4
	SLN451 raking strut / End	-11.1	0.3	-1.0	0.1	0.0	0.7
C LC_152	SLN39 column below / Begin	100.8	0.2	2.7	0.0	10.9	-0.9
	SLN185 ring / Begin	-21.0	-0.5	-14.2	0.0	0.0	1.4
	SLN264 ring / End	28.2	0.1	-13.9	0.0	0.0	0.4
	SLN451 raking strut / End	4.5	0.3	-1.0	0.2	0.0	0.7
C LC_153	SLN39 column below / Begin	145.4	0.2	1.4	0.0	5.5	-1.0
	SLN185 ring / Begin	-19.3	-0.6	-21.0	0.0	0.0	1.6
	SLN264 ring / End	27.7	0.2	-20.3	0.0	0.0	0.5
	SLN451 raking strut / End	-9.1	0.3	-1.0	0.2	0.0	0.9
C LC_154	SLN39 column below / Begin	90.9	0.2	0.0	0.0	0.0	-0.7
	SLN185 ring / Begin	-6.7	-0.4	-11.7	0.0	0.0	1.1
	SLN264 ring / End	12.4	0.1	-11.3	0.0	0.0	0.4
	SLN451 raking strut / End	-10.8	0.3	-1.0	0.1	0.0	0.7

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.4 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	29.2 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN184+SLN183 radial (continuous)-bfl 1	12.7	C LC_108	220.7	0.0	33.7	OK
SLN184+SLN183 radial (continuous)-tfl 1	12.7	C LC_135	135.4	0.0	0.0	OK
SLN184+SLN183 radial (continuous)-w 1	7.2	C LC_135	275.1	0.0	52.3	OK
SLN39 column below	10.0	C LC_111	275.3	0.2	0.0	OK
SLN185 ring-bfl 1	12.7	C LC_135	110.9	0.0	0.0	OK
SLN185 ring-tfl 1	12.7	C LC_135	108.7	0.0	0.0	OK
SLN185 ring-w 1	7.2	C LC_135	273.7	0.0	0.0	OK
SLN264 ring-bfl 1	12.7	C LC_112	60.9	0.0	0.0	OK
SLN264 ring-tfl 1	12.7	C LC_112	55.0	0.0	0.0	OK
SLN264 ring-w 1	7.2	C LC_112	174.1	0.0	0.0	OK
SLN451 raking strut	10.0	C LC_125	72.1	0.0	0.0	OK
End plate column below SLN39 column below	12.0	C LC_108	251.5	0.0	33.7	OK
End plate SLN185 ring	12.0	C LC_135	73.6	0.0	385.2	OK
End plate SLN264 ring	12.0	C LC_135	51.2	0.0	270.5	OK
Cap plate SLN451 raking strut	12.0	C LC_125	169.6	0.0	0.0	OK
Fin plate SLN451 raking strut	12.0	C LC_125	275.3	0.2	100.7	OK
Fin plate 2 SLN451 raking strut	12.0	C LC_125	275.4	0.2	141.1	OK
Gusset SLN451 raking strut	12.0	C LC_125	275.9	0.4	59.4	OK
Flange top a end plate SLN185 ring	12.0	C LC_135	147.1	0.0	0.0	OK
Flange top b end plate SLN185 ring	12.0	C LC_135	126.8	0.0	0.0	OK
Flange bottom a end plate SLN185 ring	12.0	C LC_135	128.1	0.0	0.0	OK
Flange bottom b end plate SLN185 ring	12.0	C LC_135	139.2	0.0	0.0	OK
Flange top a end plate SLN264 ring	12.0	C LC_112	71.0	0.0	0.0	OK
Flange top b end plate SLN264 ring	12.0	C LC_112	70.7	0.0	0.0	OK
Flange bottom a end plate SLN264 ring	12.0	C LC_120	68.9	0.0	0.0	OK
Flange bottom b end plate SLN264 ring	12.0	C LC_112	87.8	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

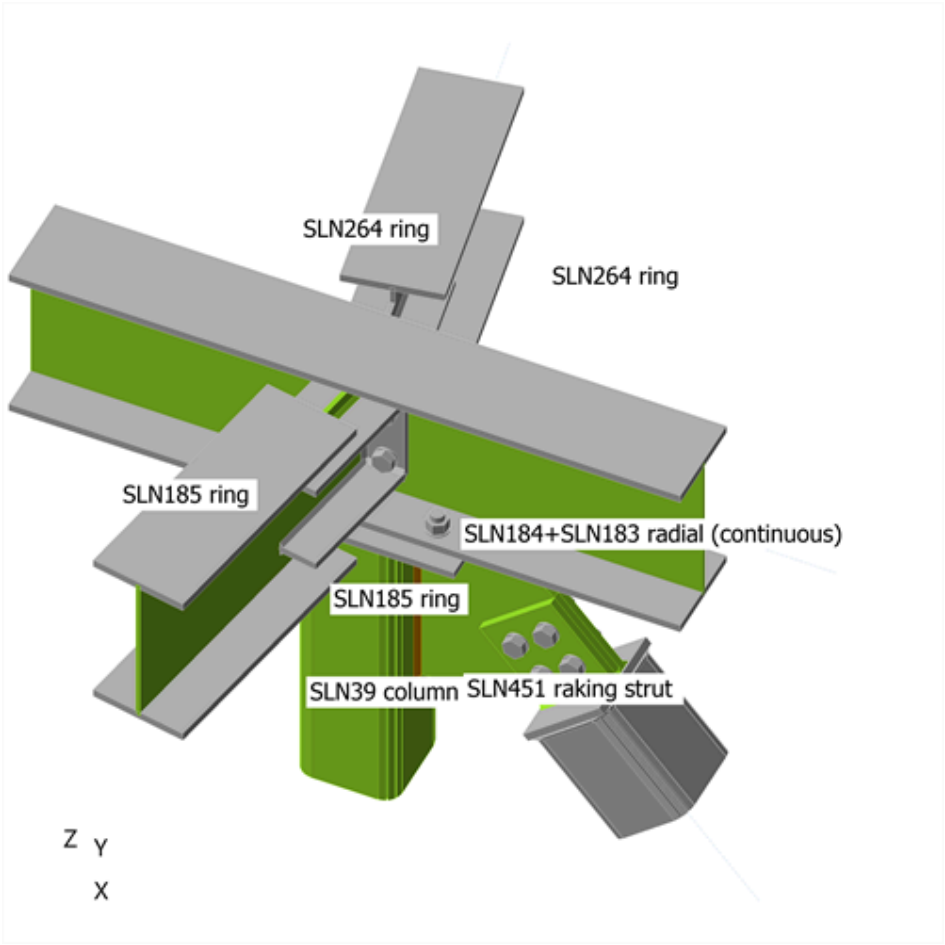
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

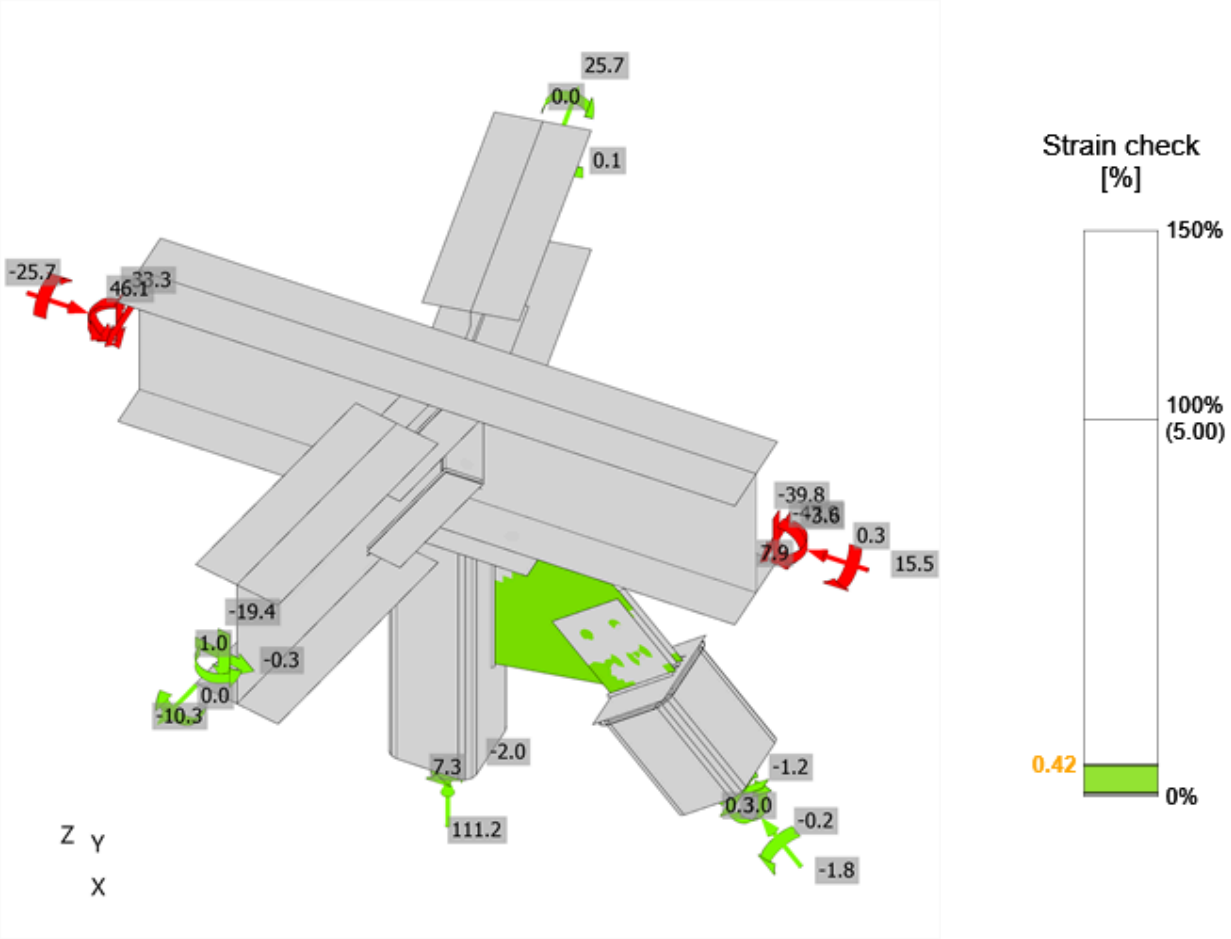
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN39 column below	150	C LC_108	0	5	0.1	OK
SLN451 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

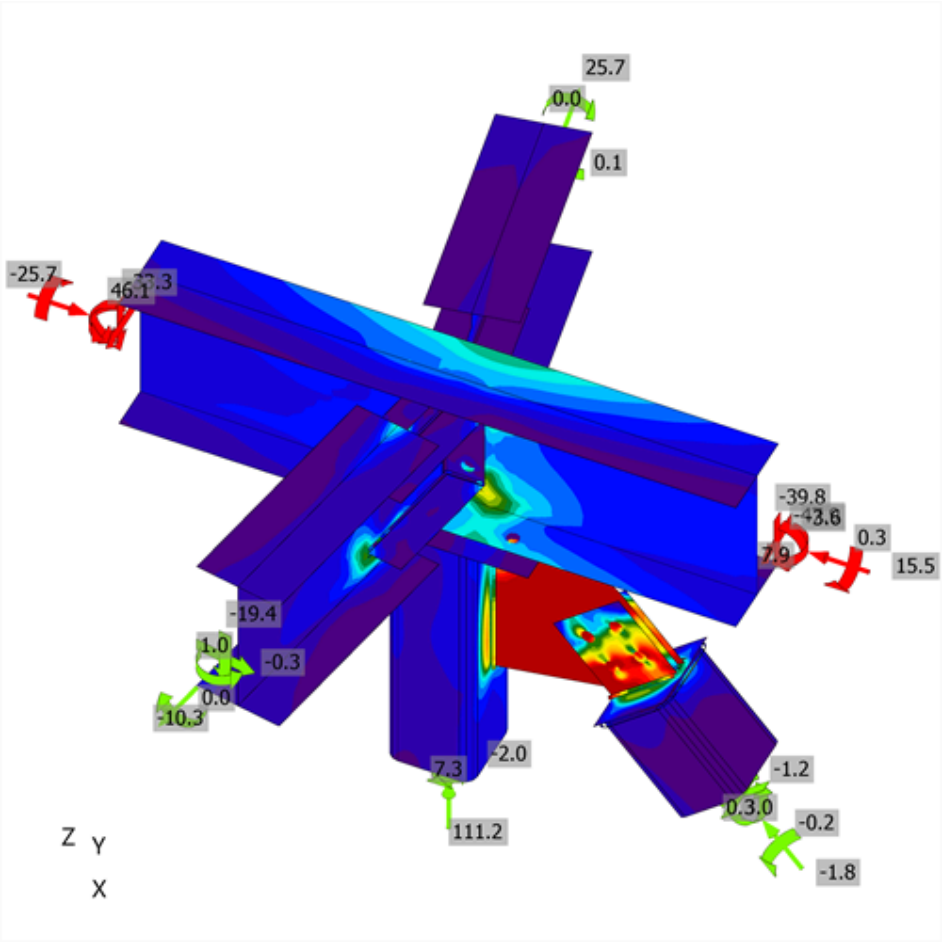
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

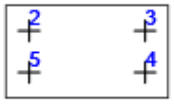
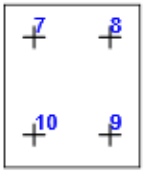
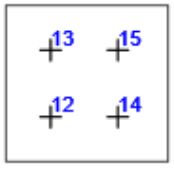


Strain check, C LC_125



Equivalent stress, C LC_125

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_107	18.5	6.4	122.6	13.1	6.9	16.2	OK	OK
	B3	M20 8.8 - 1	C LC_108	36.3	6.9	103.5	25.7	7.3	25.7	OK	OK
	B4	M20 8.8 - 1	C LC_108	35.8	6.8	111.1	25.4	7.2	25.3	OK	OK
	B5	M20 8.8 - 1	C LC_107	19.0	6.3	125.6	13.4	6.7	16.3	OK	OK
	B7	M20 8.8 - 1	C LC_135	4.4	0.6	123.8	3.1	0.8	2.8	OK	OK
	B8	M20 8.8 - 1	C LC_112	3.3	0.5	73.6	2.4	0.6	2.2	OK	OK
	B9	M20 8.8 - 1	C LC_135	4.2	1.2	123.8	3.0	1.9	3.4	OK	OK
	B10	M20 8.8 - 1	C LC_120	7.0	0.4	123.8	5.0	0.6	4.0	OK	OK
	B12	M20 8.8 - 1	C LC_125	22.7	16.7	106.0	16.1	17.7	29.2	OK	OK
	B13	M20 8.8 - 1	C LC_125	12.2	14.8	115.3	8.6	15.7	21.9	OK	OK
	B14	M20 8.8 - 1	C LC_116	1.5	10.2	106.0	1.1	16.1	11.6	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.5	7.9	115.3	0.3	11.4	8.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN451 raking strut	SLN451 raking strut	◢ 4.2	519	C LC_116	203.1	0.0	96.6	-103.0	5.7	50.2	27.4	OK	OK
SLN39 column below-w 3	Gusset SLN451 raking strut	◢ 5.6 ◣	288	C LC_111	397.4	0.5	-136.4	-131.7	-170.6	98.2	65.5	OK	OK
		◢ 5.6 ◣	288	C LC_108	397.4	0.5	135.9	-140.8	-163.3	98.2	69.3	OK	OK
SLN185 ring-w 1	Flange top b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_135	173.9	0.0	93.3	77.2	35.1	43.0	26.4	OK	OK
		◢ 4.2 ◣	197	C LC_135	74.8	0.0	-37.6	34.5	14.1	18.5	10.5	OK	OK
SLN185 ring-w 1	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	198	C LC_120	116.6	0.0	34.1	45.8	45.2	28.8	17.8	OK	OK
		◢ 4.2 ◣	198	C LC_135	231.6	0.0	106.0	-92.6	-74.6	57.2	35.8	OK	OK
SLN264 ring-w 1	Flange top a end plate SLN264 ring	◢ 4.2 ◣	198	C LC_112	112.4	0.0	-53.8	-46.2	33.4	27.8	14.3	OK	OK
		◢ 4.2 ◣	198	C LC_119	56.1	0.0	-15.5	19.7	-24.1	13.9	8.0	OK	OK
SLN264 ring-w 1	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	198	C LC_121	55.5	0.0	15.3	11.5	-28.6	13.7	9.2	OK	OK
		◢ 4.2 ◣	198	C LC_120	88.1	0.0	3.6	-13.2	-49.1	21.8	13.3	OK	OK
Cap plate SLN451 raking strut	Fin plate SLN451 raking strut	◢ 5.6 ◣	139	C LC_113	270.4	0.0	76.6	116.9	93.5	66.8	53.8	OK	OK
		◢ 5.6 ◣	139	C LC_116	187.2	0.0	93.5	-77.6	-52.4	46.3	27.8	OK	OK
Cap plate SLN451 raking strut	Fin plate 2 SLN451 raking strut	◢ 5.6 ◣	139	C LC_116	148.4	0.0	65.2	61.8	45.8	36.7	22.9	OK	OK
		◢ 5.6 ◣	139	C LC_125	264.6	0.0	-129.8	103.2	-84.2	65.4	50.8	OK	OK
End plate column below SLN39 column below	SLN39 column below	◢ 5.6 ◣	521	C LC_103	311.9	0.0	-189.3	-142.9	8.5	77.1	42.5	OK	OK
		◢ 5.6 ◣	521	C LC_152	256.5	0.0	-132.0	126.1	15.0	63.4	35.7	OK	OK
SLN184+SLN183 radial (continuous)-w 1	SLN185 ring-w 1	◢ 7.2	189	C LC_120	135.1	0.0	34.4	8.7	74.9	33.4	21.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN184+SLN183 radial (continuous)-w 1	SLN264 ring-w 1	◢ 7.2	189	C LC_120	138.4	0.0	68.0	53.7	-44.3	34.2	20.5	OK	OK
End plate SLN185 ring	SLN185 ring-w 1	◢ 4.2 ◣	149	C LC_120	139.3	0.0	-54.7	-35.7	-64.8	34.4	18.9	OK	OK
		◢ 4.2 ◣	149	C LC_120	92.4	0.0	-24.1	38.0	34.8	22.8	12.5	OK	OK
End plate SLN264 ring	SLN264 ring-w 1	◢ 4.2 ◣	149	C LC_120	85.1	0.0	-20.7	-46.1	12.2	21.0	13.6	OK	OK
		◢ 4.2 ◣	149	C LC_135	65.7	0.0	-35.1	18.3	-26.4	16.2	10.3	OK	OK
SLN185 ring-w 1	Flange top a end plate SLN185 ring	◢ 4.2 ◣	199	C LC_135	258.6	0.0	-21.3	12.0	-148.3	63.9	41.4	OK	OK
		◢ 4.2 ◣	198	C LC_135	188.2	0.0	28.7	31.0	-102.8	46.5	26.5	OK	OK
SLN185 ring-w 1	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	198	C LC_135	206.0	0.0	38.0	-39.5	110.0	50.9	28.8	OK	OK
		◢ 4.2 ◣	199	C LC_135	210.9	0.0	-31.7	-13.7	119.6	52.1	30.6	OK	OK
SLN264 ring-w 1	Flange top b end plate SLN264 ring	◢ 4.2 ◣	199	C LC_112	122.2	0.0	9.5	-4.3	-70.2	30.2	20.8	OK	OK
		◢ 4.2 ◣	198	C LC_112	97.8	0.0	-15.4	-17.6	-52.9	24.2	13.8	OK	OK
SLN264 ring-w 1	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	199	C LC_120	91.5	0.0	11.2	-44.5	-27.8	22.6	12.9	OK	OK
		◢ 4.2 ◣	199	C LC_112	151.0	0.0	76.0	35.3	66.5	37.3	26.8	OK	OK
End plate SLN185 ring	Flange top a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	232.2	0.0	-110.3	91.9	74.0	57.4	27.9	OK	OK
		◢ 4.2 ◣	58	C LC_135	192.7	0.0	-63.8	-82.2	-65.3	47.6	24.2	OK	OK
End plate SLN185 ring	Flange top b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	45.0	0.0	5.9	-0.2	-25.7	11.1	8.7	OK	OK
		◢ 4.2 ◣	58	C LC_109	24.3	0.0	-9.4	-11.2	6.5	6.0	5.9	OK	OK
End plate SLN185 ring	Flange bottom a end plate SLN185 ring	◢ 4.2 ◣	58	C LC_135	154.5	0.0	-44.7	69.7	49.2	38.2	21.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_135	221.7	0.0	-111.5	-86.5	-69.0	54.8	28.6	OK	OK
End plate SLN185 ring	Flange bottom b end plate SLN185 ring	◢ 4.2 ◣	58	C LC_120	70.1	0.0	13.6	-33.9	-20.7	17.3	11.2	OK	OK
		◢ 4.2 ◣	58	C LC_135	58.4	0.0	0.9	-5.0	33.3	14.4	9.6	OK	OK
End plate SLN264 ring	Flange top a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	179.4	0.0	-89.6	77.4	45.5	44.3	23.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	153.1	0.0	-58.4	-70.6	-41.1	37.8	20.5	OK	OK
End plate SLN264 ring	Flange top b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_110	27.0	0.0	-12.6	11.2	-8.1	6.7	6.3	OK	OK
		◢ 4.2 ◣	58	C LC_119	37.9	0.0	-6.1	-5.2	21.0	9.4	8.4	OK	OK
End plate SLN264 ring	Flange bottom a end plate SLN264 ring	◢ 4.2 ◣	58	C LC_112	108.9	0.0	-51.1	49.0	26.1	26.9	15.2	OK	OK
		◢ 4.2 ◣	58	C LC_120	150.3	0.0	-71.7	-73.3	-21.2	37.1	19.5	OK	OK
End plate SLN264 ring	Flange bottom b end plate SLN264 ring	◢ 4.2 ◣	58	C LC_135	67.1	0.0	11.4	-23.3	-30.2	16.6	9.4	OK	OK
		◢ 4.2 ◣	58	C LC_112	34.7	0.0	4.9	3.7	19.5	8.6	6.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_100
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut

Name	SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

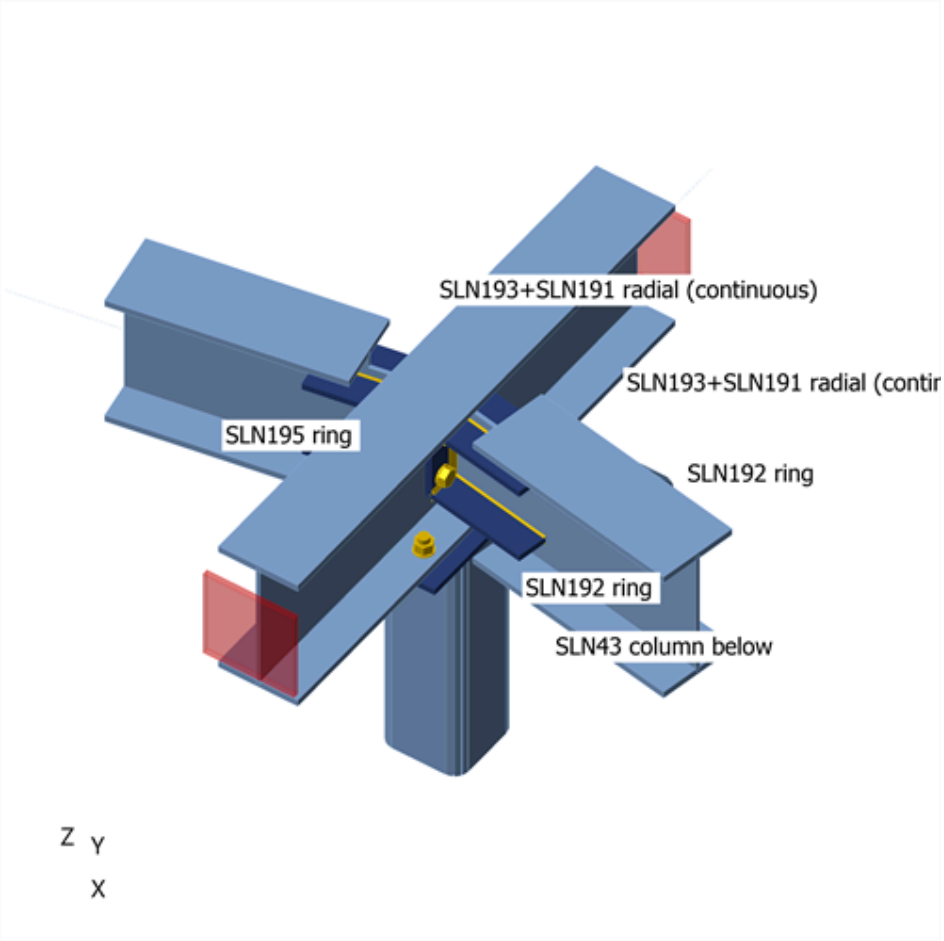
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN193+SLN191 radial (continuous)	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN43 column below	2 - SHS150x150x10	0.0	90.0	-80.0	0	0	0
SLN192 ring	1 - UB254x146x43	180.0	0.0	0.0	0	0	0
SLN195 ring	1 - UB254x146x43	-160.0	0.0	0.0	0	0	0
SLN446 raking strut	2 - SHS150x150x10	100.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN193+SLN191 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN193+SLN191 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN43 column below / begin		Position	0
SLN192 ring / begin		Position	0
SLN195 ring / end		Position	0
SLN446 raking strut / end		Position	0



SLN193+SLN191 radial (continuous) - SLN43 column below - SLN192 ring - SLN195 ring - SLN446 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN43 column below / Begin	26.5	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-2.7	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN43 column below / Begin	124.7	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.6	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	27.7	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.9	0.0	-1.0	0.1	0.0	0.0
C LC_103	SLN43 column below / Begin	116.1	3.7	-0.5	0.0	0.6	-0.9
	SLN192 ring / Begin	-17.4	-0.1	-24.0	0.0	0.0	0.4
	SLN195 ring / End	21.6	-0.1	-24.1	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.7	-0.1	-1.1	-0.2	0.0	1.1
C LC_104	SLN43 column below / Begin	110.5	0.0	0.3	0.0	1.1	-0.1
	SLN192 ring / Begin	-16.9	-0.2	-24.0	0.0	0.0	0.6
	SLN195 ring / End	16.5	-0.2	-24.1	0.0	0.0	-0.5
	SLN446 raking strut / End	-29.2	0.3	-1.0	0.1	0.0	0.3
C LC_105	SLN43 column below / Begin	138.1	-0.7	-3.9	0.0	-1.8	0.2
	SLN192 ring / Begin	-30.3	0.0	-24.0	0.0	0.0	0.2
	SLN195 ring / End	29.6	0.0	-24.1	0.0	0.0	-0.3
	SLN446 raking strut / End	7.9	0.0	-1.3	0.1	0.0	-0.2
C LC_106	SLN43 column below / Begin	98.2	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-14.5	-0.3	-24.0	0.0	0.0	0.7
	SLN195 ring / End	14.9	-0.3	-24.1	0.0	0.0	-0.6
	SLN446 raking strut / End	-45.4	-0.1	-0.8	0.0	0.0	-0.1
C LC_107	SLN43 column below / Begin	4.0	7.4	-1.4	0.0	-0.8	-2.0
	SLN192 ring / Begin	7.9	0.1	-0.5	0.0	0.0	-0.2
	SLN195 ring / End	-0.3	0.0	-0.5	0.0	0.0	-0.1
	SLN446 raking strut / End	0.4	-0.1	-0.8	-0.5	0.0	2.2
C LC_108	SLN43 column below / Begin	-6.2	0.1	0.0	0.0	0.2	-0.4
	SLN192 ring / Begin	8.6	-0.1	-0.5	0.0	0.0	0.2
	SLN195 ring / End	-9.2	0.0	-0.5	0.0	0.0	0.0
	SLN446 raking strut / End	-13.5	0.7	-0.7	0.0	0.0	0.5
C LC_109	SLN43 column below / Begin	43.7	-1.3	-8.3	0.0	-5.6	0.4
	SLN192 ring / Begin	-15.8	0.2	-0.5	0.0	0.0	-0.5
	SLN195 ring / End	14.2	0.1	-0.5	0.0	0.0	0.1
	SLN446 raking strut / End	53.7	0.1	-1.3	0.1	0.0	-0.4
C LC_110	SLN43 column below / Begin	-26.5	0.0	0.1	0.0	0.3	0.1
	SLN192 ring / Begin	11.4	-0.1	-0.5	0.0	0.0	0.3
	SLN195 ring / End	-11.0	-0.1	-0.5	0.0	0.0	-0.2
	SLN446 raking strut / End	-40.1	-0.2	-0.3	0.0	0.0	-0.1
C LC_111	SLN43 column below / Begin	79.0	7.4	-1.3	0.0	-0.2	-2.0
	SLN192 ring / Begin	-5.5	0.0	-16.2	0.0	0.0	0.1
	SLN195 ring / End	13.3	-0.1	-16.3	0.0	0.0	-0.4
	SLN446 raking strut / End	-14.4	-0.1	-1.1	-0.5	0.0	2.2
C LC_112	SLN43 column below / Begin	68.5	0.1	0.2	0.0	0.8	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN192 ring / Begin	-4.6	-0.2	-16.2	0.0	0.0	0.5
	SLN195 ring / End	4.0	-0.1	-16.3	0.0	0.0	-0.3
	SLN446 raking strut / End	-28.7	0.7	-0.9	0.1	0.0	0.6
C LC_113	SLN43 column below / Begin	120.1	-1.3	-8.1	0.0	-5.0	0.4
	SLN192 ring / Begin	-29.8	0.1	-16.2	0.0	0.0	-0.2
	SLN195 ring / End	28.3	0.0	-16.3	0.0	0.0	-0.1
	SLN446 raking strut / End	40.8	0.0	-1.5	0.1	0.0	-0.3
C LC_114	SLN43 column below / Begin	46.8	0.0	0.2	0.0	0.9	0.1
	SLN192 ring / Begin	-1.2	-0.3	-16.2	0.0	0.0	0.7
	SLN195 ring / End	1.7	-0.3	-16.3	0.0	0.0	-0.6
	SLN446 raking strut / End	-57.3	-0.2	-0.5	0.0	0.0	-0.1
C LC_115	SLN43 column below / Begin	10.9	7.4	-1.4	0.0	-0.7	-2.0
	SLN192 ring / Begin	5.6	0.0	-1.7	0.0	0.0	-0.1
	SLN195 ring / End	2.0	0.0	-1.8	0.0	0.0	-0.1
	SLN446 raking strut / End	-0.2	-0.1	-1.1	-0.5	0.0	2.2
C LC_116	SLN43 column below / Begin	0.6	0.1	0.1	0.0	0.3	-0.4
	SLN192 ring / Begin	6.4	-0.1	-1.7	0.0	0.0	0.2
	SLN195 ring / End	-7.0	0.0	-1.8	0.0	0.0	0.0
	SLN446 raking strut / End	-14.2	0.7	-0.9	0.0	0.0	0.6
C LC_117	SLN43 column below / Begin	50.7	-1.3	-8.2	0.0	-5.5	0.4
	SLN192 ring / Begin	-18.1	0.2	-1.7	0.0	0.0	-0.4
	SLN195 ring / End	16.5	0.0	-1.8	0.0	0.0	0.1
	SLN446 raking strut / End	53.3	0.0	-1.5	0.1	0.0	-0.4
C LC_118	SLN43 column below / Begin	-19.8	0.0	0.1	0.0	0.4	0.1
	SLN192 ring / Begin	9.2	-0.1	-1.7	0.0	0.0	0.3
	SLN195 ring / End	-8.8	-0.1	-1.8	0.0	0.0	-0.3
	SLN446 raking strut / End	-41.0	-0.2	-0.5	0.0	0.0	-0.1
C LC_119	SLN43 column below / Begin	122.7	3.7	-0.4	0.0	0.7	-0.9
	SLN192 ring / Begin	-20.9	-0.2	-25.6	0.0	0.0	0.5
	SLN195 ring / End	25.0	-0.2	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-24.4	-0.1	-1.1	-0.2	0.0	1.1
C LC_120	SLN43 column below / Begin	118.8	0.0	0.3	0.0	1.2	-0.1
	SLN192 ring / Begin	-20.3	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	19.9	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-29.7	0.3	-1.0	0.1	0.0	0.3
C LC_121	SLN43 column below / Begin	143.0	-0.7	-3.8	0.0	-1.6	0.2
	SLN192 ring / Begin	-34.0	0.0	-25.6	0.0	0.0	0.3
	SLN195 ring / End	33.2	-0.1	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	2.7	0.0	-1.3	0.1	0.0	-0.2
C LC_122	SLN43 column below / Begin	110.8	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-17.8	-0.3	-25.6	0.0	0.0	0.7
	SLN195 ring / End	18.2	-0.3	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-40.1	-0.1	-0.8	0.0	0.0	-0.1
C LC_123	SLN43 column below / Begin	92.3	7.4	-1.2	0.0	0.0	-2.0
	SLN192 ring / Begin	-12.4	-0.1	-19.4	0.0	0.0	0.3
	SLN195 ring / End	20.2	-0.1	-19.5	0.0	0.0	-0.5
	SLN446 raking strut / End	-19.9	-0.1	-1.1	-0.5	0.0	2.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN43 column below / Begin	85.1	0.1	0.2	0.0	1.0	-0.3
	SLN192 ring / Begin	-11.5	-0.2	-19.4	0.0	0.0	0.5
	SLN195 ring / End	10.8	-0.2	-19.5	0.0	0.0	-0.4
	SLN446 raking strut / End	-29.7	0.7	-0.9	0.1	0.0	0.6
C LC_125	SLN43 column below / Begin	129.9	-1.3	-8.0	0.0	-4.7	0.4
	SLN192 ring / Begin	-37.2	0.1	-19.4	0.0	0.0	0.0
	SLN195 ring / End	35.6	-0.1	-19.5	0.0	0.0	-0.4
	SLN446 raking strut / End	30.3	0.0	-1.5	0.2	0.0	-0.3
C LC_126	SLN43 column below / Begin	72.1	0.0	0.2	0.0	1.0	0.1
	SLN192 ring / Begin	-7.7	-0.3	-19.4	0.0	0.0	0.7
	SLN195 ring / End	8.3	-0.2	-19.5	0.0	0.0	-0.5
	SLN446 raking strut / End	-46.5	-0.2	-0.5	0.0	0.0	-0.1
C LC_127	SLN43 column below / Begin	24.2	7.4	-1.3	0.0	-0.5	-2.0
	SLN192 ring / Begin	-1.3	0.0	-4.9	0.0	0.0	0.0
	SLN195 ring / End	8.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-5.8	-0.1	-1.1	-0.5	0.0	2.2
C LC_128	SLN43 column below / Begin	17.3	0.1	0.1	0.0	0.5	-0.4
	SLN192 ring / Begin	-0.5	-0.1	-4.9	0.0	0.0	0.3
	SLN195 ring / End	-0.1	0.0	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-15.2	0.7	-0.9	0.1	0.0	0.5
C LC_129	SLN43 column below / Begin	60.5	-1.3	-8.2	0.0	-5.1	0.4
	SLN192 ring / Begin	-25.5	0.1	-4.9	0.0	0.0	-0.2
	SLN195 ring / End	23.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	42.8	0.1	-1.5	0.1	0.0	-0.3
C LC_130	SLN43 column below / Begin	5.4	0.0	0.1	0.0	0.5	0.1
	SLN192 ring / Begin	2.6	-0.1	-4.9	0.0	0.0	0.3
	SLN195 ring / End	-2.2	-0.1	-4.9	0.0	0.0	-0.2
	SLN446 raking strut / End	-30.3	-0.2	-0.5	0.0	0.0	-0.1
C LC_135	SLN43 column below / Begin	124.3	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-28.0	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	28.1	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-22.4	0.0	-1.0	0.1	0.0	0.0
C LC_141	SLN43 column below / Begin	26.6	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.9	0.0	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-2.6	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN43 column below / Begin	26.5	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-2.7	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN43 column below / Begin	27.2	0.0	0.1	0.0	0.6	0.0
	SLN192 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.1
	SLN195 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.1
	SLN446 raking strut / End	-1.8	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN43 column below / Begin	25.9	0.0	0.2	0.0	0.6	0.0
	SLN192 ring / Begin	-8.5	-0.1	-4.9	0.0	0.0	0.2
	SLN195 ring / End	8.5	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN446 raking strut / End	-3.5	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN43 column below / Begin	125.0	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.6	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	28.0	-0.2	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-21.6	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN43 column below / Begin	124.5	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-27.5	-0.2	-25.6	0.0	0.0	0.6
	SLN195 ring / End	27.4	-0.2	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-22.2	0.0	-1.0	0.1	0.0	0.0
C LC_147	SLN43 column below / Begin	127.4	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-28.8	-0.1	-25.6	0.0	0.0	0.5
	SLN195 ring / End	28.9	-0.1	-25.7	0.0	0.0	-0.5
	SLN446 raking strut / End	-18.4	0.0	-1.0	0.1	0.0	0.0
C LC_148	SLN43 column below / Begin	122.1	0.0	0.3	0.0	1.3	0.1
	SLN192 ring / Begin	-26.4	-0.3	-25.6	0.0	0.0	0.7
	SLN195 ring / End	26.6	-0.3	-25.7	0.0	0.0	-0.6
	SLN446 raking strut / End	-25.4	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN43 column below / Begin	79.9	0.0	-1.7	0.0	-6.7	0.0
	SLN192 ring / Begin	-12.3	-0.1	-11.9	0.0	0.0	0.3
	SLN195 ring / End	12.4	-0.1	-11.8	0.0	0.0	-0.3
	SLN446 raking strut / End	-13.2	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN43 column below / Begin	73.3	0.0	2.2	0.0	8.7	0.0
	SLN192 ring / Begin	-28.3	-0.2	-14.2	0.0	0.0	0.5
	SLN195 ring / End	28.4	-0.2	-14.1	0.0	0.0	-0.5
	SLN446 raking strut / End	-8.7	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN43 column below / Begin	125.1	0.0	0.3	0.0	1.2	0.1
	SLN192 ring / Begin	-27.7	-0.2	-21.2	0.0	0.0	0.6
	SLN195 ring / End	27.8	-0.2	-21.0	0.0	0.0	-0.6
	SLN446 raking strut / End	-21.8	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN43 column below / Begin	79.5	0.0	-1.6	0.0	-6.6	0.0
	SLN192 ring / Begin	-12.4	-0.1	-11.9	0.0	0.0	0.3
	SLN195 ring / End	12.5	-0.1	-11.8	0.0	0.0	-0.3
	SLN446 raking strut / End	-12.9	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	28.6 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN193+SLN191 radial (continuous)-bfl 1	12.7	C LC_109	234.7	0.0	37.6	OK
SLN193+SLN191 radial (continuous)-tfl 1	12.7	C LC_153	72.5	0.0	0.0	OK
SLN193+SLN191 radial (continuous)-w 1	7.2	C LC_122	176.6	0.0	48.7	OK
SLN43 column below	10.0	C LC_114	275.3	0.2	0.0	OK
SLN192 ring-bfl 1	12.7	C LC_122	50.5	0.0	0.0	OK
SLN192 ring-tfl 1	12.7	C LC_122	46.0	0.0	0.0	OK
SLN192 ring-w 1	7.2	C LC_122	153.5	0.0	0.0	OK
SLN195 ring-bfl 1	12.7	C LC_148	46.8	0.0	0.0	OK
SLN195 ring-tfl 1	12.7	C LC_106	42.1	0.0	0.0	OK
SLN195 ring-w 1	7.2	C LC_148	145.7	0.0	0.0	OK
SLN446 raking strut	10.0	C LC_123	64.2	0.0	0.0	OK
End plate column below SLN43 column below	12.0	C LC_109	268.4	0.0	37.6	OK
End plate SLN192 ring	12.0	C LC_122	71.4	0.0	0.0	OK
End plate SLN195 ring	12.0	C LC_148	59.2	0.0	9.5	OK
Cap plate SLN446 raking strut	12.0	C LC_123	141.7	0.0	0.0	OK
Fin plate SLN446 raking strut	12.0	C LC_115	275.1	0.1	129.4	OK
Fin plate 2 SLN446 raking strut	12.0	C LC_123	275.2	0.1	96.7	OK
Gusset SLN446 raking strut	12.0	C LC_123	275.2	0.1	66.1	OK
Flange top a end plate SLN192 ring	12.0	C LC_122	61.1	0.0	0.0	OK
Flange top b end plate SLN192 ring	12.0	C LC_122	48.2	0.0	0.0	OK
Flange bottom a end plate SLN192 ring	12.0	C LC_122	45.2	0.0	0.0	OK
Flange bottom b end plate SLN192 ring	12.0	C LC_148	71.0	0.0	0.0	OK
Flange top a end plate SLN195 ring	12.0	C LC_148	41.8	0.0	0.0	OK
Flange top b end plate SLN195 ring	12.0	C LC_106	55.6	0.0	0.0	OK
Flange bottom a end plate SLN195 ring	12.0	C LC_148	68.7	0.0	0.0	OK
Flange bottom b end plate SLN195 ring	12.0	C LC_106	40.6	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

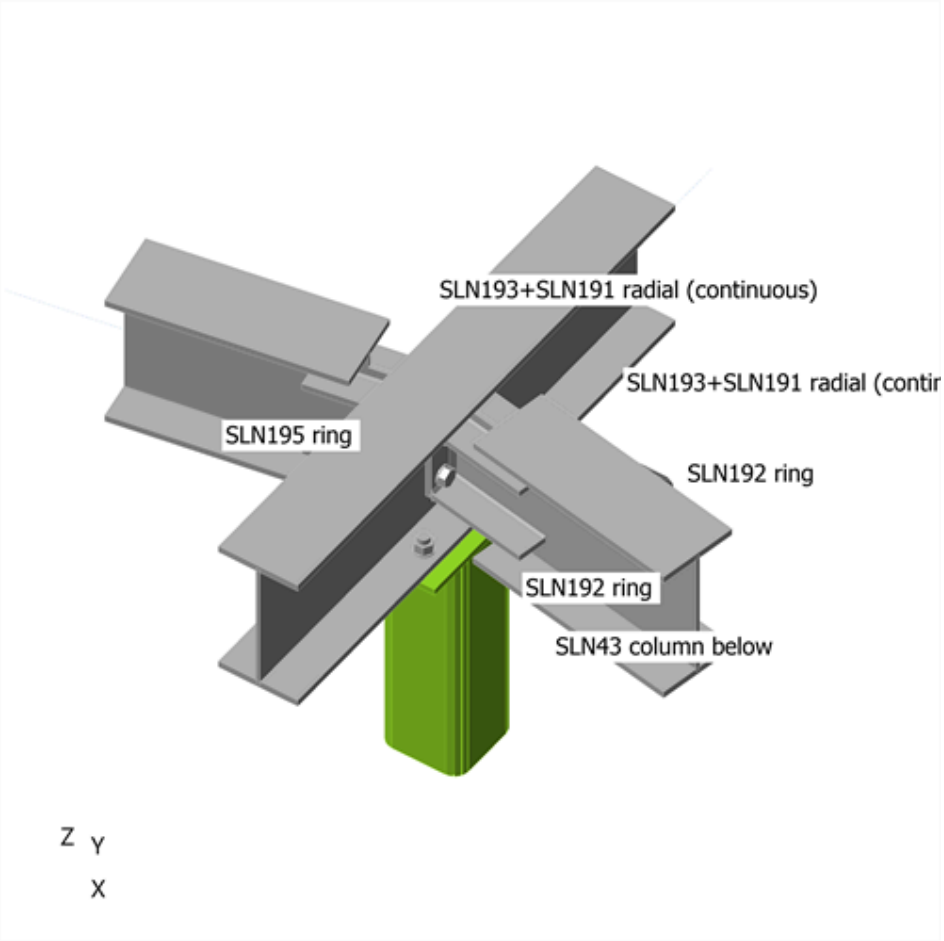
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

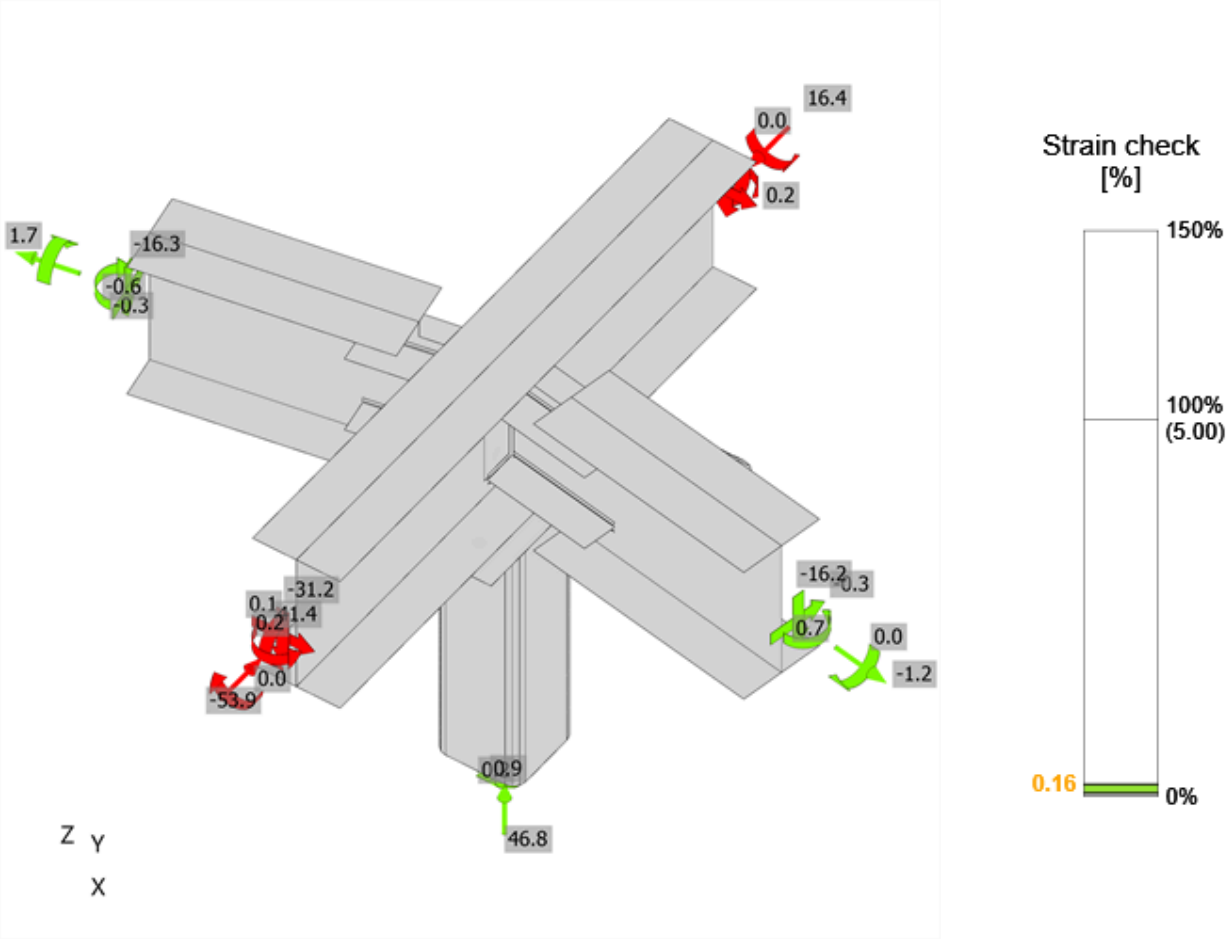
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN43 column below	150	C LC_109	0	5	0.1	OK
SLN446 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

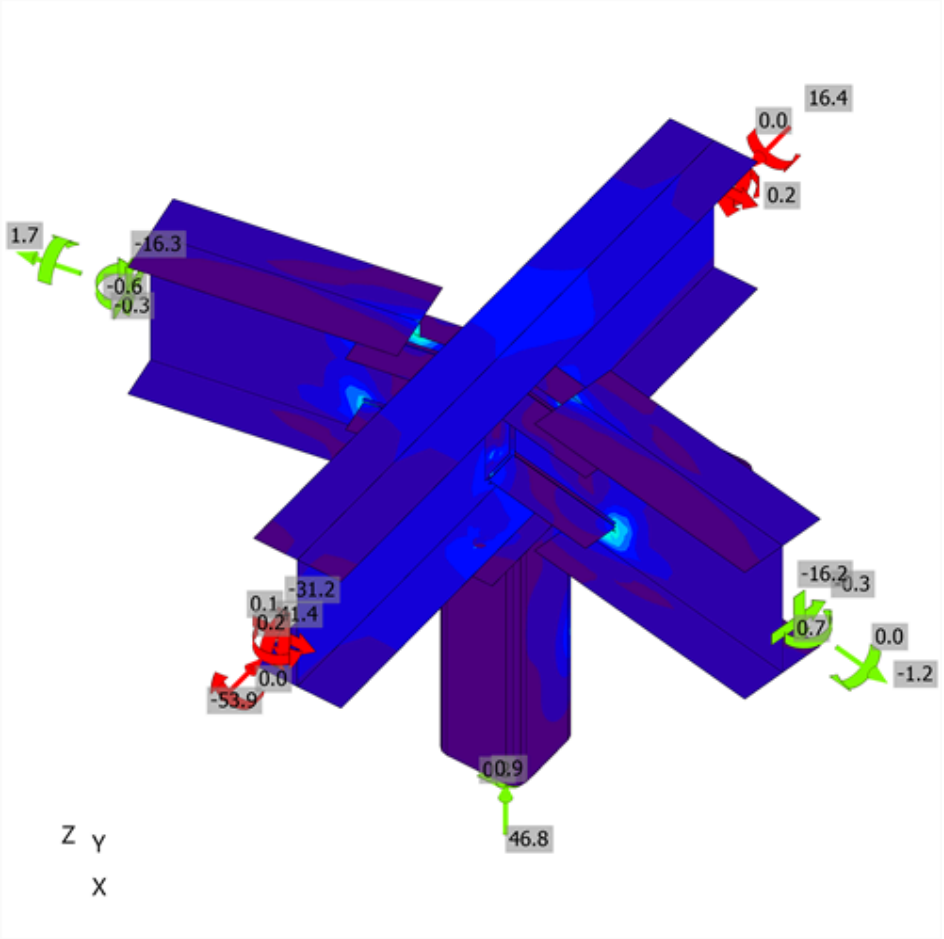
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

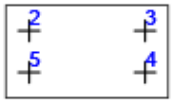
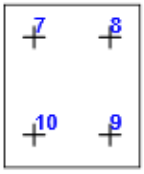
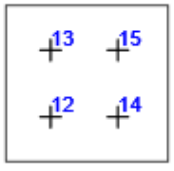


Strain check, C LC_114



Equivalent stress, C LC_114

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_110	17.1	6.3	126.4	12.1	6.7	15.4	OK	OK
	B3	M20 8.8 - 1	C LC_109	39.9	6.0	98.1	28.3	6.3	26.5	OK	OK
	B4	M20 8.8 - 1	C LC_109	40.4	6.2	103.8	28.6	6.6	27.1	OK	OK
	B5	M20 8.8 - 1	C LC_110	17.5	6.2	124.3	12.4	6.6	15.4	OK	OK
	B7	M20 8.8 - 1	C LC_122	10.1	0.8	123.8	7.2	0.9	5.9	OK	OK
	B8	M20 8.8 - 1	C LC_109	5.7	0.7	123.8	4.1	1.0	3.7	OK	OK
	B9	M20 8.8 - 1	C LC_109	5.6	0.8	123.8	4.0	1.0	3.7	OK	OK
	B10	M20 8.8 - 1	C LC_122	11.0	0.9	123.8	7.8	1.0	6.5	OK	OK
	B12	M20 8.8 - 1	C LC_123	17.5	15.5	115.3	12.4	16.4	25.3	OK	OK
	B13	M20 8.8 - 1	C LC_123	8.0	13.5	115.3	5.7	14.3	18.4	OK	OK
	B14	M20 8.8 - 1	C LC_117	1.1	9.3	106.0	0.8	15.2	10.4	OK	OK
	B15	M20 8.8 - 1	C LC_114	0.2	7.5	115.3	0.1	12.9	8.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN446 raking strut	SLN446 raking strut	◢ 4.2	519	C LC_117	189.5	0.0	90.2	-95.8	-8.7	46.8	25.5	OK	OK
SLN43 column below-w 3	Gusset SLN446 raking strut	◢ 5.6 ◣	287	C LC_114	397.5	0.6	-135.7	-136.7	-166.9	98.2	68.6	OK	OK
		◢ 5.6 ◣	287	C LC_114	397.5	0.5	-136.5	135.4	167.7	98.2	68.3	OK	OK
SLN192 ring-w 1	Flange top b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_148	55.2	0.0	30.7	24.9	9.0	13.6	9.0	OK	OK
		◢ 4.2 ◣	197	C LC_122	33.3	0.0	-16.0	14.7	8.3	8.2	6.9	OK	OK
SLN192 ring-w 1	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	198	C LC_148	91.8	0.0	27.0	31.5	39.7	22.7	13.9	OK	OK
		◢ 4.2 ◣	198	C LC_148	131.2	0.0	51.9	-47.5	-50.8	32.4	19.9	OK	OK
SLN195 ring-w 1	Flange top a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_152	52.4	0.0	25.9	22.3	-14.0	12.9	7.6	OK	OK
		◢ 4.2 ◣	197	C LC_106	29.6	0.0	-14.3	13.3	-6.9	7.3	6.2	OK	OK
SLN195 ring-w 1	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	198	C LC_121	93.6	0.0	27.4	30.4	-41.8	23.1	12.9	OK	OK
		◢ 4.2 ◣	198	C LC_148	128.6	0.0	50.2	-46.4	50.2	31.8	17.3	OK	OK
Cap plate SLN446 raking strut	Fin plate SLN446 raking strut	◢ 5.6 ◣	139	C LC_115	267.8	0.0	84.9	116.7	88.8	66.2	49.5	OK	OK
		◢ 5.6 ◣	139	C LC_117	171.6	0.0	81.9	-71.6	-49.4	42.4	25.6	OK	OK
Cap plate SLN446 raking strut	Fin plate 2 SLN446 raking strut	◢ 5.6 ◣	139	C LC_114	148.7	0.0	-57.9	-63.3	47.3	36.7	22.0	OK	OK
		◢ 5.6 ◣	139	C LC_123	259.2	0.0	-119.2	104.0	-82.7	64.1	50.4	OK	OK
End plate column below SLN43 column below	SLN43 column below	◢ 5.6 ◣	521	C LC_106	288.4	0.0	-187.4	-125.1	-19.2	71.3	39.0	OK	OK
		◢ 5.6 ◣	521	C LC_109	234.8	0.0	161.8	-78.6	-59.0	58.0	36.3	OK	OK
SLN193+SLN191 radial (continuous)-w 1	SLN192 ring-w 1	◢ 7.2	189	C LC_121	99.9	0.0	37.4	38.0	37.6	24.7	15.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN193+SLN191 radial (continuous)-w 1	SLN195 ring-w 1	◢ 7.2	189	C LC_121	96.9	0.0	36.3	37.3	-36.1	23.9	16.4	OK	OK
End plate SLN192 ring	SLN192 ring-w 1	◢ 4.2 ◣	149	C LC_121	87.5	0.0	-26.7	-30.4	-37.3	21.6	11.6	OK	OK
		◢ 4.2 ◣	149	C LC_148	67.7	0.0	-13.3	25.9	28.3	16.7	11.1	OK	OK
End plate SLN195 ring	SLN195 ring-w 1	◢ 4.2 ◣	149	C LC_121	97.0	0.0	-31.9	-29.9	43.6	24.0	14.6	OK	OK
		◢ 4.2 ◣	149	C LC_148	73.0	0.0	-19.8	24.9	-32.0	18.0	11.8	OK	OK
SLN192 ring-w 1	Flange top a end plate SLN192 ring	◢ 4.2 ◣	199	C LC_122	102.3	0.0	-12.3	8.0	-58.1	25.3	18.0	OK	OK
		◢ 4.2 ◣	198	C LC_122	66.0	0.0	11.5	12.9	-35.2	16.3	9.7	OK	OK
SLN192 ring-w 1	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	198	C LC_122	94.5	0.0	17.1	-17.1	50.8	23.3	13.9	OK	OK
		◢ 4.2 ◣	199	C LC_121	77.0	0.0	-9.4	12.6	42.3	19.0	10.2	OK	OK
SLN195 ring-w 1	Flange top b end plate SLN195 ring	◢ 4.2 ◣	199	C LC_106	94.2	0.0	-14.0	8.7	53.1	23.3	14.9	OK	OK
		◢ 4.2 ◣	198	C LC_148	59.6	0.0	10.9	11.8	31.7	14.7	8.8	OK	OK
SLN195 ring-w 1	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	198	C LC_148	89.5	0.0	15.6	-15.5	-48.5	22.1	13.0	OK	OK
		◢ 4.2 ◣	199	C LC_121	80.4	0.0	-9.5	14.8	-43.7	19.9	12.6	OK	OK
End plate SLN192 ring	Flange top a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_109	33.1	0.0	-0.7	-2.2	19.0	8.2	7.4	OK	OK
		◢ 4.2 ◣	58	C LC_109	35.4	0.0	15.3	10.9	14.9	8.8	8.3	OK	OK
End plate SLN192 ring	Flange top b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_122	46.6	0.0	-0.9	-0.4	-26.9	11.5	9.1	OK	OK
		◢ 4.2 ◣	58	C LC_122	58.0	0.0	33.0	22.6	-15.8	14.3	11.8	OK	OK
End plate SLN192 ring	Flange bottom a end plate SLN192 ring	◢ 4.2 ◣	58	C LC_121	57.9	0.0	12.5	-23.0	23.2	14.3	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_122	43.8	0.0	-11.5	-17.2	17.3	10.8	8.3	OK	OK
End plate SLN192 ring	Flange bottom b end plate SLN192 ring	◢ 4.2 ◣	58	C LC_122	70.2	0.0	39.8	-30.2	14.3	17.3	13.7	OK	OK
		◢ 4.2 ◣	58	C LC_122	56.8	0.0	-3.8	-0.1	32.7	14.0	10.0	OK	OK
End plate SLN195 ring	Flange top a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	38.5	0.0	0.9	-2.9	22.0	9.5	8.6	OK	OK
		◢ 4.2 ◣	58	C LC_106	46.8	0.0	4.9	8.9	25.4	11.6	9.5	OK	OK
End plate SLN195 ring	Flange top b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_125	48.8	0.0	-24.5	18.5	-15.8	12.1	8.1	OK	OK
		◢ 4.2 ◣	58	C LC_125	39.8	0.0	-12.0	-18.0	12.6	9.8	7.5	OK	OK
End plate SLN195 ring	Flange bottom a end plate SLN195 ring	◢ 4.2 ◣	58	C LC_148	58.3	0.0	32.4	-25.8	-10.8	14.4	11.7	OK	OK
		◢ 4.2 ◣	58	C LC_148	47.9	0.0	-2.8	1.4	-27.6	11.8	9.5	OK	OK
End plate SLN195 ring	Flange bottom b end plate SLN195 ring	◢ 4.2 ◣	58	C LC_121	60.6	0.0	13.3	-24.0	-24.3	15.0	8.4	OK	OK
		◢ 4.2 ◣	58	C LC_125	48.6	0.0	-24.4	-18.9	15.3	12.0	8.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_140
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut

Name	SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

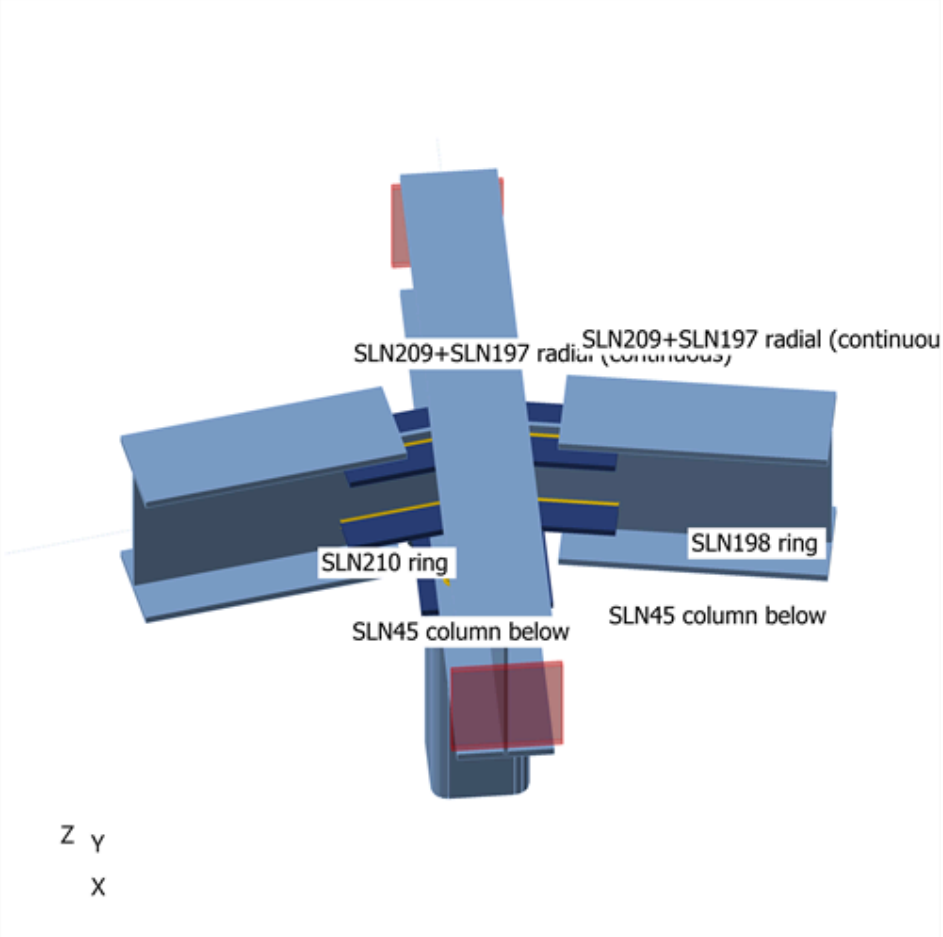
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN209+SLN197 radial (continuous)	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN45 column below	2 - SHS150x150x10	0.0	90.0	-40.0	0	0	0
SLN198 ring	1 - UB254x146x43	-140.0	0.0	0.0	0	0	0
SLN210 ring	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN445 raking strut	2 - SHS150x150x10	140.1	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN209+SLN197 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN209+SLN197 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN45 column below / begin		Position	0
SLN198 ring / begin		Position	0
SLN210 ring / end		Position	0
SLN445 raking strut / end		Position	0



SLN209+SLN197 radial (continuous) - SLN45 column below - SLN198 ring - SLN210 ring - SLN445 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN45 column below / Begin	23.7	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.8	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-4.4	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN45 column below / Begin	111.2	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-27.8	-0.2	-26.0	0.0	0.0	0.7
	SLN210 ring / End	27.8	-0.2	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-30.4	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN45 column below / Begin	117.9	2.4	-2.5	0.0	0.7	-0.6
	SLN198 ring / Begin	-25.1	-0.1	-24.4	0.0	0.0	0.4
	SLN210 ring / End	27.8	-0.1	-24.0	0.0	0.0	-0.4
	SLN445 raking strut / End	-9.6	0.0	-1.2	-0.2	0.0	0.7
C LC_104	SLN45 column below / Begin	89.9	0.0	0.7	0.0	2.9	-0.1
	SLN198 ring / Begin	-16.1	-0.3	-24.4	0.0	0.0	0.8
	SLN210 ring / End	15.7	-0.3	-24.0	0.0	0.0	-0.6
	SLN445 raking strut / End	-47.7	0.2	-0.8	0.0	0.0	0.2
C LC_105	SLN45 column below / Begin	115.4	-2.9	-2.0	0.0	1.1	0.8
	SLN198 ring / Begin	-27.6	-0.1	-24.4	0.0	0.0	0.5
	SLN210 ring / End	24.3	-0.1	-24.0	0.0	0.0	-0.4
	SLN445 raking strut / End	-13.0	0.0	-1.2	0.2	0.0	-0.8
C LC_106	SLN45 column below / Begin	90.9	0.0	0.7	0.0	2.9	0.2
	SLN198 ring / Begin	-15.3	-0.3	-24.4	0.0	0.0	0.8
	SLN210 ring / End	15.8	-0.3	-24.0	0.0	0.0	-0.6
	SLN445 raking strut / End	-46.2	-0.3	-0.9	0.0	0.0	-0.2
C LC_107	SLN45 column below / Begin	30.0	4.8	-6.4	0.0	-4.1	-1.3
	SLN198 ring / Begin	-5.7	0.1	-0.5	0.0	0.0	-0.2
	SLN210 ring / End	10.7	0.1	-0.5	0.0	0.0	0.2
	SLN445 raking strut / End	36.5	0.0	-1.1	-0.3	0.0	1.5
C LC_108	SLN45 column below / Begin	-19.7	0.1	0.1	0.0	0.3	-0.3
	SLN198 ring / Begin	9.4	-0.1	-0.5	0.0	0.0	0.3
	SLN210 ring / End	-9.8	-0.1	-0.5	0.0	0.0	-0.1
	SLN445 raking strut / End	-31.0	0.5	-0.4	0.0	0.0	0.4
C LC_109	SLN45 column below / Begin	25.3	-5.7	-5.3	0.0	-3.4	1.6
	SLN198 ring / Begin	-10.5	0.1	-0.5	0.0	0.0	-0.1
	SLN210 ring / End	4.4	0.1	-0.5	0.0	0.0	0.2
	SLN445 raking strut / End	30.0	0.1	-1.1	0.4	0.0	-1.7
C LC_110	SLN45 column below / Begin	-17.8	-0.1	0.1	0.0	0.3	0.3
	SLN198 ring / Begin	10.9	-0.1	-0.5	0.0	0.0	0.2
	SLN210 ring / End	-10.2	-0.1	-0.5	0.0	0.0	-0.1
	SLN445 raking strut / End	-28.4	-0.6	-0.4	0.0	0.0	-0.4
C LC_111	SLN45 column below / Begin	97.7	4.8	-5.9	0.0	-2.4	-1.3
	SLN198 ring / Begin	-19.7	0.0	-16.5	0.0	0.0	0.2
	SLN210 ring / End	24.8	0.0	-16.2	0.0	0.0	-0.1
	SLN445 raking strut / End	17.7	0.0	-1.4	-0.3	0.0	1.5
C LC_112	SLN45 column below / Begin	45.8	0.1	0.5	0.0	2.1	-0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN198 ring / Begin	-3.6	-0.3	-16.5	0.0	0.0	0.7
	SLN210 ring / End	3.1	-0.2	-16.2	0.0	0.0	-0.5
	SLN445 raking strut / End	-52.6	0.5	-0.6	0.0	0.0	0.4
C LC_113	SLN45 column below / Begin	92.9	-5.7	-4.9	0.0	-1.6	1.6
	SLN198 ring / Begin	-24.5	0.0	-16.5	0.0	0.0	0.2
	SLN210 ring / End	18.3	0.0	-16.2	0.0	0.0	-0.1
	SLN445 raking strut / End	11.1	0.1	-1.3	0.4	0.0	-1.7
C LC_114	SLN45 column below / Begin	47.9	-0.1	0.5	0.0	2.1	0.3
	SLN198 ring / Begin	-2.1	-0.3	-16.5	0.0	0.0	0.7
	SLN210 ring / End	2.9	-0.2	-16.2	0.0	0.0	-0.5
	SLN445 raking strut / End	-49.9	-0.6	-0.7	-0.1	0.0	-0.4
C LC_115	SLN45 column below / Begin	36.3	4.8	-6.3	0.0	-3.9	-1.3
	SLN198 ring / Begin	-8.1	0.0	-1.8	0.0	0.0	-0.1
	SLN210 ring / End	13.0	0.1	-1.7	0.0	0.0	0.2
	SLN445 raking strut / End	35.5	0.0	-1.4	-0.3	0.0	1.5
C LC_116	SLN45 column below / Begin	-13.7	0.1	0.1	0.0	0.6	-0.3
	SLN198 ring / Begin	7.1	-0.1	-1.8	0.0	0.0	0.3
	SLN210 ring / End	-7.6	-0.1	-1.7	0.0	0.0	-0.2
	SLN445 raking strut / End	-32.3	0.5	-0.6	0.0	0.0	0.4
C LC_117	SLN45 column below / Begin	31.6	-5.7	-5.3	0.0	-3.2	1.6
	SLN198 ring / Begin	-12.9	0.0	-1.8	0.0	0.0	-0.1
	SLN210 ring / End	6.7	0.0	-1.7	0.0	0.0	0.1
	SLN445 raking strut / End	29.0	0.1	-1.3	0.4	0.0	-1.7
C LC_118	SLN45 column below / Begin	-11.7	-0.1	0.1	0.0	0.6	0.3
	SLN198 ring / Begin	8.6	-0.1	-1.8	0.0	0.0	0.3
	SLN210 ring / End	-7.9	-0.1	-1.7	0.0	0.0	-0.2
	SLN445 raking strut / End	-29.7	-0.6	-0.7	0.0	0.0	-0.4
C LC_119	SLN45 column below / Begin	122.1	2.4	-2.4	0.0	1.0	-0.7
	SLN198 ring / Begin	-28.7	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	31.4	-0.1	-25.6	0.0	0.0	-0.5
	SLN445 raking strut / End	-15.3	0.0	-1.2	-0.2	0.0	0.7
C LC_120	SLN45 column below / Begin	100.4	0.0	0.8	0.0	3.1	-0.1
	SLN198 ring / Begin	-19.5	-0.3	-26.0	0.0	0.0	0.8
	SLN210 ring / End	19.1	-0.3	-25.6	0.0	0.0	-0.7
	SLN445 raking strut / End	-44.6	0.2	-0.8	0.0	0.0	0.2
C LC_121	SLN45 column below / Begin	120.3	-2.9	-1.9	0.0	1.3	0.8
	SLN198 ring / Begin	-31.2	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	27.9	-0.1	-25.6	0.0	0.0	-0.5
	SLN445 raking strut / End	-17.7	0.0	-1.2	0.2	0.0	-0.8
C LC_122	SLN45 column below / Begin	100.7	0.0	0.8	0.0	3.1	0.2
	SLN198 ring / Begin	-18.6	-0.3	-26.0	0.0	0.0	0.8
	SLN210 ring / End	19.2	-0.3	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-44.1	-0.3	-0.9	0.0	0.0	-0.2
C LC_123	SLN45 column below / Begin	106.0	4.8	-5.8	0.0	-1.9	-1.3
	SLN198 ring / Begin	-26.8	-0.1	-19.7	0.0	0.0	0.4
	SLN210 ring / End	31.9	0.0	-19.4	0.0	0.0	-0.3
	SLN445 raking strut / End	6.4	0.0	-1.4	-0.4	0.0	1.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN45 column below / Begin	66.9	0.1	0.6	0.0	2.4	-0.2
	SLN198 ring / Begin	-10.4	-0.3	-19.7	0.0	0.0	0.7
	SLN210 ring / End	9.8	-0.2	-19.4	0.0	0.0	-0.5
	SLN445 raking strut / End	-46.4	0.5	-0.6	0.0	0.0	0.4
C LC_125	SLN45 column below / Begin	102.6	-5.7	-4.8	0.0	-1.2	1.6
	SLN198 ring / Begin	-31.7	-0.1	-19.7	0.0	0.0	0.4
	SLN210 ring / End	25.4	-0.1	-19.4	0.0	0.0	-0.4
	SLN445 raking strut / End	1.7	0.1	-1.3	0.4	0.0	-1.7
C LC_126	SLN45 column below / Begin	67.4	-0.1	0.6	0.0	2.4	0.3
	SLN198 ring / Begin	-8.8	-0.3	-19.7	0.0	0.0	0.7
	SLN210 ring / End	9.7	-0.2	-19.4	0.0	0.0	-0.5
	SLN445 raking strut / End	-45.7	-0.6	-0.7	-0.1	0.0	-0.4
C LC_127	SLN45 column below / Begin	44.6	4.8	-6.2	0.0	-3.4	-1.3
	SLN198 ring / Begin	-15.1	0.0	-5.0	0.0	0.0	0.1
	SLN210 ring / End	20.2	0.0	-4.9	0.0	0.0	0.0
	SLN445 raking strut / End	24.3	0.0	-1.4	-0.3	0.0	1.5
C LC_128	SLN45 column below / Begin	7.4	0.1	0.2	0.0	0.9	-0.2
	SLN198 ring / Begin	0.3	-0.1	-5.0	0.0	0.0	0.4
	SLN210 ring / End	-0.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-26.0	0.5	-0.6	0.0	0.0	0.4
C LC_129	SLN45 column below / Begin	41.3	-5.7	-5.2	0.0	-2.7	1.6
	SLN198 ring / Begin	-20.1	0.0	-5.0	0.0	0.0	0.1
	SLN210 ring / End	13.9	0.0	-4.9	0.0	0.0	-0.1
	SLN445 raking strut / End	19.7	0.1	-1.3	0.4	0.0	-1.7
C LC_130	SLN45 column below / Begin	7.8	-0.1	0.2	0.0	0.9	0.3
	SLN198 ring / Begin	1.9	-0.1	-5.0	0.0	0.0	0.4
	SLN210 ring / End	-1.2	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-25.5	-0.6	-0.7	-0.1	0.0	-0.4
C LC_135	SLN45 column below / Begin	110.6	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.2	-0.2	-26.0	0.0	0.0	0.7
	SLN210 ring / End	28.3	-0.2	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-31.2	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN45 column below / Begin	24.2	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-9.0	0.0	-5.0	0.0	0.0	0.2
	SLN210 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-3.9	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN45 column below / Begin	23.3	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.6	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.6	-0.1	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN45 column below / Begin	24.1	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-9.0	0.0	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.9	0.0	-4.9	0.0	0.0	-0.2
	SLN445 raking strut / End	-3.9	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN45 column below / Begin	23.3	0.0	0.2	0.0	1.0	0.0
	SLN198 ring / Begin	-8.6	-0.1	-5.0	0.0	0.0	0.2
	SLN210 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN445 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN45 column below / Begin	113.0	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.6	-0.2	-26.0	0.0	0.0	0.6
	SLN210 ring / End	28.8	-0.1	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-28.0	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN45 column below / Begin	109.5	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-27.0	-0.3	-26.0	0.0	0.0	0.8
	SLN210 ring / End	26.8	-0.3	-25.6	0.0	0.0	-0.7
	SLN445 raking strut / End	-32.7	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN45 column below / Begin	112.9	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-28.6	-0.1	-26.0	0.0	0.0	0.6
	SLN210 ring / End	28.5	-0.1	-25.6	0.0	0.0	-0.6
	SLN445 raking strut / End	-28.2	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN45 column below / Begin	109.6	0.0	0.8	0.0	3.2	0.0
	SLN198 ring / Begin	-26.9	-0.3	-26.0	0.0	0.0	0.7
	SLN210 ring / End	27.2	-0.3	-25.6	0.0	0.0	-0.7
	SLN445 raking strut / End	-32.6	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN45 column below / Begin	69.1	0.0	-1.1	0.0	-4.5	0.0
	SLN198 ring / Begin	-12.4	-0.1	-11.9	0.0	0.0	0.3
	SLN210 ring / End	12.4	-0.1	-11.5	0.0	0.0	-0.3
	SLN445 raking strut / End	-15.0	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN45 column below / Begin	67.9	0.0	2.2	0.0	8.8	0.0
	SLN198 ring / Begin	-28.4	-0.2	-14.0	0.0	0.0	0.5
	SLN210 ring / End	28.4	-0.2	-14.1	0.0	0.0	-0.5
	SLN445 raking strut / End	-15.4	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN45 column below / Begin	110.9	0.0	0.8	0.0	3.1	0.0
	SLN198 ring / Begin	-27.9	-0.2	-21.0	0.0	0.0	0.7
	SLN210 ring / End	28.0	-0.2	-20.7	0.0	0.0	-0.6
	SLN445 raking strut / End	-30.0	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN45 column below / Begin	68.4	0.0	-1.1	0.0	-4.4	0.0
	SLN198 ring / Begin	-12.5	-0.1	-11.9	0.0	0.0	0.3
	SLN210 ring / End	12.6	-0.1	-11.5	0.0	0.0	-0.3
	SLN445 raking strut / End	-14.9	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	22.3 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN209+SLN197 radial (continuous)-bfl 1	12.7	C LC_107	168.2	0.0	26.8	OK
SLN209+SLN197 radial (continuous)-tfl 1	12.7	C LC_153	69.2	0.0	0.0	OK
SLN209+SLN197 radial (continuous)-w 1	7.2	C LC_122	186.6	0.0	53.7	OK
SLN45 column below	10.0	C LC_104	275.3	0.1	0.0	OK
SLN198 ring-bfl 1	12.7	C LC_120	55.8	0.0	0.0	OK
SLN198 ring-tfl 1	12.7	C LC_120	51.0	0.0	0.0	OK
SLN198 ring-w 1	7.2	C LC_120	167.0	0.0	0.0	OK
SLN210 ring-bfl 1	12.7	C LC_148	49.6	0.0	0.0	OK
SLN210 ring-tfl 1	12.7	C LC_148	43.8	0.0	0.0	OK
SLN210 ring-w 1	7.2	C LC_148	152.5	0.0	0.0	OK
SLN445 raking strut	10.0	C LC_115	54.3	0.0	0.0	OK
End plate column below SLN45 column below	12.0	C LC_107	191.4	0.0	26.8	OK
End plate SLN198 ring	12.0	C LC_120	80.3	0.0	0.0	OK
End plate SLN210 ring	12.0	C LC_146	61.9	0.0	10.8	OK
Cap plate SLN445 raking strut	12.0	C LC_109	110.8	0.0	0.0	OK
Fin plate SLN445 raking strut	12.0	C LC_115	272.2	0.0	86.4	OK
Fin plate 2 SLN445 raking strut	12.0	C LC_117	275.1	0.0	99.4	OK
Gusset SLN445 raking strut	12.0	C LC_109	275.1	0.0	49.0	OK
Flange top a end plate SLN198 ring	12.0	C LC_120	68.7	0.0	0.0	OK
Flange top b end plate SLN198 ring	12.0	C LC_120	55.8	0.0	0.0	OK
Flange bottom a end plate SLN198 ring	12.0	C LC_120	50.1	0.0	0.0	OK
Flange bottom b end plate SLN198 ring	12.0	C LC_146	76.8	0.0	0.0	OK
Flange top a end plate SLN210 ring	12.0	C LC_148	47.2	0.0	0.0	OK
Flange top b end plate SLN210 ring	12.0	C LC_120	56.9	0.0	0.0	OK
Flange bottom a end plate SLN210 ring	12.0	C LC_148	71.5	0.0	0.0	OK
Flange bottom b end plate SLN210 ring	12.0	C LC_148	41.8	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

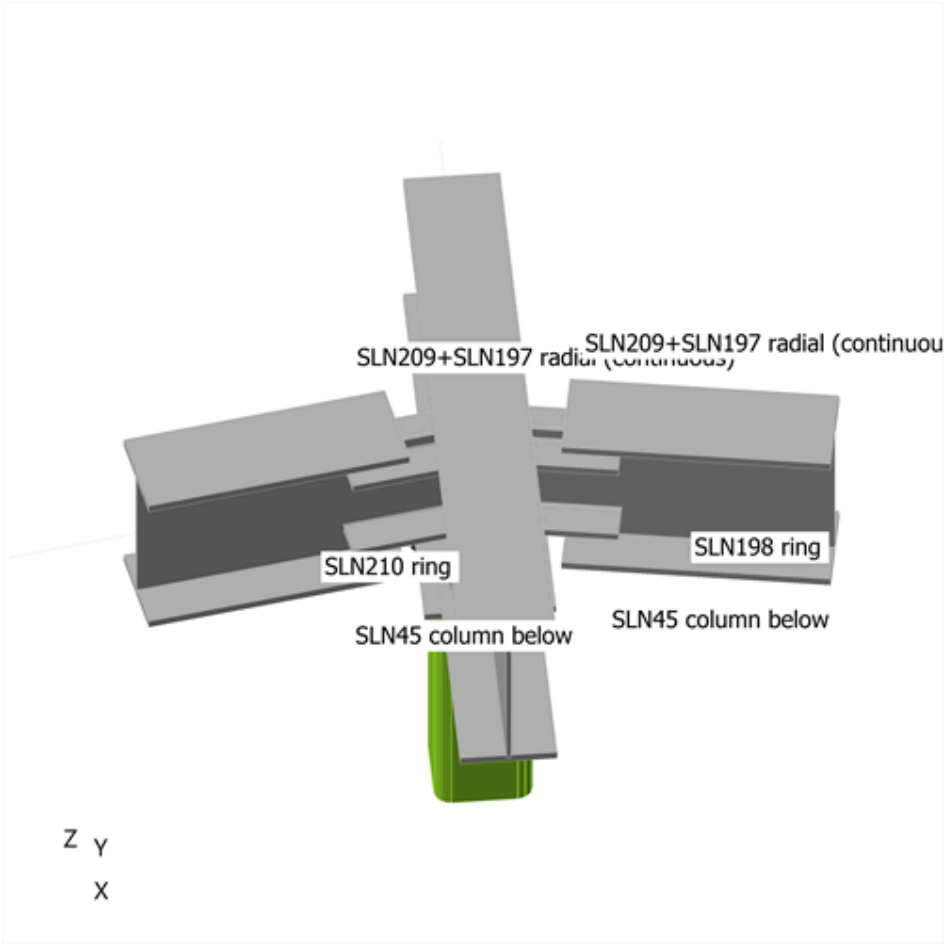
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

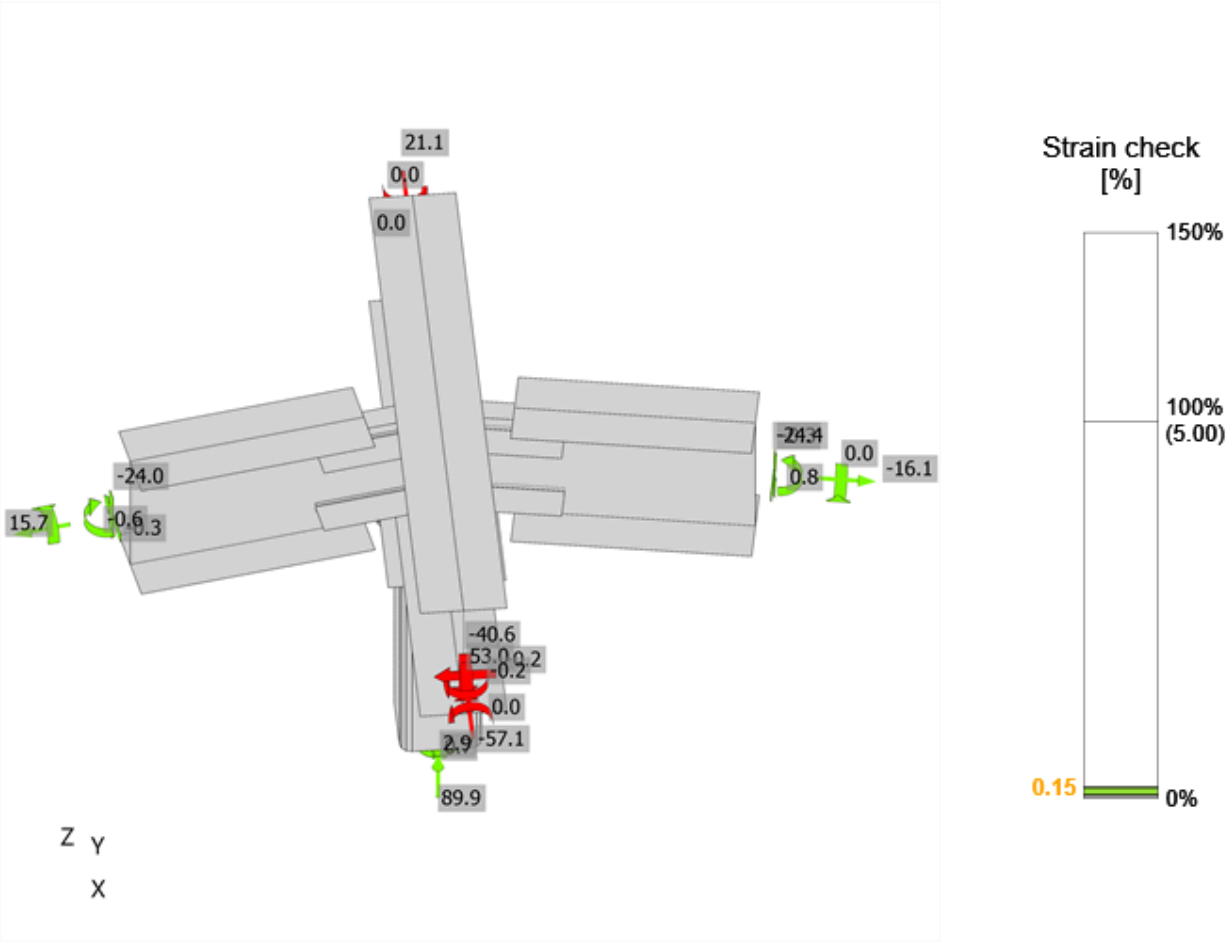
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN45 column below	150	C LC_112	0	5	0.1	OK
SLN445 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

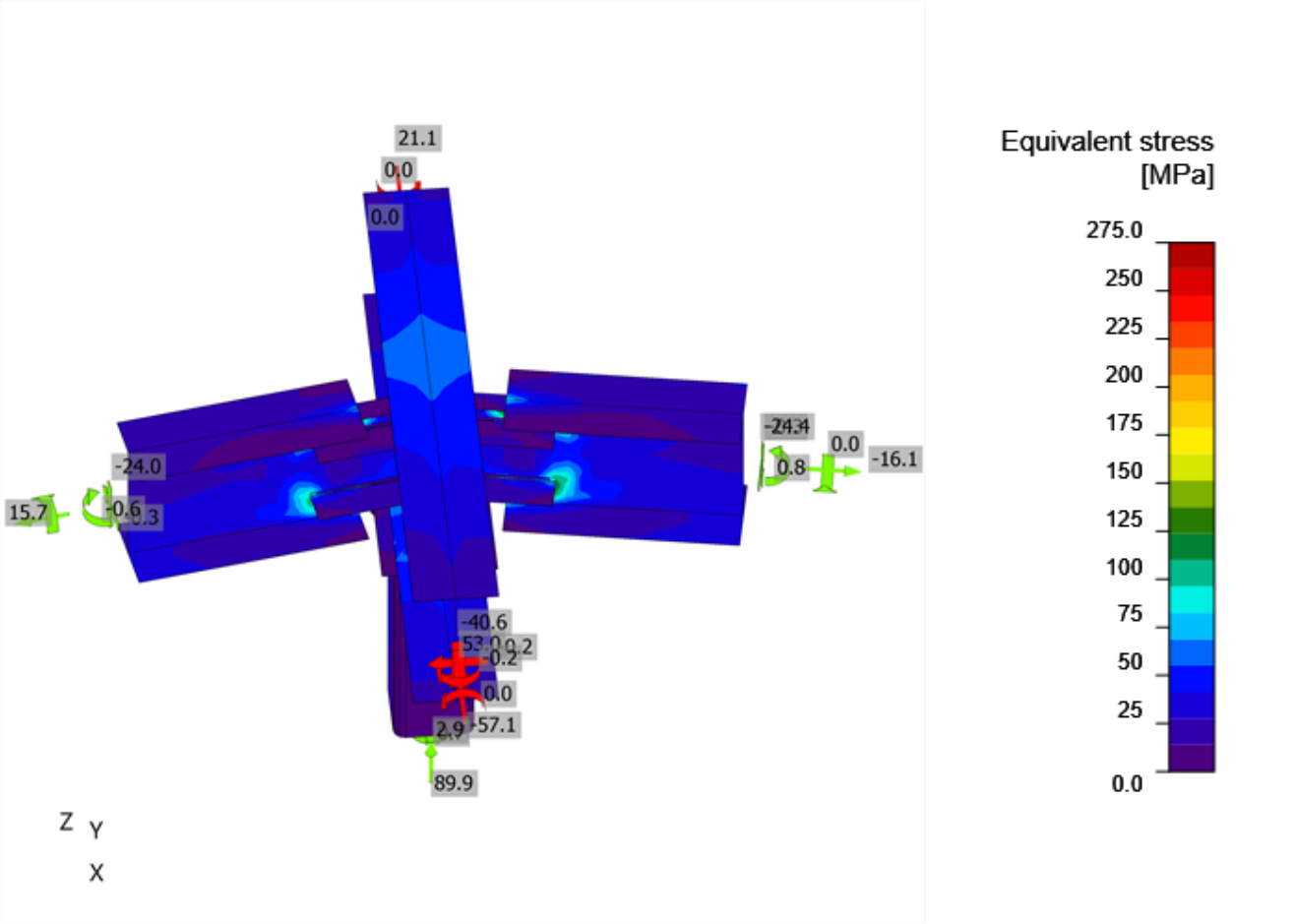
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

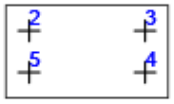
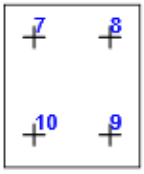
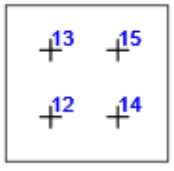


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	18.5	3.6	111.9	13.1	3.8	13.2	OK	OK
	B3	M20 8.8 - 1	C LC_107	28.5	4.4	130.4	20.2	4.7	19.1	OK	OK
	B4	M20 8.8 - 1	C LC_107	27.6	3.7	80.9	19.5	4.6	17.9	OK	OK
	B5	M20 8.8 - 1	C LC_112	5.4	10.5	171.2	3.8	11.1	13.8	OK	OK
	B7	M20 8.8 - 1	C LC_120	11.6	0.9	123.8	8.2	1.1	6.8	OK	OK
	B8	M20 8.8 - 1	C LC_104	0.0	1.0	63.4	0.0	1.6	1.1	OK	OK
	B9	M20 8.8 - 1	C LC_120	0.0	1.3	123.8	0.0	2.0	1.4	OK	OK
	B10	M20 8.8 - 1	C LC_120	12.1	1.0	123.8	8.6	1.1	7.2	OK	OK
	B12	M20 8.8 - 1	C LC_117	14.4	14.1	106.0	10.2	15.0	22.3	OK	OK
	B13	M20 8.8 - 1	C LC_109	6.6	9.7	106.0	4.7	10.3	13.6	OK	OK
	B14	M20 8.8 - 1	C LC_117	2.6	10.2	106.0	1.8	10.8	12.2	OK	OK
	B15	M20 8.8 - 1	C LC_112	0.2	7.3	115.3	0.1	12.1	7.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN445 raking strut	SLN445 raking strut	◢ 4.2	519	C LC_115	182.0	0.0	83.9	-92.0	15.2	45.0	25.0	OK	OK
SLN45 column below-w 3	Gusset SLN445 raking strut	◢ 5.6 ◣	288	C LC_112	397.3	0.4	-134.7	-132.6	-170.3	98.2	65.5	OK	OK
		◢ 5.6 ◣	288	C LC_112	397.4	0.5	-131.3	133.4	170.6	98.2	67.1	OK	OK
SLN198 ring-w 1	Flange top b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_153	69.1	0.0	36.6	30.4	14.8	17.1	10.7	OK	OK
		◢ 4.2 ◣	197	C LC_120	35.0	0.0	-16.9	15.4	8.7	8.7	7.0	OK	OK
SLN198 ring-w 1	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	198	C LC_146	95.8	0.0	28.4	33.5	40.9	23.7	14.5	OK	OK
		◢ 4.2 ◣	198	C LC_146	140.4	0.0	56.6	-51.4	-53.5	34.7	21.3	OK	OK
SLN210 ring-w 1	Flange top a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_153	60.9	0.0	32.0	27.1	-12.7	15.0	8.5	OK	OK
		◢ 4.2 ◣	197	C LC_148	29.3	0.0	-14.7	13.5	-5.8	7.2	6.6	OK	OK
SLN210 ring-w 1	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	198	C LC_148	93.2	0.0	27.9	31.8	-40.3	23.0	12.3	OK	OK
		◢ 4.2 ◣	198	C LC_148	132.4	0.0	52.3	-48.4	50.9	32.7	17.9	OK	OK
Cap plate SLN445 raking strut	Fin plate SLN445 raking strut	◢ 5.6 ◣	139	C LC_115	288.7	0.0	104.6	126.1	90.7	71.3	48.8	OK	OK
		◢ 5.6 ◣	139	C LC_117	141.4	0.0	90.9	-52.3	-34.2	34.9	23.2	OK	OK
Cap plate SLN445 raking strut	Fin plate 2 SLN445 raking strut	◢ 5.6 ◣	139	C LC_115	150.1	0.0	89.9	57.5	38.8	37.1	24.5	OK	OK
		◢ 5.6 ◣	139	C LC_117	294.9	0.0	104.6	-128.1	-94.6	72.9	50.7	OK	OK
End plate column below SLN45 column below	SLN45 column below	◢ 5.6 ◣	521	C LC_120	307.5	0.0	-196.0	-135.0	-21.9	76.0	41.6	OK	OK
		◢ 5.6 ◣	521	C LC_152	212.6	0.0	-93.7	110.0	-5.7	52.5	31.7	OK	OK
SLN209+SLN197 radial (continuous)-w 1	SLN198 ring-w 1	◢ 7.2	189	C LC_119	86.8	0.0	8.6	5.4	49.6	21.5	14.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN209+SLN197 radial (continuous)-w 1	SLN210 ring-w 1	◢ 7.2	189	C LC_121	93.0	0.0	4.4	10.4	-52.6	23.0	15.0	OK	OK
End plate SLN198 ring	SLN198 ring-w 1	◢ 4.2 ◣	149	C LC_119	88.8	0.0	-32.7	-19.5	-43.5	21.9	12.6	OK	OK
		◢ 4.2 ◣	149	C LC_148	70.3	0.0	-12.1	27.5	28.9	17.4	11.3	OK	OK
End plate SLN210 ring	SLN210 ring-w 1	◢ 4.2 ◣	149	C LC_121	87.4	0.0	-26.0	-23.5	42.0	21.6	14.1	OK	OK
		◢ 4.2 ◣	149	C LC_121	70.6	0.0	-20.2	22.5	-31.9	17.4	10.9	OK	OK
SLN198 ring-w 1	Flange top a end plate SLN198 ring	◢ 4.2 ◣	199	C LC_120	118.3	0.0	-10.8	6.6	-67.7	29.2	20.7	OK	OK
		◢ 4.2 ◣	198	C LC_120	77.7	0.0	12.1	13.7	-42.2	19.2	11.1	OK	OK
SLN198 ring-w 1	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	198	C LC_120	102.7	0.0	19.5	-19.6	54.8	25.4	15.0	OK	OK
		◢ 4.2 ◣	198	C LC_104	82.8	0.0	-11.8	-3.2	47.2	20.5	12.0	OK	OK
SLN210 ring-w 1	Flange top b end plate SLN210 ring	◢ 4.2 ◣	199	C LC_148	99.2	0.0	-10.5	6.1	56.6	24.5	15.1	OK	OK
		◢ 4.2 ◣	198	C LC_148	67.2	0.0	10.9	12.2	36.3	16.6	9.7	OK	OK
SLN210 ring-w 1	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	198	C LC_148	94.4	0.0	16.6	-16.5	-51.1	23.3	13.6	OK	OK
		◢ 4.2 ◣	199	C LC_121	69.8	0.0	-12.9	14.1	-37.0	17.2	10.9	OK	OK
End plate SLN198 ring	Flange top a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_104	35.2	0.0	0.6	2.3	-20.2	8.7	7.2	OK	OK
		◢ 4.2 ◣	58	C LC_120	23.0	0.0	-6.6	-11.6	-5.3	5.7	5.7	OK	OK
End plate SLN198 ring	Flange top b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_120	53.6	0.0	-1.4	-0.7	-30.9	13.2	9.9	OK	OK
		◢ 4.2 ◣	58	C LC_120	66.7	0.0	38.1	26.1	-17.8	16.5	13.1	OK	OK
End plate SLN198 ring	Flange bottom a end plate SLN198 ring	◢ 4.2 ◣	58	C LC_119	55.3	0.0	12.0	-19.5	24.3	13.7	9.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_122	44.8	0.0	-12.2	-17.0	18.2	11.1	8.5	OK	OK
End plate SLN198 ring	Flange bottom b end plate SLN198 ring	◢ 4.2 ◣	58	C LC_120	76.5	0.0	43.3	-32.6	16.2	18.9	14.9	OK	OK
		◢ 4.2 ◣	58	C LC_120	63.9	0.0	-4.5	0.2	36.8	15.8	10.9	OK	OK
End plate SLN210 ring	Flange top a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_148	41.8	0.0	0.7	-3.4	23.9	10.3	9.0	OK	OK
		◢ 4.2 ◣	58	C LC_120	51.0	0.0	6.3	9.9	27.5	12.6	10.0	OK	OK
End plate SLN210 ring	Flange top b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_104	27.6	0.0	-1.5	4.9	15.1	6.8	6.1	OK	OK
		◢ 4.2 ◣	58	C LC_106	27.0	0.0	-10.0	-12.4	7.5	6.7	6.4	OK	OK
End plate SLN210 ring	Flange bottom a end plate SLN210 ring	◢ 4.2 ◣	58	C LC_146	60.1	0.0	33.1	-26.6	-11.4	14.9	12.0	OK	OK
		◢ 4.2 ◣	58	C LC_148	50.9	0.0	-3.0	2.0	-29.3	12.6	9.7	OK	OK
End plate SLN210 ring	Flange bottom b end plate SLN210 ring	◢ 4.2 ◣	58	C LC_121	57.7	0.0	11.5	-21.5	-24.6	14.3	7.9	OK	OK
		◢ 4.2 ◣	58	C LC_120	42.3	0.0	-10.8	-18.6	-14.6	10.5	7.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_180
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut

Name	SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

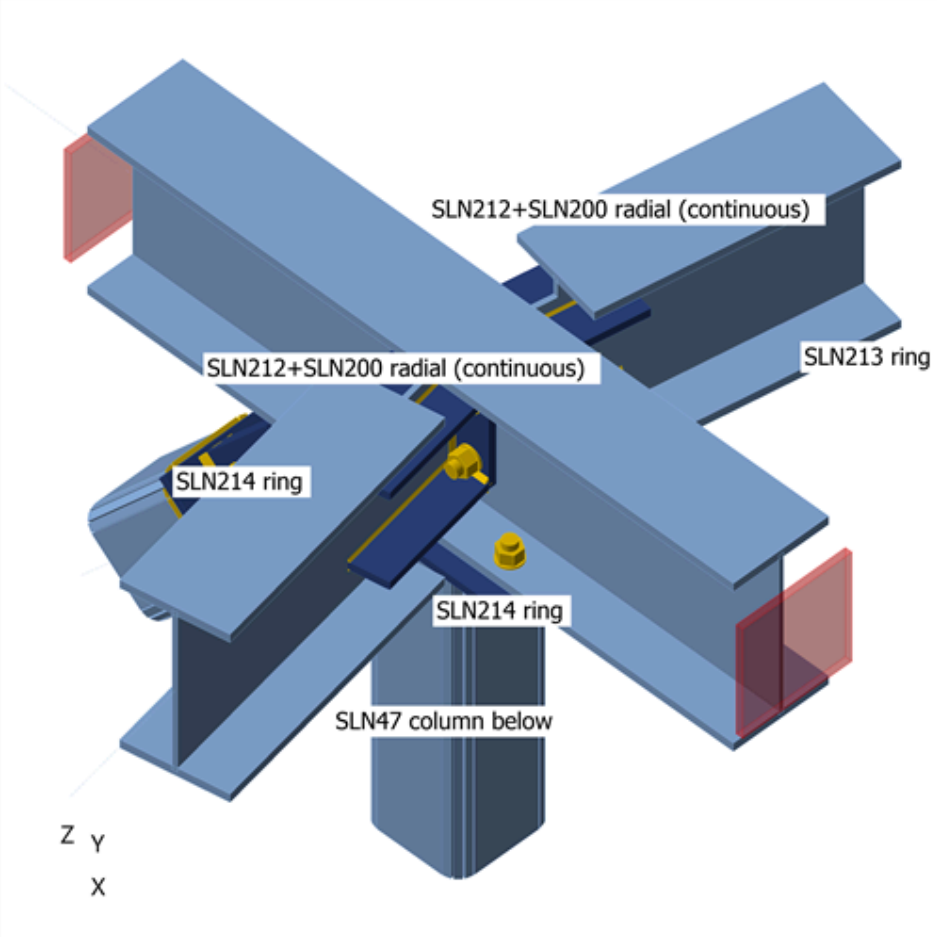
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN212+SLN200 radial (continuous)	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN47 column below	2 - SHS150x150x10	0.0	90.0	0.0	0	0	0
SLN213 ring	1 - UB254x146x43	-100.0	0.0	0.0	0	0	0
SLN214 ring	1 - UB254x146x43	-80.0	0.0	0.0	0	0	0
SLN434 raking strut	2 - SHS150x150x10	-179.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN212+SLN200 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN212+SLN200 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN47 column below / begin		Position	0
SLN213 ring / begin		Position	0
SLN214 ring / end		Position	0
SLN434 raking strut / end		Position	0



SLN212+SLN200 radial (continuous) - SLN47 column below - SLN213 ring - SLN214 ring - SLN434 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN47 column below / Begin	24.8	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN47 column below / Begin	116.5	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.6	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	27.7	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-33.2	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN47 column below / Begin	129.5	0.0	-3.9	0.0	-1.6	0.0
	SLN213 ring / Begin	-28.9	-0.1	-24.0	0.0	0.0	0.5
	SLN214 ring / End	29.0	0.0	-24.0	0.0	0.0	-0.2
	SLN434 raking strut / End	-3.9	0.0	-1.3	0.0	0.0	0.0
C LC_104	SLN47 column below / Begin	91.4	0.0	0.3	0.0	1.4	0.0
	SLN213 ring / Begin	-15.2	-0.4	-24.0	0.0	0.0	0.8
	SLN214 ring / End	15.3	-0.3	-24.0	0.0	0.0	-0.7
	SLN434 raking strut / End	-54.8	0.0	-0.8	0.0	0.0	0.0
C LC_105	SLN47 column below / Begin	105.1	-3.8	0.3	0.0	1.3	1.0
	SLN213 ring / Begin	-20.2	-0.2	-24.0	0.0	0.0	0.7
	SLN214 ring / End	16.2	-0.2	-24.0	0.0	0.0	-0.5
	SLN434 raking strut / End	-36.6	0.1	-1.0	0.3	0.0	-1.1
C LC_106	SLN47 column below / Begin	105.8	3.7	0.3	0.0	1.3	-1.0
	SLN213 ring / Begin	-16.2	-0.2	-24.0	0.0	0.0	0.4
	SLN214 ring / End	20.4	-0.2	-24.0	0.0	0.0	-0.7
	SLN434 raking strut / End	-35.9	0.0	-1.0	-0.3	0.0	1.1
C LC_107	SLN47 column below / Begin	42.2	0.0	-8.4	0.0	-5.7	0.0
	SLN213 ring / Begin	-13.1	0.0	-0.5	0.0	0.0	-0.1
	SLN214 ring / End	13.1	0.2	-0.5	0.0	0.0	0.5
	SLN434 raking strut / End	51.8	0.0	-1.3	0.0	0.0	0.0
C LC_108	SLN47 column below / Begin	-25.3	0.0	0.1	0.0	0.3	0.0
	SLN213 ring / Begin	10.0	-0.1	-0.5	0.0	0.0	0.4
	SLN214 ring / End	-10.0	-0.1	-0.5	0.0	0.0	-0.1
	SLN434 raking strut / End	-38.4	0.0	-0.2	0.0	0.0	0.0
C LC_109	SLN47 column below / Begin	-1.9	-7.5	0.1	0.0	0.2	2.0
	SLN213 ring / Begin	2.3	-0.1	-0.5	0.0	0.0	0.2
	SLN214 ring / End	-10.1	0.0	-0.5	0.0	0.0	0.1
	SLN434 raking strut / End	-7.6	0.1	-0.8	0.5	0.0	-2.2
C LC_110	SLN47 column below / Begin	-1.1	7.5	0.0	0.0	0.2	-2.0
	SLN213 ring / Begin	10.1	0.1	-0.5	0.0	0.0	-0.3
	SLN214 ring / End	-2.3	-0.1	-0.5	0.0	0.0	-0.3
	SLN434 raking strut / End	-6.7	-0.1	-0.8	-0.5	0.0	2.2
C LC_111	SLN47 column below / Begin	113.4	0.0	-8.2	0.0	-5.0	0.0
	SLN213 ring / Begin	-27.1	0.0	-16.2	0.0	0.0	0.2
	SLN214 ring / End	27.2	0.1	-16.2	0.0	0.0	0.2
	SLN434 raking strut / End	31.7	0.0	-1.5	0.0	0.0	0.0
C LC_112	SLN47 column below / Begin	43.1	0.0	0.3	0.0	1.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN213 ring / Begin	-2.6	-0.3	-16.2	0.0	0.0	0.8
	SLN214 ring / End	2.7	-0.2	-16.2	0.0	0.0	-0.5
	SLN434 raking strut / End	-62.3	0.0	-0.5	0.0	0.0	0.1
C LC_113	SLN47 column below / Begin	67.8	-7.5	0.2	0.0	1.0	2.1
	SLN213 ring / Begin	-11.1	-0.2	-16.2	0.0	0.0	0.5
	SLN214 ring / End	3.2	-0.1	-16.2	0.0	0.0	-0.2
	SLN434 raking strut / End	-29.6	0.1	-1.0	0.5	0.0	-2.2
C LC_114	SLN47 column below / Begin	68.8	7.5	0.2	0.0	0.9	-2.0
	SLN213 ring / Begin	-3.2	0.0	-16.2	0.0	0.0	0.1
	SLN214 ring / End	11.2	-0.2	-16.2	0.0	0.0	-0.7
	SLN434 raking strut / End	-28.5	-0.1	-1.0	-0.5	0.0	2.2
C LC_115	SLN47 column below / Begin	48.8	0.0	-8.4	0.0	-5.5	0.0
	SLN213 ring / Begin	-15.4	0.0	-1.7	0.0	0.0	0.0
	SLN214 ring / End	15.5	0.2	-1.7	0.0	0.0	0.4
	SLN434 raking strut / End	50.7	0.0	-1.5	0.0	0.0	0.0
C LC_116	SLN47 column below / Begin	-19.0	0.0	0.1	0.0	0.4	0.0
	SLN213 ring / Begin	7.8	-0.2	-1.7	0.0	0.0	0.4
	SLN214 ring / End	-7.8	-0.1	-1.7	0.0	0.0	-0.2
	SLN434 raking strut / End	-40.0	0.0	-0.5	0.0	0.0	0.0
C LC_117	SLN47 column below / Begin	4.5	-7.5	0.1	0.0	0.4	2.0
	SLN213 ring / Begin	0.0	-0.1	-1.7	0.0	0.0	0.2
	SLN214 ring / End	-7.8	0.0	-1.7	0.0	0.0	0.1
	SLN434 raking strut / End	-8.9	0.1	-1.0	0.5	0.0	-2.2
C LC_118	SLN47 column below / Begin	5.4	7.5	0.1	0.0	0.3	-2.0
	SLN213 ring / Begin	7.8	0.1	-1.7	0.0	0.0	-0.2
	SLN214 ring / End	0.0	-0.1	-1.7	0.0	0.0	-0.4
	SLN434 raking strut / End	-8.0	-0.1	-1.0	-0.5	0.0	2.2
C LC_119	SLN47 column below / Begin	133.5	0.0	-3.9	0.0	-1.4	0.0
	SLN213 ring / Begin	-32.5	-0.1	-25.6	0.0	0.0	0.6
	SLN214 ring / End	32.6	0.0	-25.6	0.0	0.0	-0.3
	SLN434 raking strut / End	-10.3	0.0	-1.3	0.0	0.0	0.0
C LC_120	SLN47 column below / Begin	103.8	0.0	0.4	0.0	1.4	0.0
	SLN213 ring / Begin	-18.6	-0.4	-25.6	0.0	0.0	0.8
	SLN214 ring / End	18.6	-0.3	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-49.8	0.0	-0.8	0.0	0.0	0.0
C LC_121	SLN47 column below / Begin	112.1	-3.8	0.4	0.0	1.4	1.1
	SLN213 ring / Begin	-23.7	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	19.7	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-38.9	0.1	-1.0	0.3	0.0	-1.1
C LC_122	SLN47 column below / Begin	112.9	3.7	0.4	0.0	1.4	-1.0
	SLN213 ring / Begin	-19.6	-0.2	-25.6	0.0	0.0	0.5
	SLN214 ring / End	23.8	-0.2	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-37.9	0.0	-1.0	-0.3	0.0	1.1
C LC_123	SLN47 column below / Begin	121.5	0.0	-8.1	0.0	-4.6	0.0
	SLN213 ring / Begin	-34.3	-0.1	-19.4	0.0	0.0	0.5
	SLN214 ring / End	34.4	0.0	-19.4	0.0	0.0	-0.1
	SLN434 raking strut / End	18.8	0.0	-1.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN47 column below / Begin	68.0	0.0	0.3	0.0	1.1	0.0
	SLN213 ring / Begin	-9.3	-0.3	-19.4	0.0	0.0	0.8
	SLN214 ring / End	9.3	-0.2	-19.4	0.0	0.0	-0.5
	SLN434 raking strut / End	-52.2	0.0	-0.5	0.0	0.0	0.1
C LC_125	SLN47 column below / Begin	81.8	-7.5	0.3	0.0	1.2	2.1
	SLN213 ring / Begin	-18.0	-0.2	-19.4	0.0	0.0	0.6
	SLN214 ring / End	10.1	-0.1	-19.4	0.0	0.0	-0.4
	SLN434 raking strut / End	-34.3	0.1	-1.0	0.6	0.0	-2.2
C LC_126	SLN47 column below / Begin	83.2	7.5	0.3	0.0	1.1	-2.0
	SLN213 ring / Begin	-10.0	-0.1	-19.4	0.0	0.0	0.2
	SLN214 ring / End	18.1	-0.2	-19.4	0.0	0.0	-0.8
	SLN434 raking strut / End	-32.7	-0.1	-1.0	-0.5	0.0	2.2
C LC_127	SLN47 column below / Begin	56.8	0.0	-8.3	0.0	-5.2	0.0
	SLN213 ring / Begin	-22.6	-0.1	-4.9	0.0	0.0	0.3
	SLN214 ring / End	22.7	0.1	-4.9	0.0	0.0	0.2
	SLN434 raking strut / End	37.8	0.0	-1.5	0.0	0.0	0.0
C LC_128	SLN47 column below / Begin	5.8	0.0	0.1	0.0	0.6	0.0
	SLN213 ring / Begin	1.1	-0.2	-4.9	0.0	0.0	0.4
	SLN214 ring / End	-1.1	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-29.8	0.0	-0.5	0.0	0.0	0.1
C LC_129	SLN47 column below / Begin	18.5	-7.5	0.1	0.0	0.6	2.1
	SLN213 ring / Begin	-6.9	-0.1	-4.9	0.0	0.0	0.3
	SLN214 ring / End	-1.0	0.0	-4.9	0.0	0.0	-0.1
	SLN434 raking strut / End	-13.5	0.1	-1.0	0.6	0.0	-2.2
C LC_130	SLN47 column below / Begin	19.7	7.5	0.1	0.0	0.6	-2.0
	SLN213 ring / Begin	1.0	0.0	-4.9	0.0	0.0	-0.1
	SLN214 ring / End	6.9	-0.1	-4.9	0.0	0.0	-0.5
	SLN434 raking strut / End	-12.1	-0.1	-1.0	-0.5	0.0	2.2
C LC_135	SLN47 column below / Begin	115.8	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-28.0	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	28.1	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-34.1	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN47 column below / Begin	25.4	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.2
	SLN214 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-4.2	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN47 column below / Begin	24.2	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.5	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.5	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.8	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN47 column below / Begin	24.8	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.8	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.2
	SLN434 raking strut / End	-5.1	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN47 column below / Begin	24.8	0.0	0.2	0.0	0.7	0.0
	SLN213 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN214 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN434 raking strut / End	-5.0	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN47 column below / Begin	118.9	0.0	0.4	0.0	1.6	0.0
	SLN213 ring / Begin	-28.7	-0.1	-25.6	0.0	0.0	0.6
	SLN214 ring / End	28.8	-0.1	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-30.0	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN47 column below / Begin	114.1	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-26.4	-0.3	-25.6	0.0	0.0	0.8
	SLN214 ring / End	26.5	-0.3	-25.6	0.0	0.0	-0.7
	SLN434 raking strut / End	-36.4	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN47 column below / Begin	116.4	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.7	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	27.5	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-33.4	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN47 column below / Begin	116.6	0.0	0.4	0.0	1.5	0.1
	SLN213 ring / Begin	-27.4	-0.2	-25.6	0.0	0.0	0.7
	SLN214 ring / End	27.8	-0.2	-25.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-33.0	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN47 column below / Begin	75.8	0.0	-1.6	0.0	-6.7	0.0
	SLN213 ring / Begin	-12.3	-0.1	-11.6	0.0	0.0	0.3
	SLN214 ring / End	12.3	-0.1	-12.3	0.0	0.0	-0.3
	SLN434 raking strut / End	-18.2	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN47 column below / Begin	67.7	0.0	2.2	0.0	9.0	0.0
	SLN213 ring / Begin	-28.2	-0.2	-14.2	0.0	0.0	0.5
	SLN214 ring / End	28.3	-0.2	-16.2	0.0	0.0	-0.5
	SLN434 raking strut / End	-15.8	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN47 column below / Begin	116.2	0.0	0.4	0.0	1.5	0.0
	SLN213 ring / Begin	-27.7	-0.2	-20.9	0.0	0.0	0.7
	SLN214 ring / End	27.8	-0.2	-23.6	0.0	0.0	-0.6
	SLN434 raking strut / End	-33.0	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN47 column below / Begin	75.2	0.0	-1.6	0.0	-6.6	0.0
	SLN213 ring / Begin	-12.4	-0.1	-11.6	0.0	0.0	0.3
	SLN214 ring / End	12.5	-0.1	-12.3	0.0	0.0	-0.3
	SLN434 raking strut / End	-18.0	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	28.5 < 100%	OK
Welds	98.3 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN212+SLN200 radial (continuous)-bfl 1	12.7	C LC_107	232.1	0.0	37.6	OK
SLN212+SLN200 radial (continuous)-tfl 1	12.7	C LC_121	73.4	0.0	0.0	OK
SLN212+SLN200 radial (continuous)-w 1	7.2	C LC_120	190.3	0.0	52.8	OK
SLN47 column below	10.0	C LC_112	275.5	0.2	0.0	OK
SLN213 ring-bfl 1	12.7	C LC_120	57.1	0.0	0.0	OK
SLN213 ring-tfl 1	12.7	C LC_104	52.5	0.0	0.0	OK
SLN213 ring-w 1	7.2	C LC_120	170.2	0.0	0.0	OK
SLN214 ring-bfl 1	12.7	C LC_122	53.6	0.0	0.0	OK
SLN214 ring-tfl 1	12.7	C LC_122	48.6	0.0	0.0	OK
SLN214 ring-w 1	7.2	C LC_122	162.1	0.0	0.0	OK
SLN434 raking strut	10.0	C LC_125	69.8	0.0	0.0	OK
End plate column below SLN47 column below	12.0	C LC_107	268.8	0.0	37.6	OK
End plate SLN213 ring	12.0	C LC_104	83.9	0.0	0.0	OK
End plate SLN214 ring	12.0	C LC_146	66.2	0.0	11.5	OK
Cap plate SLN434 raking strut	12.0	C LC_126	146.9	0.0	0.0	OK
Fin plate SLN434 raking strut	12.0	C LC_125	275.2	0.1	97.3	OK
Fin plate 2 SLN434 raking strut	12.0	C LC_126	275.2	0.1	96.5	OK
Gusset SLN434 raking strut	12.0	C LC_126	275.2	0.1	63.5	OK
Flange top a end plate SLN213 ring	12.0	C LC_104	71.8	0.0	0.0	OK
Flange top b end plate SLN213 ring	12.0	C LC_104	59.0	0.0	0.0	OK
Flange bottom a end plate SLN213 ring	12.0	C LC_104	54.1	0.0	0.0	OK
Flange bottom b end plate SLN213 ring	12.0	C LC_120	78.9	0.0	0.0	OK
Flange top a end plate SLN214 ring	12.0	C LC_126	52.3	0.0	0.0	OK
Flange top b end plate SLN214 ring	12.0	C LC_122	63.2	0.0	0.0	OK
Flange bottom a end plate SLN214 ring	12.0	C LC_122	74.1	0.0	0.0	OK
Flange bottom b end plate SLN214 ring	12.0	C LC_126	47.9	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

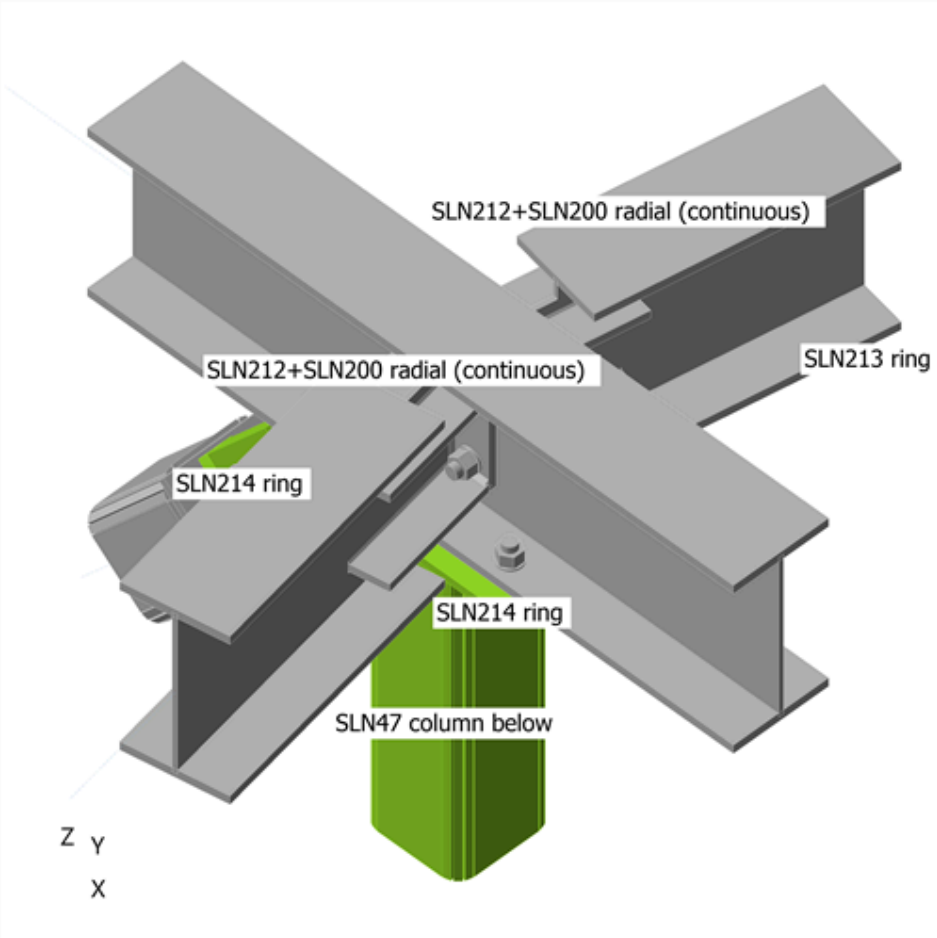
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

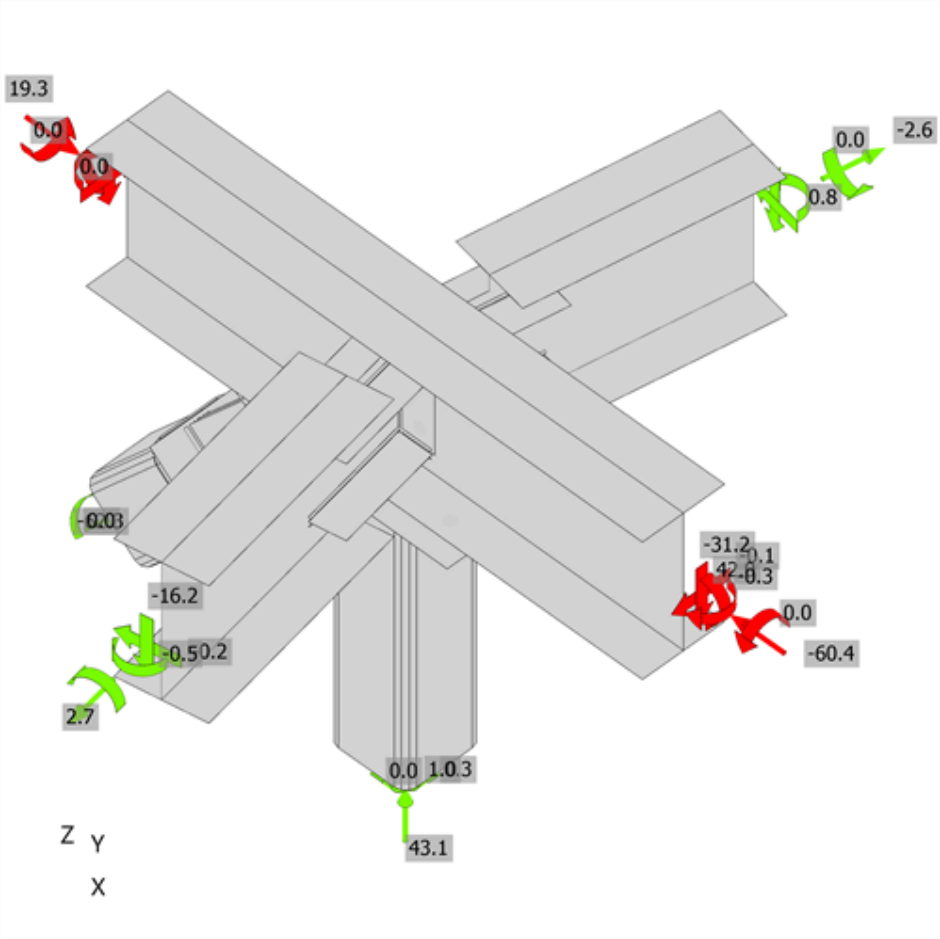
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN47 column below	150	C LC_112	0	5	0.1	OK
SLN434 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

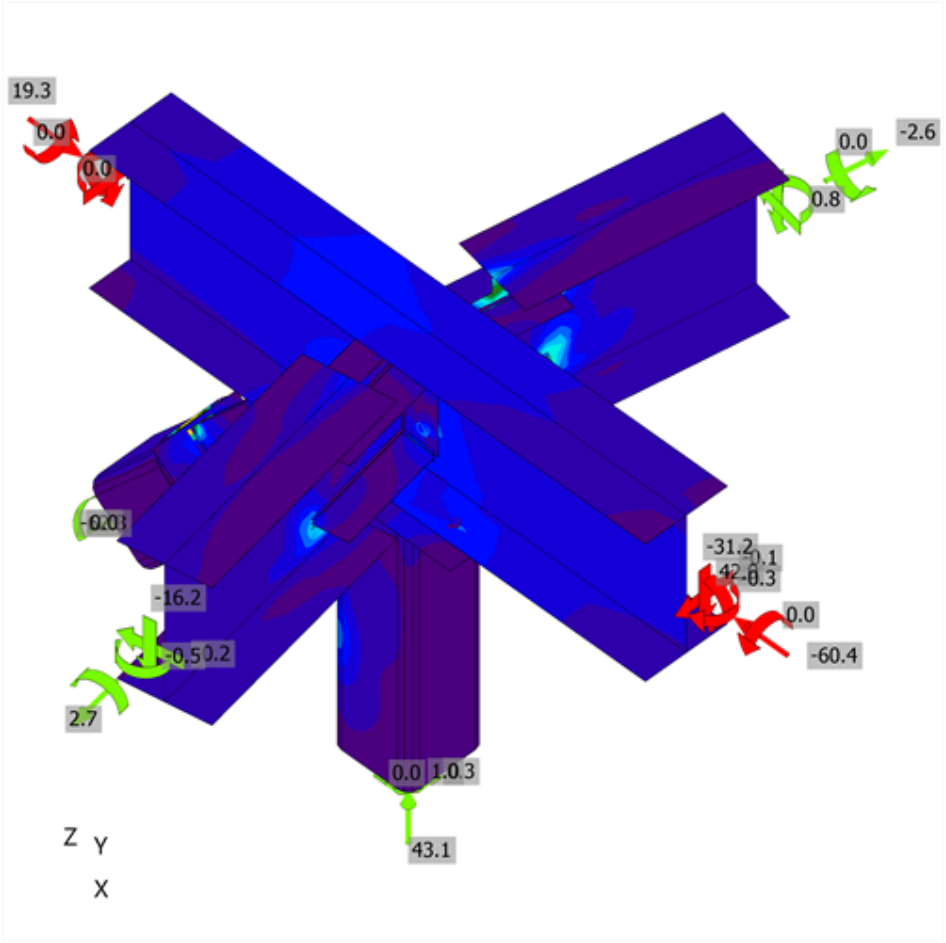
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

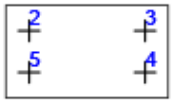
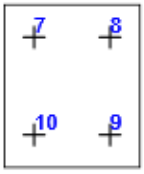
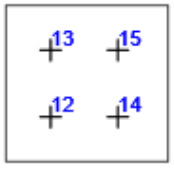


Strain check, C LC_112



Equivalent stress, C LC_112

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_116	15.4	6.4	125.3	10.9	6.8	14.6	OK	OK
	B3	M20 8.8 - 1	C LC_107	40.2	5.8	101.2	28.5	6.1	26.5	OK	OK
	B4	M20 8.8 - 1	C LC_107	39.3	5.7	97.8	27.8	6.1	25.9	OK	OK
	B5	M20 8.8 - 1	C LC_116	15.9	6.5	131.1	11.3	6.9	15.0	OK	OK
	B7	M20 8.8 - 1	C LC_104	12.2	1.0	123.8	8.6	1.1	7.2	OK	OK
	B8	M20 8.8 - 1	C LC_110	3.4	0.6	71.6	2.4	0.7	2.4	OK	OK
	B9	M20 8.8 - 1	C LC_110	3.5	0.8	63.8	2.5	0.8	2.6	OK	OK
	B10	M20 8.8 - 1	C LC_104	12.7	1.1	123.8	9.0	1.2	7.6	OK	OK
	B12	M20 8.8 - 1	C LC_125	17.1	17.1	115.3	12.1	18.1	26.8	OK	OK
	B13	M20 8.8 - 1	C LC_125	7.4	14.8	115.3	5.2	15.7	19.5	OK	OK
	B14	M20 8.8 - 1	C LC_115	0.9	7.8	106.0	0.6	14.6	8.7	OK	OK
	B15	M20 8.8 - 1	C LC_112	0.2	8.2	115.3	0.1	14.0	8.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN434 raking strut	SLN434 raking strut	◢ 4.2	519	C LC_126	205.0	0.0	-89.1	105.8	-13.1	50.6	28.7	OK	OK
SLN47 column below-w 3	Gusset SLN434 raking strut	◢ 5.6 ◣	287	C LC_112	397.8	0.7	-141.7	-141.2	-161.6	98.3	69.9	OK	OK
		◢ 5.6 ◣	287	C LC_112	397.8	0.7	-140.1	140.6	162.6	98.3	70.0	OK	OK
SLN213 ring-w 1	Flange top b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_104	71.0	0.0	39.8	32.4	10.2	17.5	11.0	OK	OK
		◢ 4.2 ◣	197	C LC_104	35.8	0.0	-17.5	16.0	8.4	8.9	7.1	OK	OK
SLN213 ring-w 1	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	198	C LC_119	96.9	0.0	28.5	33.0	42.0	23.9	14.8	OK	OK
		◢ 4.2 ◣	198	C LC_120	140.9	0.0	58.9	-53.1	-51.4	34.8	21.3	OK	OK
SLN214 ring-w 1	Flange top a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_126	68.4	0.0	36.9	31.0	-12.2	16.9	9.1	OK	OK
		◢ 4.2 ◣	197	C LC_122	32.0	0.0	-16.1	14.6	-6.5	7.9	7.0	OK	OK
SLN214 ring-w 1	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	198	C LC_146	92.3	0.0	27.6	31.5	-39.9	22.8	12.2	OK	OK
		◢ 4.2 ◣	198	C LC_122	134.5	0.0	54.1	-49.9	50.7	33.2	18.4	OK	OK
Cap plate SLN434 raking strut	Fin plate SLN434 raking strut	◢ 5.6 ◣	139	C LC_125	286.8	0.0	-128.0	-115.9	92.3	70.9	54.5	OK	OK
		◢ 5.6 ◣	139	C LC_125	180.6	0.0	-76.3	88.4	-33.5	44.6	25.7	OK	OK
Cap plate SLN434 raking strut	Fin plate 2 SLN434 raking strut	◢ 5.6 ◣	139	C LC_126	175.7	0.0	-72.3	-86.5	32.8	43.4	25.0	OK	OK
		◢ 5.6 ◣	139	C LC_126	284.8	0.0	-128.8	114.8	-91.3	70.4	54.1	OK	OK
End plate column below SLN47 column below	SLN47 column below	◢ 5.6 ◣	521	C LC_104	307.4	0.0	-201.2	-131.1	-28.6	76.0	41.4	OK	OK
		◢ 5.6 ◣	521	C LC_107	231.7	0.0	158.4	-77.1	-59.9	57.2	36.1	OK	OK
SLN212+SLN200 radial (continuous)-w 1	SLN213 ring-w 1	◢ 7.2	189	C LC_122	93.3	0.0	0.1	4.6	53.7	23.1	15.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN212+SLN200 radial (continuous)-w 1	SLN214 ring-w 1	◢ 7.2	189	C LC_121	97.8	0.0	2.4	5.4	56.2	24.2	15.3	OK	OK
End plate SLN213 ring	SLN213 ring-w 1	◢ 4.2 ◣	149	C LC_119	90.0	0.0	-36.2	-19.9	-43.2	22.2	12.8	OK	OK
		◢ 4.2 ◣	149	C LC_122	71.7	0.0	-15.6	22.1	33.9	17.7	11.2	OK	OK
End plate SLN214 ring	SLN214 ring-w 1	◢ 4.2 ◣	149	C LC_119	84.7	0.0	-23.4	-28.7	37.2	20.9	13.3	OK	OK
		◢ 4.2 ◣	149	C LC_146	72.4	0.0	-18.5	24.8	-31.9	17.9	11.5	OK	OK
SLN213 ring-w 1	Flange top a end plate SLN213 ring	◢ 4.2 ◣	199	C LC_104	124.4	0.0	-11.1	6.5	-71.3	30.7	21.7	OK	OK
		◢ 4.2 ◣	198	C LC_104	81.3	0.0	12.4	14.3	-44.2	20.1	11.7	OK	OK
SLN213 ring-w 1	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	198	C LC_120	107.6	0.0	19.8	-20.0	57.7	26.6	15.7	OK	OK
		◢ 4.2 ◣	198	C LC_112	94.3	0.0	-13.0	-5.2	53.7	23.3	14.5	OK	OK
SLN214 ring-w 1	Flange top b end plate SLN214 ring	◢ 4.2 ◣	199	C LC_122	109.7	0.0	-10.4	6.2	62.7	27.1	17.0	OK	OK
		◢ 4.2 ◣	198	C LC_126	75.6	0.0	11.9	13.5	40.9	18.7	11.2	OK	OK
SLN214 ring-w 1	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	198	C LC_122	99.7	0.0	18.6	-18.2	-53.5	24.6	14.2	OK	OK
		◢ 4.2 ◣	198	C LC_126	80.3	0.0	-11.6	-3.4	-45.7	19.8	10.8	OK	OK
End plate SLN213 ring	Flange top a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_112	37.5	0.0	-0.9	3.5	-21.4	9.3	7.3	OK	OK
		◢ 4.2 ◣	58	C LC_104	23.5	0.0	-6.4	-11.9	-5.4	5.8	5.8	OK	OK
End plate SLN213 ring	Flange top b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	56.6	0.0	-1.7	-0.7	-32.7	14.0	10.1	OK	OK
		◢ 4.2 ◣	58	C LC_104	70.4	0.0	40.1	27.7	-18.7	17.4	13.7	OK	OK
End plate SLN213 ring	Flange bottom a end plate SLN213 ring	◢ 4.2 ◣	58	C LC_119	55.0	0.0	12.5	-19.0	24.4	13.6	8.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_120	45.5	0.0	-12.6	-16.6	19.1	11.2	8.6	OK	OK
End plate SLN213 ring	Flange bottom b end plate SLN213 ring	◢ 4.2 ◣	58	C LC_120	80.4	0.0	45.5	-34.3	17.0	19.9	15.3	OK	OK
		◢ 4.2 ◣	58	C LC_120	66.5	0.0	-4.6	-0.1	38.3	16.4	11.2	OK	OK
End plate SLN214 ring	Flange top a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_107	58.2	0.0	-30.7	25.5	12.9	14.4	8.0	OK	OK
		◢ 4.2 ◣	58	C LC_104	53.2	0.0	6.0	10.0	28.9	13.1	10.3	OK	OK
End plate SLN214 ring	Flange top b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_114	64.5	0.0	-32.0	25.2	-20.3	15.9	10.2	OK	OK
		◢ 4.2 ◣	58	C LC_114	59.1	0.0	-19.2	-26.0	19.0	14.6	9.8	OK	OK
End plate SLN214 ring	Flange bottom a end plate SLN214 ring	◢ 4.2 ◣	58	C LC_146	65.2	0.0	35.8	-29.0	-12.1	16.1	12.5	OK	OK
		◢ 4.2 ◣	58	C LC_146	53.1	0.0	-3.5	1.4	-30.5	13.1	10.0	OK	OK
End plate SLN214 ring	Flange bottom b end plate SLN214 ring	◢ 4.2 ◣	58	C LC_119	59.0	0.0	12.0	-22.4	-24.7	14.6	8.0	OK	OK
		◢ 4.2 ◣	58	C LC_114	65.8	0.0	-32.5	-26.2	20.2	16.3	10.4	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_220
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut

Name	SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

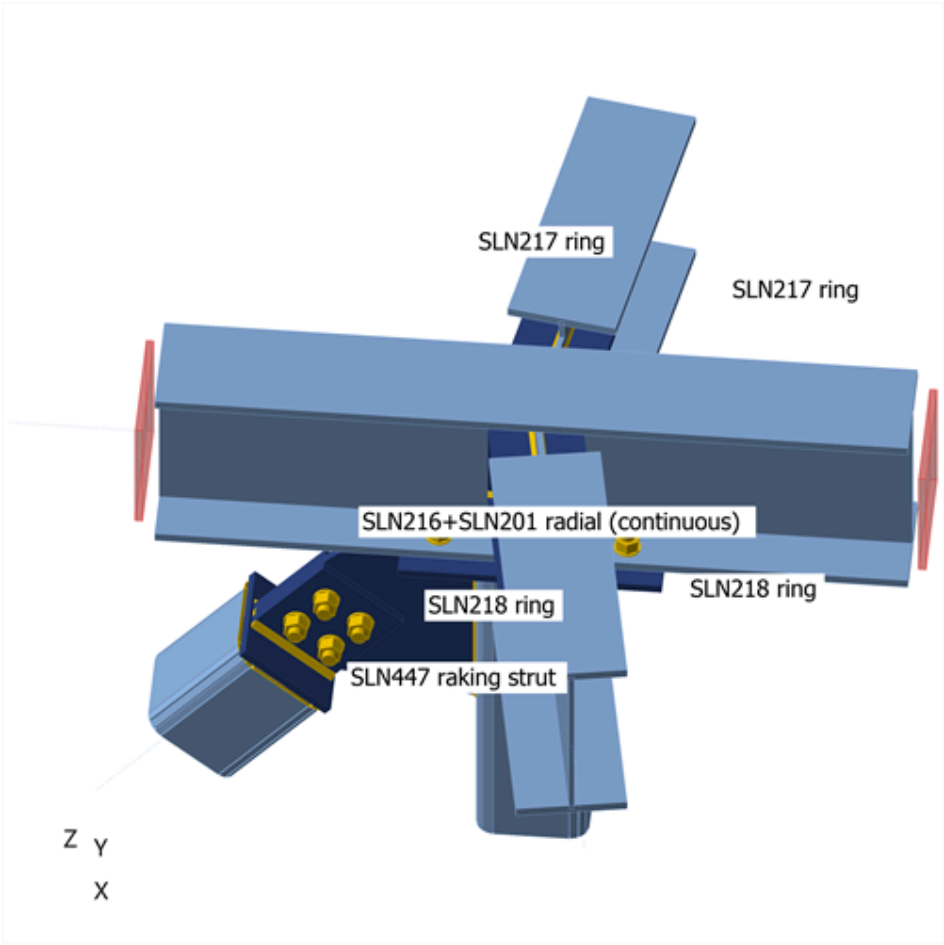
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN216+SLN201 radial (continuous)	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN49 column below	2 - SHS150x150x10	0.0	90.0	40.0	0	0	0
SLN217 ring	1 - UB254x146x43	-60.0	0.0	0.0	0	0	0
SLN218 ring	1 - UB254x146x43	-40.0	0.0	0.0	0	0	0
SLN447 raking strut	2 - SHS150x150x10	-139.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN216+SLN201 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN216+SLN201 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN49 column below / begin		Position	0
SLN217 ring / begin		Position	0
SLN218 ring / end		Position	0
SLN447 raking strut / end		Position	0



SLN216+SLN201 radial (continuous) - SLN49 column below - SLN217 ring - SLN218 ring - SLN447 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN49 column below / Begin	26.0	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.7	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-3.4	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN49 column below / Begin	122.6	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.5	-0.2	-25.6	0.0	0.0	0.6
	SLN218 ring / End	27.3	-0.2	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-24.8	0.0	-1.0	-0.1	0.0	0.0
C LC_103	SLN49 column below / Begin	129.1	-2.4	-3.0	0.0	-1.2	0.6
	SLN217 ring / Begin	-27.5	-0.1	-24.0	0.0	0.0	0.5
	SLN218 ring / End	24.7	0.0	-24.5	0.0	0.0	-0.1
	SLN447 raking strut / End	-4.2	0.0	-1.2	0.1	0.0	-0.7
C LC_104	SLN49 column below / Begin	100.5	0.0	0.3	0.0	1.1	0.1
	SLN217 ring / Begin	-15.3	-0.3	-24.0	0.0	0.0	0.7
	SLN218 ring / End	15.4	-0.3	-24.5	0.0	0.0	-0.6
	SLN447 raking strut / End	-42.3	-0.2	-0.8	-0.1	0.0	-0.1
C LC_105	SLN49 column below / Begin	101.4	0.0	0.3	0.0	1.1	-0.1
	SLN217 ring / Begin	-15.5	-0.2	-24.0	0.0	0.0	0.5
	SLN218 ring / End	14.9	-0.3	-24.5	0.0	0.0	-0.7
	SLN447 raking strut / End	-41.2	0.3	-0.9	0.0	0.0	0.2
C LC_106	SLN49 column below / Begin	126.1	2.9	-2.5	0.0	-0.8	-0.8
	SLN217 ring / Begin	-24.1	0.0	-24.0	0.0	0.0	0.3
	SLN218 ring / End	27.2	-0.1	-24.5	0.0	0.0	-0.5
	SLN447 raking strut / End	-8.1	0.0	-1.2	-0.3	0.0	0.8
C LC_107	SLN49 column below / Begin	30.8	-4.8	-6.4	0.0	-4.3	1.3
	SLN217 ring / Begin	-10.7	0.0	-0.5	0.0	0.0	0.1
	SLN218 ring / End	5.8	0.2	-0.5	0.0	0.0	0.6
	SLN447 raking strut / End	36.4	0.0	-1.1	0.3	0.0	-1.4
C LC_108	SLN49 column below / Begin	-19.4	0.0	0.1	0.0	0.3	0.2
	SLN217 ring / Begin	10.1	-0.1	-0.5	0.0	0.0	0.4
	SLN218 ring / End	-9.8	0.0	-0.5	0.0	0.0	-0.1
	SLN447 raking strut / End	-30.6	-0.5	-0.4	0.0	0.0	-0.3
C LC_109	SLN49 column below / Begin	-17.7	0.1	0.1	0.0	0.3	-0.3
	SLN217 ring / Begin	10.2	0.0	-0.5	0.0	0.0	-0.1
	SLN218 ring / End	-10.9	-0.2	-0.5	0.0	0.0	-0.4
	SLN447 raking strut / End	-28.5	0.6	-0.4	0.0	0.0	0.4
C LC_110	SLN49 column below / Begin	24.9	5.7	-5.4	0.0	-3.6	-1.5
	SLN217 ring / Begin	-4.5	0.1	-0.5	0.0	0.0	-0.4
	SLN218 ring / End	10.5	0.0	-0.5	0.0	0.0	0.0
	SLN447 raking strut / End	28.7	-0.1	-1.1	-0.4	0.0	1.7
C LC_111	SLN49 column below / Begin	105.6	-4.8	-6.3	0.0	-3.8	1.3
	SLN217 ring / Begin	-24.6	-0.1	-16.2	0.0	0.0	0.4
	SLN218 ring / End	19.5	0.1	-16.5	0.0	0.0	0.3
	SLN447 raking strut / End	21.2	0.0	-1.4	0.3	0.0	-1.4
C LC_112	SLN49 column below / Begin	53.2	0.0	0.2	0.0	0.8	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN217 ring / Begin	-2.5	-0.3	-16.2	0.0	0.0	0.7
	SLN218 ring / End	2.9	-0.2	-16.5	0.0	0.0	-0.4
	SLN447 raking strut / End	-48.8	-0.4	-0.6	0.0	0.0	-0.3
C LC_113	SLN49 column below / Begin	54.8	0.1	0.2	0.0	0.8	-0.3
	SLN217 ring / Begin	-2.6	-0.1	-16.2	0.0	0.0	0.3
	SLN218 ring / End	1.8	-0.3	-16.5	0.0	0.0	-0.8
	SLN447 raking strut / End	-46.6	0.6	-0.7	0.0	0.0	0.4
C LC_114	SLN49 column below / Begin	99.6	5.7	-5.2	0.0	-3.0	-1.5
	SLN217 ring / Begin	-18.2	0.1	-16.2	0.0	0.0	-0.1
	SLN218 ring / End	24.2	-0.1	-16.5	0.0	0.0	-0.3
	SLN447 raking strut / End	13.5	0.0	-1.3	-0.4	0.0	1.7
C LC_115	SLN49 column below / Begin	37.7	-4.8	-6.4	0.0	-4.2	1.3
	SLN217 ring / Begin	-13.0	0.0	-1.7	0.0	0.0	0.2
	SLN218 ring / End	8.1	0.2	-1.8	0.0	0.0	0.5
	SLN447 raking strut / End	35.7	0.0	-1.4	0.3	0.0	-1.4
C LC_116	SLN49 column below / Begin	-12.7	0.0	0.1	0.0	0.4	0.2
	SLN217 ring / Begin	7.9	-0.2	-1.7	0.0	0.0	0.4
	SLN218 ring / End	-7.6	0.0	-1.8	0.0	0.0	-0.1
	SLN447 raking strut / End	-31.7	-0.5	-0.6	0.0	0.0	-0.3
C LC_117	SLN49 column below / Begin	-11.1	0.1	0.1	0.0	0.4	-0.3
	SLN217 ring / Begin	8.0	0.0	-1.7	0.0	0.0	0.0
	SLN218 ring / End	-8.7	-0.2	-1.8	0.0	0.0	-0.5
	SLN447 raking strut / End	-29.5	0.6	-0.7	0.0	0.0	0.4
C LC_118	SLN49 column below / Begin	31.7	5.7	-5.3	0.0	-3.4	-1.5
	SLN217 ring / Begin	-6.8	0.1	-1.7	0.0	0.0	-0.3
	SLN218 ring / End	12.8	0.0	-1.8	0.0	0.0	-0.1
	SLN447 raking strut / End	28.0	-0.1	-1.3	-0.4	0.0	1.7
C LC_119	SLN49 column below / Begin	134.5	-2.4	-2.9	0.0	-1.1	0.6
	SLN217 ring / Begin	-31.1	-0.1	-25.6	0.0	0.0	0.6
	SLN218 ring / End	28.3	0.0	-26.1	0.0	0.0	-0.2
	SLN447 raking strut / End	-8.7	0.0	-1.2	0.1	0.0	-0.7
C LC_120	SLN49 column below / Begin	112.0	0.0	0.3	0.0	1.1	0.1
	SLN217 ring / Begin	-18.6	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	18.8	-0.3	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-38.5	-0.2	-0.8	-0.1	0.0	-0.1
C LC_121	SLN49 column below / Begin	111.4	0.0	0.3	0.0	1.1	-0.1
	SLN217 ring / Begin	-18.9	-0.2	-25.6	0.0	0.0	0.5
	SLN218 ring / End	18.2	-0.3	-26.1	0.0	0.0	-0.8
	SLN447 raking strut / End	-39.3	0.3	-0.9	0.0	0.0	0.2
C LC_122	SLN49 column below / Begin	131.5	2.9	-2.4	0.0	-0.7	-0.8
	SLN217 ring / Begin	-27.6	-0.1	-25.6	0.0	0.0	0.4
	SLN218 ring / End	30.7	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-12.5	0.0	-1.2	-0.3	0.0	0.8
C LC_123	SLN49 column below / Begin	116.4	-4.8	-6.2	0.0	-3.5	1.3
	SLN217 ring / Begin	-31.7	-0.2	-19.4	0.0	0.0	0.7
	SLN218 ring / End	26.6	0.1	-19.7	0.0	0.0	0.1
	SLN447 raking strut / End	12.2	0.0	-1.4	0.3	0.0	-1.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN49 column below / Begin	76.1	0.0	0.2	0.0	0.9	0.2
	SLN217 ring / Begin	-9.2	-0.3	-19.4	0.0	0.0	0.7
	SLN218 ring / End	9.6	-0.2	-19.7	0.0	0.0	-0.4
	SLN447 raking strut / End	-41.2	-0.4	-0.6	-0.1	0.0	-0.3
C LC_125	SLN49 column below / Begin	74.9	0.1	0.2	0.0	0.9	-0.3
	SLN217 ring / Begin	-9.4	-0.1	-19.4	0.0	0.0	0.3
	SLN218 ring / End	8.4	-0.3	-19.7	0.0	0.0	-0.8
	SLN447 raking strut / End	-42.9	0.6	-0.7	0.0	0.0	0.4
C LC_126	SLN49 column below / Begin	110.5	5.7	-5.2	0.0	-2.7	-1.5
	SLN217 ring / Begin	-25.2	0.0	-19.4	0.0	0.0	0.2
	SLN218 ring / End	31.3	-0.1	-19.7	0.0	0.0	-0.5
	SLN447 raking strut / End	4.6	0.0	-1.3	-0.5	0.0	1.7
C LC_127	SLN49 column below / Begin	48.5	-4.8	-6.3	0.0	-3.9	1.3
	SLN217 ring / Begin	-20.2	-0.1	-4.9	0.0	0.0	0.4
	SLN218 ring / End	15.2	0.1	-5.0	0.0	0.0	0.3
	SLN447 raking strut / End	26.6	0.0	-1.4	0.3	0.0	-1.4
C LC_128	SLN49 column below / Begin	10.2	0.0	0.1	0.0	0.5	0.2
	SLN217 ring / Begin	1.3	-0.2	-4.9	0.0	0.0	0.4
	SLN218 ring / End	-0.9	0.0	-5.0	0.0	0.0	-0.1
	SLN447 raking strut / End	-24.1	-0.4	-0.6	0.0	0.0	-0.3
C LC_129	SLN49 column below / Begin	9.0	0.1	0.1	0.0	0.5	-0.3
	SLN217 ring / Begin	1.3	0.0	-4.9	0.0	0.0	0.0
	SLN218 ring / End	-2.0	-0.2	-5.0	0.0	0.0	-0.5
	SLN447 raking strut / End	-25.7	0.6	-0.7	0.1	0.0	0.4
C LC_130	SLN49 column below / Begin	42.6	5.7	-5.3	0.0	-3.2	-1.5
	SLN217 ring / Begin	-13.8	0.0	-4.9	0.0	0.0	-0.1
	SLN218 ring / End	19.9	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	19.1	-0.1	-1.3	-0.4	0.0	1.7
C LC_135	SLN49 column below / Begin	122.1	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.9	-0.2	-25.6	0.0	0.0	0.6
	SLN218 ring / End	27.8	-0.2	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-25.3	0.0	-1.0	-0.1	0.0	0.0
C LC_141	SLN49 column below / Begin	26.5	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-8.9	0.0	-4.9	0.0	0.0	0.1
	SLN218 ring / End	8.9	0.0	-5.0	0.0	0.0	-0.1
	SLN447 raking strut / End	-2.8	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN49 column below / Begin	25.5	0.0	0.2	0.0	0.6	0.0
	SLN217 ring / Begin	-8.5	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.5	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-4.0	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN49 column below / Begin	25.5	0.0	0.2	0.0	0.6	0.0
	SLN217 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.5	-0.1	-5.0	0.0	0.0	-0.2
	SLN447 raking strut / End	-4.0	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN49 column below / Begin	26.4	0.0	0.1	0.0	0.6	0.0
	SLN217 ring / Begin	-8.9	0.0	-4.9	0.0	0.0	0.2
	SLN218 ring / End	8.9	0.0	-5.0	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN447 raking strut / End	-2.8	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN49 column below / Begin	124.5	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-28.5	-0.1	-25.6	0.0	0.0	0.5
	SLN218 ring / End	28.2	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-22.2	0.0	-1.0	-0.1	0.0	0.0
C LC_146	SLN49 column below / Begin	120.7	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-26.5	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	26.5	-0.3	-26.1	0.0	0.0	-0.7
	SLN447 raking strut / End	-27.3	0.0	-1.0	-0.1	0.0	0.0
C LC_147	SLN49 column below / Begin	120.7	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-26.8	-0.3	-25.6	0.0	0.0	0.7
	SLN218 ring / End	26.5	-0.3	-26.1	0.0	0.0	-0.6
	SLN447 raking strut / End	-27.3	0.0	-1.0	-0.1	0.0	0.0
C LC_148	SLN49 column below / Begin	124.5	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-28.2	-0.1	-25.6	0.0	0.0	0.5
	SLN218 ring / End	28.2	-0.1	-26.1	0.0	0.0	-0.5
	SLN447 raking strut / End	-22.3	0.0	-1.0	-0.1	0.0	0.0
C LC_151	SLN49 column below / Begin	77.4	0.0	-1.6	0.0	-6.7	0.0
	SLN217 ring / Begin	-12.3	-0.1	-11.8	0.0	0.0	0.3
	SLN218 ring / End	12.2	-0.1	-11.6	0.0	0.0	-0.3
	SLN447 raking strut / End	-15.5	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN49 column below / Begin	71.9	0.0	2.1	0.0	8.6	0.0
	SLN217 ring / Begin	-28.1	-0.2	-14.1	0.0	0.0	0.5
	SLN218 ring / End	28.0	-0.2	-14.0	0.0	0.0	-0.5
	SLN447 raking strut / End	-9.8	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN49 column below / Begin	121.8	0.0	0.3	0.0	1.2	0.0
	SLN217 ring / Begin	-27.6	-0.2	-21.0	0.0	0.0	0.6
	SLN218 ring / End	27.5	-0.2	-20.6	0.0	0.0	-0.6
	SLN447 raking strut / End	-24.4	0.0	-1.0	-0.1	0.0	0.0
C LC_154	SLN49 column below / Begin	76.9	0.0	-1.6	0.0	-6.6	0.0
	SLN217 ring / Begin	-12.4	-0.1	-11.8	0.0	0.0	0.3
	SLN218 ring / End	12.3	-0.1	-11.6	0.0	0.0	-0.3
	SLN447 raking strut / End	-15.4	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	21.8 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN216+SLN201 radial (continuous)-bfl 1	12.7	C LC_107	167.5	0.0	26.8	OK
SLN216+SLN201 radial (continuous)-tfl 1	12.7	C LC_153	71.8	0.0	0.0	OK
SLN216+SLN201 radial (continuous)-w 1	7.2	C LC_123	212.8	0.0	32.7	OK
SLN49 column below	10.0	C LC_112	275.2	0.1	0.0	OK
SLN217 ring-bfl 1	12.7	C LC_120	52.2	0.0	0.0	OK
SLN217 ring-tfl 1	12.7	C LC_120	47.3	0.0	0.0	OK
SLN217 ring-w 1	7.2	C LC_120	158.0	0.0	0.0	OK
SLN218 ring-bfl 1	12.7	C LC_121	54.2	0.0	0.0	OK
SLN218 ring-tfl 1	12.7	C LC_125	51.3	0.0	0.0	OK
SLN218 ring-w 1	7.2	C LC_121	162.7	0.0	0.0	OK
SLN447 raking strut	10.0	C LC_115	53.0	0.0	0.0	OK
End plate column below SLN49 column below	12.0	C LC_107	201.3	0.0	26.8	OK
End plate SLN217 ring	12.0	C LC_120	74.9	0.0	0.0	OK
End plate SLN218 ring	12.0	C LC_146	57.0	0.0	16.0	OK
Cap plate SLN447 raking strut	12.0	C LC_118	108.4	0.0	0.0	OK
Fin plate SLN447 raking strut	12.0	C LC_118	275.1	0.0	95.6	OK
Fin plate 2 SLN447 raking strut	12.0	C LC_115	268.6	0.0	84.2	OK
Gusset SLN447 raking strut	12.0	C LC_110	275.1	0.0	47.9	OK
Flange top a end plate SLN217 ring	12.0	C LC_120	63.6	0.0	0.0	OK
Flange top b end plate SLN217 ring	12.0	C LC_124	51.1	0.0	0.0	OK
Flange bottom a end plate SLN217 ring	12.0	C LC_112	47.5	0.0	0.0	OK
Flange bottom b end plate SLN217 ring	12.0	C LC_120	72.8	0.0	0.0	OK
Flange top a end plate SLN218 ring	12.0	C LC_125	52.4	0.0	0.0	OK
Flange top b end plate SLN218 ring	12.0	C LC_125	67.4	0.0	0.0	OK
Flange bottom a end plate SLN218 ring	12.0	C LC_121	74.9	0.0	0.0	OK
Flange bottom b end plate SLN218 ring	12.0	C LC_125	55.7	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

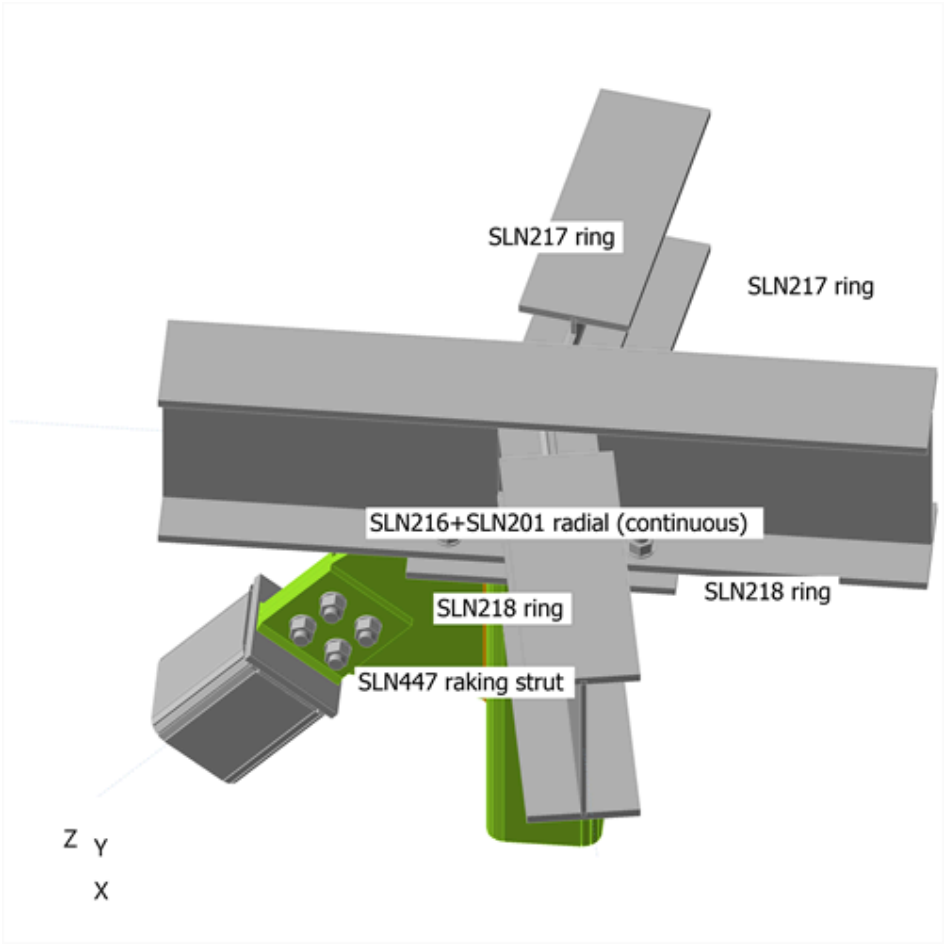
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

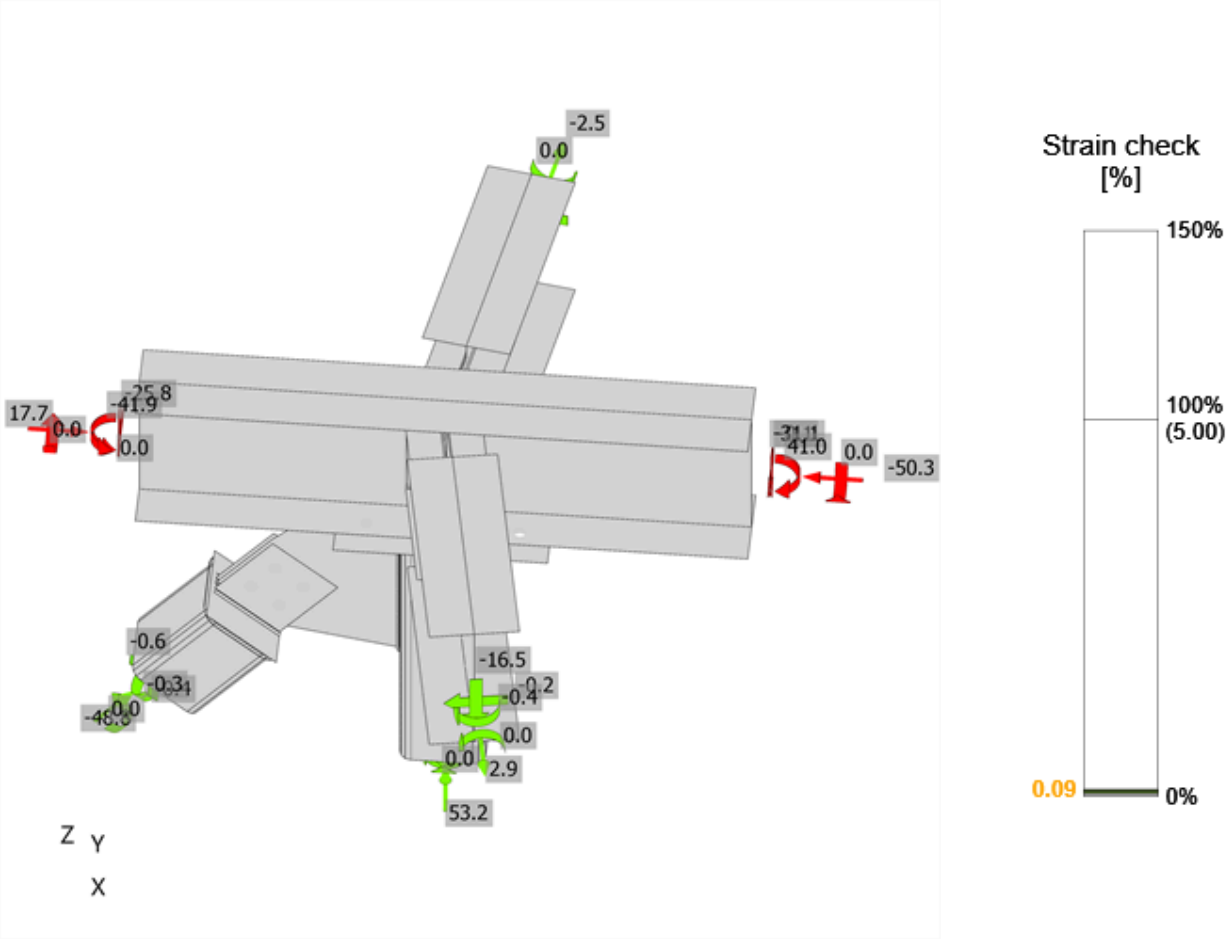
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN49 column below	150	C LC_112	0	5	0.1	OK
SLN447 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

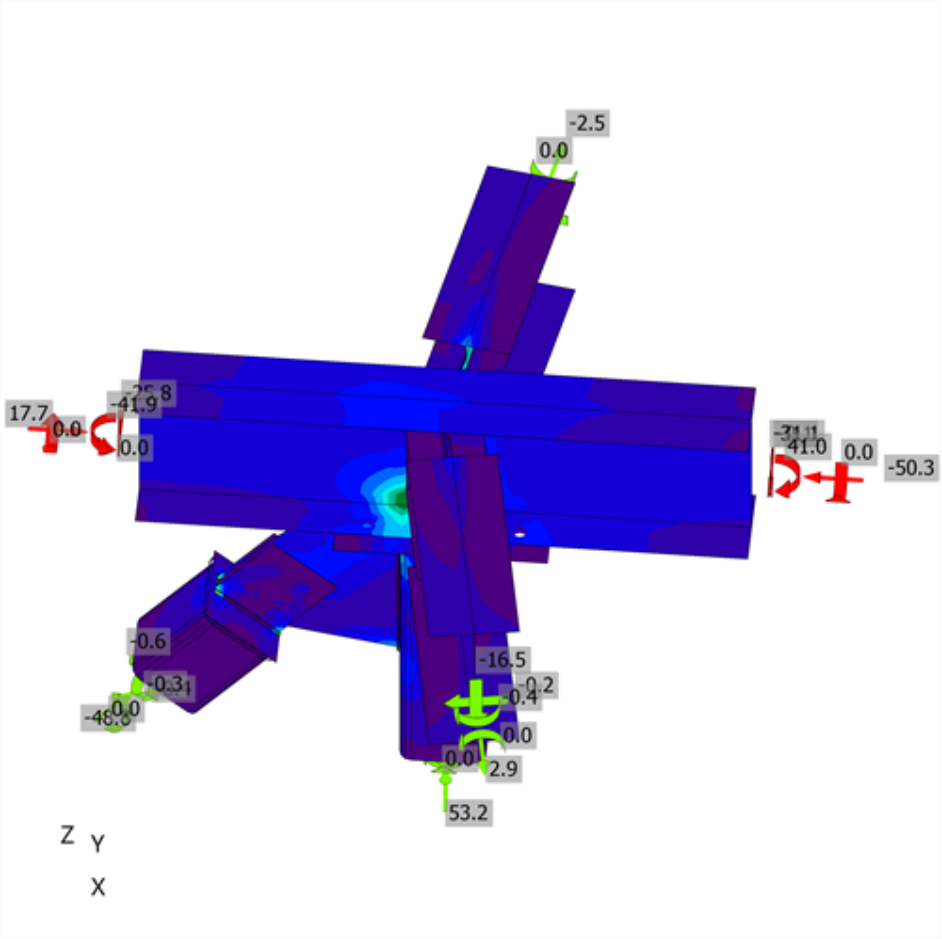
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

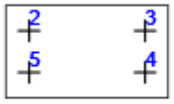
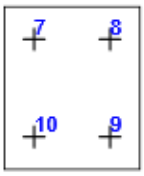
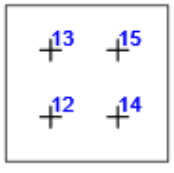


Strain check, C LC_112



Equivalent stress, C LC_112

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_108	13.0	4.9	168.9	9.2	5.2	11.8	OK	OK
	B3	M20 8.8 - 1	C LC_107	28.7	3.7	83.5	20.4	4.4	18.5	OK	OK
	B4	M20 8.8 - 1	C LC_107	28.2	4.3	123.5	20.0	4.6	18.8	OK	OK
	B5	M20 8.8 - 1	C LC_108	13.5	4.8	120.1	9.6	5.1	11.9	OK	OK
	B7	M20 8.8 - 1	C LC_120	10.5	0.7	123.8	7.5	0.8	6.1	OK	OK
	B8	M20 8.8 - 1	C LC_110	4.5	0.6	66.9	3.2	0.7	2.9	OK	OK
	B9	M20 8.8 - 1	C LC_110	4.5	0.7	91.0	3.2	0.7	3.0	OK	OK
	B10	M20 8.8 - 1	C LC_120	11.5	0.8	123.8	8.1	0.9	6.7	OK	OK
	B12	M20 8.8 - 1	C LC_118	14.1	13.8	106.0	10.0	14.6	21.8	OK	OK
	B13	M20 8.8 - 1	C LC_110	6.4	9.3	106.0	4.6	9.9	13.1	OK	OK
	B14	M20 8.8 - 1	C LC_118	2.6	9.9	106.0	1.8	10.6	11.9	OK	OK
	B15	M20 8.8 - 1	C LC_112	0.1	6.6	115.3	0.1	11.3	7.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN447 raking strut	SLN447 raking strut	◢ 4.2	519	C LC_115	180.3	0.0	82.4	-92.5	3.6	44.6	25.0	OK	OK
SLN49 column below-w 3	Gusset SLN447 raking strut	◢ 5.6 ◣	288	C LC_112	397.2	0.3	-131.9	-132.9	-170.6	98.1	62.7	OK	OK
		◢ 5.6 ◣	288	C LC_112	397.1	0.3	-134.0	132.6	170.3	98.1	61.9	OK	OK
SLN217 ring-w 1	Flange top b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_123	71.8	0.0	37.2	31.1	16.9	17.7	11.0	OK	OK
		◢ 4.2 ◣	197	C LC_120	32.1	0.0	-15.4	14.0	8.2	7.9	6.7	OK	OK
SLN217 ring-w 1	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	198	C LC_119	96.4	0.0	28.5	33.1	41.6	23.8	14.6	OK	OK
		◢ 4.2 ◣	198	C LC_119	135.5	0.0	53.5	-48.8	-52.7	33.5	20.6	OK	OK
SLN218 ring-w 1	Flange top a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_125	62.0	0.0	34.3	28.5	-8.6	15.3	8.4	OK	OK
		◢ 4.2 ◣	197	C LC_125	35.8	0.0	-18.4	16.7	-5.8	8.8	7.1	OK	OK
SLN218 ring-w 1	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	198	C LC_122	94.6	0.0	27.9	31.3	-41.7	23.4	13.0	OK	OK
		◢ 4.2 ◣	198	C LC_121	136.4	0.0	56.0	-51.3	50.2	33.7	18.4	OK	OK
Cap plate SLN447 raking strut	Fin plate SLN447 raking strut	◢ 5.6 ◣	139	C LC_118	285.2	0.0	102.5	124.3	90.4	70.5	49.4	OK	OK
		◢ 5.6 ◣	139	C LC_115	147.4	0.0	87.6	-56.6	-38.5	36.4	23.9	OK	OK
Cap plate SLN447 raking strut	Fin plate 2 SLN447 raking strut	◢ 5.6 ◣	139	C LC_118	137.7	0.0	88.9	50.9	33.0	34.0	22.5	OK	OK
		◢ 5.6 ◣	139	C LC_115	280.4	0.0	102.6	-122.3	-88.1	69.3	47.4	OK	OK
End plate column below SLN49 column below	SLN49 column below	◢ 5.6 ◣	521	C LC_104	281.0	0.0	-180.6	-121.8	24.6	69.4	37.9	OK	OK
		◢ 5.6 ◣	521	C LC_152	210.0	0.0	-94.0	106.0	-22.6	51.9	30.5	OK	OK
SLN216+SLN201 radial (continuous)-w 1	SLN217 ring-w 1	◢ 7.2	189	C LC_122	90.8	0.0	6.0	-2.1	-52.3	22.4	14.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN216+SLN201 radial (continuous)-w 1	SLN218 ring-w 1	◢ 7.2	189	C LC_122	92.3	0.0	32.4	-32.2	38.2	22.8	13.8	OK	OK
End plate SLN217 ring	SLN217 ring-w 1	◢ 4.2 ◣	149	C LC_122	89.0	0.0	-25.1	-23.9	-43.1	22.0	12.4	OK	OK
		◢ 4.2 ◣	149	C LC_119	73.8	0.0	-16.6	31.0	27.6	18.2	11.5	OK	OK
End plate SLN218 ring	SLN218 ring-w 1	◢ 4.2 ◣	149	C LC_119	78.9	0.0	-20.6	-28.1	33.8	19.5	12.4	OK	OK
		◢ 4.2 ◣	149	C LC_147	68.6	0.0	-19.5	23.3	-30.0	17.0	10.9	OK	OK
SLN217 ring-w 1	Flange top a end plate SLN217 ring	◢ 4.2 ◣	199	C LC_120	109.5	0.0	-9.9	6.0	-62.7	27.1	19.1	OK	OK
		◢ 4.2 ◣	198	C LC_123	72.8	0.0	11.1	12.3	-39.7	18.0	10.4	OK	OK
SLN217 ring-w 1	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	198	C LC_120	97.4	0.0	18.1	-17.9	52.2	24.1	14.3	OK	OK
		◢ 4.2 ◣	198	C LC_112	83.5	0.0	-11.9	-4.7	47.5	20.6	12.9	OK	OK
SLN218 ring-w 1	Flange top b end plate SLN218 ring	◢ 4.2 ◣	199	C LC_125	113.4	0.0	-58.3	25.4	50.1	28.0	17.6	OK	OK
		◢ 4.2 ◣	198	C LC_125	76.2	0.0	15.6	15.2	40.3	18.8	10.5	OK	OK
SLN218 ring-w 1	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	198	C LC_121	103.8	0.0	17.6	-17.7	-56.4	25.7	14.5	OK	OK
		◢ 4.2 ◣	198	C LC_125	95.4	0.0	-9.8	-3.0	-54.7	23.6	13.1	OK	OK
End plate SLN217 ring	Flange top a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_112	33.8	0.0	-0.6	3.0	-19.3	8.3	7.0	OK	OK
		◢ 4.2 ◣	58	C LC_110	26.6	0.0	11.3	8.5	11.0	6.6	6.6	OK	OK
End plate SLN217 ring	Flange top b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_123	52.7	0.0	-0.1	-2.2	-30.3	13.0	9.3	OK	OK
		◢ 4.2 ◣	58	C LC_120	61.2	0.0	34.7	23.9	-16.6	15.1	12.2	OK	OK
End plate SLN217 ring	Flange bottom a end plate SLN217 ring	◢ 4.2 ◣	58	C LC_122	58.0	0.0	11.6	-22.5	23.9	14.3	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_120	46.2	0.0	-11.8	-17.7	18.7	11.4	8.6	OK	OK
End plate SLN217 ring	Flange bottom b end plate SLN217 ring	◢ 4.2 ◣	58	C LC_120	73.5	0.0	41.5	-31.6	15.1	18.2	14.3	OK	OK
		◢ 4.2 ◣	58	C LC_120	59.6	0.0	-4.6	-0.4	34.3	14.7	10.4	OK	OK
End plate SLN218 ring	Flange top a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_107	64.9	0.0	-32.2	31.4	8.6	16.0	8.4	OK	OK
		◢ 4.2 ◣	58	C LC_107	52.8	0.0	-24.0	-24.7	-11.2	13.0	7.5	OK	OK
End plate SLN218 ring	Flange top b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_125	62.7	0.0	-30.7	24.8	-19.5	15.5	9.4	OK	OK
		◢ 4.2 ◣	58	C LC_125	57.2	0.0	-18.5	-24.3	19.6	14.1	9.7	OK	OK
End plate SLN218 ring	Flange bottom a end plate SLN218 ring	◢ 4.2 ◣	58	C LC_107	61.4	0.0	-27.5	28.8	13.2	15.2	8.3	OK	OK
		◢ 4.2 ◣	58	C LC_107	80.0	0.0	-39.9	-38.6	-10.5	19.8	10.0	OK	OK
End plate SLN218 ring	Flange bottom b end plate SLN218 ring	◢ 4.2 ◣	58	C LC_119	58.0	0.0	11.4	-21.6	-24.7	14.3	7.9	OK	OK
		◢ 4.2 ◣	58	C LC_125	67.0	0.0	-34.9	-26.5	19.6	16.6	11.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_300
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut

Name	SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

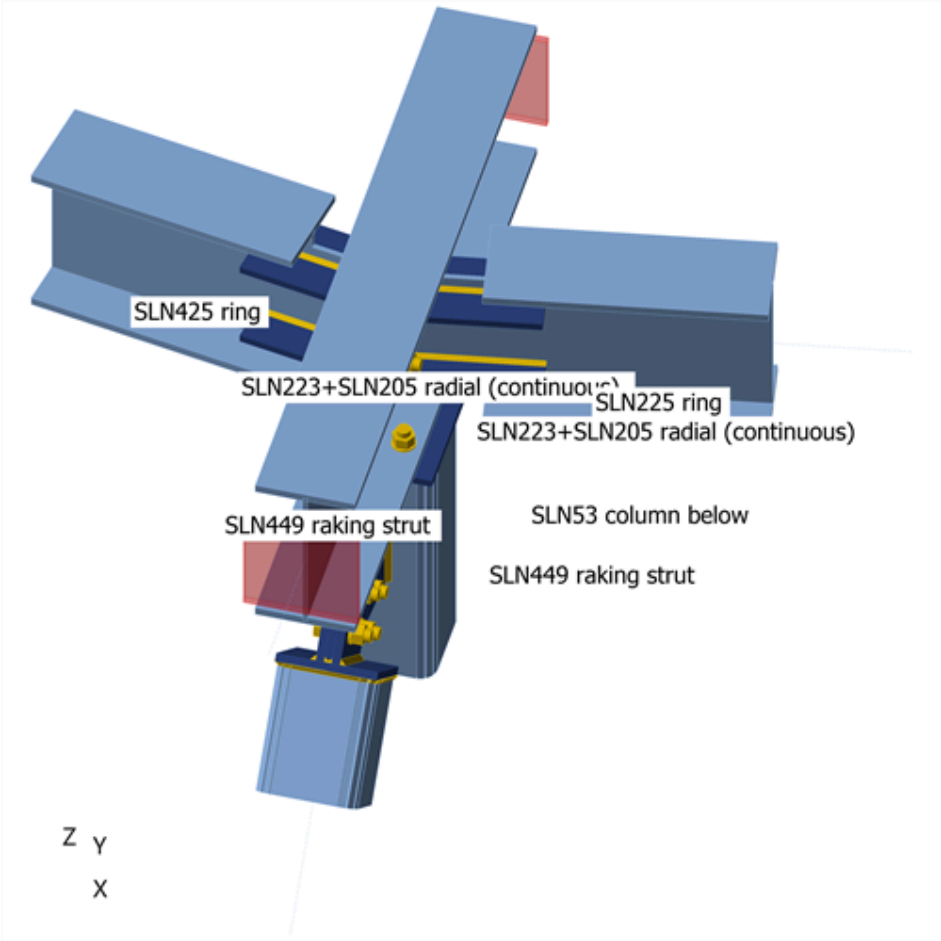
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN223+SLN205 radial (continuous)	1 - UB254x146x43	120.0	0.0	0.0	0	0	0
SLN53 column below	2 - SHS150x150x10	0.0	90.0	120.0	0	0	0
SLN425 ring	1 - UB254x146x43	20.0	0.0	0.0	0	0	0
SLN225 ring	1 - UB254x146x43	40.0	0.0	0.0	0	0	0
SLN449 raking strut	2 - SHS150x150x10	-60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN223+SLN205 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN223+SLN205 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN53 column below / begin		Position	0
SLN425 ring / begin		Position	0
SLN225 ring / end		Position	0
SLN449 raking strut / end		Position	0



SLN223+SLN205 radial (continuous) - SLN53 column below - SLN425 ring - SLN225 ring - SLN449 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN53 column below / Begin	32.3	-0.1	0.3	0.0	1.0	0.2
	SLN425 ring / Begin	-8.7	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.3	-0.1	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	-0.4	-0.1	-1.0	-0.1	0.0	-0.3
C LC_102	SLN53 column below / Begin	149.3	-0.2	1.0	0.0	4.0	0.9
	SLN425 ring / Begin	-27.3	0.1	-25.5	0.0	0.0	-0.4
	SLN225 ring / End	19.2	-0.5	-25.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-13.3	-0.3	-1.0	-0.2	0.0	-0.9
C LC_103	SLN53 column below / Begin	126.4	-0.2	0.9	0.0	3.6	0.7
	SLN425 ring / Begin	-16.4	0.1	-23.9	0.0	0.0	-0.3
	SLN225 ring / End	8.3	-0.6	-23.9	0.0	0.0	-1.5
	SLN449 raking strut / End	-29.5	0.0	-0.9	-0.2	0.0	-0.6
C LC_104	SLN53 column below / Begin	149.7	3.0	-1.1	0.0	2.6	0.0
	SLN425 ring / Begin	-20.9	0.2	-23.9	0.0	0.0	-0.5
	SLN225 ring / End	17.4	-0.4	-23.9	0.0	0.0	-1.3
	SLN449 raking strut / End	-2.0	-0.3	-1.1	-0.3	0.0	0.2
C LC_105	SLN53 column below / Begin	124.1	-0.2	1.0	0.0	4.2	0.9
	SLN425 ring / Begin	-13.9	0.0	-23.9	0.0	0.0	-0.3
	SLN225 ring / End	7.2	-0.6	-23.9	0.0	0.0	-1.6
	SLN449 raking strut / End	-35.8	-0.4	-0.8	-0.1	0.0	-0.9
C LC_106	SLN53 column below / Begin	157.8	-2.1	-2.8	0.0	1.0	1.5
	SLN425 ring / Begin	-29.6	0.3	-23.9	0.0	0.0	-0.7
	SLN225 ring / End	18.8	-0.4	-23.9	0.0	0.0	-1.3
	SLN449 raking strut / End	11.4	-0.3	-1.2	-0.1	0.0	-1.5
C LC_107	SLN53 column below / Begin	-17.1	0.1	0.1	0.0	0.2	-0.3
	SLN425 ring / Begin	8.6	0.0	-0.5	0.0	0.0	0.1
	SLN225 ring / End	-10.1	-0.1	-0.5	0.0	0.0	-0.2
	SLN449 raking strut / End	-26.7	0.6	-0.5	0.0	0.0	0.3
C LC_108	SLN53 column below / Begin	24.6	6.5	-4.0	0.0	-1.9	-1.7
	SLN425 ring / Begin	1.0	0.0	-0.5	0.0	0.0	-0.1
	SLN225 ring / End	5.6	0.0	-0.5	0.0	0.0	0.0
	SLN449 raking strut / End	21.8	-0.1	-1.0	-0.4	0.0	2.0
C LC_109	SLN53 column below / Begin	-20.7	0.0	0.3	0.0	1.3	0.1
	SLN425 ring / Begin	12.3	0.0	-0.5	0.0	0.0	0.1
	SLN225 ring / End	-11.7	-0.1	-0.5	0.0	0.0	-0.3
	SLN449 raking strut / End	-38.0	-0.3	-0.3	0.1	0.0	-0.2
C LC_110	SLN53 column below / Begin	39.9	-3.8	-7.3	0.0	-5.1	1.4
	SLN425 ring / Begin	-15.0	0.2	-0.5	0.0	0.0	-0.5
	SLN225 ring / End	8.0	0.0	-0.5	0.0	0.0	-0.1
	SLN449 raking strut / End	47.4	-0.1	-1.2	0.1	0.0	-1.5
C LC_111	SLN53 column below / Begin	71.6	-0.1	0.6	0.0	2.5	0.2
	SLN425 ring / Begin	-4.3	0.0	-16.2	0.0	0.0	-0.1
	SLN225 ring / End	-1.9	-0.4	-16.1	0.0	0.0	-1.1
	SLN449 raking strut / End	-37.9	0.4	-0.8	-0.1	0.0	-0.2
C LC_112	SLN53 column below / Begin	114.9	6.4	-3.4	0.0	0.4	-1.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN425 ring / Begin	-12.4	0.2	-16.2	0.0	0.0	-0.4
	SLN225 ring / End	14.7	-0.2	-16.1	0.0	0.0	-0.8
	SLN449 raking strut / End	12.7	-0.2	-1.3	-0.5	0.0	1.5
C LC_113	SLN53 column below / Begin	67.6	-0.2	0.9	0.0	3.6	0.7
	SLN425 ring / Begin	-0.2	0.0	-16.2	0.0	0.0	-0.1
	SLN225 ring / End	-3.7	-0.5	-16.1	0.0	0.0	-1.3
	SLN449 raking strut / End	-49.6	-0.5	-0.6	0.0	0.0	-0.7
C LC_114	SLN53 column below / Begin	130.4	-4.0	-6.7	0.0	-2.8	1.9
	SLN425 ring / Begin	-28.9	0.3	-16.2	0.0	0.0	-0.8
	SLN225 ring / End	17.2	-0.3	-16.1	0.0	0.0	-0.9
	SLN449 raking strut / End	38.7	-0.3	-1.5	0.0	0.0	-2.0
C LC_115	SLN53 column below / Begin	-8.8	0.1	0.1	0.0	0.5	-0.2
	SLN425 ring / Begin	6.3	0.0	-1.7	0.0	0.0	0.1
	SLN225 ring / End	-8.5	-0.1	-1.7	0.0	0.0	-0.3
	SLN449 raking strut / End	-27.0	0.5	-0.8	-0.1	0.0	0.2
C LC_116	SLN53 column below / Begin	33.1	6.5	-3.9	0.0	-1.7	-1.7
	SLN425 ring / Begin	-1.2	0.1	-1.7	0.0	0.0	-0.2
	SLN225 ring / End	7.3	0.0	-1.7	0.0	0.0	-0.1
	SLN449 raking strut / End	21.8	-0.1	-1.3	-0.4	0.0	1.9
C LC_117	SLN53 column below / Begin	-12.5	0.0	0.4	0.0	1.6	0.2
	SLN425 ring / Begin	10.1	0.0	-1.7	0.0	0.0	0.1
	SLN225 ring / End	-10.1	-0.2	-1.7	0.0	0.0	-0.5
	SLN449 raking strut / End	-38.3	-0.3	-0.6	0.0	0.0	-0.3
C LC_118	SLN53 column below / Begin	48.4	-3.8	-7.2	0.0	-4.8	1.4
	SLN425 ring / Begin	-17.3	0.2	-1.7	0.0	0.0	-0.5
	SLN225 ring / End	9.7	0.0	-1.7	0.0	0.0	-0.2
	SLN449 raking strut / End	47.4	-0.1	-1.5	0.1	0.0	-1.6
C LC_119	SLN53 column below / Begin	136.7	-0.2	0.9	0.0	3.6	0.7
	SLN425 ring / Begin	-19.8	0.1	-25.5	0.0	0.0	-0.3
	SLN225 ring / End	11.3	-0.6	-25.4	0.0	0.0	-1.5
	SLN449 raking strut / End	-27.5	0.0	-0.9	-0.2	0.0	-0.7
C LC_120	SLN53 column below / Begin	156.0	3.0	-1.1	0.0	2.6	0.0
	SLN425 ring / Begin	-24.4	0.2	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	20.5	-0.4	-25.4	0.0	0.0	-1.4
	SLN449 raking strut / End	-5.5	-0.3	-1.1	-0.4	0.0	0.2
C LC_121	SLN53 column below / Begin	136.9	-0.2	1.0	0.0	4.1	0.9
	SLN425 ring / Begin	-17.1	0.0	-25.5	0.0	0.0	-0.3
	SLN225 ring / End	10.3	-0.6	-25.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-30.1	-0.4	-0.8	-0.1	0.0	-0.9
C LC_122	SLN53 column below / Begin	163.3	-2.1	-2.8	0.0	1.0	1.5
	SLN425 ring / Begin	-33.3	0.3	-25.5	0.0	0.0	-0.7
	SLN225 ring / End	21.9	-0.4	-25.4	0.0	0.0	-1.5
	SLN449 raking strut / End	6.7	-0.4	-1.2	-0.1	0.0	-1.6
C LC_123	SLN53 column below / Begin	92.2	-0.1	0.6	0.0	2.4	0.3
	SLN425 ring / Begin	-11.2	0.0	-19.3	0.0	0.0	-0.2
	SLN225 ring / End	4.2	-0.4	-19.3	0.0	0.0	-1.2
	SLN449 raking strut / End	-33.8	0.3	-0.8	-0.2	0.0	-0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN53 column below / Begin	127.4	6.3	-3.4	0.0	0.4	-1.2
	SLN425 ring / Begin	-19.3	0.1	-19.3	0.0	0.0	-0.3
	SLN225 ring / End	20.9	-0.3	-19.3	0.0	0.0	-1.1
	SLN449 raking strut / End	5.9	-0.3	-1.3	-0.5	0.0	1.4
C LC_125	SLN53 column below / Begin	93.3	-0.2	0.9	0.0	3.4	0.7
	SLN425 ring / Begin	-6.6	0.0	-19.3	0.0	0.0	-0.2
	SLN225 ring / End	2.4	-0.5	-19.3	0.0	0.0	-1.3
	SLN449 raking strut / End	-38.2	-0.5	-0.6	0.0	0.0	-0.7
C LC_126	SLN53 column below / Begin	141.4	-4.0	-6.7	0.0	-2.7	2.0
	SLN425 ring / Begin	-36.2	0.3	-19.3	0.0	0.0	-0.7
	SLN225 ring / End	23.4	-0.4	-19.3	0.0	0.0	-1.2
	SLN449 raking strut / End	29.3	-0.3	-1.5	0.0	0.0	-2.1
C LC_127	SLN53 column below / Begin	11.8	0.0	0.1	0.0	0.4	-0.1
	SLN425 ring / Begin	-0.5	0.0	-4.9	0.0	0.0	0.0
	SLN225 ring / End	-2.5	-0.1	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	-22.9	0.5	-0.8	-0.1	0.0	0.1
C LC_128	SLN53 column below / Begin	45.7	6.5	-3.9	0.0	-1.7	-1.6
	SLN425 ring / Begin	-8.2	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	13.6	-0.1	-4.9	0.0	0.0	-0.3
	SLN449 raking strut / End	14.9	-0.1	-1.3	-0.4	0.0	1.9
C LC_129	SLN53 column below / Begin	13.3	0.0	0.4	0.0	1.4	0.2
	SLN425 ring / Begin	3.7	0.0	-4.9	0.0	0.0	0.0
	SLN225 ring / End	-4.1	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	-26.9	-0.3	-0.6	0.0	0.0	-0.3
C LC_130	SLN53 column below / Begin	59.4	-3.9	-7.2	0.0	-4.7	1.5
	SLN425 ring / Begin	-24.7	0.2	-4.9	0.0	0.0	-0.4
	SLN225 ring / End	15.9	-0.2	-4.9	0.0	0.0	-0.5
	SLN449 raking strut / End	38.0	-0.2	-1.5	0.1	0.0	-1.7
C LC_135	SLN53 column below / Begin	153.8	-0.2	1.0	0.0	4.1	1.0
	SLN425 ring / Begin	-27.7	0.2	-25.5	0.0	0.0	-0.7
	SLN225 ring / End	16.5	-0.6	-25.4	0.0	0.0	-1.8
	SLN449 raking strut / End	-10.9	-0.4	-1.0	-0.2	0.0	-1.1
C LC_141	SLN53 column below / Begin	32.0	-0.1	0.3	0.0	1.0	0.2
	SLN425 ring / Begin	-8.6	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.2	-0.2	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	-0.9	-0.1	-1.0	-0.1	0.0	-0.3
C LC_142	SLN53 column below / Begin	32.7	-0.1	0.3	0.0	1.0	0.2
	SLN425 ring / Begin	-8.8	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.5	-0.1	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	0.0	-0.1	-1.0	-0.1	0.0	-0.3
C LC_143	SLN53 column below / Begin	31.8	-0.1	0.3	0.0	1.0	0.2
	SLN425 ring / Begin	-8.4	0.0	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.2	-0.2	-4.9	0.0	0.0	-0.4
	SLN449 raking strut / End	-1.0	-0.1	-1.0	-0.1	0.0	-0.3
C LC_144	SLN53 column below / Begin	32.8	-0.1	0.3	0.0	1.0	0.2
	SLN425 ring / Begin	-8.9	0.1	-4.9	0.0	0.0	-0.1
	SLN225 ring / End	6.5	-0.1	-4.9	0.0	0.0	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN449 raking strut / End	0.2	-0.1	-1.0	-0.1	0.0	-0.3
C LC_145	SLN53 column below / Begin	148.0	-0.2	1.0	0.0	4.0	0.9
	SLN425 ring / Begin	-26.8	0.1	-25.5	0.0	0.0	-0.4
	SLN225 ring / End	18.5	-0.6	-25.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-15.1	-0.3	-1.0	-0.2	0.0	-0.9
C LC_146	SLN53 column below / Begin	150.7	-0.2	1.0	0.0	4.0	0.9
	SLN425 ring / Begin	-27.7	0.2	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	19.9	-0.5	-25.4	0.0	0.0	-1.5
	SLN449 raking strut / End	-11.5	-0.3	-1.0	-0.2	0.0	-0.9
C LC_147	SLN53 column below / Begin	147.4	-0.2	1.0	0.0	4.0	0.9
	SLN425 ring / Begin	-26.1	0.0	-25.5	0.0	0.0	-0.4
	SLN225 ring / End	18.3	-0.6	-25.4	0.0	0.0	-1.6
	SLN449 raking strut / End	-15.7	-0.3	-1.0	-0.2	0.0	-0.9
C LC_148	SLN53 column below / Begin	151.2	-0.2	1.0	0.0	4.0	0.9
	SLN425 ring / Begin	-28.4	0.2	-25.5	0.0	0.0	-0.5
	SLN225 ring / End	20.1	-0.4	-25.4	0.0	0.0	-1.5
	SLN449 raking strut / End	-10.8	-0.3	-1.0	-0.2	0.0	-0.9
C LC_151	SLN53 column below / Begin	96.1	-0.2	-0.5	0.0	-2.0	0.7
	SLN425 ring / Begin	-12.2	0.1	-11.5	0.0	0.0	-0.4
	SLN225 ring / End	6.5	-0.4	-11.7	0.0	0.0	-1.1
	SLN449 raking strut / End	-15.2	-0.3	-1.0	-0.1	0.0	-0.8
C LC_152	SLN53 column below / Begin	99.8	-0.2	2.6	0.0	10.7	0.8
	SLN425 ring / Begin	-27.8	0.1	-14.0	0.0	0.0	-0.3
	SLN225 ring / End	21.0	-0.4	-13.8	0.0	0.0	-1.3
	SLN449 raking strut / End	4.4	-0.3	-1.0	-0.2	0.0	-0.7
C LC_153	SLN53 column below / Begin	149.0	-0.2	0.9	0.0	3.8	0.9
	SLN425 ring / Begin	-27.4	0.1	-20.6	0.0	0.0	-0.4
	SLN225 ring / End	19.2	-0.5	-20.6	0.0	0.0	-1.6
	SLN449 raking strut / End	-12.9	-0.3	-1.0	-0.2	0.0	-0.9
C LC_154	SLN53 column below / Begin	95.7	-0.2	-0.5	0.0	-1.9	0.7
	SLN425 ring / Begin	-12.3	0.1	-11.5	0.0	0.0	-0.4
	SLN225 ring / End	6.6	-0.4	-11.7	0.0	0.0	-1.1
	SLN449 raking strut / End	-15.0	-0.3	-1.0	-0.1	0.0	-0.7

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.2 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	25.7 < 100%	OK
Welds	98.2 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN223+SLN205 radial (continuous)-bfl 1	12.7	C LC_110	212.7	0.0	34.0	OK
SLN223+SLN205 radial (continuous)-tfl 1	12.7	C LC_135	135.7	0.0	0.0	OK
SLN223+SLN205 radial (continuous)-w 1	7.2	C LC_135	275.1	0.0	49.4	OK
SLN53 column below	10.0	C LC_113	275.3	0.2	0.0	OK
SLN425 ring-bfl 1	12.7	C LC_114	49.7	0.0	0.0	OK
SLN425 ring-tfl 1	12.7	C LC_114	44.2	0.0	0.0	OK
SLN425 ring-w 1	7.2	C LC_135	151.9	0.0	0.0	OK
SLN225 ring-bfl 1	12.7	C LC_135	108.8	0.0	0.0	OK
SLN225 ring-tfl 1	12.7	C LC_135	105.8	0.0	0.0	OK
SLN225 ring-w 1	7.2	C LC_135	272.1	0.0	0.0	OK
SLN449 raking strut	10.0	C LC_114	63.4	0.0	0.0	OK
End plate column below SLN53 column below	12.0	C LC_110	245.2	0.0	34.0	OK
End plate SLN425 ring	12.0	C LC_135	42.3	0.0	99.8	OK
End plate SLN225 ring	12.0	C LC_135	74.1	0.0	397.2	OK
Cap plate SLN449 raking strut	12.0	C LC_126	125.9	0.0	0.0	OK
Fin plate SLN449 raking strut	12.0	C LC_108	275.1	0.1	64.9	OK
Fin plate 2 SLN449 raking strut	12.0	C LC_126	275.1	0.0	134.5	OK
Gusset SLN449 raking strut	12.0	C LC_126	275.1	0.1	66.0	OK
Flange top a end plate SLN425 ring	12.0	C LC_114	54.4	0.0	0.0	OK
Flange top b end plate SLN425 ring	12.0	C LC_114	52.3	0.0	0.0	OK
Flange bottom a end plate SLN425 ring	12.0	C LC_114	71.8	0.0	0.0	OK
Flange bottom b end plate SLN425 ring	12.0	C LC_135	64.0	0.0	0.0	OK
Flange top a end plate SLN225 ring	12.0	C LC_135	122.5	0.0	0.0	OK
Flange top b end plate SLN225 ring	12.0	C LC_135	143.4	0.0	0.0	OK
Flange bottom a end plate SLN225 ring	12.0	C LC_135	139.3	0.0	0.0	OK
Flange bottom b end plate SLN225 ring	12.0	C LC_135	127.9	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

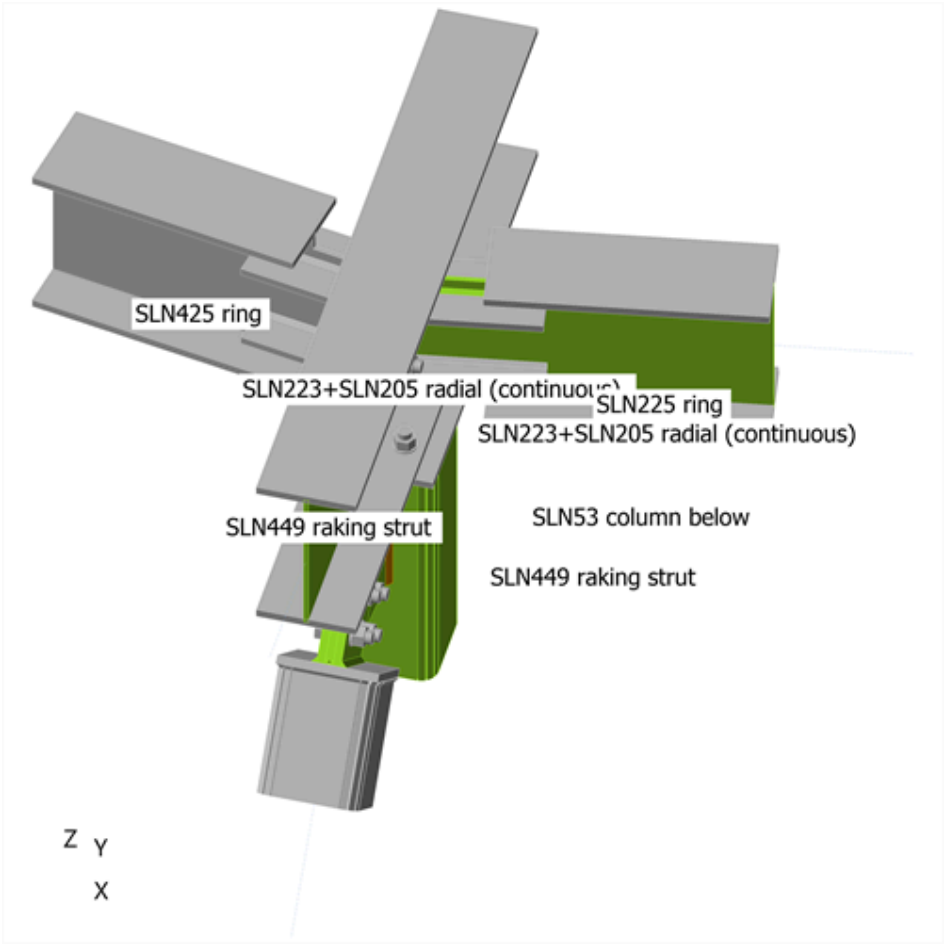
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

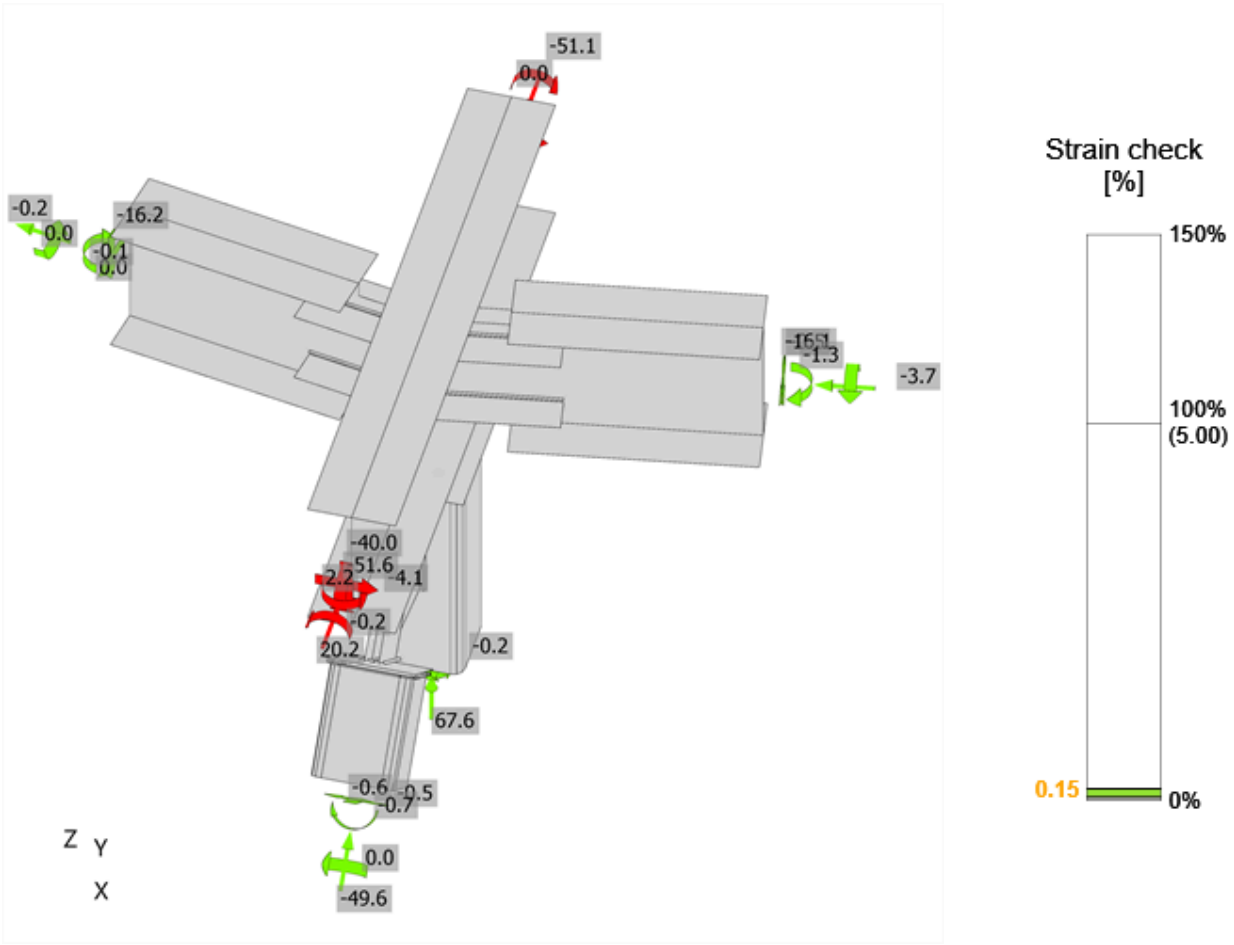
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN53 column below	150	C LC_110	0	5	0.1	OK
SLN449 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

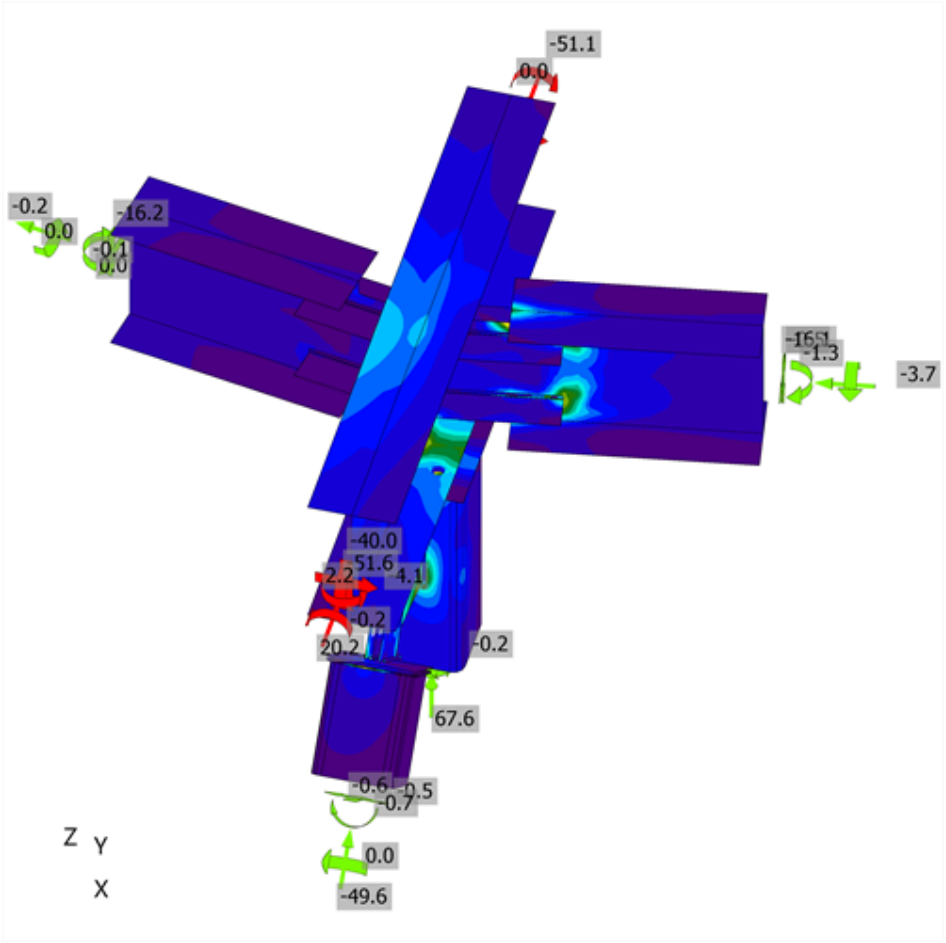
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

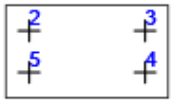
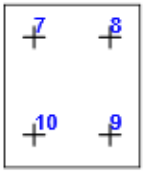
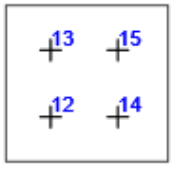


Strain check, C LC_113



Equivalent stress, C LC_113

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_109	19.6	6.1	129.0	13.9	6.5	16.4	OK	OK
	B3	M20 8.8 - 1	C LC_110	34.6	4.8	85.4	24.5	5.7	22.6	OK	OK
	B4	M20 8.8 - 1	C LC_110	36.3	6.2	123.7	25.7	6.6	24.9	OK	OK
	B5	M20 8.8 - 1	C LC_109	19.1	5.9	116.2	13.5	6.2	15.9	OK	OK
	B7	M20 8.8 - 1	C LC_135	5.0	0.5	123.8	3.5	0.7	3.0	OK	OK
	B8	M20 8.8 - 1	C LC_114	3.5	0.5	123.8	2.5	0.7	2.3	OK	OK
	B9	M20 8.8 - 1	C LC_135	4.4	1.1	123.8	3.1	1.7	3.4	OK	OK
	B10	M20 8.8 - 1	C LC_135	6.9	0.4	123.8	4.9	0.6	3.9	OK	OK
	B12	M20 8.8 - 1	C LC_114	14.4	16.0	106.0	10.2	17.0	24.3	OK	OK
	B13	M20 8.8 - 1	C LC_114	8.1	11.4	106.0	5.7	12.2	16.3	OK	OK
	B14	M20 8.8 - 1	C LC_118	2.5	11.6	106.0	1.8	14.1	13.6	OK	OK
	B15	M20 8.8 - 1	C LC_113	0.4	7.5	115.3	0.3	11.3	8.2	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN449 raking strut	SLN449 raking strut	◢ 4.2	519	C LC_118	216.2	0.0	99.6	-110.8	-1.5	53.4	29.8	OK	OK
SLN53 column below-w 3	Gusset SLN449 raking strut	◢ 5.6 ◣	288	C LC_113	397.3	0.5	-124.0	-128.7	-175.9	98.2	67.7	OK	OK
		◢ 5.6 ◣	288	C LC_113	397.3	0.4	-134.0	129.1	173.2	98.2	64.9	OK	OK
SLN425 ring-w 1	Flange top b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_135	87.9	0.0	-38.9	-34.2	-30.0	21.7	13.1	OK	OK
		◢ 4.2 ◣	198	C LC_135	46.7	0.0	-13.3	18.0	18.6	11.5	8.3	OK	OK
SLN425 ring-w 1	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	198	C LC_122	54.1	0.0	14.8	9.8	28.4	13.4	10.6	OK	OK
		◢ 4.2 ◣	198	C LC_122	88.0	0.0	2.3	-13.9	48.8	21.7	11.7	OK	OK
SLN225 ring-w 1	Flange top a end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	166.6	0.0	89.1	74.3	-32.9	41.2	21.3	OK	OK
		◢ 4.2 ◣	197	C LC_135	74.0	0.0	-37.3	34.7	-12.6	18.3	10.4	OK	OK
SLN225 ring-w 1	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	119.4	0.0	35.9	49.0	-43.8	29.5	16.2	OK	OK
		◢ 4.2 ◣	198	C LC_135	235.5	0.0	108.1	-95.0	74.6	58.2	32.1	OK	OK
Cap plate SLN449 raking strut	Fin plate SLN449 raking strut	◢ 5.6 ◣	139	C LC_116	271.9	0.0	101.4	120.7	81.6	67.2	49.4	OK	OK
		◢ 5.6 ◣	139	C LC_118	168.8	0.0	93.2	-64.8	-49.0	41.7	28.5	OK	OK
Cap plate SLN449 raking strut	Fin plate 2 SLN449 raking strut	◢ 5.6 ◣	139	C LC_110	139.1	0.0	65.1	61.6	-35.2	34.4	20.8	OK	OK
		◢ 5.6 ◣	139	C LC_114	332.5	0.0	97.0	-143.5	-114.6	82.2	56.7	OK	OK
End plate column below SLN53 column below	SLN53 column below	◢ 5.6 ◣	521	C LC_105	314.6	0.0	-191.3	-140.4	-32.6	77.7	42.8	OK	OK
		◢ 5.6 ◣	521	C LC_152	253.8	0.0	-127.1	124.1	25.9	62.7	35.3	OK	OK
SLN223+SLN205 radial (continuous)-w 1	SLN425 ring-w 1	◢ 7.2	189	C LC_135	123.5	0.0	38.1	-54.6	-40.1	30.5	18.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN223+SLN205 radial (continuous)-w 1	SLN225 ring-w 1	◢ 7.2	189	C LC_122	118.5	0.0	0.8	-24.0	64.0	29.3	20.3	OK	OK
End plate SLN425 ring	SLN425 ring-w 1	◢ 4.2 ◣	149	C LC_122	85.5	0.0	-21.0	-44.7	-17.0	21.1	12.4	OK	OK
		◢ 4.2 ◣	149	C LC_135	68.2	0.0	-33.6	17.3	29.5	16.8	10.3	OK	OK
End plate SLN225 ring	SLN225 ring-w 1	◢ 4.2 ◣	149	C LC_122	122.8	0.0	-45.3	-28.1	59.6	30.3	19.3	OK	OK
		◢ 4.2 ◣	149	C LC_135	90.5	0.0	-21.1	40.6	-30.6	22.4	13.7	OK	OK
SLN425 ring-w 1	Flange top a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_114	95.3	0.0	12.7	-6.7	54.1	23.5	16.3	OK	OK
		◢ 4.2 ◣	198	C LC_114	75.5	0.0	-13.8	-15.3	40.0	18.6	10.7	OK	OK
SLN425 ring-w 1	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	199	C LC_122	81.3	0.0	12.3	-36.2	29.0	20.1	13.3	OK	OK
		◢ 4.2 ◣	199	C LC_122	126.7	0.0	53.4	29.5	-59.5	31.3	21.1	OK	OK
SLN225 ring-w 1	Flange top b end plate SLN225 ring	◢ 4.2 ◣	199	C LC_135	253.0	0.0	-24.1	13.2	144.8	62.5	38.7	OK	OK
		◢ 4.2 ◣	198	C LC_135	182.6	0.0	27.8	31.7	99.3	45.1	25.2	OK	OK
SLN225 ring-w 1	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	198	C LC_135	207.5	0.0	34.5	-37.0	-112.2	51.3	28.5	OK	OK
		◢ 4.2 ◣	199	C LC_135	217.0	0.0	-23.0	-8.3	-124.3	53.6	30.8	OK	OK
End plate SLN425 ring	Flange top a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_126	26.9	0.0	9.6	-7.7	12.3	6.6	6.6	OK	OK
		◢ 4.2 ◣	58	C LC_105	36.3	0.0	-20.0	-13.5	-11.1	9.0	8.5	OK	OK
End plate SLN425 ring	Flange top b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	128.5	0.0	-60.9	62.6	-18.5	31.7	19.9	OK	OK
		◢ 4.2 ◣	58	C LC_135	115.4	0.0	-55.5	-53.9	22.7	28.5	17.9	OK	OK
End plate SLN425 ring	Flange bottom a end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	68.3	0.0	11.4	-24.7	30.0	16.9	10.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_114	31.7	0.0	4.1	2.8	-18.0	7.8	6.5	OK	OK
End plate SLN425 ring	Flange bottom b end plate SLN425 ring	◢ 4.2 ◣	58	C LC_135	93.3	0.0	-42.1	41.7	-23.8	23.0	17.6	OK	OK
		◢ 4.2 ◣	58	C LC_135	136.4	0.0	-65.4	-65.9	20.7	33.7	22.0	OK	OK
End plate SLN225 ring	Flange top a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	41.6	0.0	5.6	0.0	23.8	10.3	8.5	OK	OK
		◢ 4.2 ◣	58	C LC_114	23.0	0.0	-0.9	2.9	-12.9	5.7	5.7	OK	OK
End plate SLN225 ring	Flange top b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	225.1	0.0	-109.6	88.1	-71.6	55.6	32.3	OK	OK
		◢ 4.2 ◣	58	C LC_135	182.2	0.0	-58.2	-79.7	59.9	45.0	27.6	OK	OK
End plate SLN225 ring	Flange bottom a end plate SLN225 ring	◢ 4.2 ◣	58	C LC_122	65.8	0.0	13.3	-31.3	20.0	16.3	11.5	OK	OK
		◢ 4.2 ◣	58	C LC_135	56.7	0.0	0.0	-5.1	-32.3	14.0	9.9	OK	OK
End plate SLN225 ring	Flange bottom b end plate SLN225 ring	◢ 4.2 ◣	58	C LC_135	157.8	0.0	-45.9	70.1	-51.8	39.0	24.9	OK	OK
		◢ 4.2 ◣	58	C LC_135	225.3	0.0	-112.0	-87.8	70.9	55.7	33.8	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-S_340
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut

Name	SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

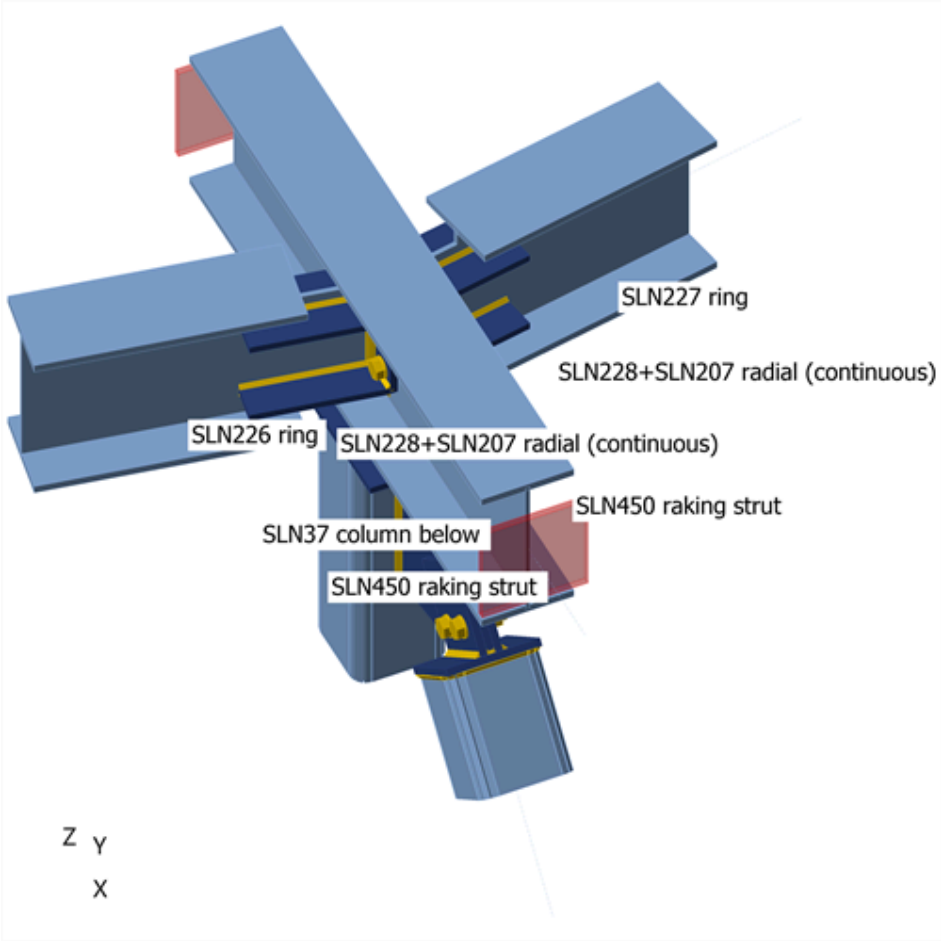
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN228+SLN207 radial (continuous)	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN37 column below	2 - SHS150x150x10	0.0	90.0	160.0	0	0	0
SLN226 ring	1 - UB254x146x43	60.0	0.0	0.0	0	0	0
SLN227 ring	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN450 raking strut	2 - SHS150x150x10	-19.9	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN228+SLN207 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN228+SLN207 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN37 column below / begin		Position	0
SLN226 ring / begin		Position	0
SLN227 ring / end		Position	0
SLN450 raking strut / end		Position	0



SLN228+SLN207 radial (continuous) - SLN37 column below - SLN226 ring - SLN227 ring - SLN450 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN37 column below / Begin	27.8	0.0	0.1	0.0	0.6	0.0
	SLN226 ring / Begin	-6.3	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.3	0.0	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-0.9	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN37 column below / Begin	133.3	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-18.9	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	18.9	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-9.6	0.0	-1.0	0.0	0.0	0.0
C LC_103	SLN37 column below / Begin	107.4	0.0	0.2	0.0	0.7	0.0
	SLN226 ring / Begin	-7.7	-0.3	-24.0	0.0	0.0	0.7
	SLN227 ring / End	7.5	-0.3	-24.1	0.0	0.0	-0.5
	SLN450 raking strut / End	-32.3	0.1	-0.8	0.0	0.0	0.1
C LC_104	SLN37 column below / Begin	144.2	1.3	-3.8	0.0	-2.1	-0.3
	SLN226 ring / Begin	-19.7	0.0	-24.0	0.0	0.0	0.2
	SLN227 ring / End	21.0	0.0	-24.1	0.0	0.0	-0.2
	SLN450 raking strut / End	16.9	0.0	-1.3	-0.1	0.0	0.4
C LC_105	SLN37 column below / Begin	115.8	0.0	0.2	0.0	0.6	0.2
	SLN226 ring / Begin	-7.3	-0.2	-24.0	0.0	0.0	0.4
	SLN227 ring / End	7.7	-0.2	-24.1	0.0	0.0	-0.6
	SLN450 raking strut / End	-21.2	-0.3	-0.9	0.0	0.0	-0.2
C LC_106	SLN37 column below / Begin	128.3	-3.5	-1.3	0.0	-0.4	1.0
	SLN226 ring / Begin	-15.5	-0.1	-24.0	0.0	0.0	0.5
	SLN227 ring / End	11.9	0.0	-24.1	0.0	0.0	-0.2
	SLN450 raking strut / End	-4.6	0.0	-1.1	0.2	0.0	-1.1
C LC_107	SLN37 column below / Begin	-25.4	0.0	0.1	0.0	0.2	-0.1
	SLN226 ring / Begin	10.1	-0.2	-0.5	0.0	0.0	0.4
	SLN227 ring / End	-10.3	0.0	-0.5	0.0	0.0	-0.1
	SLN450 raking strut / End	-38.6	0.2	-0.3	0.0	0.0	0.2
C LC_108	SLN37 column below / Begin	39.5	2.6	-7.9	0.0	-5.5	-0.7
	SLN226 ring / Begin	-10.0	0.1	-0.5	0.0	0.0	-0.2
	SLN227 ring / End	12.5	0.1	-0.5	0.0	0.0	0.3
	SLN450 raking strut / End	48.3	0.0	-1.2	-0.2	0.0	0.8
C LC_109	SLN37 column below / Begin	-11.3	-0.1	0.0	0.0	0.1	0.4
	SLN226 ring / Begin	11.9	0.0	-0.5	0.0	0.0	-0.1
	SLN227 ring / End	-11.6	-0.1	-0.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-20.0	-0.6	-0.6	0.0	0.0	-0.4
C LC_110	SLN37 column below / Begin	10.4	-7.0	-2.9	0.0	-1.9	1.9
	SLN226 ring / Begin	-2.8	-0.1	-0.5	0.0	0.0	0.2
	SLN227 ring / End	-4.2	0.1	-0.5	0.0	0.0	0.4
	SLN450 raking strut / End	8.9	0.0	-0.9	0.4	0.0	-2.2
C LC_111	SLN37 column below / Begin	53.7	0.0	0.1	0.0	0.5	-0.1
	SLN226 ring / Begin	2.3	-0.3	-16.2	0.0	0.0	0.7
	SLN227 ring / End	-2.5	-0.2	-16.3	0.0	0.0	-0.4
	SLN450 raking strut / End	-47.4	0.2	-0.5	0.0	0.0	0.2
C LC_112	SLN37 column below / Begin	121.4	2.6	-7.8	0.0	-5.2	-0.7

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN226 ring / Begin	-19.1	0.1	-16.2	0.0	0.0	0.0
	SLN227 ring / End	21.7	0.1	-16.3	0.0	0.0	0.1
	SLN450 raking strut / End	43.3	0.0	-1.5	-0.2	0.0	0.8
C LC_113	SLN37 column below / Begin	68.7	-0.1	0.1	0.0	0.4	0.4
	SLN226 ring / Begin	3.8	-0.1	-16.2	0.0	0.0	0.2
	SLN227 ring / End	-3.3	-0.2	-16.3	0.0	0.0	-0.6
	SLN450 raking strut / End	-27.6	-0.6	-0.8	0.0	0.0	-0.4
C LC_114	SLN37 column below / Begin	91.5	-7.0	-2.8	0.0	-1.6	1.9
	SLN226 ring / Begin	-11.5	-0.1	-16.2	0.0	0.0	0.5
	SLN227 ring / End	4.4	0.1	-16.3	0.0	0.0	0.1
	SLN450 raking strut / End	2.7	0.0	-1.2	0.4	0.0	-2.1
C LC_115	SLN37 column below / Begin	-18.3	0.0	0.1	0.0	0.4	-0.1
	SLN226 ring / Begin	8.6	-0.2	-1.7	0.0	0.0	0.5
	SLN227 ring / End	-8.7	-0.1	-1.8	0.0	0.0	-0.1
	SLN450 raking strut / End	-39.0	0.2	-0.5	0.0	0.0	0.2
C LC_116	SLN37 column below / Begin	46.8	2.6	-7.9	0.0	-5.3	-0.7
	SLN226 ring / Begin	-11.7	0.1	-1.7	0.0	0.0	-0.2
	SLN227 ring / End	14.2	0.1	-1.8	0.0	0.0	0.3
	SLN450 raking strut / End	48.3	0.0	-1.5	-0.2	0.0	0.8
C LC_117	SLN37 column below / Begin	-4.2	-0.1	0.1	0.0	0.3	0.4
	SLN226 ring / Begin	10.3	0.0	-1.7	0.0	0.0	0.0
	SLN227 ring / End	-10.0	-0.1	-1.8	0.0	0.0	-0.3
	SLN450 raking strut / End	-20.3	-0.6	-0.8	0.0	0.0	-0.4
C LC_118	SLN37 column below / Begin	17.7	-7.0	-2.8	0.0	-1.8	1.9
	SLN226 ring / Begin	-4.4	-0.1	-1.7	0.0	0.0	0.3
	SLN227 ring / End	-2.6	0.1	-1.8	0.0	0.0	0.3
	SLN450 raking strut / End	8.8	0.0	-1.2	0.4	0.0	-2.2
C LC_119	SLN37 column below / Begin	120.5	0.0	0.2	0.0	0.8	0.0
	SLN226 ring / Begin	-10.7	-0.3	-25.6	0.0	0.0	0.7
	SLN227 ring / End	10.5	-0.3	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-26.4	0.1	-0.8	0.0	0.0	0.1
C LC_120	SLN37 column below / Begin	149.5	1.3	-3.8	0.0	-1.9	-0.3
	SLN226 ring / Begin	-22.8	0.0	-25.6	0.0	0.0	0.3
	SLN227 ring / End	24.2	0.0	-25.7	0.0	0.0	-0.3
	SLN450 raking strut / End	12.2	0.0	-1.3	-0.1	0.0	0.4
C LC_121	SLN37 column below / Begin	125.3	0.0	0.2	0.0	0.8	0.2
	SLN226 ring / Begin	-10.3	-0.2	-25.6	0.0	0.0	0.4
	SLN227 ring / End	10.7	-0.2	-25.7	0.0	0.0	-0.6
	SLN450 raking strut / End	-20.2	-0.3	-0.9	0.0	0.0	-0.2
C LC_122	SLN37 column below / Begin	135.2	-3.5	-1.3	0.0	-0.2	1.0
	SLN226 ring / Begin	-18.6	-0.2	-25.6	0.0	0.0	0.6
	SLN227 ring / End	15.0	-0.1	-25.7	0.0	0.0	-0.2
	SLN450 raking strut / End	-7.1	0.0	-1.1	0.2	0.0	-1.1
C LC_123	SLN37 column below / Begin	80.0	0.0	0.2	0.0	0.7	-0.1
	SLN226 ring / Begin	-3.7	-0.3	-19.4	0.0	0.0	0.7
	SLN227 ring / End	3.5	-0.2	-19.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-35.5	0.2	-0.5	0.0	0.0	0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN37 column below / Begin	132.0	2.6	-7.7	0.0	-4.8	-0.7
	SLN226 ring / Begin	-25.4	0.0	-19.4	0.0	0.0	0.2
	SLN227 ring / End	28.0	0.0	-19.5	0.0	0.0	-0.2
	SLN450 raking strut / End	33.8	0.0	-1.5	-0.2	0.0	0.8
C LC_125	SLN37 column below / Begin	87.6	-0.1	0.2	0.0	0.7	0.4
	SLN226 ring / Begin	-2.3	-0.1	-19.4	0.0	0.0	0.2
	SLN227 ring / End	2.8	-0.2	-19.5	0.0	0.0	-0.7
	SLN450 raking strut / End	-25.6	-0.6	-0.8	0.0	0.0	-0.4
C LC_126	SLN37 column below / Begin	105.3	-7.1	-2.7	0.0	-1.3	2.0
	SLN226 ring / Begin	-17.7	-0.2	-19.4	0.0	0.0	0.7
	SLN227 ring / End	10.6	0.0	-19.5	0.0	0.0	0.0
	SLN450 raking strut / End	-2.3	0.0	-1.2	0.4	0.0	-2.2
C LC_127	SLN37 column below / Begin	7.9	0.0	0.2	0.0	0.6	-0.1
	SLN226 ring / Begin	2.6	-0.2	-4.9	0.0	0.0	0.4
	SLN227 ring / End	-2.7	0.0	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-27.1	0.2	-0.5	0.0	0.0	0.2
C LC_128	SLN37 column below / Begin	57.4	2.6	-7.8	0.0	-4.9	-0.7
	SLN226 ring / Begin	-18.0	0.0	-4.9	0.0	0.0	0.0
	SLN227 ring / End	20.5	0.0	-4.9	0.0	0.0	0.0
	SLN450 raking strut / End	38.8	0.0	-1.5	-0.2	0.0	0.8
C LC_129	SLN37 column below / Begin	14.7	-0.1	0.1	0.0	0.5	0.3
	SLN226 ring / Begin	4.2	0.0	-4.9	0.0	0.0	0.0
	SLN227 ring / End	-3.9	-0.1	-4.9	0.0	0.0	-0.4
	SLN450 raking strut / End	-18.3	-0.6	-0.8	0.0	0.0	-0.4
C LC_130	SLN37 column below / Begin	31.5	-7.1	-2.8	0.0	-1.5	2.0
	SLN226 ring / Begin	-10.6	-0.1	-4.9	0.0	0.0	0.4
	SLN227 ring / End	3.6	0.1	-4.9	0.0	0.0	0.2
	SLN450 raking strut / End	3.8	0.0	-1.2	0.4	0.0	-2.2
C LC_135	SLN37 column below / Begin	132.6	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-16.2	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	16.2	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-10.8	0.0	-1.0	0.0	0.0	0.0
C LC_141	SLN37 column below / Begin	27.2	0.0	0.2	0.0	0.6	0.0
	SLN226 ring / Begin	-6.1	-0.1	-4.9	0.0	0.0	0.2
	SLN227 ring / End	6.1	-0.1	-4.9	0.0	0.0	-0.2
	SLN450 raking strut / End	-1.7	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN37 column below / Begin	28.4	0.0	0.1	0.0	0.6	0.0
	SLN226 ring / Begin	-6.5	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.5	0.0	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-0.1	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN37 column below / Begin	27.6	0.0	0.2	0.0	0.6	0.0
	SLN226 ring / Begin	-6.2	-0.1	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.2	-0.1	-4.9	0.0	0.0	-0.1
	SLN450 raking strut / End	-1.2	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN37 column below / Begin	28.0	0.0	0.1	0.0	0.6	0.0
	SLN226 ring / Begin	-6.4	0.0	-4.9	0.0	0.0	0.1
	SLN227 ring / End	6.4	0.0	-4.9	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN450 raking strut / End	-0.6	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN37 column below / Begin	130.9	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-17.9	-0.3	-25.6	0.0	0.0	0.6
	SLN227 ring / End	17.8	-0.3	-25.7	0.0	0.0	-0.6
	SLN450 raking strut / End	-12.8	0.0	-1.0	0.0	0.0	0.0
C LC_146	SLN37 column below / Begin	135.7	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-20.0	-0.1	-25.6	0.0	0.0	0.4
	SLN227 ring / End	20.1	-0.1	-25.7	0.0	0.0	-0.4
	SLN450 raking strut / End	-6.4	0.0	-1.0	0.0	0.0	0.0
C LC_147	SLN37 column below / Begin	132.5	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-18.4	-0.2	-25.6	0.0	0.0	0.5
	SLN227 ring / End	18.7	-0.2	-25.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-10.8	0.0	-1.0	0.0	0.0	0.0
C LC_148	SLN37 column below / Begin	134.2	0.0	0.2	0.0	0.9	0.0
	SLN226 ring / Begin	-19.4	-0.1	-25.6	0.0	0.0	0.5
	SLN227 ring / End	19.2	-0.1	-25.7	0.0	0.0	-0.4
	SLN450 raking strut / End	-8.5	0.0	-1.0	0.0	0.0	0.0
C LC_151	SLN37 column below / Begin	79.0	0.0	-2.0	0.0	-8.0	0.0
	SLN226 ring / Begin	-6.3	-0.1	-11.8	0.0	0.0	0.3
	SLN227 ring / End	6.4	-0.1	-11.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-11.7	0.0	-1.0	0.0	0.0	0.0
C LC_152	SLN37 column below / Begin	81.1	0.0	2.1	0.0	8.4	0.0
	SLN226 ring / Begin	-20.7	-0.1	-14.1	0.0	0.0	0.4
	SLN227 ring / End	20.7	-0.1	-14.2	0.0	0.0	-0.4
	SLN450 raking strut / End	2.5	0.0	-1.0	0.0	0.0	0.0
C LC_153	SLN37 column below / Begin	133.0	0.0	0.2	0.0	0.7	0.0
	SLN226 ring / Begin	-19.0	-0.2	-21.0	0.0	0.0	0.5
	SLN227 ring / End	19.0	-0.2	-20.7	0.0	0.0	-0.5
	SLN450 raking strut / End	-9.3	0.0	-1.0	0.0	0.0	0.0
C LC_154	SLN37 column below / Begin	78.6	0.0	-1.9	0.0	-7.9	0.0
	SLN226 ring / Begin	-6.5	-0.1	-11.8	0.0	0.0	0.3
	SLN227 ring / End	6.5	-0.1	-11.5	0.0	0.0	-0.3
	SLN450 raking strut / End	-11.5	0.0	-1.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.1 < 5.0%	OK
Loc. deformation	0.1 < 3%	OK
Bolts	26.9 < 100%	OK
Welds	98.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN228+SLN207 radial (continuous)-bfl 1	12.7	C LC_108	222.0	0.0	35.6	OK
SLN228+SLN207 radial (continuous)-tfl 1	12.7	C LC_153	70.1	0.0	0.0	OK
SLN228+SLN207 radial (continuous)-w 1	7.2	C LC_126	191.7	0.0	24.1	OK
SLN37 column below	10.0	C LC_108	275.2	0.1	0.0	OK
SLN226 ring-bfl 1	12.7	C LC_119	48.4	0.0	0.0	OK
SLN226 ring-tfl 1	12.7	C LC_119	45.4	0.0	0.0	OK
SLN226 ring-w 1	7.2	C LC_119	146.7	0.0	0.0	OK
SLN227 ring-bfl 1	12.7	C LC_121	45.5	0.0	0.0	OK
SLN227 ring-tfl 1	12.7	C LC_121	43.1	0.0	0.0	OK
SLN227 ring-w 1	7.2	C LC_121	138.1	0.0	0.0	OK
SLN450 raking strut	10.0	C LC_130	54.9	0.0	0.0	OK
End plate column below SLN37 column below	12.0	C LC_108	257.3	0.0	35.6	OK
End plate SLN226 ring	12.0	C LC_111	72.3	0.0	0.0	OK
End plate SLN227 ring	12.0	C LC_145	51.1	0.0	11.5	OK
Cap plate SLN450 raking strut	12.0	C LC_126	124.0	0.0	0.0	OK
Fin plate SLN450 raking strut	12.0	C LC_130	275.1	0.0	83.7	OK
Fin plate 2 SLN450 raking strut	12.0	C LC_118	275.1	0.1	130.3	OK
Gusset SLN450 raking strut	12.0	C LC_126	275.2	0.1	64.5	OK
Flange top a end plate SLN226 ring	12.0	C LC_111	64.3	0.0	0.0	OK
Flange top b end plate SLN226 ring	12.0	C LC_111	51.2	0.0	0.0	OK
Flange bottom a end plate SLN226 ring	12.0	C LC_111	48.1	0.0	0.0	OK
Flange bottom b end plate SLN226 ring	12.0	C LC_119	66.2	0.0	0.0	OK
Flange top a end plate SLN227 ring	12.0	C LC_125	40.6	0.0	0.0	OK
Flange top b end plate SLN227 ring	12.0	C LC_125	57.4	0.0	0.0	OK
Flange bottom a end plate SLN227 ring	12.0	C LC_121	62.1	0.0	0.0	OK
Flange bottom b end plate SLN227 ring	12.0	C LC_125	43.0	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

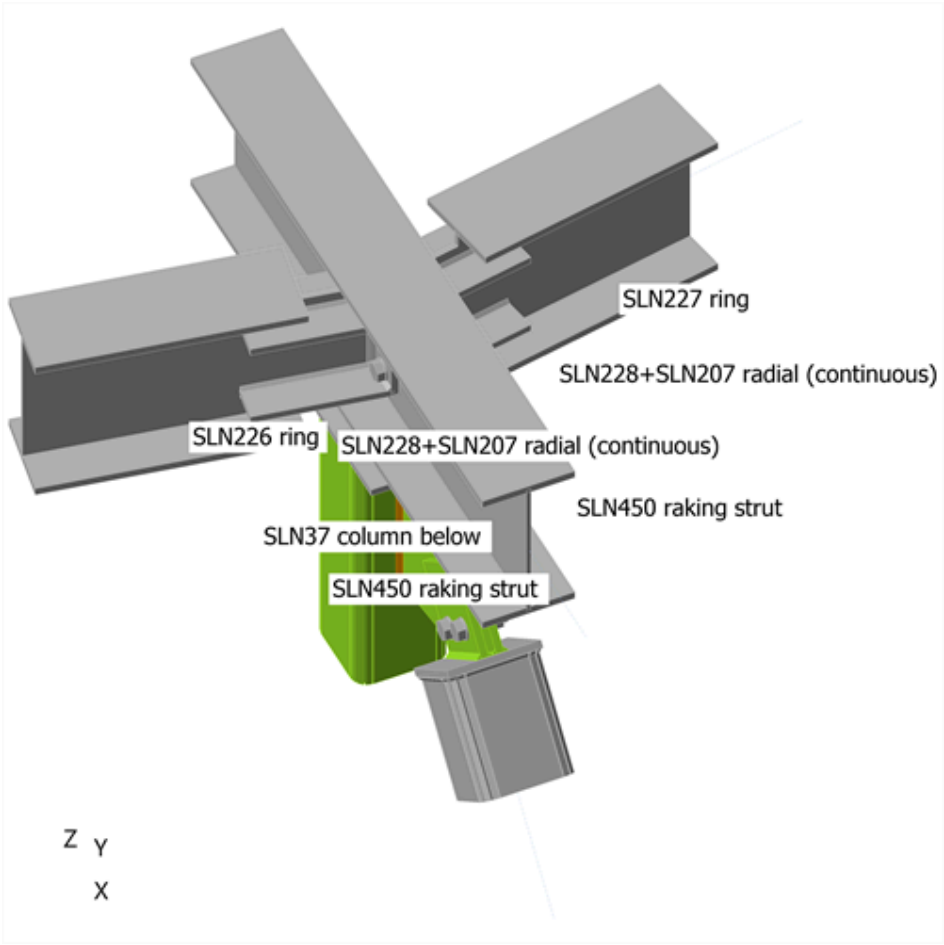
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

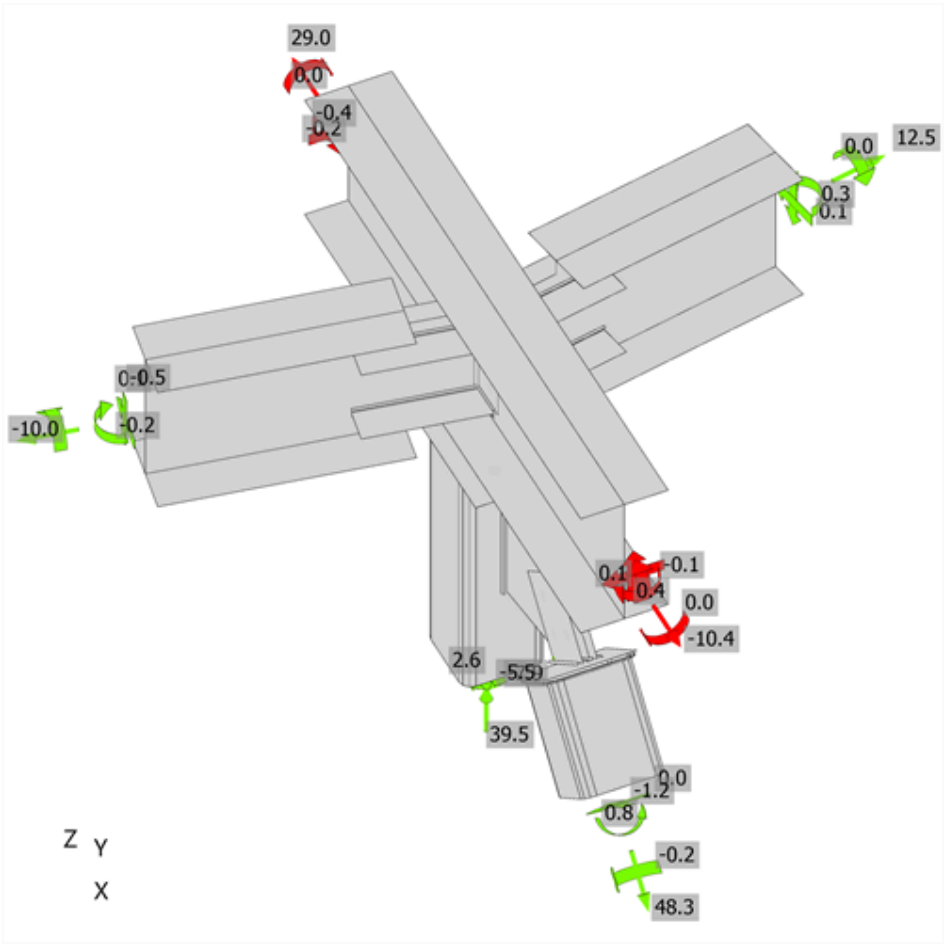
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN37 column below	150	C LC_108	0	5	0.1	OK
SLN450 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

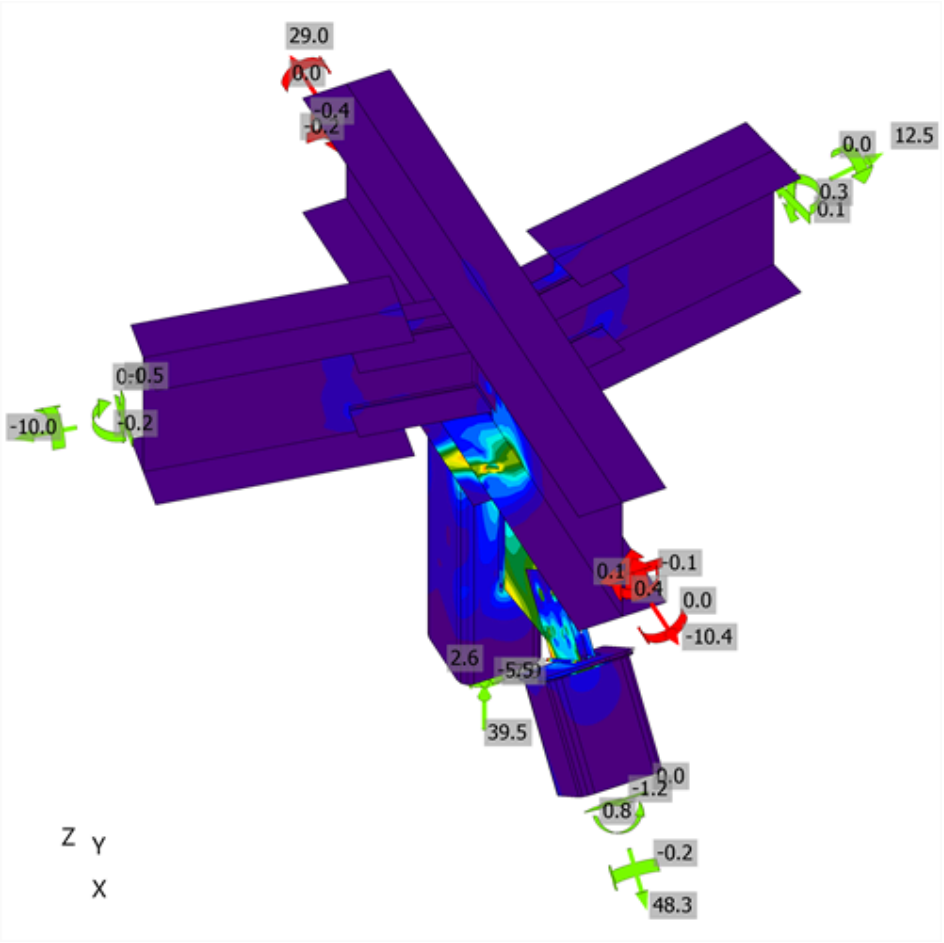
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

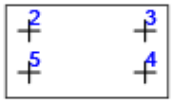
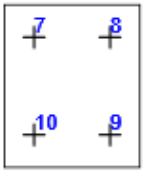
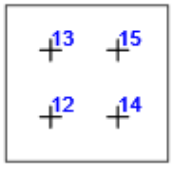


Strain check, C LC_108



Equivalent stress, C LC_108

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_107	15.9	5.9	118.9	11.3	6.3	14.4	OK	OK
	B3	M20 8.8 - 1	C LC_108	38.0	5.6	111.0	26.9	5.9	25.1	OK	OK
	B4	M20 8.8 - 1	C LC_108	37.1	5.2	89.7	26.3	5.8	24.3	OK	OK
	B5	M20 8.8 - 1	C LC_107	17.0	6.1	132.6	12.0	6.5	15.1	OK	OK
	B7	M20 8.8 - 1	C LC_111	10.6	0.8	70.2	7.5	0.9	6.2	OK	OK
	B8	M20 8.8 - 1	C LC_108	3.0	0.6	123.8	2.1	0.9	2.1	OK	OK
	B9	M20 8.8 - 1	C LC_108	3.1	0.6	123.8	2.2	0.9	2.2	OK	OK
	B10	M20 8.8 - 1	C LC_111	11.1	0.9	123.8	7.8	1.0	6.6	OK	OK
	B12	M20 8.8 - 1	C LC_118	17.1	13.8	106.0	12.1	14.7	23.4	OK	OK
	B13	M20 8.8 - 1	C LC_126	7.9	11.1	106.0	5.6	11.8	15.8	OK	OK
	B14	M20 8.8 - 1	C LC_116	1.5	9.9	106.0	1.1	14.2	11.3	OK	OK
	B15	M20 8.8 - 1	C LC_111	0.1	6.4	115.3	0.1	10.9	6.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN450 raking strut	SLN450 raking strut	◢ 4.2	519	C LC_116	191.2	0.0	90.5	-97.2	2.5	47.2	24.9	OK	OK
SLN37 column below-w 3	Gusset SLN450 raking strut	◢ 5.6 ◣	288	C LC_108	397.1	0.3	129.7	137.2	167.7	98.1	63.9	OK	OK
		◢ 5.6 ◣	288	C LC_111	397.1	0.3	-132.0	133.3	170.2	98.1	61.8	OK	OK
SLN226 ring-w 1	Flange top b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_126	59.2	0.0	32.6	26.8	10.0	14.6	9.4	OK	OK
		◢ 4.2 ◣	197	C LC_111	32.3	0.0	-15.1	13.8	9.0	8.0	7.0	OK	OK
SLN226 ring-w 1	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	198	C LC_122	82.5	0.0	24.4	28.5	35.5	20.4	12.4	OK	OK
		◢ 4.2 ◣	198	C LC_119	118.5	0.0	48.7	-44.2	-44.0	29.3	17.8	OK	OK
SLN227 ring-w 1	Flange top a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_110	43.9	0.0	-20.2	-17.7	13.9	10.8	6.7	OK	OK
		◢ 4.2 ◣	198	C LC_114	31.7	0.0	-8.8	9.5	-14.8	7.8	5.9	OK	OK
SLN227 ring-w 1	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	198	C LC_147	81.0	0.0	24.1	27.1	-35.4	20.0	10.9	OK	OK
		◢ 4.2 ◣	198	C LC_145	113.4	0.0	44.2	-41.1	44.1	28.0	15.2	OK	OK
Cap plate SLN450 raking strut	Fin plate SLN450 raking strut	◢ 5.6 ◣	139	C LC_116	252.7	0.0	97.2	110.6	76.8	62.4	40.8	OK	OK
		◢ 5.6 ◣	139	C LC_111	123.0	0.0	-47.3	52.1	-39.8	30.4	18.1	OK	OK
Cap plate SLN450 raking strut	Fin plate 2 SLN450 raking strut	◢ 5.6 ◣	139	C LC_116	164.9	0.0	84.6	67.0	46.8	40.8	25.6	OK	OK
		◢ 5.6 ◣	139	C LC_118	284.6	0.0	89.4	-123.0	-95.9	70.3	51.4	OK	OK
End plate column below SLN37 column below	SLN37 column below	◢ 5.6 ◣	521	C LC_111	250.2	0.0	-167.8	-103.7	-26.8	61.8	33.9	OK	OK
		◢ 5.6 ◣	521	C LC_108	220.0	0.0	151.1	-73.2	-56.3	54.4	33.9	OK	OK
SLN228+SLN207 radial (continuous)-w 1	SLN226 ring-w 1	◢ 7.2	189	C LC_120	85.3	0.0	27.7	-25.7	-38.9	21.1	13.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
SLN228+SLN207 radial (continuous)-w 1	SLN227 ring-w 1	◢ 7.2	189	C LC_119	85.0	0.0	0.0	8.6	48.3	21.0	12.6	OK	OK
End plate SLN226 ring	SLN226 ring-w 1	◢ 4.2 ◣	149	C LC_120	82.0	0.0	-22.6	-21.2	-40.3	20.3	11.3	OK	OK
		◢ 4.2 ◣	149	C LC_122	65.5	0.0	-11.5	26.1	26.6	16.2	10.8	OK	OK
End plate SLN227 ring	SLN227 ring-w 1	◢ 4.2 ◣	149	C LC_120	82.1	0.0	-21.4	-24.2	38.8	20.3	12.6	OK	OK
		◢ 4.2 ◣	149	C LC_145	64.6	0.0	-16.7	19.8	-30.1	16.0	10.4	OK	OK
SLN226 ring-w 1	Flange top a end plate SLN226 ring	◢ 4.2 ◣	199	C LC_111	109.9	0.0	-9.9	6.5	-62.8	27.1	19.4	OK	OK
		◢ 4.2 ◣	198	C LC_111	69.1	0.0	10.4	11.8	-37.7	17.1	10.4	OK	OK
SLN226 ring-w 1	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	198	C LC_119	91.1	0.0	16.5	-16.8	48.9	22.5	13.5	OK	OK
		◢ 4.2 ◣	198	C LC_111	87.4	0.0	-13.1	-5.7	49.5	21.6	13.6	OK	OK
SLN227 ring-w 1	Flange top b end plate SLN227 ring	◢ 4.2 ◣	199	C LC_125	97.7	0.0	-48.7	22.2	43.6	24.2	15.3	OK	OK
		◢ 4.2 ◣	198	C LC_125	57.1	0.0	10.3	11.2	30.4	14.1	8.5	OK	OK
SLN227 ring-w 1	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	198	C LC_121	84.4	0.0	14.7	-15.0	-45.6	20.9	12.1	OK	OK
		◢ 4.2 ◣	198	C LC_125	72.8	0.0	-9.1	-2.8	-41.6	18.0	10.1	OK	OK
End plate SLN226 ring	Flange top a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_111	36.9	0.0	-1.2	3.9	-21.0	9.1	7.2	OK	OK
		◢ 4.2 ◣	58	C LC_103	22.2	0.0	-7.2	-11.9	-2.3	5.5	5.5	OK	OK
End plate SLN226 ring	Flange top b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_126	47.3	0.0	-0.9	-1.1	-27.3	11.7	9.0	OK	OK
		◢ 4.2 ◣	58	C LC_111	61.6	0.0	5.1	9.4	-34.2	15.2	12.5	OK	OK
End plate SLN226 ring	Flange bottom a end plate SLN226 ring	◢ 4.2 ◣	58	C LC_120	51.5	0.0	11.1	-20.3	20.8	12.7	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	45.4	0.0	-12.2	-17.4	18.3	11.2	8.4	OK	OK
End plate SLN226 ring	Flange bottom b end plate SLN226 ring	◢ 4.2 ◣	58	C LC_123	68.7	0.0	39.0	-29.1	14.9	17.0	12.7	OK	OK
		◢ 4.2 ◣	58	C LC_126	54.4	0.0	-3.2	0.1	31.3	13.4	9.5	OK	OK
End plate SLN227 ring	Flange top a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_130	67.3	0.0	-34.4	26.9	19.8	16.6	9.3	OK	OK
		◢ 4.2 ◣	58	C LC_111	48.2	0.0	3.3	5.8	27.1	11.9	8.9	OK	OK
End plate SLN227 ring	Flange top b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_125	50.2	0.0	-24.7	19.6	-15.9	12.4	8.2	OK	OK
		◢ 4.2 ◣	58	C LC_125	46.3	0.0	-14.7	-19.8	15.8	11.4	8.8	OK	OK
End plate SLN227 ring	Flange bottom a end plate SLN227 ring	◢ 4.2 ◣	58	C LC_145	51.6	0.0	28.5	-23.0	-9.4	12.8	10.6	OK	OK
		◢ 4.2 ◣	58	C LC_126	71.0	0.0	-34.9	-31.8	-16.2	17.5	10.8	OK	OK
End plate SLN227 ring	Flange bottom b end plate SLN227 ring	◢ 4.2 ◣	58	C LC_120	52.8	0.0	11.2	-20.8	-21.3	13.1	7.4	OK	OK
		◢ 4.2 ◣	58	C LC_125	46.0	0.0	-22.8	-18.6	13.6	11.4	9.2	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-T_060
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut

Name	SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

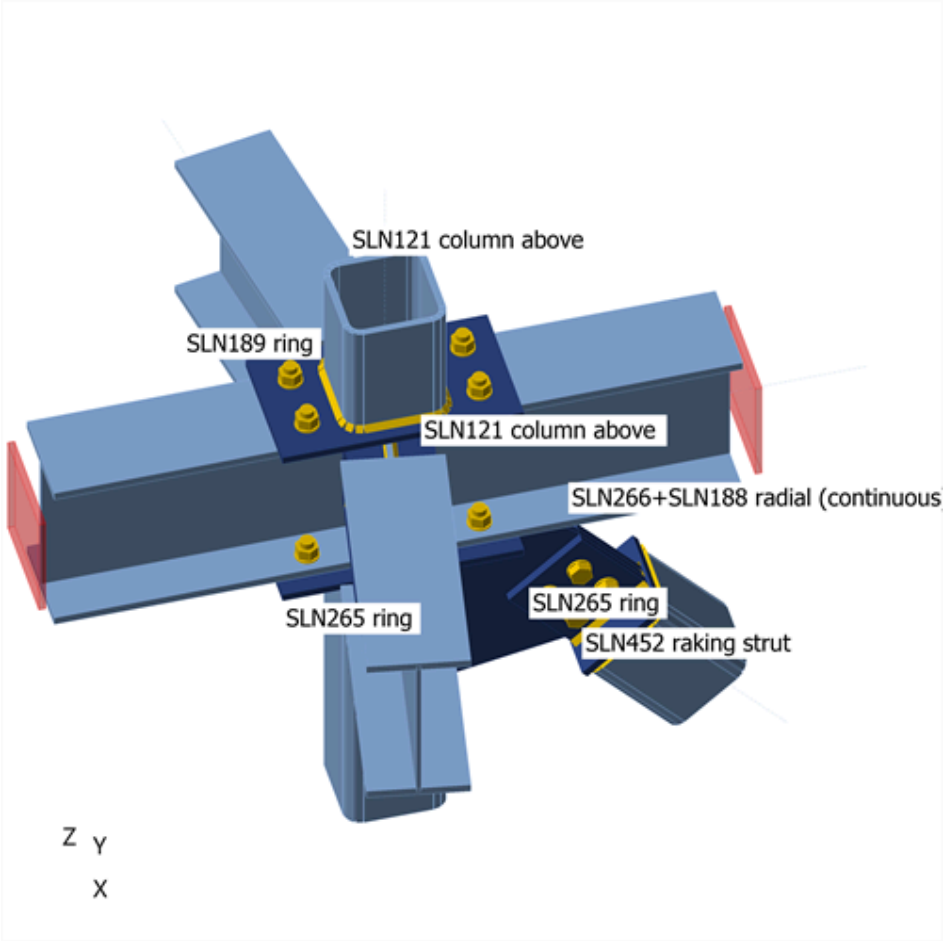
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN266+SLN188 radial (continuous)	1 - UB254x146x43	-120.0	0.0	0.0	0	0	0
SLN41 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN121 column above	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN265 ring	1 - UB254x146x43	140.0	0.0	0.0	0	0	0
SLN189 ring	1 - UB254x146x43	160.0	0.0	0.0	0	0	0
SLN452 raking strut	2 - SHS150x150x10	60.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN266+SLN188 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN266+SLN188 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN41 column below / begin		Position	0
SLN121 column above / end		Position	0
SLN265 ring / begin		Position	0
SLN189 ring / end		Position	0
SLN452 raking strut / end		Position	0



SLN266+SLN188 radial (continuous) - SLN41 column below - SLN121 column above - SLN265 ring - SLN189 ring - SLN452 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN41 column below / Begin	60.1	0.0	0.2	0.0	0.6	0.0
	SLN121 column above / End	-39.7	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.8	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.8	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.7	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN41 column below / Begin	254.3	0.0	0.3	0.0	1.4	-0.2
	SLN121 column above / End	-156.4	0.1	0.1	0.5	0.2	0.3
	SLN265 ring / Begin	-27.8	-0.3	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.0	-0.1	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-56.7	-0.1	-1.0	0.2	0.0	-0.1
C LC_103	SLN41 column below / Begin	229.8	0.0	0.3	0.0	1.2	-0.2
	SLN121 column above / End	-148.5	0.2	0.2	0.6	0.0	0.5
	SLN265 ring / Begin	-16.6	-0.4	-24.4	0.0	0.0	1.0
	SLN189 ring / End	17.4	-0.1	-24.2	0.0	0.0	-0.3
	SLN452 raking strut / End	-66.9	-0.4	-0.9	0.2	0.0	-0.4
C LC_104	SLN41 column below / Begin	243.6	-3.4	-1.8	0.0	-0.3	1.5
	SLN121 column above / End	-146.5	-0.2	0.0	0.4	0.5	-0.9
	SLN265 ring / Begin	-25.3	-0.2	-24.4	0.0	0.0	0.8
	SLN189 ring / End	21.3	0.0	-24.2	0.0	0.0	-0.1
	SLN452 raking strut / End	-45.8	0.0	-1.1	0.3	0.0	-0.6
C LC_105	SLN41 column below / Begin	253.2	1.9	-3.4	0.0	-1.4	-0.8
	SLN121 column above / End	-144.5	0.2	-0.1	0.6	0.7	0.5
	SLN265 ring / Begin	-27.8	-0.1	-24.4	0.0	0.0	0.4
	SLN189 ring / End	30.1	0.0	-24.2	0.0	0.0	-0.2
	SLN452 raking strut / End	-30.5	-0.1	-1.2	0.1	0.0	0.4
C LC_106	SLN41 column below / Begin	224.8	0.1	0.3	0.0	1.2	-0.4
	SLN121 column above / End	-150.2	0.2	0.2	0.5	-0.1	0.5
	SLN265 ring / Begin	-14.7	-0.4	-24.4	0.0	0.0	0.9
	SLN189 ring / End	14.9	-0.2	-24.2	0.0	0.0	-0.5
	SLN452 raking strut / End	-76.0	0.1	-0.8	0.2	0.0	-0.1
C LC_107	SLN41 column below / Begin	1.3	0.0	0.0	0.0	0.2	0.0
	SLN121 column above / End	-15.0	0.1	0.0	0.1	0.1	0.5
	SLN265 ring / Begin	8.5	-0.2	-0.5	0.0	0.0	0.4
	SLN189 ring / End	-7.8	0.0	-0.5	0.0	0.0	0.1
	SLN452 raking strut / End	-22.9	-0.7	-0.5	0.0	0.0	-0.5
C LC_108	SLN41 column below / Begin	25.6	-6.9	-4.2	0.0	-2.8	3.3
	SLN121 column above / End	-11.4	-0.6	-0.4	-0.2	1.0	-2.3
	SLN265 ring / Begin	-6.3	0.0	-0.5	0.0	0.0	0.1
	SLN189 ring / End	-1.6	0.1	-0.5	0.0	0.0	0.3
	SLN452 raking strut / End	14.3	0.2	-1.0	0.3	0.0	-1.1
C LC_109	SLN41 column below / Begin	43.2	3.8	-7.3	0.0	-5.1	-1.4
	SLN121 column above / End	-7.7	0.1	-0.4	0.2	1.3	0.5
	SLN265 ring / Begin	-10.8	0.2	-0.5	0.0	0.0	-0.5
	SLN189 ring / End	14.7	0.0	-0.5	0.0	0.0	0.0
	SLN452 raking strut / End	42.6	0.0	-1.2	-0.2	0.0	1.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_110	SLN41 column below / Begin	-7.2	0.1	0.1	0.0	0.2	-0.5
	SLN121 column above / End	-18.3	0.1	0.0	0.1	0.0	0.5
	SLN265 ring / Begin	11.7	-0.1	-0.5	0.0	0.0	0.3
	SLN189 ring / End	-11.6	-0.1	-0.5	0.0	0.0	-0.2
	SLN452 raking strut / End	-38.6	0.3	-0.3	0.1	0.0	0.0
C LC_111	SLN41 column below / Begin	150.8	0.0	0.2	0.0	0.9	-0.1
	SLN121 column above / End	-106.5	0.2	0.1	0.5	0.0	0.6
	SLN265 ring / Begin	-4.5	-0.4	-16.5	0.0	0.0	0.9
	SLN189 ring / End	5.5	-0.1	-16.3	0.0	0.0	-0.1
	SLN452 raking strut / End	-59.7	-0.7	-0.8	0.1	0.0	-0.6
C LC_112	SLN41 column below / Begin	176.3	-6.8	-4.0	0.0	-2.2	3.2
	SLN121 column above / End	-102.7	-0.5	-0.2	0.2	1.0	-2.1
	SLN265 ring / Begin	-20.2	-0.1	-16.5	0.0	0.0	0.6
	SLN189 ring / End	12.3	0.1	-16.3	0.0	0.0	0.2
	SLN452 raking strut / End	-20.9	0.2	-1.3	0.4	0.0	-1.1
C LC_113	SLN41 column below / Begin	194.3	3.9	-7.2	0.0	-4.4	-1.5
	SLN121 column above / End	-98.9	0.2	-0.3	0.5	1.3	0.7
	SLN265 ring / Begin	-24.9	0.1	-16.5	0.0	0.0	-0.1
	SLN189 ring / End	29.0	0.0	-16.3	0.0	0.0	-0.1
	SLN452 raking strut / End	8.2	-0.1	-1.5	0.0	0.0	0.9
C LC_114	SLN41 column below / Begin	141.8	0.1	0.2	0.0	0.9	-0.6
	SLN121 column above / End	-109.9	0.2	0.1	0.4	-0.1	0.7
	SLN265 ring / Begin	-1.2	-0.3	-16.5	0.0	0.0	0.8
	SLN189 ring / End	1.3	-0.2	-16.3	0.0	0.0	-0.4
	SLN452 raking strut / End	-76.2	0.2	-0.6	0.2	0.0	-0.1
C LC_115	SLN41 column below / Begin	16.8	0.0	0.1	0.0	0.4	0.0
	SLN121 column above / End	-25.3	0.1	-0.1	0.2	0.2	0.5
	SLN265 ring / Begin	6.3	-0.2	-1.8	0.0	0.0	0.5
	SLN189 ring / End	-5.5	0.0	-1.8	0.0	0.0	0.1
	SLN452 raking strut / End	-25.8	-0.7	-0.8	0.0	0.0	-0.6
C LC_116	SLN41 column below / Begin	41.3	-6.9	-4.2	0.0	-2.7	3.3
	SLN121 column above / End	-21.7	-0.6	-0.4	-0.2	1.2	-2.3
	SLN265 ring / Begin	-8.6	0.0	-1.8	0.0	0.0	0.2
	SLN189 ring / End	0.7	0.1	-1.8	0.0	0.0	0.3
	SLN452 raking strut / End	11.7	0.2	-1.3	0.3	0.0	-1.1
C LC_117	SLN41 column below / Begin	58.9	3.8	-7.3	0.0	-4.9	-1.4
	SLN121 column above / End	-17.9	0.1	-0.5	0.2	1.4	0.5
	SLN265 ring / Begin	-13.1	0.2	-1.8	0.0	0.0	-0.4
	SLN189 ring / End	17.1	0.0	-1.8	0.0	0.0	0.0
	SLN452 raking strut / End	40.0	0.0	-1.5	-0.2	0.0	1.0
C LC_118	SLN41 column below / Begin	8.3	0.1	0.1	0.0	0.4	-0.5
	SLN121 column above / End	-28.6	0.1	-0.1	0.1	0.1	0.5
	SLN265 ring / Begin	9.4	-0.1	-1.8	0.0	0.0	0.3
	SLN189 ring / End	-9.3	-0.1	-1.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-41.6	0.3	-0.6	0.1	0.0	0.0
C LC_119	SLN41 column below / Begin	246.7	0.1	0.3	0.0	1.3	-0.2
	SLN121 column above / End	-156.4	0.2	0.3	0.6	-0.1	0.6

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN265 ring / Begin	-19.9	-0.4	-26.0	0.0	0.0	1.0
	SLN189 ring / End	20.8	-0.1	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-66.4	-0.4	-0.9	0.2	0.0	-0.4
C LC_120	SLN41 column below / Begin	256.8	-3.4	-1.8	0.0	-0.2	1.5
	SLN121 column above / End	-154.7	-0.2	-0.1	0.5	0.7	-1.0
	SLN265 ring / Begin	-28.9	-0.2	-26.0	0.0	0.0	0.8
	SLN189 ring / End	24.8	0.0	-25.8	0.0	0.0	-0.2
	SLN452 raking strut / End	-50.9	0.1	-1.1	0.3	0.0	-0.6
C LC_121	SLN41 column below / Begin	265.4	2.0	-3.4	0.0	-1.3	-0.8
	SLN121 column above / End	-152.8	0.2	-0.2	0.6	1.0	0.5
	SLN265 ring / Begin	-31.5	-0.1	-26.0	0.0	0.0	0.5
	SLN189 ring / End	33.8	0.0	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-37.1	-0.1	-1.2	0.1	0.0	0.4
C LC_122	SLN41 column below / Begin	243.6	0.1	0.3	0.0	1.3	-0.4
	SLN121 column above / End	-158.1	0.2	0.3	0.5	-0.2	0.6
	SLN265 ring / Begin	-18.0	-0.4	-26.0	0.0	0.0	0.9
	SLN189 ring / End	18.2	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-72.8	0.1	-0.8	0.2	0.0	-0.1
C LC_123	SLN41 column below / Begin	184.7	0.1	0.3	0.0	1.1	-0.3
	SLN121 column above / End	-122.3	0.3	0.3	0.5	-0.2	0.8
	SLN265 ring / Begin	-11.2	-0.4	-19.7	0.0	0.0	1.0
	SLN189 ring / End	12.3	-0.1	-19.5	0.0	0.0	-0.1
	SLN452 raking strut / End	-58.8	-0.7	-0.8	0.1	0.0	-0.7
C LC_124	SLN41 column below / Begin	202.6	-6.9	-4.0	0.0	-1.9	3.3
	SLN121 column above / End	-119.2	-0.5	-0.4	0.2	1.2	-2.2
	SLN265 ring / Begin	-27.3	-0.2	-19.7	0.0	0.0	0.7
	SLN189 ring / End	19.3	0.0	-19.5	0.0	0.0	0.0
	SLN452 raking strut / End	-31.1	0.2	-1.3	0.4	0.0	-1.1
C LC_125	SLN41 column below / Begin	218.8	3.9	-7.1	0.0	-4.2	-1.5
	SLN121 column above / End	-115.5	0.2	-0.6	0.5	1.8	0.7
	SLN265 ring / Begin	-32.3	0.0	-19.7	0.0	0.0	0.2
	SLN189 ring / End	36.4	-0.1	-19.5	0.0	0.0	-0.4
	SLN452 raking strut / End	-4.9	-0.1	-1.5	-0.1	0.0	0.9
C LC_126	SLN41 column below / Begin	179.5	0.1	0.3	0.0	1.1	-0.6
	SLN121 column above / End	-125.7	0.2	0.3	0.4	-0.4	0.7
	SLN265 ring / Begin	-7.7	-0.3	-19.7	0.0	0.0	0.8
	SLN189 ring / End	7.9	-0.2	-19.5	0.0	0.0	-0.4
	SLN452 raking strut / End	-69.9	0.2	-0.6	0.2	0.0	-0.1
C LC_127	SLN41 column below / Begin	50.7	0.1	0.1	0.0	0.6	-0.2
	SLN121 column above / End	-41.2	0.2	0.1	0.1	0.0	0.7
	SLN265 ring / Begin	-0.4	-0.2	-5.0	0.0	0.0	0.6
	SLN189 ring / End	1.3	0.0	-4.9	0.0	0.0	0.1
	SLN452 raking strut / End	-24.8	-0.7	-0.8	0.0	0.0	-0.6
C LC_128	SLN41 column below / Begin	67.6	-6.9	-4.1	0.0	-2.4	3.4
	SLN121 column above / End	-38.2	-0.6	-0.5	-0.1	1.4	-2.4
	SLN265 ring / Begin	-15.8	-0.1	-5.0	0.0	0.0	0.3
	SLN189 ring / End	7.8	0.0	-4.9	0.0	0.0	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN452 raking strut / End	1.4	0.2	-1.3	0.3	0.0	-1.0
C LC_129	SLN41 column below / Begin	83.3	3.8	-7.2	0.0	-4.7	-1.4
	SLN121 column above / End	-34.6	0.2	-0.8	0.2	1.9	0.6
	SLN265 ring / Begin	-20.6	0.1	-5.0	0.0	0.0	-0.2
	SLN189 ring / End	24.5	-0.1	-4.9	0.0	0.0	-0.3
	SLN452 raking strut / End	26.9	0.0	-1.5	-0.2	0.0	1.0
C LC_130	SLN41 column below / Begin	46.0	0.1	0.1	0.0	0.6	-0.5
	SLN121 column above / End	-44.4	0.2	0.1	0.1	-0.1	0.6
	SLN265 ring / Begin	2.9	-0.1	-5.0	0.0	0.0	0.4
	SLN189 ring / End	-2.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN452 raking strut / End	-35.3	0.2	-0.6	0.1	0.0	-0.1
C LC_135	SLN41 column below / Begin	251.3	0.0	0.4	0.0	1.4	-0.1
	SLN121 column above / End	-152.3	0.1	0.0	0.4	0.3	0.3
	SLN265 ring / Begin	-28.2	-0.3	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.4	-0.1	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-55.4	-0.1	-1.0	0.2	0.0	-0.1
C LC_141	SLN41 column below / Begin	59.9	0.0	0.2	0.0	0.6	0.0
	SLN121 column above / End	-39.7	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.6	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.7	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-11.1	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN41 column below / Begin	60.3	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-39.7	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.9	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.9	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.4	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN41 column below / Begin	60.5	0.0	0.1	0.0	0.6	0.0
	SLN121 column above / End	-39.6	0.0	-0.2	0.1	0.5	0.1
	SLN265 ring / Begin	-9.0	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	9.1	0.0	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-10.0	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN41 column below / Begin	59.6	0.0	0.2	0.0	0.6	0.0
	SLN121 column above / End	-39.8	0.0	-0.1	0.1	0.4	0.1
	SLN265 ring / Begin	-8.6	-0.1	-5.0	0.0	0.0	0.2
	SLN189 ring / End	8.6	-0.1	-4.9	0.0	0.0	-0.1
	SLN452 raking strut / End	-11.4	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN41 column below / Begin	253.3	0.0	0.3	0.0	1.4	-0.2
	SLN121 column above / End	-156.5	0.2	0.1	0.5	0.1	0.3
	SLN265 ring / Begin	-27.1	-0.3	-26.0	0.0	0.0	0.9
	SLN189 ring / End	27.5	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-58.1	-0.1	-1.0	0.2	0.0	-0.1
C LC_146	SLN41 column below / Begin	255.2	0.0	0.3	0.0	1.4	-0.2
	SLN121 column above / End	-156.3	0.1	0.1	0.5	0.2	0.3
	SLN265 ring / Begin	-28.5	-0.2	-26.0	0.0	0.0	0.8
	SLN189 ring / End	28.4	-0.1	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-55.4	-0.1	-1.0	0.2	0.0	-0.1
C LC_147	SLN41 column below / Begin	256.0	0.0	0.3	0.0	1.4	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN121 column above / End	-156.1	0.1	0.1	0.5	0.3	0.3
	SLN265 ring / Begin	-28.8	-0.2	-26.0	0.0	0.0	0.8
	SLN189 ring / End	29.1	0.0	-25.8	0.0	0.0	-0.3
	SLN452 raking strut / End	-54.1	-0.1	-1.0	0.2	0.0	-0.1
C LC_148	SLN41 column below / Begin	252.5	0.0	0.3	0.0	1.4	-0.2
	SLN121 column above / End	-156.6	0.1	0.1	0.5	0.1	0.3
	SLN265 ring / Begin	-26.8	-0.4	-26.0	0.0	0.0	0.9
	SLN189 ring / End	26.9	-0.2	-25.8	0.0	0.0	-0.4
	SLN452 raking strut / End	-59.4	-0.1	-1.0	0.2	0.0	-0.1
C LC_151	SLN41 column below / Begin	174.8	0.0	-1.4	0.0	-5.7	0.0
	SLN121 column above / End	-121.2	0.1	3.8	0.4	-9.3	0.2
	SLN265 ring / Begin	-12.5	-0.2	-11.2	0.0	0.0	0.5
	SLN189 ring / End	12.6	-0.1	-11.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-44.2	-0.1	-1.0	0.2	0.0	-0.1
C LC_152	SLN41 column below / Begin	153.6	0.0	1.8	0.0	7.4	-0.1
	SLN121 column above / End	-94.6	0.1	-3.9	0.6	9.7	0.3
	SLN265 ring / Begin	-28.3	-0.2	-14.0	0.0	0.0	0.7
	SLN189 ring / End	28.5	-0.1	-14.1	0.0	0.0	-0.3
	SLN452 raking strut / End	-27.4	-0.1	-1.0	0.2	0.0	-0.1
C LC_153	SLN41 column below / Begin	251.6	0.0	0.3	0.0	1.3	-0.2
	SLN121 column above / End	-154.8	0.1	0.1	0.5	0.1	0.3
	SLN265 ring / Begin	-27.9	-0.3	-20.2	0.0	0.0	0.8
	SLN189 ring / End	28.1	-0.1	-20.6	0.0	0.0	-0.4
	SLN452 raking strut / End	-56.1	-0.1	-1.0	0.2	0.0	-0.1
C LC_154	SLN41 column below / Begin	176.0	0.0	-1.4	0.0	-5.6	0.0
	SLN121 column above / End	-123.1	0.1	3.8	0.4	-9.3	0.2
	SLN265 ring / Begin	-12.6	-0.2	-11.2	0.0	0.0	0.5
	SLN189 ring / End	12.7	-0.1	-11.4	0.0	0.0	-0.2
	SLN452 raking strut / End	-44.3	-0.1	-1.0	0.2	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	1.1 < 5.0%	OK
Loc. deformation	0.2 < 3%	OK
Bolts	23.4 < 100%	OK
Welds	98.9 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN266+SLN188 radial (continuous)-bfl 1	12.7	C LC_122	244.3	0.0	78.9	OK
SLN266+SLN188 radial (continuous)-tfl 1	12.7	C LC_124	179.7	0.0	28.9	OK
SLN266+SLN188 radial (continuous)-w 1	7.2	C LC_124	275.8	0.4	52.7	OK
SLN41 column below	10.0	C LC_122	277.3	1.1	0.0	OK
SLN121 column above	10.0	C LC_152	188.1	0.0	0.0	OK
SLN265 ring-bfl 1	12.7	C LC_119	67.0	0.0	0.0	OK
SLN265 ring-tfl 1	12.7	C LC_119	61.2	0.0	0.0	OK
SLN265 ring-w 1	7.2	C LC_119	196.7	0.0	0.0	OK
SLN189 ring-bfl 1	12.7	C LC_148	37.5	0.0	0.0	OK
SLN189 ring-tfl 1	12.7	C LC_106	32.6	0.0	0.0	OK
SLN189 ring-w 1	7.2	C LC_124	126.4	0.0	0.0	OK
SLN452 raking strut	10.0	C LC_106	64.2	0.0	0.0	OK
End plate column below SLN41 column below	12.0	C LC_122	247.2	0.0	66.7	OK
End plate column above SLN121 column above	12.0	C LC_154	128.3	0.0	47.7	OK
End plate SLN265 ring	12.0	C LC_119	101.3	0.0	0.0	OK
End plate SLN189 ring	12.0	C LC_120	75.3	0.0	113.1	OK
Cap plate SLN452 raking strut	12.0	C LC_124	80.9	0.0	0.0	OK
Fin plate SLN452 raking strut	12.0	C LC_109	226.8	0.0	60.0	OK
Fin plate 2 SLN452 raking strut	12.0	C LC_116	226.4	0.0	67.0	OK
Gusset SLN452 raking strut	12.0	C LC_106	238.2	0.0	8.0	OK
Flange top a end plate SLN265 ring	12.0	C LC_119	82.7	0.0	0.0	OK
Flange top b end plate SLN265 ring	12.0	C LC_119	72.9	0.0	0.0	OK
Flange bottom a end plate SLN265 ring	12.0	C LC_123	64.4	0.0	0.0	OK
Flange bottom b end plate SLN265 ring	12.0	C LC_119	90.8	0.0	0.0	OK
Flange top a end plate SLN189 ring	12.0	C LC_120	47.5	0.0	0.0	OK
Flange top b end plate SLN189 ring	12.0	C LC_106	43.0	0.0	0.0	OK
Flange bottom a end plate SLN189 ring	12.0	C LC_120	68.8	0.0	0.0	OK
Flange bottom b end plate SLN189 ring	12.0	C LC_120	52.2	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

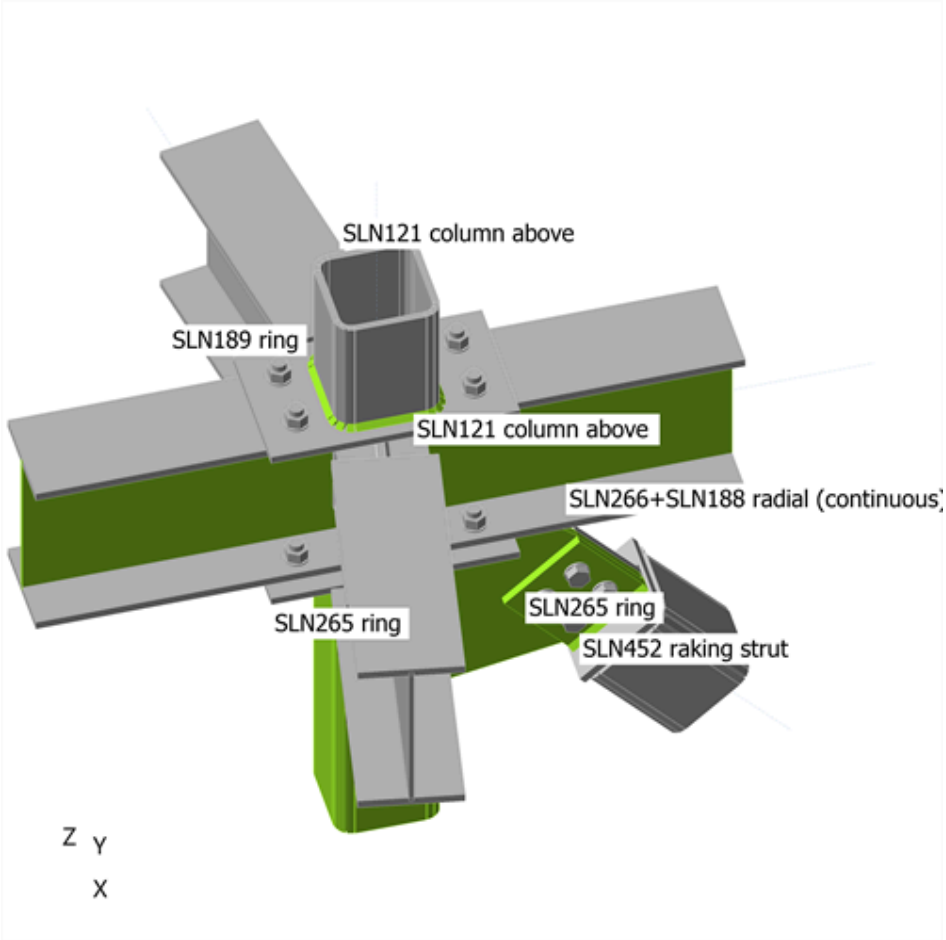
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

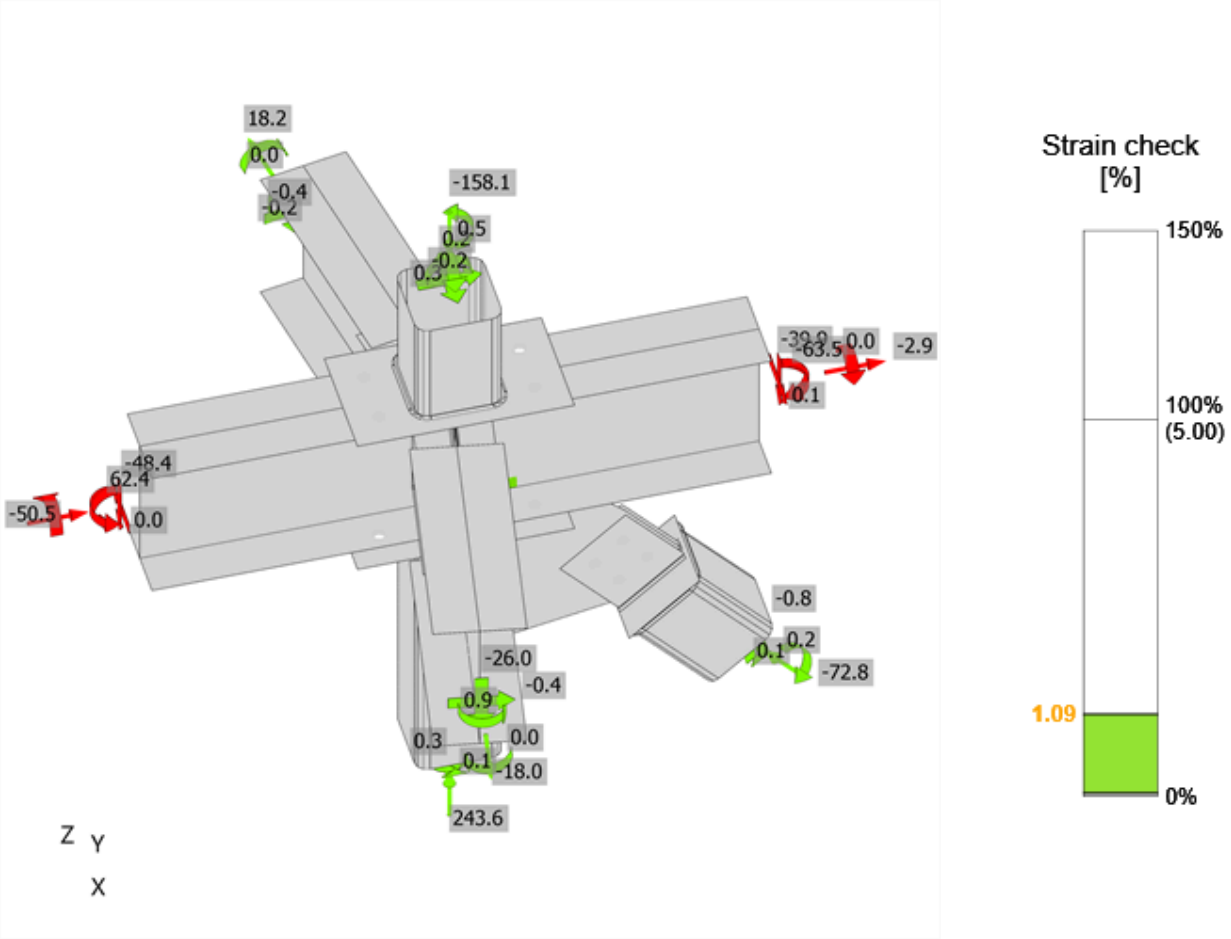
Name	d_0 [mm]	Loads	δ [mm]	$δ_{lim}$ [mm]	$δ/d_0$ [%]	Check status
SLN41 column below	150	C LC_106	0	5	0.2	OK
SLN121 column above	150	C LC_154	0	5	0.0	OK
SLN452 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

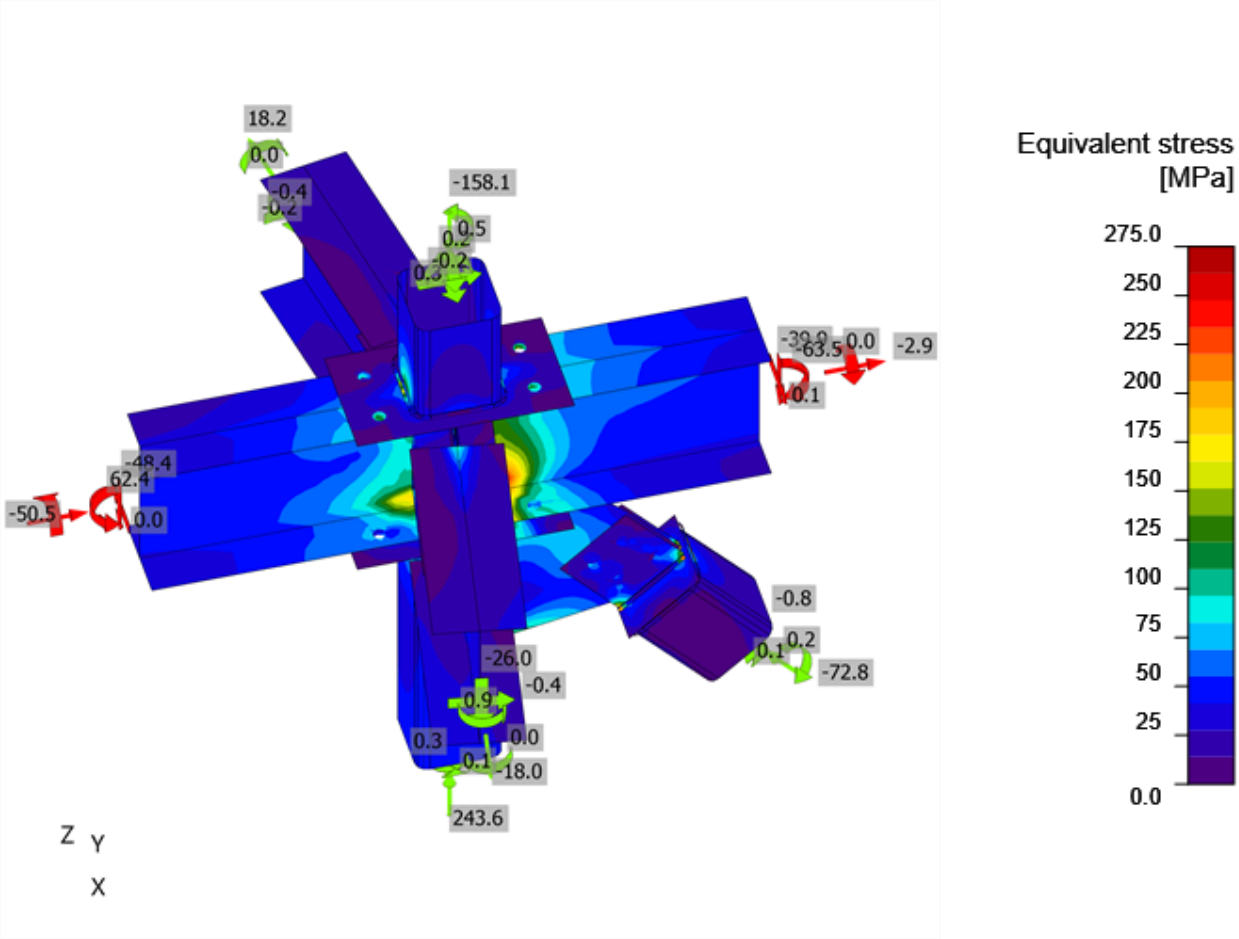
d_0	Cross-section size
δ	Local cross-section deformation
$δ_{lim}$	Allowed deformation
$δ/d_0$	Relative deformation



Overall check

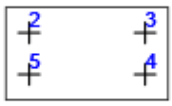
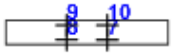
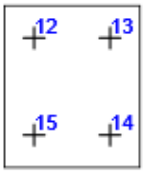
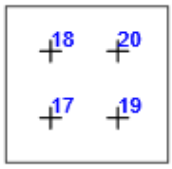


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_106	0.4	16.6	171.5	0.3	17.7	17.9	OK	OK
	B3	M20 8.8 - 1	C LC_109	29.5	5.2	110.2	20.9	5.6	20.5	OK	OK
	B4	M20 8.8 - 1	C LC_109	33.1	4.2	91.9	23.4	4.6	21.2	OK	OK
	B5	M20 8.8 - 1	C LC_106	1.1	15.6	168.6	0.8	16.6	17.1	OK	OK
	B7	M20 8.8 - 1	C LC_151	3.5	1.8	156.4	2.5	1.9	3.6	OK	OK
	B8	M20 8.8 - 1	C LC_152	9.6	3.0	83.2	6.8	3.6	8.1	OK	OK
	B9	M20 8.8 - 1	C LC_108	12.1	1.9	120.2	8.6	2.0	8.2	OK	OK
	B10	M20 8.8 - 1	C LC_124	8.6	2.8	156.4	6.1	2.9	7.3	OK	OK
	B12	M20 8.8 - 1	C LC_119	14.3	1.4	123.8	10.1	1.9	8.7	OK	OK
	B13	M20 8.8 - 1	C LC_109	6.0	0.5	123.8	4.3	0.8	3.6	OK	OK
	B14	M20 8.8 - 1	C LC_109	5.7	0.5	123.8	4.1	0.7	3.5	OK	OK
	B15	M20 8.8 - 1	C LC_119	15.2	1.3	123.8	10.7	1.3	9.0	OK	OK
	B17	M20 8.8 - 1	C LC_109	8.5	11.6	115.3	6.1	12.4	16.7	OK	OK
	B18	M20 8.8 - 1	C LC_124	3.4	9.9	115.3	2.4	10.6	12.3	OK	OK
	B19	M20 8.8 - 1	C LC_114	0.5	9.5	115.3	0.3	15.5	10.4	OK	OK
	B20	M20 8.8 - 1	C LC_106	0.2	10.0	115.3	0.2	17.3	10.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN452 raking strut	SLN452 raking strut	◢ 4.2	519	C LC_106	222.5	0.0	-106.3	112.7	-4.0	55.0	29.8	OK	OK
SLN41 column below-w 3	Gusset SLN452 raking strut	◢ 5.6 ◣	287	C LC_106	400.3	2.3	-133.4	-136.0	-170.3	98.9	83.4	OK	OK
		◢ 5.6 ◣	287	C LC_106	400.1	2.1	-133.5	130.8	174.1	98.9	82.4	OK	OK
SLN265 ring-w 1	Flange top b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_119	89.1	0.0	48.9	39.9	15.9	22.0	13.4	OK	OK
		◢ 4.2 ◣	197	C LC_119	42.5	0.0	-21.0	19.3	9.0	10.5	7.9	OK	OK
SLN265 ring-w 1	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	198	C LC_120	101.8	0.0	30.0	36.2	43.0	25.2	15.4	OK	OK
		◢ 4.2 ◣	198	C LC_119	160.5	0.0	68.0	-60.8	-57.9	39.7	24.3	OK	OK
SLN189 ring-w 1	Flange top a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_124	87.3	0.0	-30.3	6.4	46.8	21.6	13.1	OK	OK
		◢ 4.2 ◣	198	C LC_124	33.3	0.0	23.1	13.6	2.9	8.2	6.9	OK	OK
SLN189 ring-w 1	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	198	C LC_121	90.9	0.0	26.1	29.3	-40.9	22.5	12.3	OK	OK
		◢ 4.2 ◣	198	C LC_124	142.9	0.0	-34.4	-16.0	-78.5	35.3	21.6	OK	OK
Cap plate SLN452 raking strut	Fin plate SLN452 raking strut	◢ 5.6 ◣	139	C LC_109	248.9	0.0	92.8	108.8	77.1	61.5	41.1	OK	OK
		◢ 5.6 ◣	139	C LC_106	212.7	0.0	-87.4	91.8	-64.1	52.5	28.5	OK	OK
Cap plate SLN452 raking strut	Fin plate 2 SLN452 raking strut	◢ 5.6 ◣	139	C LC_106	190.2	0.0	-70.6	-80.6	62.4	47.0	29.1	OK	OK
		◢ 5.6 ◣	139	C LC_106	212.9	0.0	-99.2	89.2	-62.2	52.6	29.8	OK	OK
End plate column below SLN41 column below	SLN41 column below	◢ 5.6 ◣	521	C LC_122	396.3	0.0	-253.7	-172.8	-32.2	97.9	53.1	OK	OK
		◢ 5.6 ◣	521	C LC_147	325.8	0.0	-106.2	175.2	30.6	80.5	43.9	OK	OK
End plate column above SLN121 column above	SLN121 column above	◢ 5.6 ◣	521	C LC_154	227.6	0.0	-111.8	-114.3	-5.9	56.2	30.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_154	245.0	0.0	-120.9	121.0	22.5	60.5	33.2	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN265 ring-w 1	◢ 7.2	189	C LC_119	259.2	0.0	6.9	-15.4	148.8	64.0	34.7	OK	OK
SLN266+SLN188 radial (continuous)-w 1	SLN189 ring-w 1	◢ 7.2	189	C LC_124	361.6	0.0	5.7	27.5	-206.9	89.3	63.9	OK	OK
End plate SLN265 ring	SLN265 ring-w 1	◢ 4.2 ◣	149	C LC_119	233.4	0.0	-62.6	-25.3	-127.3	57.7	34.2	OK	OK
		◢ 4.2 ◣	149	C LC_119	166.9	0.0	-20.9	49.3	81.9	41.2	26.3	OK	OK
End plate SLN189 ring	SLN189 ring-w 1	◢ 4.2 ◣	149	C LC_124	292.2	0.0	-60.6	-72.5	148.3	72.2	44.2	OK	OK
		◢ 4.2 ◣	149	C LC_124	217.6	0.0	-60.7	52.5	-108.6	53.8	34.1	OK	OK
SLN265 ring-w 1	Flange top a end plate SLN265 ring	◢ 4.2 ◣	199	C LC_119	143.5	0.0	-12.9	7.5	-82.2	35.5	25.2	OK	OK
		◢ 4.2 ◣	198	C LC_119	97.4	0.0	15.7	17.4	-52.7	24.1	13.7	OK	OK
SLN265 ring-w 1	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	198	C LC_119	125.4	0.0	23.6	-23.5	67.1	31.0	18.5	OK	OK
		◢ 4.2 ◣	199	C LC_121	125.8	0.0	-29.4	24.6	66.2	31.1	16.4	OK	OK
SLN189 ring-w 1	Flange top b end plate SLN189 ring	◢ 4.2 ◣	199	C LC_124	92.4	0.0	-34.6	-11.6	48.1	22.8	13.8	OK	OK
		◢ 4.2 ◣	199	C LC_124	56.9	0.0	20.9	-25.4	17.0	14.1	8.9	OK	OK
SLN189 ring-w 1	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	198	C LC_148	68.8	0.0	12.5	-12.2	-37.1	17.0	10.9	OK	OK
		◢ 4.2 ◣	199	C LC_120	146.4	0.0	-37.5	23.5	-78.3	36.2	22.3	OK	OK
End plate SLN265 ring	Flange top a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	74.3	0.0	-11.2	26.2	-33.4	18.4	11.4	OK	OK
		◢ 4.2 ◣	58	C LC_125	51.0	0.0	-1.4	20.4	-21.3	12.6	8.3	OK	OK
End plate SLN265 ring	Flange top b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	82.7	0.0	-9.6	46.5	-9.3	20.4	12.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_119	84.7	0.0	47.2	34.7	-21.0	20.9	16.0	OK	OK
End plate SLN265 ring	Flange bottom a end plate SLN265 ring	◢ 4.2 ◣	58	C LC_121	98.1	0.0	11.8	-39.7	39.8	24.2	14.7	OK	OK
		◢ 4.2 ◣	58	C LC_119	108.7	0.0	-22.8	-47.2	39.2	26.9	17.0	OK	OK
End plate SLN265 ring	Flange bottom b end plate SLN265 ring	◢ 4.2 ◣	58	C LC_119	97.5	0.0	54.3	-42.5	19.5	24.1	19.8	OK	OK
		◢ 4.2 ◣	58	C LC_119	115.8	0.0	-22.7	-65.5	-2.2	28.6	16.2	OK	OK
End plate SLN189 ring	Flange top a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_123	98.4	0.0	-45.1	39.5	31.5	24.3	13.7	OK	OK
		◢ 4.2 ◣	58	C LC_123	56.9	0.0	-19.2	-24.7	-18.5	14.1	10.9	OK	OK
End plate SLN189 ring	Flange top b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_124	85.7	0.0	-9.7	41.5	26.4	21.2	11.2	OK	OK
		◢ 4.2 ◣	58	C LC_124	60.5	0.0	-7.3	24.4	24.6	15.0	8.1	OK	OK
End plate SLN189 ring	Flange bottom a end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	88.2	0.0	6.8	-46.2	21.0	21.8	15.0	OK	OK
		◢ 4.2 ◣	58	C LC_124	132.9	0.0	-30.9	-71.7	20.7	32.8	23.9	OK	OK
End plate SLN189 ring	Flange bottom b end plate SLN189 ring	◢ 4.2 ◣	58	C LC_120	110.4	0.0	11.0	-46.2	-43.5	27.3	13.8	OK	OK
		◢ 4.2 ◣	58	C LC_124	108.8	0.0	-16.5	-53.9	-30.8	26.9	14.1	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L1-R8-T_260
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut

Name	SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut
Description	
Analysis	Stress, strain/ simplified loading

Design

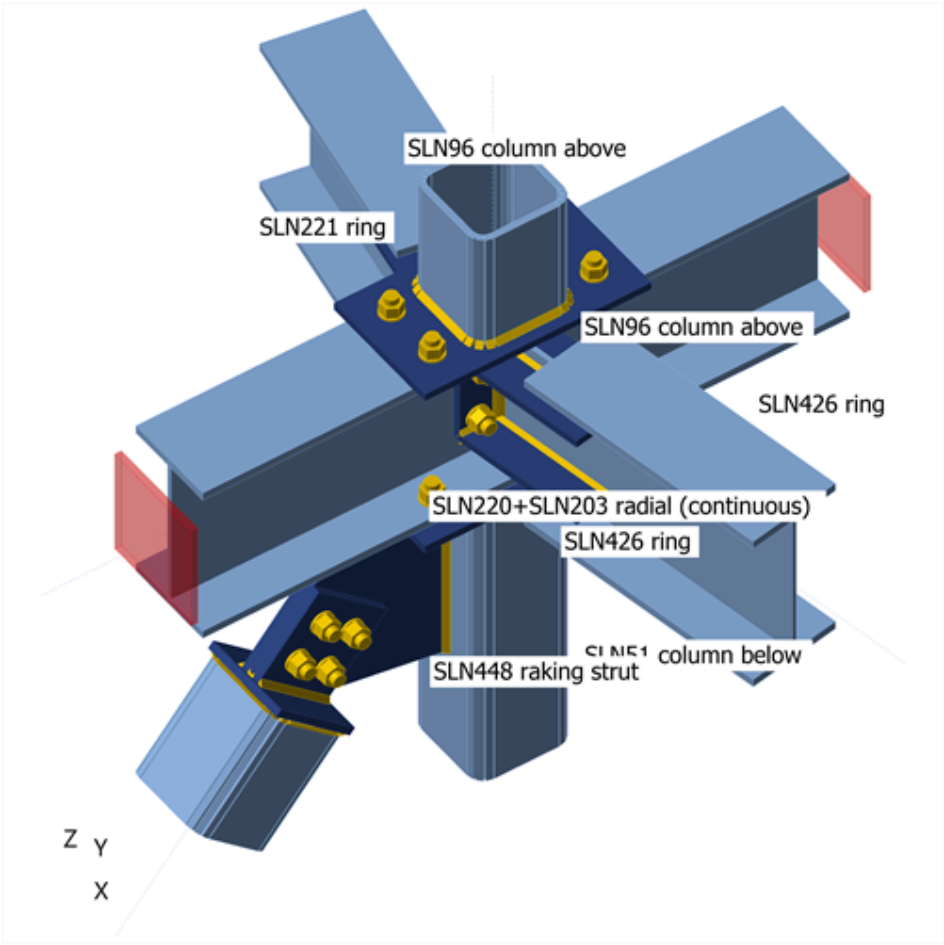
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN220+SLN203 radial (continuous)	1 - UB254x146x43	80.0	0.0	0.0	0	0	0
SLN51 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN96 column above	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN221 ring	1 - UB254x146x43	-20.0	0.0	0.0	0	0	0
SLN426 ring	1 - UB254x146x43	0.0	0.0	0.0	0	0	0
SLN448 raking strut	2 - SHS150x150x10	-100.0	-49.2	0.0	0	0	0

Supports and forces

Name	Support	Forces in	X [mm]
SLN220+SLN203 radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN220+SLN203 radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN51 column below / begin		Position	0
SLN96 column above / end		Position	0
SLN221 ring / begin		Position	0
SLN426 ring / end		Position	0
SLN448 raking strut / end		Position	0



SLN220+SLN203 radial (continuous) - SLN51 column below - SLN96 column above - SLN221 ring - SLN426 ring - SLN448 raking strut, Axonometric view

Cross-sections

Name	Material
1 - UB254x146x43	S 275
2 - SHS150x150x10	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN51 column below / Begin	54.5	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-35.5	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.8	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.7	0.0	-1.0	0.0	0.0	0.0
C LC_102	SLN51 column below / Begin	233.2	0.0	0.7	0.0	2.9	0.0
	SLN96 column above / End	-142.7	0.0	-1.1	-0.6	2.9	-0.2
	SLN221 ring / Begin	-27.8	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	27.6	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-57.0	0.1	-1.0	-0.2	0.0	0.0
C LC_103	SLN51 column below / Begin	218.9	-3.9	-0.1	0.0	2.1	1.7
	SLN96 column above / End	-134.4	-0.3	-1.1	-0.7	2.8	-1.3
	SLN221 ring / Begin	-21.7	-0.1	-24.1	0.0	0.0	0.5
	SLN426 ring / End	17.0	-0.2	-23.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-53.6	0.2	-1.1	0.0	0.0	-0.6
C LC_104	SLN51 column below / Begin	213.4	0.0	0.7	0.0	2.6	0.2
	SLN96 column above / End	-136.7	0.0	-1.0	-0.5	2.7	-0.2
	SLN221 ring / Begin	-16.2	-0.2	-24.1	0.0	0.0	0.5
	SLN426 ring / End	16.5	-0.2	-23.9	0.0	0.0	-0.7
	SLN448 raking strut / End	-64.2	-0.3	-1.0	-0.2	0.0	-0.2
C LC_105	SLN51 column below / Begin	205.1	-0.1	0.7	0.0	2.6	0.3
	SLN96 column above / End	-139.2	-0.2	-0.9	-0.6	2.3	-0.6
	SLN221 ring / Begin	-14.9	-0.2	-24.1	0.0	0.0	0.4
	SLN426 ring / End	14.3	-0.4	-23.9	0.0	0.0	-1.0
	SLN448 raking strut / End	-78.7	0.2	-0.8	-0.2	0.0	0.2
C LC_106	SLN51 column below / Begin	237.3	0.7	-3.5	0.0	-0.3	-0.5
	SLN96 column above / End	-131.8	0.1	-1.3	-0.5	3.5	0.3
	SLN221 ring / Begin	-29.7	0.0	-24.1	0.0	0.0	0.2
	SLN426 ring / End	30.5	-0.1	-23.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-25.3	0.0	-1.3	-0.2	0.0	0.0
C LC_107	SLN51 column below / Begin	14.0	-7.8	-1.4	0.0	-0.8	3.5
	SLN96 column above / End	-10.3	-0.6	-0.3	-0.3	0.7	-2.3
	SLN221 ring / Begin	0.3	-0.1	-0.5	0.0	0.0	0.3
	SLN426 ring / End	-8.9	0.1	-0.5	0.0	0.0	0.3
	SLN448 raking strut / End	0.7	0.2	-0.8	0.3	0.0	-1.4
C LC_108	SLN51 column below / Begin	4.5	-0.1	0.1	0.0	0.2	0.3
	SLN96 column above / End	-14.7	0.0	-0.2	0.0	0.5	-0.1
	SLN221 ring / Begin	9.9	-0.1	-0.5	0.0	0.0	0.2
	SLN426 ring / End	-9.5	0.0	-0.5	0.0	0.0	0.1
	SLN448 raking strut / End	-18.1	-0.7	-0.7	-0.1	0.0	-0.4
C LC_109	SLN51 column below / Begin	-9.5	-0.1	0.1	0.0	0.2	0.5
	SLN96 column above / End	-19.3	-0.2	0.1	-0.1	-0.1	-0.8
	SLN221 ring / Begin	11.0	0.0	-0.5	0.0	0.0	0.0
	SLN426 ring / End	-11.7	-0.2	-0.5	0.0	0.0	-0.5
	SLN448 raking strut / End	-42.9	0.2	-0.3	-0.1	0.0	0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_110	SLN51 column below / Begin	48.0	1.4	-8.3	0.0	-5.6	-0.9
	SLN96 column above / End	-5.5	0.2	-0.7	0.1	1.8	0.9
	SLN221 ring / Begin	-14.2	0.1	-0.5	0.0	0.0	-0.2
	SLN426 ring / End	16.2	0.1	-0.5	0.0	0.0	0.3
	SLN448 raking strut / End	53.2	-0.1	-1.3	0.0	0.0	0.0
C LC_111	SLN51 column below / Begin	152.4	-7.8	-1.0	0.0	0.8	3.5
	SLN96 column above / End	-94.3	-0.6	-0.8	-0.6	2.2	-2.4
	SLN221 ring / Begin	-13.4	-0.2	-16.3	0.0	0.0	0.5
	SLN426 ring / End	4.5	0.0	-16.2	0.0	0.0	-0.1
	SLN448 raking strut / End	-35.1	0.3	-1.1	0.2	0.0	-1.3
C LC_112	SLN51 column below / Begin	142.5	-0.1	0.5	0.0	1.8	0.3
	SLN96 column above / End	-98.8	0.0	-0.8	-0.3	2.0	-0.1
	SLN221 ring / Begin	-3.4	-0.2	-16.3	0.0	0.0	0.5
	SLN426 ring / End	3.8	-0.1	-16.2	0.0	0.0	-0.4
	SLN448 raking strut / End	-54.7	-0.6	-0.9	-0.2	0.0	-0.4
C LC_113	SLN51 column below / Begin	127.6	-0.1	0.4	0.0	1.8	0.5
	SLN96 column above / End	-103.5	-0.2	-0.5	-0.5	1.4	-0.9
	SLN221 ring / Begin	-1.7	-0.2	-16.3	0.0	0.0	0.3
	SLN426 ring / End	0.9	-0.4	-16.2	0.0	0.0	-1.0
	SLN448 raking strut / End	-80.8	0.3	-0.5	-0.2	0.0	0.4
C LC_114	SLN51 column below / Begin	187.3	1.4	-7.9	0.0	-4.0	-1.0
	SLN96 column above / End	-89.4	0.2	-1.3	-0.3	3.4	0.8
	SLN221 ring / Begin	-28.4	0.1	-16.3	0.0	0.0	0.0
	SLN426 ring / End	30.2	0.0	-16.2	0.0	0.0	-0.1
	SLN448 raking strut / End	18.7	-0.1	-1.5	-0.2	0.0	0.0
C LC_115	SLN51 column below / Begin	28.1	-7.8	-1.4	0.0	-0.6	3.5
	SLN96 column above / End	-19.5	-0.6	-0.4	-0.3	0.9	-2.3
	SLN221 ring / Begin	-2.0	-0.1	-1.7	0.0	0.0	0.3
	SLN426 ring / End	-6.6	0.1	-1.7	0.0	0.0	0.3
	SLN448 raking strut / End	-2.0	0.2	-1.1	0.3	0.0	-1.4
C LC_116	SLN51 column below / Begin	18.6	-0.1	0.1	0.0	0.5	0.3
	SLN96 column above / End	-23.9	0.0	-0.3	0.0	0.7	-0.1
	SLN221 ring / Begin	7.6	-0.1	-1.7	0.0	0.0	0.3
	SLN426 ring / End	-7.2	0.0	-1.7	0.0	0.0	0.0
	SLN448 raking strut / End	-20.9	-0.7	-0.9	-0.1	0.0	-0.4
C LC_117	SLN51 column below / Begin	4.5	-0.1	0.1	0.0	0.4	0.5
	SLN96 column above / End	-28.5	-0.2	-0.1	-0.1	0.1	-0.8
	SLN221 ring / Begin	8.8	0.0	-1.7	0.0	0.0	0.0
	SLN426 ring / End	-9.5	-0.2	-1.7	0.0	0.0	-0.6
	SLN448 raking strut / End	-45.8	0.2	-0.5	-0.1	0.0	0.4
C LC_118	SLN51 column below / Begin	62.2	1.4	-8.2	0.0	-5.3	-1.0
	SLN96 column above / End	-14.7	0.2	-0.8	0.0	2.1	0.9
	SLN221 ring / Begin	-16.6	0.1	-1.7	0.0	0.0	-0.2
	SLN426 ring / End	18.5	0.1	-1.7	0.0	0.0	0.2
	SLN448 raking strut / End	50.6	-0.1	-1.5	-0.1	0.0	0.0
C LC_119	SLN51 column below / Begin	231.4	-3.9	0.0	0.0	2.3	1.8
	SLN96 column above / End	-141.4	-0.3	-1.2	-0.6	3.1	-1.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN221 ring / Begin	-25.2	-0.2	-25.7	0.0	0.0	0.6
	SLN426 ring / End	20.5	-0.2	-25.5	0.0	0.0	-0.6
	SLN448 raking strut / End	-57.4	0.2	-1.1	0.0	0.0	-0.6
C LC_120	SLN51 column below / Begin	226.8	0.0	0.7	0.0	2.8	0.1
	SLN96 column above / End	-143.3	0.0	-1.1	-0.6	2.9	-0.1
	SLN221 ring / Begin	-19.6	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	19.9	-0.3	-25.5	0.0	0.0	-0.7
	SLN448 raking strut / End	-66.1	-0.3	-1.0	-0.2	0.0	-0.2
C LC_121	SLN51 column below / Begin	222.3	-0.1	0.7	0.0	2.8	0.3
	SLN96 column above / End	-144.7	-0.2	-0.8	-0.6	2.3	-0.6
	SLN221 ring / Begin	-18.3	-0.2	-25.7	0.0	0.0	0.4
	SLN426 ring / End	17.6	-0.4	-25.5	0.0	0.0	-1.0
	SLN448 raking strut / End	-73.9	0.2	-0.8	-0.2	0.0	0.3
C LC_122	SLN51 column below / Begin	248.6	0.7	-3.4	0.0	0.0	-0.5
	SLN96 column above / End	-140.1	0.1	-1.5	-0.5	3.8	0.4
	SLN221 ring / Begin	-33.3	0.0	-25.7	0.0	0.0	0.3
	SLN426 ring / End	34.2	-0.1	-25.5	0.0	0.0	-0.6
	SLN448 raking strut / End	-32.6	0.0	-1.3	-0.2	0.0	0.0
C LC_123	SLN51 column below / Begin	177.4	-7.8	-0.9	0.0	1.2	3.6
	SLN96 column above / End	-108.4	-0.6	-1.1	-0.6	2.8	-2.5
	SLN221 ring / Begin	-20.5	-0.2	-19.5	0.0	0.0	0.7
	SLN426 ring / End	11.5	-0.1	-19.3	0.0	0.0	-0.2
	SLN448 raking strut / End	-42.8	0.3	-1.1	0.2	0.0	-1.3
C LC_124	SLN51 column below / Begin	169.3	-0.1	0.5	0.0	2.2	0.2
	SLN96 column above / End	-111.9	0.0	-1.0	-0.4	2.4	0.0
	SLN221 ring / Begin	-10.1	-0.2	-19.5	0.0	0.0	0.5
	SLN426 ring / End	10.6	-0.2	-19.3	0.0	0.0	-0.5
	SLN448 raking strut / End	-58.5	-0.6	-0.9	-0.2	0.0	-0.4
C LC_125	SLN51 column below / Begin	162.1	-0.1	0.5	0.0	2.2	0.6
	SLN96 column above / End	-114.6	-0.3	-0.4	-0.5	1.4	-1.0
	SLN221 ring / Begin	-8.4	-0.1	-19.5	0.0	0.0	0.2
	SLN426 ring / End	7.3	-0.4	-19.3	0.0	0.0	-1.0
	SLN448 raking strut / End	-71.3	0.3	-0.5	-0.2	0.0	0.5
C LC_126	SLN51 column below / Begin	209.9	1.5	-7.7	0.0	-3.5	-1.0
	SLN96 column above / End	-106.1	0.2	-1.6	-0.3	4.1	0.9
	SLN221 ring / Begin	-35.6	0.0	-19.5	0.0	0.0	0.2
	SLN426 ring / End	37.6	-0.1	-19.3	0.0	0.0	-0.4
	SLN448 raking strut / End	4.1	-0.1	-1.5	-0.2	0.0	0.0
C LC_127	SLN51 column below / Begin	53.2	-7.8	-1.3	0.0	-0.2	3.6
	SLN96 column above / End	-33.6	-0.6	-0.6	-0.3	1.5	-2.4
	SLN221 ring / Begin	-9.1	-0.2	-4.9	0.0	0.0	0.5
	SLN426 ring / End	0.4	0.1	-4.9	0.0	0.0	0.1
	SLN448 raking strut / End	-9.7	0.2	-1.1	0.3	0.0	-1.3
C LC_128	SLN51 column below / Begin	45.4	-0.1	0.2	0.0	0.8	0.2
	SLN96 column above / End	-37.0	0.0	-0.5	-0.1	1.1	0.1
	SLN221 ring / Begin	0.9	-0.1	-4.9	0.0	0.0	0.3
	SLN426 ring / End	-0.4	0.0	-4.9	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN448 raking strut / End	-24.7	-0.7	-0.9	-0.1	0.0	-0.4
C LC_129	SLN51 column below / Begin	39.0	-0.1	0.2	0.0	0.8	0.6
	SLN96 column above / End	-39.6	-0.2	0.0	-0.1	0.1	-0.9
	SLN221 ring / Begin	2.2	0.0	-4.9	0.0	0.0	0.0
	SLN426 ring / End	-3.0	-0.2	-4.9	0.0	0.0	-0.6
	SLN448 raking strut / End	-36.3	0.3	-0.5	-0.1	0.0	0.5
C LC_130	SLN51 column below / Begin	84.8	1.5	-8.1	0.0	-4.8	-1.0
	SLN96 column above / End	-31.4	0.3	-1.1	0.0	2.8	1.0
	SLN221 ring / Begin	-23.8	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	25.9	0.0	-4.9	0.0	0.0	0.0
	SLN448 raking strut / End	36.0	-0.1	-1.5	-0.1	0.0	0.0
C LC_135	SLN51 column below / Begin	230.4	0.0	0.7	0.0	3.0	0.0
	SLN96 column above / End	-138.8	0.0	-1.2	-0.4	3.0	-0.1
	SLN221 ring / Begin	-28.3	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	28.1	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-55.7	0.1	-1.0	-0.2	0.0	0.1
C LC_141	SLN51 column below / Begin	54.6	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-35.5	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.9	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.8	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.6	0.0	-1.0	0.0	0.0	0.0
C LC_142	SLN51 column below / Begin	54.4	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-35.6	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.7	-0.1	-4.9	0.0	0.0	0.1
	SLN426 ring / End	8.7	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.9	0.0	-1.0	0.0	0.0	0.0
C LC_143	SLN51 column below / Begin	54.0	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-35.6	0.0	-0.4	-0.1	1.0	0.0
	SLN221 ring / Begin	-8.6	-0.1	-4.9	0.0	0.0	0.2
	SLN426 ring / End	8.5	-0.1	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-11.5	0.0	-1.0	0.0	0.0	0.0
C LC_144	SLN51 column below / Begin	54.9	0.0	0.2	0.0	0.9	0.0
	SLN96 column above / End	-35.5	0.0	-0.4	-0.1	1.1	0.0
	SLN221 ring / Begin	-9.0	0.0	-4.9	0.0	0.0	0.1
	SLN426 ring / End	9.0	0.0	-4.9	0.0	0.0	-0.2
	SLN448 raking strut / End	-10.0	0.0	-1.0	0.0	0.0	0.0
C LC_145	SLN51 column below / Begin	233.6	0.0	0.7	0.0	2.9	0.0
	SLN96 column above / End	-142.6	-0.1	-1.2	-0.6	2.9	-0.2
	SLN221 ring / Begin	-28.1	-0.1	-25.7	0.0	0.0	0.5
	SLN426 ring / End	27.7	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-56.3	0.1	-1.0	-0.2	0.0	0.0
C LC_146	SLN51 column below / Begin	232.8	0.0	0.7	0.0	2.9	0.0
	SLN96 column above / End	-142.7	0.0	-1.1	-0.6	2.9	-0.2
	SLN221 ring / Begin	-27.4	-0.2	-25.7	0.0	0.0	0.5
	SLN426 ring / End	27.5	-0.3	-25.5	0.0	0.0	-0.8
	SLN448 raking strut / End	-57.7	0.1	-1.0	-0.2	0.0	0.0
C LC_147	SLN51 column below / Begin	231.3	0.0	0.7	0.0	2.9	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN96 column above / End	-143.0	-0.1	-1.1	-0.6	2.8	-0.2
	SLN221 ring / Begin	-26.6	-0.3	-25.7	0.0	0.0	0.5
	SLN426 ring / End	26.4	-0.4	-25.5	0.0	0.0	-0.9
	SLN448 raking strut / End	-60.0	0.1	-1.0	-0.2	0.0	0.0
C LC_148	SLN51 column below / Begin	235.1	0.0	0.7	0.0	2.9	0.0
	SLN96 column above / End	-142.4	0.0	-1.2	-0.6	3.0	-0.1
	SLN221 ring / Begin	-28.9	0.0	-25.7	0.0	0.0	0.4
	SLN426 ring / End	28.8	-0.2	-25.5	0.0	0.0	-0.7
	SLN448 raking strut / End	-54.0	0.1	-1.0	-0.2	0.0	0.0
C LC_151	SLN51 column below / Begin	153.4	0.0	-1.0	0.0	-4.0	-0.1
	SLN96 column above / End	-105.8	0.0	2.5	-0.4	-6.4	-0.1
	SLN221 ring / Begin	-12.5	-0.1	-11.7	0.0	0.0	0.3
	SLN426 ring / End	12.4	-0.2	-11.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-42.3	0.1	-1.0	-0.2	0.0	0.0
C LC_152	SLN51 column below / Begin	148.3	0.0	1.9	0.0	7.6	0.1
	SLN96 column above / End	-91.0	-0.1	-4.1	-0.6	10.2	-0.2
	SLN221 ring / Begin	-28.3	-0.1	-13.9	0.0	0.0	0.3
	SLN426 ring / End	28.0	-0.2	-14.1	0.0	0.0	-0.7
	SLN448 raking strut / End	-29.9	0.1	-1.0	-0.2	0.0	0.0
C LC_153	SLN51 column below / Begin	232.6	0.0	0.7	0.0	2.8	0.0
	SLN96 column above / End	-141.8	0.0	-1.1	-0.6	2.8	-0.2
	SLN221 ring / Begin	-27.9	-0.2	-20.6	0.0	0.0	0.5
	SLN426 ring / End	27.7	-0.3	-21.1	0.0	0.0	-0.8
	SLN448 raking strut / End	-56.7	0.1	-1.0	-0.2	0.0	0.0
C LC_154	SLN51 column below / Begin	153.1	0.0	-1.0	0.0	-3.9	-0.1
	SLN96 column above / End	-106.1	0.0	2.5	-0.4	-6.2	-0.1
	SLN221 ring / Begin	-12.6	-0.1	-11.7	0.0	0.0	0.3
	SLN426 ring / End	12.5	-0.2	-11.9	0.0	0.0	-0.5
	SLN448 raking strut / End	-42.2	0.1	-1.0	-0.2	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	1.2 < 5.0%	OK
Loc. deformation	0.2 < 3%	OK
Bolts	29.3 < 100%	OK
Welds	99.0 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN220+SLN203 radial (continuous)-bfl 1	12.7	C LC_110	247.6	0.0	39.0	OK
SLN220+SLN203 radial (continuous)-tfl 1	12.7	C LC_123	199.5	0.0	30.0	OK
SLN220+SLN203 radial (continuous)-w 1	7.2	C LC_123	275.7	0.3	59.9	OK
SLN51 column below	10.0	C LC_105	277.5	1.2	0.0	OK
SLN96 column above	10.0	C LC_152	196.0	0.0	0.0	OK
SLN221 ring-bfl 1	12.7	C LC_123	45.1	0.0	0.0	OK
SLN221 ring-tfl 1	12.7	C LC_123	42.4	0.0	0.0	OK
SLN221 ring-w 1	7.2	C LC_119	137.7	0.0	0.0	OK
SLN426 ring-bfl 1	12.7	C LC_121	67.3	0.0	0.0	OK
SLN426 ring-tfl 1	12.7	C LC_121	63.9	0.0	0.0	OK
SLN426 ring-w 1	7.2	C LC_121	196.3	0.0	0.0	OK
SLN448 raking strut	10.0	C LC_113	67.0	0.0	0.0	OK
End plate column below SLN51 column below	12.0	C LC_110	275.0	0.0	39.0	OK
End plate column above SLN96 column above	12.0	C LC_152	116.1	0.0	41.4	OK
End plate SLN221 ring	12.0	C LC_127	73.6	0.0	159.0	OK
End plate SLN426 ring	12.0	C LC_123	72.4	0.0	61.0	OK
Cap plate SLN448 raking strut	12.0	C LC_123	104.3	0.0	0.0	OK
Fin plate SLN448 raking strut	12.0	C LC_123	259.1	0.0	51.5	OK
Fin plate 2 SLN448 raking strut	12.0	C LC_107	252.7	0.0	84.4	OK
Gusset SLN448 raking strut	12.0	C LC_105	240.0	0.0	9.5	OK
Flange top a end plate SLN221 ring	12.0	C LC_123	83.1	0.0	0.0	OK
Flange top b end plate SLN221 ring	12.0	C LC_123	56.7	0.0	0.0	OK
Flange bottom a end plate SLN221 ring	12.0	C LC_127	65.6	0.0	0.0	OK
Flange bottom b end plate SLN221 ring	12.0	C LC_119	64.4	0.0	0.0	OK
Flange top a end plate SLN426 ring	12.0	C LC_121	69.3	0.0	0.0	OK
Flange top b end plate SLN426 ring	12.0	C LC_121	85.2	0.0	0.0	OK
Flange bottom a end plate SLN426 ring	12.0	C LC_121	88.3	0.0	0.0	OK
Flange bottom b end plate SLN426 ring	12.0	C LC_113	68.2	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

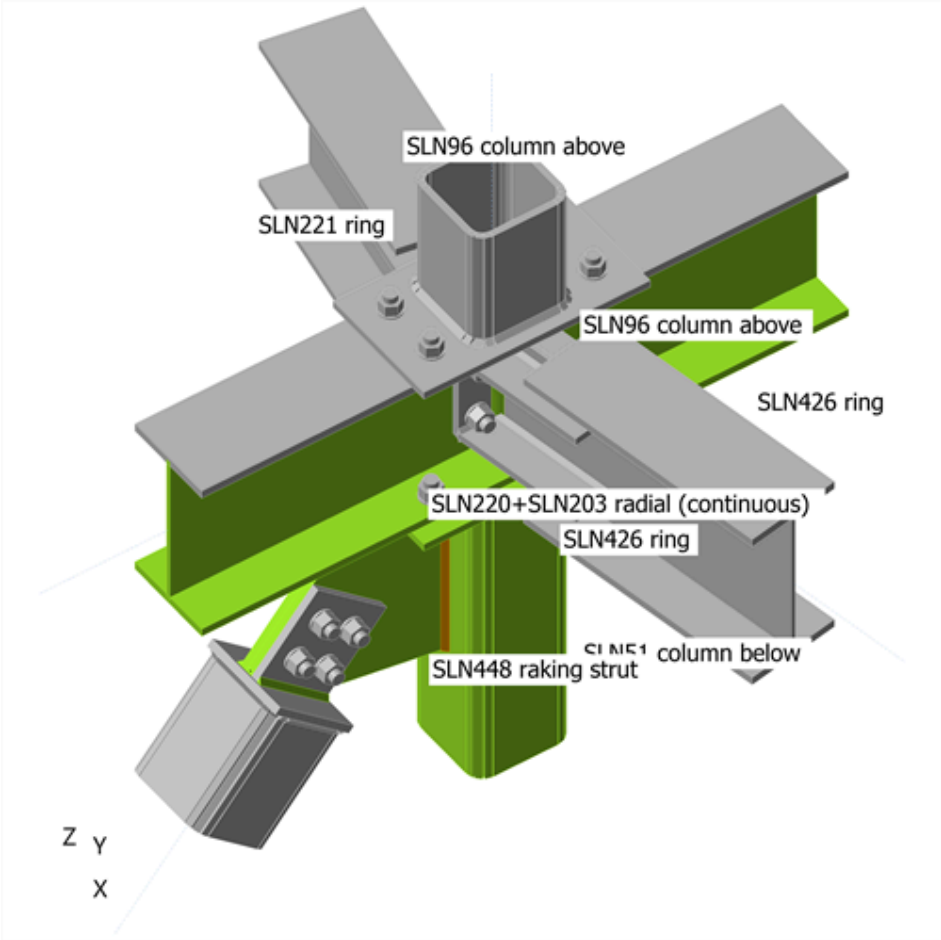
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

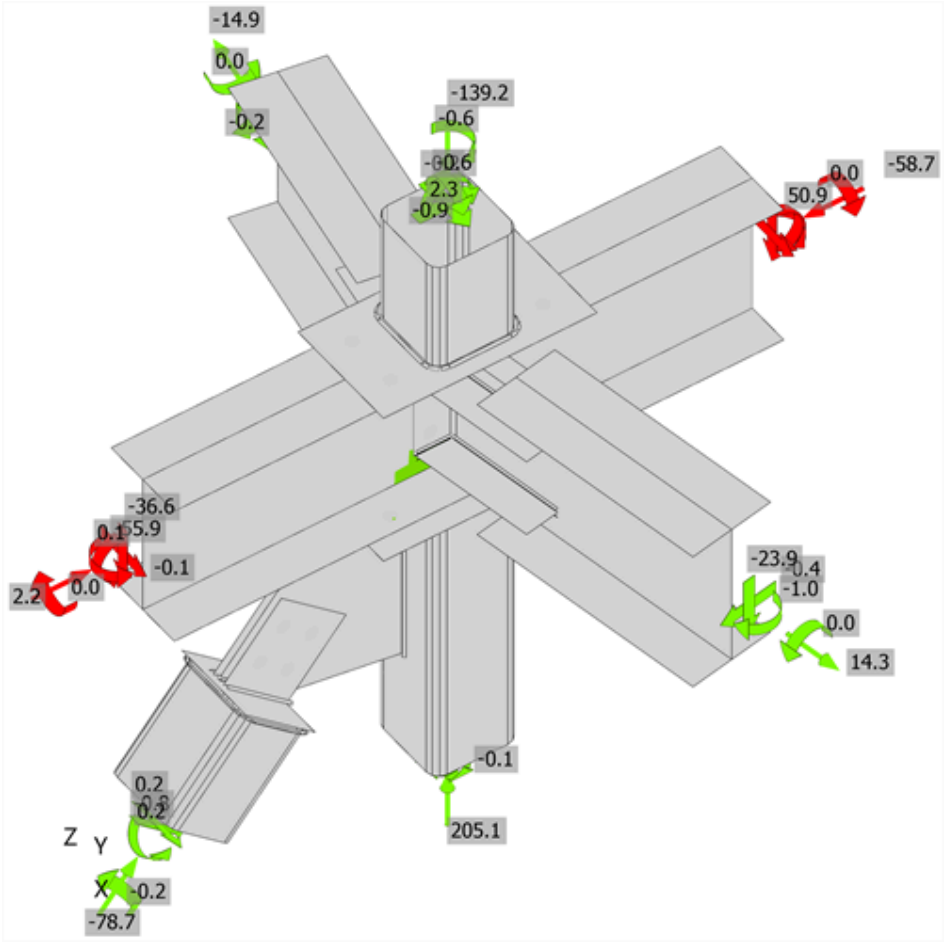
Name	d_0 [mm]	Loads	δ [mm]	$δ_{lim}$ [mm]	$δ/d_0$ [%]	Check status
SLN51 column below	150	C LC_105	0	5	0.2	OK
SLN96 column above	150	C LC_154	0	5	0.0	OK
SLN448 raking strut	150	C LC_154	0	5	0.0	OK

Symbol explanation

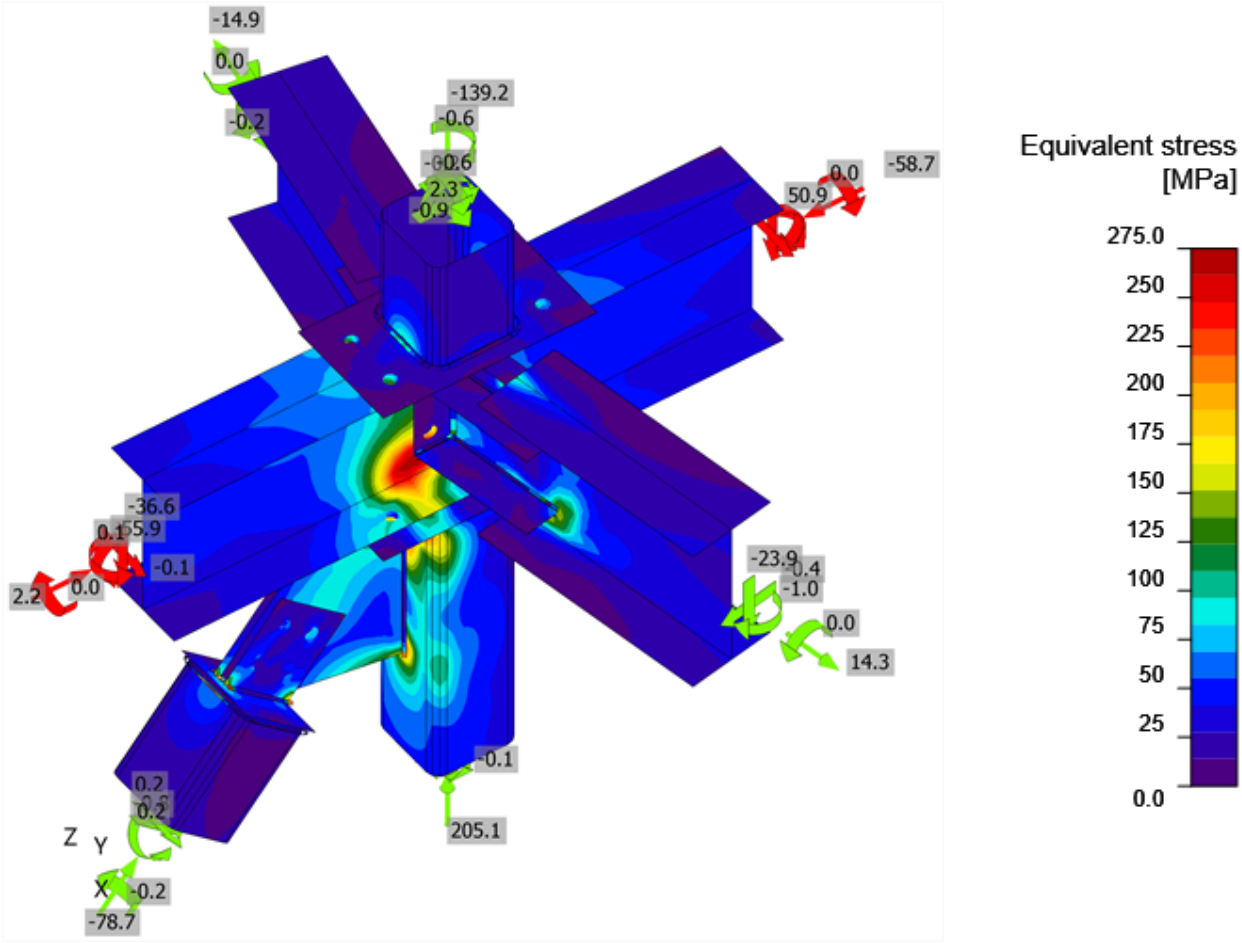
d_0	Cross-section size
δ	Local cross-section deformation
$δ_{lim}$	Allowed deformation
$δ/d_0$	Relative deformation



Overall check

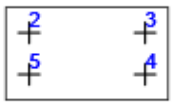
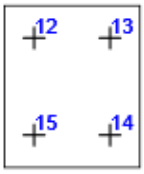
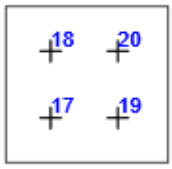


Strain check, C LC_105



Equivalent stress, C LC_105

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_105	1.2	15.6	170.0	0.8	16.6	17.1	OK	OK
	B3	M20 8.8 - 1	C LC_110	35.4	6.6	99.9	25.1	7.0	24.9	OK	OK
	B4	M20 8.8 - 1	C LC_110	41.4	5.5	104.6	29.3	5.8	26.7	OK	OK
	B5	M20 8.8 - 1	C LC_105	0.4	16.8	170.1	0.3	17.8	18.1	OK	OK
	B7	M20 8.8 - 1	C LC_126	3.2	1.4	156.4	2.2	1.4	3.0	OK	OK
	B8	M20 8.8 - 1	C LC_152	11.4	1.6	156.4	8.1	1.7	7.4	OK	OK
	B9	M20 8.8 - 1	C LC_152	12.5	3.2	81.6	8.8	4.0	9.7	OK	OK
	B10	M20 8.8 - 1	C LC_123	9.2	2.5	158.0	6.5	2.7	7.3	OK	OK
	B12	M20 8.8 - 1	C LC_123	5.1	1.0	123.8	3.6	1.5	3.7	OK	OK
	B13	M20 8.8 - 1	C LC_107	5.1	0.3	63.7	3.6	0.4	3.0	OK	OK
	B14	M20 8.8 - 1	C LC_152	0.6	2.3	123.8	0.4	3.7	2.7	OK	OK
	B15	M20 8.8 - 1	C LC_123	6.2	0.6	110.5	4.4	0.6	3.8	OK	OK
	B17	M20 8.8 - 1	C LC_123	9.5	12.9	115.3	6.7	13.7	18.5	OK	OK
	B18	M20 8.8 - 1	C LC_123	4.5	12.6	115.3	3.2	13.4	15.7	OK	OK
	B19	M20 8.8 - 1	C LC_113	0.4	10.6	115.3	0.3	16.6	11.4	OK	OK
	B20	M20 8.8 - 1	C LC_113	0.3	10.7	115.3	0.2	17.9	11.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Cap plate SLN448 raking strut	SLN448 raking strut	◢ 4.2	519	C LC_113	232.4	0.0	-110.3	117.6	-10.5	57.4	31.5	OK	OK
SLN51 column below-w 3	Gusset SLN448 raking strut	◢ 5.6 ◣	288	C LC_105	400.4	2.4	-135.0	-132.2	-172.9	98.9	83.3	OK	OK
		◢ 5.6 ◣	287	C LC_105	400.7	2.5	-133.9	137.1	169.5	99.0	83.9	OK	OK
SLN221 ring-w 1	Flange top b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_126	67.8	0.0	-26.1	4.7	-35.9	16.8	9.2	OK	OK
		◢ 4.2 ◣	198	C LC_127	67.6	0.0	-12.3	-12.0	36.4	16.7	8.9	OK	OK
SLN221 ring-w 1	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	198	C LC_122	88.2	0.0	25.2	27.4	40.4	21.8	13.6	OK	OK
		◢ 4.2 ◣	198	C LC_122	124.7	0.0	-28.9	-13.9	68.7	30.8	17.9	OK	OK
SLN426 ring-w 1	Flange top a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_123	91.1	0.0	-40.5	10.9	45.9	22.5	13.5	OK	OK
		◢ 4.2 ◣	198	C LC_123	61.6	0.0	29.2	22.2	22.1	15.2	9.9	OK	OK
SLN426 ring-w 1	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	198	C LC_122	94.7	0.0	27.6	31.6	-41.7	23.4	13.2	OK	OK
		◢ 4.2 ◣	198	C LC_121	152.9	0.0	65.1	-58.5	54.3	37.8	22.4	OK	OK
Cap plate SLN448 raking strut	Fin plate SLN448 raking strut	◢ 5.6 ◣	139	C LC_123	250.8	0.0	-113.7	-102.8	78.1	62.0	43.9	OK	OK
		◢ 5.6 ◣	139	C LC_113	201.1	0.0	-82.7	84.5	63.8	49.7	30.3	OK	OK
Cap plate SLN448 raking strut	Fin plate 2 SLN448 raking strut	◢ 5.6 ◣	139	C LC_113	223.4	0.0	-90.4	-96.4	67.9	55.2	30.5	OK	OK
		◢ 5.6 ◣	139	C LC_113	248.0	0.0	-109.3	103.3	-76.5	61.3	38.7	OK	OK
End plate column below SLN51 column below	SLN51 column below	◢ 5.6 ◣	521	C LC_121	396.6	0.0	-253.9	-172.3	-35.3	98.0	53.2	OK	OK
		◢ 5.6 ◣	521	C LC_148	335.0	0.0	-106.7	175.5	52.9	82.8	43.8	OK	OK
End plate column above SLN96 column above	SLN96 column above	◢ 5.6 ◣	521	C LC_152	192.3	0.0	-90.1	-98.0	-4.3	47.5	26.9	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 5.6 ◣	521	C LC_152	225.8	0.0	-115.9	111.5	9.7	55.8	32.4	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN221 ring-w 1	◢ 7.2	189	C LC_107	263.7	0.0	27.0	-32.9	-147.8	65.1	38.1	OK	OK
SLN220+SLN203 radial (continuous)-w 1	SLN426 ring-w 1	◢ 7.2	189	C LC_123	377.1	0.0	-11.1	16.5	217.0	93.2	62.9	OK	OK
End plate SLN221 ring	SLN221 ring-w 1	◢ 4.2 ◣	149	C LC_122	208.3	0.0	-43.8	-47.8	-107.4	51.5	28.4	OK	OK
		◢ 4.2 ◣	149	C LC_122	153.8	0.0	-39.2	36.7	77.6	38.0	22.7	OK	OK
End plate SLN426 ring	SLN426 ring-w 1	◢ 4.2 ◣	149	C LC_123	269.2	0.0	-50.5	-52.7	143.3	66.5	40.2	OK	OK
		◢ 4.2 ◣	149	C LC_123	194.3	0.0	-43.2	42.1	-101.0	48.0	30.0	OK	OK
SLN221 ring-w 1	Flange top a end plate SLN221 ring	◢ 4.2 ◣	199	C LC_123	95.5	0.0	-50.5	22.0	-41.3	23.6	15.5	OK	OK
		◢ 4.2 ◣	199	C LC_107	75.6	0.0	-5.5	24.3	36.1	18.7	10.1	OK	OK
SLN221 ring-w 1	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	198	C LC_123	84.3	0.0	15.5	-15.8	45.1	20.8	12.4	OK	OK
		◢ 4.2 ◣	199	C LC_122	130.4	0.0	-30.1	20.1	70.4	32.2	16.8	OK	OK
SLN426 ring-w 1	Flange top b end plate SLN426 ring	◢ 4.2 ◣	199	C LC_121	144.2	0.0	-76.5	32.9	62.4	35.6	24.3	OK	OK
		◢ 4.2 ◣	198	C LC_125	95.8	0.0	17.3	20.0	50.6	23.7	13.5	OK	OK
SLN426 ring-w 1	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	198	C LC_121	124.8	0.0	23.6	-23.5	-66.7	30.8	18.3	OK	OK
		◢ 4.2 ◣	199	C LC_123	135.1	0.0	-40.8	21.7	-71.1	33.4	20.4	OK	OK
End plate SLN221 ring	Flange top a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_127	122.4	0.0	-45.0	56.7	33.3	30.2	17.9	OK	OK
		◢ 4.2 ◣	58	C LC_123	159.2	0.0	-73.2	-59.1	-56.3	39.3	21.1	OK	OK
End plate SLN221 ring	Flange top b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_125	92.5	0.0	-38.5	42.0	-24.3	22.8	14.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	58	C LC_107	101.8	0.0	-57.8	-40.8	26.0	25.2	17.6	OK	OK
End plate SLN221 ring	Flange bottom a end plate SLN221 ring	◢ 4.2 ◣	58	C LC_127	144.3	0.0	-63.4	52.5	53.4	35.7	19.4	OK	OK
		◢ 4.2 ◣	58	C LC_127	119.3	0.0	-43.4	-54.2	-34.3	29.5	17.5	OK	OK
End plate SLN221 ring	Flange bottom b end plate SLN221 ring	◢ 4.2 ◣	58	C LC_122	89.7	0.0	13.9	-44.1	-25.9	22.2	13.2	OK	OK
		◢ 4.2 ◣	58	C LC_122	93.9	0.0	-19.2	-49.5	-19.2	23.2	12.1	OK	OK
End plate SLN426 ring	Flange top a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	119.6	0.0	-23.9	60.5	-30.4	29.5	18.0	OK	OK
		◢ 4.2 ◣	58	C LC_123	63.5	0.0	-7.7	28.9	-22.2	15.7	11.8	OK	OK
End plate SLN426 ring	Flange top b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	121.8	0.0	-32.2	64.5	20.9	30.1	15.0	OK	OK
		◢ 4.2 ◣	58	C LC_123	76.0	0.0	-9.7	22.7	37.2	18.8	10.4	OK	OK
End plate SLN426 ring	Flange bottom a end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	99.2	0.0	9.4	-44.9	35.1	24.5	18.2	OK	OK
		◢ 4.2 ◣	58	C LC_123	135.2	0.0	-34.2	-69.7	29.0	33.4	20.6	OK	OK
End plate SLN426 ring	Flange bottom b end plate SLN426 ring	◢ 4.2 ◣	58	C LC_123	111.8	0.0	5.6	-52.0	-38.1	27.6	14.7	OK	OK
		◢ 4.2 ◣	58	C LC_123	129.3	0.0	-20.1	-66.5	-32.0	32.0	16.5	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring

Name	SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

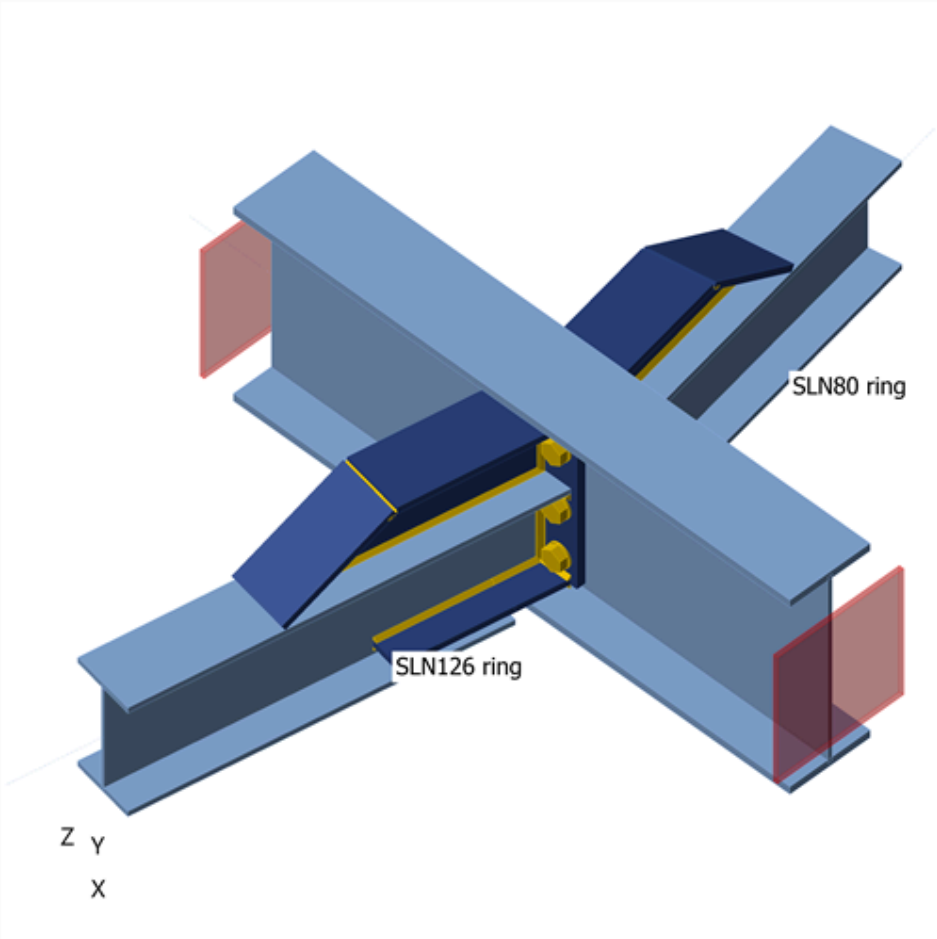
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73b+SLN73a radial (continuous)	1 - UB457x191x89	0.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN73b+SLN73a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN73b+SLN73a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN126 ring / end		Position	0



SLN73b+SLN73a radial (continuous) - SLN80 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN80 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN80 ring / End	-2.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.7	0.1	-25.6	0.0	0.0	0.1
C LC_103	SLN80 ring / End	-8.9	-0.2	-24.0	0.0	0.0	-0.2
	SLN126 ring / End	-8.7	0.2	-24.0	0.0	0.0	0.2
C LC_104	SLN80 ring / End	-4.4	0.1	-24.0	0.0	0.0	0.0
	SLN126 ring / End	-4.4	0.0	-24.0	0.0	0.0	0.1
C LC_105	SLN80 ring / End	-6.8	0.0	-24.0	0.0	0.0	-0.1
	SLN126 ring / End	-7.1	0.1	-24.0	0.0	0.0	0.1
C LC_106	SLN80 ring / End	-6.4	-0.2	-24.0	0.0	0.0	-0.2
	SLN126 ring / End	-6.1	-0.1	-24.0	0.0	0.0	0.0
C LC_107	SLN80 ring / End	-5.9	0.0	-0.5	0.0	0.0	0.0
	SLN126 ring / End	-5.7	0.1	-0.5	0.0	0.0	0.1
C LC_108	SLN80 ring / End	0.3	0.2	-0.5	0.0	0.0	0.3
	SLN126 ring / End	0.1	0.1	-0.5	0.0	0.0	0.1
C LC_109	SLN80 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.2
	SLN126 ring / End	-3.8	0.1	-0.5	0.0	0.0	0.1
C LC_110	SLN80 ring / End	-2.0	-0.2	-0.5	0.0	0.0	-0.2
	SLN126 ring / End	-1.8	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN80 ring / End	-10.2	-0.1	-16.2	0.0	0.0	-0.2
	SLN126 ring / End	-9.9	0.1	-16.2	0.0	0.0	0.2
C LC_112	SLN80 ring / End	-3.1	0.2	-16.2	0.0	0.0	0.2
	SLN126 ring / End	-3.1	0.1	-16.2	0.0	0.0	0.1
C LC_113	SLN80 ring / End	-7.3	0.1	-16.2	0.0	0.0	0.1
	SLN126 ring / End	-7.6	0.1	-16.2	0.0	0.0	0.1
C LC_114	SLN80 ring / End	-6.0	-0.2	-16.2	0.0	0.0	-0.3
	SLN126 ring / End	-5.6	-0.2	-16.2	0.0	0.0	-0.2
C LC_115	SLN80 ring / End	-5.7	0.0	-1.7	0.0	0.0	-0.1
	SLN126 ring / End	-5.5	0.1	-1.7	0.0	0.0	0.1
C LC_116	SLN80 ring / End	0.5	0.2	-1.7	0.0	0.0	0.3
	SLN126 ring / End	0.4	0.1	-1.7	0.0	0.0	0.1
C LC_117	SLN80 ring / End	-3.4	0.1	-1.7	0.0	0.0	0.1
	SLN126 ring / End	-3.6	0.1	-1.7	0.0	0.0	0.1
C LC_118	SLN80 ring / End	-1.8	-0.2	-1.7	0.0	0.0	-0.2
	SLN126 ring / End	-1.6	-0.2	-1.7	0.0	0.0	-0.3
C LC_119	SLN80 ring / End	-6.6	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-6.5	0.2	-25.6	0.0	0.0	0.2
C LC_120	SLN80 ring / End	-3.6	0.1	-25.6	0.0	0.0	0.0
	SLN126 ring / End	-3.6	0.0	-25.6	0.0	0.0	0.1
C LC_121	SLN80 ring / End	-4.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-5.1	0.0	-25.6	0.0	0.0	0.1
C LC_122	SLN80 ring / End	-4.6	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.2	0.0	-25.6	0.0	0.0	0.1
C LC_123	SLN80 ring / End	-5.6	-0.1	-19.4	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN126 ring / End	-5.4	0.2	-19.4	0.0	0.0	0.2
C LC_124	SLN80 ring / End	-1.5	0.1	-19.4	0.0	0.0	0.1
	SLN126 ring / End	-1.6	0.1	-19.4	0.0	0.0	0.2
C LC_125	SLN80 ring / End	-3.3	-0.1	-19.4	0.0	0.0	-0.2
	SLN126 ring / End	-3.6	0.0	-19.4	0.0	0.0	0.0
C LC_126	SLN80 ring / End	-2.2	-0.2	-19.4	0.0	0.0	-0.2
	SLN126 ring / End	-1.8	0.0	-19.4	0.0	0.0	0.0
C LC_127	SLN80 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	-1.0	0.1	-4.9	0.0	0.0	0.1
C LC_128	SLN80 ring / End	2.1	0.1	-4.9	0.0	0.0	0.1
	SLN126 ring / End	2.0	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN80 ring / End	0.6	-0.1	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN80 ring / End	1.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN126 ring / End	2.2	0.0	-4.9	0.0	0.0	-0.1
C LC_135	SLN80 ring / End	0.5	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	0.6	0.1	-25.6	0.0	0.0	0.1
C LC_141	SLN80 ring / End	0.6	-0.1	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.1
C LC_142	SLN80 ring / End	1.0	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	1.0	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN80 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN80 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.1
C LC_145	SLN80 ring / End	-3.6	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-3.5	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN80 ring / End	-2.0	0.0	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-1.9	0.0	-25.6	0.0	0.0	0.1
C LC_147	SLN80 ring / End	-2.6	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.7	0.1	-25.6	0.0	0.0	0.1
C LC_148	SLN80 ring / End	-3.0	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.7	0.1	-25.6	0.0	0.0	0.2
C LC_151	SLN80 ring / End	-0.9	-0.1	-11.7	0.0	0.0	-0.1
	SLN126 ring / End	-0.9	0.1	-11.7	0.0	0.0	0.1
C LC_152	SLN80 ring / End	-5.7	-0.1	-14.1	0.0	0.0	-0.2
	SLN126 ring / End	-5.7	0.1	-14.2	0.0	0.0	0.1
C LC_153	SLN80 ring / End	-2.9	-0.1	-20.9	0.0	0.0	-0.2
	SLN126 ring / End	-2.9	0.1	-20.9	0.0	0.0	0.1
C LC_154	SLN80 ring / End	-0.9	-0.1	-11.7	0.0	0.0	-0.2
	SLN126 ring / End	-0.8	0.1	-11.7	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	18.0 < 100%	OK
Buckling	Not calculated	

Plates

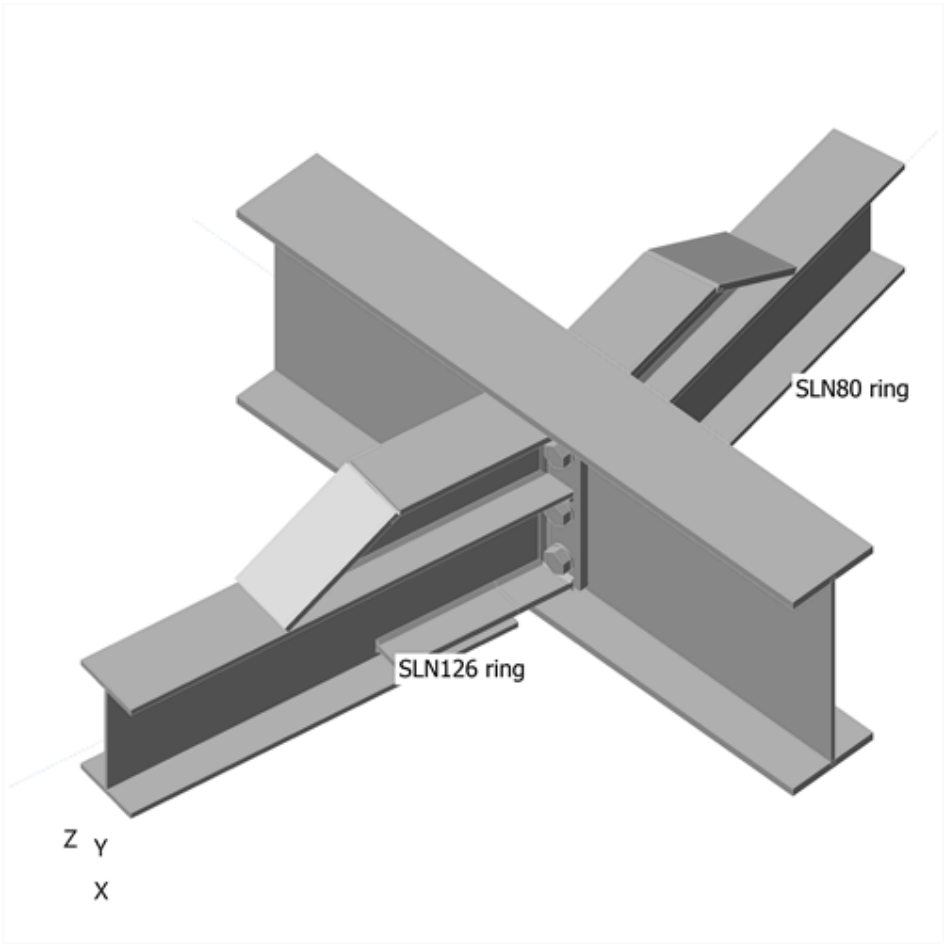
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73b+SLN73a radial (continuous)-bfl 1	17.7	C LC_121	22.2	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-tfl 1	17.7	C LC_122	18.7	0.0	0.0	OK
SLN73b+SLN73a radial (continuous)-w 1	10.5	C LC_122	26.2	0.0	0.4	OK
SLN80 ring-bfl 1	12.7	C LC_135	46.6	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_121	47.8	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_119	46.8	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_135	46.7	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_119	47.6	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_145	6.8	0.0	1.3	OK
Bottom flange plate SLN80 ring	15.0	C LC_135	13.7	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_135	13.3	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_135	12.9	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_119	31.5	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_145	5.6	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_119	6.8	0.0	0.9	OK
Bottom flange plate SLN126 ring	15.0	C LC_135	13.7	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_135	13.0	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_135	13.2	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_119	32.4	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_145	5.4	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_119	14.3	0.0	0.0	OK

Design data

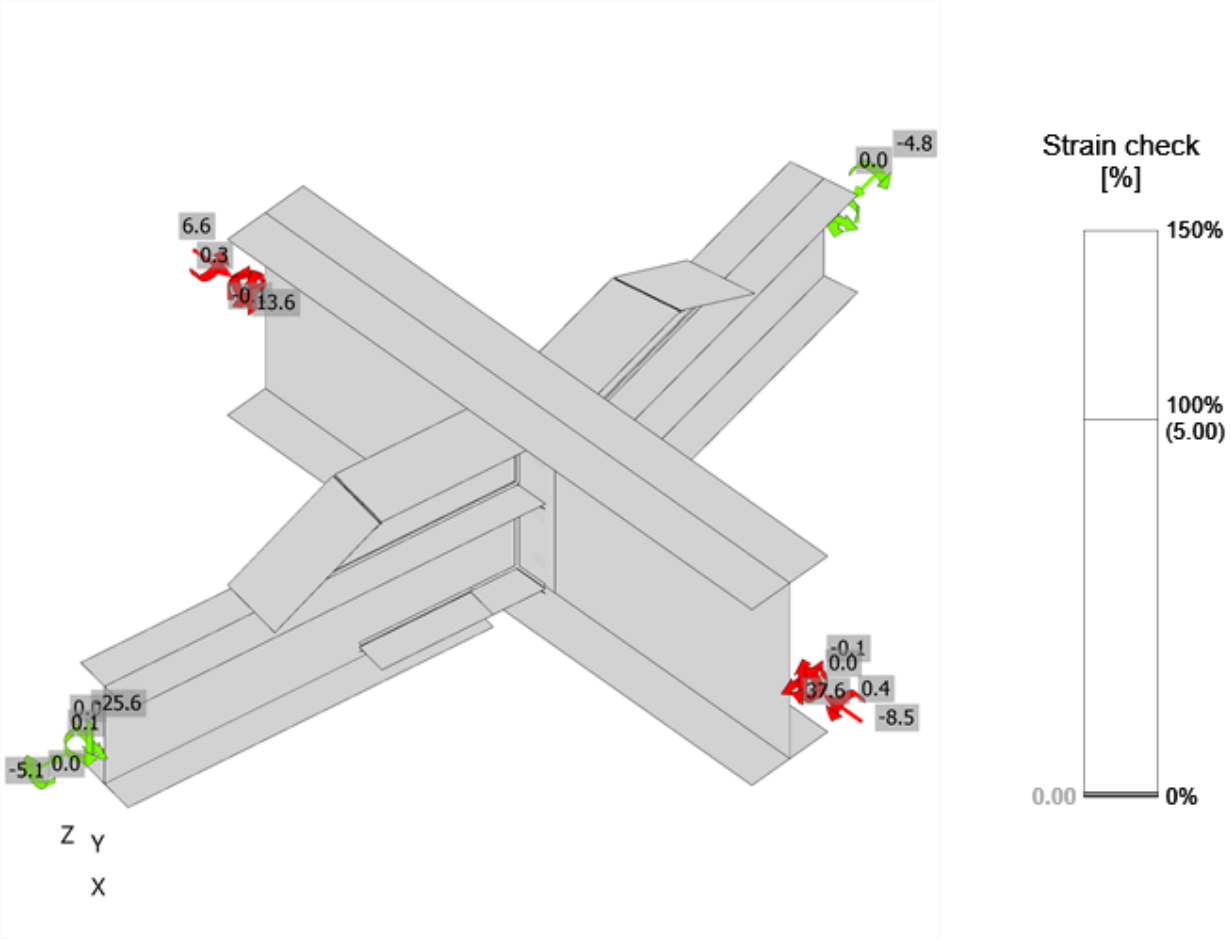
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

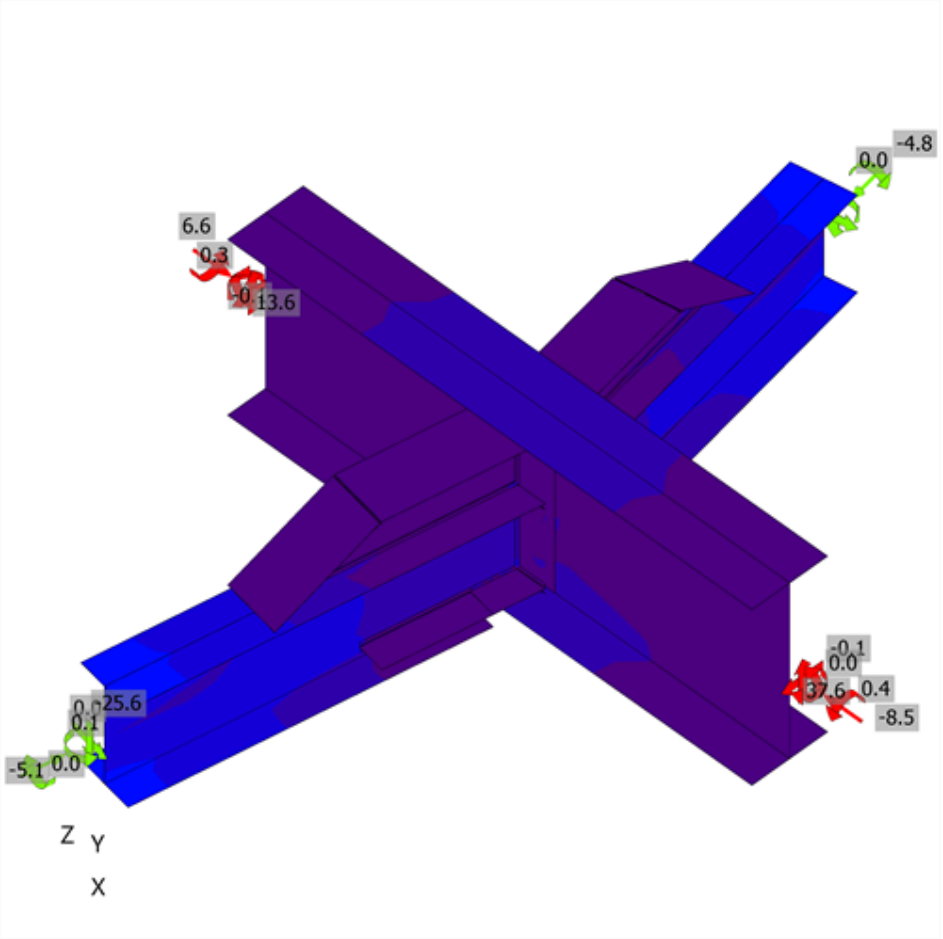
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

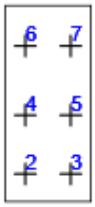


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_145	0.9	0.4	176.5	0.3	0.5	0.4	OK	OK
	B3	M30 8.8 - 1	C LC_135	0.8	0.5	176.5	0.3	0.6	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_122	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_148	2.4	0.8	229.5	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_120	1.6	0.9	229.5	0.5	0.8	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 10.5	149	C LC_119	11.7	0.0	-10.6	-2.8	-0.4	3.4	3.4	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 10.5	149	C LC_119	12.3	0.0	-5.2	4.7	4.3	3.4	3.4	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_135	10.4	0.0	0.7	-0.6	5.9	2.6	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	8.6	0.0	-0.9	0.0	-4.9	2.1	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.4	0.0	1.6	1.3	11.7	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	21.3	0.0	2.1	-2.4	-12.0	5.3	5.3	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_135	16.7	0.0	4.1	3.9	8.5	4.1	4.1	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	6.5	2.1	-10.8	5.0	5.0	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_135	17.3	0.0	3.9	4.1	-8.8	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.9	0.0	5.5	2.6	10.7	4.9	4.9	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_119	72.4	0.0	-17.3	-18.2	36.3	17.9	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.0	0.0	-17.9	16.9	-34.6	17.1	9.8	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_135	8.2	0.0	-0.8	0.1	4.7	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	10.6	0.0	0.7	0.6	-6.1	2.6	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_135	21.3	0.0	2.0	1.9	12.1	5.3	5.3	OK	OK
		◢ 6.0 ◣	100	C LC_147	20.3	0.0	1.7	-1.9	-11.5	5.0	5.0	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_135	17.7	0.0	4.5	4.2	9.0	4.4	4.4	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.5	0.0	6.0	2.4	-10.4	4.8	4.8	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_135	16.3	0.0	3.6	3.9	-8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.6	0.0	6.0	2.4	11.1	5.1	5.1	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_119	72.9	0.0	-19.2	-19.1	35.8	18.0	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	68.7	0.0	-16.0	16.1	-35.1	17.0	9.8	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_118	11.8	0.0	-3.4	-4.8	4.4	2.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	16.3	0.0	-7.3	5.1	-6.8	4.0	4.0	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_120	6.8	0.0	0.7	0.7	3.9	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	10.5	0.0	1.8	-1.7	-5.7	2.6	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_110	7.1	0.0	-3.5	-2.4	2.6	1.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	5.3	0.0	-1.5	2.5	-1.5	1.3	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_145	4.0	0.0	1.5	1.4	1.6	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_145	3.9	0.0	-0.2	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_119	57.6	0.0	-10.1	-8.9	31.5	14.2	8.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	55.7	0.0	-9.7	10.9	-29.7	13.8	8.5	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	12.5	0.0	-5.6	-6.2	1.7	3.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	17.1	0.0	-8.9	7.8	-3.1	4.2	4.2	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_120	10.3	0.0	1.8	1.8	5.6	2.5	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.5	0.0	0.7	-0.6	-4.3	1.9	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	6.6	0.0	-3.5	-2.8	1.6	1.6	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_128	5.3	0.0	-1.6	2.4	1.6	1.3	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_145	3.5	0.0	-0.2	-0.2	-2.0	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.7	0.0	0.0	0.0	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_119	60.0	0.0	-11.3	-11.3	32.1	14.8	9.0	OK	OK
		◢ 6.0 ◣	271	C LC_119	53.2	0.0	-8.4	8.4	-29.2	13.2	8.4	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_135	40.0	0.0	9.6	10.3	-19.9	9.9	9.1	OK	OK
SLN73b+SLN73a radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_135	38.9	0.0	9.6	7.7	-20.3	9.6	9.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_145	2.7	0.0	-0.6	1.4	0.6	0.7	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_121	7.0	0.0	-5.0	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_145	2.5	0.0	-0.5	1.3	-0.6	0.6	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_020
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring

Name	SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

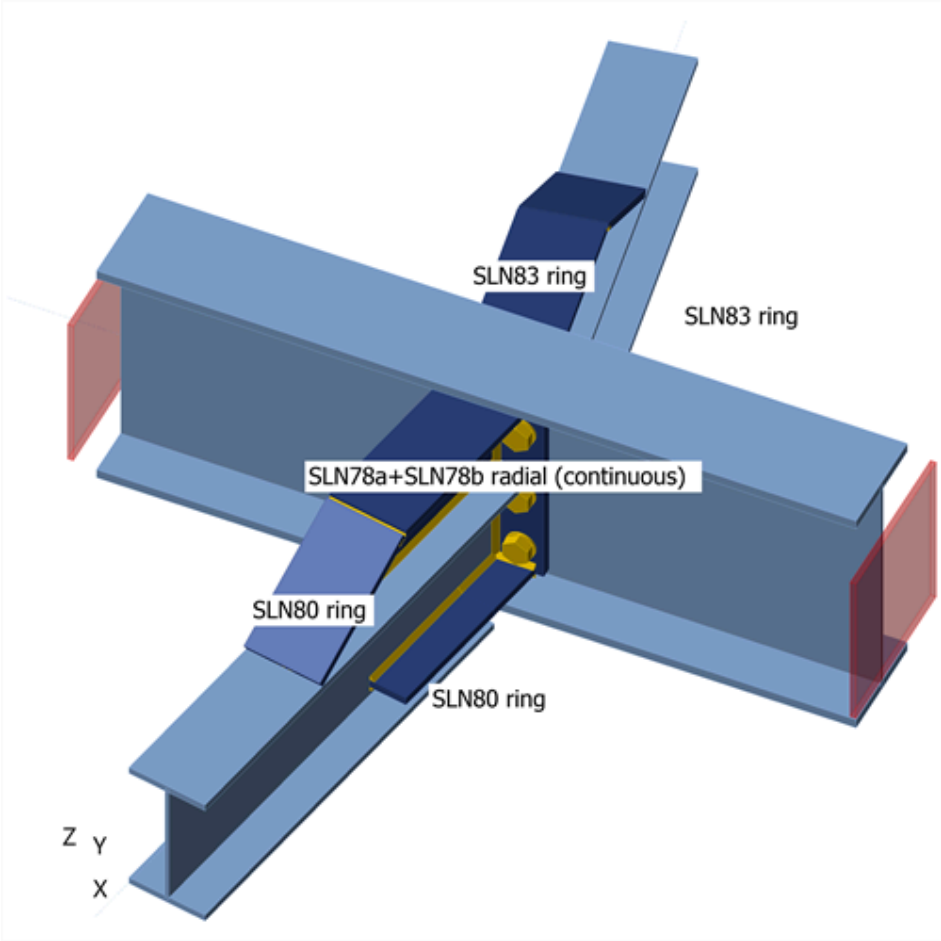
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN78a+SLN78b radial (continuous)	1 - UB457x191x89	20.0	0.0	0.0	0	0	0
SLN80 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102
SLN83 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN78a+SLN78b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN78a+SLN78b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN80 ring / end		Position	0
SLN83 ring / end		Position	0



SLN78a+SLN78b radial (continuous) - SLN80 ring - SLN83 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN80 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.8	0.0	-2.2	0.0	0.0	0.0
C LC_102	SLN80 ring / End	-2.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-2.8	0.0	-8.8	0.0	0.0	0.0
C LC_103	SLN80 ring / End	-8.9	0.0	-24.0	0.0	0.0	-0.1
	SLN83 ring / End	-8.9	-0.1	-8.3	0.0	0.0	0.0
C LC_104	SLN80 ring / End	-4.4	-0.1	-24.0	0.0	0.0	0.0
	SLN83 ring / End	-4.6	0.0	-8.3	0.0	0.0	0.0
C LC_105	SLN80 ring / End	-6.7	-0.1	-24.0	0.0	0.0	0.0
	SLN83 ring / End	-6.0	-0.1	-8.3	0.0	0.0	0.0
C LC_106	SLN80 ring / End	-6.6	-0.1	-24.0	0.0	0.0	-0.2
	SLN83 ring / End	-7.1	-0.1	-8.3	0.0	0.0	-0.1
C LC_107	SLN80 ring / End	-5.9	0.0	-0.5	0.0	0.0	0.0
	SLN83 ring / End	-6.0	0.0	-0.6	0.0	0.0	0.1
C LC_108	SLN80 ring / End	0.3	0.2	-0.5	0.0	0.0	0.2
	SLN83 ring / End	0.0	0.0	-0.6	0.0	0.0	0.0
C LC_109	SLN80 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.3
	SLN83 ring / End	-2.3	0.0	-0.6	0.0	0.0	0.1
C LC_110	SLN80 ring / End	-2.0	-0.2	-0.5	0.0	0.0	-0.3
	SLN83 ring / End	-2.7	-0.1	-0.6	0.0	0.0	-0.2
C LC_111	SLN80 ring / End	-10.2	0.0	-16.2	0.0	0.0	-0.1
	SLN83 ring / End	-10.3	0.0	-5.8	0.0	0.0	0.0
C LC_112	SLN80 ring / End	-3.1	0.1	-16.2	0.0	0.0	0.1
	SLN83 ring / End	-3.4	0.0	-5.8	0.0	0.0	0.0
C LC_113	SLN80 ring / End	-7.2	0.1	-16.2	0.0	0.0	0.2
	SLN83 ring / End	-5.8	0.0	-5.8	0.0	0.0	0.1
C LC_114	SLN80 ring / End	-6.1	-0.2	-16.2	0.0	0.0	-0.3
	SLN83 ring / End	-6.9	-0.1	-5.8	0.0	0.0	-0.2
C LC_115	SLN80 ring / End	-5.7	0.0	-1.7	0.0	0.0	0.0
	SLN83 ring / End	-5.9	0.0	-1.1	0.0	0.0	0.1
C LC_116	SLN80 ring / End	0.5	0.1	-1.7	0.0	0.0	0.2
	SLN83 ring / End	0.3	0.0	-1.1	0.0	0.0	0.0
C LC_117	SLN80 ring / End	-3.4	0.1	-1.7	0.0	0.0	0.3
	SLN83 ring / End	-2.0	0.0	-1.1	0.0	0.0	0.1
C LC_118	SLN80 ring / End	-1.9	-0.2	-1.7	0.0	0.0	-0.3
	SLN83 ring / End	-2.5	-0.1	-1.1	0.0	0.0	-0.2
C LC_119	SLN80 ring / End	-6.6	0.0	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-6.6	-0.1	-8.8	0.0	0.0	0.0
C LC_120	SLN80 ring / End	-3.6	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-3.6	0.0	-8.8	0.0	0.0	0.1
C LC_121	SLN80 ring / End	-4.7	-0.2	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-4.7	-0.1	-8.8	0.0	0.0	-0.1
C LC_122	SLN80 ring / End	-4.7	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-4.9	-0.1	-8.8	0.0	0.0	0.0
C LC_123	SLN80 ring / End	-5.5	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN83 ring / End	-5.6	0.0	-6.8	0.0	0.0	0.0
C LC_124	SLN80 ring / End	-1.5	0.0	-19.4	0.0	0.0	0.1
	SLN83 ring / End	-1.4	0.1	-6.8	0.0	0.0	0.2
C LC_125	SLN80 ring / End	-3.2	-0.1	-19.4	0.0	0.0	-0.2
	SLN83 ring / End	-3.2	-0.1	-6.8	0.0	0.0	-0.1
C LC_126	SLN80 ring / End	-2.3	-0.1	-19.4	0.0	0.0	-0.2
	SLN83 ring / End	-2.5	-0.1	-6.8	0.0	0.0	0.0
C LC_127	SLN80 ring / End	-1.1	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	-1.2	0.0	-2.2	0.0	0.0	0.1
C LC_128	SLN80 ring / End	2.1	0.1	-4.9	0.0	0.0	0.1
	SLN83 ring / End	2.3	0.2	-2.2	0.0	0.0	0.2
C LC_129	SLN80 ring / End	0.6	-0.1	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
C LC_130	SLN80 ring / End	1.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	1.9	0.0	-2.2	0.0	0.0	0.0
C LC_135	SLN80 ring / End	0.5	-0.1	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	0.6	0.0	-8.8	0.0	0.0	0.1
C LC_141	SLN80 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
C LC_142	SLN80 ring / End	1.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.9	0.0	-2.2	0.0	0.0	0.0
C LC_143	SLN80 ring / End	0.9	0.0	-4.9	0.0	0.0	-0.1
	SLN83 ring / End	0.9	0.0	-2.2	0.0	0.0	0.0
C LC_144	SLN80 ring / End	0.7	0.0	-4.9	0.0	0.0	0.0
	SLN83 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
C LC_145	SLN80 ring / End	-3.6	0.0	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-3.5	-0.1	-8.8	0.0	0.0	0.0
C LC_146	SLN80 ring / End	-2.0	-0.2	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-2.2	0.0	-8.8	0.0	0.0	0.0
C LC_147	SLN80 ring / End	-2.4	-0.2	-25.6	0.0	0.0	-0.2
	SLN83 ring / End	-2.3	-0.1	-8.8	0.0	0.0	0.0
C LC_148	SLN80 ring / End	-3.2	-0.1	-25.6	0.0	0.0	-0.1
	SLN83 ring / End	-3.3	0.0	-8.8	0.0	0.0	0.0
C LC_151	SLN80 ring / End	-0.9	-0.1	-11.7	0.0	0.0	-0.1
	SLN83 ring / End	-0.9	0.0	-8.8	0.0	0.0	0.0
C LC_152	SLN80 ring / End	-5.7	-0.1	-14.2	0.0	0.0	-0.1
	SLN83 ring / End	-5.8	-0.1	-2.2	0.0	0.0	0.0
C LC_153	SLN80 ring / End	-2.9	-0.1	-21.0	0.0	0.0	-0.2
	SLN83 ring / End	-3.0	0.0	-8.8	0.0	0.0	0.0
C LC_154	SLN80 ring / End	-0.9	-0.1	-11.7	0.0	0.0	-0.1
	SLN83 ring / End	-0.9	0.0	-8.8	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.6 < 100%	OK
Welds	17.5 < 100%	OK
Buckling	Not calculated	

Plates

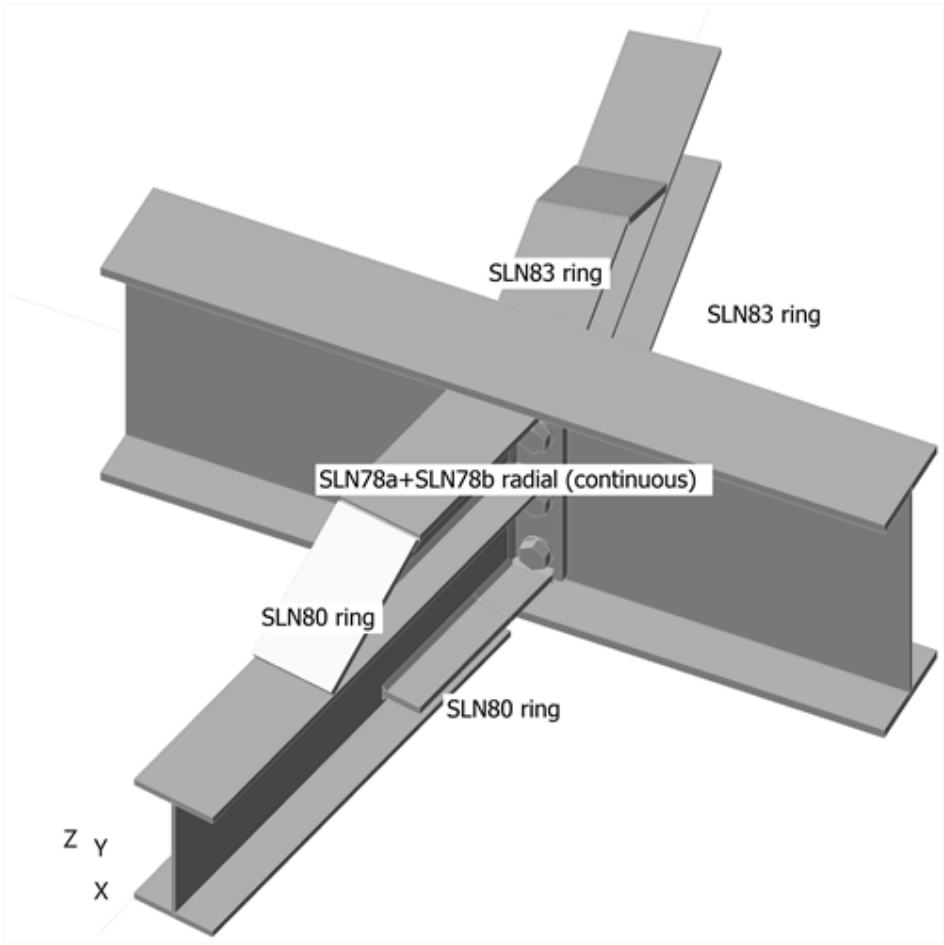
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN78a+SLN78b radial (continuous)-bfl 1	17.7	C LC_147	15.0	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-tfl 1	17.7	C LC_147	12.6	0.0	0.0	OK
SLN78a+SLN78b radial (continuous)-w 1	10.5	C LC_121	18.4	0.0	0.6	OK
SLN80 ring-bfl 1	12.7	C LC_135	46.5	0.0	0.0	OK
SLN80 ring-tfl 1	12.7	C LC_119	47.9	0.0	0.0	OK
SLN80 ring-w 1	7.2	C LC_119	47.0	0.0	0.0	OK
SLN83 ring-bfl 1	12.7	C LC_135	16.3	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_119	17.5	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_119	16.9	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_114	5.1	0.0	5.4	OK
Bottom flange plate SLN83 ring	15.0	C LC_114	12.0	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_135	4.8	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_135	4.8	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_119	11.6	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_110	4.5	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_119	5.3	0.0	0.0	OK
End plate SLN80 ring	20.0	C LC_109	4.6	0.0	5.2	OK
Bottom flange plate SLN80 ring	15.0	C LC_135	13.6	0.0	0.0	OK
Bottom flange half plate L SLN80 ring	15.0	C LC_135	13.1	0.0	0.0	OK
Bottom flange half plate R SLN80 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Haunch web SLN80 ring	12.0	C LC_119	31.3	0.0	0.0	OK
Haunch top plate SLN80 ring	15.0	C LC_121	5.7	0.0	0.0	OK
Haunch sloping plate SLN80 ring	15.0	C LC_119	14.2	0.0	0.0	OK

Design data

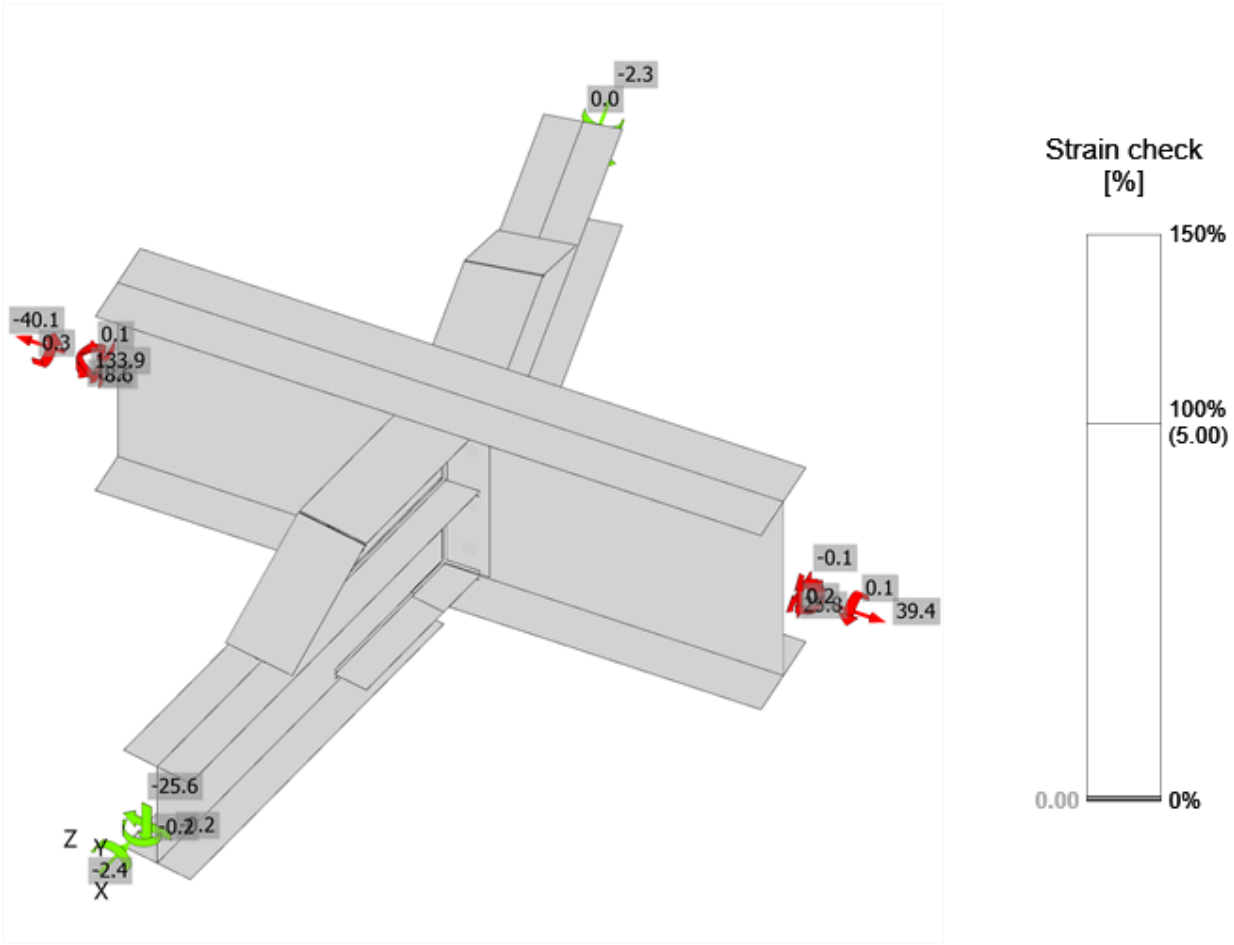
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

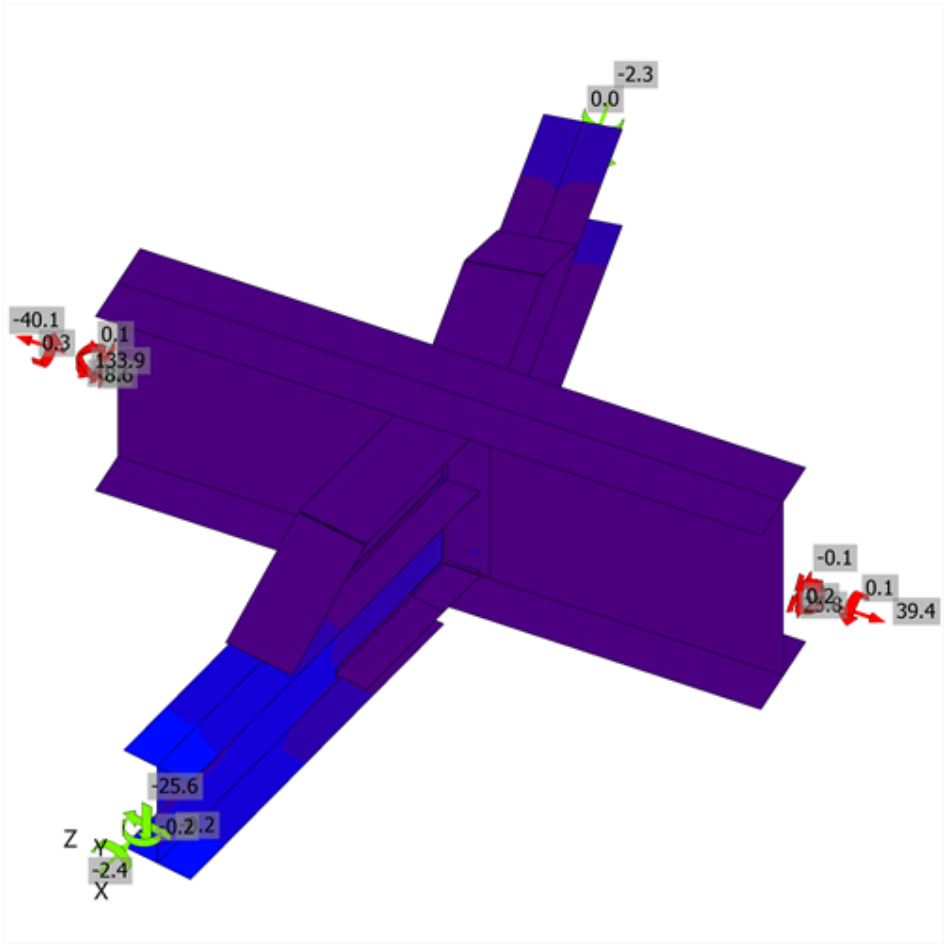
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

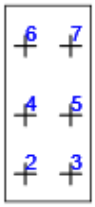


Strain check, C LC_147



Equivalent stress, C LC_147

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	0.5	0.3	176.5	0.2	0.3	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_135	1.0	0.4	176.5	0.3	0.4	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.3	213.9	0.0	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.3	213.9	0.0	0.3	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.5	0.6	229.5	0.5	0.5	0.6	OK	OK
	B7	M30 8.8 - 1	C LC_147	1.4	0.6	229.5	0.4	0.5	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN80 ring	Bottom flange plate SLN80 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-tfl 1	◢ 10.5	149	C LC_119	8.1	0.0	-8.0	-0.4	0.2	2.6	2.6	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-tfl 1	◢ 10.5	149	C LC_119	7.1	0.0	-5.6	-2.5	-0.1	1.8	0.0	OK	OK
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	208	C LC_135	6.9	0.0	0.1	-1.0	3.9	1.7	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_120	4.9	0.0	-0.9	0.2	-2.8	1.2	0.0	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_135	7.4	0.0	0.7	0.8	4.1	1.8	1.8	OK	OK
		◢ 6.0 ◣	100	C LC_121	7.8	0.0	0.6	-1.1	-4.4	1.9	1.9	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_135	6.0	0.0	1.3	1.4	3.0	1.5	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	7.3	0.0	2.0	0.9	-3.9	1.8	0.0	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_135	6.2	0.0	1.6	1.5	-3.1	1.5	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	7.0	0.0	2.3	0.7	3.7	1.7	0.0	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_119	25.5	0.0	-6.3	-6.1	12.9	6.3	6.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	26.2	0.0	-6.4	6.6	-13.1	6.5	6.0	OK	OK
End plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	208	C LC_135	6.5	0.0	-3.5	-3.1	0.7	1.6	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	7.7	0.0	0.6	0.4	-4.4	1.9	0.0	OK	OK
Bottom flange plate SLN80 ring	SLN80 ring-w 1	◢ 6.0 ◣	100	C LC_135	20.4	0.0	1.7	1.3	11.7	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_147	21.8	0.0	1.9	-2.5	-12.3	5.4	5.4	OK	OK
SLN80 ring-w 1	Bottom flange half plate L SLN80 ring	◢ 6.0 ◣	243	C LC_135	16.2	0.0	3.9	3.8	8.2	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.1	0.0	6.6	2.1	-10.8	5.0	5.0	OK	OK
SLN80 ring-w 1	Bottom flange half plate R SLN80 ring	◢ 6.0 ◣	243	C LC_135	17.7	0.0	4.1	4.2	-9.0	4.4	4.4	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.6	0.0	5.4	2.7	10.6	4.8	4.8	OK	OK
SLN80 ring-tfl 1	Haunch web SLN80 ring	◢ 6.0 ◣	572	C LC_121	70.8	0.0	-16.7	-17.4	35.7	17.5	10.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	71.0	0.0	-18.0	17.9	-35.4	17.5	10.0	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	13.7	0.0	-3.7	-5.8	5.0	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_114	17.0	0.0	-7.5	5.4	-7.0	4.2	4.2	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_135	3.3	0.0	-0.4	0.3	-1.9	0.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	5.0	0.0	0.8	-0.9	-2.7	1.2	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_114	7.9	0.0	-3.8	-2.5	3.1	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	5.7	0.0	-1.4	2.7	-1.7	1.4	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_114	1.7	0.0	0.5	0.4	0.8	0.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_124	1.5	0.0	0.6	-0.5	-0.7	0.4	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_119	20.3	0.0	-3.4	-3.6	11.0	5.0	5.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_119	21.0	0.0	-3.7	3.5	-11.4	5.2	5.2	OK	OK
End plate SLN80 ring	Bottom flange plate SLN80 ring	◢ 6.0 ◣	149	C LC_109	14.7	0.0	-4.5	-6.0	-5.4	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_109	16.5	0.0	-7.1	5.6	6.6	4.1	4.1	OK	OK
End plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	145	C LC_120	9.2	0.0	1.1	1.3	5.1	2.3	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	6.6	0.0	0.8	-0.7	-3.7	1.6	0.0	OK	OK
End plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	149	C LC_109	7.9	0.0	-3.8	-2.8	-2.8	1.9	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_109	5.9	0.0	-1.7	2.8	1.7	1.5	0.0	OK	OK
Haunch top plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	342	C LC_121	4.2	0.0	1.5	1.4	1.7	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	3.7	0.0	-0.2	0.2	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch web SLN80 ring	◢ 6.0 ◣	271	C LC_119	56.0	0.0	-9.6	-9.3	30.5	13.8	8.6	OK	OK
		◢ 6.0 ◣	271	C LC_119	57.2	0.0	-10.1	10.4	-30.8	14.1	8.7	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_135	22.4	0.0	3.5	4.6	-11.9	5.5	5.5	OK	OK
SLN78a+SLN78b radial (continuous)-w 1	SLN80 ring-w 1	◢ 7.2	208	C LC_135	35.1	0.0	9.8	-11.9	15.3	8.7	8.7	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_121	2.4	0.0	-1.8	0.9	0.1	0.7	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	1.5	0.0	-0.3	0.8	-0.2	0.4	0.0	OK	OK
Haunch sloping plate SLN80 ring	Haunch top plate SLN80 ring	◢ 6.0 ◣	146	C LC_121	7.4	0.0	-5.3	3.0	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	2.7	0.0	-0.6	1.5	0.4	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_080
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring

Name	SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

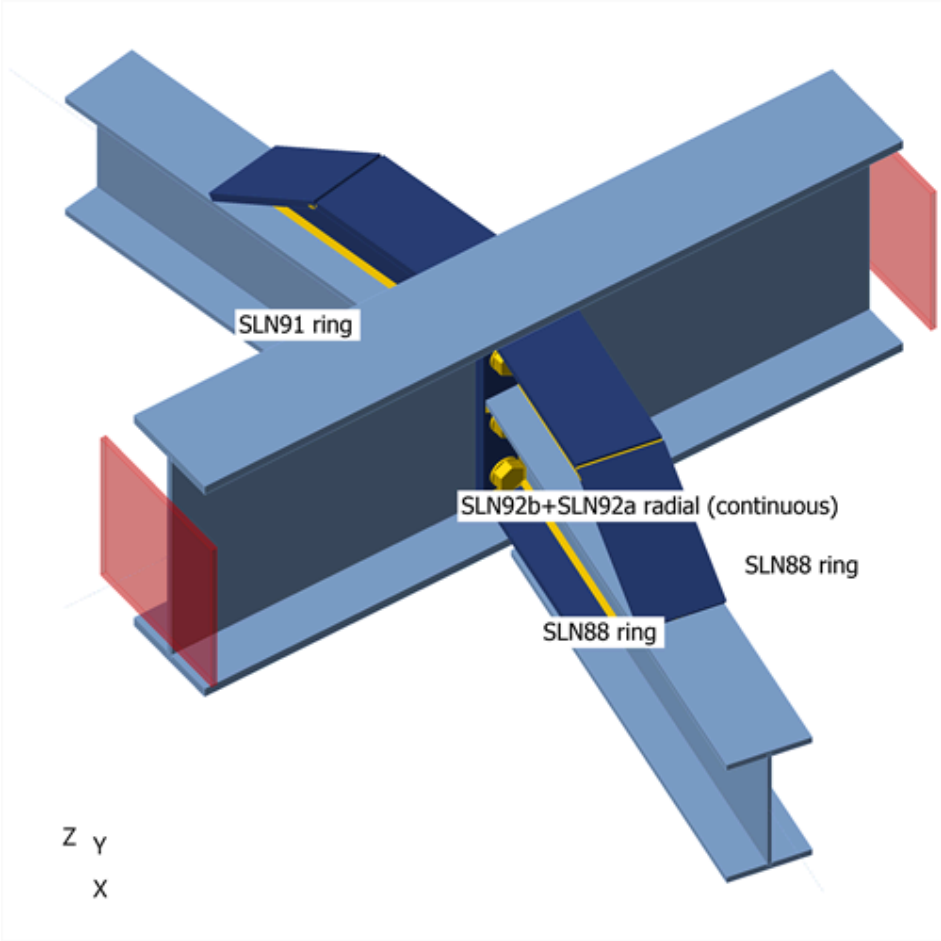
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN92b+SLN92a radial (continuous)	1 - UB457x191x89	80.0	0.0	0.0	0	0	0
SLN88 ring	2 - UB254x146x43	-20.0	0.0	0.0	0	0	-102
SLN91 ring	2 - UB254x146x43	180.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN92b+SLN92a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN92b+SLN92a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN88 ring / end		Position	0
SLN91 ring / end		Position	0



SLN92b+SLN92a radial (continuous) - SLN88 ring - SLN91 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN88 ring / End	4.9	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
C LC_102	SLN88 ring / End	19.2	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	19.2	-0.3	-25.6	0.0	0.0	-0.4
C LC_103	SLN88 ring / End	16.3	1.0	-24.2	0.0	0.0	0.9
	SLN91 ring / End	16.3	-0.2	-24.0	0.0	0.0	-0.4
C LC_104	SLN88 ring / End	13.4	0.9	-24.2	0.0	0.0	0.9
	SLN91 ring / End	13.3	-0.1	-24.0	0.0	0.0	-0.2
C LC_105	SLN88 ring / End	15.4	0.9	-24.2	0.0	0.0	0.8
	SLN91 ring / End	15.7	0.0	-24.0	0.0	0.0	-0.1
C LC_106	SLN88 ring / End	14.4	1.0	-24.2	0.0	0.0	0.8
	SLN91 ring / End	14.1	-0.4	-24.0	0.0	0.0	-0.6
C LC_107	SLN88 ring / End	2.9	0.3	-0.5	0.0	0.0	0.2
	SLN91 ring / End	2.5	0.0	-0.5	0.0	0.0	-0.1
C LC_108	SLN88 ring / End	-3.9	0.0	-0.5	0.0	0.0	0.1
	SLN91 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.3
C LC_109	SLN88 ring / End	-1.0	0.1	-0.5	0.0	0.0	0.1
	SLN91 ring / End	-0.5	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN88 ring / End	0.2	0.0	-0.5	0.0	0.0	-0.1
	SLN91 ring / End	-0.3	-0.3	-0.5	0.0	0.0	-0.4
C LC_111	SLN88 ring / End	12.4	0.9	-16.3	0.0	0.0	0.7
	SLN91 ring / End	12.1	-0.2	-16.2	0.0	0.0	-0.3
C LC_112	SLN88 ring / End	6.0	0.6	-16.3	0.0	0.0	0.6
	SLN91 ring / End	6.1	0.1	-16.2	0.0	0.0	0.1
C LC_113	SLN88 ring / End	9.2	0.7	-16.3	0.0	0.0	0.7
	SLN91 ring / End	9.8	0.1	-16.2	0.0	0.0	0.1
C LC_114	SLN88 ring / End	9.3	0.6	-16.3	0.0	0.0	0.4
	SLN91 ring / End	8.8	-0.5	-16.2	0.0	0.0	-0.7
C LC_115	SLN88 ring / End	4.1	0.4	-1.8	0.0	0.0	0.3
	SLN91 ring / End	3.7	0.0	-1.7	0.0	0.0	-0.1
C LC_116	SLN88 ring / End	-2.6	0.0	-1.8	0.0	0.0	0.1
	SLN91 ring / End	-2.3	0.1	-1.7	0.0	0.0	0.3
C LC_117	SLN88 ring / End	0.3	0.2	-1.8	0.0	0.0	0.2
	SLN91 ring / End	0.8	0.2	-1.7	0.0	0.0	0.3
C LC_118	SLN88 ring / End	1.4	0.0	-1.8	0.0	0.0	-0.1
	SLN91 ring / End	0.9	-0.3	-1.7	0.0	0.0	-0.4
C LC_119	SLN88 ring / End	18.8	1.1	-25.8	0.0	0.0	0.9
	SLN91 ring / End	18.8	-0.3	-25.6	0.0	0.0	-0.5
C LC_120	SLN88 ring / End	15.7	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	15.6	-0.1	-25.6	0.0	0.0	-0.2
C LC_121	SLN88 ring / End	17.1	0.9	-25.8	0.0	0.0	0.9
	SLN91 ring / End	17.4	-0.1	-25.6	0.0	0.0	-0.2
C LC_122	SLN88 ring / End	17.1	1.0	-25.8	0.0	0.0	0.8
	SLN91 ring / End	16.8	-0.4	-25.6	0.0	0.0	-0.6
C LC_123	SLN88 ring / End	17.4	0.9	-19.5	0.0	0.0	0.8

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN91 ring / End	17.1	-0.3	-19.4	0.0	0.0	-0.5
C LC_124	SLN88 ring / End	10.6	0.7	-19.5	0.0	0.0	0.8
	SLN91 ring / End	10.7	0.1	-19.4	0.0	0.0	0.1
C LC_125	SLN88 ring / End	12.7	0.8	-19.5	0.0	0.0	0.8
	SLN91 ring / End	13.2	0.0	-19.4	0.0	0.0	-0.1
C LC_126	SLN88 ring / End	14.7	0.7	-19.5	0.0	0.0	0.5
	SLN91 ring / End	14.2	-0.5	-19.4	0.0	0.0	-0.7
C LC_127	SLN88 ring / End	9.1	0.4	-4.9	0.0	0.0	0.3
	SLN91 ring / End	8.8	-0.2	-4.9	0.0	0.0	-0.3
C LC_128	SLN88 ring / End	2.1	0.2	-4.9	0.0	0.0	0.3
	SLN91 ring / End	2.2	0.2	-4.9	0.0	0.0	0.2
C LC_129	SLN88 ring / End	3.8	0.3	-4.9	0.0	0.0	0.3
	SLN91 ring / End	4.3	0.1	-4.9	0.0	0.0	0.1
C LC_130	SLN88 ring / End	6.8	0.1	-4.9	0.0	0.0	0.0
	SLN91 ring / End	6.3	-0.3	-4.9	0.0	0.0	-0.4
C LC_135	SLN88 ring / End	16.0	0.7	-25.8	0.0	0.0	0.7
	SLN91 ring / End	16.0	-0.2	-25.6	0.0	0.0	-0.3
C LC_141	SLN88 ring / End	4.9	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
C LC_142	SLN88 ring / End	4.9	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN88 ring / End	5.1	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	5.1	0.0	-4.9	0.0	0.0	-0.1
C LC_144	SLN88 ring / End	4.7	0.2	-4.9	0.0	0.0	0.2
	SLN91 ring / End	4.7	-0.1	-4.9	0.0	0.0	-0.1
C LC_145	SLN88 ring / End	18.9	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	19.1	-0.3	-25.6	0.0	0.0	-0.5
C LC_146	SLN88 ring / End	19.4	1.0	-25.8	0.0	0.0	0.9
	SLN91 ring / End	19.2	-0.2	-25.6	0.0	0.0	-0.4
C LC_147	SLN88 ring / End	20.0	0.9	-25.8	0.0	0.0	0.9
	SLN91 ring / End	20.1	-0.2	-25.6	0.0	0.0	-0.4
C LC_148	SLN88 ring / End	18.3	1.1	-25.8	0.0	0.0	1.0
	SLN91 ring / End	18.2	-0.3	-25.6	0.0	0.0	-0.5
C LC_151	SLN88 ring / End	12.2	0.6	-11.6	0.0	0.0	0.6
	SLN91 ring / End	12.1	-0.2	-11.8	0.0	0.0	-0.3
C LC_152	SLN88 ring / End	19.1	1.1	-14.0	0.0	0.0	1.0
	SLN91 ring / End	19.1	-0.3	-14.2	0.0	0.0	-0.4
C LC_153	SLN88 ring / End	19.2	1.0	-20.7	0.0	0.0	0.9
	SLN91 ring / End	19.2	-0.3	-21.1	0.0	0.0	-0.4
C LC_154	SLN88 ring / End	12.3	0.6	-11.6	0.0	0.0	0.6
	SLN91 ring / End	12.2	-0.2	-11.8	0.0	0.0	-0.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.3 < 100%	OK
Welds	21.1 < 100%	OK
Buckling	Not calculated	

Plates

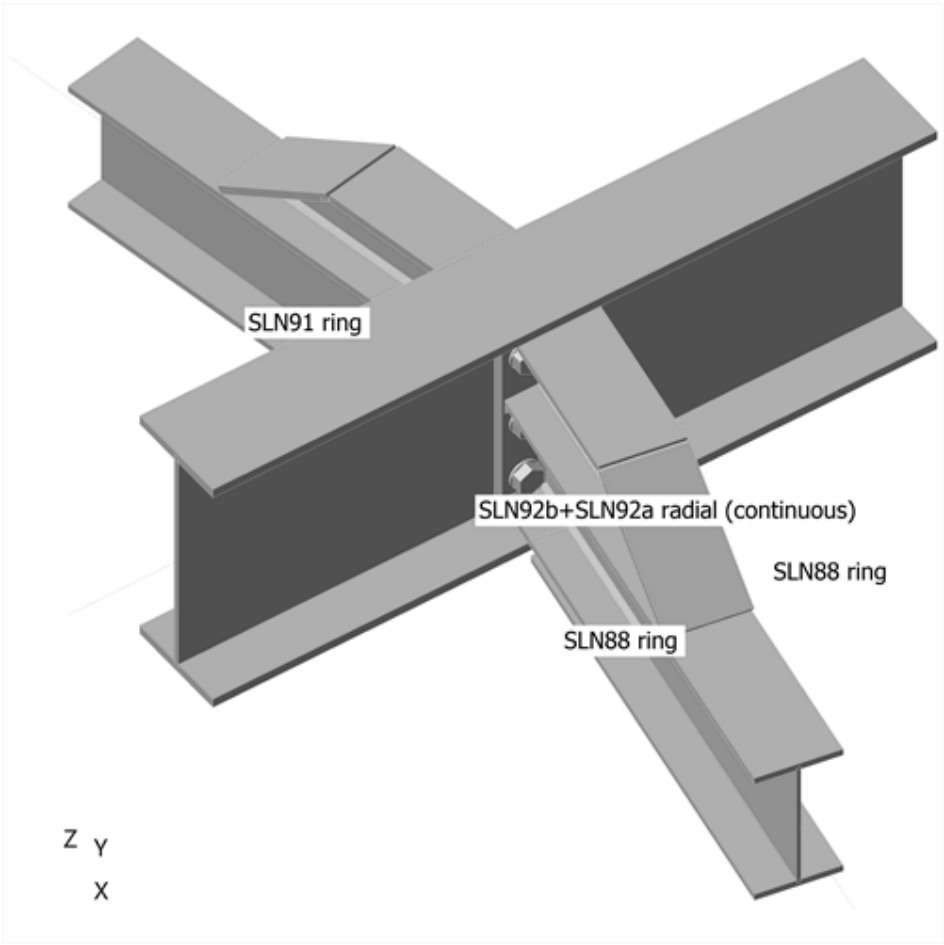
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN92b+SLN92a radial (continuous)-bfl 1	17.7	C LC_120	22.6	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-tfl 1	17.7	C LC_120	19.2	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-w 1	10.5	C LC_121	29.9	0.0	1.4	OK
SLN88 ring-bfl 1	12.7	C LC_119	51.2	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_120	44.9	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
SLN91 ring-bfl 1	12.7	C LC_147	51.3	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_122	45.5	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	49.1	0.0	0.0	OK
End plate SLN91 ring	20.0	C LC_146	12.3	0.0	5.0	OK
Bottom flange plate SLN91 ring	15.0	C LC_147	20.7	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_147	20.1	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_147	19.7	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_122	28.3	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_124	6.4	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN88 ring	20.0	C LC_148	14.0	0.0	6.5	OK
Bottom flange plate SLN88 ring	15.0	C LC_148	24.3	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_147	19.6	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_147	20.7	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_120	29.2	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_120	9.1	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_120	12.8	0.0	0.0	OK

Design data

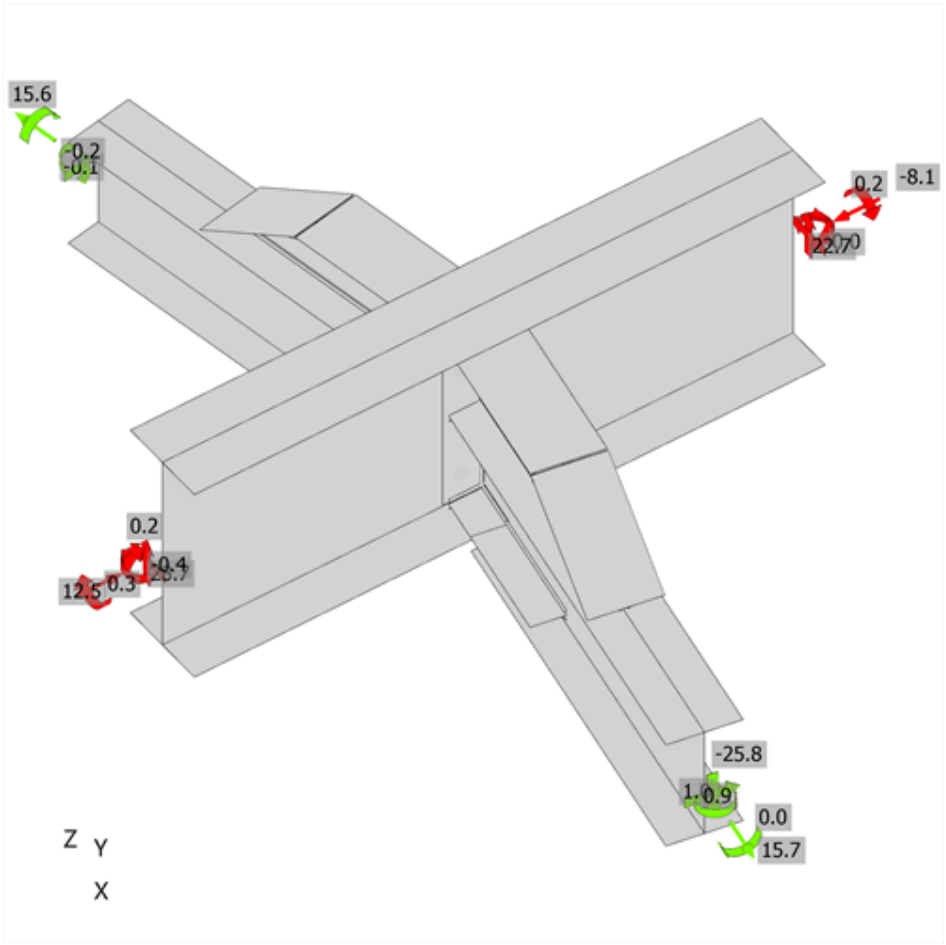
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

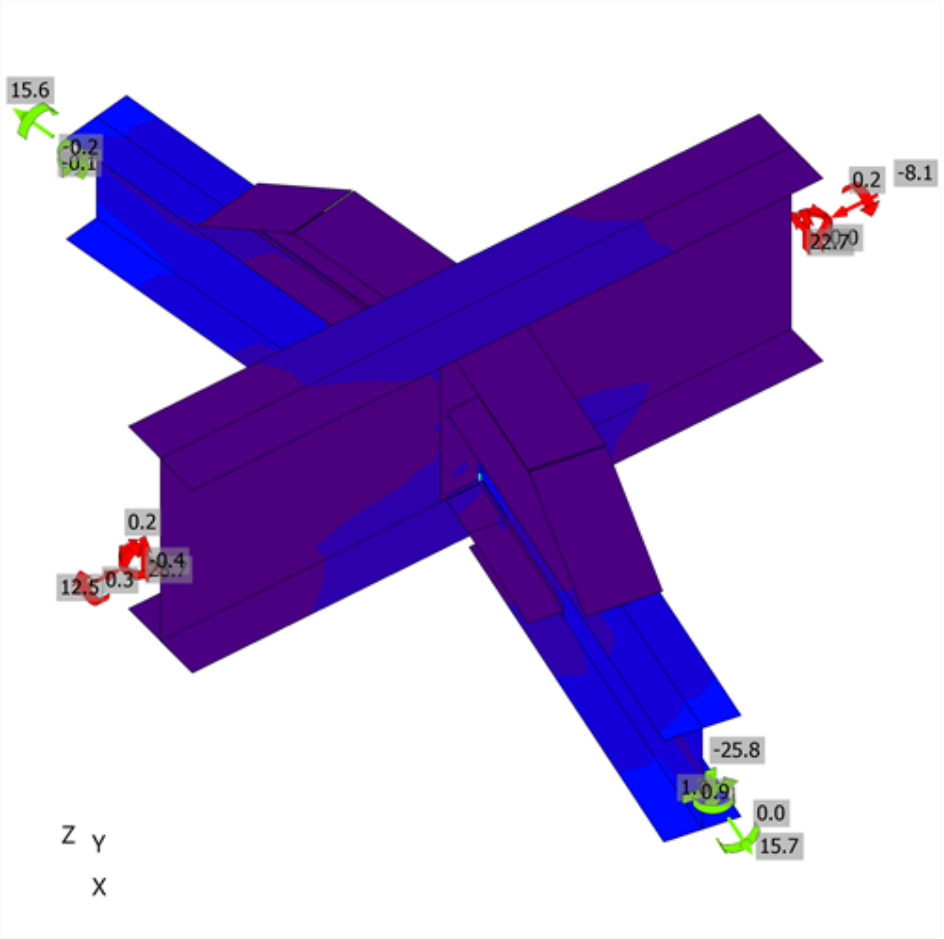
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

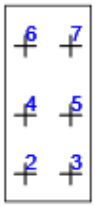


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_148	4.3	0.5	176.5	1.3	0.5	1.2	OK	OK
	B3	M30 8.8 - 1	C LC_105	1.4	0.6	176.5	0.4	0.6	0.6	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.2	0.5	213.9	0.1	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_120	3.0	0.8	229.5	0.9	0.7	1.0	OK	OK
	B7	M30 8.8 - 1	C LC_147	1.2	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN88 ring-tfl 1	◢ 10.5	149	C LC_152	16.2	0.0	7.8	-8.2	0.8	4.0	4.0	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 10.5	149	C LC_122	13.5	0.0	-6.2	5.8	-3.8	3.3	3.3	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	25.6	0.0	-10.9	-13.0	-3.2	6.3	6.1	OK	OK
		◢ 6.0 ◣	208	C LC_147	21.4	0.0	-11.5	10.2	-1.7	5.3	5.3	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_120	24.5	0.0	1.9	2.1	13.9	6.0	6.0	OK	OK
		◢ 6.0 ◣	100	C LC_119	25.9	0.0	3.0	-4.0	-14.3	6.4	6.4	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_147	24.2	0.0	6.1	5.9	12.2	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.8	0.0	8.9	2.3	-15.1	6.9	6.3	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	243	C LC_119	25.4	0.0	5.9	6.2	-12.9	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_122	28.0	0.0	7.1	-6.7	14.2	6.9	6.6	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_122	66.5	0.0	-14.5	-16.9	33.5	16.4	9.5	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.2	0.0	-16.2	15.5	-31.0	15.4	9.0	OK	OK
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	208	C LC_147	23.2	0.0	-11.1	-11.7	0.2	5.7	5.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	29.3	0.0	-14.1	13.4	6.2	7.2	6.3	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	100	C LC_146	28.6	0.0	-0.8	-3.2	16.2	7.1	7.1	OK	OK
		◢ 6.0 ◣	100	C LC_147	23.2	0.0	1.3	0.3	-13.4	5.7	5.7	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	243	C LC_146	28.2	0.0	6.1	6.9	14.3	7.0	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_146	33.3	0.0	8.9	-8.1	-16.7	8.2	7.4	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_147	21.0	0.0	5.9	5.1	-10.5	5.2	5.2	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.7	0.0	10.8	1.3	15.3	7.1	6.2	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_135	60.4	0.0	-17.7	-15.2	29.7	14.9	8.8	OK	OK
		◢ 6.0 ◣	572	C LC_120	68.7	0.0	-13.6	17.2	-34.8	17.0	9.7	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	149	C LC_110	17.6	0.0	-5.7	-6.8	6.8	4.4	4.4	OK	OK
		◢ 6.0 ◣	149	C LC_124	23.5	0.0	-12.6	10.1	5.4	5.8	5.8	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	8.5	0.0	1.1	1.3	4.7	2.1	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	9.4	0.0	2.6	-2.3	-4.7	2.3	0.0	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_124	9.7	0.0	-5.6	-3.5	-2.9	2.4	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_118	7.4	0.0	-2.4	3.4	-2.3	1.8	0.0	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_122	4.8	0.0	1.9	1.7	2.0	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	3.6	0.0	-0.4	0.3	2.0	0.9	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_122	52.3	0.0	-9.2	-5.8	29.2	12.9	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_135	51.2	0.0	-9.3	10.9	-26.9	12.6	8.1	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_120	27.5	0.0	-8.5	-9.3	-11.9	6.8	6.8	OK	OK
		◢ 6.0 ◣	149	C LC_120	32.8	0.0	-12.3	11.5	13.3	8.1	6.7	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_120	11.3	0.0	2.7	2.8	5.7	2.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	6.4	0.0	0.8	-0.8	-3.6	1.6	0.0	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_124	13.9	0.0	-6.2	-5.2	-4.9	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_124	12.6	0.0	-4.3	5.3	4.3	3.1	3.1	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_120	4.9	0.0	-0.3	-0.4	-2.8	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	6.6	0.0	2.3	-1.9	-3.0	1.6	0.0	OK	OK
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	52.0	0.0	-9.4	-14.4	25.8	12.8	8.2	OK	OK
		◢ 6.0 ◣	271	C LC_148	53.3	0.0	-9.3	4.3	-30.0	13.2	8.6	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_147	85.2	0.0	34.4	39.1	-22.3	21.1	14.4	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_147	79.6	0.0	39.8	-37.8	12.4	19.7	13.6	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_120	7.1	0.0	-5.1	2.9	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	3.5	0.0	-0.9	1.8	0.8	0.9	0.0	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_120	7.6	0.0	-6.9	-1.9	-0.2	2.2	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	5.0	0.0	-1.2	2.8	-0.2	1.2	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_100
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring

Name	SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

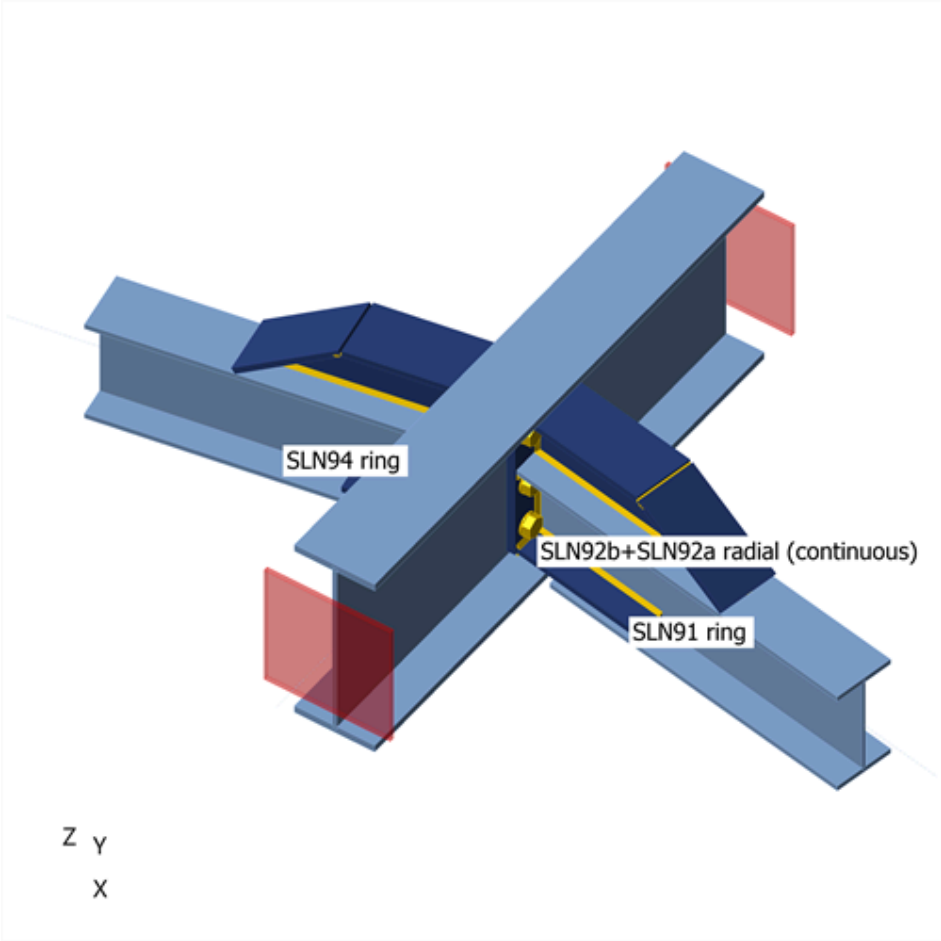
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN92b+SLN92a radial (continuous)	1 - UB457x191x89	100.0	0.0	0.0	0	0	0
SLN91 ring	2 - UB254x146x43	0.0	0.0	0.0	0	0	-102
SLN94 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN92b+SLN92a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN92b+SLN92a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN91 ring / end		Position	0
SLN94 ring / end		Position	0



SLN92b+SLN92a radial (continuous) - SLN91 ring - SLN94 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN91 ring / End	19.2	-0.3	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	19.0	0.1	-25.7	0.0	0.0	0.1
C LC_103	SLN91 ring / End	16.5	-0.2	-24.1	0.0	0.0	-0.3
	SLN94 ring / End	16.8	0.1	-24.1	0.0	0.0	0.1
C LC_104	SLN91 ring / End	13.2	-0.1	-24.1	0.0	0.0	-0.1
	SLN94 ring / End	12.7	0.1	-24.1	0.0	0.0	0.2
C LC_105	SLN91 ring / End	15.7	-0.2	-24.1	0.0	0.0	-0.1
	SLN94 ring / End	15.8	0.2	-24.1	0.0	0.0	0.2
C LC_106	SLN91 ring / End	14.1	-0.3	-24.1	0.0	0.0	-0.4
	SLN94 ring / End	13.6	-0.1	-24.1	0.0	0.0	-0.1
C LC_107	SLN91 ring / End	2.5	0.0	-0.5	0.0	0.0	0.1
	SLN94 ring / End	3.1	0.0	-0.5	0.0	0.0	0.1
C LC_108	SLN91 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.2
	SLN94 ring / End	-3.9	0.1	-0.5	0.0	0.0	0.2
C LC_109	SLN91 ring / End	-0.5	0.2	-0.5	0.0	0.0	0.3
	SLN94 ring / End	-0.1	0.1	-0.5	0.0	0.0	0.1
C LC_110	SLN91 ring / End	-0.3	-0.3	-0.5	0.0	0.0	-0.4
	SLN94 ring / End	-0.9	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN91 ring / End	12.3	-0.2	-16.3	0.0	0.0	-0.1
	SLN94 ring / End	12.8	0.0	-16.3	0.0	0.0	0.1
C LC_112	SLN91 ring / End	6.0	0.1	-16.3	0.0	0.0	0.1
	SLN94 ring / End	5.5	0.2	-16.3	0.0	0.0	0.2
C LC_113	SLN91 ring / End	9.8	0.0	-16.3	0.0	0.0	0.1
	SLN94 ring / End	10.1	0.2	-16.3	0.0	0.0	0.2
C LC_114	SLN91 ring / End	8.8	-0.3	-16.3	0.0	0.0	-0.5
	SLN94 ring / End	8.1	-0.2	-16.3	0.0	0.0	-0.3
C LC_115	SLN91 ring / End	3.8	0.0	-1.7	0.0	0.0	0.0
	SLN94 ring / End	4.4	0.0	-1.7	0.0	0.0	0.1
C LC_116	SLN91 ring / End	-2.4	0.1	-1.7	0.0	0.0	0.2
	SLN94 ring / End	-2.7	0.1	-1.7	0.0	0.0	0.2
C LC_117	SLN91 ring / End	0.8	0.2	-1.7	0.0	0.0	0.3
	SLN94 ring / End	1.2	0.1	-1.7	0.0	0.0	0.1
C LC_118	SLN91 ring / End	0.9	-0.3	-1.7	0.0	0.0	-0.4
	SLN94 ring / End	0.3	-0.2	-1.7	0.0	0.0	-0.3
C LC_119	SLN91 ring / End	19.0	-0.3	-25.7	0.0	0.0	-0.4
	SLN94 ring / End	18.7	0.0	-25.7	0.0	0.0	0.0
C LC_120	SLN91 ring / End	15.5	-0.1	-25.7	0.0	0.0	-0.1
	SLN94 ring / End	15.4	0.2	-25.7	0.0	0.0	0.3
C LC_121	SLN91 ring / End	17.4	-0.2	-25.7	0.0	0.0	-0.2
	SLN94 ring / End	17.6	0.2	-25.7	0.0	0.0	0.3
C LC_122	SLN91 ring / End	16.8	-0.3	-25.7	0.0	0.0	-0.4
	SLN94 ring / End	16.3	-0.1	-25.7	0.0	0.0	-0.1
C LC_123	SLN91 ring / End	17.3	-0.3	-19.4	0.0	0.0	-0.4

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN94 ring / End	16.6	0.0	-19.5	0.0	0.0	-0.1
C LC_124	SLN91 ring / End	10.5	0.1	-19.4	0.0	0.0	0.1
	SLN94 ring / End	10.8	0.3	-19.5	0.0	0.0	0.4
C LC_125	SLN91 ring / End	13.2	-0.1	-19.4	0.0	0.0	-0.1
	SLN94 ring / End	13.7	0.3	-19.5	0.0	0.0	0.4
C LC_126	SLN91 ring / End	14.2	-0.3	-19.4	0.0	0.0	-0.5
	SLN94 ring / End	13.5	-0.2	-19.5	0.0	0.0	-0.3
C LC_127	SLN91 ring / End	8.8	-0.2	-4.9	0.0	0.0	-0.3
	SLN94 ring / End	8.1	0.0	-4.9	0.0	0.0	-0.1
C LC_128	SLN91 ring / End	2.2	0.2	-4.9	0.0	0.0	0.3
	SLN94 ring / End	2.6	0.2	-4.9	0.0	0.0	0.4
C LC_129	SLN91 ring / End	4.3	0.0	-4.9	0.0	0.0	0.1
	SLN94 ring / End	4.8	0.2	-4.9	0.0	0.0	0.3
C LC_130	SLN91 ring / End	6.3	-0.3	-4.9	0.0	0.0	-0.4
	SLN94 ring / End	5.7	-0.2	-4.9	0.0	0.0	-0.3
C LC_135	SLN91 ring / End	16.0	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	16.0	0.1	-25.7	0.0	0.0	0.2
C LC_141	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	5.0	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN91 ring / End	4.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
C LC_143	SLN91 ring / End	5.1	-0.1	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	5.1	0.1	-4.9	0.0	0.0	0.1
C LC_144	SLN91 ring / End	4.7	0.0	-4.9	0.0	0.0	-0.1
	SLN94 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN91 ring / End	19.3	-0.3	-25.7	0.0	0.0	-0.4
	SLN94 ring / End	19.3	0.1	-25.7	0.0	0.0	0.1
C LC_146	SLN91 ring / End	19.0	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	18.7	0.1	-25.7	0.0	0.0	0.2
C LC_147	SLN91 ring / End	20.1	-0.3	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	20.0	0.2	-25.7	0.0	0.0	0.2
C LC_148	SLN91 ring / End	18.2	-0.2	-25.7	0.0	0.0	-0.3
	SLN94 ring / End	18.1	0.0	-25.7	0.0	0.0	0.1
C LC_151	SLN91 ring / End	12.1	-0.2	-11.8	0.0	0.0	-0.2
	SLN94 ring / End	12.0	0.1	-11.6	0.0	0.0	0.1
C LC_152	SLN91 ring / End	19.1	-0.3	-14.3	0.0	0.0	-0.3
	SLN94 ring / End	18.9	0.1	-14.0	0.0	0.0	0.1
C LC_153	SLN91 ring / End	19.2	-0.3	-21.2	0.0	0.0	-0.3
	SLN94 ring / End	19.0	0.1	-20.7	0.0	0.0	0.1
C LC_154	SLN91 ring / End	12.2	-0.2	-11.8	0.0	0.0	-0.2
	SLN94 ring / End	12.1	0.1	-11.6	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.2 < 100%	OK
Buckling	Not calculated	

Plates

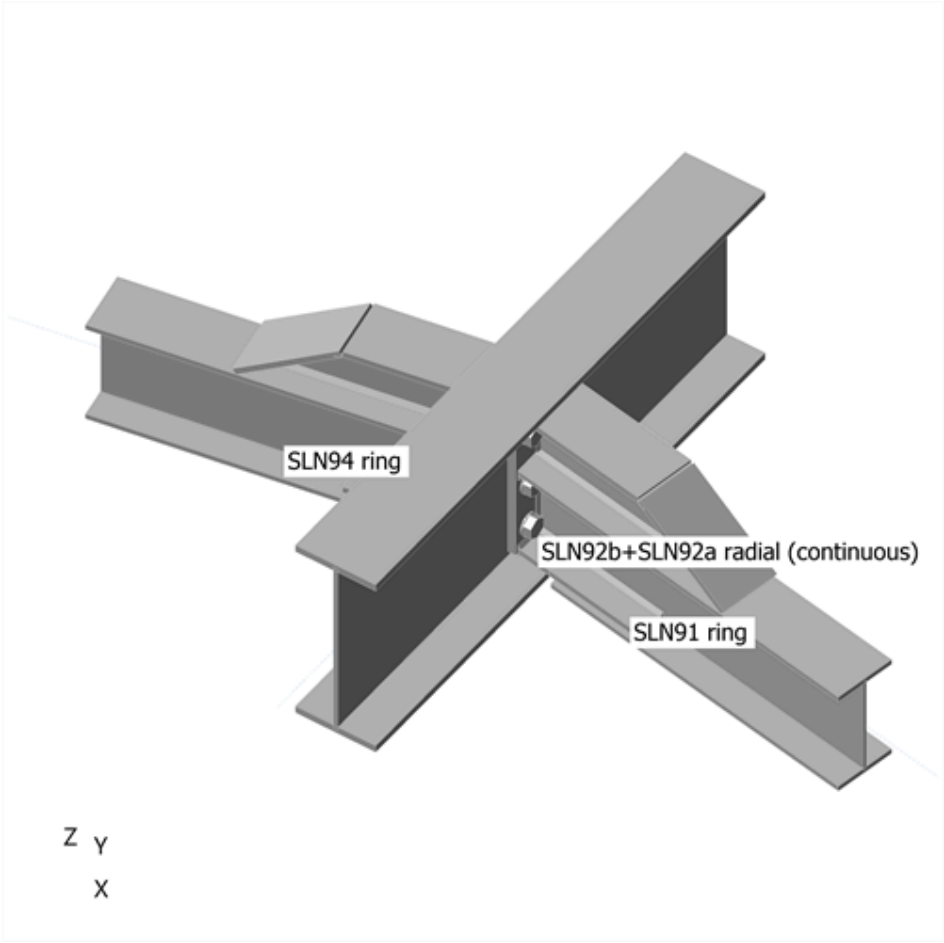
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN92b+SLN92a radial (continuous)-bfl 1	17.7	C LC_122	22.2	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-tfl 1	17.7	C LC_122	19.0	0.0	0.0	OK
SLN92b+SLN92a radial (continuous)-w 1	10.5	C LC_122	29.1	0.0	0.9	OK
SLN91 ring-bfl 1	12.7	C LC_122	50.5	0.0	0.0	OK
SLN91 ring-tfl 1	12.7	C LC_122	45.1	0.0	0.0	OK
SLN91 ring-w 1	7.2	C LC_147	49.2	0.0	0.0	OK
SLN94 ring-bfl 1	12.7	C LC_146	50.3	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_120	45.0	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_126	8.7	0.0	8.2	OK
Bottom flange plate SLN94 ring	15.0	C LC_147	20.6	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_120	28.5	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_126	7.4	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN91 ring	20.0	C LC_147	9.1	0.0	0.8	OK
Bottom flange plate SLN91 ring	15.0	C LC_147	20.6	0.0	0.0	OK
Bottom flange half plate L SLN91 ring	15.0	C LC_147	20.2	0.0	0.0	OK
Bottom flange half plate R SLN91 ring	15.0	C LC_147	19.9	0.0	0.0	OK
Haunch web SLN91 ring	12.0	C LC_120	28.4	0.0	0.0	OK
Haunch top plate SLN91 ring	15.0	C LC_126	6.8	0.0	0.0	OK
Haunch sloping plate SLN91 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

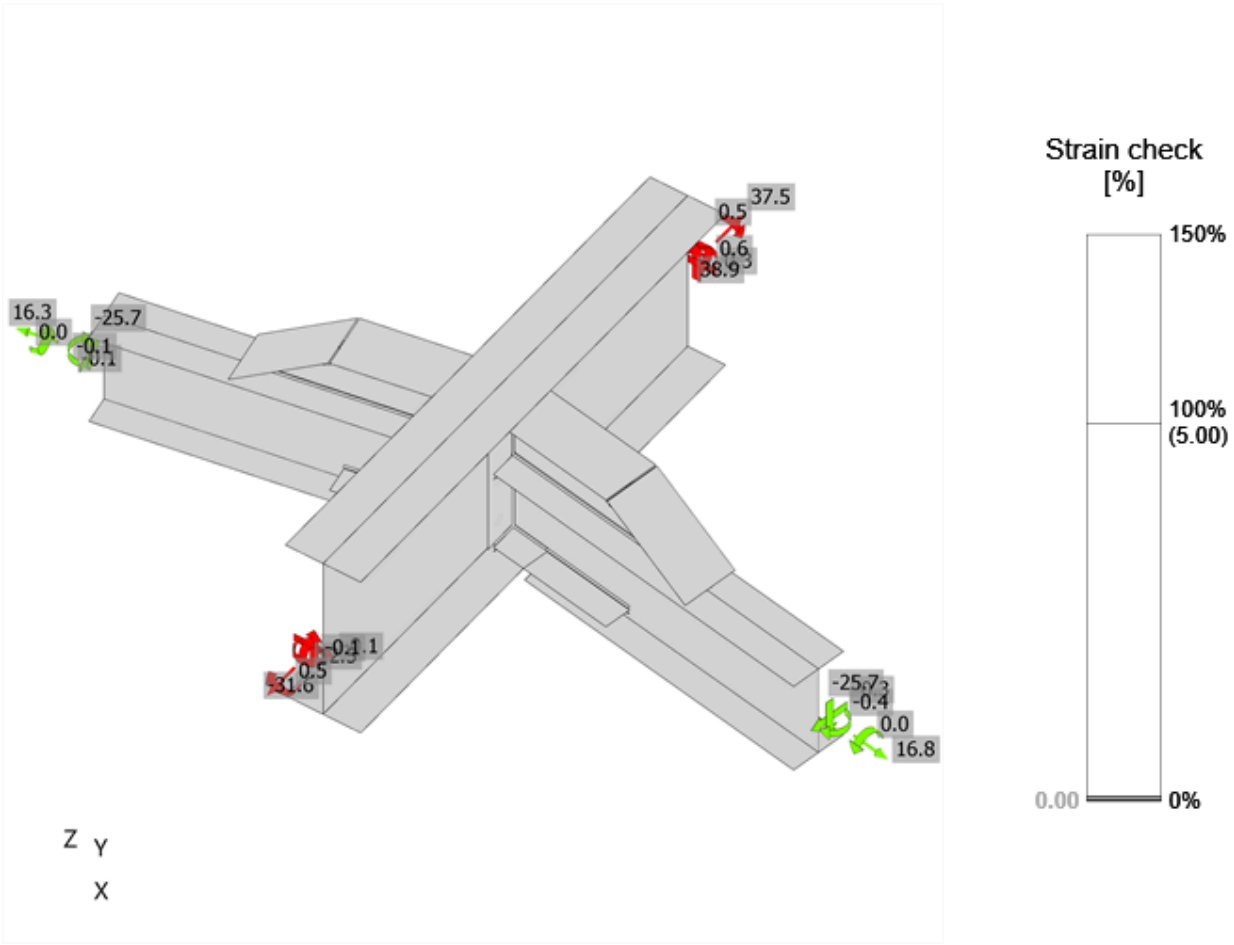
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

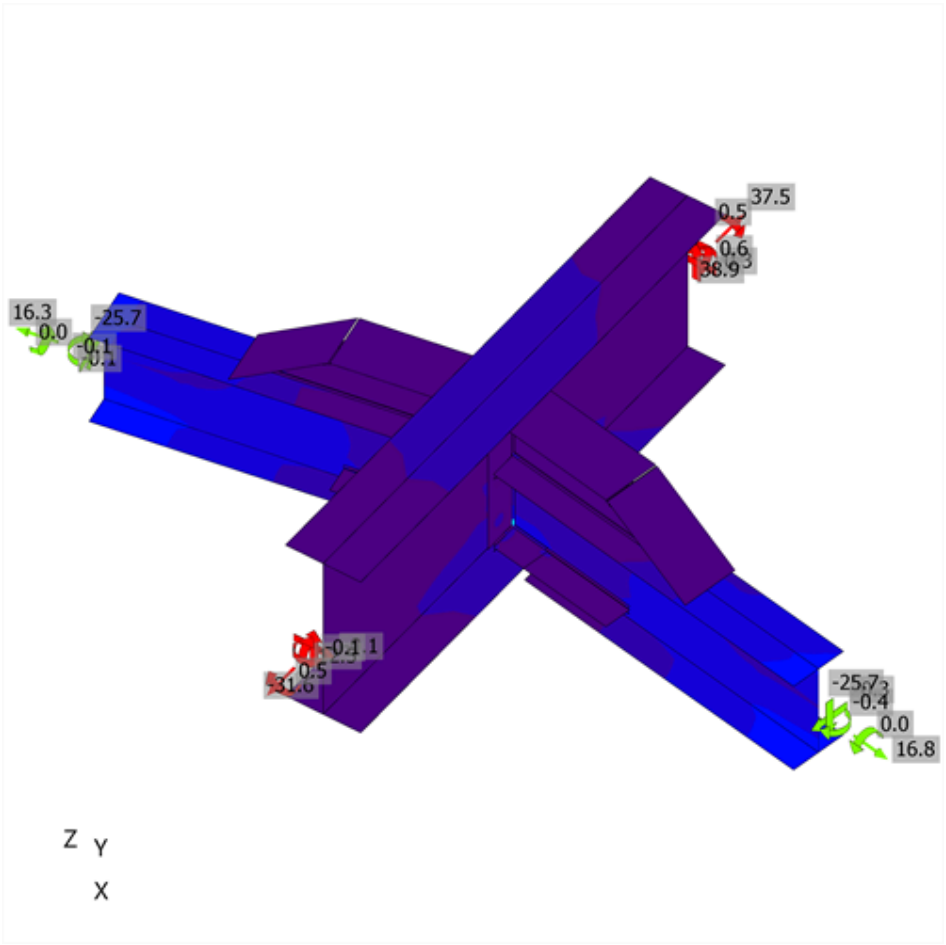
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

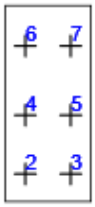


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	2.5	0.5	176.5	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_147	2.5	0.7	176.5	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	1.9	0.9	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.4	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN91 ring	Bottom flange plate SLN91 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-tfl 1	◢ 10.5	149	C LC_147	12.1	0.0	-5.9	5.3	-3.1	3.0	0.0	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 10.5	149	C LC_121	11.6	0.0	-5.7	5.1	2.8	2.9	2.9	OK	OK
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_147	25.0	0.0	-10.7	-12.8	-2.8	6.2	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_147	20.4	0.0	-11.2	9.8	0.3	5.0	5.0	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_120	24.6	0.0	2.5	2.9	13.8	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.4	0.0	2.5	-3.5	-14.2	6.3	6.3	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	25.4	0.0	6.4	6.2	12.8	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.6	0.0	8.0	2.7	-15.0	6.8	6.4	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_119	24.8	0.0	5.7	6.1	-12.6	6.1	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.9	0.0	8.2	2.6	15.2	6.9	6.5	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	65.8	0.0	-16.3	-17.1	32.6	16.3	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.7	0.0	-14.3	15.4	-32.3	15.7	9.2	OK	OK
End plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	208	C LC_147	20.9	0.0	-11.9	-9.9	-0.6	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	25.4	0.0	-10.2	13.2	2.4	6.3	6.1	OK	OK
Bottom flange plate SLN91 ring	SLN91 ring-w 1	◢ 6.0 ◣	100	C LC_120	23.9	0.0	2.4	2.8	13.4	5.9	5.9	OK	OK
		◢ 6.0 ◣	100	C LC_122	26.0	0.0	2.6	-3.5	-14.5	6.4	6.4	OK	OK
SLN91 ring-w 1	Bottom flange half plate L SLN91 ring	◢ 6.0 ◣	243	C LC_147	23.9	0.0	5.9	5.9	12.0	5.9	5.9	OK	OK
		◢ 6.0 ◣	243	C LC_119	28.0	0.0	9.2	2.0	-15.1	6.9	6.2	OK	OK
SLN91 ring-w 1	Bottom flange half plate R SLN91 ring	◢ 6.0 ◣	243	C LC_119	26.2	0.0	6.3	6.4	-13.2	6.5	5.9	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.6	0.0	7.6	2.9	15.0	6.8	6.5	OK	OK
SLN91 ring-tfl 1	Haunch web SLN91 ring	◢ 6.0 ◣	572	C LC_122	63.9	0.0	-13.7	-15.4	32.6	15.8	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_120	64.8	0.0	-16.8	16.9	-31.9	16.0	9.2	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	149	C LC_126	20.8	0.0	-5.4	-7.9	8.5	5.1	5.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	28.1	0.0	-11.5	9.0	-11.7	6.9	5.9	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	7.6	0.0	1.8	1.5	4.0	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.1	0.0	2.0	-2.5	-5.8	2.8	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_126	12.5	0.0	-6.3	-3.7	5.0	3.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	8.4	0.0	-1.8	4.2	-2.2	2.1	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_126	4.2	0.0	1.3	1.2	2.0	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	4.5	0.0	1.7	-1.5	-1.9	1.1	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_122	53.5	0.0	-9.9	-8.8	29.0	13.2	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	49.5	0.0	-8.1	6.6	-27.4	12.2	8.1	OK	OK
End plate SLN91 ring	Bottom flange plate SLN91 ring	◢ 6.0 ◣	149	C LC_110	17.1	0.0	-7.5	-8.5	2.6	4.2	4.2	OK	OK
		◢ 6.0 ◣	149	C LC_114	24.9	0.0	-13.1	11.2	-4.9	6.2	5.6	OK	OK
End plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	145	C LC_120	10.7	0.0	1.8	2.3	5.7	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.1	0.0	1.9	-1.7	-4.2	2.0	0.0	OK	OK
End plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	149	C LC_114	11.5	0.0	-5.8	-4.9	3.0	2.8	2.8	OK	OK
		◢ 6.0 ◣	149	C LC_126	9.7	0.0	-3.5	4.3	-3.0	2.4	0.0	OK	OK
Haunch top plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	342	C LC_122	5.1	0.0	1.8	1.6	2.3	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.2	0.3	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch web SLN91 ring	◢ 6.0 ◣	271	C LC_135	49.6	0.0	-8.3	-7.0	27.3	12.3	8.1	OK	OK
		◢ 6.0 ◣	271	C LC_120	53.2	0.0	-9.9	9.8	-28.5	13.2	8.3	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_147	78.3	0.0	32.4	37.5	-17.0	19.3	13.4	OK	OK
SLN92b+SLN92a radial (continuous)-w 1	SLN91 ring-w 1	◢ 7.2	208	C LC_119	81.7	0.0	31.7	-39.2	18.8	20.2	13.9	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.3	2.8	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_124	3.3	0.0	-0.7	1.9	0.1	0.8	0.0	OK	OK
Haunch sloping plate SLN91 ring	Haunch top plate SLN91 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.7	2.7	0.4	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.9	0.0	-0.8	2.2	-0.3	1.0	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_120
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring

Name	SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

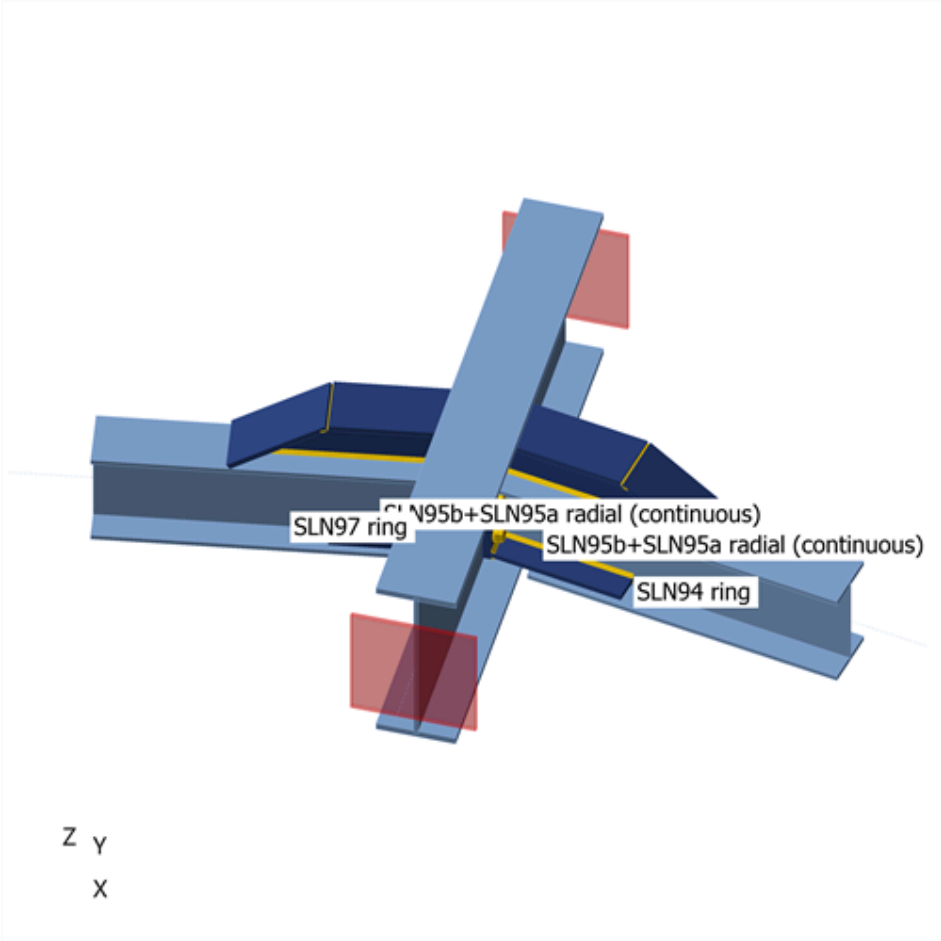
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN95b+SLN95a radial (continuous)	1 - UB457x191x89	120.0	0.0	0.0	0	0	0
SLN94 ring	2 - UB254x146x43	20.0	0.0	0.0	0	0	-102
SLN97 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN95b+SLN95a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN95b+SLN95a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN94 ring / end		Position	0
SLN97 ring / end		Position	0



SLN95b+SLN95a radial (continuous) - SLN94 ring - SLN97 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN94 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.9	0.0	-5.0	0.0	0.0	-0.1
C LC_102	SLN94 ring / End	19.0	0.1	-25.7	0.0	0.0	0.1
	SLN97 ring / End	19.0	-0.2	-26.0	0.0	0.0	-0.2
C LC_103	SLN94 ring / End	17.0	0.0	-24.1	0.0	0.0	0.0
	SLN97 ring / End	16.9	0.0	-24.4	0.0	0.0	-0.1
C LC_104	SLN94 ring / End	12.6	0.2	-24.1	0.0	0.0	0.3
	SLN97 ring / End	12.5	-0.1	-24.4	0.0	0.0	-0.1
C LC_105	SLN94 ring / End	15.7	0.0	-24.1	0.0	0.0	0.1
	SLN97 ring / End	15.8	0.0	-24.4	0.0	0.0	0.0
C LC_106	SLN94 ring / End	13.7	0.0	-24.1	0.0	0.0	0.0
	SLN97 ring / End	13.5	-0.3	-24.4	0.0	0.0	-0.4
C LC_107	SLN94 ring / End	3.1	0.0	-0.5	0.0	0.0	-0.1
	SLN97 ring / End	2.8	0.1	-0.5	0.0	0.0	0.1
C LC_108	SLN94 ring / End	-3.9	0.1	-0.5	0.0	0.0	0.2
	SLN97 ring / End	-3.7	0.1	-0.5	0.0	0.0	0.2
C LC_109	SLN94 ring / End	-0.1	0.0	-0.5	0.0	0.0	0.1
	SLN97 ring / End	0.3	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN94 ring / End	-0.9	-0.2	-0.5	0.0	0.0	-0.3
	SLN97 ring / End	-1.4	-0.3	-0.5	0.0	0.0	-0.4
C LC_111	SLN94 ring / End	13.0	0.0	-16.3	0.0	0.0	0.0
	SLN97 ring / End	12.8	0.0	-16.5	0.0	0.0	0.0
C LC_112	SLN94 ring / End	5.4	0.2	-16.3	0.0	0.0	0.3
	SLN97 ring / End	5.5	0.0	-16.5	0.0	0.0	0.1
C LC_113	SLN94 ring / End	10.0	0.0	-16.3	0.0	0.0	0.1
	SLN97 ring / End	10.4	0.1	-16.5	0.0	0.0	0.2
C LC_114	SLN94 ring / End	8.2	-0.1	-16.3	0.0	0.0	-0.2
	SLN97 ring / End	7.7	-0.4	-16.5	0.0	0.0	-0.5
C LC_115	SLN94 ring / End	4.4	0.0	-1.8	0.0	0.0	0.0
	SLN97 ring / End	4.1	0.1	-1.8	0.0	0.0	0.1
C LC_116	SLN94 ring / End	-2.7	0.2	-1.8	0.0	0.0	0.3
	SLN97 ring / End	-2.5	0.1	-1.8	0.0	0.0	0.2
C LC_117	SLN94 ring / End	1.2	0.1	-1.8	0.0	0.0	0.1
	SLN97 ring / End	1.6	0.1	-1.8	0.0	0.0	0.2
C LC_118	SLN94 ring / End	0.3	-0.2	-1.8	0.0	0.0	-0.3
	SLN97 ring / End	-0.1	-0.3	-1.8	0.0	0.0	-0.4
C LC_119	SLN94 ring / End	18.8	0.0	-25.7	0.0	0.0	0.0
	SLN97 ring / End	18.9	-0.1	-26.0	0.0	0.0	-0.2
C LC_120	SLN94 ring / End	15.2	0.2	-25.7	0.0	0.0	0.3
	SLN97 ring / End	15.1	-0.1	-26.0	0.0	0.0	-0.1
C LC_121	SLN94 ring / End	17.5	0.1	-25.7	0.0	0.0	0.2
	SLN97 ring / End	17.6	0.0	-26.0	0.0	0.0	0.0
C LC_122	SLN94 ring / End	16.4	0.0	-25.7	0.0	0.0	0.0
	SLN97 ring / End	16.2	-0.4	-26.0	0.0	0.0	-0.4
C LC_123	SLN94 ring / End	16.8	0.0	-19.5	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN97 ring / End	16.7	-0.1	-19.7	0.0	0.0	-0.2
C LC_124	SLN94 ring / End	10.7	0.3	-19.5	0.0	0.0	0.4
	SLN97 ring / End	10.7	0.0	-19.7	0.0	0.0	0.1
C LC_125	SLN94 ring / End	13.6	0.2	-19.5	0.0	0.0	0.3
	SLN97 ring / End	13.9	0.1	-19.7	0.0	0.0	0.1
C LC_126	SLN94 ring / End	13.6	-0.1	-19.5	0.0	0.0	-0.2
	SLN97 ring / End	13.1	-0.4	-19.7	0.0	0.0	-0.6
C LC_127	SLN94 ring / End	8.2	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	8.0	-0.1	-5.0	0.0	0.0	-0.1
C LC_128	SLN94 ring / End	2.6	0.2	-4.9	0.0	0.0	0.3
	SLN97 ring / End	2.8	0.1	-5.0	0.0	0.0	0.2
C LC_129	SLN94 ring / End	4.8	0.2	-4.9	0.0	0.0	0.2
	SLN97 ring / End	5.1	0.1	-5.0	0.0	0.0	0.1
C LC_130	SLN94 ring / End	5.7	-0.2	-4.9	0.0	0.0	-0.3
	SLN97 ring / End	5.3	-0.3	-5.0	0.0	0.0	-0.4
C LC_135	SLN94 ring / End	16.0	0.1	-25.7	0.0	0.0	0.2
	SLN97 ring / End	15.9	-0.1	-26.0	0.0	0.0	-0.2
C LC_141	SLN94 ring / End	5.0	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	5.0	0.0	-5.0	0.0	0.0	-0.1
C LC_142	SLN94 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
	SLN97 ring / End	4.7	-0.1	-5.0	0.0	0.0	-0.1
C LC_143	SLN94 ring / End	5.1	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	5.0	0.0	-5.0	0.0	0.0	0.0
C LC_144	SLN94 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
	SLN97 ring / End	4.7	-0.1	-5.0	0.0	0.0	-0.1
C LC_145	SLN94 ring / End	19.5	0.0	-25.7	0.0	0.0	0.1
	SLN97 ring / End	19.6	-0.1	-26.0	0.0	0.0	-0.2
C LC_146	SLN94 ring / End	18.5	0.2	-25.7	0.0	0.0	0.2
	SLN97 ring / End	18.3	-0.2	-26.0	0.0	0.0	-0.2
C LC_147	SLN94 ring / End	19.9	0.0	-25.7	0.0	0.0	0.1
	SLN97 ring / End	19.8	-0.1	-26.0	0.0	0.0	-0.2
C LC_148	SLN94 ring / End	18.1	0.2	-25.7	0.0	0.0	0.1
	SLN97 ring / End	18.2	-0.3	-26.0	0.0	0.0	-0.3
C LC_151	SLN94 ring / End	12.0	0.1	-11.7	0.0	0.0	0.1
	SLN97 ring / End	12.0	-0.1	-11.6	0.0	0.0	-0.2
C LC_152	SLN94 ring / End	18.9	0.1	-14.1	0.0	0.0	0.1
	SLN97 ring / End	18.9	-0.1	-14.0	0.0	0.0	-0.2
C LC_153	SLN94 ring / End	19.0	0.1	-20.8	0.0	0.0	0.1
	SLN97 ring / End	19.0	-0.2	-20.6	0.0	0.0	-0.2
C LC_154	SLN94 ring / End	12.1	0.1	-11.7	0.0	0.0	0.1
	SLN97 ring / End	12.1	-0.1	-11.6	0.0	0.0	-0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.2 < 100%	OK
Buckling	Not calculated	

Plates

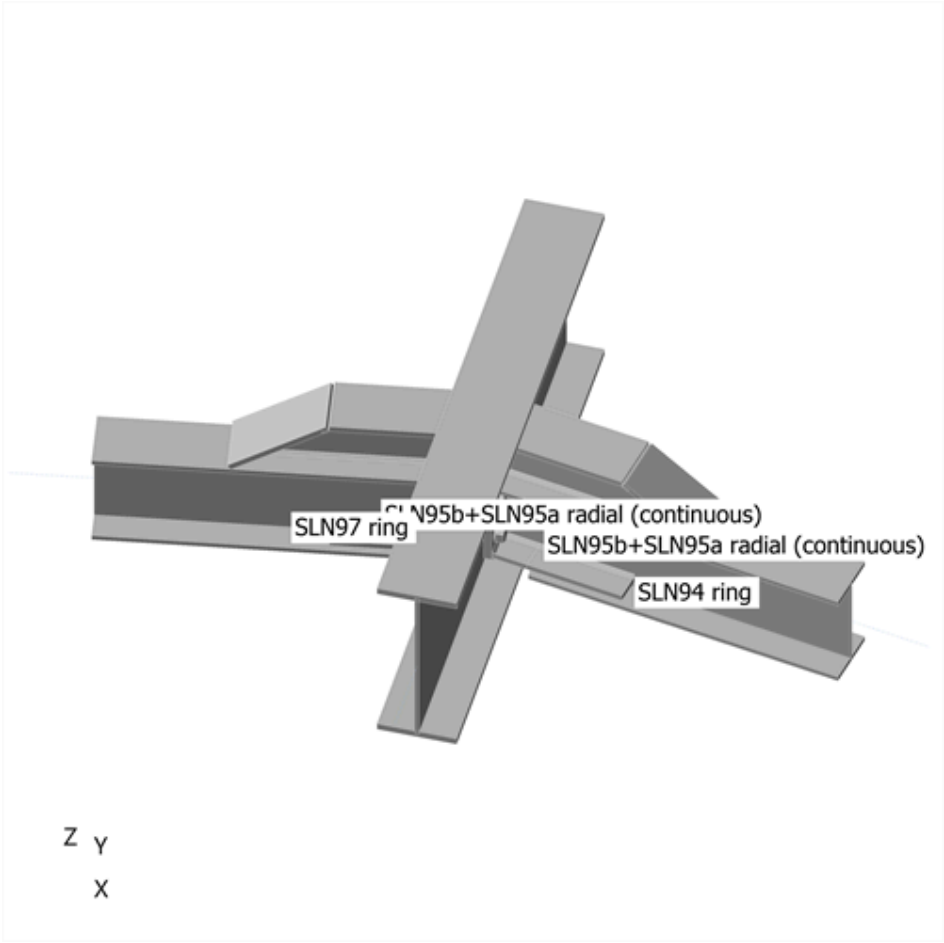
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN95b+SLN95a radial (continuous)-bfl 1	17.7	C LC_145	22.3	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-tfl 1	17.7	C LC_122	18.9	0.0	0.0	OK
SLN95b+SLN95a radial (continuous)-w 1	10.5	C LC_122	29.4	0.0	0.9	OK
SLN94 ring-bfl 1	12.7	C LC_147	50.8	0.0	0.0	OK
SLN94 ring-tfl 1	12.7	C LC_120	44.6	0.0	0.0	OK
SLN94 ring-w 1	7.2	C LC_147	49.3	0.0	0.0	OK
SLN97 ring-bfl 1	12.7	C LC_145	51.1	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	45.3	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_147	49.6	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_148	9.6	0.0	2.0	OK
Bottom flange plate SLN97 ring	15.0	C LC_145	20.7	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_145	20.3	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_122	28.8	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_126	6.6	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_120	12.9	0.0	0.0	OK
End plate SLN94 ring	20.0	C LC_122	8.3	0.0	6.1	OK
Bottom flange plate SLN94 ring	15.0	C LC_147	20.5	0.0	0.0	OK
Bottom flange half plate L SLN94 ring	15.0	C LC_147	19.9	0.0	0.0	OK
Bottom flange half plate R SLN94 ring	15.0	C LC_147	20.2	0.0	0.0	OK
Haunch web SLN94 ring	12.0	C LC_135	28.8	0.0	0.0	OK
Haunch top plate SLN94 ring	15.0	C LC_130	5.6	0.0	0.0	OK
Haunch sloping plate SLN94 ring	15.0	C LC_120	12.8	0.0	0.0	OK

Design data

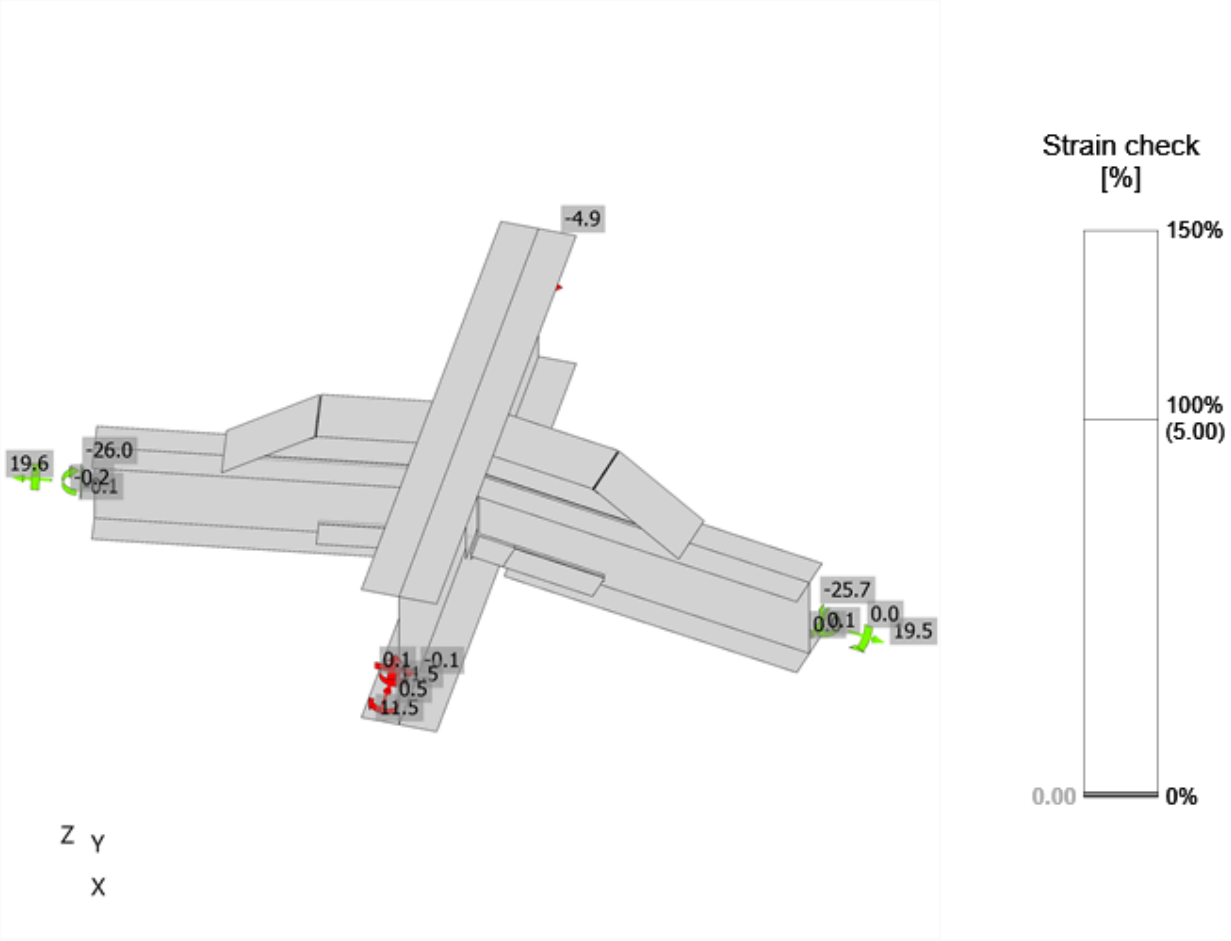
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

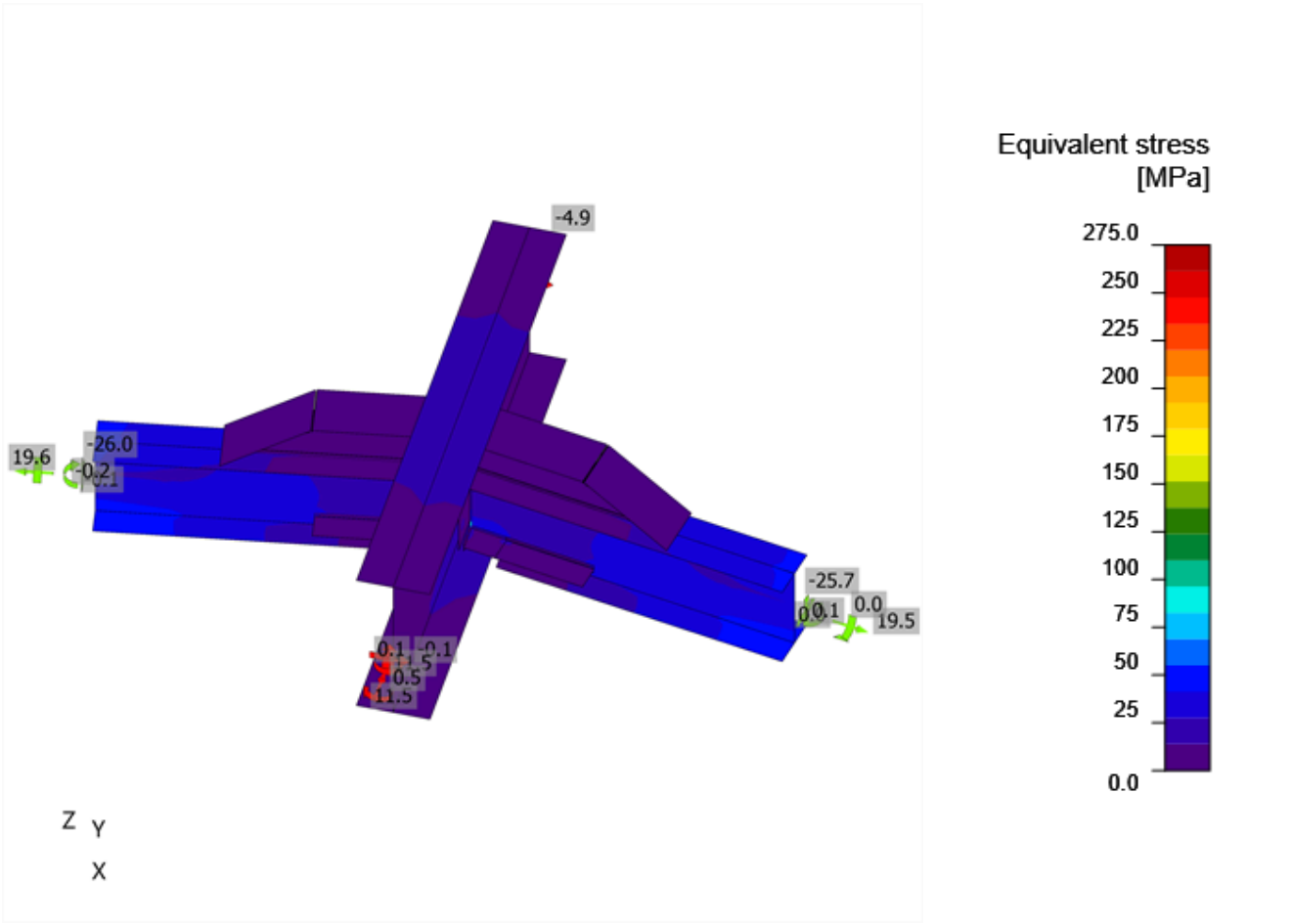
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

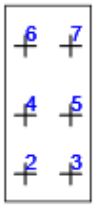


Strain check, C LC_145



Equivalent stress, C LC_145

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_148	2.7	0.5	176.5	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_119	2.2	0.6	176.5	0.7	0.7	0.8	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.4	0.9	229.5	0.7	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_145	1.3	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN94 ring	Bottom flange plate SLN94 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-tfl 1	◢ 10.5	149	C LC_121	10.6	0.0	-7.3	-4.4	-0.4	2.6	2.6	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 10.5	149	C LC_119	10.5	0.0	-7.2	-4.4	0.4	2.6	0.0	OK	OK
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	25.4	0.0	-11.3	-12.6	-3.8	6.3	6.1	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.9	0.0	-11.0	10.2	-0.5	5.2	5.2	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.0	0.0	2.3	2.3	13.6	5.9	5.9	OK	OK
		◢ 6.0 ◣	100	C LC_122	26.2	0.0	2.8	-4.1	-14.5	6.5	6.5	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	24.8	0.0	6.4	6.1	12.5	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.2	0.0	9.1	2.1	-15.2	7.0	6.2	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_122	25.6	0.0	5.7	6.2	-13.0	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_122	29.0	0.0	7.5	-6.9	14.6	7.2	6.7	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	67.8	0.0	-15.3	-17.4	33.9	16.8	9.6	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.8	0.0	-15.8	15.4	-31.5	15.5	9.1	OK	OK
End plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	208	C LC_145	20.7	0.0	-11.8	-9.8	0.2	5.1	5.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_147	25.4	0.0	-10.2	13.1	2.9	6.3	6.0	OK	OK
Bottom flange plate SLN94 ring	SLN94 ring-w 1	◢ 6.0 ◣	100	C LC_147	25.0	0.0	2.6	2.7	14.1	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.8	0.0	2.3	-3.1	-13.9	6.1	6.1	OK	OK
SLN94 ring-w 1	Bottom flange half plate L SLN94 ring	◢ 6.0 ◣	243	C LC_147	25.8	0.0	6.6	6.3	12.9	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.0	0.0	8.2	2.6	-14.6	6.7	6.2	OK	OK
SLN94 ring-w 1	Bottom flange half plate R SLN94 ring	◢ 6.0 ◣	243	C LC_119	24.3	0.0	5.6	6.0	-12.3	6.0	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_147	28.6	0.0	8.4	2.5	15.6	7.1	6.6	OK	OK
SLN94 ring-tfl 1	Haunch web SLN94 ring	◢ 6.0 ◣	572	C LC_122	65.9	0.0	-16.6	-17.3	32.5	16.3	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.1	0.0	-14.3	15.1	-32.1	15.6	9.1	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	22.6	0.0	-6.3	-8.3	9.3	5.6	5.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	29.1	0.0	-11.7	9.7	-12.0	7.2	6.0	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_120	7.2	0.0	1.3	1.2	3.9	1.8	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.0	0.0	2.4	-2.5	-5.7	2.7	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	12.3	0.0	-5.7	-4.3	4.6	3.0	3.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	10.3	0.0	-3.2	4.5	-3.4	2.5	0.0	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_122	5.0	0.0	1.8	1.6	2.2	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.3	0.0	-0.1	0.2	2.5	1.1	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_122	53.8	0.0	-9.7	-6.7	29.8	13.3	8.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	50.4	0.0	-8.6	9.2	-27.2	12.5	8.1	OK	OK
End plate SLN94 ring	Bottom flange plate SLN94 ring	◢ 6.0 ◣	148	C LC_119	14.4	0.0	7.5	4.4	5.6	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	23.9	0.0	-12.3	10.6	-5.3	5.9	5.5	OK	OK
End plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	145	C LC_120	10.4	0.0	2.3	2.4	5.3	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.1	0.0	1.4	-1.3	-4.4	2.0	0.0	OK	OK
End plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	149	C LC_130	9.3	0.0	-5.0	-3.8	2.4	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_124	7.9	0.0	-2.7	3.3	2.7	1.9	0.0	OK	OK
Haunch top plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	342	C LC_126	3.9	0.0	1.3	1.1	1.8	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	4.3	0.0	1.6	-1.4	-1.8	1.1	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch web SLN94 ring	◢ 6.0 ◣	271	C LC_135	54.1	0.0	-10.3	-10.3	28.9	13.4	8.4	OK	OK
		◢ 6.0 ◣	271	C LC_120	49.0	0.0	-7.9	6.8	-27.1	12.1	8.1	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_147	78.9	0.0	32.4	37.6	-17.8	19.5	13.7	OK	OK
SLN95b+SLN95a radial (continuous)-w 1	SLN94 ring-w 1	◢ 7.2	208	C LC_145	81.6	0.0	31.8	-39.4	18.3	20.2	14.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_122	7.4	0.0	-5.7	2.7	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.7	0.0	-0.8	2.1	0.0	0.9	0.0	OK	OK
Haunch sloping plate SLN94 ring	Haunch top plate SLN94 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.2	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_124	3.1	0.0	-0.7	1.7	-0.1	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_140
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring

Name	SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

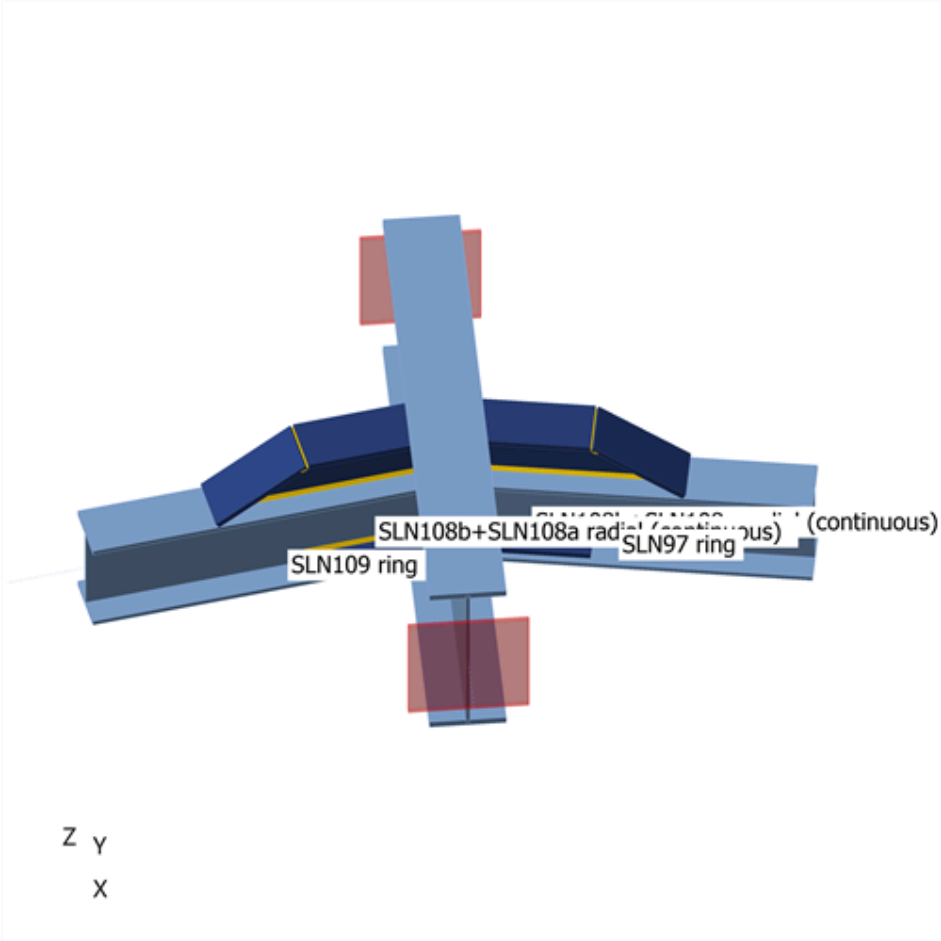
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN108b+SLN108a radial (continuous)	1 - UB457x191x89	140.0	0.0	0.0	0	0	0
SLN97 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN109 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN108b+SLN108a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN108b+SLN108a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN97 ring / end		Position	0
SLN109 ring / end		Position	0



SLN108b+SLN108a radial (continuous) - SLN97 ring - SLN109 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN97 ring / End	4.9	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
C LC_102	SLN97 ring / End	19.0	-0.2	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	18.9	0.1	-25.6	0.0	0.0	0.2
C LC_103	SLN97 ring / End	17.1	-0.2	-24.4	0.0	0.0	-0.2
	SLN109 ring / End	17.3	0.1	-24.0	0.0	0.0	0.1
C LC_104	SLN97 ring / End	12.3	0.0	-24.4	0.0	0.0	0.0
	SLN109 ring / End	12.3	0.1	-24.0	0.0	0.0	0.2
C LC_105	SLN97 ring / End	15.7	-0.1	-24.4	0.0	0.0	-0.1
	SLN109 ring / End	15.5	0.2	-24.0	0.0	0.0	0.2
C LC_106	SLN97 ring / End	13.6	-0.2	-24.4	0.0	0.0	-0.4
	SLN109 ring / End	13.4	-0.1	-24.0	0.0	0.0	-0.1
C LC_107	SLN97 ring / End	2.8	0.1	-0.5	0.0	0.0	0.2
	SLN109 ring / End	3.3	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN97 ring / End	-3.7	0.1	-0.5	0.0	0.0	0.2
	SLN109 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.3
C LC_109	SLN97 ring / End	0.3	0.1	-0.5	0.0	0.0	0.2
	SLN109 ring / End	0.1	0.1	-0.5	0.0	0.0	0.0
C LC_110	SLN97 ring / End	-1.3	-0.3	-0.5	0.0	0.0	-0.4
	SLN109 ring / End	-1.9	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN97 ring / End	12.9	0.0	-16.5	0.0	0.0	0.0
	SLN109 ring / End	13.3	0.1	-16.2	0.0	0.0	0.1
C LC_112	SLN97 ring / End	5.4	0.1	-16.5	0.0	0.0	0.1
	SLN109 ring / End	5.4	0.2	-16.2	0.0	0.0	0.3
C LC_113	SLN97 ring / End	10.3	0.0	-16.5	0.0	0.0	0.1
	SLN109 ring / End	10.0	0.2	-16.2	0.0	0.0	0.2
C LC_114	SLN97 ring / End	7.8	-0.3	-16.5	0.0	0.0	-0.5
	SLN109 ring / End	7.3	-0.2	-16.2	0.0	0.0	-0.3
C LC_115	SLN97 ring / End	4.1	0.1	-1.8	0.0	0.0	0.1
	SLN109 ring / End	4.6	0.0	-1.7	0.0	0.0	0.0
C LC_116	SLN97 ring / End	-2.5	0.1	-1.8	0.0	0.0	0.2
	SLN109 ring / End	-2.4	0.2	-1.7	0.0	0.0	0.3
C LC_117	SLN97 ring / End	1.6	0.1	-1.8	0.0	0.0	0.2
	SLN109 ring / End	1.4	0.1	-1.7	0.0	0.0	0.1
C LC_118	SLN97 ring / End	-0.1	-0.3	-1.8	0.0	0.0	-0.4
	SLN109 ring / End	-0.7	-0.2	-1.7	0.0	0.0	-0.3
C LC_119	SLN97 ring / End	19.0	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	18.8	0.1	-25.6	0.0	0.0	0.1
C LC_120	SLN97 ring / End	15.0	0.0	-26.0	0.0	0.0	0.0
	SLN109 ring / End	14.9	0.1	-25.6	0.0	0.0	0.2
C LC_121	SLN97 ring / End	17.5	-0.1	-26.0	0.0	0.0	-0.1
	SLN109 ring / End	17.7	0.3	-25.6	0.0	0.0	0.3
C LC_122	SLN97 ring / End	16.3	-0.2	-26.0	0.0	0.0	-0.4
	SLN109 ring / End	16.0	-0.1	-25.6	0.0	0.0	-0.1
C LC_123	SLN97 ring / End	16.8	-0.2	-19.7	0.0	0.0	-0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN109 ring / End	16.4	0.1	-19.4	0.0	0.0	0.1
C LC_124	SLN97 ring / End	10.6	0.1	-19.7	0.0	0.0	0.1
	SLN109 ring / End	10.8	0.2	-19.4	0.0	0.0	0.3
C LC_125	SLN97 ring / End	13.8	0.0	-19.7	0.0	0.0	0.1
	SLN109 ring / End	14.4	0.3	-19.4	0.0	0.0	0.5
C LC_126	SLN97 ring / End	13.2	-0.3	-19.7	0.0	0.0	-0.6
	SLN109 ring / End	12.5	-0.2	-19.4	0.0	0.0	-0.3
C LC_127	SLN97 ring / End	8.0	-0.1	-5.0	0.0	0.0	-0.2
	SLN109 ring / End	7.6	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN97 ring / End	2.7	0.1	-5.0	0.0	0.0	0.2
	SLN109 ring / End	3.0	0.2	-4.9	0.0	0.0	0.3
C LC_129	SLN97 ring / End	5.1	0.1	-5.0	0.0	0.0	0.2
	SLN109 ring / End	5.9	0.2	-4.9	0.0	0.0	0.4
C LC_130	SLN97 ring / End	5.3	-0.3	-5.0	0.0	0.0	-0.5
	SLN109 ring / End	4.6	-0.2	-4.9	0.0	0.0	-0.4
C LC_135	SLN97 ring / End	15.9	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
C LC_141	SLN97 ring / End	5.0	-0.1	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	5.0	0.0	-4.9	0.0	0.0	0.1
C LC_142	SLN97 ring / End	4.7	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.6	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN97 ring / End	5.0	-0.1	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	5.0	0.1	-4.9	0.0	0.0	0.1
C LC_144	SLN97 ring / End	4.7	0.0	-5.0	0.0	0.0	-0.1
	SLN109 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN97 ring / End	19.8	-0.2	-26.0	0.0	0.0	-0.3
	SLN109 ring / End	19.8	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN97 ring / End	18.2	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	18.0	0.1	-25.6	0.0	0.0	0.1
C LC_147	SLN97 ring / End	19.7	-0.2	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	19.5	0.2	-25.6	0.0	0.0	0.2
C LC_148	SLN97 ring / End	18.3	-0.1	-26.0	0.0	0.0	-0.2
	SLN109 ring / End	18.3	0.0	-25.6	0.0	0.0	0.1
C LC_151	SLN97 ring / End	12.0	-0.1	-11.7	0.0	0.0	-0.2
	SLN109 ring / End	11.9	0.1	-11.5	0.0	0.0	0.1
C LC_152	SLN97 ring / End	18.9	-0.1	-14.0	0.0	0.0	-0.2
	SLN109 ring / End	18.8	0.1	-14.0	0.0	0.0	0.1
C LC_153	SLN97 ring / End	19.0	-0.2	-20.7	0.0	0.0	-0.2
	SLN109 ring / End	18.9	0.1	-20.6	0.0	0.0	0.2
C LC_154	SLN97 ring / End	12.1	-0.1	-11.7	0.0	0.0	-0.2
	SLN109 ring / End	12.0	0.1	-11.5	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.5 < 100%	OK
Buckling	Not calculated	

Plates

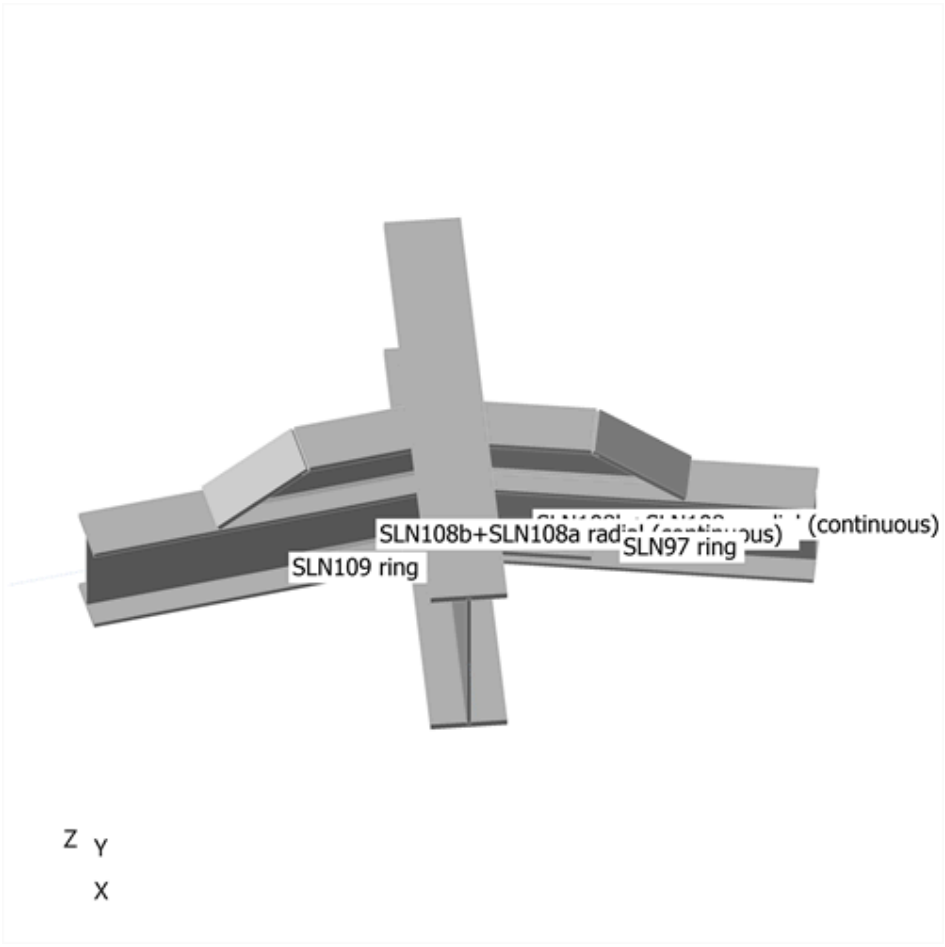
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN108b+SLN108a radial (continuous)-bfl 1	17.7	C LC_122	22.2	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-tfl 1	17.7	C LC_122	18.9	0.0	0.0	OK
SLN108b+SLN108a radial (continuous)-w 1	10.5	C LC_122	29.5	0.0	1.1	OK
SLN97 ring-bfl 1	12.7	C LC_122	51.2	0.0	0.0	OK
SLN97 ring-tfl 1	12.7	C LC_122	46.0	0.0	0.0	OK
SLN97 ring-w 1	7.2	C LC_145	49.7	0.0	0.0	OK
SLN109 ring-bfl 1	12.7	C LC_146	50.1	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_120	44.7	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_145	48.9	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_126	9.8	0.0	8.8	OK
Bottom flange plate SLN109 ring	15.0	C LC_126	21.4	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_147	19.9	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_120	28.2	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_126	7.6	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN97 ring	20.0	C LC_145	10.3	0.0	2.1	OK
Bottom flange plate SLN97 ring	15.0	C LC_145	20.7	0.0	0.0	OK
Bottom flange half plate L SLN97 ring	15.0	C LC_145	20.3	0.0	0.0	OK
Bottom flange half plate R SLN97 ring	15.0	C LC_147	20.0	0.0	0.0	OK
Haunch web SLN97 ring	12.0	C LC_135	28.8	0.0	0.0	OK
Haunch top plate SLN97 ring	15.0	C LC_126	6.9	0.0	0.0	OK
Haunch sloping plate SLN97 ring	15.0	C LC_120	12.9	0.0	0.0	OK

Design data

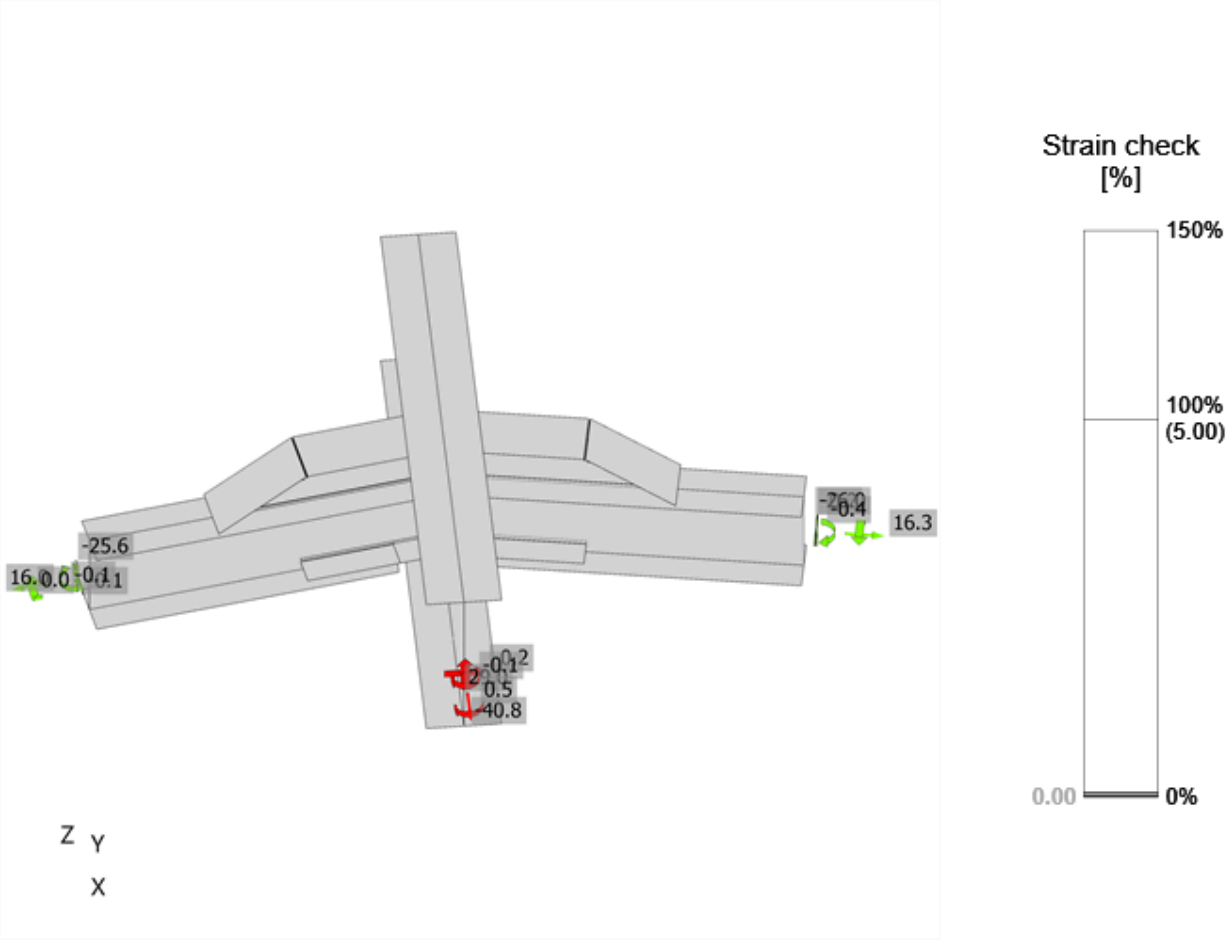
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

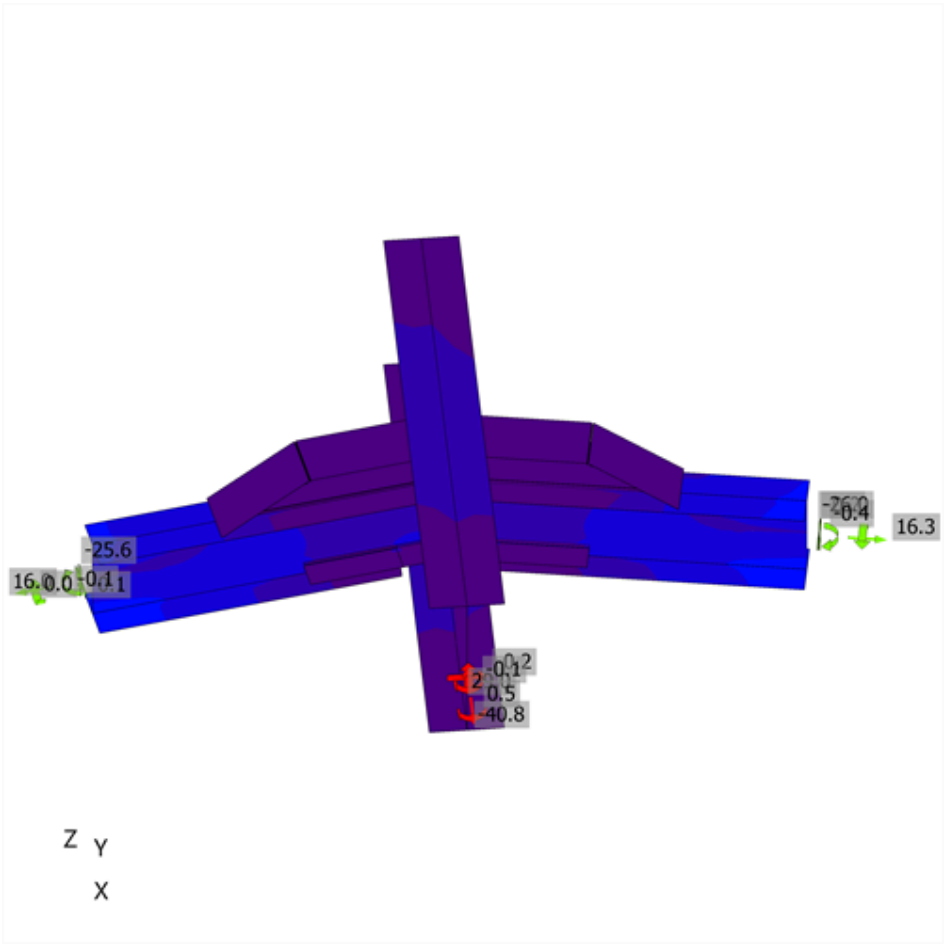
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

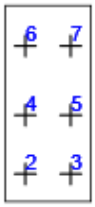


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	2.1	0.5	176.5	0.6	0.5	0.7	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.9	0.6	176.5	0.9	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.0	0.8	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.2	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN97 ring	Bottom flange plate SLN97 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-tfl 1	◢ 10.5	149	C LC_121	10.6	0.0	-7.0	-4.6	-0.5	2.6	2.6	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 10.5	149	C LC_145	10.6	0.0	-6.8	-4.7	0.4	2.6	2.6	OK	OK
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.7	0.0	-10.4	-12.7	-2.4	6.1	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.2	0.0	-11.2	9.7	0.4	5.0	5.0	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.1	0.0	2.7	3.4	14.0	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.2	0.0	2.4	-3.3	-14.1	6.2	6.2	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	25.5	0.0	6.2	6.2	12.9	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.4	0.0	7.9	2.7	-14.9	6.8	6.4	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_122	24.5	0.0	5.6	6.0	-12.4	6.0	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_147	27.7	0.0	8.7	2.2	15.0	6.8	6.4	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	65.1	0.0	-15.9	-16.7	32.4	16.1	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.6	0.0	-14.6	15.6	-32.2	15.7	9.1	OK	OK
End plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	208	C LC_145	21.1	0.0	-12.3	-9.9	-1.1	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	25.9	0.0	-10.0	13.6	2.2	6.4	6.1	OK	OK
Bottom flange plate SLN97 ring	SLN97 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.6	0.0	2.4	2.4	14.0	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	26.5	0.0	2.7	-4.1	-14.6	6.5	6.5	OK	OK
SLN97 ring-w 1	Bottom flange half plate L SLN97 ring	◢ 6.0 ◣	243	C LC_147	24.6	0.0	6.4	6.0	12.3	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.9	0.0	9.2	2.1	-15.0	6.9	6.1	OK	OK
SLN97 ring-w 1	Bottom flange half plate R SLN97 ring	◢ 6.0 ◣	243	C LC_145	25.9	0.0	5.9	6.3	-13.1	6.4	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_122	28.9	0.0	7.4	-6.9	14.6	7.1	6.7	OK	OK
SLN97 ring-tfl 1	Haunch web SLN97 ring	◢ 6.0 ◣	572	C LC_122	67.5	0.0	-15.2	-17.2	33.8	16.7	9.6	OK	OK
		◢ 6.0 ◣	572	C LC_120	63.4	0.0	-15.9	15.7	-31.8	15.7	9.1	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	21.7	0.0	-5.4	-8.4	8.7	5.4	5.4	OK	OK
		◢ 6.0 ◣	149	C LC_126	29.7	0.0	-12.4	9.4	-12.5	7.3	6.0	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	7.5	0.0	1.8	1.6	3.9	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.0	0.0	2.1	-2.5	-5.7	2.7	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	13.5	0.0	-6.6	-4.2	5.3	3.3	3.3	OK	OK
		◢ 6.0 ◣	149	C LC_126	9.6	0.0	-2.3	4.7	-2.7	2.4	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_126	4.5	0.0	1.4	1.2	2.1	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.6	0.0	1.7	-1.5	-2.0	1.1	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_120	52.5	0.0	-9.7	-11.0	27.7	13.0	8.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	49.9	0.0	-8.4	7.2	-27.5	12.3	8.1	OK	OK
End plate SLN97 ring	Bottom flange plate SLN97 ring	◢ 6.0 ◣	149	C LC_130	18.4	0.0	-7.8	-9.1	3.2	4.6	4.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	28.9	0.0	-14.8	13.1	-5.7	7.1	5.8	OK	OK
End plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	145	C LC_121	10.7	0.0	2.0	2.4	5.6	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.2	0.0	1.8	-1.6	-4.3	2.0	0.0	OK	OK
End plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	149	C LC_126	11.8	0.0	-6.1	-4.9	3.1	2.9	2.9	OK	OK
		◢ 6.0 ◣	149	C LC_126	9.2	0.0	-3.0	4.2	-2.8	2.3	0.0	OK	OK
Haunch top plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	342	C LC_126	5.1	0.0	1.7	1.4	2.4	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.2	0.0	-0.2	0.2	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch web SLN97 ring	◢ 6.0 ◣	271	C LC_135	53.8	0.0	-9.8	-8.0	29.5	13.3	8.4	OK	OK
		◢ 6.0 ◣	271	C LC_120	50.9	0.0	-8.8	9.1	-27.5	12.6	8.1	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_147	79.3	0.0	30.7	38.2	-18.1	19.6	13.5	OK	OK
SLN108b+SLN108a radial (continuous)-w 1	SLN97 ring-w 1	◢ 7.2	208	C LC_145	82.8	0.0	31.6	-40.5	17.8	20.5	14.1	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.3	2.7	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_125	3.3	0.0	-0.7	1.9	0.0	0.8	0.0	OK	OK
Haunch sloping plate SLN97 ring	Haunch top plate SLN97 ring	◢ 6.0 ◣	146	C LC_122	7.4	0.0	-5.7	2.7	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.0	0.0	-0.8	2.2	-0.2	1.0	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_160
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring

Name	SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

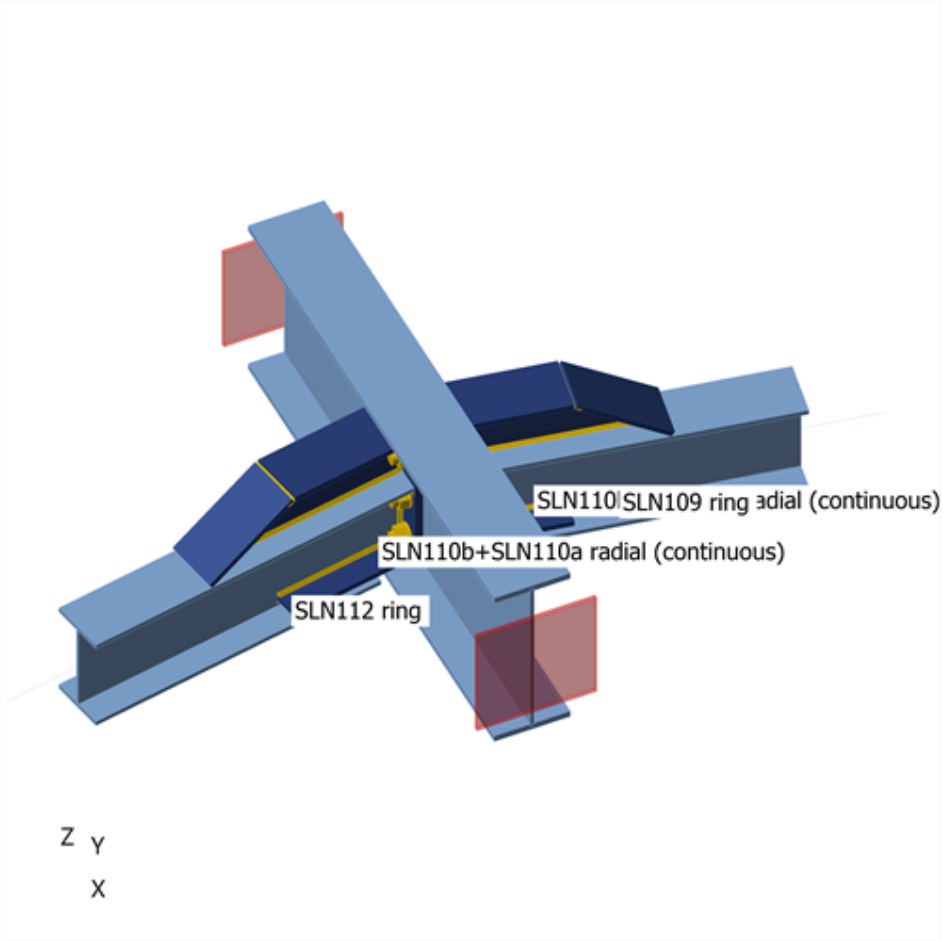
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN110b+SLN110a radial (continuous)	1 - UB457x191x89	160.0	0.0	0.0	0	0	0
SLN109 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102
SLN112 ring	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN110b+SLN110a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN110b+SLN110a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN109 ring / end		Position	0
SLN112 ring / end		Position	0



SLN110b+SLN110a radial (continuous) - SLN109 ring - SLN112 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN109 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
	SLN112 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
C LC_102	SLN109 ring / End	18.9	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	18.9	-0.2	-25.5	0.0	0.0	-0.2
C LC_103	SLN109 ring / End	17.4	0.0	-24.0	0.0	0.0	0.0
	SLN112 ring / End	17.4	0.0	-24.0	0.0	0.0	-0.1
C LC_104	SLN109 ring / End	12.2	0.2	-24.0	0.0	0.0	0.3
	SLN112 ring / End	12.2	-0.1	-24.0	0.0	0.0	-0.1
C LC_105	SLN109 ring / End	15.3	0.1	-24.0	0.0	0.0	0.2
	SLN112 ring / End	15.4	0.0	-24.0	0.0	0.0	0.0
C LC_106	SLN109 ring / End	13.5	0.0	-24.0	0.0	0.0	-0.1
	SLN112 ring / End	13.4	-0.3	-24.0	0.0	0.0	-0.4
C LC_107	SLN109 ring / End	3.3	0.0	-0.5	0.0	0.0	-0.1
	SLN112 ring / End	3.1	0.1	-0.5	0.0	0.0	0.1
C LC_108	SLN109 ring / End	-3.6	0.2	-0.5	0.0	0.0	0.3
	SLN112 ring / End	-3.5	0.1	-0.5	0.0	0.0	0.1
C LC_109	SLN109 ring / End	0.1	0.1	-0.5	0.0	0.0	0.1
	SLN112 ring / End	0.7	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN109 ring / End	-1.9	-0.2	-0.5	0.0	0.0	-0.4
	SLN112 ring / End	-2.5	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN109 ring / End	13.4	0.0	-16.2	0.0	0.0	0.0
	SLN112 ring / End	13.3	0.1	-16.2	0.0	0.0	0.0
C LC_112	SLN109 ring / End	5.4	0.3	-16.2	0.0	0.0	0.4
	SLN112 ring / End	5.4	0.0	-16.2	0.0	0.0	0.0
C LC_113	SLN109 ring / End	9.9	0.1	-16.2	0.0	0.0	0.2
	SLN112 ring / End	10.4	0.1	-16.2	0.0	0.0	0.2
C LC_114	SLN109 ring / End	7.4	-0.2	-16.2	0.0	0.0	-0.3
	SLN112 ring / End	6.9	-0.3	-16.2	0.0	0.0	-0.5
C LC_115	SLN109 ring / End	4.6	0.0	-1.7	0.0	0.0	-0.1
	SLN112 ring / End	4.4	0.1	-1.7	0.0	0.0	0.1
C LC_116	SLN109 ring / End	-2.4	0.2	-1.7	0.0	0.0	0.3
	SLN112 ring / End	-2.3	0.1	-1.7	0.0	0.0	0.1
C LC_117	SLN109 ring / End	1.4	0.1	-1.7	0.0	0.0	0.2
	SLN112 ring / End	2.0	0.1	-1.7	0.0	0.0	0.3
C LC_118	SLN109 ring / End	-0.7	-0.2	-1.7	0.0	0.0	-0.3
	SLN112 ring / End	-1.2	-0.2	-1.7	0.0	0.0	-0.4
C LC_119	SLN109 ring / End	18.9	0.0	-25.6	0.0	0.0	0.1
	SLN112 ring / End	18.9	-0.1	-25.5	0.0	0.0	-0.2
C LC_120	SLN109 ring / End	14.8	0.3	-25.6	0.0	0.0	0.3
	SLN112 ring / End	14.8	-0.1	-25.5	0.0	0.0	-0.1
C LC_121	SLN109 ring / End	17.5	0.2	-25.6	0.0	0.0	0.3
	SLN112 ring / End	17.6	0.0	-25.5	0.0	0.0	0.0
C LC_122	SLN109 ring / End	16.1	0.0	-25.6	0.0	0.0	-0.1
	SLN112 ring / End	16.0	-0.3	-25.5	0.0	0.0	-0.4
C LC_123	SLN109 ring / End	16.5	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN112 ring / End	16.4	-0.1	-19.4	0.0	0.0	-0.1
C LC_124	SLN109 ring / End	10.7	0.3	-19.4	0.0	0.0	0.4
	SLN112 ring / End	10.8	0.0	-19.4	0.0	0.0	0.0
C LC_125	SLN109 ring / End	14.3	0.2	-19.4	0.0	0.0	0.4
	SLN112 ring / End	14.7	0.2	-19.4	0.0	0.0	0.2
C LC_126	SLN109 ring / End	12.6	-0.2	-19.4	0.0	0.0	-0.3
	SLN112 ring / End	12.2	-0.4	-19.4	0.0	0.0	-0.6
C LC_127	SLN109 ring / End	7.7	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	7.5	0.0	-4.9	0.0	0.0	-0.1
C LC_128	SLN109 ring / End	3.0	0.2	-4.9	0.0	0.0	0.3
	SLN112 ring / End	3.1	0.1	-4.9	0.0	0.0	0.1
C LC_129	SLN109 ring / End	5.8	0.2	-4.9	0.0	0.0	0.3
	SLN112 ring / End	6.3	0.2	-4.9	0.0	0.0	0.2
C LC_130	SLN109 ring / End	4.6	-0.2	-4.9	0.0	0.0	-0.3
	SLN112 ring / End	4.1	-0.3	-4.9	0.0	0.0	-0.5
C LC_135	SLN109 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	15.9	-0.2	-25.5	0.0	0.0	-0.2
C LC_141	SLN109 ring / End	5.1	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	5.1	0.0	-4.9	0.0	0.0	-0.1
C LC_142	SLN109 ring / End	4.6	0.1	-4.9	0.0	0.0	0.1
	SLN112 ring / End	4.6	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN109 ring / End	4.9	0.0	-4.9	0.0	0.0	0.1
	SLN112 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN109 ring / End	4.7	0.0	-4.9	0.0	0.0	0.0
	SLN112 ring / End	4.8	-0.1	-4.9	0.0	0.0	-0.1
C LC_145	SLN109 ring / End	19.9	0.0	-25.6	0.0	0.0	0.1
	SLN112 ring / End	19.9	-0.1	-25.5	0.0	0.0	-0.2
C LC_146	SLN109 ring / End	17.9	0.2	-25.6	0.0	0.0	0.2
	SLN112 ring / End	17.8	-0.2	-25.5	0.0	0.0	-0.3
C LC_147	SLN109 ring / End	19.3	0.1	-25.6	0.0	0.0	0.2
	SLN112 ring / End	19.1	-0.1	-25.5	0.0	0.0	-0.2
C LC_148	SLN109 ring / End	18.4	0.1	-25.6	0.0	0.0	0.1
	SLN112 ring / End	18.6	-0.2	-25.5	0.0	0.0	-0.3
C LC_151	SLN109 ring / End	11.9	0.1	-11.6	0.0	0.0	0.1
	SLN112 ring / End	11.9	-0.1	-11.8	0.0	0.0	-0.1
C LC_152	SLN109 ring / End	18.8	0.1	-14.0	0.0	0.0	0.1
	SLN112 ring / End	18.8	-0.1	-14.1	0.0	0.0	-0.2
C LC_153	SLN109 ring / End	18.9	0.1	-20.7	0.0	0.0	0.2
	SLN112 ring / End	18.9	-0.2	-21.0	0.0	0.0	-0.2
C LC_154	SLN109 ring / End	12.0	0.1	-11.6	0.0	0.0	0.1
	SLN112 ring / End	12.0	-0.1	-11.8	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	20.1 < 100%	OK
Buckling	Not calculated	

Plates

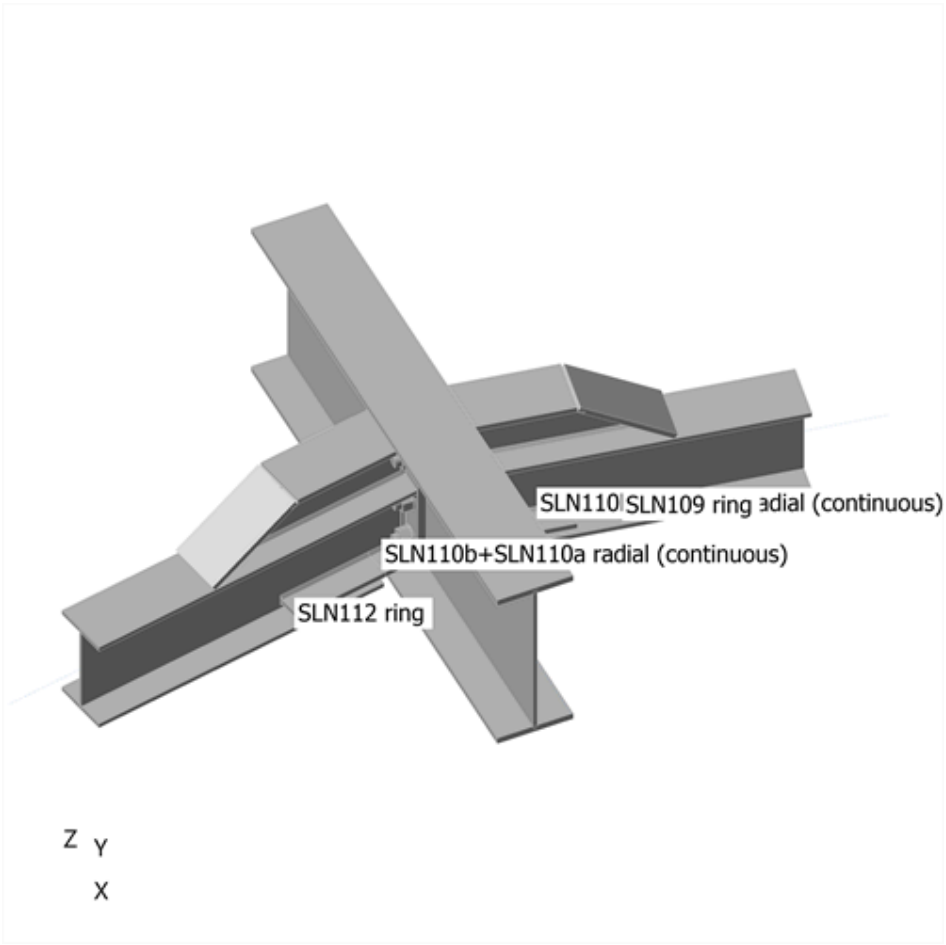
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN110b+SLN110a radial (continuous)-bfl 1	17.7	C LC_148	22.1	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-tfl 1	17.7	C LC_122	18.7	0.0	0.0	OK
SLN110b+SLN110a radial (continuous)-w 1	10.5	C LC_122	29.1	0.0	1.0	OK
SLN109 ring-bfl 1	12.7	C LC_145	50.4	0.0	0.0	OK
SLN109 ring-tfl 1	12.7	C LC_120	44.2	0.0	0.0	OK
SLN109 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
SLN112 ring-bfl 1	12.7	C LC_145	50.5	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	44.8	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_145	48.8	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_148	9.2	0.0	1.0	OK
Bottom flange plate SLN112 ring	15.0	C LC_126	21.5	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_145	20.2	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_145	19.7	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	28.2	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_130	7.3	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN109 ring	20.0	C LC_147	8.5	0.0	0.9	OK
Bottom flange plate SLN109 ring	15.0	C LC_145	20.4	0.0	0.0	OK
Bottom flange half plate L SLN109 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN109 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN109 ring	12.0	C LC_120	28.1	0.0	0.0	OK
Haunch top plate SLN109 ring	15.0	C LC_126	6.6	0.0	0.0	OK
Haunch sloping plate SLN109 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

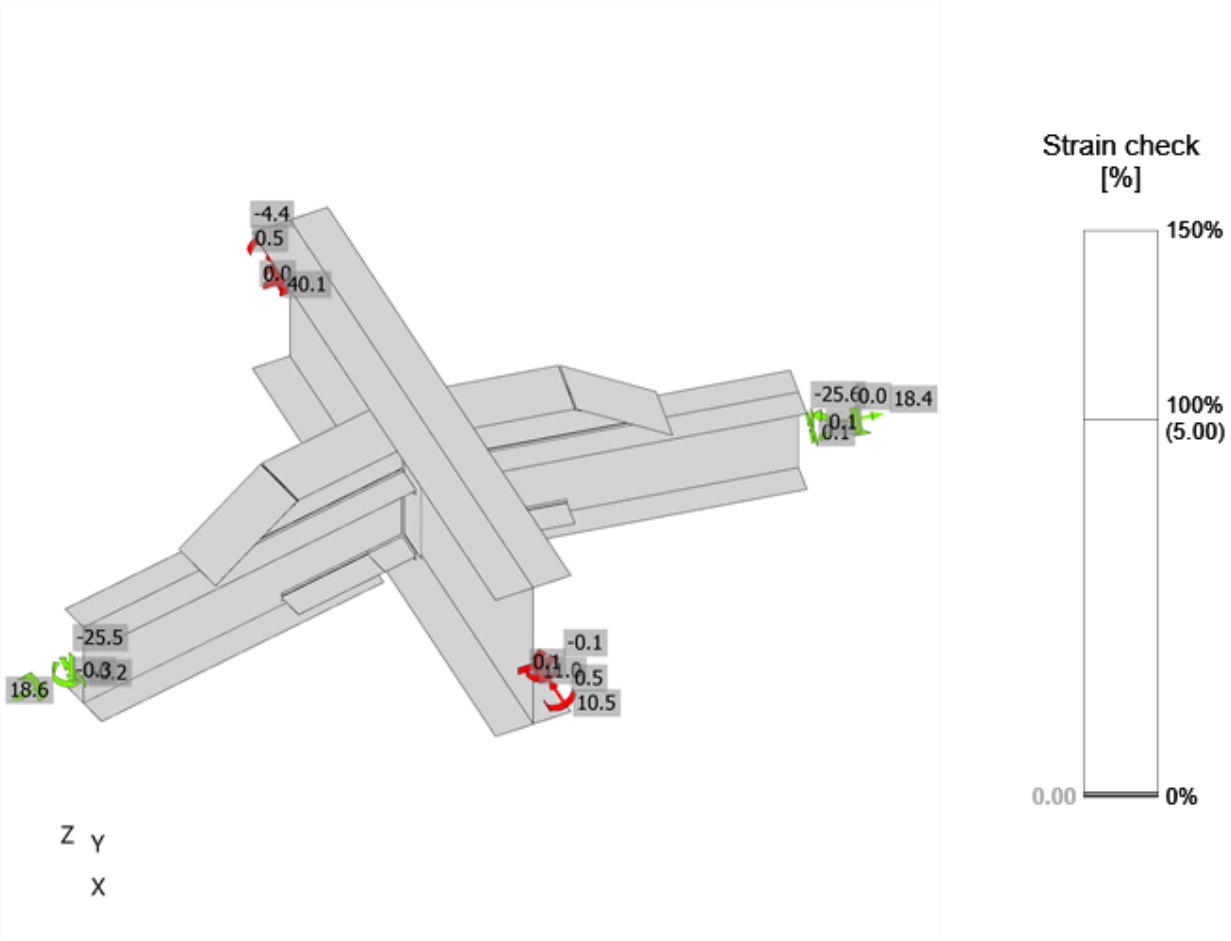
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

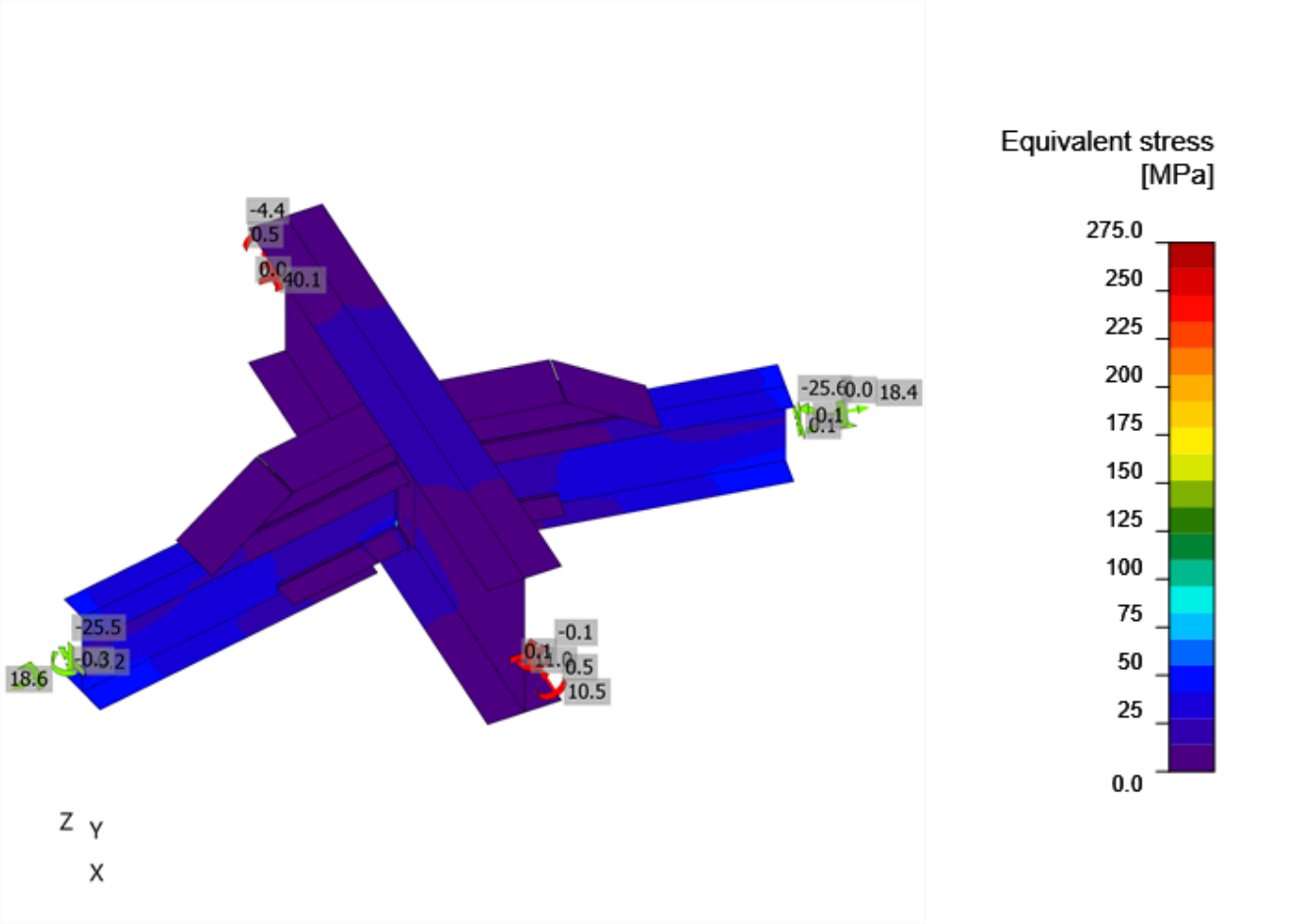
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

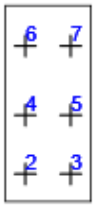


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	2.6	0.5	176.5	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_119	2.1	0.6	176.5	0.7	0.7	0.8	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_147	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.4	0.8	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_122	1.7	0.9	229.5	0.5	0.8	0.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN109 ring	Bottom flange plate SLN109 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-tfl 1	◢ 10.5	149	C LC_122	12.1	0.0	-7.8	-5.2	0.9	3.0	3.0	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 10.5	149	C LC_121	12.1	0.0	-7.8	-5.2	-0.8	3.0	3.0	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_145	25.5	0.0	-10.9	-12.8	-3.5	6.3	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.6	0.0	-11.2	10.0	-0.1	5.1	5.1	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.1	0.0	2.4	2.6	13.6	6.0	6.0	OK	OK
		◢ 6.0 ◣	100	C LC_122	26.2	0.0	2.7	-4.0	-14.5	6.5	6.5	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_121	24.3	0.0	5.9	5.9	12.3	6.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.1	0.0	8.8	2.2	-15.2	6.9	6.3	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_122	25.4	0.0	5.8	6.2	-12.9	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_122	28.2	0.0	7.2	-6.7	14.3	7.0	6.6	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	65.7	0.0	-14.6	-16.4	33.1	16.2	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_121	62.5	0.0	-15.6	15.8	-31.2	15.5	9.0	OK	OK
End plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	208	C LC_145	20.5	0.0	-11.7	-9.7	0.0	5.1	5.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	25.1	0.0	-10.0	13.0	2.5	6.2	6.0	OK	OK
Bottom flange plate SLN109 ring	SLN109 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.2	0.0	2.7	3.3	14.1	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.0	0.0	2.3	-3.0	-14.0	6.2	6.2	OK	OK
SLN109 ring-w 1	Bottom flange half plate L SLN109 ring	◢ 6.0 ◣	243	C LC_147	25.3	0.0	6.2	6.2	12.7	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.2	0.0	7.8	2.8	-14.8	6.7	6.3	OK	OK
SLN109 ring-w 1	Bottom flange half plate R SLN109 ring	◢ 6.0 ◣	243	C LC_122	24.3	0.0	5.6	5.9	-12.3	6.0	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.1	0.0	8.3	2.5	15.3	6.9	6.5	OK	OK
SLN109 ring-tfl 1	Haunch web SLN109 ring	◢ 6.0 ◣	572	C LC_122	64.6	0.0	-15.9	-16.5	32.2	16.0	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_120	64.3	0.0	-14.8	15.9	-32.4	15.9	9.2	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	23.7	0.0	-6.4	-8.9	9.7	5.9	5.8	OK	OK
		◢ 6.0 ◣	149	C LC_126	30.8	0.0	-12.6	10.1	-12.7	7.6	6.1	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	7.5	0.0	1.4	1.3	4.1	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.9	0.0	2.4	-2.6	-5.6	2.7	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_126	14.4	0.0	-6.7	-5.1	5.4	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	12.0	0.0	-3.7	5.3	-3.9	3.0	0.0	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_126	5.2	0.0	1.7	1.4	2.4	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.2	0.0	-0.1	0.2	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_122	51.5	0.0	-8.9	-6.2	28.6	12.7	8.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	50.8	0.0	-9.1	11.9	-26.3	12.6	8.0	OK	OK
End plate SLN109 ring	Bottom flange plate SLN109 ring	◢ 6.0 ◣	149	C LC_110	16.6	0.0	-7.2	-8.3	2.3	4.1	4.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	26.0	0.0	-13.6	11.5	-5.6	6.4	5.6	OK	OK
End plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	145	C LC_121	10.6	0.0	2.3	2.4	5.4	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	1.4	-1.4	-4.3	2.0	0.0	OK	OK
End plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	149	C LC_126	11.1	0.0	-6.1	-4.4	3.1	2.7	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	8.1	0.0	-2.5	3.6	2.7	2.0	0.0	OK	OK
Haunch top plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	342	C LC_126	4.2	0.0	1.4	1.2	2.0	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.4	0.0	1.6	-1.4	-1.9	1.1	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch web SLN109 ring	◢ 6.0 ◣	271	C LC_122	52.0	0.0	-9.4	-8.4	28.3	12.9	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	50.6	0.0	-8.6	7.2	-27.9	12.5	8.2	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_147	78.1	0.0	31.4	36.9	-18.4	19.3	13.4	OK	OK
SLN110b+SLN110a radial (continuous)-w 1	SLN109 ring-w 1	◢ 7.2	208	C LC_145	81.2	0.0	31.5	-38.8	18.9	20.1	13.9	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.6	2.6	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.9	0.0	-0.9	2.2	-0.1	1.0	0.0	OK	OK
Haunch sloping plate SLN109 ring	Haunch top plate SLN109 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.2	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.1	0.0	-0.6	1.7	-0.2	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_180
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring

Name	SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

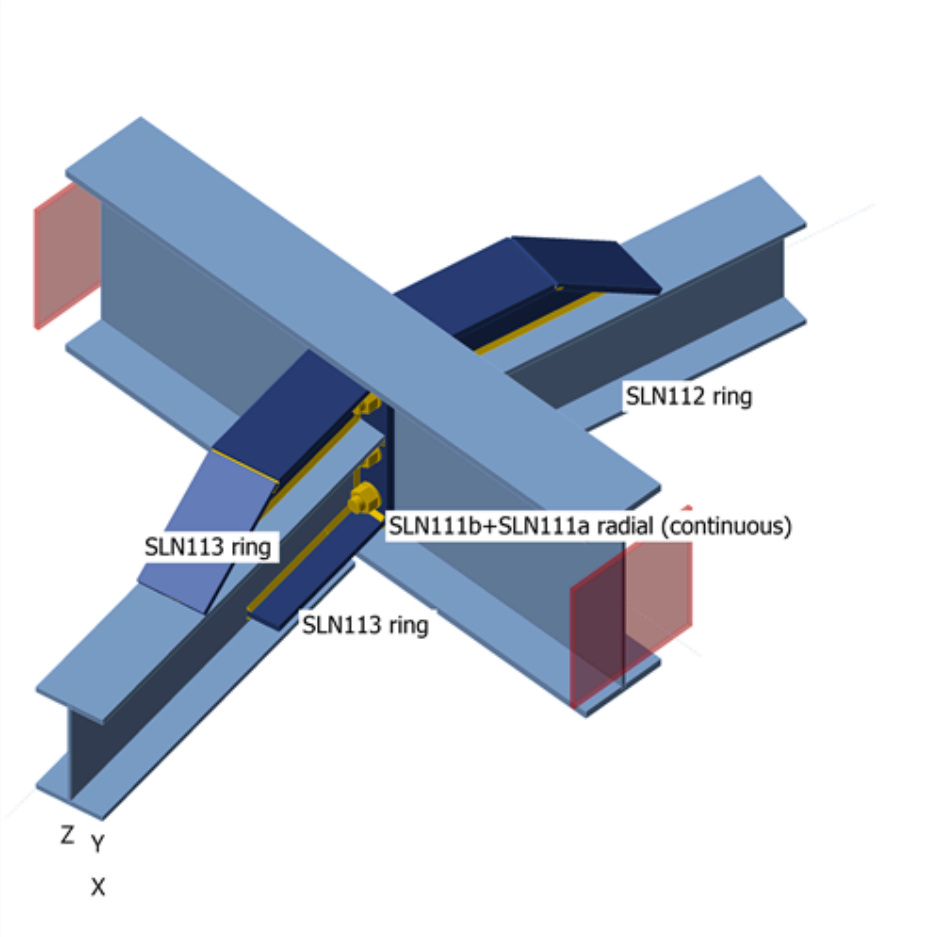
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN111b+SLN111a radial (continuous)	1 - UB457x191x89	-180.0	0.0	0.0	0	0	0
SLN112 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102
SLN113 ring	2 - UB254x146x43	-80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN111b+SLN111a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN111b+SLN111a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN112 ring / end		Position	0
SLN113 ring / end		Position	0



SLN111b+SLN111a radial (continuous) - SLN112 ring - SLN113 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN112 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
C LC_102	SLN112 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	18.9	0.1	-25.6	0.0	0.0	0.2
C LC_103	SLN112 ring / End	17.4	-0.2	-24.0	0.0	0.0	-0.2
	SLN113 ring / End	17.3	0.1	-24.0	0.0	0.0	0.1
C LC_104	SLN112 ring / End	12.1	0.0	-24.0	0.0	0.0	0.0
	SLN113 ring / End	12.3	0.1	-24.0	0.0	0.0	0.2
C LC_105	SLN112 ring / End	15.3	0.0	-24.0	0.0	0.0	0.0
	SLN113 ring / End	15.1	0.2	-24.0	0.0	0.0	0.3
C LC_106	SLN112 ring / End	13.5	-0.2	-24.0	0.0	0.0	-0.3
	SLN113 ring / End	13.7	0.0	-24.0	0.0	0.0	0.0
C LC_107	SLN112 ring / End	3.1	0.1	-0.5	0.0	0.0	0.1
	SLN113 ring / End	2.9	-0.1	-0.5	0.0	0.0	-0.1
C LC_108	SLN112 ring / End	-3.5	0.1	-0.5	0.0	0.0	0.2
	SLN113 ring / End	-3.3	0.2	-0.5	0.0	0.0	0.3
C LC_109	SLN112 ring / End	0.7	0.1	-0.5	0.0	0.0	0.2
	SLN113 ring / End	0.6	0.1	-0.5	0.0	0.0	0.2
C LC_110	SLN112 ring / End	-2.4	-0.2	-0.5	0.0	0.0	-0.2
	SLN113 ring / End	-2.3	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN112 ring / End	13.3	-0.1	-16.2	0.0	0.0	0.0
	SLN113 ring / End	13.0	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN112 ring / End	5.4	0.1	-16.2	0.0	0.0	0.1
	SLN113 ring / End	5.6	0.2	-16.2	0.0	0.0	0.3
C LC_113	SLN112 ring / End	10.2	0.1	-16.2	0.0	0.0	0.1
	SLN113 ring / End	10.1	0.2	-16.2	0.0	0.0	0.3
C LC_114	SLN112 ring / End	7.0	-0.3	-16.2	0.0	0.0	-0.4
	SLN113 ring / End	7.2	-0.2	-16.2	0.0	0.0	-0.2
C LC_115	SLN112 ring / End	4.4	0.0	-1.7	0.0	0.0	0.1
	SLN113 ring / End	4.2	0.0	-1.7	0.0	0.0	-0.1
C LC_116	SLN112 ring / End	-2.3	0.1	-1.7	0.0	0.0	0.1
	SLN113 ring / End	-2.1	0.2	-1.7	0.0	0.0	0.3
C LC_117	SLN112 ring / End	1.9	0.1	-1.7	0.0	0.0	0.2
	SLN113 ring / End	1.9	0.1	-1.7	0.0	0.0	0.2
C LC_118	SLN112 ring / End	-1.2	-0.2	-1.7	0.0	0.0	-0.3
	SLN113 ring / End	-1.1	-0.2	-1.7	0.0	0.0	-0.2
C LC_119	SLN112 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	18.8	0.2	-25.6	0.0	0.0	0.2
C LC_120	SLN112 ring / End	14.8	0.0	-25.6	0.0	0.0	0.0
	SLN113 ring / End	14.9	0.1	-25.6	0.0	0.0	0.2
C LC_121	SLN112 ring / End	17.4	0.0	-25.6	0.0	0.0	0.0
	SLN113 ring / End	17.8	0.2	-25.6	0.0	0.0	0.4
C LC_122	SLN112 ring / End	16.2	-0.3	-25.6	0.0	0.0	-0.5
	SLN113 ring / End	15.9	-0.1	-25.6	0.0	0.0	-0.1
C LC_123	SLN112 ring / End	16.4	-0.2	-19.4	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN113 ring / End	16.1	0.2	-19.4	0.0	0.0	0.2
C LC_124	SLN112 ring / End	10.7	0.1	-19.4	0.0	0.0	0.1
	SLN113 ring / End	10.9	0.1	-19.4	0.0	0.0	0.3
C LC_125	SLN112 ring / End	14.6	0.1	-19.4	0.0	0.0	0.2
	SLN113 ring / End	15.5	0.3	-19.4	0.0	0.0	0.5
C LC_126	SLN112 ring / End	12.3	-0.4	-19.4	0.0	0.0	-0.6
	SLN113 ring / End	11.6	-0.2	-19.4	0.0	0.0	-0.3
C LC_127	SLN112 ring / End	7.5	-0.1	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	7.3	0.1	-4.9	0.0	0.0	0.1
C LC_128	SLN112 ring / End	3.1	0.1	-4.9	0.0	0.0	0.1
	SLN113 ring / End	3.3	0.1	-4.9	0.0	0.0	0.2
C LC_129	SLN112 ring / End	6.3	0.2	-4.9	0.0	0.0	0.3
	SLN113 ring / End	7.2	0.3	-4.9	0.0	0.0	0.4
C LC_130	SLN112 ring / End	4.1	-0.3	-4.9	0.0	0.0	-0.5
	SLN113 ring / End	3.3	-0.2	-4.9	0.0	0.0	-0.4
C LC_135	SLN112 ring / End	15.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
C LC_141	SLN112 ring / End	5.1	-0.1	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	5.1	0.1	-4.9	0.0	0.0	0.1
C LC_142	SLN112 ring / End	4.6	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.6	0.0	-4.9	0.0	0.0	0.1
C LC_143	SLN112 ring / End	4.9	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
C LC_144	SLN112 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
	SLN113 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN112 ring / End	19.9	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	19.9	0.2	-25.6	0.0	0.0	0.2
C LC_146	SLN112 ring / End	17.8	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	17.8	0.0	-25.6	0.0	0.0	0.1
C LC_147	SLN112 ring / End	19.0	-0.1	-25.6	0.0	0.0	-0.2
	SLN113 ring / End	18.9	0.2	-25.6	0.0	0.0	0.2
C LC_148	SLN112 ring / End	18.8	-0.2	-25.6	0.0	0.0	-0.3
	SLN113 ring / End	18.9	0.1	-25.6	0.0	0.0	0.1
C LC_151	SLN112 ring / End	11.9	-0.1	-11.8	0.0	0.0	-0.1
	SLN113 ring / End	11.9	0.1	-11.5	0.0	0.0	0.1
C LC_152	SLN112 ring / End	18.8	-0.1	-14.2	0.0	0.0	-0.2
	SLN113 ring / End	18.9	0.1	-14.1	0.0	0.0	0.2
C LC_153	SLN112 ring / End	18.9	-0.2	-21.1	0.0	0.0	-0.2
	SLN113 ring / End	18.9	0.1	-20.8	0.0	0.0	0.2
C LC_154	SLN112 ring / End	12.0	-0.1	-11.8	0.0	0.0	-0.1
	SLN113 ring / End	12.0	0.1	-11.5	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	19.8 < 100%	OK
Buckling	Not calculated	

Plates

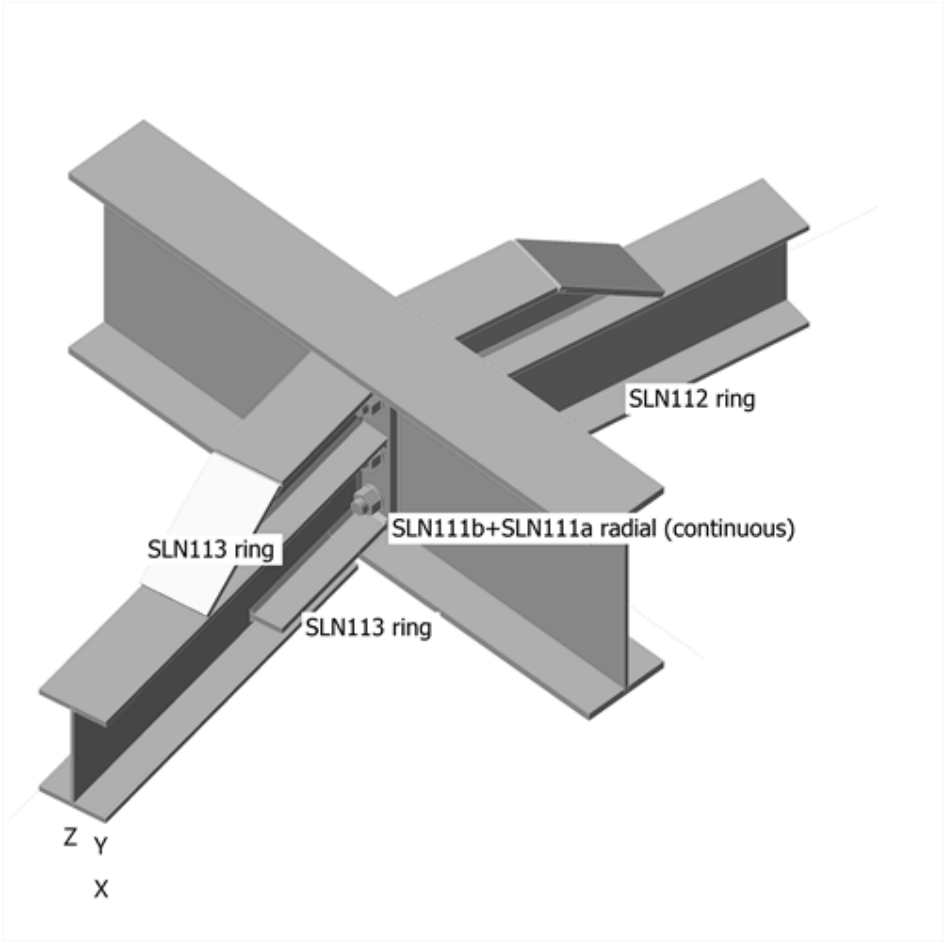
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN111b+SLN111a radial (continuous)-bfl 1	17.7	C LC_148	22.1	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-tfl 1	17.7	C LC_121	18.8	0.0	0.0	OK
SLN111b+SLN111a radial (continuous)-w 1	10.5	C LC_122	29.4	0.0	1.1	OK
SLN112 ring-bfl 1	12.7	C LC_122	50.2	0.0	0.0	OK
SLN112 ring-tfl 1	12.7	C LC_122	45.2	0.0	0.0	OK
SLN112 ring-w 1	7.2	C LC_145	48.9	0.0	0.0	OK
SLN113 ring-bfl 1	12.7	C LC_121	50.5	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_120	44.7	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_145	49.2	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_126	10.1	0.0	9.8	OK
Bottom flange plate SLN113 ring	15.0	C LC_126	22.9	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_120	29.4	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_130	8.2	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN112 ring	20.0	C LC_145	9.2	0.0	1.3	OK
Bottom flange plate SLN112 ring	15.0	C LC_126	21.3	0.0	0.0	OK
Bottom flange half plate L SLN112 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Bottom flange half plate R SLN112 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Haunch web SLN112 ring	12.0	C LC_122	28.3	0.0	0.0	OK
Haunch top plate SLN112 ring	15.0	C LC_125	7.5	0.0	0.0	OK
Haunch sloping plate SLN112 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

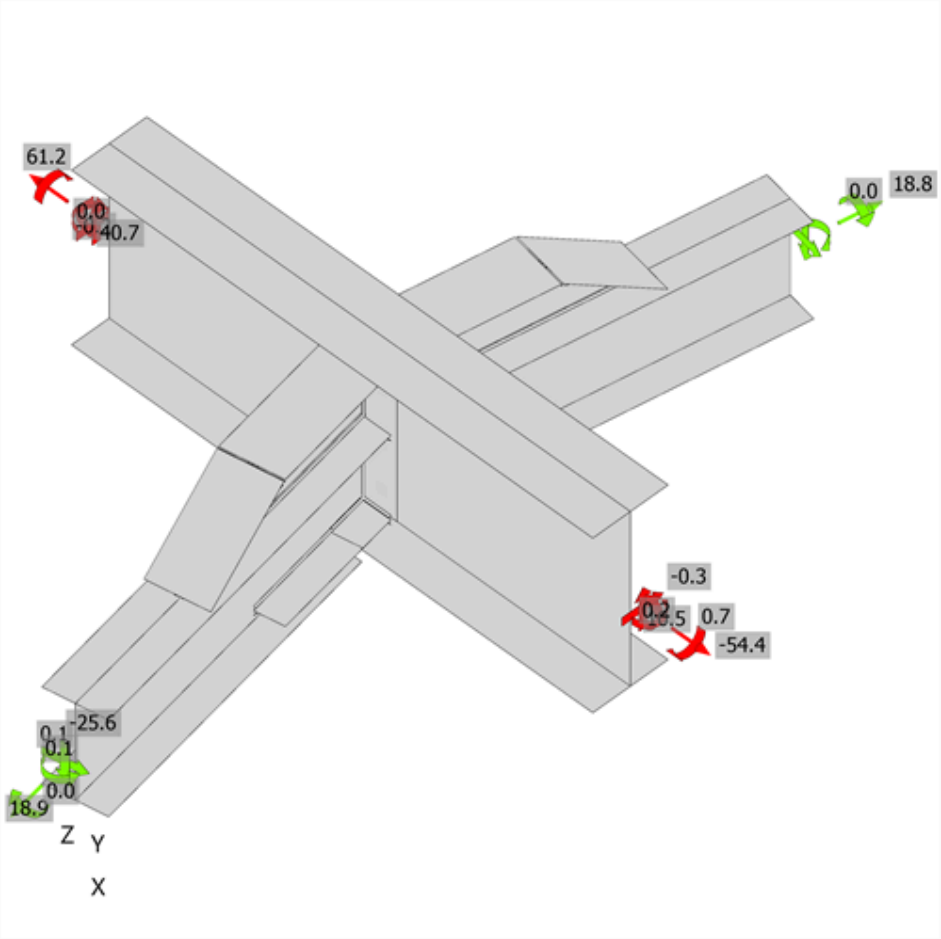
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

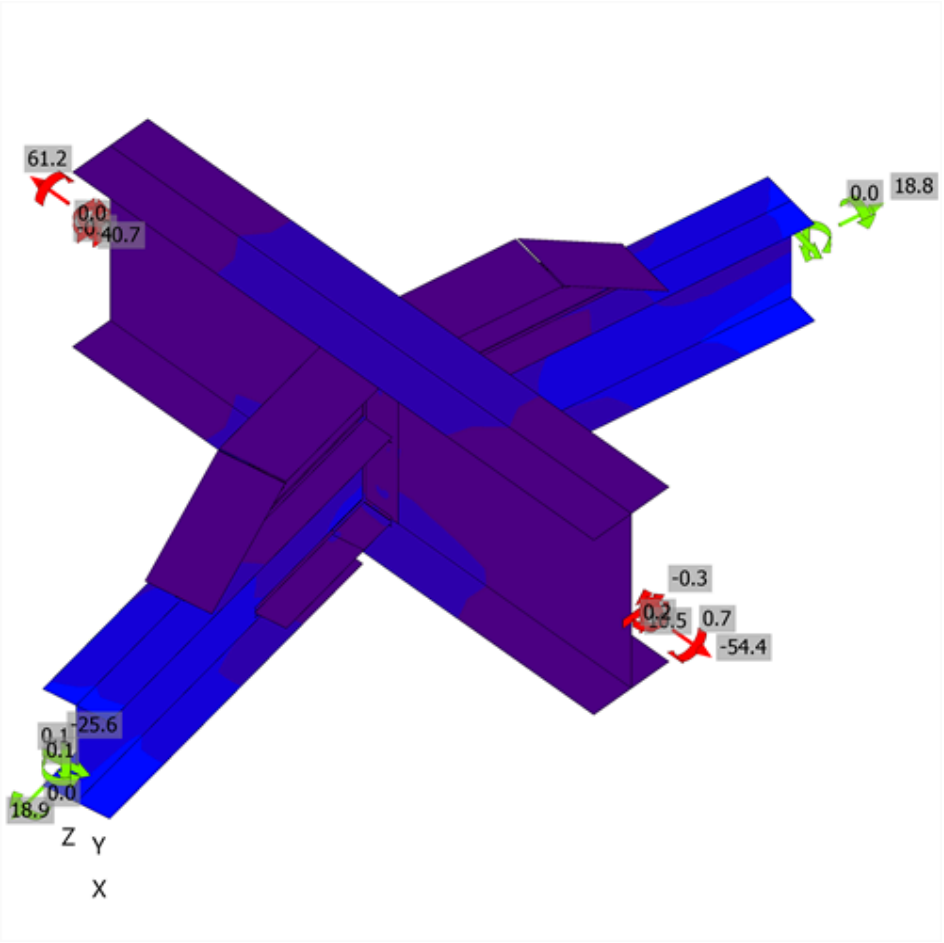
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

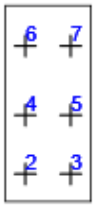


Strain check, C LC_148



Equivalent stress, C LC_148

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_122	2.5	0.5	176.5	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.6	0.6	176.5	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.9	0.9	229.5	0.6	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_121	2.4	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN112 ring	Bottom flange plate SLN112 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-tfl 1	◢ 10.5	149	C LC_121	14.2	0.0	-8.5	-6.5	-0.9	3.5	3.5	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 10.5	149	C LC_121	14.7	0.0	-8.2	-7.0	-0.4	3.6	3.6	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.3	0.0	-10.2	-12.3	-3.2	6.0	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	20.0	0.0	-11.0	9.6	0.3	4.9	4.9	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.7	0.0	2.6	3.2	14.4	6.3	6.3	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.3	0.0	2.5	-3.6	-14.1	6.3	6.3	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_121	26.0	0.0	6.4	6.3	13.1	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.1	0.0	8.0	2.7	-14.7	6.7	6.3	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_122	24.3	0.0	5.2	5.9	-12.3	6.0	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_122	28.2	0.0	7.4	-6.7	14.2	7.0	6.7	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	67.0	0.0	-16.8	-17.8	32.9	16.6	9.5	OK	OK
		◢ 6.0 ◣	572	C LC_121	61.7	0.0	-13.1	14.4	-31.7	15.2	9.0	OK	OK
End plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	208	C LC_145	20.2	0.0	-11.6	-9.5	-0.6	5.0	5.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	24.9	0.0	-9.9	13.1	2.0	6.2	6.0	OK	OK
Bottom flange plate SLN112 ring	SLN112 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.1	0.0	2.5	3.0	14.1	6.2	6.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	26.1	0.0	2.6	-3.9	-14.5	6.4	6.4	OK	OK
SLN112 ring-w 1	Bottom flange half plate L SLN112 ring	◢ 6.0 ◣	243	C LC_121	24.9	0.0	6.0	6.1	12.6	6.2	6.2	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.4	0.0	8.6	2.4	-14.8	6.8	6.2	OK	OK
SLN112 ring-w 1	Bottom flange half plate R SLN112 ring	◢ 6.0 ◣	243	C LC_122	25.5	0.0	5.7	6.2	-12.9	6.3	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_122	28.3	0.0	7.2	-6.7	14.3	7.0	6.6	OK	OK
SLN112 ring-tfl 1	Haunch web SLN112 ring	◢ 6.0 ◣	572	C LC_122	66.2	0.0	-14.5	-16.7	33.3	16.4	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.7	0.0	-16.0	15.7	-31.3	15.5	9.0	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	149	C LC_126	25.1	0.0	-6.7	-9.7	10.0	6.2	5.9	OK	OK
		◢ 6.0 ◣	149	C LC_126	32.9	0.0	-13.6	10.6	-13.7	8.1	6.3	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	7.8	0.0	1.9	1.7	4.0	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.1	0.0	2.0	-2.4	-5.8	2.8	0.0	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_130	13.8	0.0	-6.8	-4.5	5.2	3.4	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	10.0	0.0	-3.7	4.4	3.0	2.5	0.0	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_126	4.5	0.0	1.4	1.2	2.1	1.1	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_125	5.0	0.0	1.7	-1.4	-2.4	1.2	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_120	55.2	0.0	-10.8	-11.3	29.2	13.6	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_120	47.1	0.0	-7.3	6.7	-26.0	11.6	7.9	OK	OK
End plate SLN112 ring	Bottom flange plate SLN112 ring	◢ 6.0 ◣	149	C LC_130	20.4	0.0	-8.7	-10.1	3.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	30.5	0.0	-15.7	13.8	-6.0	7.5	5.9	OK	OK
End plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	145	C LC_121	10.7	0.0	1.9	2.5	5.6	2.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.4	0.0	1.9	-1.7	-4.4	2.1	0.0	OK	OK
End plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	149	C LC_125	12.9	0.0	-6.4	-3.9	-5.1	3.2	3.2	OK	OK
		◢ 6.0 ◣	149	C LC_126	10.3	0.0	-3.6	4.6	-3.1	2.5	0.0	OK	OK
Haunch top plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	342	C LC_126	5.4	0.0	1.8	1.5	2.5	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.2	0.0	-0.2	0.3	2.4	1.0	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch web SLN112 ring	◢ 6.0 ◣	271	C LC_135	52.2	0.0	-9.4	-7.5	28.7	12.9	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	50.7	0.0	-8.9	9.4	-27.3	12.5	8.1	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_145	68.3	0.0	35.1	-29.6	16.3	16.9	12.0	OK	OK
SLN111b+SLN111a radial (continuous)-w 1	SLN112 ring-w 1	◢ 7.2	208	C LC_145	80.1	0.0	31.0	-38.6	18.1	19.8	13.7	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.2	2.8	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_125	4.0	0.0	-0.8	2.3	0.3	1.0	0.0	OK	OK
Haunch sloping plate SLN112 ring	Haunch top plate SLN112 ring	◢ 6.0 ◣	146	C LC_121	7.1	0.0	-5.1	2.8	-0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.3	0.0	-0.9	2.4	-0.3	1.1	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_200
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring

Name	SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

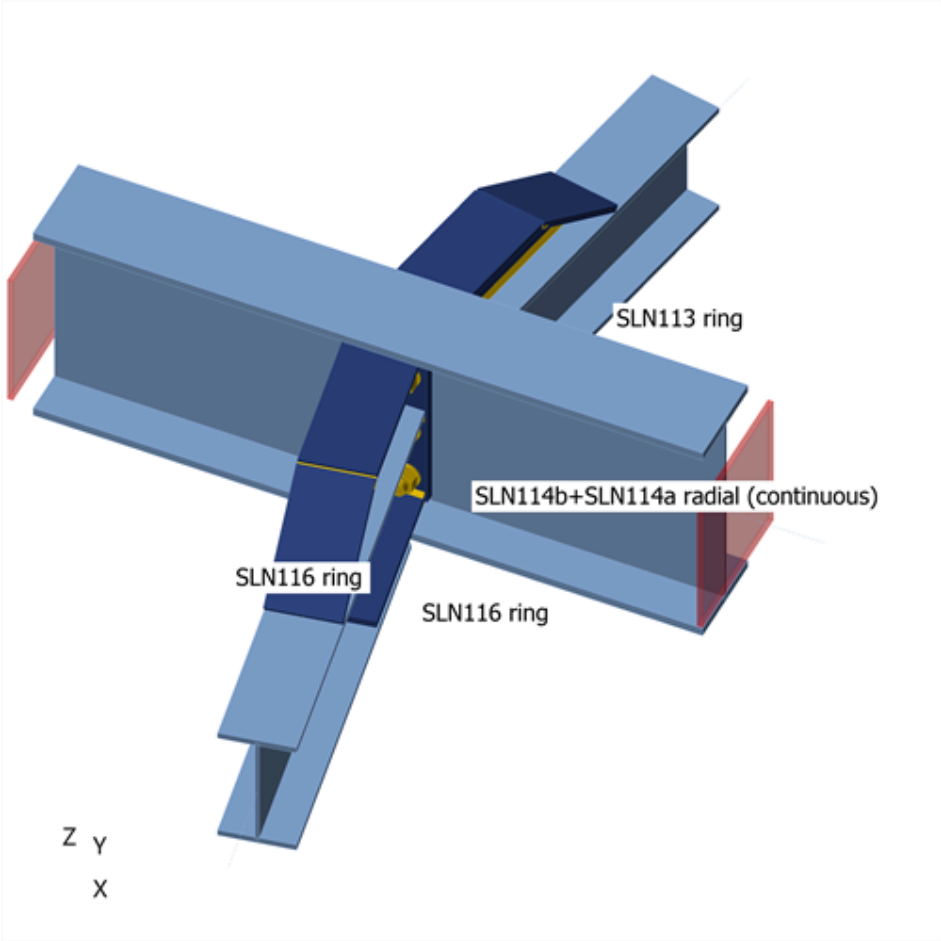
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN114b+SLN114a radial (continuous)	1 - UB457x191x89	-160.0	0.0	0.0	0	0	0
SLN113 ring	2 - UB254x146x43	100.0	0.0	0.0	0	0	-102
SLN116 ring	2 - UB254x146x43	-60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN114b+SLN114a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN114b+SLN114a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN113 ring / end		Position	0
SLN116 ring / end		Position	0



SLN114b+SLN114a radial (continuous) - SLN113 ring - SLN116 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN113 ring / End	4.8	0.0	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.9	0.0	-4.9	0.0	0.0	-0.1
C LC_102	SLN113 ring / End	18.9	0.1	-25.6	0.0	0.0	0.2
	SLN116 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.2
C LC_103	SLN113 ring / End	17.2	0.0	-24.0	0.0	0.0	0.0
	SLN116 ring / End	17.0	0.0	-24.0	0.0	0.0	-0.1
C LC_104	SLN113 ring / End	12.3	0.3	-24.0	0.0	0.0	0.3
	SLN116 ring / End	12.5	-0.2	-24.0	0.0	0.0	-0.1
C LC_105	SLN113 ring / End	15.0	0.2	-24.0	0.0	0.0	0.3
	SLN116 ring / End	15.1	0.0	-24.0	0.0	0.0	0.0
C LC_106	SLN113 ring / End	13.9	-0.1	-24.0	0.0	0.0	-0.1
	SLN116 ring / End	13.8	-0.2	-24.0	0.0	0.0	-0.3
C LC_107	SLN113 ring / End	2.9	-0.1	-0.5	0.0	0.0	-0.1
	SLN116 ring / End	2.6	0.0	-0.5	0.0	0.0	0.1
C LC_108	SLN113 ring / End	-3.2	0.2	-0.5	0.0	0.0	0.3
	SLN116 ring / End	-3.1	0.1	-0.5	0.0	0.0	0.1
C LC_109	SLN113 ring / End	0.6	0.1	-0.5	0.0	0.0	0.2
	SLN116 ring / End	1.2	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN113 ring / End	-2.3	-0.2	-0.5	0.0	0.0	-0.4
	SLN116 ring / End	-2.8	-0.1	-0.5	0.0	0.0	-0.2
C LC_111	SLN113 ring / End	13.0	-0.1	-16.2	0.0	0.0	-0.1
	SLN116 ring / End	12.7	0.0	-16.2	0.0	0.0	0.0
C LC_112	SLN113 ring / End	5.7	0.3	-16.2	0.0	0.0	0.4
	SLN116 ring / End	5.9	0.0	-16.2	0.0	0.0	0.0
C LC_113	SLN113 ring / End	10.0	0.2	-16.2	0.0	0.0	0.4
	SLN116 ring / End	10.5	0.1	-16.2	0.0	0.0	0.2
C LC_114	SLN113 ring / End	7.4	-0.2	-16.2	0.0	0.0	-0.3
	SLN116 ring / End	6.9	-0.2	-16.2	0.0	0.0	-0.4
C LC_115	SLN113 ring / End	4.2	-0.1	-1.7	0.0	0.0	-0.1
	SLN116 ring / End	3.9	0.0	-1.7	0.0	0.0	0.1
C LC_116	SLN113 ring / End	-2.1	0.2	-1.7	0.0	0.0	0.3
	SLN116 ring / End	-1.9	0.1	-1.7	0.0	0.0	0.1
C LC_117	SLN113 ring / End	1.8	0.1	-1.7	0.0	0.0	0.3
	SLN116 ring / End	2.5	0.1	-1.7	0.0	0.0	0.2
C LC_118	SLN113 ring / End	-1.0	-0.2	-1.7	0.0	0.0	-0.3
	SLN116 ring / End	-1.5	-0.1	-1.7	0.0	0.0	-0.3
C LC_119	SLN113 ring / End	18.8	0.0	-25.6	0.0	0.0	0.1
	SLN116 ring / End	18.6	0.0	-25.6	0.0	0.0	-0.1
C LC_120	SLN113 ring / End	15.0	0.2	-25.6	0.0	0.0	0.3
	SLN116 ring / End	15.1	-0.2	-25.6	0.0	0.0	-0.2
C LC_121	SLN113 ring / End	17.6	0.3	-25.6	0.0	0.0	0.4
	SLN116 ring / End	17.8	0.0	-25.6	0.0	0.0	0.0
C LC_122	SLN113 ring / End	16.1	-0.1	-25.6	0.0	0.0	-0.1
	SLN116 ring / End	16.1	-0.2	-25.6	0.0	0.0	-0.4
C LC_123	SLN113 ring / End	16.1	0.0	-19.4	0.0	0.0	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN116 ring / End	15.7	0.0	-19.4	0.0	0.0	-0.1
C LC_124	SLN113 ring / End	11.0	0.3	-19.4	0.0	0.0	0.3
	SLN116 ring / End	11.1	-0.1	-19.4	0.0	0.0	-0.1
C LC_125	SLN113 ring / End	15.3	0.4	-19.4	0.0	0.0	0.5
	SLN116 ring / End	15.8	0.1	-19.4	0.0	0.0	0.1
C LC_126	SLN113 ring / End	11.7	-0.2	-19.4	0.0	0.0	-0.3
	SLN116 ring / End	11.4	-0.3	-19.4	0.0	0.0	-0.5
C LC_127	SLN113 ring / End	7.3	0.0	-4.9	0.0	0.0	0.1
	SLN116 ring / End	7.0	0.0	-4.9	0.0	0.0	-0.1
C LC_128	SLN113 ring / End	3.3	0.2	-4.9	0.0	0.0	0.2
	SLN116 ring / End	3.4	0.0	-4.9	0.0	0.0	0.1
C LC_129	SLN113 ring / End	7.2	0.3	-4.9	0.0	0.0	0.4
	SLN116 ring / End	7.8	0.2	-4.9	0.0	0.0	0.2
C LC_130	SLN113 ring / End	3.3	-0.2	-4.9	0.0	0.0	-0.3
	SLN116 ring / End	3.0	-0.3	-4.9	0.0	0.0	-0.4
C LC_135	SLN113 ring / End	15.9	0.1	-25.6	0.0	0.0	0.2
	SLN116 ring / End	15.9	-0.2	-25.6	0.0	0.0	-0.2
C LC_141	SLN113 ring / End	5.1	0.0	-4.9	0.0	0.0	0.0
	SLN116 ring / End	5.1	0.0	-4.9	0.0	0.0	-0.1
C LC_142	SLN113 ring / End	4.6	0.1	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.6	-0.1	-4.9	0.0	0.0	-0.1
C LC_143	SLN113 ring / End	4.8	0.1	-4.9	0.0	0.0	0.1
	SLN116 ring / End	4.8	0.0	-4.9	0.0	0.0	-0.1
C LC_144	SLN113 ring / End	4.9	0.0	-4.9	0.0	0.0	0.0
	SLN116 ring / End	4.9	0.0	-4.9	0.0	0.0	-0.1
C LC_145	SLN113 ring / End	19.9	0.0	-25.6	0.0	0.0	0.1
	SLN116 ring / End	19.9	-0.1	-25.6	0.0	0.0	-0.2
C LC_146	SLN113 ring / End	17.9	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	18.0	-0.2	-25.6	0.0	0.0	-0.3
C LC_147	SLN113 ring / End	18.7	0.2	-25.6	0.0	0.0	0.2
	SLN116 ring / End	18.5	-0.2	-25.6	0.0	0.0	-0.2
C LC_148	SLN113 ring / End	19.1	0.1	-25.6	0.0	0.0	0.1
	SLN116 ring / End	19.3	-0.1	-25.6	0.0	0.0	-0.2
C LC_151	SLN113 ring / End	11.9	0.1	-11.6	0.0	0.0	0.1
	SLN116 ring / End	11.9	-0.1	-11.5	0.0	0.0	-0.1
C LC_152	SLN113 ring / End	18.9	0.1	-14.2	0.0	0.0	0.2
	SLN116 ring / End	18.9	-0.1	-14.0	0.0	0.0	-0.2
C LC_153	SLN113 ring / End	18.9	0.1	-20.8	0.0	0.0	0.2
	SLN116 ring / End	18.9	-0.2	-20.6	0.0	0.0	-0.2
C LC_154	SLN113 ring / End	12.0	0.1	-11.6	0.0	0.0	0.1
	SLN116 ring / End	12.0	-0.1	-11.5	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	16.7 < 100%	OK
Buckling	Not calculated	

Plates

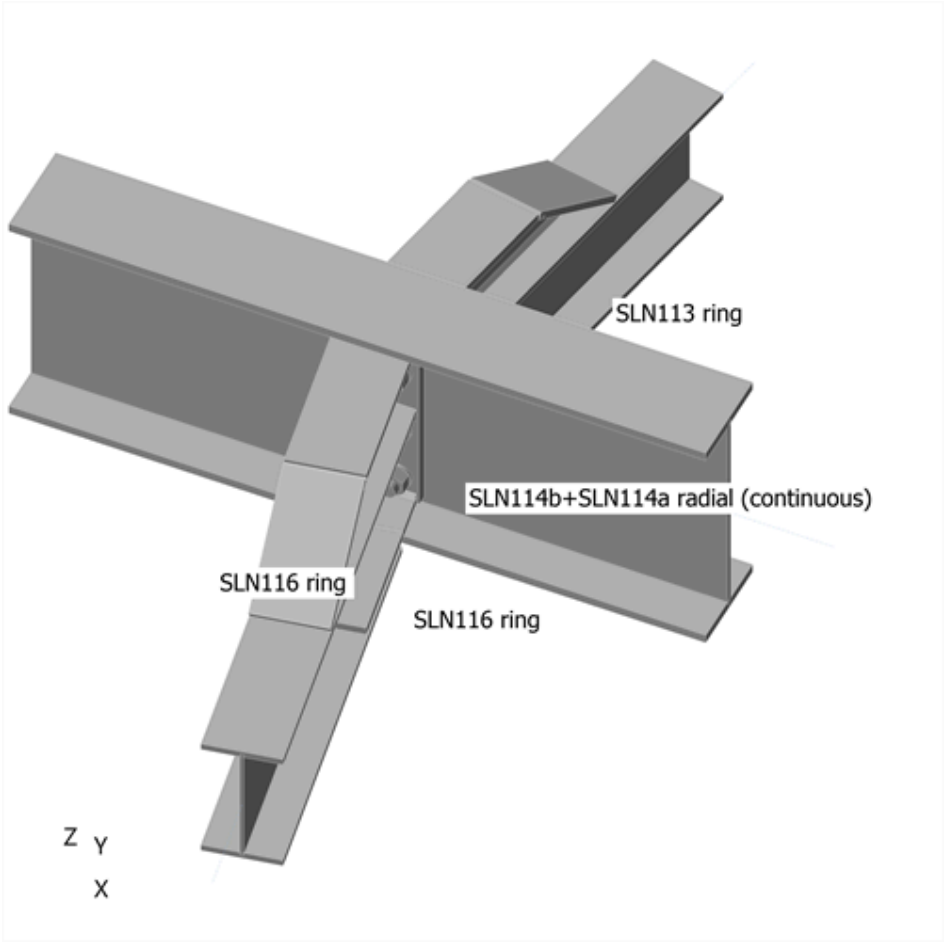
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN114b+SLN114a radial (continuous)-bfl 1	17.7	C LC_122	22.1	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-tfl 1	17.7	C LC_122	18.7	0.0	0.0	OK
SLN114b+SLN114a radial (continuous)-w 1	10.5	C LC_122	28.9	0.0	0.9	OK
SLN113 ring-bfl 1	12.7	C LC_145	50.6	0.0	0.0	OK
SLN113 ring-tfl 1	12.7	C LC_121	44.3	0.0	0.0	OK
SLN113 ring-w 1	7.2	C LC_145	49.2	0.0	0.0	OK
SLN116 ring-bfl 1	12.7	C LC_145	50.4	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	44.8	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_145	48.9	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_121	8.7	0.0	3.5	OK
Bottom flange plate SLN116 ring	15.0	C LC_145	20.4	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_145	19.7	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_122	28.1	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_126	6.4	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_120	12.7	0.0	0.0	OK
End plate SLN113 ring	20.0	C LC_103	9.5	0.0	0.5	OK
Bottom flange plate SLN113 ring	15.0	C LC_145	20.4	0.0	0.0	OK
Bottom flange half plate L SLN113 ring	15.0	C LC_145	19.7	0.0	0.0	OK
Bottom flange half plate R SLN113 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN113 ring	12.0	C LC_135	29.1	0.0	0.0	OK
Haunch top plate SLN113 ring	15.0	C LC_125	6.2	0.0	0.0	OK
Haunch sloping plate SLN113 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

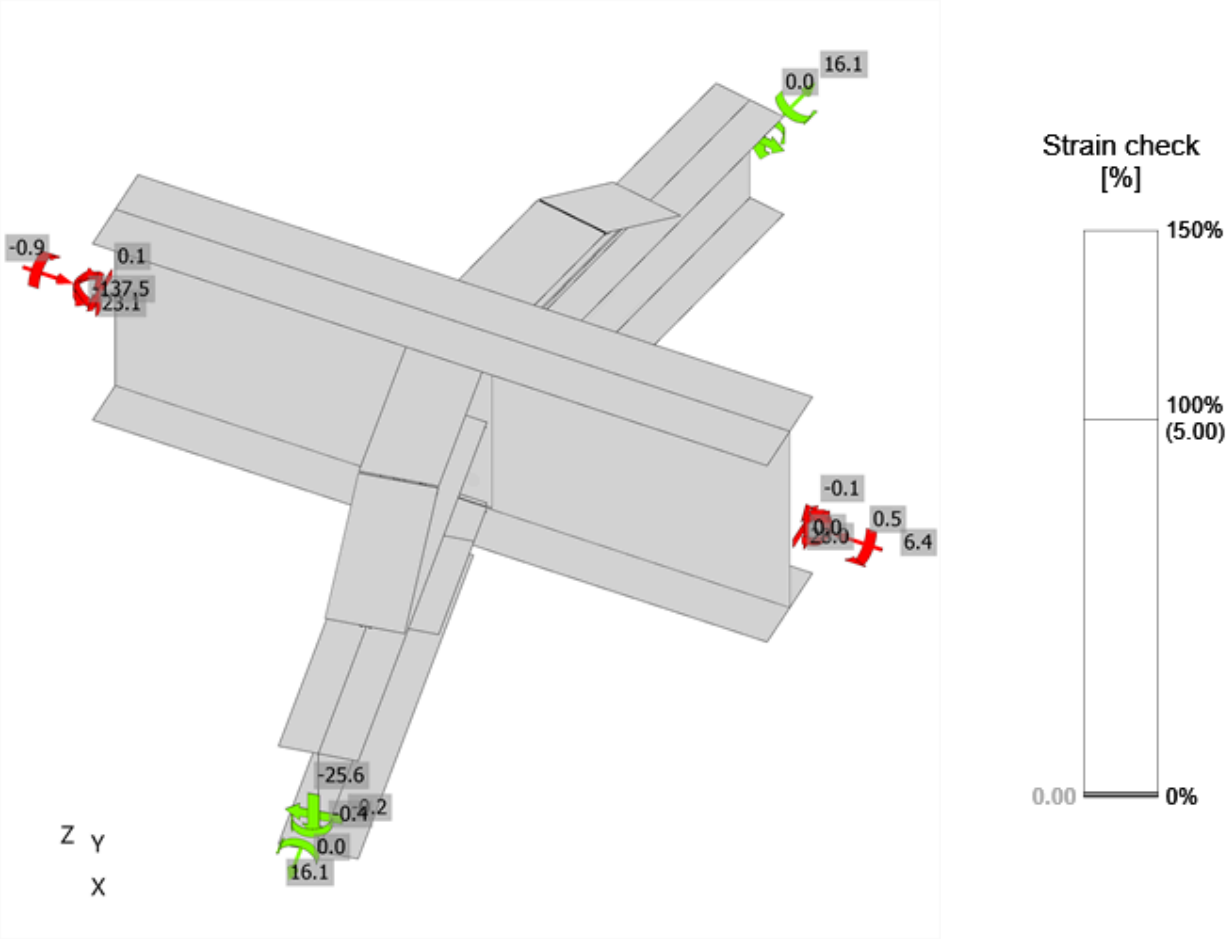
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

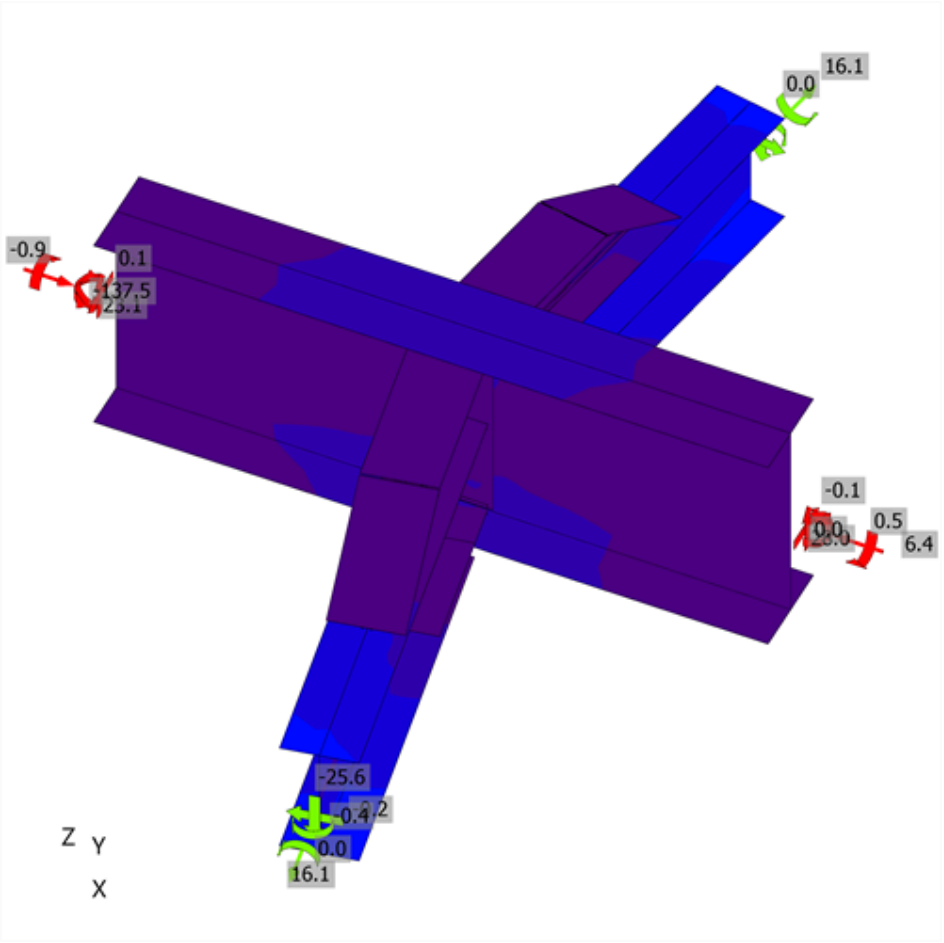
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

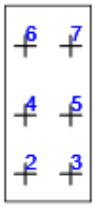


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	2.3	0.5	176.5	0.7	0.5	0.7	OK	OK
	B3	M30 8.8 - 1	C LC_122	2.7	0.6	176.5	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_121	2.5	0.8	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_145	1.2	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN113 ring	Bottom flange plate SLN113 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-tfl 1	◢ 10.5	149	C LC_122	12.5	0.0	-7.9	-5.6	0.7	3.1	3.1	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 10.5	149	C LC_121	12.7	0.0	-7.8	-5.7	-0.8	3.1	3.1	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_145	24.1	0.0	-9.8	-12.4	-2.9	6.0	6.0	OK	OK
		◢ 6.0 ◣	208	C LC_145	19.3	0.0	-10.9	9.2	0.2	4.8	4.8	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.5	0.0	2.4	2.8	13.8	6.0	6.0	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.6	0.0	2.6	-3.5	-14.3	6.3	6.3	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_121	24.6	0.0	5.8	6.0	12.4	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.0	0.0	8.3	2.5	-15.2	6.9	6.4	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_148	25.0	0.0	6.1	6.1	-12.6	6.2	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.2	0.0	7.9	2.7	14.8	6.7	6.4	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	64.3	0.0	-14.2	-15.7	32.6	15.9	9.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.3	0.0	-16.1	16.2	-31.4	15.6	9.1	OK	OK
End plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	208	C LC_145	19.2	0.0	-11.3	-8.9	-0.5	4.7	4.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	24.7	0.0	-9.5	12.9	2.6	6.1	6.0	OK	OK
Bottom flange plate SLN113 ring	SLN113 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.4	0.0	2.7	3.1	14.3	6.3	6.3	OK	OK
		◢ 6.0 ◣	100	C LC_122	24.9	0.0	2.4	-3.4	-13.9	6.1	6.1	OK	OK
SLN113 ring-w 1	Bottom flange half plate L SLN113 ring	◢ 6.0 ◣	243	C LC_121	25.6	0.0	6.4	6.2	12.9	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_145	26.7	0.0	8.1	2.6	-14.5	6.6	6.2	OK	OK
SLN113 ring-w 1	Bottom flange half plate R SLN113 ring	◢ 6.0 ◣	243	C LC_122	24.2	0.0	5.3	5.9	-12.3	6.0	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.5	0.0	8.1	2.6	15.6	7.0	6.7	OK	OK
SLN113 ring-tfl 1	Haunch web SLN113 ring	◢ 6.0 ◣	572	C LC_122	66.9	0.0	-16.9	-17.9	32.8	16.5	9.5	OK	OK
		◢ 6.0 ◣	572	C LC_120	62.0	0.0	-13.9	14.6	-31.7	15.3	9.0	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	149	C LC_130	18.6	0.0	-5.3	-7.4	7.2	4.6	4.6	OK	OK
		◢ 6.0 ◣	149	C LC_126	25.8	0.0	-10.8	8.1	-10.8	6.4	5.8	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	7.6	0.0	1.3	1.3	4.1	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.7	0.0	2.4	-2.5	-5.5	2.6	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_126	12.8	0.0	-5.8	-4.6	4.7	3.2	3.2	OK	OK
		◢ 6.0 ◣	149	C LC_126	11.2	0.0	-3.6	4.8	-3.7	2.8	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	4.9	0.0	1.7	1.5	2.1	1.2	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.1	0.2	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_135	50.3	0.0	-8.6	-7.3	27.7	12.4	8.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	52.2	0.0	-9.7	11.9	-27.1	12.9	8.2	OK	OK
End plate SLN113 ring	Bottom flange plate SLN113 ring	◢ 6.0 ◣	149	C LC_129	16.3	0.0	-4.6	-6.3	-6.4	4.0	4.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	25.6	0.0	-13.0	11.6	-5.3	6.3	5.6	OK	OK
End plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	145	C LC_121	10.4	0.0	2.4	2.5	5.3	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.0	0.0	1.3	-1.3	-4.4	2.0	0.0	OK	OK
End plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	149	C LC_125	12.1	0.0	-5.4	-4.4	-4.4	3.0	3.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	10.9	0.0	-3.7	4.7	3.7	2.7	2.7	OK	OK
Haunch top plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	342	C LC_126	4.0	0.0	1.3	1.2	1.9	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.8	0.0	1.7	-1.5	-2.1	1.2	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch web SLN113 ring	◢ 6.0 ◣	271	C LC_122	55.0	0.0	-10.6	-9.2	29.8	13.6	8.5	OK	OK
		◢ 6.0 ◣	271	C LC_120	47.8	0.0	-7.5	6.7	-26.4	11.8	8.0	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_145	67.2	0.0	34.9	-28.6	16.8	16.6	11.8	OK	OK
SLN114b+SLN114a radial (continuous)-w 1	SLN113 ring-w 1	◢ 7.2	208	C LC_145	67.5	0.0	35.9	28.1	-17.2	16.7	11.9	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_122	7.3	0.0	-5.5	2.7	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	3.7	0.0	-0.8	2.1	-0.1	0.9	0.0	OK	OK
Haunch sloping plate SLN113 ring	Haunch top plate SLN113 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.2	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_125	3.5	0.0	-0.8	2.0	0.1	0.9	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_220
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring

Name	SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

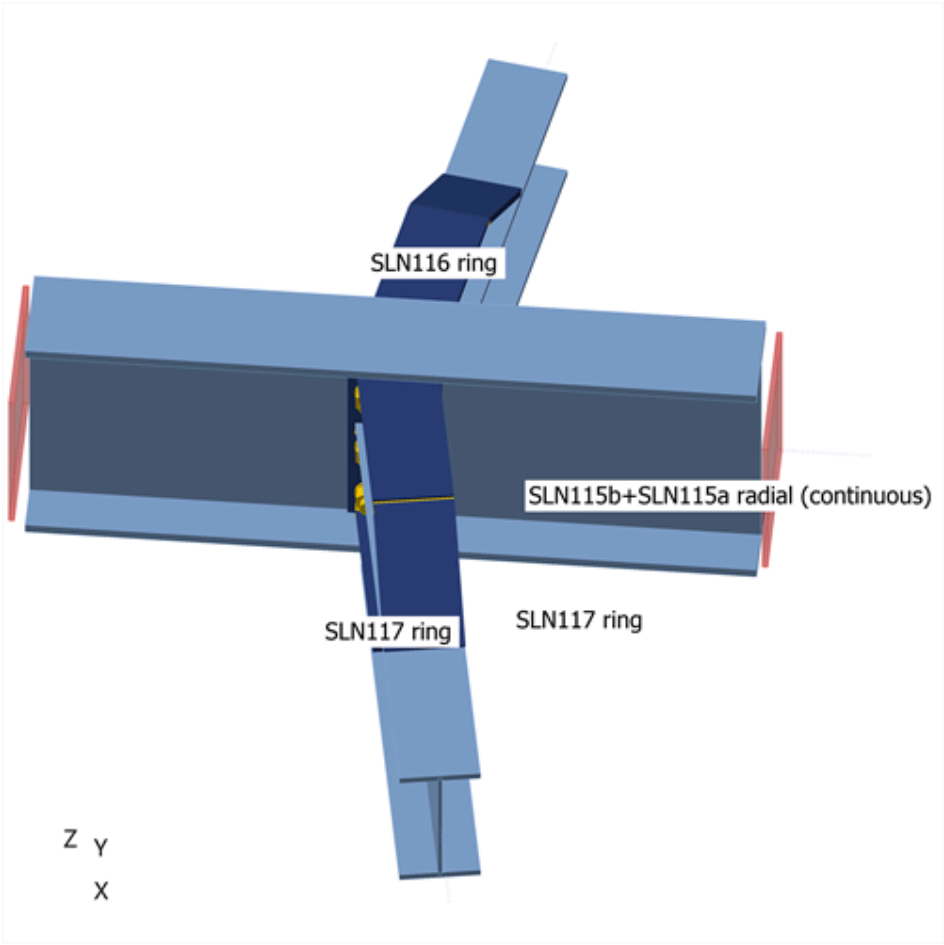
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN115b+SLN115a radial (continuous)	1 - UB457x191x89	-140.0	0.0	0.0	0	0	0
SLN116 ring	2 - UB254x146x43	120.0	0.0	0.0	0	0	-102
SLN117 ring	2 - UB254x146x43	-40.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN115b+SLN115a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN115b+SLN115a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN116 ring / end		Position	0
SLN117 ring / end		Position	0



SLN115b+SLN115a radial (continuous) - SLN116 ring - SLN117 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN116 ring / End	4.9	0.0	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
C LC_102	SLN116 ring / End	18.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.9	0.2	-26.0	0.0	0.0	0.3
C LC_103	SLN116 ring / End	17.0	-0.2	-24.0	0.0	0.0	-0.2
	SLN117 ring / End	16.4	0.2	-24.3	0.0	0.0	0.2
C LC_104	SLN116 ring / End	12.6	0.0	-24.0	0.0	0.0	-0.1
	SLN117 ring / End	12.8	0.2	-24.3	0.0	0.0	0.2
C LC_105	SLN116 ring / End	15.0	0.0	-24.0	0.0	0.0	0.0
	SLN117 ring / End	15.2	0.2	-24.3	0.0	0.0	0.3
C LC_106	SLN116 ring / End	14.0	-0.3	-24.0	0.0	0.0	-0.3
	SLN117 ring / End	14.2	0.1	-24.3	0.0	0.0	0.0
C LC_107	SLN116 ring / End	2.6	0.0	-0.5	0.0	0.0	0.0
	SLN117 ring / End	1.7	0.0	-0.5	0.0	0.0	-0.1
C LC_108	SLN116 ring / End	-3.1	0.1	-0.5	0.0	0.0	0.1
	SLN117 ring / End	-2.8	0.1	-0.5	0.0	0.0	0.2
C LC_109	SLN116 ring / End	1.2	0.2	-0.5	0.0	0.0	0.2
	SLN117 ring / End	1.8	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN116 ring / End	-2.8	-0.1	-0.5	0.0	0.0	-0.1
	SLN117 ring / End	-2.5	-0.2	-0.5	0.0	0.0	-0.3
C LC_111	SLN116 ring / End	12.6	-0.1	-16.2	0.0	0.0	-0.2
	SLN117 ring / End	11.6	0.1	-16.4	0.0	0.0	0.0
C LC_112	SLN116 ring / End	6.0	0.1	-16.2	0.0	0.0	0.1
	SLN117 ring / End	6.3	0.2	-16.4	0.0	0.0	0.3
C LC_113	SLN116 ring / End	10.4	0.1	-16.2	0.0	0.0	0.2
	SLN117 ring / End	10.9	0.3	-16.4	0.0	0.0	0.4
C LC_114	SLN116 ring / End	7.1	-0.3	-16.2	0.0	0.0	-0.3
	SLN117 ring / End	7.4	-0.1	-16.4	0.0	0.0	-0.2
C LC_115	SLN116 ring / End	3.9	0.0	-1.7	0.0	0.0	0.0
	SLN117 ring / End	3.0	0.0	-1.8	0.0	0.0	-0.1
C LC_116	SLN116 ring / End	-1.9	0.1	-1.7	0.0	0.0	0.1
	SLN117 ring / End	-1.6	0.1	-1.8	0.0	0.0	0.2
C LC_117	SLN116 ring / End	2.4	0.2	-1.7	0.0	0.0	0.2
	SLN117 ring / End	3.0	0.2	-1.8	0.0	0.0	0.3
C LC_118	SLN116 ring / End	-1.5	-0.2	-1.7	0.0	0.0	-0.2
	SLN117 ring / End	-1.2	-0.2	-1.8	0.0	0.0	-0.2
C LC_119	SLN116 ring / End	18.5	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.3	0.3	-26.0	0.0	0.0	0.3
C LC_120	SLN116 ring / End	15.2	0.0	-25.6	0.0	0.0	-0.1
	SLN117 ring / End	15.4	0.1	-26.0	0.0	0.0	0.2
C LC_121	SLN116 ring / End	17.6	0.0	-25.6	0.0	0.0	0.0
	SLN117 ring / End	18.0	0.2	-26.0	0.0	0.0	0.4
C LC_122	SLN116 ring / End	16.2	-0.3	-25.6	0.0	0.0	-0.5
	SLN117 ring / End	16.0	0.1	-26.0	0.0	0.0	0.0
C LC_123	SLN116 ring / End	15.7	-0.1	-19.4	0.0	0.0	-0.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN117 ring / End	15.5	0.3	-19.7	0.0	0.0	0.3
C LC_124	SLN116 ring / End	11.2	0.0	-19.4	0.0	0.0	0.0
	SLN117 ring / End	11.4	0.2	-19.7	0.0	0.0	0.2
C LC_125	SLN116 ring / End	15.7	0.1	-19.4	0.0	0.0	0.2
	SLN117 ring / End	16.5	0.3	-19.7	0.0	0.0	0.5
C LC_126	SLN116 ring / End	11.6	-0.4	-19.4	0.0	0.0	-0.6
	SLN117 ring / End	11.0	0.0	-19.7	0.0	0.0	-0.2
C LC_127	SLN116 ring / End	6.9	0.0	-4.9	0.0	0.0	0.0
	SLN117 ring / End	6.8	0.1	-5.0	0.0	0.0	0.2
C LC_128	SLN116 ring / End	3.4	0.1	-4.9	0.0	0.0	0.1
	SLN117 ring / End	3.5	0.1	-5.0	0.0	0.0	0.2
C LC_129	SLN116 ring / End	7.8	0.2	-4.9	0.0	0.0	0.2
	SLN117 ring / End	8.5	0.3	-5.0	0.0	0.0	0.4
C LC_130	SLN116 ring / End	3.0	-0.3	-4.9	0.0	0.0	-0.4
	SLN117 ring / End	2.3	-0.2	-5.0	0.0	0.0	-0.3
C LC_135	SLN116 ring / End	15.9	-0.2	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	15.9	0.2	-26.0	0.0	0.0	0.2
C LC_141	SLN116 ring / End	5.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	5.0	0.1	-5.0	0.0	0.0	0.1
C LC_142	SLN116 ring / End	4.7	0.0	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	4.7	0.0	-5.0	0.0	0.0	0.1
C LC_143	SLN116 ring / End	4.8	0.0	-4.9	0.0	0.0	0.0
	SLN117 ring / End	4.7	0.0	-5.0	0.0	0.0	0.1
C LC_144	SLN116 ring / End	5.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN117 ring / End	5.0	0.1	-5.0	0.0	0.0	0.1
C LC_145	SLN116 ring / End	19.8	-0.2	-25.6	0.0	0.0	-0.3
	SLN117 ring / End	19.7	0.3	-26.0	0.0	0.0	0.3
C LC_146	SLN116 ring / End	18.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.2	0.1	-26.0	0.0	0.0	0.2
C LC_147	SLN116 ring / End	18.4	-0.1	-25.6	0.0	0.0	-0.2
	SLN117 ring / End	18.3	0.2	-26.0	0.0	0.0	0.3
C LC_148	SLN116 ring / End	19.4	-0.2	-25.6	0.0	0.0	-0.3
	SLN117 ring / End	19.6	0.2	-26.0	0.0	0.0	0.2
C LC_151	SLN116 ring / End	11.9	-0.1	-11.6	0.0	0.0	-0.2
	SLN117 ring / End	11.9	0.1	-11.9	0.0	0.0	0.2
C LC_152	SLN116 ring / End	18.9	-0.1	-14.0	0.0	0.0	-0.2
	SLN117 ring / End	19.1	0.2	-14.0	0.0	0.0	0.3
C LC_153	SLN116 ring / End	18.9	-0.2	-20.7	0.0	0.0	-0.2
	SLN117 ring / End	19.0	0.2	-20.9	0.0	0.0	0.3
C LC_154	SLN116 ring / End	12.0	-0.1	-11.6	0.0	0.0	-0.2
	SLN117 ring / End	12.0	0.1	-11.9	0.0	0.0	0.2

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	16.7 < 100%	OK
Buckling	Not calculated	

Plates

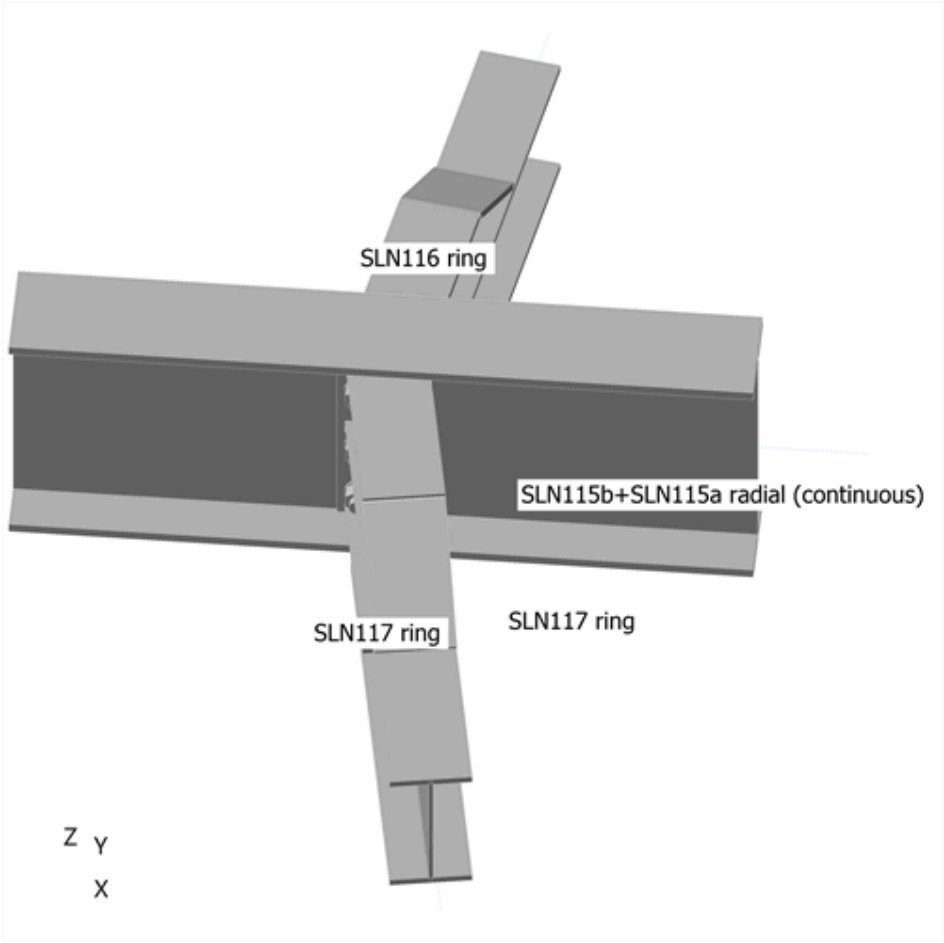
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN115b+SLN115a radial (continuous)-bfl 1	17.7	C LC_119	22.2	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-tfl 1	17.7	C LC_121	18.9	0.0	0.0	OK
SLN115b+SLN115a radial (continuous)-w 1	10.5	C LC_122	28.8	0.0	0.8	OK
SLN116 ring-bfl 1	12.7	C LC_146	50.2	0.0	0.0	OK
SLN116 ring-tfl 1	12.7	C LC_122	45.0	0.0	0.0	OK
SLN116 ring-w 1	7.2	C LC_145	49.0	0.0	0.0	OK
SLN117 ring-bfl 1	12.7	C LC_121	51.0	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	45.1	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	49.7	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_145	9.0	0.0	0.6	OK
Bottom flange plate SLN117 ring	15.0	C LC_145	20.6	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	20.0	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_145	20.1	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_120	29.3	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_130	6.7	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_120	12.9	0.0	0.0	OK
End plate SLN116 ring	20.0	C LC_148	9.1	0.0	0.6	OK
Bottom flange plate SLN116 ring	15.0	C LC_145	20.4	0.0	0.0	OK
Bottom flange half plate L SLN116 ring	15.0	C LC_145	19.9	0.0	0.0	OK
Bottom flange half plate R SLN116 ring	15.0	C LC_145	19.8	0.0	0.0	OK
Haunch web SLN116 ring	12.0	C LC_122	28.2	0.0	0.0	OK
Haunch top plate SLN116 ring	15.0	C LC_126	6.9	0.0	0.0	OK
Haunch sloping plate SLN116 ring	15.0	C LC_120	12.7	0.0	0.0	OK

Design data

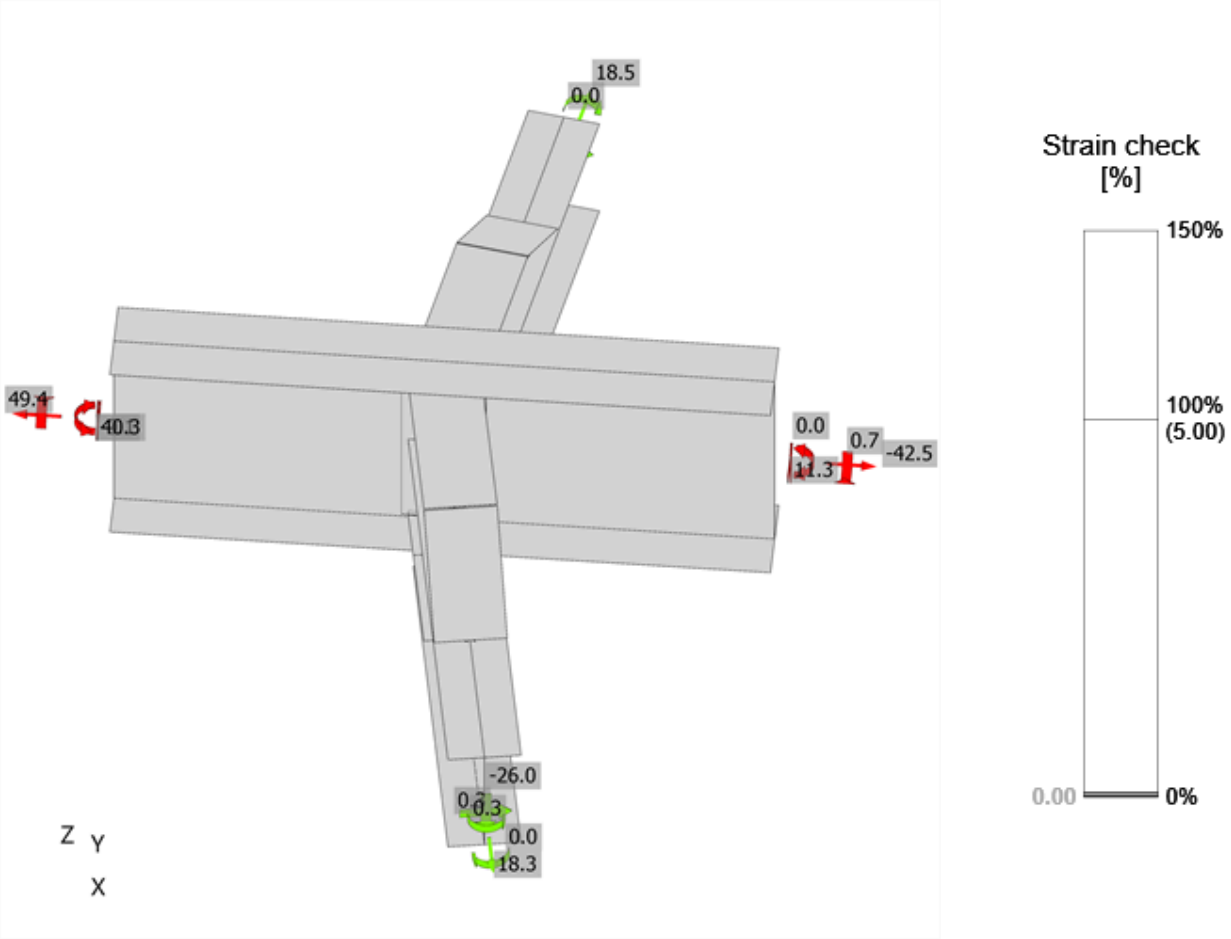
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

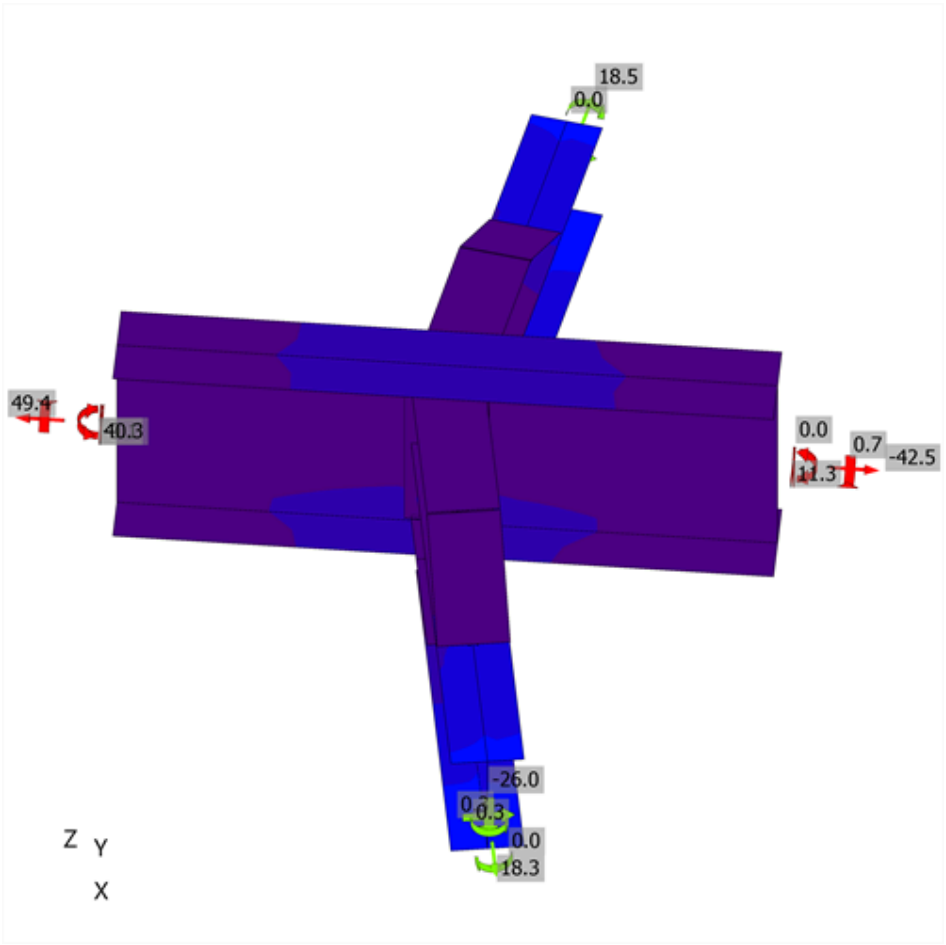
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

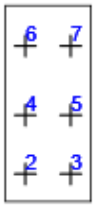


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_121	2.6	0.5	176.5	0.8	0.5	0.8	OK	OK
	B3	M30 8.8 - 1	C LC_145	2.6	0.7	176.5	0.8	0.7	0.9	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.1	0.4	213.9	0.0	0.4	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_145	0.1	0.5	213.9	0.0	0.4	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_121	1.7	0.9	229.5	0.5	0.8	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.3	0.9	229.5	0.7	0.8	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN116 ring	Bottom flange plate SLN116 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-tfl 1	◢ 10.5	149	C LC_122	12.5	0.0	-6.0	5.5	-3.1	3.1	3.1	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 10.5	149	C LC_145	12.2	0.0	-7.3	-5.7	0.3	3.0	3.0	OK	OK
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_145	23.9	0.0	-9.7	-12.3	-2.6	5.9	5.9	OK	OK
		◢ 6.0 ◣	208	C LC_148	19.5	0.0	-10.7	9.5	0.0	4.8	4.8	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_121	25.8	0.0	2.6	3.2	14.4	6.4	6.4	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.1	0.0	2.4	-3.2	-14.1	6.2	6.2	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_121	26.0	0.0	6.4	6.3	13.1	6.4	6.4	OK	OK
		◢ 6.0 ◣	243	C LC_148	27.4	0.0	8.1	2.6	-14.9	6.8	6.3	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_122	23.9	0.0	5.4	5.9	-12.1	5.9	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_145	28.2	0.0	8.5	2.4	15.3	7.0	6.5	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	66.5	0.0	-17.1	-17.5	32.7	16.4	9.4	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.1	0.0	-13.6	15.0	-32.3	15.6	9.1	OK	OK
End plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	208	C LC_145	19.5	0.0	-10.8	-9.3	-0.2	4.8	4.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_145	23.9	0.0	-9.9	12.2	2.9	5.9	5.9	OK	OK
Bottom flange plate SLN116 ring	SLN116 ring-w 1	◢ 6.0 ◣	100	C LC_121	24.9	0.0	2.5	3.0	14.0	6.1	6.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.6	0.0	2.6	-3.6	-14.3	6.3	6.3	OK	OK
SLN116 ring-w 1	Bottom flange half plate L SLN116 ring	◢ 6.0 ◣	243	C LC_121	24.7	0.0	5.8	6.0	12.5	6.1	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.6	0.0	8.4	2.5	-15.0	6.8	6.3	OK	OK
SLN116 ring-w 1	Bottom flange half plate R SLN116 ring	◢ 6.0 ◣	243	C LC_122	25.2	0.0	5.9	6.1	-12.8	6.2	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.5	0.0	7.8	2.7	15.0	6.8	6.5	OK	OK
SLN116 ring-tfl 1	Haunch web SLN116 ring	◢ 6.0 ◣	572	C LC_122	65.2	0.0	-14.2	-16.2	33.0	16.1	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_121	63.5	0.0	-16.0	16.2	-31.5	15.7	9.1	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	149	C LC_130	18.3	0.0	-5.0	-7.4	6.9	4.5	4.5	OK	OK
		◢ 6.0 ◣	149	C LC_126	25.6	0.0	-10.7	8.0	-10.8	6.3	5.8	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_121	7.7	0.0	1.9	1.7	4.0	1.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.9	0.0	1.9	-2.4	-5.7	2.7	0.0	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_130	11.2	0.0	-5.6	-3.6	4.3	2.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_125	8.9	0.0	-3.3	3.9	2.7	2.2	0.0	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_126	3.9	0.0	1.2	1.1	1.8	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	5.0	0.0	1.8	-1.6	-2.2	1.2	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_120	55.0	0.0	-10.5	-11.2	29.1	13.6	8.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	48.8	0.0	-8.0	8.6	-26.4	12.1	7.9	OK	OK
End plate SLN116 ring	Bottom flange plate SLN116 ring	◢ 6.0 ◣	149	C LC_130	16.5	0.0	-7.0	-8.1	2.8	4.1	4.1	OK	OK
		◢ 6.0 ◣	149	C LC_126	24.9	0.0	-12.9	11.2	-5.0	6.1	5.6	OK	OK
End plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	145	C LC_121	10.7	0.0	1.9	2.4	5.6	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.9	0.0	1.9	-1.7	-4.1	1.9	0.0	OK	OK
End plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	149	C LC_125	10.9	0.0	-5.5	-3.2	-4.4	2.7	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_126	9.1	0.0	-3.4	4.0	-2.9	2.3	0.0	OK	OK
Haunch top plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	342	C LC_122	5.1	0.0	1.8	1.6	2.2	1.3	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	4.0	0.0	-0.3	0.3	2.3	1.0	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch web SLN116 ring	◢ 6.0 ◣	271	C LC_122	50.8	0.0	-8.6	-5.9	28.3	12.6	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_120	52.1	0.0	-9.5	10.1	-27.8	12.9	8.2	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_145	67.5	0.0	35.1	-28.3	17.5	16.7	12.0	OK	OK
SLN115b+SLN115a radial (continuous)-w 1	SLN116 ring-w 1	◢ 7.2	208	C LC_145	67.3	0.0	34.6	28.8	-16.8	16.6	11.9	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.1	2.8	0.3	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_125	3.8	0.0	-0.8	2.1	0.2	0.9	0.0	OK	OK
Haunch sloping plate SLN116 ring	Haunch top plate SLN116 ring	◢ 6.0 ◣	146	C LC_121	7.1	0.0	-5.1	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_126	4.0	0.0	-0.8	2.2	-0.2	1.0	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_240
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring

Name	SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

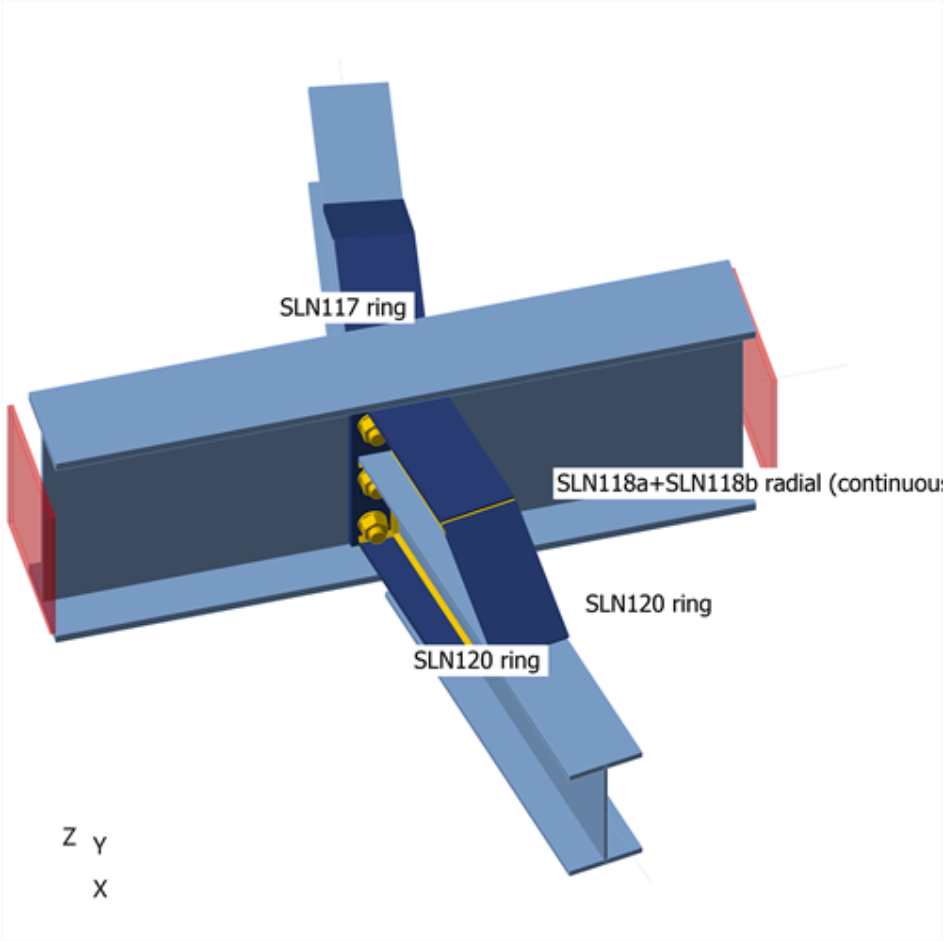
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN118a+SLN118b radial (continuous)	1 - UB457x191x89	-120.0	0.0	0.0	0	0	0
SLN117 ring	2 - UB254x146x43	140.0	0.0	0.0	0	0	-102
SLN120 ring	2 - UB254x146x43	-20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN118a+SLN118b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN118a+SLN118b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN117 ring / end		Position	0
SLN120 ring / end		Position	0



SLN118a+SLN118b radial (continuous) - SLN117 ring - SLN120 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN117 ring / End	4.9	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.2
C LC_102	SLN117 ring / End	18.9	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	18.9	-1.0	-25.7	0.0	0.0	-1.0
C LC_103	SLN117 ring / End	16.2	0.1	-24.4	0.0	0.0	0.3
	SLN120 ring / End	16.0	-1.0	-24.1	0.0	0.0	-0.9
C LC_104	SLN117 ring / End	13.0	0.3	-24.4	0.0	0.0	0.4
	SLN120 ring / End	13.2	-0.9	-24.1	0.0	0.0	-0.8
C LC_105	SLN117 ring / End	15.1	0.4	-24.4	0.0	0.0	0.5
	SLN120 ring / End	15.2	-1.1	-24.1	0.0	0.0	-0.9
C LC_106	SLN117 ring / End	14.3	0.0	-24.4	0.0	0.0	0.0
	SLN120 ring / End	14.2	-0.9	-24.1	0.0	0.0	-0.9
C LC_107	SLN117 ring / End	1.7	-0.1	-0.5	0.0	0.0	0.0
	SLN120 ring / End	1.5	-0.2	-0.5	0.0	0.0	-0.1
C LC_108	SLN117 ring / End	-2.8	0.1	-0.5	0.0	0.0	0.2
	SLN120 ring / End	-2.7	0.1	-0.5	0.0	0.0	0.2
C LC_109	SLN117 ring / End	1.8	0.2	-0.5	0.0	0.0	0.3
	SLN120 ring / End	2.2	-0.2	-0.5	0.0	0.0	-0.1
C LC_110	SLN117 ring / End	-2.5	-0.2	-0.5	0.0	0.0	-0.4
	SLN120 ring / End	-2.9	0.0	-0.5	0.0	0.0	0.0
C LC_111	SLN117 ring / End	11.5	0.0	-16.5	0.0	0.0	0.2
	SLN120 ring / End	11.2	-0.8	-16.3	0.0	0.0	-0.6
C LC_112	SLN117 ring / End	6.4	0.3	-16.5	0.0	0.0	0.4
	SLN120 ring / End	6.6	-0.5	-16.3	0.0	0.0	-0.4
C LC_113	SLN117 ring / End	10.9	0.4	-16.5	0.0	0.0	0.6
	SLN120 ring / End	11.2	-0.9	-16.3	0.0	0.0	-0.7
C LC_114	SLN117 ring / End	7.5	-0.2	-16.5	0.0	0.0	-0.2
	SLN120 ring / End	7.2	-0.5	-16.3	0.0	0.0	-0.6
C LC_115	SLN117 ring / End	3.0	-0.1	-1.8	0.0	0.0	0.0
	SLN120 ring / End	2.8	-0.2	-1.7	0.0	0.0	-0.1
C LC_116	SLN117 ring / End	-1.6	0.2	-1.8	0.0	0.0	0.2
	SLN120 ring / End	-1.5	0.1	-1.7	0.0	0.0	0.1
C LC_117	SLN117 ring / End	3.0	0.3	-1.8	0.0	0.0	0.4
	SLN120 ring / End	3.5	-0.3	-1.7	0.0	0.0	-0.2
C LC_118	SLN117 ring / End	-1.2	-0.2	-1.8	0.0	0.0	-0.3
	SLN120 ring / End	-1.6	0.0	-1.7	0.0	0.0	-0.1
C LC_119	SLN117 ring / End	18.1	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	17.8	-1.0	-25.7	0.0	0.0	-0.9
C LC_120	SLN117 ring / End	15.5	0.3	-26.0	0.0	0.0	0.4
	SLN120 ring / End	15.7	-1.0	-25.7	0.0	0.0	-0.9
C LC_121	SLN117 ring / End	17.8	0.4	-26.0	0.0	0.0	0.5
	SLN120 ring / End	18.0	-1.1	-25.7	0.0	0.0	-0.9
C LC_122	SLN117 ring / End	16.1	0.0	-26.0	0.0	0.0	0.1
	SLN120 ring / End	16.1	-0.9	-25.7	0.0	0.0	-0.9
C LC_123	SLN117 ring / End	15.3	0.2	-19.7	0.0	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN120 ring / End	14.9	-0.8	-19.5	0.0	0.0	-0.7
C LC_124	SLN117 ring / End	11.5	0.2	-19.7	0.0	0.0	0.4
	SLN120 ring / End	11.7	-0.6	-19.5	0.0	0.0	-0.6
C LC_125	SLN117 ring / End	16.4	0.4	-19.7	0.0	0.0	0.6
	SLN120 ring / End	16.8	-0.9	-19.5	0.0	0.0	-0.7
C LC_126	SLN117 ring / End	11.1	-0.1	-19.7	0.0	0.0	-0.1
	SLN120 ring / End	10.9	-0.7	-19.5	0.0	0.0	-0.8
C LC_127	SLN117 ring / End	6.8	0.1	-5.0	0.0	0.0	0.2
	SLN120 ring / End	6.5	-0.3	-4.9	0.0	0.0	-0.2
C LC_128	SLN117 ring / End	3.5	0.1	-5.0	0.0	0.0	0.2
	SLN120 ring / End	3.6	0.0	-4.9	0.0	0.0	0.0
C LC_129	SLN117 ring / End	8.5	0.3	-5.0	0.0	0.0	0.4
	SLN120 ring / End	9.0	-0.3	-4.9	0.0	0.0	-0.2
C LC_130	SLN117 ring / End	2.4	-0.2	-5.0	0.0	0.0	-0.2
	SLN120 ring / End	2.2	-0.2	-4.9	0.0	0.0	-0.3
C LC_135	SLN117 ring / End	15.9	0.2	-26.0	0.0	0.0	0.3
	SLN120 ring / End	15.9	-0.7	-25.7	0.0	0.0	-0.7
C LC_141	SLN117 ring / End	5.0	0.0	-5.0	0.0	0.0	0.1
	SLN120 ring / End	5.0	-0.2	-4.9	0.0	0.0	-0.2
C LC_142	SLN117 ring / End	4.7	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.8	-0.2	-4.9	0.0	0.0	-0.2
C LC_143	SLN117 ring / End	4.7	0.1	-5.0	0.0	0.0	0.1
	SLN120 ring / End	4.7	-0.2	-4.9	0.0	0.0	-0.2
C LC_144	SLN117 ring / End	5.0	0.0	-5.0	0.0	0.0	0.1
	SLN120 ring / End	5.0	-0.2	-4.9	0.0	0.0	-0.2
C LC_145	SLN117 ring / End	19.5	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	19.3	-1.0	-25.7	0.0	0.0	-0.9
C LC_146	SLN117 ring / End	18.4	0.2	-26.0	0.0	0.0	0.4
	SLN120 ring / End	18.6	-1.1	-25.7	0.0	0.0	-1.0
C LC_147	SLN117 ring / End	18.2	0.3	-26.0	0.0	0.0	0.4
	SLN120 ring / End	18.1	-1.1	-25.7	0.0	0.0	-1.0
C LC_148	SLN117 ring / End	19.7	0.1	-26.0	0.0	0.0	0.3
	SLN120 ring / End	19.8	-1.0	-25.7	0.0	0.0	-0.9
C LC_151	SLN117 ring / End	11.9	0.1	-11.9	0.0	0.0	0.2
	SLN120 ring / End	11.9	-0.6	-11.8	0.0	0.0	-0.6
C LC_152	SLN117 ring / End	19.1	0.2	-14.0	0.0	0.0	0.4
	SLN120 ring / End	19.1	-1.1	-13.8	0.0	0.0	-1.0
C LC_153	SLN117 ring / End	19.0	0.2	-20.9	0.0	0.0	0.4
	SLN120 ring / End	18.9	-1.0	-20.7	0.0	0.0	-1.0
C LC_154	SLN117 ring / End	12.0	0.1	-11.9	0.0	0.0	0.2
	SLN120 ring / End	12.0	-0.6	-11.8	0.0	0.0	-0.6

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	1.5 < 100%	OK
Welds	19.2 < 100%	OK
Buckling	Not calculated	

Plates

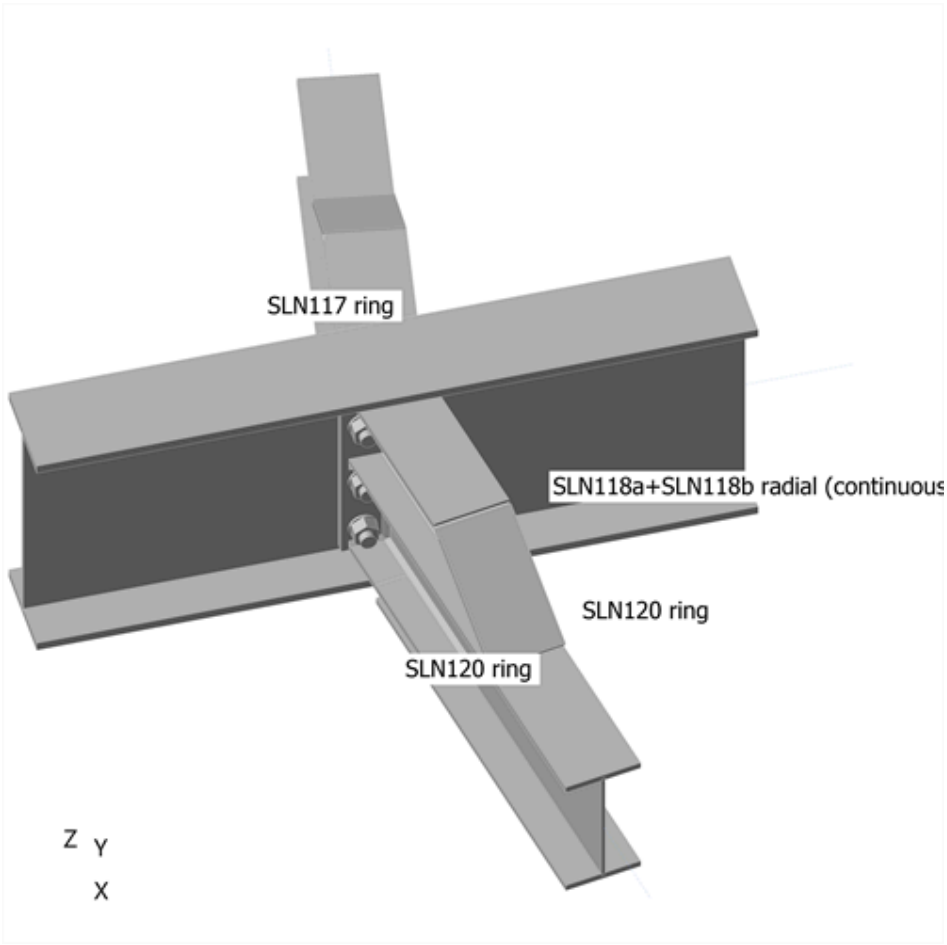
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN118a+SLN118b radial (continuous)-bfl 1	17.7	C LC_146	22.8	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-tfl 1	17.7	C LC_122	19.2	0.0	0.0	OK
SLN118a+SLN118b radial (continuous)-w 1	10.5	C LC_122	30.8	0.0	1.7	OK
SLN117 ring-bfl 1	12.7	C LC_145	51.7	0.0	0.0	OK
SLN117 ring-tfl 1	12.7	C LC_121	45.8	0.0	0.0	OK
SLN117 ring-w 1	7.2	C LC_148	49.7	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_121	51.1	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_122	45.0	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	49.1	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_146	15.3	0.0	8.6	OK
Bottom flange plate SLN120 ring	15.0	C LC_121	26.9	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	20.9	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_146	19.7	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_120	29.5	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_122	9.3	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_120	12.9	0.0	0.0	OK
End plate SLN117 ring	20.0	C LC_146	13.1	0.0	6.3	OK
Bottom flange plate SLN117 ring	15.0	C LC_122	21.4	0.0	0.0	OK
Bottom flange half plate L SLN117 ring	15.0	C LC_148	19.7	0.0	0.0	OK
Bottom flange half plate R SLN117 ring	15.0	C LC_145	20.0	0.0	0.0	OK
Haunch web SLN117 ring	12.0	C LC_135	28.5	0.0	0.0	OK
Haunch top plate SLN117 ring	15.0	C LC_126	6.5	0.0	0.0	OK
Haunch sloping plate SLN117 ring	15.0	C LC_120	12.9	0.0	0.0	OK

Design data

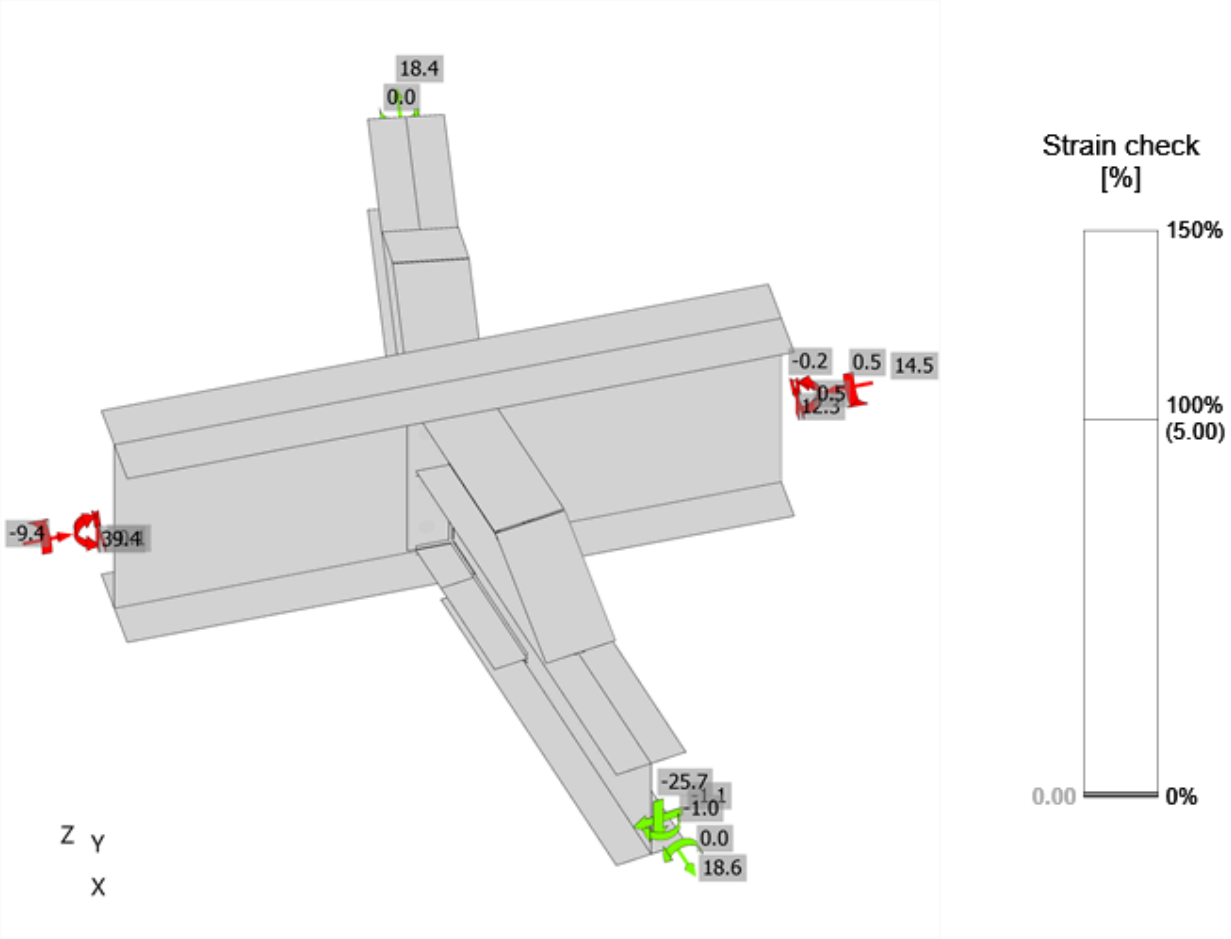
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

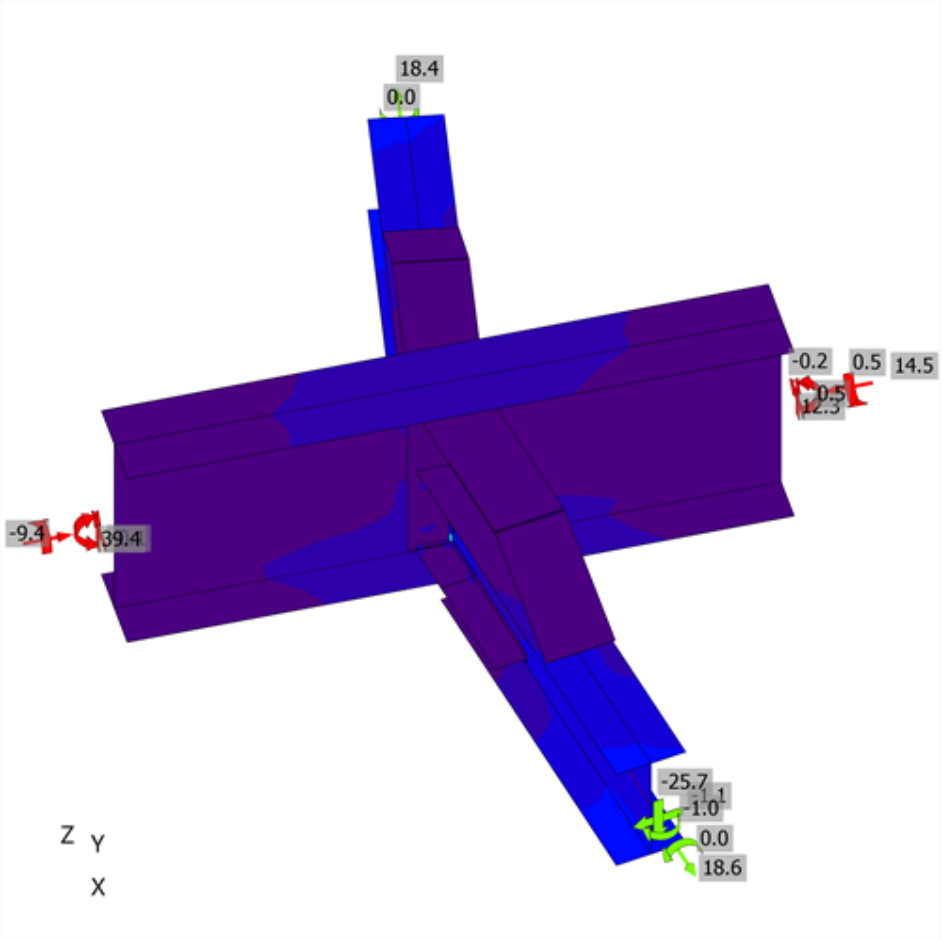
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

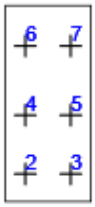


Strain check, C LC_146



Equivalent stress, C LC_146

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_146	4.8	0.5	176.5	1.5	0.6	1.3	OK	OK
	B3	M30 8.8 - 1	C LC_122	1.6	0.6	176.5	0.5	0.7	0.6	OK	OK
	B4	M30 8.8 - 1	C LC_135	0.2	0.4	213.9	0.1	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_148	0.3	0.5	213.9	0.1	0.5	0.3	OK	OK
	B6	M30 8.8 - 1	C LC_122	2.9	0.9	229.5	0.9	0.7	1.0	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.3	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN117 ring	Bottom flange plate SLN117 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-tfl 1	◢ 10.5	149	C LC_121	13.2	0.0	-7.0	6.4	1.1	3.3	3.3	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 10.5	149	C LC_152	16.2	0.0	7.8	-8.2	-0.5	4.0	4.0	OK	OK
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	27.2	0.0	-13.3	-11.9	-6.8	6.7	6.2	OK	OK
		◢ 6.0 ◣	208	C LC_148	24.2	0.0	-11.1	12.4	-0.7	6.0	5.9	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_145	23.4	0.0	1.2	-0.9	13.5	5.8	5.8	OK	OK
		◢ 6.0 ◣	100	C LC_148	28.5	0.0	-0.6	3.0	-16.2	7.0	7.0	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_145	20.9	0.0	6.2	5.1	10.4	5.2	5.2	OK	OK
		◢ 6.0 ◣	243	C LC_148	28.4	0.0	11.5	0.9	-15.0	7.0	6.1	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	28.4	0.0	5.8	7.0	-14.5	7.0	6.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	35.1	0.0	9.7	-8.4	17.6	8.7	7.2	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_146	70.7	0.0	-14.4	-18.8	35.3	17.5	9.9	OK	OK
		◢ 6.0 ◣	572	C LC_135	57.5	0.0	-16.9	13.7	-28.6	14.2	8.6	OK	OK
End plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	208	C LC_148	21.3	0.0	-11.5	-10.2	2.1	5.3	5.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_148	25.2	0.0	-10.4	12.6	4.0	6.2	6.0	OK	OK
Bottom flange plate SLN117 ring	SLN117 ring-w 1	◢ 6.0 ◣	100	C LC_121	26.1	0.0	3.1	3.9	14.4	6.5	6.5	OK	OK
		◢ 6.0 ◣	100	C LC_122	25.2	0.0	1.9	-2.4	-14.3	6.2	6.2	OK	OK
SLN117 ring-w 1	Bottom flange half plate L SLN117 ring	◢ 6.0 ◣	243	C LC_121	25.7	0.0	6.0	6.3	13.0	6.3	6.3	OK	OK
		◢ 6.0 ◣	243	C LC_121	27.8	0.0	7.0	-6.7	-14.0	6.9	6.7	OK	OK
SLN117 ring-w 1	Bottom flange half plate R SLN117 ring	◢ 6.0 ◣	243	C LC_148	24.5	0.0	6.0	6.0	-12.3	6.0	5.8	OK	OK
		◢ 6.0 ◣	243	C LC_145	27.9	0.0	8.8	2.4	15.1	6.9	6.4	OK	OK
SLN117 ring-tfl 1	Haunch web SLN117 ring	◢ 6.0 ◣	572	C LC_122	64.8	0.0	-16.7	-16.5	32.1	16.0	9.3	OK	OK
		◢ 6.0 ◣	572	C LC_121	65.9	0.0	-14.5	16.5	-33.2	16.3	9.4	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_122	31.7	0.0	-9.9	-10.9	13.6	7.8	6.8	OK	OK
		◢ 6.0 ◣	149	C LC_122	37.2	0.0	-14.0	13.0	-15.1	9.2	6.5	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_135	5.7	0.0	0.9	0.8	3.1	1.4	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	11.3	0.0	2.7	-2.7	-5.7	2.8	0.0	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_126	13.9	0.0	-6.3	-5.1	5.0	3.4	3.4	OK	OK
		◢ 6.0 ◣	149	C LC_122	12.3	0.0	-4.4	5.1	-4.3	3.1	3.1	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_122	6.7	0.0	2.4	1.9	3.1	1.7	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_122	5.1	0.0	-0.3	0.4	2.9	1.3	0.0	OK	OK
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_121	56.1	0.0	-10.5	-4.9	31.4	13.9	8.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	49.2	0.0	-8.3	14.0	-24.3	12.2	8.0	OK	OK
End plate SLN117 ring	Bottom flange plate SLN117 ring	◢ 6.0 ◣	148	C LC_148	16.0	0.0	10.2	7.1	-0.4	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_126	25.4	0.0	-13.5	11.0	-5.8	6.3	5.6	OK	OK
End plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	145	C LC_135	9.1	0.0	2.3	2.2	4.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	8.9	0.0	1.3	-1.3	-4.9	2.2	0.0	OK	OK
End plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	149	C LC_126	9.9	0.0	-5.7	-3.7	2.9	2.5	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	5.8	0.0	-1.2	2.9	-1.5	1.4	0.0	OK	OK
Haunch top plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	342	C LC_122	4.0	0.0	1.2	1.2	1.8	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	4.6	0.0	1.8	-1.6	-1.8	1.1	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch web SLN117 ring	◢ 6.0 ◣	271	C LC_135	53.4	0.0	-9.9	-11.0	28.2	13.2	8.3	OK	OK
		◢ 6.0 ◣	271	C LC_121	51.7	0.0	-8.9	6.1	-28.7	12.8	8.3	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	77.8	0.0	34.7	-37.9	13.5	19.2	13.4	OK	OK
SLN118a+SLN118b radial (continuous)-w 1	SLN117 ring-w 1	◢ 7.2	208	C LC_148	75.0	0.0	36.8	31.8	-20.3	18.5	13.0	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_122	7.7	0.0	-7.0	-1.9	0.1	2.3	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_122	5.1	0.0	-1.2	2.9	0.2	1.3	0.0	OK	OK
Haunch sloping plate SLN117 ring	Haunch top plate SLN117 ring	◢ 6.0 ◣	146	C LC_122	7.2	0.0	-5.2	2.8	0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_121	3.3	0.0	-0.9	1.6	-0.9	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_300
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring

Name	SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

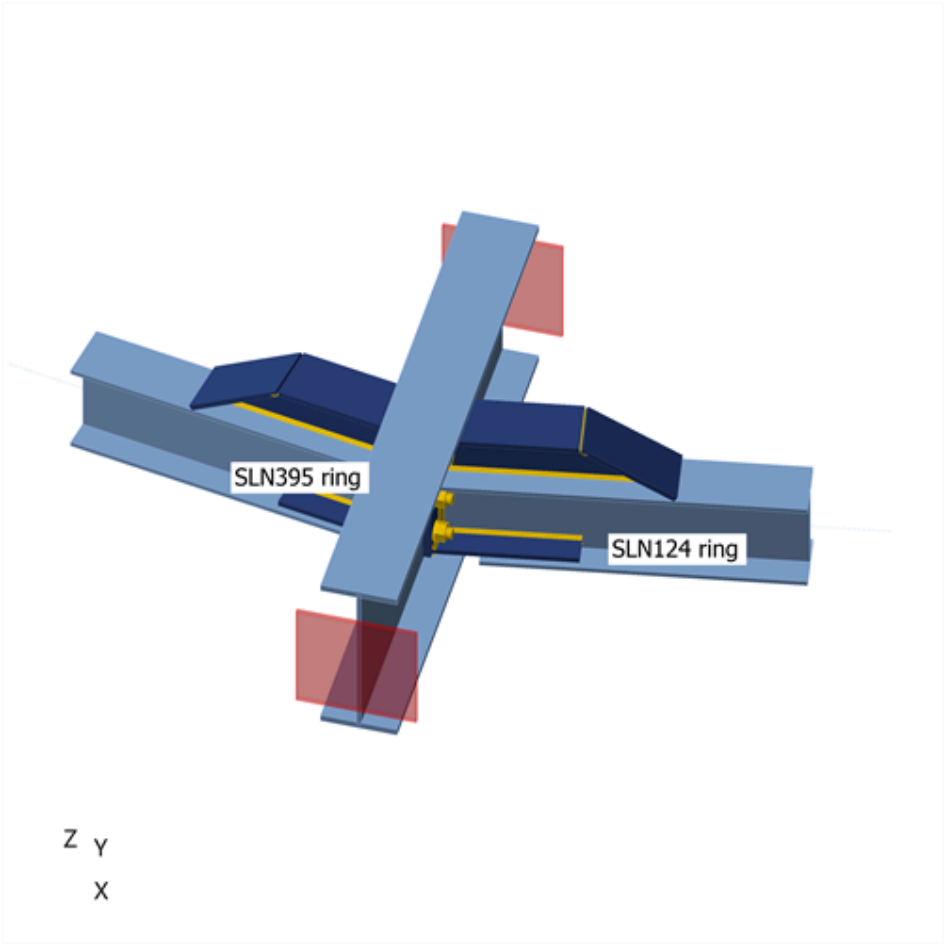
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN122a+SLN122b radial (continuous)	1 - UB457x191x89	-60.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	40.0	0.0	0.0	0	0	-102
SLN395 ring	2 - UB254x146x43	-160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN122a+SLN122b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN122a+SLN122b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN395 ring / end		Position	0



SLN122a+SLN122b radial (continuous) - SLN124 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN124 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.8	0.0	-2.2	0.0	0.0	0.0
C LC_102	SLN124 ring / End	-2.8	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-2.9	0.0	-8.7	0.0	0.0	0.0
C LC_103	SLN124 ring / End	-8.2	0.1	-24.0	0.0	0.0	0.1
	SLN395 ring / End	-7.9	0.1	-8.2	0.0	0.0	0.1
C LC_104	SLN124 ring / End	-5.3	0.2	-24.0	0.0	0.0	0.3
	SLN395 ring / End	-6.1	0.1	-8.2	0.0	0.0	0.2
C LC_105	SLN124 ring / End	-8.5	0.0	-24.0	0.0	0.0	0.1
	SLN395 ring / End	-8.8	0.1	-8.2	0.0	0.0	0.0
C LC_106	SLN124 ring / End	-4.4	0.0	-24.0	0.0	0.0	-0.1
	SLN395 ring / End	-4.0	0.0	-8.2	0.0	0.0	-0.1
C LC_107	SLN124 ring / End	-5.2	0.0	-0.5	0.0	0.0	0.0
	SLN395 ring / End	-4.9	0.1	-0.6	0.0	0.0	0.1
C LC_108	SLN124 ring / End	-0.7	0.2	-0.5	0.0	0.0	0.3
	SLN395 ring / End	-1.9	0.2	-0.6	0.0	0.0	0.4
C LC_109	SLN124 ring / End	-5.1	0.1	-0.5	0.0	0.0	0.2
	SLN395 ring / End	-5.5	0.0	-0.6	0.0	0.0	0.0
C LC_110	SLN124 ring / End	0.1	-0.3	-0.5	0.0	0.0	-0.4
	SLN395 ring / End	1.1	-0.1	-0.6	0.0	0.0	-0.2
C LC_111	SLN124 ring / End	-9.2	0.1	-16.2	0.0	0.0	0.1
	SLN395 ring / End	-8.8	0.1	-5.8	0.0	0.0	0.1
C LC_112	SLN124 ring / End	-4.3	0.2	-16.2	0.0	0.0	0.4
	SLN395 ring / End	-5.6	0.2	-5.8	0.0	0.0	0.4
C LC_113	SLN124 ring / End	-9.4	0.1	-16.2	0.0	0.0	0.2
	SLN395 ring / End	-9.9	0.1	-5.8	0.0	0.0	0.1
C LC_114	SLN124 ring / End	-3.2	-0.2	-16.2	0.0	0.0	-0.3
	SLN395 ring / End	-2.2	-0.1	-5.8	0.0	0.0	-0.2
C LC_115	SLN124 ring / End	-5.0	0.0	-1.7	0.0	0.0	0.1
	SLN395 ring / End	-4.7	0.1	-1.1	0.0	0.0	0.1
C LC_116	SLN124 ring / End	-0.5	0.2	-1.7	0.0	0.0	0.3
	SLN395 ring / End	-1.7	0.2	-1.1	0.0	0.0	0.4
C LC_117	SLN124 ring / End	-4.9	0.1	-1.7	0.0	0.0	0.2
	SLN395 ring / End	-5.4	0.0	-1.1	0.0	0.0	0.0
C LC_118	SLN124 ring / End	0.4	-0.3	-1.7	0.0	0.0	-0.4
	SLN395 ring / End	1.4	-0.1	-1.1	0.0	0.0	-0.2
C LC_119	SLN124 ring / End	-6.0	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-6.0	0.1	-8.7	0.0	0.0	0.1
C LC_120	SLN124 ring / End	-4.3	0.2	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-4.6	0.0	-8.7	0.0	0.0	0.0
C LC_121	SLN124 ring / End	-6.1	0.0	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-6.2	0.0	-8.7	0.0	0.0	0.0
C LC_122	SLN124 ring / End	-3.2	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-3.1	0.0	-8.7	0.0	0.0	-0.1
C LC_123	SLN124 ring / End	-5.0	0.1	-19.4	0.0	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN395 ring / End	-5.0	0.2	-6.8	0.0	0.0	0.2
C LC_124	SLN124 ring / End	-2.4	0.2	-19.4	0.0	0.0	0.3
	SLN395 ring / End	-2.6	0.0	-6.8	0.0	0.0	0.0
C LC_125	SLN124 ring / End	-4.5	0.0	-19.4	0.0	0.0	0.1
	SLN395 ring / End	-4.7	0.0	-6.8	0.0	0.0	0.0
C LC_126	SLN124 ring / End	-0.7	0.0	-19.4	0.0	0.0	0.0
	SLN395 ring / End	-0.4	-0.1	-6.8	0.0	0.0	-0.2
C LC_127	SLN124 ring / End	-0.8	0.1	-4.9	0.0	0.0	0.2
	SLN395 ring / End	-0.9	0.1	-2.2	0.0	0.0	0.2
C LC_128	SLN124 ring / End	1.4	0.2	-4.9	0.0	0.0	0.2
	SLN395 ring / End	1.3	0.0	-2.2	0.0	0.0	0.1
C LC_129	SLN124 ring / End	-0.1	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	-0.1	0.0	-2.2	0.0	0.0	-0.1
C LC_130	SLN124 ring / End	2.9	-0.1	-4.9	0.0	0.0	-0.1
	SLN395 ring / End	3.2	-0.1	-2.2	0.0	0.0	-0.2
C LC_135	SLN124 ring / End	0.5	0.1	-25.6	0.0	0.0	0.1
	SLN395 ring / End	0.5	0.0	-8.7	0.0	0.0	-0.1
C LC_141	SLN124 ring / End	0.7	0.0	-4.9	0.0	0.0	0.1
	SLN395 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0
C LC_142	SLN124 ring / End	0.9	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.8	0.0	-2.2	0.0	0.0	0.0
C LC_143	SLN124 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN395 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
C LC_144	SLN124 ring / End	0.9	0.1	-4.9	0.0	0.0	0.1
	SLN395 ring / End	0.9	0.0	-2.2	0.0	0.0	0.0
C LC_145	SLN124 ring / End	-3.1	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-3.1	0.1	-8.7	0.0	0.0	0.0
C LC_146	SLN124 ring / End	-2.4	0.1	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-2.7	0.0	-8.7	0.0	0.0	-0.1
C LC_147	SLN124 ring / End	-3.6	0.0	-25.6	0.0	0.0	0.1
	SLN395 ring / End	-3.7	0.0	-8.7	0.0	0.0	0.0
C LC_148	SLN124 ring / End	-2.0	0.2	-25.6	0.0	0.0	0.2
	SLN395 ring / End	-2.1	0.0	-8.7	0.0	0.0	0.0
C LC_151	SLN124 ring / End	-0.9	0.1	-11.6	0.0	0.0	0.1
	SLN395 ring / End	-0.8	0.0	-8.7	0.0	0.0	-0.1
C LC_152	SLN124 ring / End	-5.8	0.1	-13.9	0.0	0.0	0.2
	SLN395 ring / End	-5.9	0.1	-2.2	0.0	0.0	0.0
C LC_153	SLN124 ring / End	-2.9	0.1	-20.6	0.0	0.0	0.2
	SLN395 ring / End	-3.0	0.0	-8.7	0.0	0.0	0.0
C LC_154	SLN124 ring / End	-0.8	0.1	-11.6	0.0	0.0	0.1
	SLN395 ring / End	-0.8	0.0	-8.7	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.6 < 100%	OK
Welds	18.2 < 100%	OK
Buckling	Not calculated	

Plates

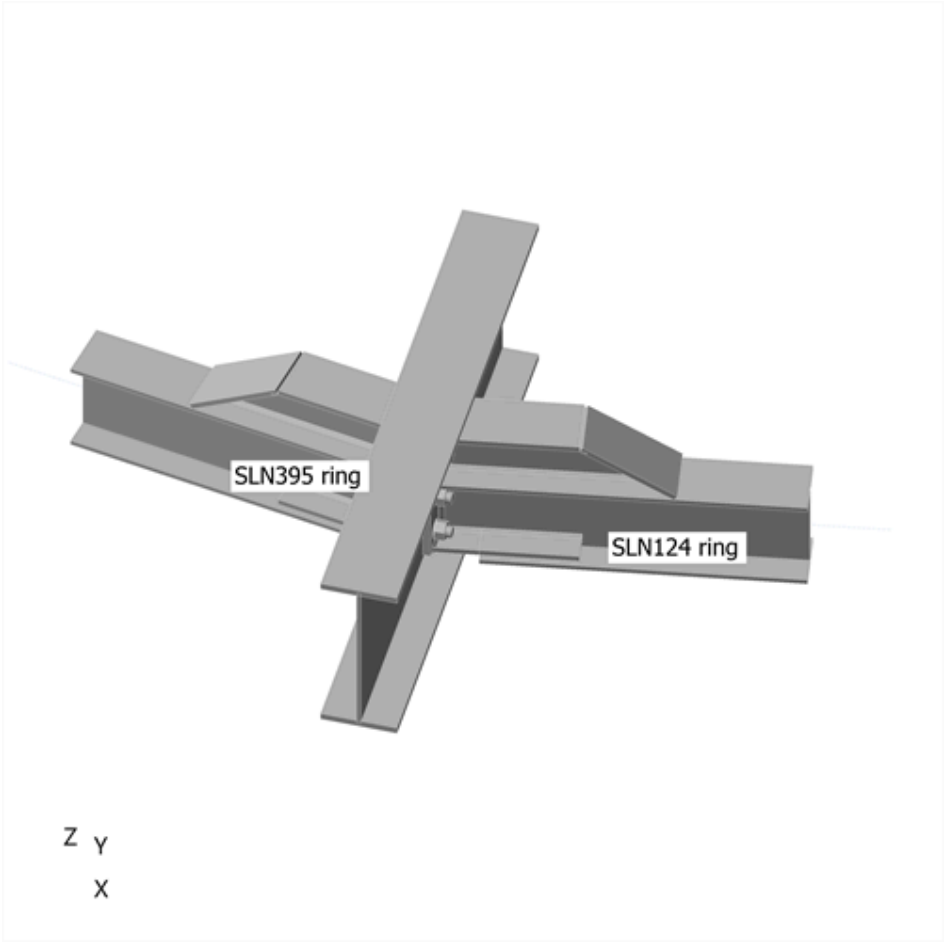
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN122a+SLN122b radial (continuous)-bfl 1	17.7	C LC_104	15.0	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-tfl 1	17.7	C LC_104	12.7	0.0	0.0	OK
SLN122a+SLN122b radial (continuous)-w 1	10.5	C LC_119	18.5	0.0	0.7	OK
SLN124 ring-bfl 1	12.7	C LC_135	46.6	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_119	48.1	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_135	16.1	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_121	17.5	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_119	16.7	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_110	6.5	0.0	7.1	OK
Bottom flange plate SLN124 ring	15.0	C LC_114	14.9	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_135	12.8	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_135	13.0	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_119	32.8	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_110	6.5	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN395 ring	20.0	C LC_112	7.2	0.0	7.7	OK
Bottom flange plate SLN395 ring	15.0	C LC_112	17.0	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_135	4.7	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_112	5.3	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_119	11.7	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_112	7.0	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_121	5.2	0.0	0.0	OK

Design data

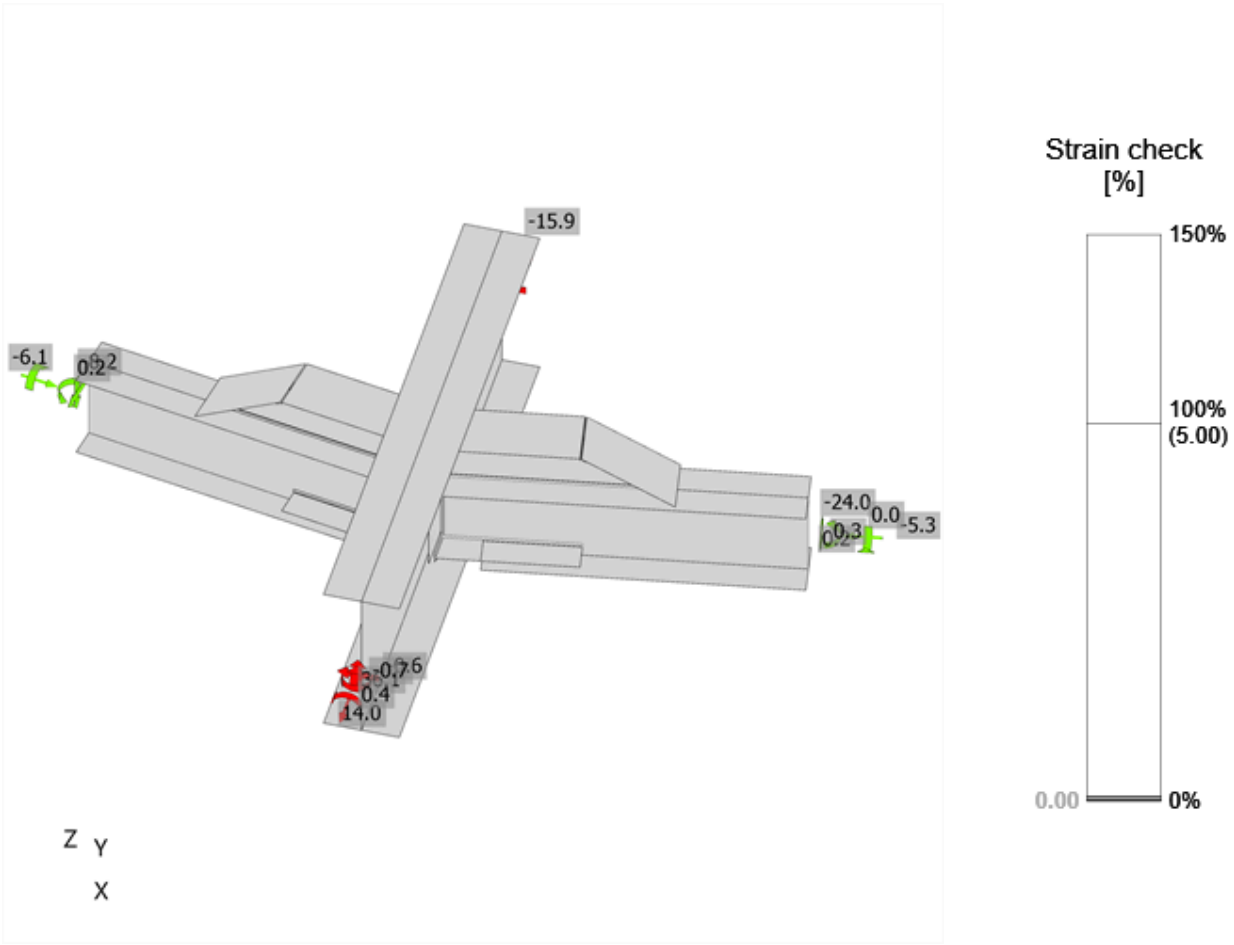
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

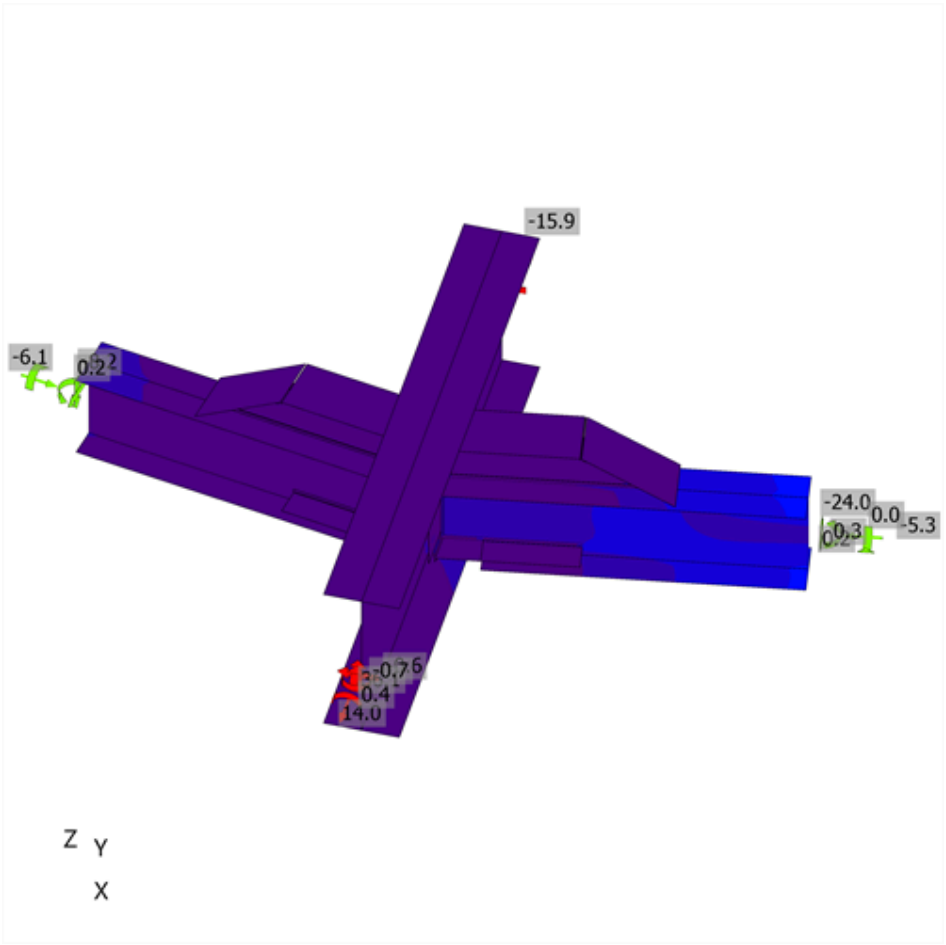
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

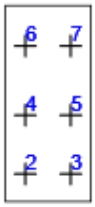


Strain check, C LC_104



Equivalent stress, C LC_104

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_114	1.1	0.2	176.5	0.3	0.2	0.4	OK	OK
	B3	M30 8.8 - 1	C LC_148	0.6	0.4	176.5	0.2	0.4	0.3	OK	OK
	B4	M30 8.8 - 1	C LC_121	0.1	0.3	213.9	0.0	0.2	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.3	213.9	0.0	0.3	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_119	1.4	0.6	229.5	0.4	0.5	0.6	OK	OK
	B7	M30 8.8 - 1	C LC_148	1.6	0.6	229.5	0.5	0.5	0.6	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 10.5	149	C LC_121	9.3	0.0	-8.8	-1.9	-0.1	2.8	2.8	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-tfl 1	◢ 10.5	149	C LC_135	9.3	0.0	-5.8	-4.2	0.2	2.3	0.0	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_135	7.4	0.0	0.4	-0.6	4.2	1.8	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	6.4	0.0	-3.0	3.0	-1.1	1.6	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_119	21.3	0.0	1.6	1.9	12.1	5.3	5.3	OK	OK
		◢ 6.0 ◣	100	C LC_135	20.2	0.0	1.8	-1.9	-11.5	5.0	5.0	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_135	17.8	0.0	4.5	4.2	9.0	4.4	4.4	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.2	0.0	5.9	2.4	-10.3	4.7	4.7	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_135	16.0	0.0	3.5	3.8	-8.1	3.9	3.9	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.5	0.0	6.0	2.3	11.1	5.1	5.1	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_121	73.6	0.0	-19.2	-19.4	36.1	18.2	10.3	OK	OK
		◢ 6.0 ◣	572	C LC_119	68.1	0.0	-15.5	15.6	-35.0	16.8	9.7	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	208	C LC_135	4.9	0.0	-0.9	0.0	2.8	1.2	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	7.3	0.0	-0.1	1.2	-4.1	1.8	0.0	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_112	8.0	0.0	0.5	2.1	4.1	2.0	2.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	7.1	0.0	0.7	-0.8	-4.0	1.8	1.8	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_135	6.2	0.0	1.6	1.5	3.2	1.5	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_112	7.4	0.0	2.1	-1.7	-3.7	1.8	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_135	5.7	0.0	1.3	1.4	-2.9	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	7.2	0.0	2.1	0.8	3.9	1.8	0.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_121	26.7	0.0	-6.2	-6.8	13.3	6.6	6.0	OK	OK
		◢ 6.0 ◣	572	C LC_119	25.2	0.0	-5.8	5.9	-12.9	6.2	6.0	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	21.1	0.0	-6.7	-8.3	8.1	5.2	5.2	OK	OK
		◢ 6.0 ◣	149	C LC_110	23.3	0.0	-9.6	8.0	-9.3	5.8	5.5	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_119	6.3	0.0	0.8	0.7	3.5	1.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	9.4	0.0	1.1	-1.3	-5.3	2.3	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	11.1	0.0	-5.4	-3.9	4.0	2.8	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_110	8.5	0.0	-2.5	4.0	-2.5	2.1	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_114	3.5	0.0	1.2	1.1	1.6	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_119	4.2	0.0	1.5	-1.4	-1.8	1.0	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_119	61.1	0.0	-11.8	-11.9	32.5	15.1	9.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_121	52.3	0.0	-8.1	8.7	-28.5	12.9	8.3	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	20.5	0.0	-5.8	-8.5	-7.6	5.1	5.1	OK	OK
		◢ 6.0 ◣	149	C LC_112	24.7	0.0	-10.8	8.1	10.0	6.1	6.0	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_119	5.0	0.0	0.7	0.9	2.7	1.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_135	3.4	0.0	-0.3	-0.4	1.9	0.8	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_112	12.2	0.0	-6.0	-4.0	-4.7	3.0	3.0	OK	OK
		◢ 6.0 ◣	149	C LC_112	8.8	0.0	-2.3	4.3	2.5	2.2	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_110	1.8	0.0	-0.2	0.0	1.0	0.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_108	2.5	0.0	0.2	-0.4	-1.4	0.6	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_119	21.4	0.0	-4.0	-4.2	11.4	5.3	5.3	OK	OK
		◢ 6.0 ◣	271	C LC_119	19.5	0.0	-3.1	2.9	-10.7	4.8	4.8	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_135	30.5	0.0	8.9	-8.2	14.8	7.5	7.5	OK	OK
SLN122a+SLN122b radial (continuous)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_135	18.6	0.0	2.7	2.5	-10.4	4.6	4.6	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_119	7.4	0.0	-5.3	3.0	-0.4	2.0	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_112	3.1	0.0	-0.6	1.7	0.2	0.8	0.0	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_112	3.1	0.0	-2.8	-0.7	0.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_112	2.2	0.0	-0.4	1.2	0.3	0.5	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_320
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring

Name	SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

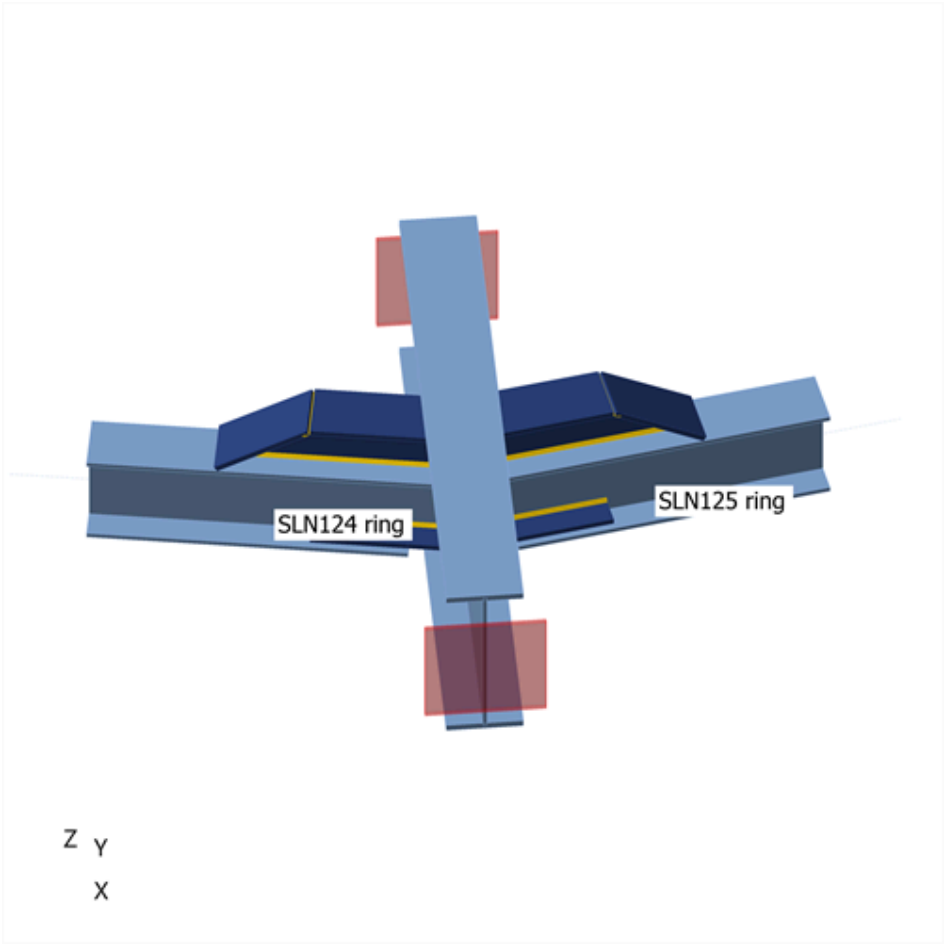
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN123b+SLN123a radial (continuous)	1 - UB457x191x89	-40.0	0.0	0.0	0	0	0
SLN124 ring	2 - UB254x146x43	-140.0	0.0	0.0	0	0	-102
SLN125 ring	2 - UB254x146x43	60.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN123b+SLN123a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN123b+SLN123a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN124 ring / end		Position	0
SLN125 ring / end		Position	0



SLN123b+SLN123a radial (continuous) - SLN124 ring - SLN125 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN124 ring / End	0.8	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
C LC_102	SLN124 ring / End	-2.8	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-2.7	-0.1	-25.6	0.0	0.0	-0.2
C LC_103	SLN124 ring / End	-8.3	0.2	-24.0	0.0	0.0	0.2
	SLN125 ring / End	-8.4	-0.1	-24.0	0.0	0.0	-0.1
C LC_104	SLN124 ring / End	-5.1	0.1	-24.0	0.0	0.0	0.1
	SLN125 ring / End	-4.8	0.1	-24.0	0.0	0.0	0.1
C LC_105	SLN124 ring / End	-8.4	0.2	-24.0	0.0	0.0	0.2
	SLN125 ring / End	-7.9	-0.1	-24.0	0.0	0.0	-0.1
C LC_106	SLN124 ring / End	-4.5	-0.1	-24.0	0.0	0.0	-0.1
	SLN125 ring / End	-4.8	-0.1	-24.0	0.0	0.0	-0.2
C LC_107	SLN124 ring / End	-5.2	0.1	-0.5	0.0	0.0	0.1
	SLN125 ring / End	-5.4	0.0	-0.5	0.0	0.0	0.0
C LC_108	SLN124 ring / End	-0.7	0.2	-0.5	0.0	0.0	0.2
	SLN125 ring / End	-0.5	0.2	-0.5	0.0	0.0	0.3
C LC_109	SLN124 ring / End	-5.1	0.1	-0.5	0.0	0.0	0.2
	SLN125 ring / End	-4.7	0.1	-0.5	0.0	0.0	0.1
C LC_110	SLN124 ring / End	0.1	-0.3	-0.5	0.0	0.0	-0.4
	SLN125 ring / End	-0.3	-0.2	-0.5	0.0	0.0	-0.2
C LC_111	SLN124 ring / End	-9.3	0.1	-16.2	0.0	0.0	0.2
	SLN125 ring / End	-9.6	-0.1	-16.2	0.0	0.0	-0.1
C LC_112	SLN124 ring / End	-4.2	0.2	-16.2	0.0	0.0	0.2
	SLN125 ring / End	-3.8	0.2	-16.2	0.0	0.0	0.2
C LC_113	SLN124 ring / End	-9.3	0.2	-16.2	0.0	0.0	0.2
	SLN125 ring / End	-8.7	0.0	-16.2	0.0	0.0	0.0
C LC_114	SLN124 ring / End	-3.3	-0.3	-16.2	0.0	0.0	-0.3
	SLN125 ring / End	-3.7	-0.2	-16.2	0.0	0.0	-0.3
C LC_115	SLN124 ring / End	-5.1	0.1	-1.7	0.0	0.0	0.1
	SLN125 ring / End	-5.3	0.0	-1.7	0.0	0.0	0.0
C LC_116	SLN124 ring / End	-0.5	0.2	-1.7	0.0	0.0	0.2
	SLN125 ring / End	-0.2	0.2	-1.7	0.0	0.0	0.3
C LC_117	SLN124 ring / End	-4.9	0.1	-1.7	0.0	0.0	0.2
	SLN125 ring / End	-4.5	0.1	-1.7	0.0	0.0	0.1
C LC_118	SLN124 ring / End	0.4	-0.3	-1.7	0.0	0.0	-0.3
	SLN125 ring / End	-0.1	-0.2	-1.7	0.0	0.0	-0.2
C LC_119	SLN124 ring / End	-6.2	0.2	-25.6	0.0	0.0	0.3
	SLN125 ring / End	-6.3	-0.1	-25.6	0.0	0.0	-0.1
C LC_120	SLN124 ring / End	-4.2	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-3.8	0.0	-25.6	0.0	0.0	0.0
C LC_121	SLN124 ring / End	-6.0	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-5.6	-0.2	-25.6	0.0	0.0	-0.2
C LC_122	SLN124 ring / End	-3.3	0.0	-25.6	0.0	0.0	0.0
	SLN125 ring / End	-3.6	-0.1	-25.6	0.0	0.0	-0.2
C LC_123	SLN124 ring / End	-5.1	0.2	-19.4	0.0	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN125 ring / End	-5.2	-0.1	-19.4	0.0	0.0	0.0
C LC_124	SLN124 ring / End	-2.3	0.1	-19.4	0.0	0.0	0.3
	SLN125 ring / End	-1.9	0.1	-19.4	0.0	0.0	0.1
C LC_125	SLN124 ring / End	-4.4	0.1	-19.4	0.0	0.0	0.1
	SLN125 ring / End	-4.0	-0.1	-19.4	0.0	0.0	-0.2
C LC_126	SLN124 ring / End	-0.7	-0.1	-19.4	0.0	0.0	-0.1
	SLN125 ring / End	-1.2	-0.2	-19.4	0.0	0.0	-0.3
C LC_127	SLN124 ring / End	-0.8	0.1	-4.9	0.0	0.0	0.2
	SLN125 ring / End	-0.9	0.0	-4.9	0.0	0.0	0.0
C LC_128	SLN124 ring / End	1.4	0.1	-4.9	0.0	0.0	0.2
	SLN125 ring / End	1.8	0.1	-4.9	0.0	0.0	0.1
C LC_129	SLN124 ring / End	0.0	0.1	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.2	-0.1	-4.9	0.0	0.0	-0.1
C LC_130	SLN124 ring / End	2.9	-0.1	-4.9	0.0	0.0	-0.2
	SLN125 ring / End	2.4	-0.2	-4.9	0.0	0.0	-0.2
C LC_135	SLN124 ring / End	0.5	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	0.6	-0.1	-25.6	0.0	0.0	-0.2
C LC_141	SLN124 ring / End	0.7	0.1	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.7	0.0	-4.9	0.0	0.0	-0.1
C LC_142	SLN124 ring / End	0.9	0.0	-4.9	0.0	0.0	0.0
	SLN125 ring / End	0.9	0.0	-4.9	0.0	0.0	0.0
C LC_143	SLN124 ring / End	0.6	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.7	-0.1	-4.9	0.0	0.0	-0.1
C LC_144	SLN124 ring / End	0.9	0.0	-4.9	0.0	0.0	0.1
	SLN125 ring / End	0.9	0.0	-4.9	0.0	0.0	0.0
C LC_145	SLN124 ring / End	-3.3	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-3.3	-0.2	-25.6	0.0	0.0	-0.2
C LC_146	SLN124 ring / End	-2.3	0.0	-25.6	0.0	0.0	0.1
	SLN125 ring / End	-2.1	-0.1	-25.6	0.0	0.0	-0.1
C LC_147	SLN124 ring / End	-3.5	0.2	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-3.2	-0.2	-25.6	0.0	0.0	-0.2
C LC_148	SLN124 ring / End	-2.1	0.1	-25.6	0.0	0.0	0.2
	SLN125 ring / End	-2.2	-0.1	-25.6	0.0	0.0	-0.1
C LC_151	SLN124 ring / End	-0.9	0.1	-11.6	0.0	0.0	0.1
	SLN125 ring / End	-0.8	-0.1	-11.5	0.0	0.0	-0.1
C LC_152	SLN124 ring / End	-5.8	0.1	-13.9	0.0	0.0	0.2
	SLN125 ring / End	-5.7	-0.1	-14.0	0.0	0.0	-0.1
C LC_153	SLN124 ring / End	-2.9	0.1	-20.6	0.0	0.0	0.2
	SLN125 ring / End	-2.9	-0.1	-20.6	0.0	0.0	-0.2
C LC_154	SLN124 ring / End	-0.8	0.1	-11.6	0.0	0.0	0.1
	SLN125 ring / End	-0.8	-0.1	-11.5	0.0	0.0	-0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	17.8 < 100%	OK
Buckling	Not calculated	

Plates

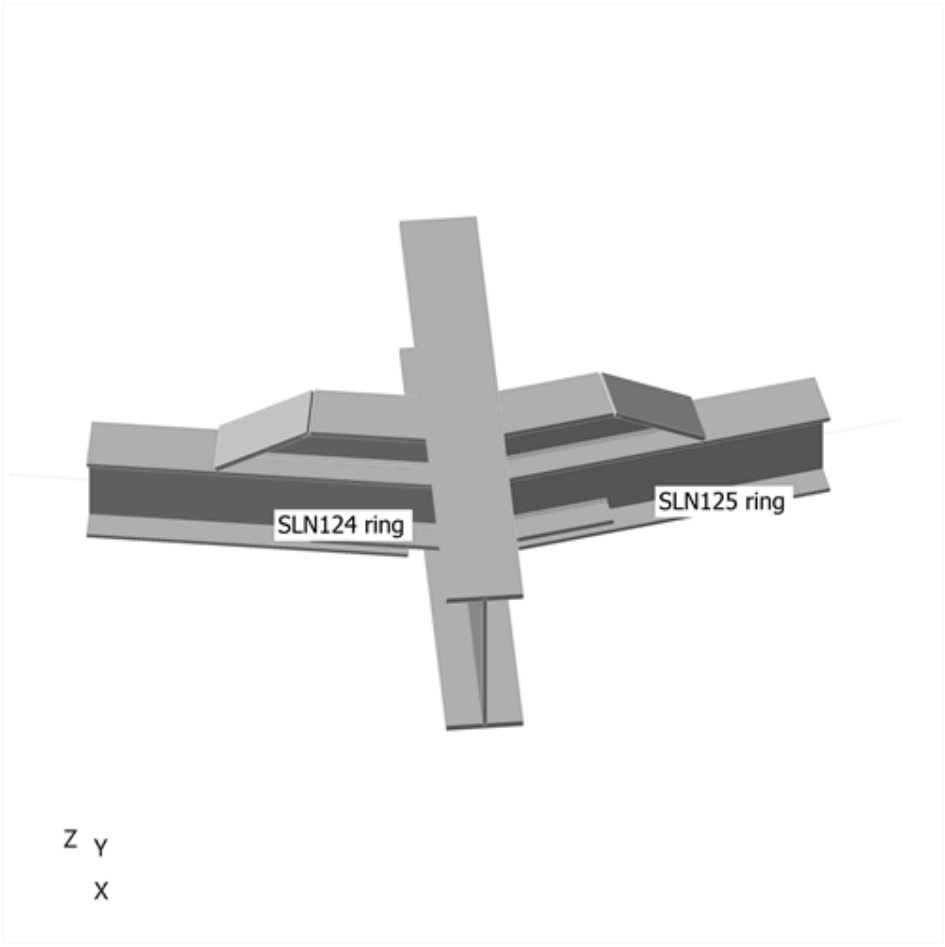
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN123b+SLN123a radial (continuous)-bfl 1	17.7	C LC_121	22.3	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-tfl 1	17.7	C LC_121	18.7	0.0	0.0	OK
SLN123b+SLN123a radial (continuous)-w 1	10.5	C LC_122	26.3	0.0	0.4	OK
SLN124 ring-bfl 1	12.7	C LC_135	46.7	0.0	0.0	OK
SLN124 ring-tfl 1	12.7	C LC_119	47.9	0.0	0.0	OK
SLN124 ring-w 1	7.2	C LC_119	47.2	0.0	0.0	OK
SLN125 ring-bfl 1	12.7	C LC_135	46.6	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_122	47.7	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_119	47.1	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_147	7.2	0.0	0.6	OK
Bottom flange plate SLN125 ring	15.0	C LC_114	13.9	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_135	13.2	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_135	13.0	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_119	31.6	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_110	5.6	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN124 ring	20.0	C LC_119	7.3	0.0	1.3	OK
Bottom flange plate SLN124 ring	15.0	C LC_135	13.7	0.0	0.0	OK
Bottom flange half plate L SLN124 ring	15.0	C LC_135	13.0	0.0	0.0	OK
Bottom flange half plate R SLN124 ring	15.0	C LC_135	13.1	0.0	0.0	OK
Haunch web SLN124 ring	12.0	C LC_119	32.3	0.0	0.0	OK
Haunch top plate SLN124 ring	15.0	C LC_119	5.5	0.0	0.0	OK
Haunch sloping plate SLN124 ring	15.0	C LC_119	14.3	0.0	0.0	OK

Design data

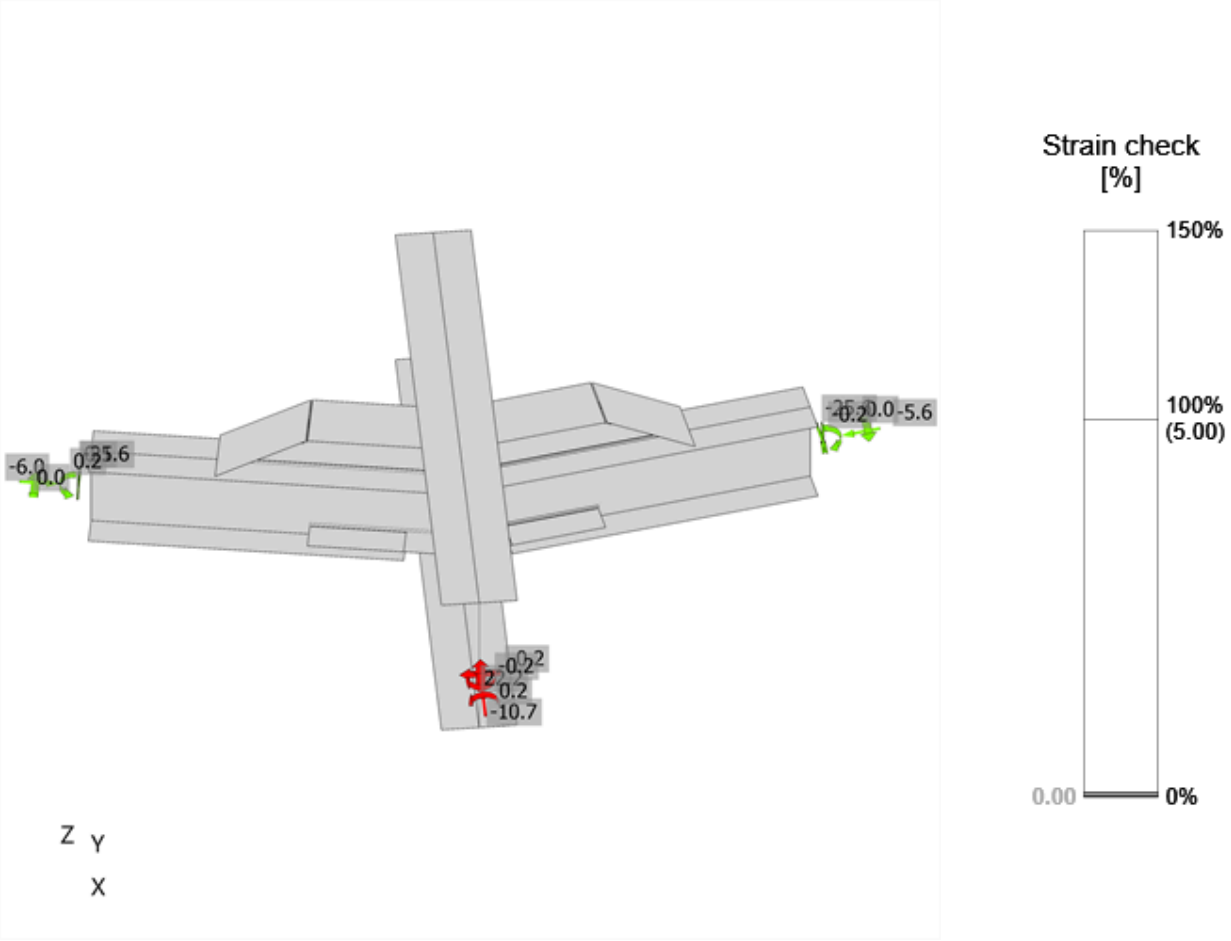
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

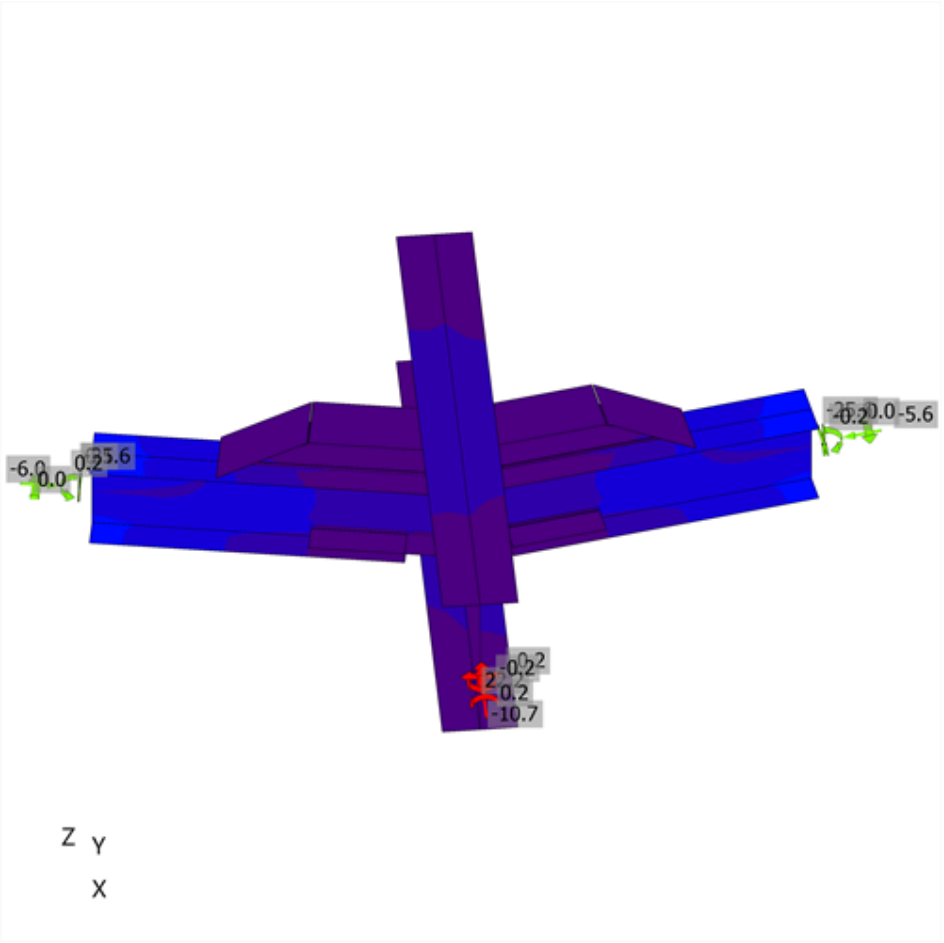
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

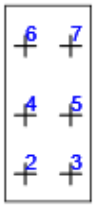


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_119	0.5	0.4	176.5	0.1	0.5	0.3	OK	OK
	B3	M30 8.8 - 1	C LC_135	1.1	0.5	176.5	0.3	0.6	0.5	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.1	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.5	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_147	2.5	0.9	229.5	0.8	0.7	0.9	OK	OK
	B7	M30 8.8 - 1	C LC_135	1.4	0.9	229.5	0.4	0.8	0.7	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN124 ring	Bottom flange plate SLN124 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-tfl 1	◢ 10.5	149	C LC_119	13.0	0.0	-5.5	5.1	4.5	3.5	3.5	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 10.5	149	C LC_119	12.3	0.0	-10.8	-3.4	-0.4	3.5	3.5	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_135	10.4	0.0	0.6	-0.6	5.9	2.6	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_120	8.5	0.0	-0.9	0.1	-4.9	2.1	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_120	20.5	0.0	1.8	2.2	11.6	5.1	5.1	OK	OK
		◢ 6.0 ◣	100	C LC_122	21.2	0.0	1.8	-2.0	-12.0	5.2	5.2	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_135	16.6	0.0	3.8	4.0	8.5	4.1	4.1	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.4	0.0	6.0	2.3	-11.0	5.1	5.1	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_135	17.4	0.0	4.3	4.1	-8.9	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.6	0.0	5.9	2.4	10.5	4.8	4.8	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_119	70.1	0.0	-16.6	-17.0	35.5	17.3	9.9	OK	OK
		◢ 6.0 ◣	572	C LC_119	71.2	0.0	-18.6	18.2	-35.3	17.6	10.0	OK	OK
End plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	208	C LC_122	8.5	0.0	-0.8	0.0	4.9	2.1	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_135	10.5	0.0	0.6	0.5	-6.0	2.6	0.0	OK	OK
Bottom flange plate SLN124 ring	SLN124 ring-w 1	◢ 6.0 ◣	100	C LC_135	21.2	0.0	2.0	1.9	12.0	5.2	5.2	OK	OK
		◢ 6.0 ◣	100	C LC_122	20.5	0.0	1.8	-2.3	-11.6	5.1	5.1	OK	OK
SLN124 ring-w 1	Bottom flange half plate L SLN124 ring	◢ 6.0 ◣	243	C LC_135	17.6	0.0	4.4	4.1	8.9	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.4	0.0	6.0	2.4	-10.4	4.8	4.8	OK	OK
SLN124 ring-w 1	Bottom flange half plate R SLN124 ring	◢ 6.0 ◣	243	C LC_135	16.3	0.0	3.6	3.9	-8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.5	0.0	6.0	2.3	11.1	5.1	5.1	OK	OK
SLN124 ring-tfl 1	Haunch web SLN124 ring	◢ 6.0 ◣	572	C LC_119	72.2	0.0	-19.3	-18.9	35.4	17.8	10.1	OK	OK
		◢ 6.0 ◣	572	C LC_119	69.1	0.0	-15.8	16.2	-35.3	17.1	9.8	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	14.7	0.0	-4.1	-6.1	5.4	3.6	3.6	OK	OK
		◢ 6.0 ◣	149	C LC_118	17.6	0.0	-7.6	5.6	-7.2	4.3	4.3	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_120	7.0	0.0	0.6	0.6	4.0	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	10.2	0.0	1.9	-1.7	-5.5	2.5	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_118	9.4	0.0	-4.6	-3.2	3.5	2.3	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	7.1	0.0	-2.3	3.1	-2.4	1.8	0.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_147	3.9	0.0	1.5	1.4	1.5	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_147	3.7	0.0	-0.2	0.2	2.1	0.9	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_119	55.0	0.0	-9.1	-8.6	30.1	13.6	8.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_122	58.4	0.0	-10.9	11.4	-31.1	14.4	8.8	OK	OK
End plate SLN124 ring	Bottom flange plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	15.8	0.0	-7.1	-7.8	2.4	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_114	20.5	0.0	-10.5	9.5	-3.6	5.1	5.1	OK	OK
End plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	145	C LC_119	10.4	0.0	1.9	1.7	5.7	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.3	0.0	0.7	-0.6	-4.1	1.8	0.0	OK	OK
End plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	149	C LC_110	8.6	0.0	-4.5	-3.6	2.1	2.1	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_110	6.0	0.0	-1.9	2.8	-1.7	1.5	0.0	OK	OK
Haunch top plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	342	C LC_119	3.6	0.0	-0.2	-0.2	-2.1	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_119	3.9	0.0	1.5	-1.4	-1.6	1.0	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch web SLN124 ring	◢ 6.0 ◣	271	C LC_119	59.9	0.0	-11.3	-11.8	31.8	14.8	9.0	OK	OK
		◢ 6.0 ◣	271	C LC_121	53.6	0.0	-8.6	8.3	-29.4	13.2	8.5	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_135	37.2	0.0	9.2	-7.1	19.5	9.2	8.9	OK	OK
SLN123b+SLN123a radial (continuous)-w 1	SLN124 ring-w 1	◢ 7.2	208	C LC_135	37.6	0.0	9.3	6.9	-19.9	9.3	8.9	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_122	7.1	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	146	C LC_147	2.6	0.0	-0.6	1.4	0.5	0.6	0.0	OK	OK
Haunch sloping plate SLN124 ring	Haunch top plate SLN124 ring	◢ 6.0 ◣	146	C LC_120	7.1	0.0	-5.2	2.8	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_119	2.7	0.0	-0.6	1.4	-0.5	0.7	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-A_340
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring

Name	SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

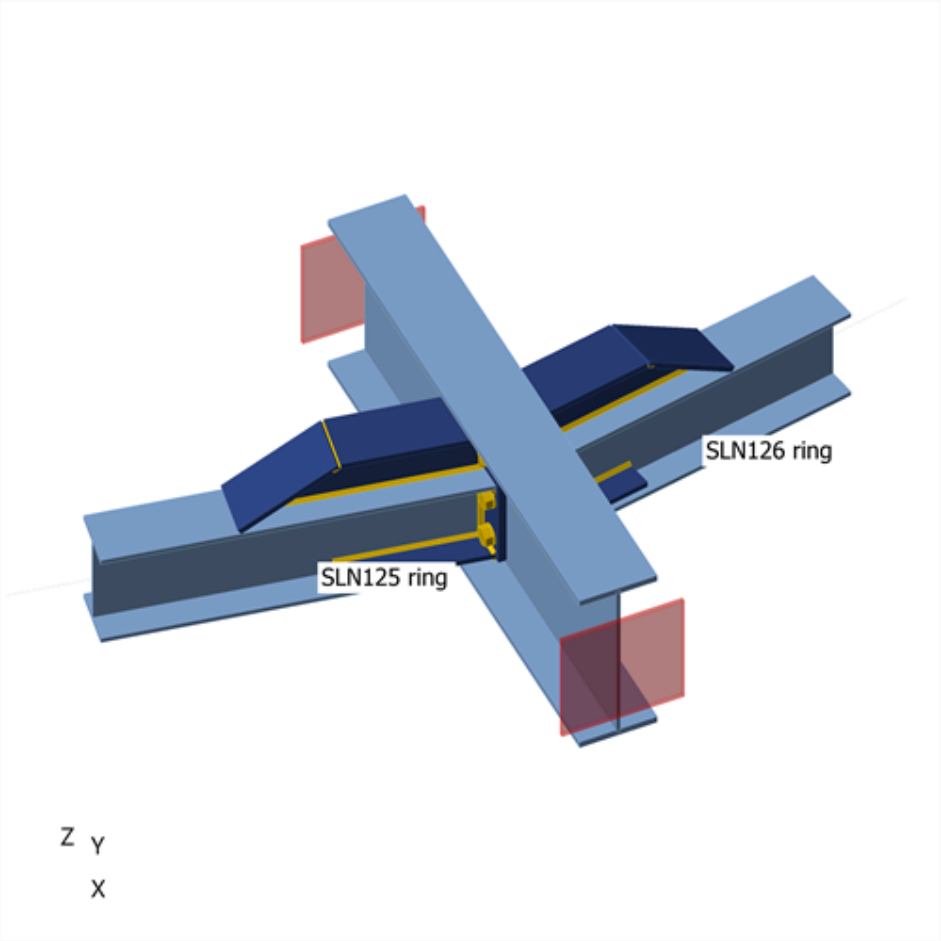
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN127a+SLN127b radial (continuous)	1 - UB457x191x89	-20.0	0.0	0.0	0	0	0
SLN125 ring	2 - UB254x146x43	-120.0	0.0	0.0	0	0	-102
SLN126 ring	2 - UB254x146x43	80.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN127a+SLN127b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN127a+SLN127b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN125 ring / end		Position	0
SLN126 ring / end		Position	0



SLN127a+SLN127b radial (continuous) - SLN125 ring - SLN126 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN125 ring / End	0.8	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
C LC_102	SLN125 ring / End	-2.7	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.7	0.1	-25.6	0.0	0.0	0.1
C LC_103	SLN125 ring / End	-8.5	0.0	-24.0	0.0	0.0	-0.1
	SLN126 ring / End	-8.7	0.0	-24.0	0.0	0.0	0.1
C LC_104	SLN125 ring / End	-4.7	-0.1	-24.0	0.0	0.0	0.0
	SLN126 ring / End	-4.4	0.2	-24.0	0.0	0.0	0.2
C LC_105	SLN125 ring / End	-7.8	0.0	-24.0	0.0	0.0	-0.1
	SLN126 ring / End	-7.3	0.1	-24.0	0.0	0.0	0.1
C LC_106	SLN125 ring / End	-5.0	-0.2	-24.0	0.0	0.0	-0.3
	SLN126 ring / End	-5.9	0.0	-24.0	0.0	0.0	-0.1
C LC_107	SLN125 ring / End	-5.5	0.0	-0.5	0.0	0.0	0.0
	SLN126 ring / End	-5.7	0.0	-0.5	0.0	0.0	0.1
C LC_108	SLN125 ring / End	-0.5	0.2	-0.5	0.0	0.0	0.4
	SLN126 ring / End	0.1	0.1	-0.5	0.0	0.0	0.2
C LC_109	SLN125 ring / End	-4.6	0.1	-0.5	0.0	0.0	0.2
	SLN126 ring / End	-3.9	0.1	-0.5	0.0	0.0	0.2
C LC_110	SLN125 ring / End	-0.3	-0.2	-0.5	0.0	0.0	-0.4
	SLN126 ring / End	-1.8	-0.2	-0.5	0.0	0.0	-0.4
C LC_111	SLN125 ring / End	-9.6	0.0	-16.2	0.0	0.0	-0.1
	SLN126 ring / End	-9.9	0.0	-16.2	0.0	0.0	0.1
C LC_112	SLN125 ring / End	-3.8	0.1	-16.2	0.0	0.0	0.2
	SLN126 ring / End	-3.2	0.2	-16.2	0.0	0.0	0.2
C LC_113	SLN125 ring / End	-8.6	0.1	-16.2	0.0	0.0	0.1
	SLN126 ring / End	-7.7	0.1	-16.2	0.0	0.0	0.2
C LC_114	SLN125 ring / End	-3.9	-0.3	-16.2	0.0	0.0	-0.5
	SLN126 ring / End	-5.4	-0.2	-16.2	0.0	0.0	-0.4
C LC_115	SLN125 ring / End	-5.3	0.0	-1.7	0.0	0.0	0.0
	SLN126 ring / End	-5.5	0.0	-1.7	0.0	0.0	0.1
C LC_116	SLN125 ring / End	-0.2	0.2	-1.7	0.0	0.0	0.3
	SLN126 ring / End	0.4	0.1	-1.7	0.0	0.0	0.2
C LC_117	SLN125 ring / End	-4.5	0.1	-1.7	0.0	0.0	0.1
	SLN126 ring / End	-3.7	0.1	-1.7	0.0	0.0	0.2
C LC_118	SLN125 ring / End	-0.1	-0.2	-1.7	0.0	0.0	-0.4
	SLN126 ring / End	-1.6	-0.2	-1.7	0.0	0.0	-0.4
C LC_119	SLN125 ring / End	-6.3	0.0	-25.6	0.0	0.0	0.0
	SLN126 ring / End	-6.5	0.0	-25.6	0.0	0.0	0.1
C LC_120	SLN125 ring / End	-3.7	-0.1	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-3.6	0.2	-25.6	0.0	0.0	0.2
C LC_121	SLN125 ring / End	-5.4	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-5.2	0.0	-25.6	0.0	0.0	0.0
C LC_122	SLN125 ring / End	-3.7	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-4.0	0.1	-25.6	0.0	0.0	0.1
C LC_123	SLN125 ring / End	-5.3	0.0	-19.4	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN126 ring / End	-5.4	0.1	-19.4	0.0	0.0	0.1
C LC_124	SLN125 ring / End	-1.8	0.0	-19.4	0.0	0.0	0.0
	SLN126 ring / End	-1.6	0.2	-19.4	0.0	0.0	0.3
C LC_125	SLN125 ring / End	-3.9	-0.1	-19.4	0.0	0.0	-0.1
	SLN126 ring / End	-3.7	0.0	-19.4	0.0	0.0	0.0
C LC_126	SLN125 ring / End	-1.4	-0.2	-19.4	0.0	0.0	-0.3
	SLN126 ring / End	-1.7	0.0	-19.4	0.0	0.0	0.0
C LC_127	SLN125 ring / End	-0.9	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	-1.0	0.1	-4.9	0.0	0.0	0.1
C LC_128	SLN125 ring / End	1.8	0.1	-4.9	0.0	0.0	0.1
	SLN126 ring / End	2.0	0.2	-4.9	0.0	0.0	0.3
C LC_129	SLN125 ring / End	0.3	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.4	0.0	-4.9	0.0	0.0	0.0
C LC_130	SLN125 ring / End	2.4	-0.2	-4.9	0.0	0.0	-0.2
	SLN126 ring / End	2.2	0.0	-4.9	0.0	0.0	-0.1
C LC_135	SLN125 ring / End	0.6	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	0.6	0.1	-25.6	0.0	0.0	0.1
C LC_141	SLN125 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
	SLN126 ring / End	0.6	0.0	-4.9	0.0	0.0	0.0
C LC_142	SLN125 ring / End	1.0	-0.1	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	1.0	0.0	-4.9	0.0	0.0	0.1
C LC_143	SLN125 ring / End	0.7	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.0
C LC_144	SLN125 ring / End	0.9	0.0	-4.9	0.0	0.0	-0.1
	SLN126 ring / End	0.8	0.0	-4.9	0.0	0.0	0.1
C LC_145	SLN125 ring / End	-3.4	0.0	-25.6	0.0	0.0	-0.1
	SLN126 ring / End	-3.5	0.0	-25.6	0.0	0.0	0.1
C LC_146	SLN125 ring / End	-2.0	-0.2	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.0	0.2	-25.6	0.0	0.0	0.2
C LC_147	SLN125 ring / End	-3.1	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.9	0.0	-25.6	0.0	0.0	0.1
C LC_148	SLN125 ring / End	-2.3	-0.1	-25.6	0.0	0.0	-0.2
	SLN126 ring / End	-2.6	0.1	-25.6	0.0	0.0	0.2
C LC_151	SLN125 ring / End	-0.8	-0.1	-11.6	0.0	0.0	-0.1
	SLN126 ring / End	-0.9	0.1	-11.6	0.0	0.0	0.1
C LC_152	SLN125 ring / End	-5.7	-0.1	-14.0	0.0	0.0	-0.1
	SLN126 ring / End	-5.7	0.1	-14.1	0.0	0.0	0.1
C LC_153	SLN125 ring / End	-2.9	-0.1	-20.7	0.0	0.0	-0.2
	SLN126 ring / End	-2.9	0.1	-20.8	0.0	0.0	0.1
C LC_154	SLN125 ring / End	-0.8	-0.1	-11.6	0.0	0.0	-0.1
	SLN126 ring / End	-0.8	0.1	-11.6	0.0	0.0	0.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Bolts	0.9 < 100%	OK
Welds	17.9 < 100%	OK
Buckling	Not calculated	

Plates

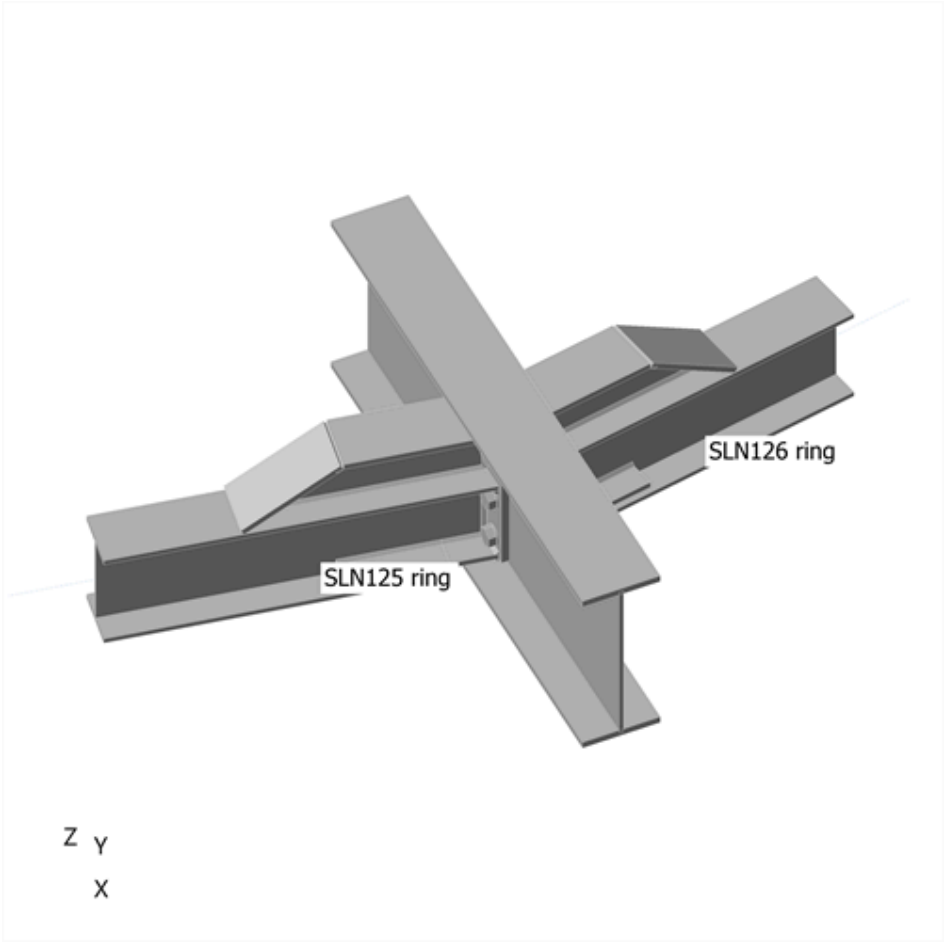
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN127a+SLN127b radial (continuous)-bfl 1	17.7	C LC_121	22.1	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-tfl 1	17.7	C LC_121	18.6	0.0	0.0	OK
SLN127a+SLN127b radial (continuous)-w 1	10.5	C LC_121	26.0	0.0	0.4	OK
SLN125 ring-bfl 1	12.7	C LC_135	46.7	0.0	0.0	OK
SLN125 ring-tfl 1	12.7	C LC_121	48.1	0.0	0.0	OK
SLN125 ring-w 1	7.2	C LC_119	47.3	0.0	0.0	OK
SLN126 ring-bfl 1	12.7	C LC_135	46.6	0.0	0.0	OK
SLN126 ring-tfl 1	12.7	C LC_119	48.1	0.0	0.0	OK
SLN126 ring-w 1	7.2	C LC_119	47.2	0.0	0.0	OK
End plate SLN126 ring	20.0	C LC_114	8.4	0.0	9.0	OK
Bottom flange plate SLN126 ring	15.0	C LC_114	20.5	0.0	0.0	OK
Bottom flange half plate L SLN126 ring	15.0	C LC_135	13.1	0.0	0.0	OK
Bottom flange half plate R SLN126 ring	15.0	C LC_135	13.1	0.0	0.0	OK
Haunch web SLN126 ring	12.0	C LC_119	32.3	0.0	0.0	OK
Haunch top plate SLN126 ring	15.0	C LC_110	8.1	0.0	0.0	OK
Haunch sloping plate SLN126 ring	15.0	C LC_119	14.3	0.0	0.0	OK
End plate SLN125 ring	20.0	C LC_122	6.8	0.0	1.6	OK
Bottom flange plate SLN125 ring	15.0	C LC_114	16.1	0.0	0.0	OK
Bottom flange half plate L SLN125 ring	15.0	C LC_135	13.2	0.0	0.0	OK
Bottom flange half plate R SLN125 ring	15.0	C LC_135	13.1	0.0	0.0	OK
Haunch web SLN125 ring	12.0	C LC_119	31.8	0.0	0.0	OK
Haunch top plate SLN125 ring	15.0	C LC_110	6.3	0.0	0.0	OK
Haunch sloping plate SLN125 ring	15.0	C LC_119	14.3	0.0	0.0	OK

Design data

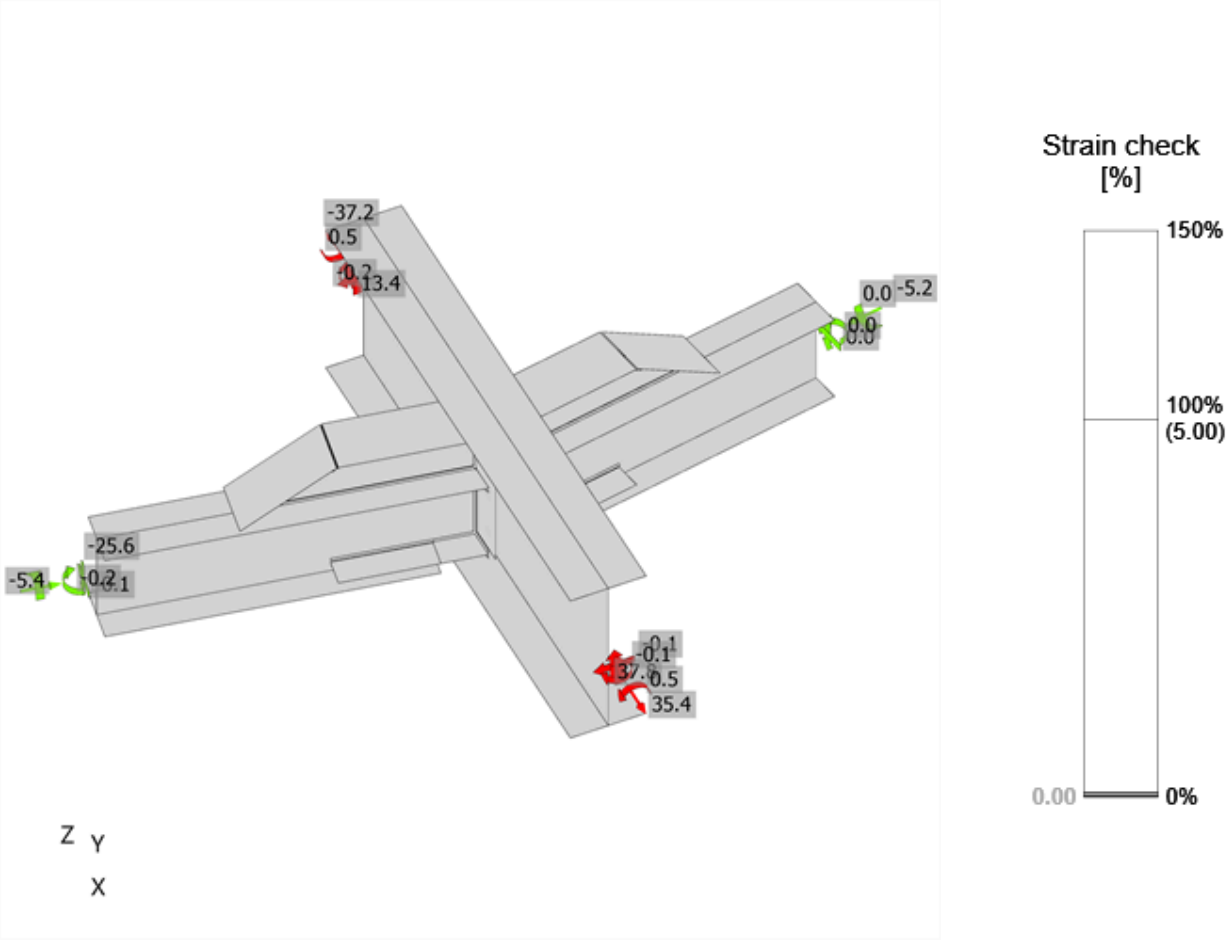
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

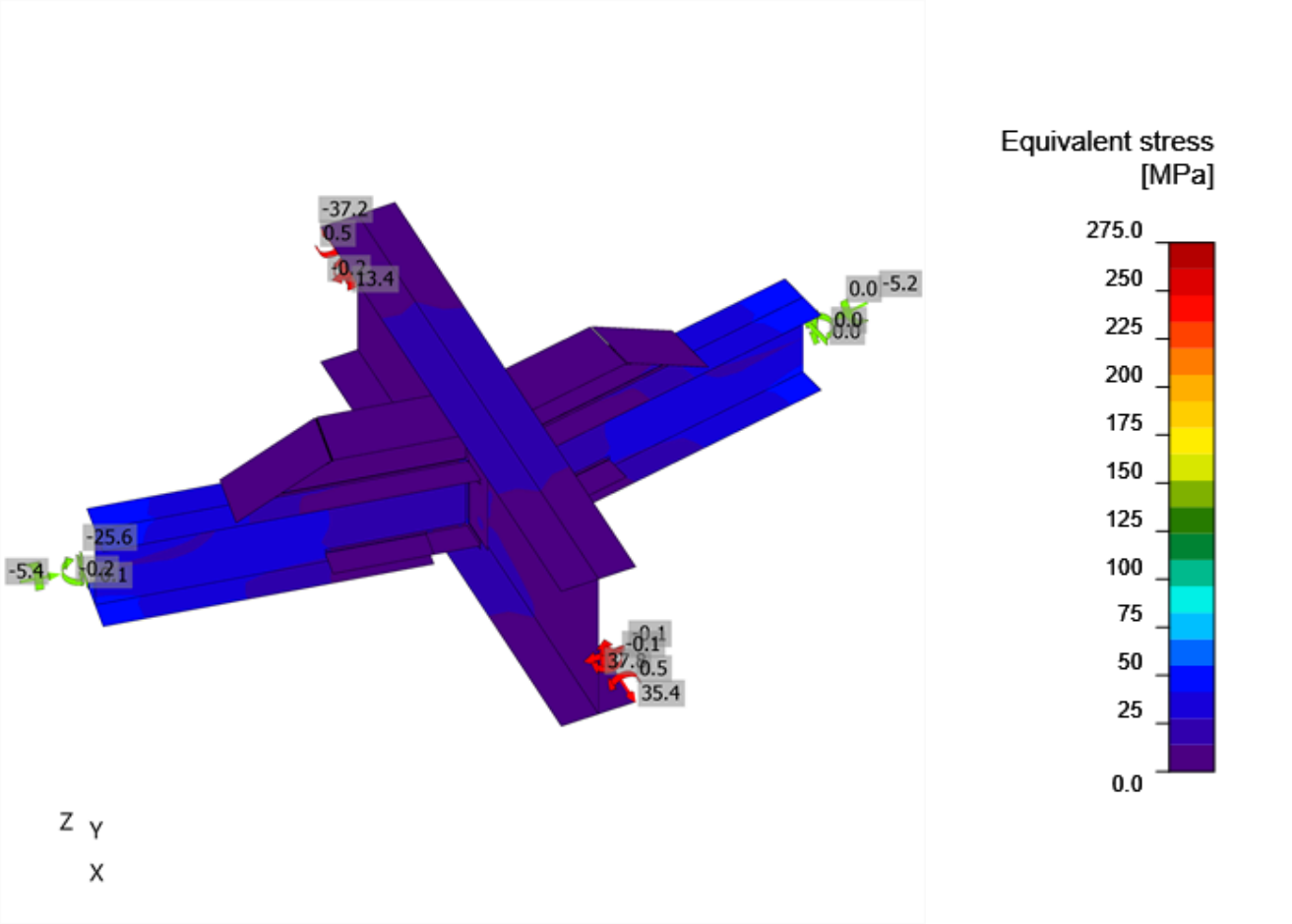
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

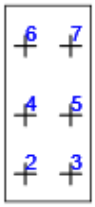


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M30 8.8 - 1	C LC_147	1.1	0.4	176.5	0.3	0.5	0.4	OK	OK
	B3	M30 8.8 - 1	C LC_106	0.7	0.5	176.5	0.2	0.5	0.4	OK	OK
	B4	M30 8.8 - 1	C LC_119	0.0	0.4	213.9	0.0	0.3	0.2	OK	OK
	B5	M30 8.8 - 1	C LC_135	0.0	0.4	213.9	0.0	0.4	0.2	OK	OK
	B6	M30 8.8 - 1	C LC_119	1.7	0.9	229.5	0.5	0.7	0.8	OK	OK
	B7	M30 8.8 - 1	C LC_122	2.2	0.9	229.5	0.7	0.7	0.9	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M30 8.8 - 1	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN126 ring	Bottom flange plate SLN126 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN125 ring	Bottom flange plate SLN125 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-tfl 1	◢ 10.5	149	C LC_119	13.4	0.0	-11.2	-4.3	0.2	3.6	3.6	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-tfl 1	◢ 10.5	149	C LC_119	13.4	0.0	-11.2	-4.3	-0.4	3.6	3.6	OK	OK
End plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	208	C LC_135	11.0	0.0	0.5	-1.3	6.2	2.7	0.0	OK	OK
		◢ 6.0 ◣	208	C LC_135	8.0	0.0	-0.8	-0.6	-4.5	2.0	0.0	OK	OK
Bottom flange plate SLN126 ring	SLN126 ring-w 1	◢ 6.0 ◣	100	C LC_135	21.0	0.0	1.8	1.7	11.9	5.2	5.2	OK	OK
		◢ 6.0 ◣	100	C LC_106	20.6	0.0	1.7	-2.8	-11.5	5.1	5.1	OK	OK
SLN126 ring-w 1	Bottom flange half plate L SLN126 ring	◢ 6.0 ◣	243	C LC_135	17.4	0.0	4.4	4.1	8.8	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.6	0.0	6.1	2.3	-10.5	4.8	4.8	OK	OK
SLN126 ring-w 1	Bottom flange half plate R SLN126 ring	◢ 6.0 ◣	243	C LC_135	16.6	0.0	3.7	4.0	-8.4	4.1	4.1	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.4	0.0	5.9	2.4	11.0	5.0	5.0	OK	OK
SLN126 ring-tfl 1	Haunch web SLN126 ring	◢ 6.0 ◣	572	C LC_119	72.5	0.0	-19.1	-18.9	35.7	17.9	10.2	OK	OK
		◢ 6.0 ◣	572	C LC_119	68.9	0.0	-16.0	16.2	-35.1	17.0	9.8	OK	OK
End plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	208	C LC_135	8.0	0.0	-0.7	0.6	4.6	2.0	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	208	C LC_120	10.8	0.0	0.4	1.4	-6.1	2.7	0.0	OK	OK
Bottom flange plate SLN125 ring	SLN125 ring-w 1	◢ 6.0 ◣	100	C LC_120	20.4	0.0	1.8	2.1	11.5	5.0	5.0	OK	OK
		◢ 6.0 ◣	100	C LC_135	21.2	0.0	1.9	-1.9	-12.0	5.2	5.2	OK	OK
SLN125 ring-w 1	Bottom flange half plate L SLN125 ring	◢ 6.0 ◣	243	C LC_135	16.6	0.0	3.8	4.0	8.4	4.1	4.1	OK	OK
		◢ 6.0 ◣	243	C LC_135	20.4	0.0	6.1	2.3	-11.0	5.0	5.0	OK	OK
SLN125 ring-w 1	Bottom flange half plate R SLN125 ring	◢ 6.0 ◣	243	C LC_135	17.5	0.0	4.3	4.1	-8.9	4.3	4.3	OK	OK
		◢ 6.0 ◣	243	C LC_135	19.6	0.0	6.0	2.3	10.5	4.9	4.9	OK	OK
SLN125 ring-tfl 1	Haunch web SLN125 ring	◢ 6.0 ◣	572	C LC_119	69.7	0.0	-16.5	-16.8	35.3	17.2	9.9	OK	OK
		◢ 6.0 ◣	572	C LC_119	71.8	0.0	-18.6	18.4	-35.6	17.7	10.1	OK	OK
End plate SLN126 ring	Bottom flange plate SLN126 ring	◢ 6.0 ◣	149	C LC_114	23.5	0.0	-6.6	-9.6	8.8	5.8	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_114	29.4	0.0	-12.7	9.6	-11.9	7.3	6.0	OK	OK
End plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	145	C LC_120	7.0	0.0	1.1	0.9	3.8	1.7	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_147	10.5	0.0	1.5	-1.7	-5.8	2.6	0.0	OK	OK
End plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	149	C LC_110	13.6	0.0	-6.7	-4.7	5.0	3.4	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_110	10.0	0.0	-2.8	4.8	-2.8	2.5	0.0	OK	OK
Haunch top plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	342	C LC_114	3.8	0.0	1.2	1.1	1.8	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	3.9	0.0	1.5	-1.4	-1.6	1.0	0.0	OK	OK
Haunch sloping plate SLN126 ring	Haunch web SLN126 ring	◢ 6.0 ◣	271	C LC_119	59.7	0.0	-11.2	-11.3	31.9	14.7	9.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	271	C LC_119	53.4	0.0	-8.5	8.4	-29.3	13.2	8.4	OK	OK
End plate SLN125 ring	Bottom flange plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	17.8	0.0	-7.7	-8.8	2.7	4.4	4.4	OK	OK
		◢ 6.0 ◣	149	C LC_114	21.9	0.0	-11.9	9.8	-4.1	5.4	5.4	OK	OK
End plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	145	C LC_148	10.5	0.0	1.4	1.7	5.8	2.6	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_122	7.0	0.0	1.1	-0.9	-3.9	1.7	0.0	OK	OK
End plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	149	C LC_110	10.9	0.0	-5.8	-4.6	2.7	2.7	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_114	8.0	0.0	-2.9	3.5	-2.6	2.0	0.0	OK	OK
Haunch top plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	342	C LC_106	4.1	0.0	1.5	1.4	1.7	1.0	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_146	3.5	0.0	-0.3	0.3	2.0	0.9	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch web SLN125 ring	◢ 6.0 ◣	271	C LC_119	54.6	0.0	-9.0	-8.8	29.8	13.5	8.5	OK	OK
		◢ 6.0 ◣	271	C LC_121	58.7	0.0	-10.9	11.4	-31.3	14.5	8.9	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN126 ring-w 1	◢ 7.2	208	C LC_135	36.1	0.0	8.2	-8.4	18.5	8.9	8.8	OK	OK
SLN127a+SLN127b radial (continuous)-w 1	SLN125 ring-w 1	◢ 7.2	208	C LC_135	36.6	0.0	8.6	7.7	-19.0	9.0	8.8	OK	OK
Haunch sloping plate SLN126 ring	Haunch top plate SLN126 ring	◢ 6.0 ◣	146	C LC_119	7.1	0.0	-5.1	2.9	-0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	2.8	0.0	-0.6	1.6	0.0	0.7	0.0	OK	OK
Haunch sloping plate SLN125 ring	Haunch top plate SLN125 ring	◢ 6.0 ◣	146	C LC_121	7.1	0.0	-5.1	2.8	0.3	1.9	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_114	3.3	0.0	-0.6	1.9	-0.2	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_040
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring

Name	SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

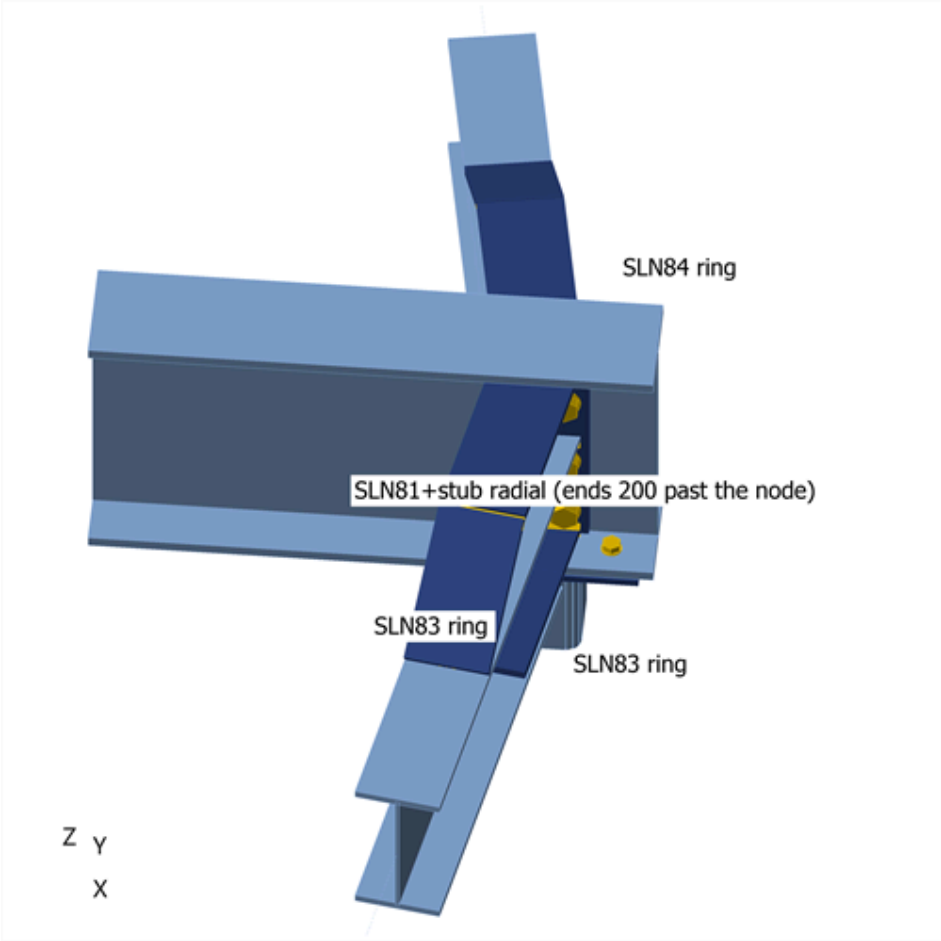
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN81+stub radial (ends 200 past the node)	1 - UB457x191x89	40.0	0.0	0.0	200	0	0
SLN121 column below	2 - SHS150x150x10	0.0	90.0	-140.0	0	0	0
SLN83 ring	3 - UB254x146x43	-60.0	0.0	0.0	0	0	-102
SLN84 ring	3 - UB254x146x43	140.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN81+stub radial (ends 200 past the node) / begin		Position	0
SLN121 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN83 ring / end		Position	0
SLN84 ring / end		Position	0



SLN81+stub radial (ends 200 past the node) - SLN121 column below - SLN83 ring - SLN84 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.0	-10.7	0.0	-1.1	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.8	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.8	0.1	-2.2	0.0	0.0	0.1
C LC_102	SLN81+stub radial (ends 200 past the node) / Begin	1.5	0.1	-45.7	0.0	-5.2	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.8	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-2.8	0.8	-8.8	0.0	0.0	0.6
C LC_103	SLN81+stub radial (ends 200 past the node) / Begin	-0.5	0.1	-42.8	0.0	-5.0	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-8.9	0.0	-8.3	0.0	0.0	-0.1
	SLN84 ring / End	-8.8	0.9	-8.3	0.0	0.0	0.7
C LC_104	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.2	-43.1	0.0	-4.6	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.7	-0.1	-8.3	0.0	0.0	-0.1
	SLN84 ring / End	-4.5	0.8	-8.3	0.0	0.0	0.7
C LC_105	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.0	-43.1	0.0	-4.8	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.9	-0.1	-8.3	0.0	0.0	-0.3
	SLN84 ring / End	-6.0	0.9	-8.3	0.0	0.0	0.7
C LC_106	SLN81+stub radial (ends 200 past the node) / Begin	-0.3	0.2	-42.8	0.0	-4.7	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-7.2	-0.1	-8.3	0.0	0.0	-0.1
	SLN84 ring / End	-7.3	0.8	-8.3	0.0	0.0	0.6
C LC_107	SLN81+stub radial (ends 200 past the node) / Begin	-1.7	0.1	-2.2	0.0	-0.3	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.0	0.0	-0.6	0.0	0.0	0.1
	SLN84 ring / End	-6.1	0.3	-0.6	0.0	0.0	0.3
C LC_108	SLN81+stub radial (ends 200 past the node) / Begin	-0.4	0.1	-2.7	0.0	0.3	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.0	0.0	-0.6	0.0	0.0	0.0
	SLN84 ring / End	0.5	0.0	-0.6	0.0	0.0	0.1
C LC_109	SLN81+stub radial (ends 200 past the node) / Begin	-0.7	0.0	-2.8	0.0	-0.2	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.3	0.0	-0.6	0.0	0.0	-0.1
	SLN84 ring / End	-2.7	0.2	-0.6	0.0	0.0	0.1
C LC_110	SLN81+stub radial (ends 200 past the node) / Begin	-1.3	0.0	-2.1	0.0	0.4	0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.7	-0.1	-0.6	0.0	0.0	-0.1
	SLN84 ring / End	-2.8	0.0	-0.6	0.0	0.0	0.0
C LC_111	SLN81+stub radial (ends 200 past the node) / Begin	-1.6	0.1	-29.5	0.0	-3.5	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-10.3	0.0	-5.8	0.0	0.0	-0.1
	SLN84 ring / End	-10.3	0.8	-5.8	0.0	0.0	0.7
C LC_112	SLN81+stub radial (ends 200 past the node) / Begin	0.1	0.2	-30.0	0.0	-2.8	-0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.4	0.0	-5.8	0.0	0.0	-0.1
	SLN84 ring / End	-3.0	0.6	-5.8	0.0	0.0	0.6
C LC_113	SLN81+stub radial (ends 200 past the node) / Begin	-0.3	0.0	-30.1	0.0	-3.3	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.8	-0.1	-5.8	0.0	0.0	-0.2
	SLN84 ring / End	-6.1	0.7	-5.8	0.0	0.0	0.5
C LC_114	SLN81+stub radial (ends 200 past the node) / Begin	-1.2	0.1	-29.4	0.0	-2.8	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.9	-0.1	-5.8	0.0	0.0	-0.2
	SLN84 ring / End	-7.0	0.5	-5.8	0.0	0.0	0.4
C LC_115	SLN81+stub radial (ends 200 past the node) / Begin	-1.6	0.1	-5.0	0.0	-0.6	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.9	0.0	-1.1	0.0	0.0	0.1
	SLN84 ring / End	-6.0	0.3	-1.1	0.0	0.0	0.3
C LC_116	SLN81+stub radial (ends 200 past the node) / Begin	-0.2	0.1	-5.5	0.0	0.0	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.3	0.0	-1.1	0.0	0.0	0.0
	SLN84 ring / End	0.7	0.1	-1.1	0.0	0.0	0.1
C LC_117	SLN81+stub radial (ends 200 past the node) / Begin	-0.5	0.0	-5.6	0.0	-0.5	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.0	0.0	-1.1	0.0	0.0	-0.1
	SLN84 ring / End	-2.5	0.2	-1.1	0.0	0.0	0.2
C LC_118	SLN81+stub radial (ends 200 past the node) / Begin	-1.2	0.0	-4.9	0.0	0.1	0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.6	-0.1	-1.1	0.0	0.0	-0.1
	SLN84 ring / End	-2.7	0.0	-1.1	0.0	0.0	0.0
C LC_119	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.1	-45.9	0.0	-5.8	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-6.5	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-6.5	0.9	-8.8	0.0	0.0	0.7
C LC_120	SLN81+stub radial (ends 200 past the node) / Begin	1.0	0.2	-45.5	0.0	-4.7	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.7	0.0	-8.9	0.0	0.0	0.0
	SLN84 ring / End	-3.5	0.8	-8.8	0.0	0.0	0.7
C LC_121	SLN81+stub radial (ends 200 past the node) / Begin	0.9	0.0	-45.6	0.0	-4.9	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-4.6	-0.1	-8.9	0.0	0.0	-0.3
	SLN84 ring / End	-4.7	0.8	-8.8	0.0	0.0	0.6
C LC_122	SLN81+stub radial (ends 200 past the node) / Begin	0.8	0.2	-45.7	0.0	-5.3	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.0	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-5.1	0.8	-8.8	0.0	0.0	0.6
C LC_123	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.1	-35.5	0.0	-5.0	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.5	0.0	-6.9	0.0	0.0	-0.1
	SLN84 ring / End	-5.6	0.8	-6.8	0.0	0.0	0.7

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN81+stub radial (ends 200 past the node) / Begin	0.8	0.2	-34.9	0.0	-3.1	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.4	0.1	-6.9	0.0	0.0	0.1
	SLN84 ring / End	-1.0	0.6	-6.8	0.0	0.0	0.5
C LC_125	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.0	-35.1	0.0	-3.6	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.2	-0.1	-6.9	0.0	0.0	-0.2
	SLN84 ring / End	-3.5	0.6	-6.8	0.0	0.0	0.4
C LC_126	SLN81+stub radial (ends 200 past the node) / Begin	1.0	0.1	-35.3	0.0	-4.2	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.5	0.0	-6.9	0.0	0.0	-0.1
	SLN84 ring / End	-2.6	0.5	-6.8	0.0	0.0	0.4
C LC_127	SLN81+stub radial (ends 200 past the node) / Begin	0.8	0.1	-11.1	0.0	-2.1	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-1.1	0.1	-2.2	0.0	0.0	0.1
	SLN84 ring / End	-1.3	0.3	-2.2	0.0	0.0	0.3
C LC_128	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.1	-10.5	0.0	-0.3	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	2.3	0.1	-2.2	0.0	0.0	0.2
	SLN84 ring / End	2.7	0.1	-2.2	0.0	0.0	0.1
C LC_129	SLN81+stub radial (ends 200 past the node) / Begin	0.5	0.0	-10.6	0.0	-0.8	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
	SLN84 ring / End	0.2	0.1	-2.2	0.0	0.0	0.1
C LC_130	SLN81+stub radial (ends 200 past the node) / Begin	1.0	0.0	-10.9	0.0	-1.3	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	1.9	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	1.8	0.1	-2.2	0.0	0.0	0.0
C LC_135	SLN81+stub radial (ends 200 past the node) / Begin	2.2	0.1	-45.7	0.0	-4.9	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.6	0.0	-8.9	0.0	0.0	0.0
	SLN84 ring / End	0.6	0.5	-8.8	0.0	0.0	0.4
C LC_141	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-10.7	0.0	-1.2	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.6	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.1
C LC_142	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.0	-10.7	0.0	-1.1	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.9	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.9	0.1	-2.2	0.0	0.0	0.1
C LC_143	SLN81+stub radial (ends 200 past the node) / Begin	0.7	0.0	-10.7	0.0	-1.1	0.0
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.9	0.0	-2.2	0.0	0.0	0.0
	SLN84 ring / End	0.9	0.1	-2.2	0.0	0.0	0.1
C LC_144	SLN81+stub radial (ends 200 past the node) / Begin	0.6	0.0	-10.7	0.0	-1.2	-0.1
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	0.7	0.0	-2.2	0.0	0.0	0.0

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.1
C LC_145	SLN81+stub radial (ends 200 past the node) / Begin	1.3	0.0	-45.7	0.0	-5.3	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.5	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-3.4	0.7	-8.8	0.0	0.0	0.6
C LC_146	SLN81+stub radial (ends 200 past the node) / Begin	1.8	0.2	-45.7	0.0	-5.1	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.2	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-2.2	0.8	-8.8	0.0	0.0	0.7
C LC_147	SLN81+stub radial (ends 200 past the node) / Begin	1.8	0.0	-45.7	0.0	-5.1	-0.3
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-2.3	-0.1	-8.9	0.0	0.0	-0.2
	SLN84 ring / End	-2.1	0.8	-8.8	0.0	0.0	0.6
C LC_148	SLN81+stub radial (ends 200 past the node) / Begin	1.3	0.2	-45.7	0.0	-5.3	-0.5
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.4	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-3.5	0.8	-8.8	0.0	0.0	0.6
C LC_151	SLN81+stub radial (ends 200 past the node) / Begin	3.5	0.1	-46.2	0.0	-7.5	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-0.9	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-0.9	0.5	-8.8	0.0	0.0	0.4
C LC_152	SLN81+stub radial (ends 200 past the node) / Begin	-3.6	0.1	-10.0	0.0	2.7	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-5.8	-0.1	-2.2	0.0	0.0	-0.2
	SLN84 ring / End	-5.8	0.9	-2.2	0.0	0.0	0.7
C LC_153	SLN81+stub radial (ends 200 past the node) / Begin	1.5	0.1	-45.0	0.0	-5.0	-0.4
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-3.0	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-3.0	0.8	-8.8	0.0	0.0	0.6
C LC_154	SLN81+stub radial (ends 200 past the node) / Begin	3.6	0.1	-47.0	0.0	-7.8	-0.2
	SLN81+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN83 ring / End	-0.9	0.0	-8.9	0.0	0.0	-0.1
	SLN84 ring / End	-0.9	0.5	-8.8	0.0	0.0	0.4

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	8.7 < 100%	OK
Welds	37.0 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN81+stub radial (ends 200 past the node)-bfl 1	17.7	C LC_121	49.6	0.0	19.3	OK
SLN81+stub radial (ends 200 past the node)-tfl 1	17.7	C LC_120	22.2	0.0	0.0	OK
SLN81+stub radial (ends 200 past the node)-w 1	10.5	C LC_120	66.6	0.0	19.7	OK
SLN121 column below	10.0	C LC_152	90.2	0.0	0.0	OK
SLN83 ring-bfl 1	12.7	C LC_135	16.2	0.0	0.0	OK
SLN83 ring-tfl 1	12.7	C LC_119	18.4	0.0	0.0	OK
SLN83 ring-w 1	7.2	C LC_146	17.1	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_147	17.1	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_121	17.7	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_119	16.9	0.0	0.0	OK
End plate column below SLN121 column below	12.0	C LC_152	86.1	0.0	12.7	OK
End plate SLN84 ring	20.0	C LC_146	4.8	0.0	8.2	OK
Bottom flange plate SLN84 ring	15.0	C LC_104	16.0	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_120	6.5	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_111	6.9	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_119	12.3	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_120	5.0	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_119	5.2	0.0	0.0	OK
End plate SLN83 ring	20.0	C LC_120	7.4	0.0	4.2	OK
Bottom flange plate SLN83 ring	15.0	C LC_123	12.0	0.0	0.0	OK
Bottom flange half plate L SLN83 ring	15.0	C LC_135	4.8	0.0	0.0	OK
Bottom flange half plate R SLN83 ring	15.0	C LC_135	5.0	0.0	0.0	OK
Haunch web SLN83 ring	12.0	C LC_119	12.7	0.0	0.0	OK
Haunch top plate SLN83 ring	15.0	C LC_120	5.4	0.0	0.0	OK
Haunch sloping plate SLN83 ring	15.0	C LC_119	5.3	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

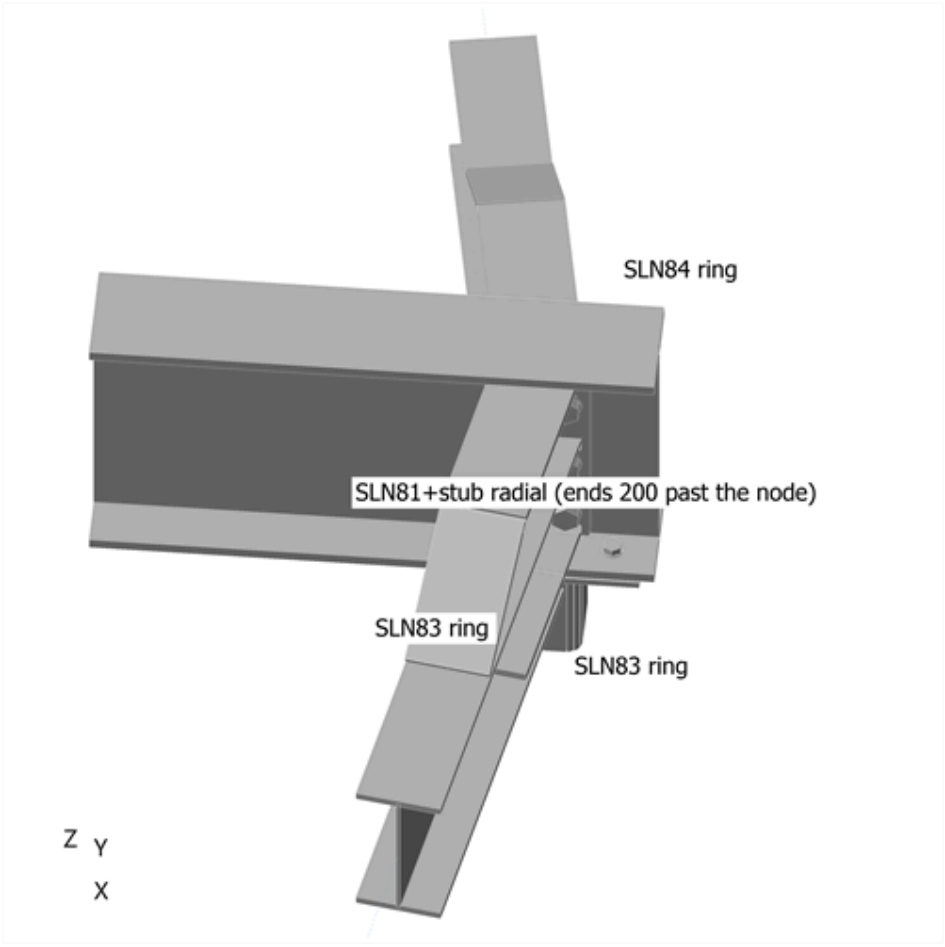
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

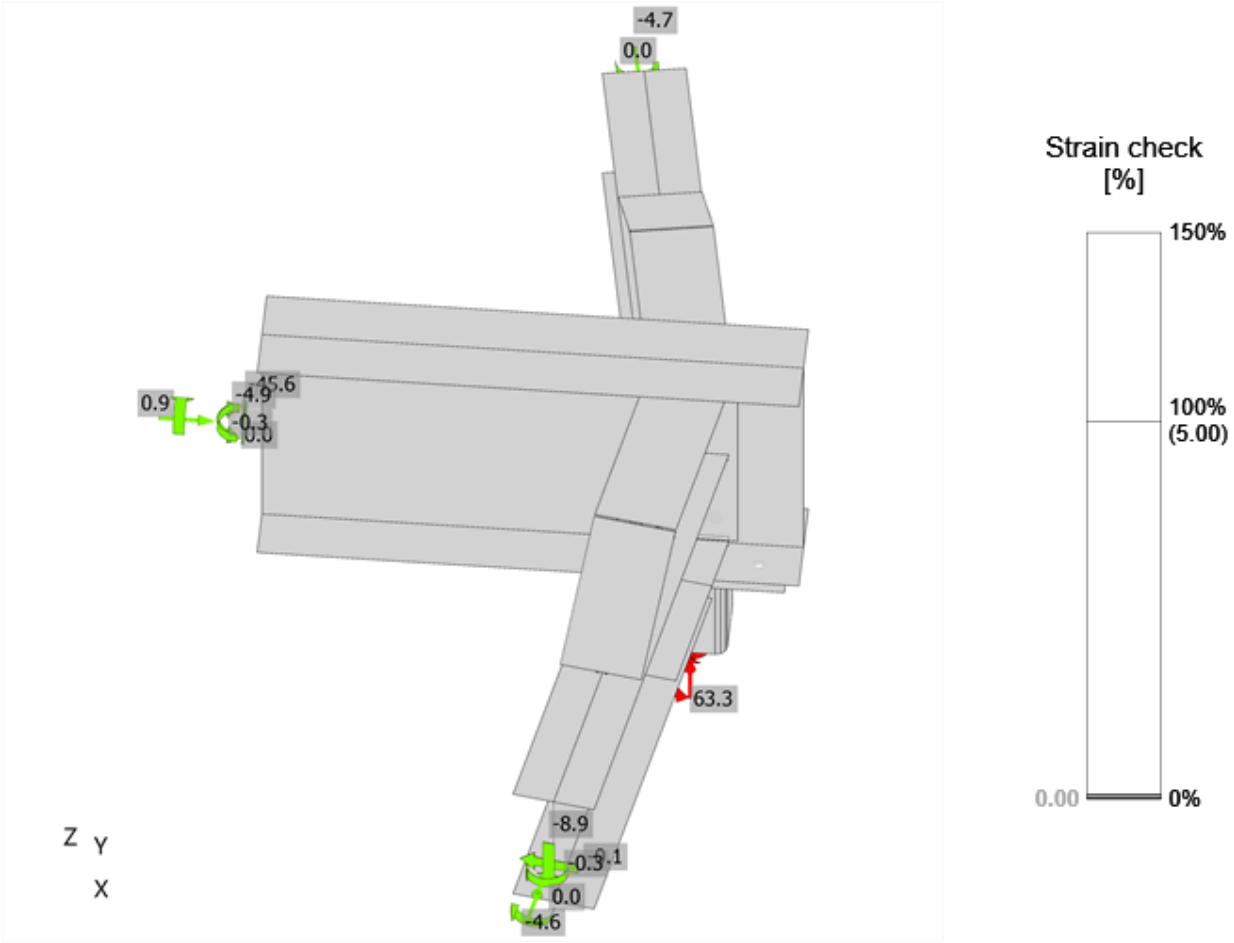
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN121 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

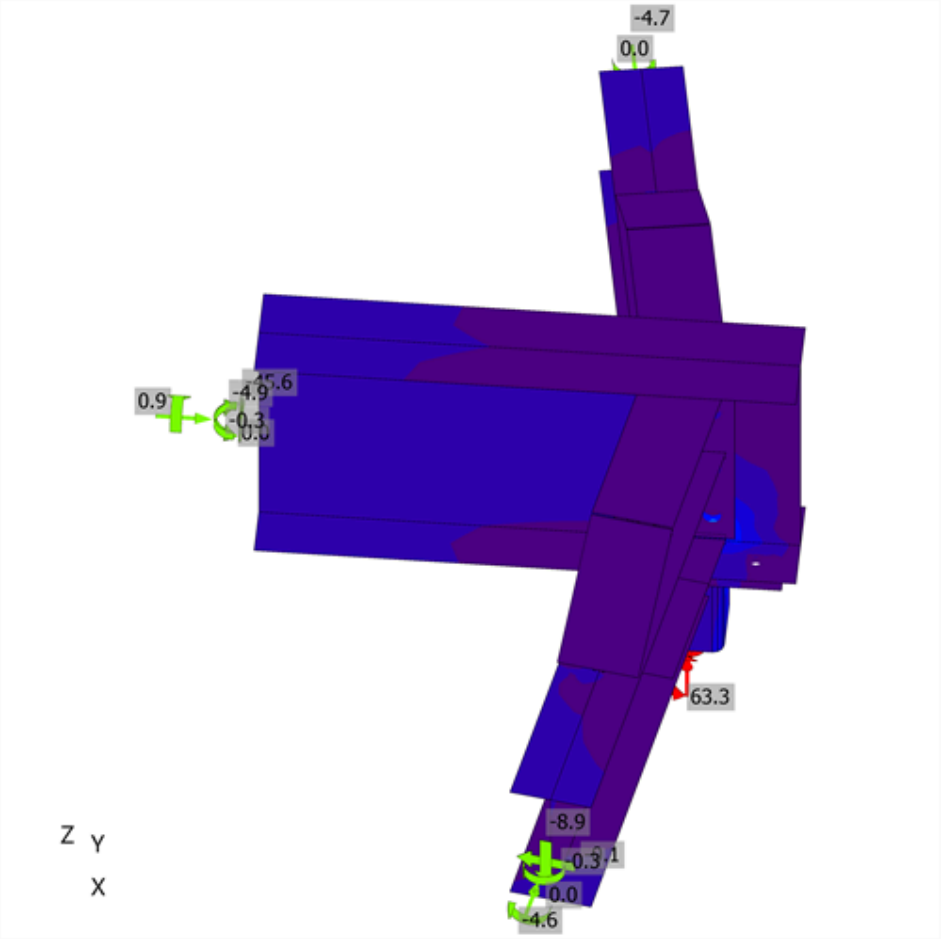
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

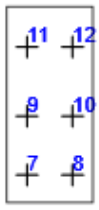


Strain check, C LC_121



Equivalent stress, C LC_121

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	11.7	0.8	156.4	8.3	0.9	6.8	OK	OK
	B3	M20 8.8 - 1	C LC_154	0.6	0.8	156.4	0.4	0.8	1.1	OK	OK
	B4	M20 8.8 - 1	C LC_120	0.7	1.0	172.0	0.5	1.0	1.4	OK	OK
	B5	M20 8.8 - 1	C LC_152	12.2	1.1	156.4	8.7	1.2	7.3	OK	OK
	B7	M30 8.8 - 2	C LC_120	0.1	0.7	213.9	0.0	0.6	0.3	OK	OK
	B8	M30 8.8 - 2	C LC_120	1.6	1.6	176.5	0.5	1.7	1.1	OK	OK
	B9	M30 8.8 - 2	C LC_120	0.2	0.7	229.5	0.1	0.5	0.4	OK	OK
	B10	M30 8.8 - 2	C LC_135	0.1	0.6	213.9	0.0	0.5	0.3	OK	OK
	B11	M30 8.8 - 2	C LC_120	0.7	0.8	229.5	0.2	0.6	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_120	1.2	0.3	142.3	0.4	0.2	0.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN83 ring	Bottom flange plate SLN83 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN83 ring-tfl 1	◢ 10.5	149	C LC_105	8.3	0.0	-3.6	4.3	0.6	2.1	0.0	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN84 ring-tfl 1	◢ 10.5	149	C LC_103	11.0	0.0	-5.1	5.6	0.3	2.7	2.7	OK	OK
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	207	C LC_154	20.4	0.0	-1.7	0.3	11.7	5.0	5.0	OK	OK
		◢ 6.0 ◣	207	C LC_154	15.4	0.0	0.5	1.0	-8.8	3.8	3.8	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_154	11.4	0.0	5.9	5.2	2.2	2.8	2.8	OK	OK
		◢ 6.0 ◣	100	C LC_135	13.4	0.0	5.6	-6.6	-2.3	3.3	3.3	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_146	8.6	0.0	1.5	2.0	4.5	2.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_120	12.1	0.0	3.4	-2.9	-6.1	3.0	3.0	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	243	C LC_152	5.5	0.0	-0.8	-1.4	2.8	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_152	9.4	0.0	-3.0	2.4	-4.6	2.3	0.0	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_119	24.2	0.0	-8.3	-6.4	11.5	6.0	5.8	OK	OK
		◢ 6.0 ◣	572	C LC_119	28.1	0.0	-4.7	6.6	-14.5	6.9	6.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	207	C LC_154	14.8	0.0	0.8	-0.9	8.5	3.6	3.6	OK	OK
		◢ 6.0 ◣	207	C LC_119	22.4	0.0	-2.7	-1.1	-12.8	5.5	5.5	OK	OK
Bottom flange plate SLN83 ring	SLN83 ring-w 1	◢ 6.0 ◣	100	C LC_146	14.8	0.0	5.9	7.7	1.4	3.7	3.7	OK	OK
		◢ 6.0 ◣	100	C LC_154	11.1	0.0	6.1	-5.1	-1.8	2.7	2.7	OK	OK
SLN83 ring-w 1	Bottom flange half plate L SLN83 ring	◢ 6.0 ◣	243	C LC_135	6.8	0.0	1.7	1.6	3.4	1.7	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.8	0.0	2.0	1.1	-3.6	1.7	0.0	OK	OK
SLN83 ring-w 1	Bottom flange half plate R SLN83 ring	◢ 6.0 ◣	243	C LC_135	5.1	0.0	1.1	1.2	-2.6	1.3	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	8.0	0.0	2.5	0.9	4.3	2.0	0.0	OK	OK
SLN83 ring-tfl 1	Haunch web SLN83 ring	◢ 6.0 ◣	572	C LC_119	29.0	0.0	-7.3	-8.1	14.1	7.2	6.1	OK	OK
		◢ 6.0 ◣	572	C LC_119	23.2	0.0	-5.7	4.9	-12.0	5.7	5.7	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	23.2	0.0	-11.2	-11.4	-2.7	5.7	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_104	25.7	0.0	-13.2	12.3	3.3	6.4	6.1	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_154	3.9	0.0	-0.2	-0.4	2.2	1.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	8.0	0.0	-0.8	0.8	-4.5	2.0	0.0	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_120	10.4	0.0	-5.1	-4.5	-2.6	2.6	2.6	OK	OK
		◢ 6.0 ◣	149	C LC_120	9.1	0.0	-3.4	3.9	2.9	2.2	0.0	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_119	2.6	0.0	-0.7	-1.3	-0.7	0.6	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	3.4	0.0	1.1	-0.9	-1.6	0.8	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_119	21.9	0.0	-4.1	-6.7	10.4	5.4	5.4	OK	OK
		◢ 6.0 ◣	271	C LC_119	20.7	0.0	-3.0	0.3	-11.8	5.1	5.1	OK	OK
End plate SLN83 ring	Bottom flange plate SLN83 ring	◢ 6.0 ◣	149	C LC_124	15.8	0.0	-4.4	-5.0	-7.2	3.9	3.9	OK	OK
		◢ 6.0 ◣	149	C LC_124	18.6	0.0	-6.4	5.8	8.3	4.6	4.6	OK	OK
End plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	145	C LC_121	8.8	0.0	-0.7	-0.5	5.0	2.2	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_154	3.3	0.0	-0.2	0.4	-1.9	0.8	0.0	OK	OK
End plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	149	C LC_120	8.7	0.0	-4.4	-2.5	-3.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_124	5.6	0.0	-1.2	2.8	1.4	1.4	0.0	OK	OK
Haunch top plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	342	C LC_121	1.5	0.0	0.7	0.6	0.5	0.4	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_120	1.9	0.0	0.5	-0.4	-1.0	0.5	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch web SLN83 ring	◢ 6.0 ◣	271	C LC_119	23.7	0.0	-4.7	-3.6	12.9	5.9	5.8	OK	OK
		◢ 6.0 ◣	271	C LC_151	17.6	0.0	-2.8	3.2	-9.5	4.4	4.4	OK	OK
End plate column below SLN121 column below	SLN121 column below	◢ 4.2 ◣	521	C LC_135	138.2	0.0	-68.9	-69.1	-2.2	34.1	18.2	OK	OK
		◢ 4.2 ◣	521	C LC_120	149.9	0.0	-75.0	74.9	1.8	37.0	20.4	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_154	60.8	0.0	8.4	8.3	-33.7	15.0	9.3	OK	OK
SLN81+stub radial (ends 200 past the node)-w 1	SLN83 ring-w 1	◢ 7.2	208	C LC_154	63.3	0.0	8.0	-6.0	35.7	15.6	9.6	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_120	4.2	0.0	-3.9	-0.9	0.1	1.3	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_120	3.1	0.0	-0.6	1.7	0.4	0.8	0.0	OK	OK
Haunch sloping plate SLN83 ring	Haunch top plate SLN83 ring	◢ 6.0 ◣	146	C LC_120	3.0	0.0	-1.9	1.3	-0.1	0.7	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_124	1.3	0.0	-0.2	0.7	0.2	0.3	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_060
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring

Name	SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

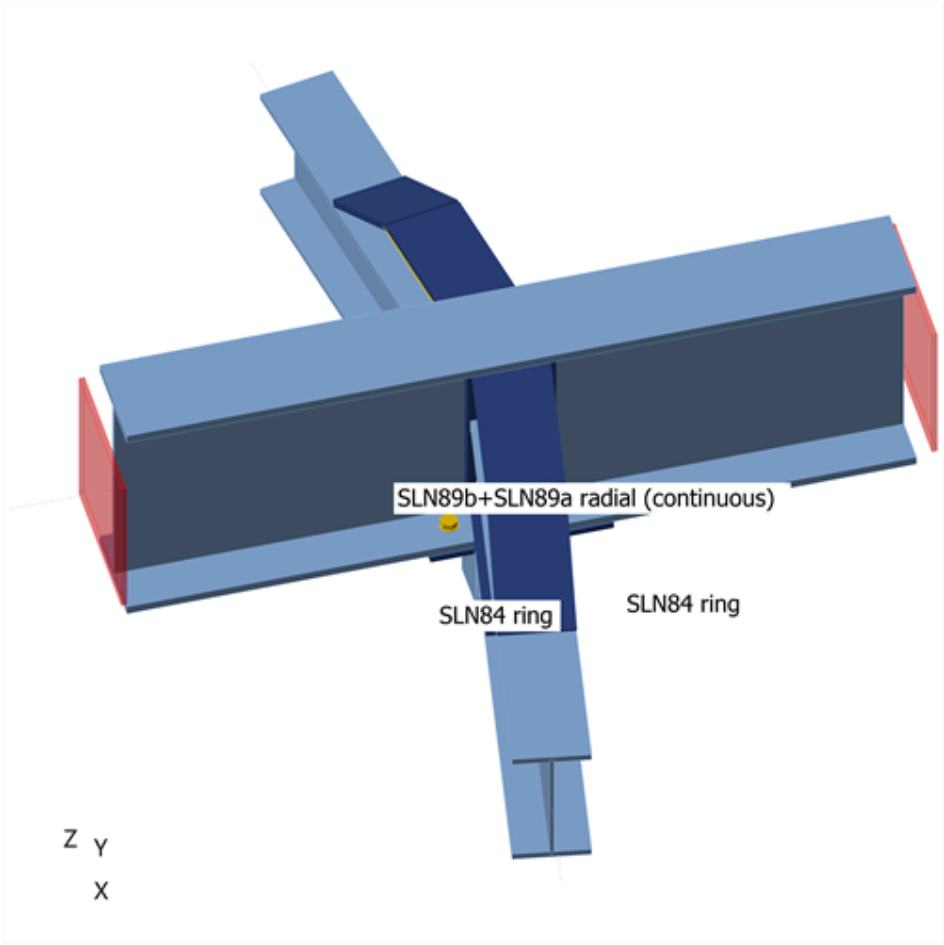
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN89b+SLN89a radial (continuous)	1 - UB457x191x89	60.0	0.0	0.0	0	0	0
SLN131 column below	2 - SHS150x150x10	0.0	90.0	-120.0	0	0	0
SLN84 ring	3 - UB254x146x43	-40.0	0.0	0.0	0	0	-102
SLN88 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN89b+SLN89a radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN89b+SLN89a radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN131 column below / begin		Position	0
SLN84 ring / end		Position	0
SLN88 ring / end		Position	0



SLN89b+SLN89a radial (continuous) - SLN131 column below - SLN84 ring - SLN88 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN131 column below / Begin	37.4	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.8	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.9	0.2	-4.9	0.0	0.0	0.4
C LC_102	SLN131 column below / Begin	154.1	-0.1	-0.1	-0.5	-0.6	0.3
	SLN84 ring / End	-2.8	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	19.2	1.0	-25.8	0.0	0.0	2.0
C LC_103	SLN131 column below / Begin	146.1	-0.2	-0.2	-0.6	-0.7	0.3
	SLN84 ring / End	-8.8	0.9	-8.4	0.0	0.0	2.0
	SLN88 ring / End	16.1	1.0	-24.2	0.0	0.0	2.1
C LC_104	SLN131 column below / Begin	144.1	0.2	0.0	-0.4	-0.4	0.3
	SLN84 ring / End	-4.5	0.8	-8.4	0.0	0.0	1.7
	SLN88 ring / End	13.6	0.9	-24.2	0.0	0.0	1.8
C LC_105	SLN131 column below / Begin	142.1	-0.2	0.1	-0.6	-0.4	0.3
	SLN84 ring / End	-6.0	0.8	-8.4	0.0	0.0	1.8
	SLN88 ring / End	15.3	1.1	-24.2	0.0	0.0	2.0
C LC_106	SLN131 column below / Begin	147.9	-0.2	-0.2	-0.5	-0.7	0.3
	SLN84 ring / End	-7.3	0.8	-8.4	0.0	0.0	1.7
	SLN88 ring / End	14.5	0.8	-24.2	0.0	0.0	1.8
C LC_107	SLN131 column below / Begin	13.3	-0.1	0.0	-0.1	0.1	0.0
	SLN84 ring / End	-6.1	0.3	-0.6	0.0	0.0	0.7
	SLN88 ring / End	2.8	0.3	-0.5	0.0	0.0	0.7
C LC_108	SLN131 column below / Begin	9.7	0.6	0.4	0.2	0.5	0.0
	SLN84 ring / End	0.5	0.0	-0.6	0.0	0.0	-0.1
	SLN88 ring / End	-3.8	0.0	-0.5	0.0	0.0	-0.2
C LC_109	SLN131 column below / Begin	5.9	-0.1	0.4	-0.2	0.5	0.0
	SLN84 ring / End	-2.7	0.2	-0.6	0.0	0.0	0.4
	SLN88 ring / End	-1.0	0.2	-0.5	0.0	0.0	0.3
C LC_110	SLN131 column below / Begin	16.6	-0.1	0.0	-0.1	0.1	0.0
	SLN84 ring / End	-2.8	0.0	-0.6	0.0	0.0	0.0
	SLN88 ring / End	0.2	-0.1	-0.5	0.0	0.0	0.0
C LC_111	SLN131 column below / Begin	104.1	-0.2	-0.1	-0.5	-0.5	0.2
	SLN84 ring / End	-10.3	0.8	-5.8	0.0	0.0	1.8
	SLN88 ring / End	12.3	0.9	-16.3	0.0	0.0	1.8
C LC_112	SLN131 column below / Begin	100.4	0.5	0.2	-0.2	0.0	0.2
	SLN84 ring / End	-3.0	0.5	-5.8	0.0	0.0	1.1
	SLN88 ring / End	6.1	0.6	-16.3	0.0	0.0	1.1
C LC_113	SLN131 column below / Begin	96.6	-0.2	0.3	-0.5	0.0	0.2
	SLN84 ring / End	-6.1	0.7	-5.8	0.0	0.0	1.5
	SLN88 ring / End	9.1	0.8	-16.3	0.0	0.0	1.5
C LC_114	SLN131 column below / Begin	107.5	-0.2	-0.1	-0.4	-0.4	0.2
	SLN84 ring / End	-7.1	0.5	-5.8	0.0	0.0	1.1
	SLN88 ring / End	9.4	0.5	-16.3	0.0	0.0	1.2
C LC_115	SLN131 column below / Begin	23.0	-0.1	0.1	-0.2	0.1	0.0
	SLN84 ring / End	-6.0	0.4	-1.1	0.0	0.0	0.7
	SLN88 ring / End	4.1	0.3	-1.8	0.0	0.0	0.8

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN131 column below / Begin	19.4	0.6	0.4	0.2	0.5	0.0
	SLN84 ring / End	0.7	0.0	-1.1	0.0	0.0	0.0
	SLN88 ring / End	-2.5	0.0	-1.8	0.0	0.0	-0.1
C LC_117	SLN131 column below / Begin	15.6	-0.1	0.5	-0.2	0.5	0.0
	SLN84 ring / End	-2.4	0.2	-1.1	0.0	0.0	0.5
	SLN88 ring / End	0.3	0.2	-1.8	0.0	0.0	0.4
C LC_118	SLN131 column below / Begin	26.3	-0.1	0.1	-0.1	0.1	0.0
	SLN84 ring / End	-2.7	0.0	-1.1	0.0	0.0	0.1
	SLN88 ring / End	1.4	0.0	-1.8	0.0	0.0	0.1
C LC_119	SLN131 column below / Begin	154.1	-0.2	-0.3	-0.6	-0.9	0.3
	SLN84 ring / End	-6.4	0.9	-8.9	0.0	0.0	2.0
	SLN88 ring / End	18.6	1.0	-25.8	0.0	0.0	2.1
C LC_120	SLN131 column below / Begin	152.4	0.2	0.1	-0.5	-0.3	0.3
	SLN84 ring / End	-3.5	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	15.9	1.0	-25.8	0.0	0.0	2.0
C LC_121	SLN131 column below / Begin	150.5	-0.2	0.2	-0.6	-0.1	0.3
	SLN84 ring / End	-4.6	0.8	-8.9	0.0	0.0	1.8
	SLN88 ring / End	17.1	1.1	-25.8	0.0	0.0	2.1
C LC_122	SLN131 column below / Begin	155.8	-0.2	-0.3	-0.5	-1.0	0.3
	SLN84 ring / End	-5.1	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	17.1	0.8	-25.8	0.0	0.0	1.9
C LC_123	SLN131 column below / Begin	120.0	-0.3	-0.3	-0.5	-0.9	0.2
	SLN84 ring / End	-5.6	0.8	-6.9	0.0	0.0	1.7
	SLN88 ring / End	17.2	0.9	-19.5	0.0	0.0	1.9
C LC_124	SLN131 column below / Begin	116.9	0.5	0.4	-0.2	0.2	0.2
	SLN84 ring / End	-1.0	0.6	-6.9	0.0	0.0	1.2
	SLN88 ring / End	10.7	0.8	-19.5	0.0	0.0	1.4
C LC_125	SLN131 column below / Begin	113.2	-0.2	0.6	-0.5	0.6	0.2
	SLN84 ring / End	-3.5	0.6	-6.9	0.0	0.0	1.4
	SLN88 ring / End	12.6	0.9	-19.5	0.0	0.0	1.7
C LC_126	SLN131 column below / Begin	123.3	-0.2	-0.3	-0.4	-1.0	0.2
	SLN84 ring / End	-2.7	0.6	-6.9	0.0	0.0	1.2
	SLN88 ring / End	14.7	0.5	-19.5	0.0	0.0	1.3
C LC_127	SLN131 column below / Begin	38.9	-0.2	-0.1	-0.1	-0.3	0.1
	SLN84 ring / End	-1.3	0.4	-2.2	0.0	0.0	0.7
	SLN88 ring / End	9.1	0.4	-4.9	0.0	0.0	0.8
C LC_128	SLN131 column below / Begin	35.8	0.6	0.5	0.1	0.8	0.0
	SLN84 ring / End	2.7	0.1	-2.2	0.0	0.0	0.1
	SLN88 ring / End	2.1	0.2	-4.9	0.0	0.0	0.3
C LC_129	SLN131 column below / Begin	32.2	-0.1	0.7	-0.2	1.1	0.0
	SLN84 ring / End	0.2	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	3.8	0.3	-4.9	0.0	0.0	0.5
C LC_130	SLN131 column below / Begin	42.1	-0.2	-0.1	-0.1	-0.4	0.1
	SLN84 ring / End	1.7	0.1	-2.2	0.0	0.0	0.2
	SLN88 ring / End	6.8	0.0	-4.9	0.0	0.0	0.2
C LC_135	SLN131 column below / Begin	150.0	-0.1	0.0	-0.4	-0.5	0.3
	SLN84 ring / End	0.6	0.5	-8.9	0.0	0.0	1.2

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN88 ring / End	16.0	0.7	-25.8	0.0	0.0	1.4
C LC_141	SLN131 column below / Begin	37.4	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.7	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.8	0.2	-4.9	0.0	0.0	0.4
C LC_142	SLN131 column below / Begin	37.3	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.9	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	5.0	0.2	-4.9	0.0	0.0	0.4
C LC_143	SLN131 column below / Begin	37.3	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.9	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	5.1	0.2	-4.9	0.0	0.0	0.4
C LC_144	SLN131 column below / Begin	37.4	0.0	0.1	-0.1	0.1	0.1
	SLN84 ring / End	0.6	0.1	-2.2	0.0	0.0	0.3
	SLN88 ring / End	4.7	0.2	-4.9	0.0	0.0	0.4
C LC_145	SLN131 column below / Begin	154.2	-0.1	-0.1	-0.5	-0.6	0.3
	SLN84 ring / End	-3.3	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	18.8	0.9	-25.8	0.0	0.0	2.0
C LC_146	SLN131 column below / Begin	153.9	-0.1	-0.1	-0.5	-0.5	0.3
	SLN84 ring / End	-2.3	0.8	-8.9	0.0	0.0	1.8
	SLN88 ring / End	19.6	1.0	-25.8	0.0	0.0	2.1
C LC_147	SLN131 column below / Begin	153.8	-0.1	-0.1	-0.5	-0.5	0.3
	SLN84 ring / End	-2.1	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	20.0	1.1	-25.8	0.0	0.0	2.1
C LC_148	SLN131 column below / Begin	154.3	-0.2	-0.1	-0.5	-0.7	0.3
	SLN84 ring / End	-3.5	0.8	-8.9	0.0	0.0	1.8
	SLN88 ring / End	18.4	0.9	-25.8	0.0	0.0	2.0
C LC_151	SLN131 column below / Begin	118.9	-0.1	-3.8	-0.4	-6.0	0.1
	SLN84 ring / End	-0.9	0.5	-8.9	0.0	0.0	1.1
	SLN88 ring / End	12.2	0.6	-11.6	0.0	0.0	1.2
C LC_152	SLN131 column below / Begin	92.3	-0.1	3.9	-0.6	5.8	0.2
	SLN84 ring / End	-5.8	0.9	-2.2	0.0	0.0	2.0
	SLN88 ring / End	19.1	1.1	-14.1	0.0	0.0	2.2
C LC_153	SLN131 column below / Begin	152.5	-0.1	-0.1	-0.5	-0.5	0.3
	SLN84 ring / End	-3.0	0.8	-8.9	0.0	0.0	1.7
	SLN88 ring / End	19.2	1.0	-20.7	0.0	0.0	2.0
C LC_154	SLN131 column below / Begin	120.8	-0.1	-3.8	-0.4	-6.1	0.1
	SLN84 ring / End	-0.9	0.5	-8.9	0.0	0.0	1.1
	SLN88 ring / End	12.3	0.6	-11.6	0.0	0.0	1.3

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	5.6 < 100%	OK
Welds	55.4 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN89b+SLN89a radial (continuous)-bfl 1	17.7	C LC_119	134.2	0.0	17.1	OK
SLN89b+SLN89a radial (continuous)-tfl 1	17.7	C LC_119	80.1	0.0	0.0	OK
SLN89b+SLN89a radial (continuous)-w 1	10.5	C LC_152	126.5	0.0	17.3	OK
SLN131 column below	10.0	C LC_154	122.8	0.0	0.0	OK
SLN84 ring-bfl 1	12.7	C LC_146	24.4	0.0	0.0	OK
SLN84 ring-tfl 1	12.7	C LC_119	27.9	0.0	0.0	OK
SLN84 ring-w 1	7.2	C LC_119	59.3	0.0	0.0	OK
SLN88 ring-bfl 1	12.7	C LC_119	58.9	0.0	0.0	OK
SLN88 ring-tfl 1	12.7	C LC_119	55.6	0.0	0.0	OK
SLN88 ring-w 1	7.2	C LC_147	54.0	0.0	0.0	OK
End plate column below SLN131 column below	12.0	C LC_154	78.7	0.0	37.9	OK
End plate SLN88 ring	20.0	C LC_119	24.8	0.0	28.7	OK
Bottom flange plate SLN88 ring	15.0	C LC_120	50.0	0.0	0.0	OK
Bottom flange half plate L SLN88 ring	15.0	C LC_119	29.2	0.0	0.0	OK
Bottom flange half plate R SLN88 ring	15.0	C LC_119	27.5	0.0	0.0	OK
Haunch web SLN88 ring	12.0	C LC_119	33.1	0.0	0.0	OK
Haunch top plate SLN88 ring	15.0	C LC_152	25.8	0.0	0.0	OK
Haunch sloping plate SLN88 ring	15.0	C LC_119	17.7	0.0	0.0	OK
End plate SLN84 ring	20.0	C LC_152	57.2	0.0	40.8	OK
Bottom flange plate SLN84 ring	15.0	C LC_152	110.8	0.0	0.0	OK
Bottom flange half plate L SLN84 ring	15.0	C LC_152	21.2	0.0	0.0	OK
Bottom flange half plate R SLN84 ring	15.0	C LC_152	31.2	0.0	0.0	OK
Haunch web SLN84 ring	12.0	C LC_119	25.3	0.0	0.0	OK
Haunch top plate SLN84 ring	15.0	C LC_152	49.3	0.0	0.0	OK
Haunch sloping plate SLN84 ring	15.0	C LC_119	16.8	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

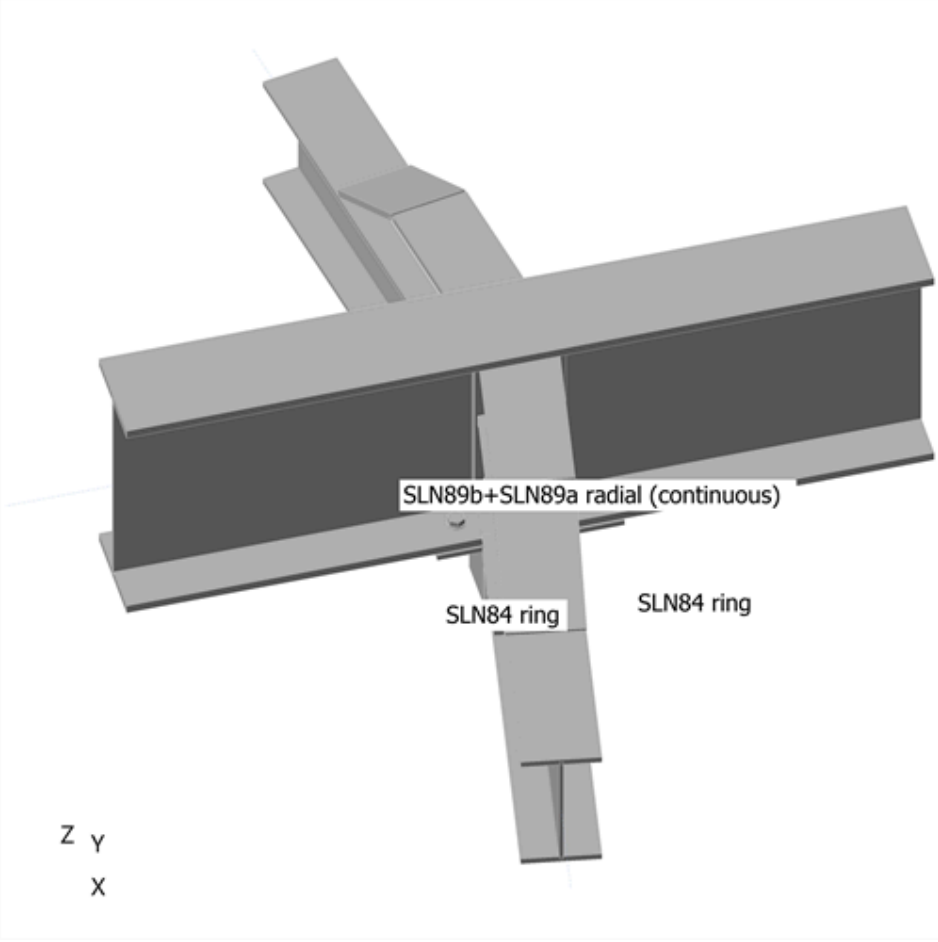
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

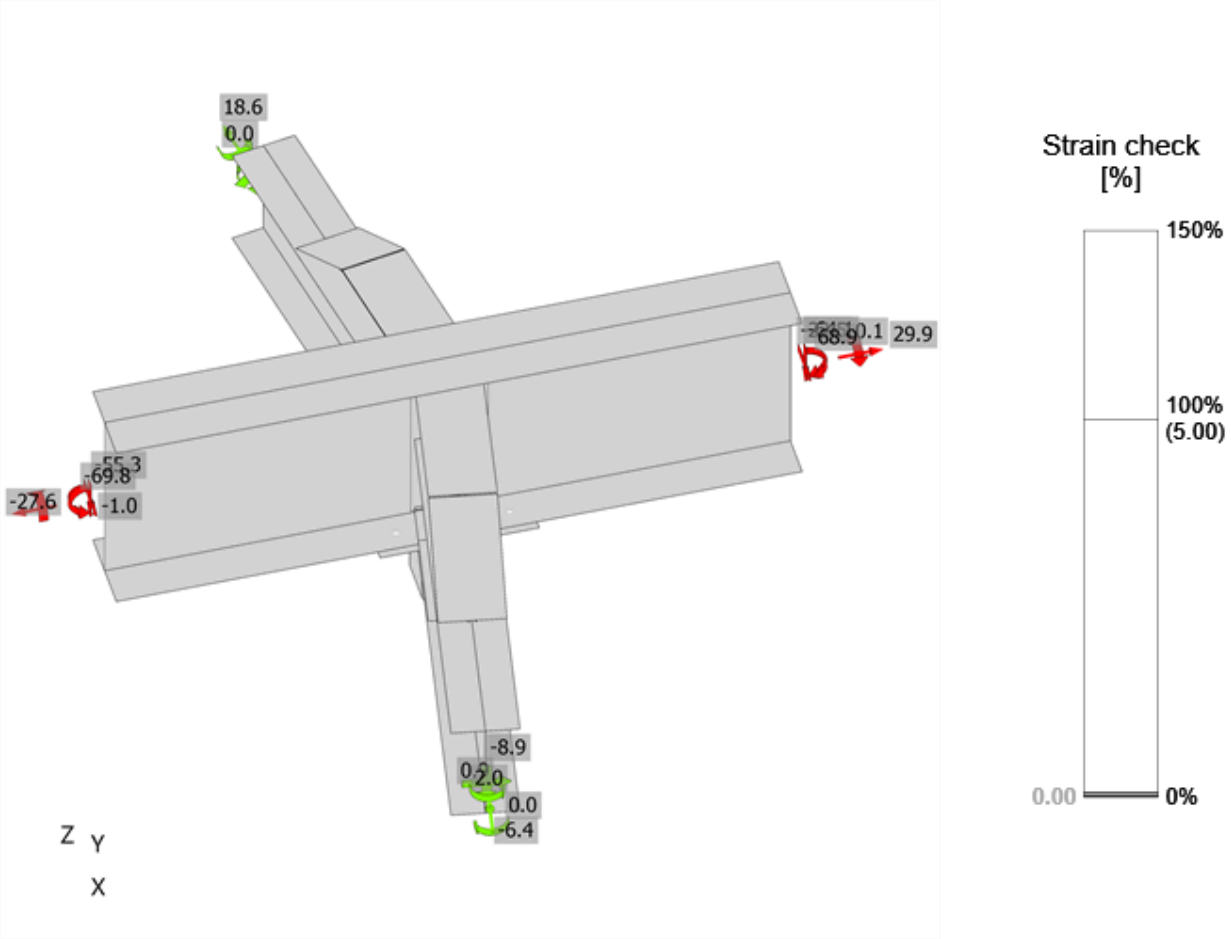
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN131 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

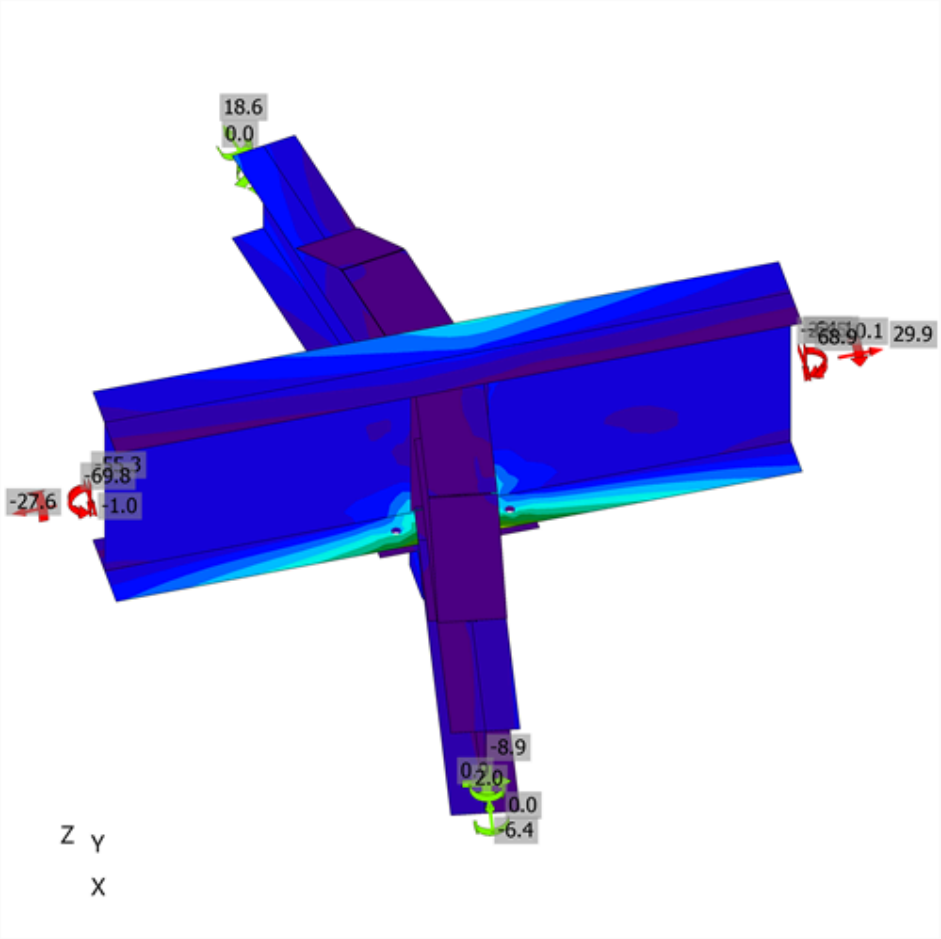
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

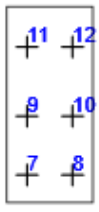


Strain check, C LC_119



Equivalent stress, C LC_119

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_119	0.5	5.0	206.4	0.4	5.3	5.6	OK	OK
	B3	M20 8.8 - 1	C LC_152	0.6	4.0	206.4	0.5	4.2	4.6	OK	OK
	B4	M20 8.8 - 1	C LC_121	0.6	2.1	172.3	0.4	2.2	2.5	OK	OK
	B5	M20 8.8 - 1	C LC_152	0.9	1.8	156.4	0.6	1.9	2.4	OK	OK
	B7	M30 8.8 - 2	C LC_154	0.4	1.4	176.5	0.1	1.4	0.7	OK	OK
	B8	M30 8.8 - 2	C LC_152	10.1	2.1	208.4	3.1	1.5	3.2	OK	OK
	B9	M30 8.8 - 2	C LC_152	1.2	0.7	229.5	0.4	0.4	0.6	OK	OK
	B10	M30 8.8 - 2	C LC_119	1.9	0.4	229.5	0.6	0.3	0.6	OK	OK
	B11	M30 8.8 - 2	C LC_119	3.4	1.4	150.0	1.1	0.9	1.4	OK	OK
	B12	M30 8.8 - 2	C LC_119	6.8	1.4	229.5	2.1	1.0	2.1	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN88 ring	Bottom flange plate SLN88 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN84 ring	Bottom flange plate SLN84 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN84 ring-tfl 1	◢ 10.5	149	C LC_119	17.5	0.0	3.2	-2.8	9.5	4.3	4.3	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-tfl 1	◢ 10.5	149	C LC_152	33.2	0.0	9.3	-10.0	15.4	8.2	7.0	OK	OK
End plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	208	C LC_147	44.7	0.0	-17.5	-21.4	10.3	11.1	8.1	OK	OK
		◢ 6.0 ◣	208	C LC_146	37.6	0.0	-19.6	17.2	-6.8	9.3	7.3	OK	OK
Bottom flange plate SLN88 ring	SLN88 ring-w 1	◢ 6.0 ◣	100	C LC_119	42.5	0.0	5.1	12.5	20.9	10.5	9.7	OK	OK
		◢ 6.0 ◣	100	C LC_147	37.5	0.0	12.9	-16.1	-12.4	9.3	8.1	OK	OK
SLN88 ring-w 1	Bottom flange half plate L SLN88 ring	◢ 6.0 ◣	243	C LC_119	37.9	0.0	6.9	9.2	19.5	9.4	7.2	OK	OK
		◢ 6.0 ◣	243	C LC_119	52.5	0.0	15.2	-12.9	-26.0	13.0	9.8	OK	OK
SLN88 ring-w 1	Bottom flange half plate R SLN88 ring	◢ 6.0 ◣	243	C LC_119	19.6	0.0	2.9	1.9	-11.1	4.9	4.9	OK	OK
		◢ 6.0 ◣	243	C LC_119	35.7	0.0	20.3	-3.1	16.7	8.8	7.2	OK	OK
SLN88 ring-tfl 1	Haunch web SLN88 ring	◢ 6.0 ◣	572	C LC_135	56.8	0.0	-21.4	-15.1	26.3	14.0	8.3	OK	OK
		◢ 6.0 ◣	572	C LC_119	76.9	0.0	-8.7	18.6	-40.0	19.0	10.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	208	C LC_119	35.4	0.0	-14.7	-17.6	6.2	8.8	7.6	OK	OK
		◢ 6.0 ◣	208	C LC_119	53.5	0.0	-22.5	18.7	-20.9	13.2	9.2	OK	OK
Bottom flange plate SLN84 ring	SLN84 ring-w 1	◢ 6.0 ◣	100	C LC_119	42.7	0.0	12.9	21.2	10.1	10.5	9.0	OK	OK
		◢ 6.0 ◣	100	C LC_119	40.2	0.0	1.8	1.8	-23.1	9.9	8.2	OK	OK
SLN84 ring-w 1	Bottom flange half plate L SLN84 ring	◢ 6.0 ◣	243	C LC_146	22.0	0.0	3.5	5.0	11.5	5.4	5.4	OK	OK
		◢ 6.0 ◣	243	C LC_119	33.9	0.0	9.8	-8.1	-16.9	8.4	7.5	OK	OK
SLN84 ring-w 1	Bottom flange half plate R SLN84 ring	◢ 6.0 ◣	243	C LC_152	25.8	0.0	-4.5	-6.4	13.2	6.4	5.7	OK	OK
		◢ 6.0 ◣	243	C LC_152	38.1	0.0	-11.5	9.6	-18.6	9.4	7.4	OK	OK
SLN84 ring-tfl 1	Haunch web SLN84 ring	◢ 6.0 ◣	572	C LC_119	29.7	0.0	-17.7	-10.6	8.8	7.3	6.1	OK	OK
		◢ 6.0 ◣	572	C LC_119	34.8	0.0	2.5	4.5	-19.5	8.6	6.5	OK	OK
End plate SLN88 ring	Bottom flange plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	78.1	0.0	-36.9	-36.5	-15.7	19.3	12.4	OK	OK
		◢ 6.0 ◣	149	C LC_152	86.6	0.0	-41.8	42.2	11.6	21.4	13.1	OK	OK
End plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	145	C LC_119	13.4	0.0	0.1	0.2	7.7	3.3	3.3	OK	OK
		◢ 6.0 ◣	145	C LC_135	9.8	0.0	-1.3	1.3	-5.5	2.4	0.0	OK	OK
End plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	149	C LC_152	49.1	0.0	-24.8	-21.2	-12.2	12.1	8.5	OK	OK
		◢ 6.0 ◣	149	C LC_152	39.2	0.0	-13.8	17.5	12.0	9.7	7.2	OK	OK
Haunch top plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	342	C LC_152	10.6	0.0	-1.5	-2.3	-5.6	2.6	2.6	OK	OK
		◢ 6.0 ◣	342	C LC_119	13.4	0.0	4.4	-3.2	-6.6	3.3	3.3	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN88 ring	Haunch web SLN88 ring	◢ 6.0 ◣	271	C LC_120	58.0	0.0	-10.6	-23.1	23.5	14.3	9.4	OK	OK
		◢ 6.0 ◣	271	C LC_119	58.4	0.0	-7.8	-6.0	-32.9	14.4	9.5	OK	OK
End plate SLN84 ring	Bottom flange plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	115.8	0.0	-33.8	-52.1	-37.1	28.6	17.9	OK	OK
		◢ 6.0 ◣	149	C LC_152	146.7	0.0	-68.4	50.1	55.7	36.2	21.0	OK	OK
End plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	145	C LC_119	15.7	0.0	3.5	6.1	6.4	3.9	3.9	OK	OK
		◢ 6.0 ◣	145	C LC_152	15.6	0.0	4.8	-3.1	8.0	3.9	3.9	OK	OK
End plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	149	C LC_152	82.1	0.0	-40.5	-27.1	-31.1	20.3	12.1	OK	OK
		◢ 6.0 ◣	149	C LC_152	58.4	0.0	-15.0	28.4	15.8	14.4	9.3	OK	OK
Haunch top plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	342	C LC_119	10.5	0.0	-1.4	-5.1	-3.2	2.6	2.6	OK	OK
		◢ 6.0 ◣	342	C LC_152	17.1	0.0	-1.8	0.5	-9.8	4.2	4.2	OK	OK
Haunch sloping plate SLN84 ring	Haunch web SLN84 ring	◢ 6.0 ◣	271	C LC_119	41.6	0.0	-9.2	-19.5	12.9	10.3	7.9	OK	OK
		◢ 6.0 ◣	271	C LC_119	28.7	0.0	1.2	-11.4	-11.9	7.1	6.8	OK	OK
End plate column below SLN131 column below	SLN131 column below	◢ 4.2 ◣	520	C LC_154	210.5	0.0	-105.0	-105.0	-8.5	52.0	28.9	OK	OK
		◢ 4.2 ◣	521	C LC_154	224.1	0.0	-111.7	111.7	9.9	55.4	30.9	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN88 ring-w 1	◢ 7.2	208	C LC_146	153.7	0.0	51.7	64.0	-53.6	38.0	23.1	OK	OK
SLN89b+SLN89a radial (continuous)-w 1	SLN84 ring-w 1	◢ 7.2	208	C LC_119	160.6	0.0	55.9	-52.4	69.3	39.7	24.4	OK	OK
Haunch sloping plate SLN88 ring	Haunch top plate SLN88 ring	◢ 6.0 ◣	146	C LC_119	17.9	0.0	-16.6	-3.7	0.7	5.4	5.4	OK	OK
		◢ 6.0 ◣	147	C LC_119	12.8	0.0	-2.3	7.0	2.1	3.2	3.2	OK	OK
Haunch sloping plate SLN84 ring	Haunch top plate SLN84 ring	◢ 6.0 ◣	146	C LC_119	16.2	0.0	-15.0	-3.6	0.7	4.8	4.8	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_119	12.2	0.0	-2.2	6.6	2.1	3.0	3.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_260
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring

Name	SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

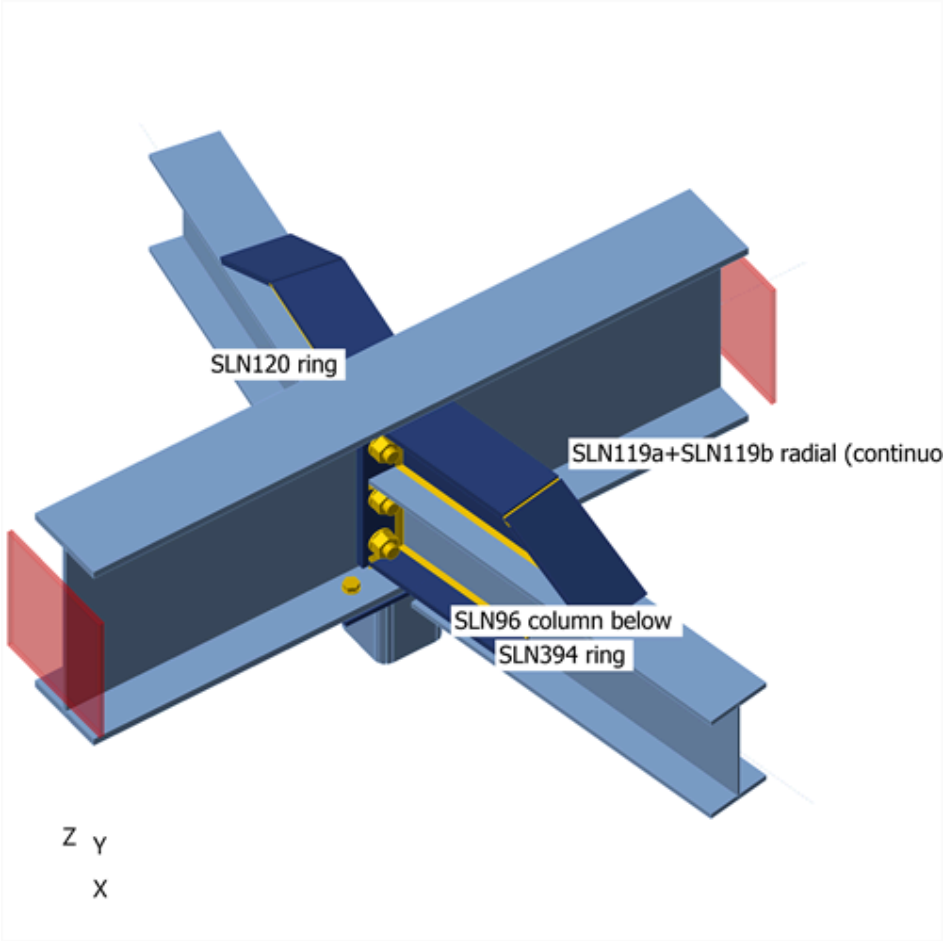
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN119a+SLN119b radial (continuous)	1 - UB457x191x89	-100.0	0.0	0.0	0	0	0
SLN96 column below	2 - SHS150x150x10	0.0	90.0	80.0	0	0	0
SLN120 ring	3 - UB254x146x43	160.0	0.0	0.0	0	0	-102
SLN394 ring	3 - UB254x146x43	0.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN119a+SLN119b radial (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN119a+SLN119b radial (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0
SLN96 column below / begin		Position	0
SLN120 ring / end		Position	0
SLN394 ring / end		Position	0



SLN119a+SLN119b radial (continuous) - SLN96 column below - SLN120 ring - SLN394 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN96 column below / Begin	33.2	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.3
C LC_102	SLN96 column below / Begin	140.3	0.1	1.1	0.6	1.6	0.0
	SLN120 ring / End	18.9	-1.0	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-2.9	-0.8	-9.0	0.0	0.0	-1.8
C LC_103	SLN96 column below / Begin	132.0	0.3	1.1	0.7	1.5	-0.1
	SLN120 ring / End	15.8	-1.0	-24.2	0.0	0.0	-2.1
	SLN394 ring / End	-7.6	-0.9	-8.5	0.0	0.0	-1.9
C LC_104	SLN96 column below / Begin	134.4	0.1	1.0	0.5	1.4	0.0
	SLN120 ring / End	13.3	-0.9	-24.2	0.0	0.0	-1.8
	SLN394 ring / End	-5.9	-0.8	-8.5	0.0	0.0	-1.6
C LC_105	SLN96 column below / Begin	136.9	0.2	0.9	0.6	1.1	0.0
	SLN120 ring / End	15.2	-0.9	-24.2	0.0	0.0	-2.1
	SLN394 ring / End	-9.0	-0.9	-8.5	0.0	0.0	-1.9
C LC_106	SLN96 column below / Begin	129.4	-0.1	1.3	0.5	1.8	0.0
	SLN120 ring / End	14.3	-1.0	-24.2	0.0	0.0	-1.9
	SLN394 ring / End	-4.0	-0.8	-8.5	0.0	0.0	-1.7
C LC_107	SLN96 column below / Begin	8.6	0.6	0.3	0.3	0.4	0.0
	SLN120 ring / End	1.5	-0.2	-0.5	0.0	0.0	-0.5
	SLN394 ring / End	-4.6	-0.2	-0.6	0.0	0.0	-0.5
C LC_108	SLN96 column below / Begin	12.9	0.0	0.2	0.0	0.4	0.0
	SLN120 ring / End	-2.7	0.1	-0.5	0.0	0.0	0.3
	SLN394 ring / End	-1.0	0.1	-0.6	0.0	0.0	0.2
C LC_109	SLN96 column below / Begin	17.6	0.2	0.0	0.1	-0.1	0.0
	SLN120 ring / End	2.2	-0.2	-0.5	0.0	0.0	-0.5
	SLN394 ring / End	-5.7	-0.2	-0.6	0.0	0.0	-0.4
C LC_110	SLN96 column below / Begin	3.8	-0.2	0.7	-0.1	0.8	0.0
	SLN120 ring / End	-2.9	0.0	-0.5	0.0	0.0	0.1
	SLN394 ring / End	1.0	-0.1	-0.6	0.0	0.0	-0.1
C LC_111	SLN96 column below / Begin	92.0	0.6	0.8	0.6	1.2	-0.1
	SLN120 ring / End	11.1	-0.8	-16.3	0.0	0.0	-1.7
	SLN394 ring / End	-8.5	-0.7	-5.9	0.0	0.0	-1.6
C LC_112	SLN96 column below / Begin	96.4	0.0	0.8	0.3	1.2	0.0
	SLN120 ring / End	6.7	-0.5	-16.3	0.0	0.0	-1.0
	SLN394 ring / End	-4.9	-0.4	-5.9	0.0	0.0	-0.9
C LC_113	SLN96 column below / Begin	101.2	0.2	0.5	0.5	0.6	0.0
	SLN120 ring / End	11.2	-0.8	-16.3	0.0	0.0	-1.7
	SLN394 ring / End	-10.1	-0.8	-5.9	0.0	0.0	-1.6
C LC_114	SLN96 column below / Begin	87.0	-0.2	1.3	0.3	1.6	0.0
	SLN120 ring / End	7.2	-0.7	-16.3	0.0	0.0	-1.2
	SLN394 ring / End	-2.3	-0.6	-5.9	0.0	0.0	-1.2
C LC_115	SLN96 column below / Begin	17.2	0.6	0.4	0.3	0.6	0.0
	SLN120 ring / End	2.7	-0.2	-1.8	0.0	0.0	-0.6
	SLN394 ring / End	-4.4	-0.2	-1.2	0.0	0.0	-0.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_116	SLN96 column below / Begin	21.6	0.0	0.3	0.0	0.6	0.0
	SLN120 ring / End	-1.4	0.1	-1.8	0.0	0.0	0.2
	SLN394 ring / End	-0.8	0.1	-1.2	0.0	0.0	0.1
C LC_117	SLN96 column below / Begin	26.2	0.2	0.1	0.1	0.1	0.0
	SLN120 ring / End	3.4	-0.2	-1.8	0.0	0.0	-0.6
	SLN394 ring / End	-5.6	-0.2	-1.2	0.0	0.0	-0.5
C LC_118	SLN96 column below / Begin	12.3	-0.2	0.8	0.0	1.0	0.0
	SLN120 ring / End	-1.5	0.0	-1.8	0.0	0.0	0.0
	SLN394 ring / End	1.3	-0.1	-1.2	0.0	0.0	-0.2
C LC_119	SLN96 column below / Begin	139.1	0.3	1.2	0.6	1.7	-0.1
	SLN120 ring / End	17.6	-1.0	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-5.7	-0.8	-9.0	0.0	0.0	-1.8
C LC_120	SLN96 column below / Begin	140.9	0.0	1.1	0.6	1.6	0.0
	SLN120 ring / End	15.9	-0.9	-25.8	0.0	0.0	-1.9
	SLN394 ring / End	-4.4	-0.8	-9.0	0.0	0.0	-1.7
C LC_121	SLN96 column below / Begin	142.4	0.2	0.8	0.6	1.0	0.0
	SLN120 ring / End	17.9	-0.9	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-6.4	-0.9	-9.0	0.0	0.0	-1.9
C LC_122	SLN96 column below / Begin	137.8	-0.1	1.5	0.5	2.1	0.0
	SLN120 ring / End	16.1	-1.1	-25.8	0.0	0.0	-2.0
	SLN394 ring / End	-3.1	-0.8	-9.0	0.0	0.0	-1.7
C LC_123	SLN96 column below / Begin	106.1	0.6	1.1	0.6	1.6	-0.1
	SLN120 ring / End	14.8	-0.8	-19.5	0.0	0.0	-1.7
	SLN394 ring / End	-4.7	-0.6	-7.0	0.0	0.0	-1.4
C LC_124	SLN96 column below / Begin	109.6	0.0	1.0	0.4	1.4	0.0
	SLN120 ring / End	11.8	-0.6	-19.5	0.0	0.0	-1.3
	SLN394 ring / End	-1.9	-0.5	-7.0	0.0	0.0	-1.1
C LC_125	SLN96 column below / Begin	112.2	0.3	0.4	0.5	0.4	0.0
	SLN120 ring / End	16.7	-0.8	-19.5	0.0	0.0	-1.7
	SLN394 ring / End	-4.9	-0.8	-7.0	0.0	0.0	-1.6
C LC_126	SLN96 column below / Begin	103.7	-0.2	1.6	0.3	2.4	0.0
	SLN120 ring / End	10.9	-0.8	-19.5	0.0	0.0	-1.5
	SLN394 ring / End	-0.5	-0.5	-7.0	0.0	0.0	-1.2
C LC_127	SLN96 column below / Begin	31.3	0.6	0.6	0.3	1.0	0.0
	SLN120 ring / End	6.4	-0.3	-4.9	0.0	0.0	-0.6
	SLN394 ring / End	-0.6	-0.2	-2.2	0.0	0.0	-0.4
C LC_128	SLN96 column below / Begin	34.7	0.0	0.5	0.1	0.8	0.0
	SLN120 ring / End	3.6	0.0	-4.9	0.0	0.0	-0.1
	SLN394 ring / End	2.2	0.0	-2.2	0.0	0.0	0.0
C LC_129	SLN96 column below / Begin	37.3	0.2	0.0	0.1	-0.2	0.0
	SLN120 ring / End	9.0	-0.3	-4.9	0.0	0.0	-0.6
	SLN394 ring / End	-0.3	-0.3	-2.2	0.0	0.0	-0.6
C LC_130	SLN96 column below / Begin	29.0	-0.2	1.1	0.0	1.7	0.0
	SLN120 ring / End	2.2	-0.2	-4.9	0.0	0.0	-0.3
	SLN394 ring / End	3.1	-0.1	-2.2	0.0	0.0	-0.1
C LC_135	SLN96 column below / Begin	136.4	0.0	1.2	0.4	1.7	0.0
	SLN120 ring / End	15.9	-0.7	-25.8	0.0	0.0	-1.5

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN394 ring / End	0.5	-0.5	-9.0	0.0	0.0	-1.2
C LC_141	SLN96 column below / Begin	33.2	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.9	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.3
C LC_142	SLN96 column below / Begin	33.2	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.8	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.3
C LC_143	SLN96 column below / Begin	33.3	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	4.7	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.3
C LC_144	SLN96 column below / Begin	33.1	0.0	0.4	0.1	0.7	0.0
	SLN120 ring / End	5.0	-0.2	-4.9	0.0	0.0	-0.4
	SLN394 ring / End	1.0	-0.1	-2.2	0.0	0.0	-0.3
C LC_145	SLN96 column below / Begin	140.2	0.0	1.2	0.6	1.7	0.0
	SLN120 ring / End	19.2	-1.0	-25.8	0.0	0.0	-2.0
	SLN394 ring / End	-2.8	-0.8	-9.0	0.0	0.0	-1.7
C LC_146	SLN96 column below / Begin	140.4	0.1	1.1	0.6	1.6	0.0
	SLN120 ring / End	18.7	-1.0	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-2.9	-0.8	-9.0	0.0	0.0	-1.8
C LC_147	SLN96 column below / Begin	140.6	0.1	1.1	0.6	1.5	0.0
	SLN120 ring / End	18.1	-0.9	-25.8	0.0	0.0	-2.0
	SLN394 ring / End	-3.7	-0.8	-9.0	0.0	0.0	-1.8
C LC_148	SLN96 column below / Begin	140.0	0.0	1.2	0.6	1.7	0.0
	SLN120 ring / End	19.8	-1.1	-25.8	0.0	0.0	-2.1
	SLN394 ring / End	-2.0	-0.8	-9.0	0.0	0.0	-1.8
C LC_151	SLN96 column below / Begin	103.4	0.0	-2.5	0.4	-3.7	0.0
	SLN120 ring / End	11.9	-0.6	-11.9	0.0	0.0	-1.3
	SLN394 ring / End	-0.8	-0.5	-9.0	0.0	0.0	-1.1
C LC_152	SLN96 column below / Begin	88.7	0.1	4.1	0.6	6.3	-0.1
	SLN120 ring / End	19.1	-1.1	-13.9	0.0	0.0	-2.3
	SLN394 ring / End	-6.0	-0.9	-2.2	0.0	0.0	-2.0
C LC_153	SLN96 column below / Begin	139.5	0.0	1.1	0.6	1.6	0.0
	SLN120 ring / End	18.9	-1.0	-20.9	0.0	0.0	-2.1
	SLN394 ring / End	-3.0	-0.8	-9.0	0.0	0.0	-1.8
C LC_154	SLN96 column below / Begin	103.8	0.0	-2.5	0.4	-3.6	0.0
	SLN120 ring / End	12.0	-0.6	-11.9	0.0	0.0	-1.3
	SLN394 ring / End	-0.8	-0.5	-9.0	0.0	0.0	-1.1

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	4.9 < 100%	OK
Welds	46.1 < 100%	OK
Buckling	Not calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN119a+SLN119b radial (continuous)-bfl 1	17.7	C LC_152	135.2	0.0	17.5	OK
SLN119a+SLN119b radial (continuous)-tfl 1	17.7	C LC_121	76.6	0.0	0.0	OK
SLN119a+SLN119b radial (continuous)-w 1	10.5	C LC_152	130.8	0.0	17.5	OK
SLN96 column below	10.0	C LC_152	106.2	0.0	0.0	OK
SLN120 ring-bfl 1	12.7	C LC_121	58.8	0.0	0.0	OK
SLN120 ring-tfl 1	12.7	C LC_121	55.8	0.0	0.0	OK
SLN120 ring-w 1	7.2	C LC_148	54.6	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_148	24.8	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_121	27.9	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_121	59.2	0.0	0.0	OK
End plate column below SLN96 column below	12.0	C LC_122	69.5	0.0	33.8	OK
End plate SLN394 ring	20.0	C LC_152	59.2	0.0	42.0	OK
Bottom flange plate SLN394 ring	15.0	C LC_152	114.2	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_152	32.0	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_152	21.4	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_121	25.1	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_152	50.6	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_152	16.7	0.0	0.0	OK
End plate SLN120 ring	20.0	C LC_148	25.6	0.0	27.4	OK
Bottom flange plate SLN120 ring	15.0	C LC_152	47.3	0.0	0.0	OK
Bottom flange half plate L SLN120 ring	15.0	C LC_148	27.5	0.0	0.0	OK
Bottom flange half plate R SLN120 ring	15.0	C LC_148	30.3	0.0	0.0	OK
Haunch web SLN120 ring	12.0	C LC_121	32.7	0.0	0.0	OK
Haunch top plate SLN120 ring	15.0	C LC_152	25.9	0.0	0.0	OK
Haunch sloping plate SLN120 ring	15.0	C LC_152	17.5	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

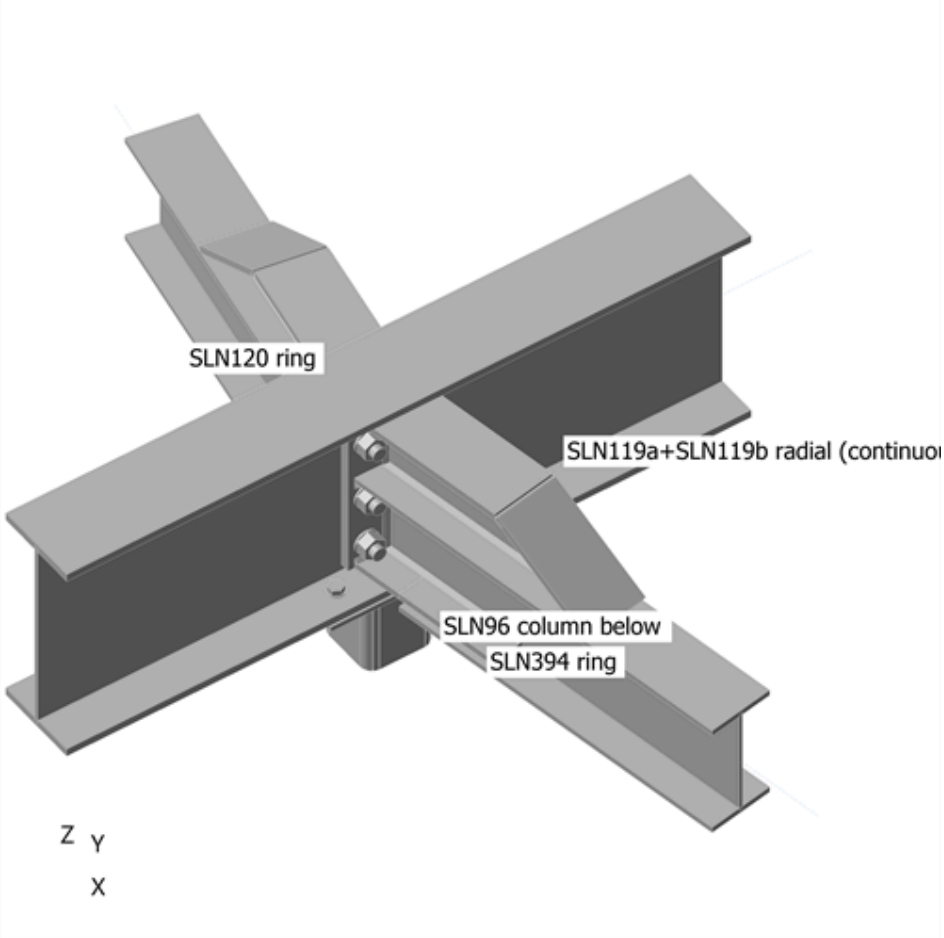
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

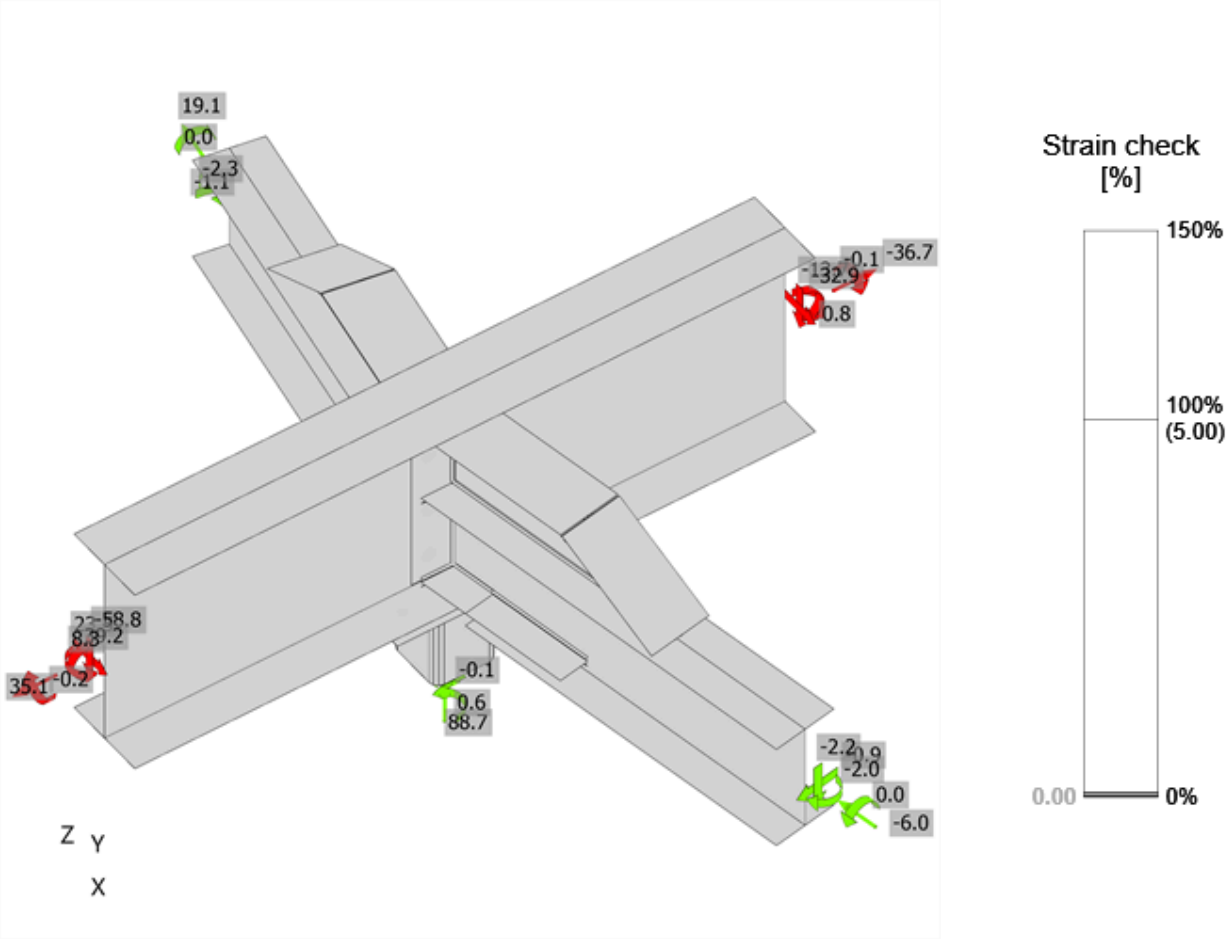
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN96 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

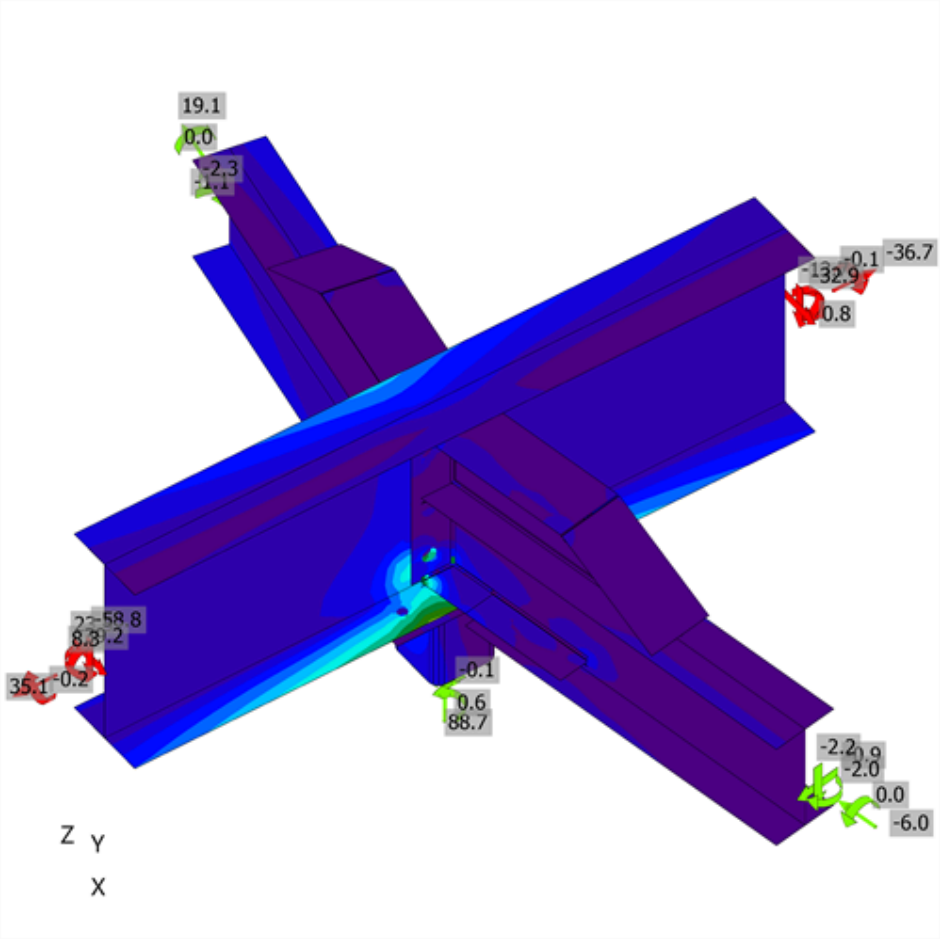
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

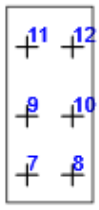


Strain check, C LC_152



Equivalent stress, C LC_152

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	1.5	2.0	156.4	1.1	2.1	2.9	OK	OK
	B3	M20 8.8 - 1	C LC_122	0.5	2.2	174.0	0.4	2.3	2.6	OK	OK
	B4	M20 8.8 - 1	C LC_152	0.9	4.0	206.4	0.6	4.2	4.7	OK	OK
	B5	M20 8.8 - 1	C LC_121	0.4	4.5	206.4	0.3	4.7	4.9	OK	OK
	B7	M30 8.8 - 2	C LC_151	0.6	1.1	176.5	0.2	1.1	0.6	OK	OK
	B8	M30 8.8 - 2	C LC_152	10.6	2.1	176.5	3.3	1.8	3.3	OK	OK
	B9	M30 8.8 - 2	C LC_152	1.4	0.7	229.5	0.4	0.4	0.6	OK	OK
	B10	M30 8.8 - 2	C LC_152	1.6	0.3	213.9	0.5	0.2	0.5	OK	OK
	B11	M30 8.8 - 2	C LC_121	3.3	1.3	150.0	1.0	0.9	1.3	OK	OK
	B12	M30 8.8 - 2	C LC_152	6.6	0.8	144.5	2.0	0.5	1.8	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN120 ring	Bottom flange plate SLN120 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-tfl 1	◢ 10.5	149	C LC_152	36.1	0.0	10.2	-10.8	-16.8	8.9	8.0	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-tfl 1	◢ 10.5	149	C LC_121	17.8	0.0	3.4	-2.9	-9.7	4.4	4.4	OK	OK
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	208	C LC_121	48.5	0.0	-21.6	-17.5	18.0	12.0	8.8	OK	OK
		◢ 6.0 ◣	208	C LC_121	36.8	0.0	-15.3	18.8	-4.6	9.1	7.7	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	100	C LC_121	40.9	0.0	0.4	-3.0	23.4	10.1	8.2	OK	OK
		◢ 6.0 ◣	100	C LC_121	41.1	0.0	11.8	-19.9	-10.9	10.2	8.9	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_152	25.6	0.0	-4.6	-6.4	-13.0	6.3	6.1	OK	OK
		◢ 6.0 ◣	243	C LC_152	37.9	0.0	-11.6	9.9	18.4	9.4	8.2	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_148	22.8	0.0	3.6	5.1	-12.0	5.6	5.5	OK	OK
		◢ 6.0 ◣	243	C LC_121	34.1	0.0	9.6	-7.9	17.1	8.4	7.1	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_121	35.2	0.0	2.3	-4.8	19.7	8.7	6.5	OK	OK
		◢ 6.0 ◣	572	C LC_119	29.8	0.0	-17.2	10.6	-9.2	7.4	6.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	208	C LC_148	35.0	0.0	-18.9	-16.5	4.4	8.7	7.1	OK	OK
		◢ 6.0 ◣	208	C LC_148	42.3	0.0	-17.2	21.1	-7.4	10.5	7.8	OK	OK
Bottom flange plate SLN120 ring	SLN120 ring-w 1	◢ 6.0 ◣	100	C LC_148	35.4	0.0	11.5	15.6	11.5	8.7	8.0	OK	OK
		◢ 6.0 ◣	100	C LC_148	42.3	0.0	5.0	-12.6	-20.7	10.5	9.7	OK	OK
SLN120 ring-w 1	Bottom flange half plate L SLN120 ring	◢ 6.0 ◣	243	C LC_148	19.5	0.0	3.0	2.1	10.9	4.8	4.8	OK	OK
		◢ 6.0 ◣	243	C LC_148	35.4	0.0	20.3	-3.1	-16.5	8.8	6.6	OK	OK
SLN120 ring-w 1	Bottom flange half plate R SLN120 ring	◢ 6.0 ◣	243	C LC_148	38.3	0.0	6.5	9.4	-19.7	9.5	6.7	OK	OK
		◢ 6.0 ◣	243	C LC_148	54.3	0.0	15.9	-13.0	27.0	13.4	8.8	OK	OK
SLN120 ring-tfl 1	Haunch web SLN120 ring	◢ 6.0 ◣	572	C LC_119	79.2	0.0	-10.1	-20.1	40.7	19.6	10.6	OK	OK
		◢ 6.0 ◣	572	C LC_135	53.8	0.0	-20.4	13.5	-25.4	13.3	8.0	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	113.5	0.0	-32.1	-52.5	34.6	28.0	16.8	OK	OK
		◢ 6.0 ◣	149	C LC_152	146.8	0.0	-70.2	49.8	-55.3	36.3	21.0	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_152	16.1	0.0	4.9	3.1	-8.3	4.0	4.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	15.6	0.0	3.4	-6.0	-6.4	3.9	3.9	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	84.1	0.0	-41.6	-27.6	31.9	20.8	10.7	OK	OK
		◢ 6.0 ◣	149	C LC_152	59.4	0.0	-15.1	29.1	-16.0	14.7	8.1	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_152	17.6	0.0	-1.7	-0.5	10.1	4.4	4.4	OK	OK
		◢ 6.0 ◣	342	C LC_152	10.2	0.0	-1.2	4.9	3.2	2.5	2.5	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_121	28.7	0.0	1.0	11.2	12.2	7.1	6.8	OK	OK
		◢ 6.0 ◣	271	C LC_121	41.5	0.0	-9.2	19.4	-13.0	10.3	7.9	OK	OK
End plate SLN120 ring	Bottom flange plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	83.1	0.0	-40.1	-38.2	17.5	20.5	12.4	OK	OK
		◢ 6.0 ◣	149	C LC_152	89.0	0.0	-41.9	43.7	-11.9	22.0	12.1	OK	OK
End plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	145	C LC_135	9.1	0.0	-1.2	-1.2	5.1	2.3	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	11.9	0.0	4.7	-4.0	-4.9	2.9	0.0	OK	OK
End plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	149	C LC_152	49.6	0.0	-24.9	-21.4	12.4	12.3	7.7	OK	OK
		◢ 6.0 ◣	149	C LC_152	39.8	0.0	-14.1	17.5	-12.4	9.8	6.8	OK	OK
Haunch top plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	342	C LC_148	13.2	0.0	4.3	3.1	6.5	3.3	3.3	OK	OK
		◢ 6.0 ◣	342	C LC_152	10.7	0.0	-1.5	2.3	5.7	2.7	2.7	OK	OK
Haunch sloping plate SLN120 ring	Haunch web SLN120 ring	◢ 6.0 ◣	271	C LC_119	61.0	0.0	-9.2	5.0	34.5	15.1	9.7	OK	OK
		◢ 6.0 ◣	271	C LC_122	54.9	0.0	-9.2	22.6	-21.6	13.6	9.0	OK	OK
End plate column below SLN96 column below	SLN96 column below	◢ 4.2 ◣	520	C LC_122	179.4	0.0	-90.8	-89.2	-5.3	44.3	24.9	OK	OK
		◢ 4.2 ◣	520	C LC_152	186.7	0.0	-93.6	92.4	12.7	46.1	25.0	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_121	148.7	0.0	47.0	-52.6	62.2	36.7	22.6	OK	OK
SLN119a+SLN119b radial (continuous)-w 1	SLN120 ring-w 1	◢ 7.2	208	C LC_148	125.4	0.0	58.0	45.7	-45.0	31.0	18.7	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_152	16.3	0.0	-15.0	-3.5	-0.7	4.9	4.9	OK	OK
		◢ 6.0 ◣	147	C LC_152	12.1	0.0	-2.2	6.5	-2.2	3.0	3.0	OK	OK
Haunch sloping plate SLN120 ring	Haunch top plate SLN120 ring	◢ 6.0 ◣	146	C LC_152	17.4	0.0	-16.1	-3.6	-0.9	5.2	5.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_152	12.8	0.0	-2.3	6.9	-2.4	3.2	3.2	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5

Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R8-B_280
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring

Name	SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring
Description	
Analysis	Stress, strain/ simplified loading

Design

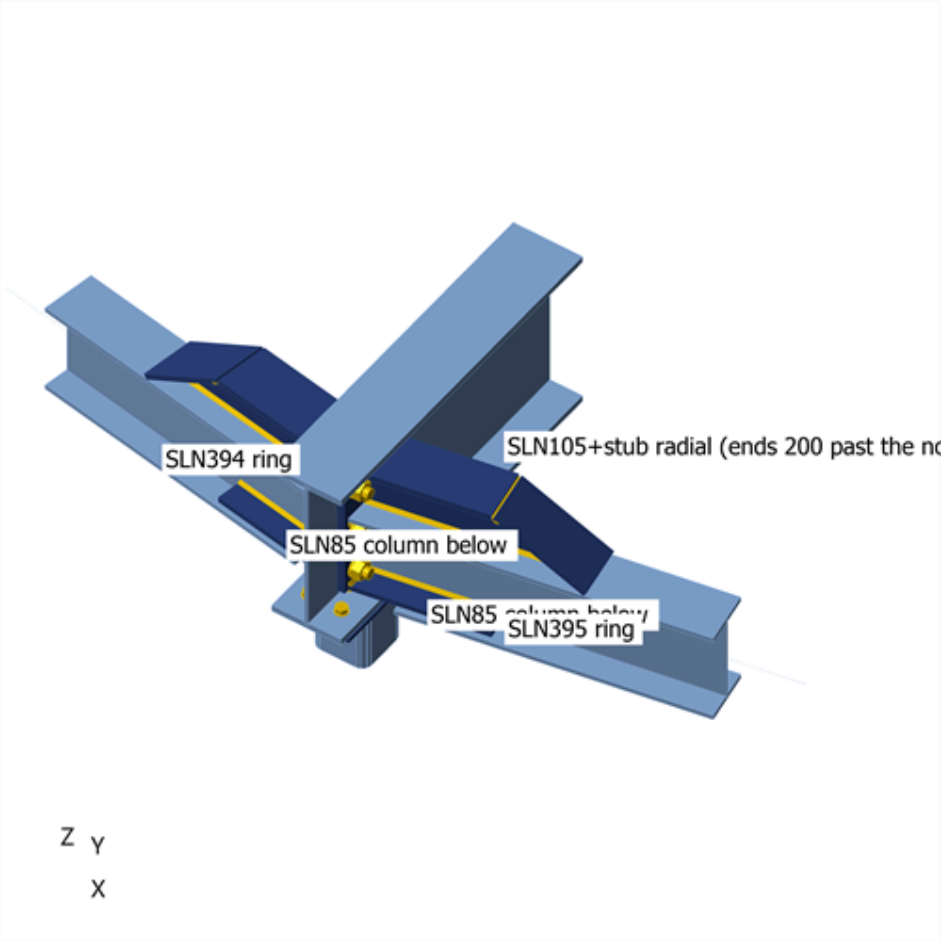
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN105+stub radial (ends 200 past the node)	1 - UB457x191x89	-80.0	0.0	0.0	200	0	0
SLN85 column below	2 - SHS150x150x10	0.0	90.0	100.0	0	0	0
SLN394 ring	3 - UB254x146x43	180.0	0.0	0.0	0	0	-102
SLN395 ring	3 - UB254x146x43	20.0	0.0	0.0	0	0	-102

Supports and forces

Name	Support	Forces in	X [mm]
SLN105+stub radial (ends 200 past the node) / begin		Position	0
SLN85 column below / begin	N-Vy-Vz-Mx-My-Mz	Position	0
SLN394 ring / end		Position	0
SLN395 ring / end		Position	0



SLN105+stub radial (ends 200 past the node) - SLN85 column below - SLN394 ring - SLN395 ring, Axonometric view

Cross-sections

Name	Material
1 - UB457x191x89	S 275
2 - SHS150x150x10	S 275
3 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M20 8.8	20	640.0	800.0	314
M30 8.8	30	640.0	800.0	707

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN105+stub radial (ends 200 past the node) / Begin	0.8	0.0	-11.4	0.0	-1.4	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.8	0.0	-2.1	0.0	0.0	0.0
C LC_102	SLN105+stub radial (ends 200 past the node) / Begin	1.8	-0.1	-48.2	0.0	-6.1	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.9	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-2.9	0.0	-8.7	0.0	0.0	0.1
C LC_103	SLN105+stub radial (ends 200 past the node) / Begin	0.2	0.0	-45.3	0.0	-5.8	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-7.6	-0.9	-8.5	0.0	0.0	-0.6
	SLN395 ring / End	-7.9	0.1	-8.2	0.0	0.0	0.3
C LC_104	SLN105+stub radial (ends 200 past the node) / Begin	0.4	-0.2	-45.3	0.0	-5.3	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.8	-0.8	-8.5	0.0	0.0	-0.6
	SLN395 ring / End	-6.1	0.1	-8.2	0.0	0.0	0.2
C LC_105	SLN105+stub radial (ends 200 past the node) / Begin	-0.4	-0.2	-45.1	0.0	-5.7	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-9.0	-0.9	-8.5	0.0	0.0	-0.7
	SLN395 ring / End	-8.9	0.0	-8.2	0.0	0.0	0.1
C LC_106	SLN105+stub radial (ends 200 past the node) / Begin	1.3	-0.1	-45.5	0.0	-5.6	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.0	-0.9	-8.5	0.0	0.0	-0.7
	SLN395 ring / End	-4.0	0.1	-8.2	0.0	0.0	0.1
C LC_107	SLN105+stub radial (ends 200 past the node) / Begin	-1.2	0.0	-2.7	0.0	-0.4	0.0
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.6	-0.2	-0.6	0.0	0.0	-0.1
	SLN395 ring / End	-4.9	0.1	-0.6	0.0	0.0	0.2
C LC_108	SLN105+stub radial (ends 200 past the node) / Begin	-1.0	0.1	-2.6	0.0	0.5	-0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-1.0	0.1	-0.6	0.0	0.0	0.1
	SLN395 ring / End	-1.9	0.2	-0.6	0.0	0.0	0.2
C LC_109	SLN105+stub radial (ends 200 past the node) / Begin	-1.9	-0.1	-2.2	0.0	0.1	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.7	-0.2	-0.6	0.0	0.0	-0.2
	SLN395 ring / End	-5.5	0.0	-0.6	0.0	0.0	0.0
C LC_110	SLN105+stub radial (ends 200 past the node) / Begin	0.2	-0.1	-3.1	0.0	-0.2	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.0	-0.1	-0.6	0.0	0.0	-0.1
	SLN395 ring / End	1.1	-0.1	-0.6	0.0	0.0	0.0
C LC_111	SLN105+stub radial (ends 200 past the node) / Begin	-0.8	0.0	-31.4	0.0	-4.0	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-8.5	-0.7	-6.0	0.0	0.0	-0.5
	SLN395 ring / End	-8.8	0.1	-5.7	0.0	0.0	0.3
C LC_112	SLN105+stub radial (ends 200 past the node) / Begin	-0.5	-0.1	-31.3	0.0	-3.2	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.9	-0.4	-6.0	0.0	0.0	-0.4
	SLN395 ring / End	-5.7	0.2	-5.7	0.0	0.0	0.3
C LC_113	SLN105+stub radial (ends 200 past the node) / Begin	-1.7	-0.1	-31.0	0.0	-3.7	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-10.1	-0.7	-6.0	0.0	0.0	-0.6
	SLN395 ring / End	-9.9	0.0	-5.7	0.0	0.0	0.0
C LC_114	SLN105+stub radial (ends 200 past the node) / Begin	0.9	-0.1	-31.8	0.0	-3.8	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.3	-0.6	-6.0	0.0	0.0	-0.6
	SLN395 ring / End	-2.2	0.0	-5.7	0.0	0.0	0.1
C LC_115	SLN105+stub radial (ends 200 past the node) / Begin	-1.0	0.0	-5.6	0.0	-0.7	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.4	-0.2	-1.2	0.0	0.0	-0.1
	SLN395 ring / End	-4.7	0.1	-1.1	0.0	0.0	0.2
C LC_116	SLN105+stub radial (ends 200 past the node) / Begin	-0.8	0.1	-5.6	0.0	0.2	-0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.8	0.1	-1.2	0.0	0.0	0.1
	SLN395 ring / End	-1.7	0.2	-1.1	0.0	0.0	0.2
C LC_117	SLN105+stub radial (ends 200 past the node) / Begin	-1.8	-0.1	-5.2	0.0	-0.3	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.6	-0.2	-1.2	0.0	0.0	-0.2
	SLN395 ring / End	-5.4	0.0	-1.1	0.0	0.0	0.0
C LC_118	SLN105+stub radial (ends 200 past the node) / Begin	0.5	-0.1	-6.0	0.0	-0.5	0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.3	-0.1	-1.2	0.0	0.0	-0.2
	SLN395 ring / End	1.4	-0.1	-1.1	0.0	0.0	0.0
C LC_119	SLN105+stub radial (ends 200 past the node) / Begin	1.0	0.0	-48.3	0.0	-6.3	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-5.8	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-6.0	0.1	-8.7	0.0	0.0	0.3
C LC_120	SLN105+stub radial (ends 200 past the node) / Begin	1.0	-0.2	-48.1	0.0	-5.7	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.3	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-4.6	0.0	-8.7	0.0	0.0	0.1
C LC_121	SLN105+stub radial (ends 200 past the node) / Begin	0.9	-0.2	-48.4	0.0	-6.6	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-6.4	-0.9	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-6.2	0.0	-8.7	0.0	0.0	0.1
C LC_122	SLN105+stub radial (ends 200 past the node) / Begin	1.5	-0.1	-48.0	0.0	-5.7	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.1	-0.8	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-3.1	0.0	-8.7	0.0	0.0	0.1
C LC_123	SLN105+stub radial (ends 200 past the node) / Begin	0.9	0.1	-37.3	0.0	-5.0	0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.7	-0.6	-7.0	0.0	0.0	-0.4
	SLN395 ring / End	-5.0	0.2	-6.7	0.0	0.0	0.3

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_124	SLN105+stub radial (ends 200 past the node) / Begin	0.8	-0.1	-37.0	0.0	-3.9	0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-1.9	-0.5	-7.0	0.0	0.0	-0.4
	SLN395 ring / End	-2.7	0.1	-6.7	0.0	0.0	0.1
C LC_125	SLN105+stub radial (ends 200 past the node) / Begin	1.0	-0.2	-37.5	0.0	-5.5	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-4.9	-0.7	-7.0	0.0	0.0	-0.6
	SLN395 ring / End	-4.7	0.0	-6.7	0.0	0.0	0.0
C LC_126	SLN105+stub radial (ends 200 past the node) / Begin	1.5	-0.1	-36.8	0.0	-4.0	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.5	-0.6	-7.0	0.0	0.0	-0.5
	SLN395 ring / End	-0.4	0.0	-6.7	0.0	0.0	0.0
C LC_127	SLN105+stub radial (ends 200 past the node) / Begin	0.7	0.1	-11.5	0.0	-1.7	0.0
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.7	-0.2	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	-0.9	0.1	-2.1	0.0	0.0	0.2
C LC_128	SLN105+stub radial (ends 200 past the node) / Begin	0.5	0.0	-11.2	0.0	-0.6	-0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	2.2	0.0	-2.2	0.0	0.0	0.1
	SLN395 ring / End	1.3	0.0	-2.1	0.0	0.0	0.1
C LC_129	SLN105+stub radial (ends 200 past the node) / Begin	0.9	-0.1	-11.7	0.0	-2.2	0.2
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.3	-0.3	-2.2	0.0	0.0	-0.2
	SLN395 ring / End	-0.1	0.0	-2.1	0.0	0.0	-0.1
C LC_130	SLN105+stub radial (ends 200 past the node) / Begin	1.0	-0.1	-11.1	0.0	-0.8	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	3.1	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	3.2	-0.1	-2.1	0.0	0.0	-0.1
C LC_135	SLN105+stub radial (ends 200 past the node) / Begin	2.5	-0.1	-48.2	0.0	-5.9	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.5	-0.5	-9.1	0.0	0.0	-0.4
	SLN395 ring / End	0.5	0.0	-8.7	0.0	0.0	0.0
C LC_141	SLN105+stub radial (ends 200 past the node) / Begin	0.7	0.0	-11.4	0.0	-1.4	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.7	0.0	-2.1	0.0	0.0	0.0
C LC_142	SLN105+stub radial (ends 200 past the node) / Begin	0.8	-0.1	-11.4	0.0	-1.4	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.8	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.8	0.0	-2.1	0.0	0.0	0.0
C LC_143	SLN105+stub radial (ends 200 past the node) / Begin	0.7	0.0	-11.4	0.0	-1.4	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	0.6	-0.1	-2.2	0.0	0.0	-0.1
	SLN395 ring / End	0.6	0.0	-2.1	0.0	0.0	0.0
C LC_144	SLN105+stub radial (ends 200 past the node) / Begin	0.8	0.0	-11.4	0.0	-1.3	0.1
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	1.0	-0.1	-2.2	0.0	0.0	-0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN395 ring / End	1.0	0.0	-2.1	0.0	0.0	0.0
C LC_145	SLN105+stub radial (ends 200 past the node) / Begin	1.7	0.0	-48.2	0.0	-6.1	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.9	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-3.0	0.0	-8.7	0.0	0.0	0.1
C LC_146	SLN105+stub radial (ends 200 past the node) / Begin	1.8	-0.3	-48.2	0.0	-6.1	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.9	-0.8	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-2.7	0.0	-8.7	0.0	0.0	0.0
C LC_147	SLN105+stub radial (ends 200 past the node) / Begin	1.5	-0.2	-48.3	0.0	-6.2	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.7	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-3.7	0.0	-8.7	0.0	0.0	0.0
C LC_148	SLN105+stub radial (ends 200 past the node) / Begin	2.1	-0.1	-48.2	0.0	-6.0	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-2.0	-0.8	-9.1	0.0	0.0	-0.7
	SLN395 ring / End	-2.0	0.1	-8.7	0.0	0.0	0.1
C LC_151	SLN105+stub radial (ends 200 past the node) / Begin	3.7	-0.1	-48.0	0.0	-8.1	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.8	-0.5	-9.1	0.0	0.0	-0.4
	SLN395 ring / End	-0.8	0.0	-8.7	0.0	0.0	0.0
C LC_152	SLN105+stub radial (ends 200 past the node) / Begin	-3.6	-0.1	-11.3	0.0	2.3	0.5
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-6.0	-0.9	-2.2	0.0	0.0	-0.7
	SLN395 ring / End	-5.9	0.1	-2.1	0.0	0.0	0.2
C LC_153	SLN105+stub radial (ends 200 past the node) / Begin	1.7	-0.1	-47.5	0.0	-5.9	0.4
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-3.0	-0.8	-9.1	0.0	0.0	-0.6
	SLN395 ring / End	-3.0	0.0	-8.7	0.0	0.0	0.1
C LC_154	SLN105+stub radial (ends 200 past the node) / Begin	3.8	-0.1	-48.8	0.0	-8.4	0.3
	SLN105+stub radial (ends 200 past the node) / End	0.0	0.0	0.0	0.0	0.0	0.0
	SLN394 ring / End	-0.8	-0.5	-9.1	0.0	0.0	-0.4
	SLN395 ring / End	-0.8	0.0	-8.7	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	0.0 < 5.0%	OK
Loc. deformation	0.0 < 3%	OK
Bolts	8.0 < 100%	OK
Welds	36.2 < 100%	OK
Buckling	Not calculated	
GMNA	Calculated	

Plates

Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN105+stub radial (ends 200 past the node)-bfl 1	17.7	C LC_120	53.1	0.0	19.2	OK
SLN105+stub radial (ends 200 past the node)-tfl 1	17.7	C LC_122	23.2	0.0	0.0	OK
SLN105+stub radial (ends 200 past the node)-w 1	10.5	C LC_120	64.7	0.0	19.2	OK
SLN85 column below	10.0	C LC_152	84.6	0.0	0.0	OK
SLN394 ring-bfl 1	12.7	C LC_148	17.3	0.0	0.0	OK
SLN394 ring-tfl 1	12.7	C LC_119	18.4	0.0	0.0	OK
SLN394 ring-w 1	7.2	C LC_121	17.4	0.0	0.0	OK
SLN395 ring-bfl 1	12.7	C LC_135	15.8	0.0	0.0	OK
SLN395 ring-tfl 1	12.7	C LC_119	18.3	0.0	0.0	OK
SLN395 ring-w 1	7.2	C LC_121	16.8	0.0	0.0	OK
End plate column below SLN85 column below	12.0	C LC_152	77.0	0.0	12.4	OK
End plate SLN395 ring	20.0	C LC_146	7.6	0.0	4.6	OK
Bottom flange plate SLN395 ring	15.0	C LC_121	12.8	0.0	0.0	OK
Bottom flange half plate L SLN395 ring	15.0	C LC_135	5.0	0.0	0.0	OK
Bottom flange half plate R SLN395 ring	15.0	C LC_135	4.6	0.0	0.0	OK
Haunch web SLN395 ring	12.0	C LC_121	12.9	0.0	0.0	OK
Haunch top plate SLN395 ring	15.0	C LC_146	5.5	0.0	0.0	OK
Haunch sloping plate SLN395 ring	15.0	C LC_121	5.2	0.0	0.0	OK
End plate SLN394 ring	20.0	C LC_146	4.8	0.0	9.0	OK
Bottom flange plate SLN394 ring	15.0	C LC_121	16.4	0.0	0.0	OK
Bottom flange half plate L SLN394 ring	15.0	C LC_152	7.0	0.0	0.0	OK
Bottom flange half plate R SLN394 ring	15.0	C LC_146	6.4	0.0	0.0	OK
Haunch web SLN394 ring	12.0	C LC_121	12.8	0.0	0.0	OK
Haunch top plate SLN394 ring	15.0	C LC_152	5.2	0.0	0.0	OK
Haunch sloping plate SLN394 ring	15.0	C LC_121	5.3	0.0	0.0	OK

Design data

Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

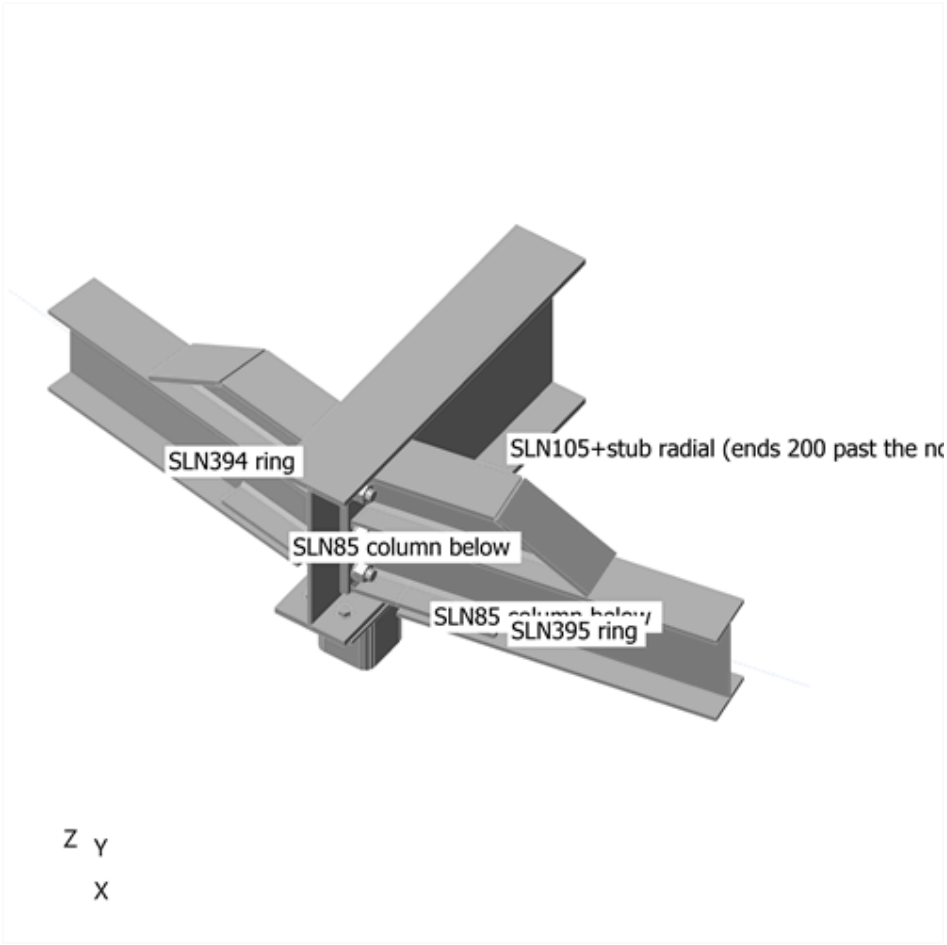
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain

Loc. deformation

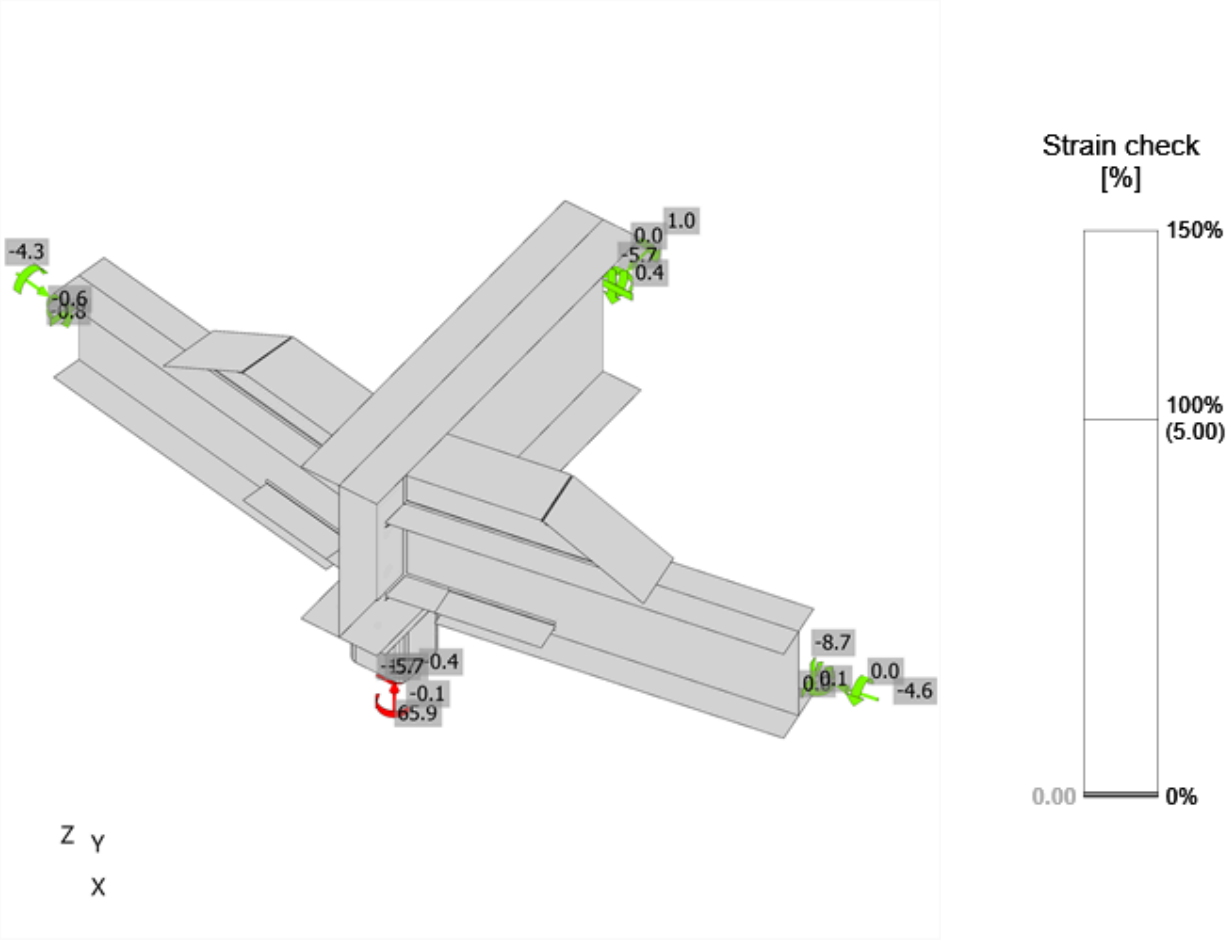
Name	d_0 [mm]	Loads	δ [mm]	δ_{lim} [mm]	δ/d_0 [%]	Check status
SLN85 column below	150	C LC_154	0	5	0.0	OK

Symbol explanation

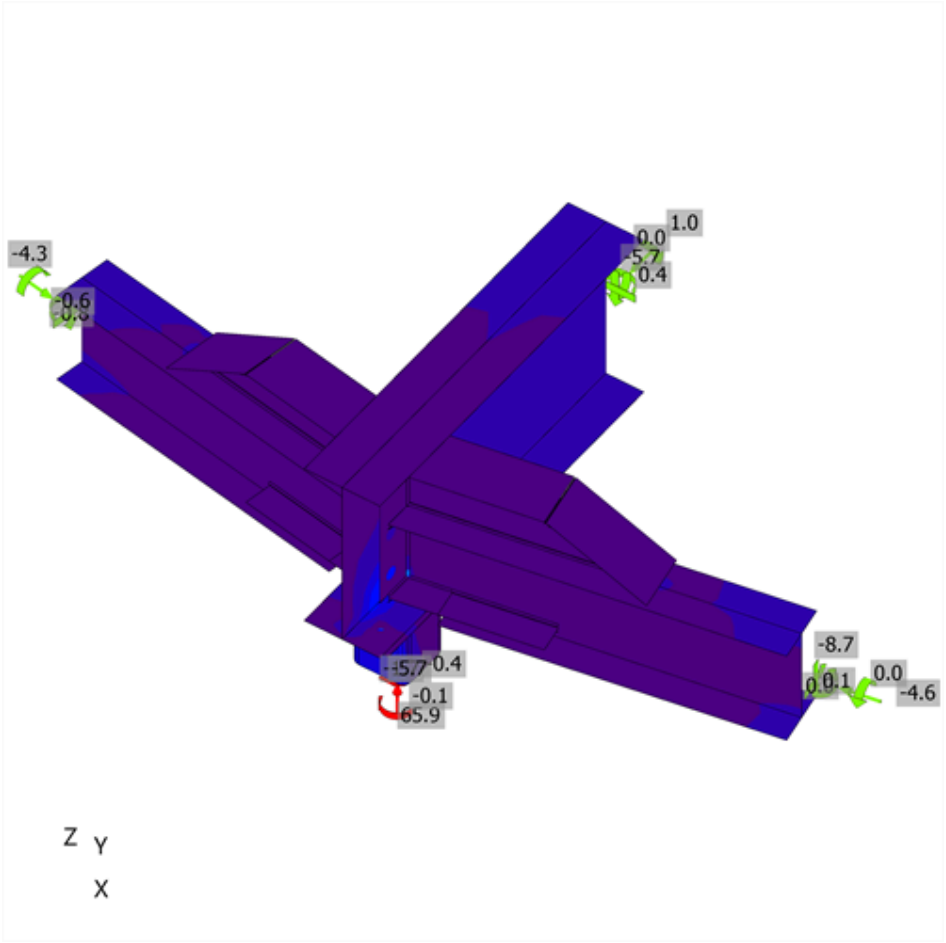
d_0	Cross-section size
δ	Local cross-section deformation
δ_{lim}	Allowed deformation
δ/d_0	Relative deformation



Overall check

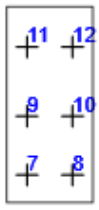


Strain check, C LC_120



Equivalent stress, C LC_120

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M20 8.8 - 1	C LC_152	11.3	1.0	156.4	8.0	1.1	6.8	OK	OK
	B3	M20 8.8 - 1	C LC_120	0.5	0.9	172.0	0.4	0.9	1.2	OK	OK
	B4	M20 8.8 - 1	C LC_154	0.7	0.8	156.4	0.5	0.8	1.2	OK	OK
	B5	M20 8.8 - 1	C LC_152	10.7	0.8	156.4	7.6	0.8	6.2	OK	OK
	B7	M30 8.8 - 2	C LC_122	0.1	0.6	213.9	0.0	0.5	0.3	OK	OK
	B8	M30 8.8 - 2	C LC_122	1.5	1.6	176.5	0.5	1.7	1.1	OK	OK
	B9	M30 8.8 - 2	C LC_135	0.2	0.7	229.5	0.1	0.5	0.3	OK	OK
	B10	M30 8.8 - 2	C LC_135	0.1	0.6	213.9	0.0	0.5	0.3	OK	OK
	B11	M30 8.8 - 2	C LC_122	0.7	0.8	229.5	0.2	0.7	0.5	OK	OK
	B12	M30 8.8 - 2	C LC_146	1.2	0.3	141.5	0.4	0.2	0.4	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M20 8.8 - 1	141.1	247.2	94.1
M30 8.8 - 2	323.1	628.0	215.4

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Bottom flange half plate R SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate L SLN395 ring	Bottom flange plate SLN395 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
Bottom flange half plate R SLN394 ring	Bottom flange plate SLN394 ring	-	70	-	-	-	-	-	-	-	-	OK	OK
		-	70	-	-	-	-	-	-	-	-	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-tfl 1	◢ 10.5	149	C LC_105	10.9	0.0	-5.0	5.5	-0.3	2.7	2.7	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-tfl 1	◢ 10.5	149	C LC_103	8.5	0.0	-3.7	4.3	-1.0	2.1	0.0	OK	OK
End plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	207	C LC_121	21.4	0.0	-2.2	2.0	12.1	5.3	5.3	OK	OK
		◢ 6.0 ◣	208	C LC_135	14.3	0.0	-2.6	3.6	-7.3	3.5	3.5	OK	OK
Bottom flange plate SLN395 ring	SLN395 ring-w 1	◢ 6.0 ◣	100	C LC_154	10.5	0.0	5.9	4.8	1.5	2.6	2.6	OK	OK
		◢ 6.0 ◣	100	C LC_148	14.7	0.0	5.8	-7.7	-1.4	3.6	3.6	OK	OK
SLN395 ring-w 1	Bottom flange half plate L SLN395 ring	◢ 6.0 ◣	243	C LC_135	4.6	0.0	1.0	1.1	2.3	1.1	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	8.0	0.0	2.6	0.8	-4.3	2.0	0.0	OK	OK
SLN395 ring-w 1	Bottom flange half plate R SLN395 ring	◢ 6.0 ◣	243	C LC_135	7.0	0.0	1.8	1.6	-3.6	1.7	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_135	6.5	0.0	1.3	-1.5	3.4	1.6	0.0	OK	OK
SLN395 ring-tfl 1	Haunch web SLN395 ring	◢ 6.0 ◣	572	C LC_121	22.5	0.0	-5.3	-4.6	11.8	5.6	5.6	OK	OK
		◢ 6.0 ◣	572	C LC_121	28.9	0.0	-7.5	8.3	-13.8	7.1	6.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
End plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	208	C LC_135	15.0	0.0	-3.2	-3.8	7.5	3.7	3.7	OK	OK
		◢ 6.0 ◣	207	C LC_119	19.1	0.0	-1.5	-1.9	-10.8	4.7	4.7	OK	OK
Bottom flange plate SLN394 ring	SLN394 ring-w 1	◢ 6.0 ◣	100	C LC_148	13.6	0.0	5.7	6.8	2.0	3.4	3.4	OK	OK
		◢ 6.0 ◣	100	C LC_154	11.3	0.0	5.8	-5.3	-1.9	2.8	2.8	OK	OK
SLN394 ring-w 1	Bottom flange half plate L SLN394 ring	◢ 6.0 ◣	243	C LC_152	5.9	0.0	-0.8	-1.5	-3.0	1.4	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_152	9.9	0.0	-3.2	2.5	4.8	2.4	2.4	OK	OK
SLN394 ring-w 1	Bottom flange half plate R SLN394 ring	◢ 6.0 ◣	243	C LC_146	8.9	0.0	1.6	2.1	-4.6	2.2	0.0	OK	OK
		◢ 6.0 ◣	243	C LC_146	12.1	0.0	3.3	-2.9	6.1	3.0	3.0	OK	OK
SLN394 ring-tfl 1	Haunch web SLN394 ring	◢ 6.0 ◣	572	C LC_121	28.1	0.0	-4.6	-6.4	14.7	7.0	6.2	OK	OK
		◢ 6.0 ◣	572	C LC_121	25.4	0.0	-8.7	6.9	-11.9	6.3	5.8	OK	OK
End plate SLN395 ring	Bottom flange plate SLN395 ring	◢ 6.0 ◣	149	C LC_105	11.4	0.0	-2.3	-4.8	4.2	2.8	2.8	OK	OK
		◢ 6.0 ◣	149	C LC_121	16.8	0.0	-7.4	4.7	-7.3	4.2	4.2	OK	OK
End plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	145	C LC_154	3.5	0.0	-0.2	-0.5	2.0	0.9	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_119	8.7	0.0	-0.8	0.5	-5.0	2.2	0.0	OK	OK
End plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	149	C LC_146	9.0	0.0	-4.5	-2.6	3.7	2.2	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_146	5.9	0.0	-1.1	3.1	-1.4	1.5	0.0	OK	OK
Haunch top plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	342	C LC_146	1.9	0.0	0.5	0.4	1.0	0.5	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_119	1.6	0.0	0.7	-0.6	-0.5	0.4	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
Haunch sloping plate SLN395 ring	Haunch web SLN395 ring	◢ 6.0 ◣	271	C LC_119	17.1	0.0	-2.3	-4.0	9.0	4.2	4.2	OK	OK
		◢ 6.0 ◣	271	C LC_121	24.0	0.0	-5.0	3.9	-13.0	5.9	5.8	OK	OK
End plate SLN394 ring	Bottom flange plate SLN394 ring	◢ 6.0 ◣	149	C LC_152	24.8	0.0	-11.9	-12.3	2.8	6.1	5.7	OK	OK
		◢ 6.0 ◣	149	C LC_121	27.0	0.0	-13.7	12.9	-3.5	6.7	5.8	OK	OK
End plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	145	C LC_119	8.2	0.0	-0.9	-0.8	4.7	2.0	0.0	OK	OK
		◢ 6.0 ◣	145	C LC_121	4.1	0.0	-0.3	0.5	-2.3	1.0	0.0	OK	OK
End plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	149	C LC_146	10.7	0.0	-5.2	-4.7	2.7	2.6	0.0	OK	OK
		◢ 6.0 ◣	149	C LC_121	9.5	0.0	-3.7	4.0	-3.1	2.4	0.0	OK	OK
Haunch top plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	342	C LC_121	3.5	0.0	1.1	0.9	1.7	0.9	0.0	OK	OK
		◢ 6.0 ◣	342	C LC_121	2.8	0.0	-0.7	1.4	0.7	0.7	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch web SLN394 ring	◢ 6.0 ◣	271	C LC_119	20.5	0.0	-3.1	-1.0	11.7	5.1	5.1	OK	OK
		◢ 6.0 ◣	271	C LC_121	23.2	0.0	-4.5	7.1	-11.1	5.7	5.7	OK	OK
End plate column below SLN85 column below	SLN85 column below	◢ 4.2 ◣	521	C LC_122	136.0	0.0	-66.7	-67.9	-8.6	33.6	18.3	OK	OK
		◢ 4.2 ◣	521	C LC_122	146.4	0.0	-73.5	72.4	10.0	36.2	19.5	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN395 ring-w 1	◢ 7.2	208	C LC_154	58.5	0.0	4.1	-6.6	33.0	14.5	9.1	OK	OK
SLN105+stub radial (ends 200 past the node)-w 1	SLN394 ring-w 1	◢ 7.2	208	C LC_154	56.8	0.0	5.9	7.0	-31.9	14.0	8.9	OK	OK
Haunch sloping plate SLN395 ring	Haunch top plate SLN395 ring	◢ 6.0 ◣	146	C LC_121	3.1	0.0	-1.9	1.4	0.1	0.8	0.0	OK	OK
		◢ 6.0 ◣	147	C LC_146	1.2	0.0	-0.2	0.7	-0.2	0.3	0.0	OK	OK
Haunch sloping plate SLN394 ring	Haunch top plate SLN394 ring	◢ 6.0 ◣	146	C LC_121	4.4	0.0	-4.1	-0.9	-0.1	1.3	0.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 6.0 ◣	147	C LC_121	3.1	0.0	-0.5	1.7	-0.5	0.8	0.0	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		

Item	Value	Unit	Reference
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5